

UNIVERSITY OF CALIFORNIA SAN DIEGO

Community Detection Problems on General Random Hypergraphs

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requirements for the degree Doctor of Philosophy

in

Mathematics

by

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University of California San Diego

2025

DEDICATION

Dedicated to my parents

献给我的父亲母亲

EPIGRAPH

History teaches the continuity of the development of science. We know that every age has its own problems, which the following age either solves or casts aside as profitless and replaces by new ones. **If we would obtain an idea of the probable development of mathematical knowledge in the immediate future, we must let the unsettled questions pass before our minds and look over the problems which the science of today sets and whose solution we expect from the future.** To such a review of problems the present day, lying at the meeting of the centuries, seems to me well adapted. For the close of a great epoch not only invites us to look back into the past, but also directs our thoughts to the unknown future.

The deep significance of certain problems for the advance of mathematical science in general and the important role which they play in the work of individual investigator are not to be denied. **As long as a branch of science offers an abundance of problems, so long is it alive; a lack of problems foreshadows extinction or the cessation of independent development.** Just as every human undertaking pursues certain objects, so also mathematical research requires its problems. It is by the solution of problems that the investigator tests the temper of his steel; he finds new methods and new outlooks, and gains a wider and freer horizon.

It is difficult and often impossible to judge the value of a problem correctly in advance; for the final award depends upon the gain which science obtains from the problem. Nevertheless we can ask whether there are general criteria which mark a good mathematical problem. An old French mathematician said: **“A mathematical theory is not to be considered complete until you have made it so clear that you can explain it to the first man whom you meet on the street.”** This clearness and ease of comprehension, here insisted on for a mathematical theory, I should still more demand for a mathematical problem if it is to be perfect; for what is clear and easily comprehended attracts, the complicated repels us.

Moreover, a mathematical problem should be difficult in order to entice us, yet not completely inaccessible, lest it mock at our efforts. **It should be to us a guide post on the mazy paths to hidden truths, and ultimately a reminder of our pleasure in the successful solution.**

— David Hilbert, *Mathematical Problems* [125, 126, 124]¹

— Lecture delivered before the International Congress of Mathematicians at Paris in 1900

¹Originally delivered in German as “Mathematische Probleme”. Translated by Mary Winston Newson.

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Chapter 3, in full, is a reprint of the material “Ioana Dumitriu and Hai-Xiao Wang. Optimal and exact recovery on the general non-uniform hypergraph stochastic block model. In *arXiv preprint* (2023). arXiv: 2304.13139. To appear in *Annals of Statistics*”. The dissertation author is the co-author of this paper.

Chapter 4, in full, is a reprint of the material “Ioana Dumitriu, Hai-Xiao Wang and Yizhe Zhu. Partial recovery and weak consistency in the non-uniform hypergraph stochastic block model, *Combinatorics, Probability and Computing* 34.1 (2025): pp. 1-51. arXiv: 2112.11671”. The dissertation author is the co-author of this paper.

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Hai-Xiao Wang. Information-theoretic limits and strong consistency on the binary non-uniform hypergraph stochastic block model. In: *arXiv preprint* (2023). arXiv: 2306.06845.

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ABSTRACT OF THE DISSERTATION

Community Detection Problems on General Random Hypergraphs

by

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This dissertation concerns the community detection problems under the *Hypergraph Stochastic Block Model* (HSBM), where the sizes of edges may vary, and each edge appears independently with some given probability, purely determined by the labels of its vertices.

For regimes where the expected degrees grow with the number of vertices, for the first time in the literature, we prove a wide-ranging, *Information-Theoretic* (IT) lower bound on the number of misclassified vertices *for any algorithm*, where the bound is characterized by the

generalized *Chernoff-Hellinger* divergence involving model parameters. Besides that, when the expected degrees grow logarithmically, we establish a sharp threshold for exact recovery for the non-uniform, multiple-community setting, subject to only minor constraints. A key insight reveals that aggregating information across uniform layers enables exact recovery even in cases where this is impossible if each layer were considered alone. We present two efficient algorithms, for minimal and full information scenarios, which successfully achieve exact recovery when above the threshold, and attain the lowest possible mismatch ratio when below the threshold where exact recovery is impossible, confirming their optimality.

In the regime with bounded expected degrees, we develop a spectral algorithm that achieves partial recovery, correctly classifying a constant fraction of vertices. This fraction is determined by the *Signal-to-Noise Ratio* (SNR) of the model, and approaches 1 as the SNR grows slowly with the number of vertices. Our algorithm employs a three-stage approach: first, preprocessing the hypergraph to maximize SNR; second, applying spectral methods to obtain an initial partition; and finally, implementing tensor-based refinement to enhance classification accuracy.

Additionally, we provide novel concentration results for adjacency matrices of sparse random hypergraphs, serving as the foundations of the theoretical analysis of our algorithms, which could be of independent interest. We also address several open problems concerning incomplete model information and parameter estimations.

Chapter 1

Introduction

1.1 Random graphs models and sparse random matrices

Random graph models are powerful tools for studying complex systems with inherent randomness and underlying structure [39]. By capturing the probabilistic nature of connections between entities, these models enable rigorous analysis of emergent properties in large networks, including connectivity patterns and community structures. They serve as both mathematical objects for rigorous study, and practical tools for developing algorithms being able to extract valuable insights from intricate networked datasets.

The study of random graphs has a long history, dating back to the 1950s when Erdős and Rényi introduced the *Erdős-Rényi* graph [98]. Initially, it was introduced to demonstrate the existence of graphs with specific properties. Since then, its simple but mathematically rich structure made it an ideal starting point for exploring more complex random graph models. In terms of applications, it has provided a bevy of theoretical uses in computer science, physics, and engineering: as the stepping stone or building block model in the study of networks, as an expander, as a basic paradigm for studying random matrices.

Meanwhile, in real-world applications, large matrices encountered in science, engineering, and mathematics are often predominantly zero, or *sparse*. The study of sparse matrices resides at the intersection of numerical linear algebra, graph theory and probability theory, which are bridged together by the Erdős-Rényi graph model. Sparse matrices has gained significant

attention in recent years due to their applications in various fields, including and network science [170, 185, 186], machine learning [144, 24] and sparse Transformers [62]. Theoretical techniques that were developed for studying sparse random matrices have been widely used to analyze the structure of random graphs and provide the theoretical foundation for applications in theoretical computer science, statistics and data science, including community detection [1], matrix completion [145, 184], matrix sketching [205], error-correcting codes [129], etc. Many efficient algorithms that achieve the information-theoretical limits are spectral methods [92, 4, 3], and their analysis is based on random matrix theory [163, 44].

1.1.1 Random graph models

Erdős and Rényi graph and its bipartite variant

Let $\mathcal{G}(N, p)$ denote the Erdős-Rényi graph with N vertices, where each edge $\{i, j\} \subset [N]$ with $i \neq j$ is included independently with probability p . Let $\mathbf{A} \in \{0, 1\}^{N \times N}$ denote the adjacency matrix of $\mathcal{G}(N, p)$, which is symmetric by construction and the upper half entries are independent and identically distributed Bernoulli random variables with expectation p , i.e., $\mathbb{P}(\mathbf{A}_{ij} = 1) = p$ for $1 \leq i < j \leq N$. It becomes sparse when p vanishes as N grows. The spectral properties of the adjacency matrix $\mathbf{A}$ (as well as its Laplacian operator, both scaled and combinatorial) are particularly important, since they provide insights into the structure and behavior of the graph, which can be used to identify communities and clusters within the graph.

Let $\mathbf{H} := (\mathbf{A} - \mathbb{E}\mathbf{A})/\sqrt{pN}$ be the centered and scaled¹ adjacency matrix, with $\lambda_1(\mathbf{H}) \geq \dots \geq \lambda_N(\mathbf{H})$ denoting its eigenvalues. It has long been known that the *Empirical Spectral Distribution* (ESD) of $\mathbf{H}$, defined as $\mu_N(\mathbf{H}) = N^{-1} \sum_{j=1}^N \delta_{\lambda_j(\mathbf{H})}$, converges to the *semicircle law* μ_{SC} [204] if $Np(1-p) \rightarrow \infty$ [81, Corollary 1.2] as shown in Figure 1.1 (a), where the probability

¹The bulk of the eigenvalues lies in an interval of order 1 after this scaling.

²This condition can't be relaxed to $Np \rightarrow \infty$. For example, when $p = 1$, the adjacency matrix has an eigenvalue $N - 1$ with multiplicity 1, and another eigenvalue -1 with multiplicity $N - 1$.

measure μ_{SC} is given by

$$d\mu_{\text{SC}}(x) = \frac{1}{2\pi} \sqrt{4 - x^2} \mathbb{1}_{\{x \in [-2, 2]\}} dx \quad (1.1.1)$$

Besides the global convergence of the ESD, other spectral properties of $\mathbf{H}$ (as well as its Laplacian operators [50, 81, 139, 131]) have also been studied in the literature, including the *local law* [189, 188, 95, 94, 194, 121], *edge statistics* [32, 33, 133, 19, 192, 18, 123], *universality* [96, 189, 189, 132, 134] and *localization-delocalization* transition of eigenvectors [194, 87, 16, 20, 35, 17]. Readers may refer to [21, 26, 187, 34, 97, 117] for a more detailed review of the methods and results in random matrix theory and random graphs.

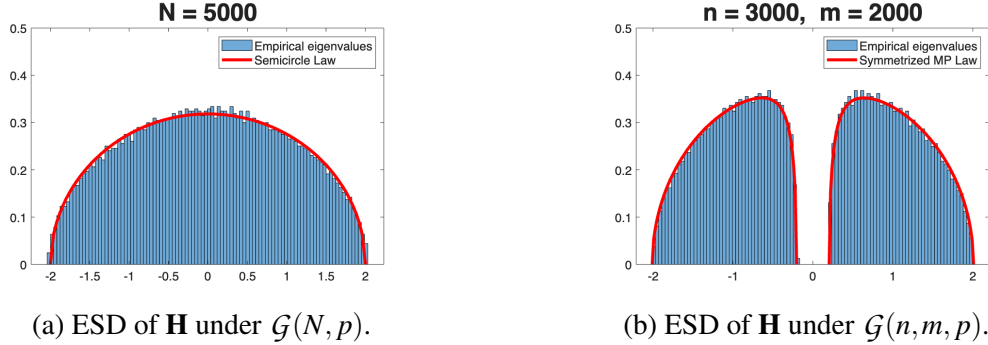


Figure 1.1. The left plot (a) shows the semicircle law (1.1.1), while the right plot (b) shows a symmetrized version of the MP law.

Another frequently studied model is the *bipartite* variant of the Erdős-Rényi graph, denoted as $\mathcal{G}(n, m, p)$, where the vertex set $\mathcal{V} = [N]$ is a disjoint union of $\mathcal{V}_1 = \{1, \dots, n\} = [n]$ and $\mathcal{V}_2 = \{n+1, \dots, n+m\} = [N] \setminus [n]$ with $N = n + m$. Each edge of $\mathcal{G}(n, m, p)$, containing one vertex in $\mathcal{V}_1$ and the other one in $\mathcal{V}_2$, is sampled with probability p , independently of all other edges. Let $\tilde{\mathbf{A}} \in \{0, 1\}^{n \times m}$ denote the *bi-adjacency* matrix of the $\mathcal{G}(n, m, p)$, where $\tilde{A}_{jk} \sim \text{Bernoulli}(p)$ independently for each $j \in \mathcal{V}_1$ and $k \in \mathcal{V}_2$, then the adjacency matrix admits

the following block structure

$$\mathbf{A} := \begin{bmatrix} \mathbf{0} & \tilde{\mathbf{A}} \\ \tilde{\mathbf{A}}^\top & \mathbf{0} \end{bmatrix}. \quad (1.1.2)$$

We denote $\mathbf{X} = (\tilde{\mathbf{A}} - \mathbb{E}\tilde{\mathbf{A}})/\sqrt{pn}$. The *centered and scaled adjacency* matrix admits a block structure as well:

$$\mathbf{H} := \frac{1}{\sqrt{d}}(\mathbf{A} - \mathbb{E}\mathbf{A}) = \begin{bmatrix} \mathbf{0} & \mathbf{X} \\ \mathbf{X}^\top & \mathbf{0} \end{bmatrix}. \quad (1.1.3)$$

One can verify that $\mathbf{H}$ has a zero eigenvalue of multiplicity at least $n - m$, and that the rest of the eigenvalues of $\mathbf{H}$ come in pairs $\pm\lambda$, with λ being a singular value of $\mathbf{X}$. Let $\mu_m(\mathbf{X}^\top \mathbf{X}) = m^{-1} \sum_{j=1}^m \delta_{\lambda_j(\mathbf{X}^\top \mathbf{X})}$ denote the ESD of the *Wishart* matrix $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{m \times m}$, where $\lambda_1(\mathbf{X}^\top \mathbf{X}) \geq \dots \geq \lambda_m(\mathbf{X}^\top \mathbf{X})$ are the eigenvalues. Suppose that $n/m = \gamma$ for some $\gamma \geq 1$ as $n, m \rightarrow \infty$, the ESD of $\mathbf{H}$ will converge weakly to the *Marčenko-Pastur* (MP) law μ_{MP} [162, 193, 172], where

$$d\mu_{\text{MP}}(x) = \frac{\gamma \sqrt{(\lambda_+ - x)(x - \lambda_-)}}{2\pi x} \mathbb{1}_{\{x \in [\lambda_-, \lambda_+]\}} dx, \quad (1.1.4)$$

and $\lambda_+ = (1 + \gamma^{-1/2})^2$ and $\lambda_- = (1 - \gamma^{-1/2})^2$ denote the right and left edges of the support, respectively. It is a quick linear algebra / calculus exercise to check that the ESD of $\mathbf{H}$ converges to a symmetrized version of the MP law as shown in Figure 1.1 (b). Notably, The singular values of $\mathbf{X}$ are the square roots of the eigenvalues of the positive semi-definite matrix $\mathbf{X}^\top \mathbf{X}$ and the nonnegative paired eigenvalues of $\mathbf{H}$.

An intriguing feature of the Wishart matrix, setting it apart from the Wigner matrix, is the behavior of its least singular values when the aspect ratio $\gamma \neq 1$. It is worth noting that both ESDs in Figure 1.1 exhibit symmetry about the origin. In essence, analyzing the right edge behavior of the MP law (1.1.4) bears similarities to examining the edge behavior of the semicircle law (1.1.1).

However, the left edge $\lambda_- = (1 - \gamma^{-1/2})^2$ of the MP law (1.1.4) is bounded away from the origin when $\gamma \neq 1$. The gap between 0 and λ_- , introduced by the rectangular structure of Wishart matrices, is absent when analyzing the edge behavior of the Wigner matrices. Besides that, the least singular value plays an important role in many applications, including numerical linear algebra, machine learning, and statistics. It can be used to assess the stability and robustness of algorithms, as well as to identify potential issues with data quality and structure. A line of research [48, 86, 84] concerns the emergence of the left outliers (smaller than the left edge λ_-) of bipartite Erdős-Rényi graphs at different regimes of sparsity, which is phenomena that is not present in the Wigner matrices.

Stochastic block model

The *Stochastic Block Model* (SBM) is a natural generalization of the Erdős-Rényi graph, where vertices are densely connected within each community but sparsely connected across different communities (*assortative* case). We introduce the formal definition below.

Definition 1.1.1 (Stochastic Block Model). *Let N and K be the number of vertices and communities respectively, where the vertex $\mathcal{V} = \mathcal{V}_1 \cup \dots \cup \mathcal{V}_K$ is a union of disjoint sets of vertices. There is a symmetric matrix $\mathbf{Q} \in [0, 1]^{K \times K}$ containing the probabilities of sampling edges, that is, $\mathbb{P}(A_{uv} = 1) = Q_{jk}$ for any pair of distinct vertices $u \in \mathcal{V}_j$ and $v \in \mathcal{V}_k$ with $(j, k) \in [K] \times [K]$.*

SBM was first introduced in the pioneering work [128] for sociology research. Over the past several decades, it has been extensively studied in [52, 42, 89, 182, 70, 164, 37, 68, 38, 178, 66, 23, 196], driven by exploring the phase transition behaviors in various connectivity regimes. In the regime where the expected degrees grow logarithmically with respect to the number of vertices, a major breakthrough was the establishment of the exact recovery thresholds for binary [2, 169] and multi-block case [5, 212, 9], where both necessary and sufficient conditions were provided. For the case where the expected degrees are of constant order, phase transition behavior for detection was discovered in [76, 75, 73], and later it was connected to the *Kesten-Stigum*

(KS) threshold, with the necessity [170] and sufficiency [163, 167] rigorously proved for the binary block case. For the multi-block case, it was shown that a spectral clustering algorithm based on the non-backtracking operator [44], and the acyclic belief propagation method [6], succeed all the way down to the KS threshold, proving a conjecture in [147]. The necessity for $K = 3, 4$ was partly established in [171]. However, the threshold is not known when the number of communities is more than 4, although it is known to be below the KS threshold [28, 6, 171].

The literature abounds with different methods of approach, like spectral methods [164, 68, 206, 211, 212, 64, 153, 197, 154, 4], sphere comparison [5, 8], as well as *semidefinite programming* (SDP) [116, 118, 136, 137, 166, 9, 177, 27]. Recently, some more general variants of SBM have been introduced, such as degree-corrected SBM [142, 105, 10, 141], contextual SBM [79, 159, 3], labeled SBM [122, 206, 211, 212], and multilayer SBM [57, 10, 161]. Readers may refer to [1] for a more detailed review.

Inhomogeneous Erdős-Rényi random graphs

The *inhomogeneous Erdős and Rényi* random graph $\mathcal{G}(N, [P_{ij}]_{i,j \in [N]})$, where edges are sampled independently with given probabilities $P_{ij} \in [0, 1]$ for each pair of vertices $i \neq j$, is a more flexible model that allows for varying connection probabilities among vertices, making it suitable for modeling real-world networks with heterogeneous structures. Many popular graph models arise as special cases of the inhomogeneous Erdős-Rényi random graph $\mathcal{G}(N, [P_{ij}]_{i,j \in [N]})$. These include the Stochastic Block Model [128], random graphs with given expected degrees [67], and W-random graphs [158, 55].

The study of inhomogeneous random graphs has attracted attention across multiple disciplines. In random matrix theory, researchers have investigated the limiting spectral distribution and extensions of universality phenomena beyond the classical Wigner matrices [14, 54, 220]. In random graph theory, these models have been central to developing the theory of graph limits and generalizing results beyond the homogeneous Erdős-Rényi case [40, 157, 46, 45]. Furthermore, in statistics and machine learning, these models have provided theoretical

foundations for community detection algorithms and network analysis techniques [1, 103].

1.1.2 Random hypergraph models

Hypergraph and tensors

Graphs are usually used to depict pairwise relationships among data. Hypergraphs, as a generalization of graphs, have gained increasing attention in recent years due to their ability to capture higher-order interactions among vertices [36, 31]. They are particularly useful in modeling complex systems where relationships involve more than two entities, including but not limited to biological networks [165, 190] with multi-way interactions, as well as recommendation systems [51, 155] with multiple users and items. Empirically, they have been shown to be better than graph models [218]. Besides that, hypergraphs and their spectral theory found applications in data science [135, 219, 120], combinatorics [102, 183, 88] and statistical physics [53, 180]. The study of random hypergraphs has led to new insights into the structure and dynamics of complex systems, as well as the development of novel algorithms for community detection and clustering.

Definition 1.1.2 (Hypergraph). *A hypergraph $\mathcal{H}$ is a pair $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{E}$ denotes the set of non-empty subsets of vertex set $\mathcal{V}$. We say that $\mathcal{H}$ is ℓ -uniform if every $e \in \mathcal{E}$ is an ℓ -subset of $\mathcal{V}$. The degree of a vertex $v \in \mathcal{V}$ is the number of hyperedges in $\mathcal{E}$ containing v .*

A faithful representation of an ℓ -uniform hypergraph is to associate it to a tensor.

Definition 1.1.3 (Adjacency tensor). *One can associate an ℓ -uniform hypergraph $\mathcal{H}_\ell = (\mathcal{V}, \mathcal{E}_\ell)$ to an order- ℓ symmetric tensor $\mathcal{A}^{(\ell)}$, where the tensor entry $\mathcal{A}_e^{(\ell)}$ denotes the presence of some ℓ -hyperedge e , i.e., $\mathcal{A}_e^{(\ell)} := \mathcal{A}_{i_1, \dots, i_\ell}^{(\ell)} = \mathbb{1}_{\{e \in \mathcal{E}_\ell\}}$ for $i_1, \dots, i_\ell \in \mathcal{V}$, and $\mathcal{A}_{i_1, \dots, i_\ell}^{(\ell)} = \mathcal{A}_{i_{\pi(1)}, \dots, i_{\pi(\ell)}}^{(\ell)}$ for any permutation π on $[\ell]$.*

Hypergraph stochastic block model

The *Hypergraph Stochastic Block Model* (HSBM), as a generalization of graph SBM, was first introduced in [109], where each edge of the uniform hypergraph appears independently

with some given probability. The non-uniform HSBM was first studied in [108], which can be treated as a superposition of several uniform HSBMs with different model parameters. It is a more realistic model to study higher-order interaction on networks. We introduce the rigorous definition of uniform HSBM first, and extend it to non-uniform hypergraphs.

Let $N, K \in \mathbb{N}_+$ denote the number of vertices and communities, respectively. Let the random variable Y_v denote the membership assignment of v . The vertex set $\mathcal{V}$ is partitioned into classes $\mathcal{V}_1, \dots, \mathcal{V}_K$ probabilistically according to a *probability vector* $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_K)$ with $\|\boldsymbol{\alpha}\|_1 = 1$, i.e., $\mathbb{P}(v \in \mathcal{V}_k) = \mathbb{P}(Y_v = k) = \alpha_k$ with $\mathcal{V}_k := \{v \in \mathcal{V} \mid Y_v = k\}$ for each $k \in [K]$. We focus on the scenario where the sizes of different communities are comparable.

Assumption 1.1.4 (Comparable sizes). *Assume that $\alpha_1 \geq \dots \geq \alpha_K \geq C$ for some universal constant $C \in (0, 1)$, i.e., no entry of $\boldsymbol{\alpha}$ vanishes.*

Assumption 1.1.4 is adopted throughout the dissertation. We first introduce the uniform HSBM, which is characterized by a partition of vertices into different classes, such that all vertices in a given class are interchangeable.

Definition 1.1.5 (Uniform HSBM). *Let $N, K, \boldsymbol{\alpha}$ be defined as above, and let each entry in the membership vector $\mathbf{y}$ be sampled independently under $\boldsymbol{\alpha}$. Let $Q^{(\ell)} \in (\mathbb{R}^K)^{\otimes \ell}$ be an order- ℓ symmetric tensor such that $Q_{\mathbf{y}_1, \dots, \mathbf{y}_\ell}^{(\ell)} = Q_{\mathbf{y}_{\pi(1)}, \dots, \mathbf{y}_{\pi(\ell)}}^{(\ell)}$ for any permutation π on $[\ell]$. Each possible edge $e = \{i_1, \dots, i_\ell\}$ is generated with probability $\mathbb{P}(\mathcal{A}_e^{(\ell)} = 1) = Q_{\mathbf{y}(e)}^{(\ell)}$, where $\mathbf{y}(e) = \{y_{i_1}, \dots, y_{i_\ell}\}$ represents the membership sequence of edge e . We denote this distribution on the set of ℓ -uniform hypergraphs by*

$$(\mathbf{y}, \mathcal{H}_\ell) \sim \text{HSBM}_\ell(N, \boldsymbol{\alpha}, Q^{(\ell)}), \quad \ell \in \mathcal{L}. \quad (1.1.5)$$

Due to the symmetry of $Q^{(\ell)}$ shown in Figure 1.2, the probability of an edge $e \in \mathcal{E}$ being present will depend only on the *membership counts* of $\mathbf{y}(e)$, denoted by $\mathbf{w} = \mathbf{w}(e) = (w_1, \dots, w_K)$, i.e., w_k nodes belong to $\mathcal{V}_k$, making $\mathbf{w}$ a *weak composition* of ℓ .

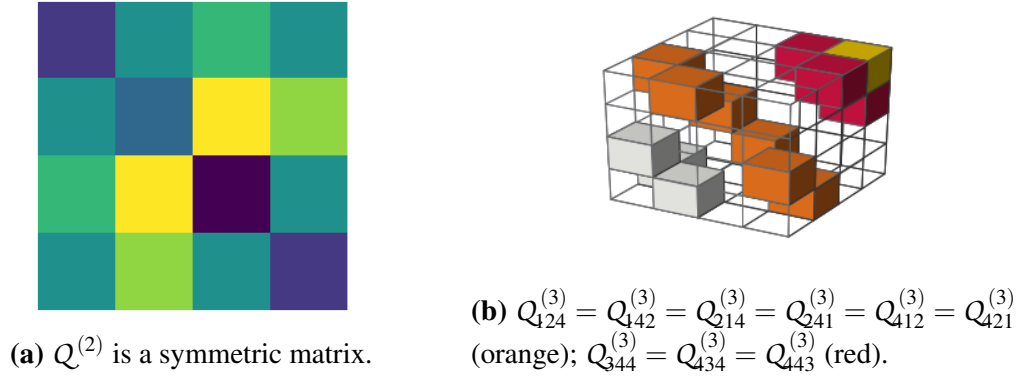


Figure 1.2. Symmetry of the two probability tensors $Q^{(2)}$ and $Q^{(3)}$.

Definition 1.1.6 (Weak composition [41]). A sequence of non-negative integers $\mathbf{w} := (w_1, \dots, w_K)$, fulfilling $\sum_{k=1}^K w_k = \ell$, is called a weak composition of ℓ . Let $\mathcal{WC}_K^{(\ell)}$ denote the set of weak compositions of ℓ into K parts, then the cardinality is $|\mathcal{WC}_K^{(\ell)}| = \binom{\ell+K-1}{K-1}$.

One can build $Q^{(\ell)}$ from the set of probabilities $\{Q_{\mathbf{w}}^{(\ell)}\}_{\mathbf{w} \in \mathcal{WC}_K^{(\ell)}}$ for each $\ell \in \mathcal{L}$, since the probability of an edge, by symmetry, depends only on its weak composition formed by membership counts. We will use $Q^{(\ell)}$ and $Q_{\mathbf{w}}^{(\ell)}$ alternately to facilitate presentation.

Remark 1.1.7. For an ℓ -hyperedge $e = \{i_1, \dots, i_\ell\}$, let w_k denote the number of vertices in e with $Y_v = k$, i.e., $w_k = |\{v \in e \mid Y_v = k\}|$, then the sequence $\mathbf{w}(e) := (w_1, \dots, w_K)$ constitutes a weak composition of ℓ . Due to the symmetry of the probability tensor, i.e., $Q_{\mathbf{y}_1, \dots, \mathbf{y}_\ell}^{(\ell)} = Q_{\mathbf{y}_{\pi(1)}, \dots, \mathbf{y}_{\pi(\ell)}}^{(\ell)}$ for any $\pi \in S_\ell$, there exists a bijection, proved by counting, between $\mathbf{w}(e)$ and $\mathbf{y}(e)$. Recall that ℓ -hyperedges e_1 and e_2 share the same generating probability if $\mathbf{y}(e_1) = \mathbf{y}(e_2)$, then the argument holds as well if $\mathbf{w}(e_1) = \mathbf{w}(e_2)$. Therefore, the set of ℓ -hyperedges can be classified into different categories according to the weak compositions constituted from nodes in each edge.

The non-uniform HSBM can be built from uniform ones as illustrated in [108].

Definition 1.1.8 (Non-uniform HSBM). Let $\mathcal{L} = \{\ell \mid \ell \geq 2, \ell \in \mathbb{N}\}$ be a set of integers with finite cardinality. The membership vector $\mathbf{y}$ is first sampled under $\boldsymbol{\alpha}$, then for each $\ell \in \mathcal{L}$, $\mathcal{H}_\ell$ is independently drawn from $\text{HSBM}_\ell(N, \boldsymbol{\alpha}, Q^{(\ell)})$. The non-uniform hypergraph $\mathcal{H}$ is a collection of ℓ -uniform hypergraphs, i.e., $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$.

Examples of 2-uniform and 3-uniform HSBM, and an example of non-uniform HSBM with $\mathcal{L} = \{2, 3\}$ and $K = 4$ can be seen in Figure 1.3 (a), Figure 1.3 (b) and Figure 1.3 (c) respectively.

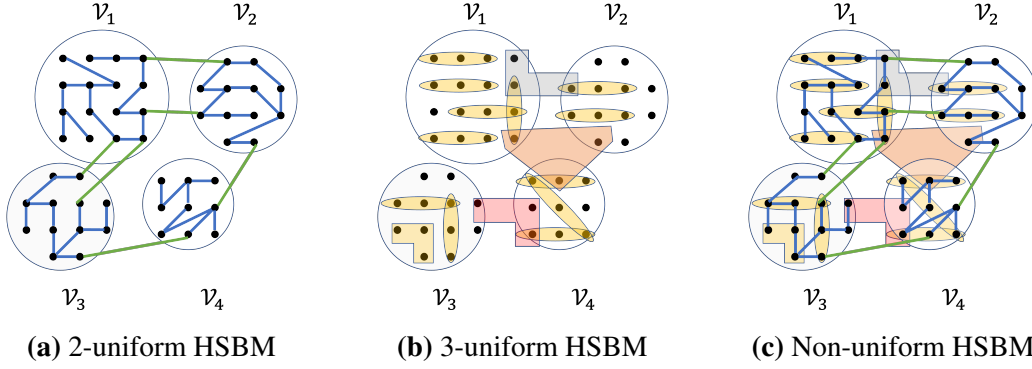


Figure 1.3. Uniform HSBM and non-uniform HSBM.

Most of the computations involving tensors are NP-hard [127]. Instead, our analysis is based on the following key concept, the *adjacency matrix*.

Definition 1.1.9 (Adjacency matrix). *For the non-uniform hypergraph $\mathcal{H}$, let $\mathcal{A}^{(\ell)}$ be the order- ℓ adjacency tensor corresponding to each underlying ℓ -uniform hypergraph for $\ell \in \mathcal{L}$. The adjacency matrix $\mathbf{A} := [\mathbf{A}_{ij}]_{N \times N}$ of $\mathcal{H}$ is defined by*

$$\mathbf{A}_{ij} := \mathbb{1}_{\{i \neq j\}} \cdot \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell, e \supset \{i, j\}} \mathcal{A}_e^{(\ell)}, \quad (1.1.6)$$

where $\mathbf{A}_{ii} = 0$ for $i \in \mathcal{V}$ since in our model each edge of size ℓ contains ℓ distinct vertices.

Inhomogeneous Erdős-Rényi random hypergraph

The *inhomogeneous Erdős and Rényi* random hypergraph is a natural generalization of standard random hypergraph models, where each hyperedge is included independently with probabilities that can vary across different vertex subsets. This flexibility enables more accurate modeling of complex networks with heterogeneous interaction patterns among vertices.

Definition 1.1.10 (Inhomogeneous Erdős-Rényi hypergraph). *Let $Q^{(\ell)} \in ([0, 1]^N)^{\otimes \ell}$ be a symmetric probability tensor, i.e., $Q_{i_1, \dots, i_\ell} = Q_{i_{\pi(1)}, \dots, i_{\pi(\ell)}}$ for any permutation π on $[\ell]$, where $\ell \geq 2$ is some finite integer. Let $\mathcal{H}_\ell = (\mathcal{V}, \mathcal{E}_\ell)$ denote inhomogeneous ℓ -uniform Erdős-Rényi hypergraph associated with the probability tensor $Q^{(\ell)}$. Let $\mathcal{A}^{(\ell)}$ denote the adjacency tensor of $\mathcal{H}_\ell$, where each ℓ -hyperedge $e = \{i_1, \dots, i_\ell\} \subset \mathcal{V}$ appears with probability $\mathbb{P}(\mathcal{A}_e^{(\ell)} = 1) = Q_{i_1, \dots, i_\ell}^{(\ell)}$. Let $\mathcal{L} = \{\ell \mid \ell \geq 2, \ell \in \mathbb{N}\}$ be a finite set of integers, where L denotes its maximum element. An inhomogeneous non-uniform Erdős-Rényi hypergraph is the union of uniform ones, i.e., $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$.*

Concentration inequalities of the adjacency matrices of inhomogeneous Erdős-Rényi hypergraphs are established in Chapter 3 and Chapter 4, which could be of independent interest.

1.1.3 Challenges of sparse random matrices and tensors

Concentration results for the spectral norm of adjacency matrices serve as the foundation of the theoretical analysis of our algorithms. However, the convergence of ESDs does not say anything about the convergence of extremal eigenvalues to the edges of the limiting distributions, both semicircle (1.1.1) and MP law (1.1.4), especially in the sparse regime. The analysis of sparse random matrices presents several challenges, primarily due to the inherent randomness and sparsity of the data.

Slower decay of moments

We first review the classical results on the convergence of extremal spectral statistics. Consider $\mathbf{X} = \mathbf{Y}/\sqrt{n}$, where the entries of $\mathbf{Y} \in \mathbb{R}^{n \times m}$ are independent and identically distributed with mean 0 and variance 1. In [207, 25, 26], the Bai-Yin Theorem states that the largest eigenvalue $\lambda_1(\mathbf{X}^\top \mathbf{X})$ converges to the right edge λ_+ of the MP law when the matrix entries of $\mathbf{Y}$ have finite fourth moments. In [191], Tikhomirov proved that the smallest eigenvalue $\lambda_m(\mathbf{X}^\top \mathbf{X})$ converges to the left edge λ_- when the entries of $\mathbf{Y}$ have finite second moments.

Consider the bipartite Erdős-Rényi random graph $\tilde{\mathbf{A}} \sim \mathcal{G}(n, m, p)$ and let $\mathbf{X} = (\tilde{\mathbf{A}} -$

$\mathbb{E}\tilde{\mathbf{A}})/\sqrt{pn}$ where the k -th moment ($k \geq 2$) of a matrix entry decays as $n^{-1}(pn)^{-(k-2)/2}$. In the dense regime where p is a constant in $(0, 1)$, the moments decay as $n^{-k/2}$, matching the rate observed in the previous case. This behavior aligns with that of Wishart (or Wigner, in the case of $\mathcal{G}(N, p)$) random matrices with similar properties. However, in the sparse regime where $p \ll 1$, the decay in k becomes significantly slower, causing all moments higher than 2 to diverge. Consequently, classical results such as the Bai-Yin Theorem and the Tikhomirov Theorem are not directly applicable to sparse bipartite Erdős-Rényi model, presenting a key challenge in their analysis. Further work is needed to extend the outlier picture to sparse matrices (including adjacency matrices of Erdős-Rényi graphs), and the last few years have seen a flurry of results which, taken together, paint an almost complete picture.

Dependence on the matrix entries

The independence assumption plays a crucial role in many classical random matrix theory results. However, one of the key challenges in analyzing the spectral properties of $\mathbf{A}$ (1.1.6) is that its upper half entries are not independent. The entries of $\mathbf{A}$ are determined by the presence or absence of hyperedges, which are themselves dependent on the underlying hypergraph structure. This dependence complicates the analysis and requires new techniques to establish concentration inequalities and limit theorems for the spectral norm of $\mathbf{A}$.

To that end, we follow the Kahn-Szemerédi approach [101, 153] and leverage the Bennett-type tail estimates developed in [71] for the study of the second largest eigenvalue (in absolute value) of random d -regular graphs [101, 43], which allows us to get rid of the dependence on the matrix entries.

1.2 Community detection problems

1.2.1 Various regimes of recovery

The task of *community detection*, or *clustering*, consists in partitioning the vertices of a graph into groups that are similarly connected [1]. It has become one of the central problems

in network analysis and machine learning [181, 173, 174, 23]. Random graph models, which generate community structure with a specified ground truth, have been frequently employed in many clustering algorithms, for benchmarking and proving theoretical guarantees. Mathematically, let $G = (\mathcal{V}, \mathcal{E})$ be a graph, where the vertex set $\mathcal{V} = [N]$ is composed of K disjoint blocks, i.e., $\mathcal{V} = \cup_{k=1}^K \mathcal{V}_k$. The proportion of each block is denoted by $\alpha_k = |\mathcal{V}_k|/|\mathcal{V}|$ and we define the vector $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_K)$ with $\|\boldsymbol{\alpha}\|_1 = 1$. Let $\mathbf{y} \in [K]^N$ denote the membership vector of the vertices, i.e., $y(v) = y_v = k$ if the vertex v belongs to block $\mathcal{V}_k$. Let $\hat{\mathbf{y}}$ denote some estimation of $\mathbf{y}$. To evaluate the accuracy of $\hat{\mathbf{y}}$, we define the *mismatch* ratio, counting the proportion of incorrectly clustered nodes:

$$\eta_N := \eta_N(\mathbf{y}, \hat{\mathbf{y}}) = N^{-1} \inf_{\pi \in \mathcal{S}_K} \mathbb{D}_{\text{HD}}(\pi \circ \mathbf{y}, \hat{\mathbf{y}}), \quad (1.2.1)$$

where the *Hamming distance* $\mathbb{D}_{\text{HD}}(\mathbf{y}, \hat{\mathbf{y}})$ counts the number of entries having different values in $\mathbf{y}$ and $\hat{\mathbf{y}}$. Here, $\pi \circ \mathbf{y}$, defined by $(\pi \circ \mathbf{y})(v) = \pi(\mathbf{y}(v))$ entrywisely, is the same assignment as $\mathbf{y}$ up to some permutation π , and $\mathcal{S}_K$ denotes the group of all permutations on $[K]$. Note that a random guess estimator has expected³ accuracy $\|\boldsymbol{\alpha}\|_2^2$, thus the output $\hat{\mathbf{y}}$ is meaningful only if $\eta_N \leq 1 - \|\boldsymbol{\alpha}\|_2^2$. Then, the recovery problem can be divided into several different regimes according to η_N [1].

1. Exact recovery (strong consistency): $\mathbb{P}(\eta_N = 0) \geq 1 - o(1)$.
2. Almost exact recovery (weak consistency): $\mathbb{P}(\eta_N = o(1)) \geq 1 - o(1)$.
3. Partial recovery: $\mathbb{P}(\eta_N \leq 1 - \gamma) \geq 1 - o(1)$ for $\gamma \in (\|\boldsymbol{\alpha}\|_2^2, 1)$.
4. Weak recovery (detection): $\mathbb{P}(\eta_N \leq 1 - \|\boldsymbol{\alpha}\|_2^2 - \Omega(1)) \geq 1 - o(1)$ ⁴.

³By Cauchy-Schwarz inequality, $\|\boldsymbol{\alpha}\|_2^2 \geq 1/K$, which is the accuracy lower bound without knowing $\boldsymbol{\alpha}$.

⁴The difference between partial and weak is that, the algorithm does better than random guess in weak recovery, but cannot quantify how much better as in partial recovery.

1.2.2 Information-Theoretic limits on clustering accuracy

We start with a simple example to illustrate the concept of the *Information-Theoretic* (IT) limits on clustering accuracy. Consider the binary symmetric stochastic block model $\text{SBM}(N, 2, \alpha, \beta)$, where the vertex set $\mathcal{V} = [N]$ is partitioned into two equi-sized disjoint communities $\mathcal{V}_1$ and $\mathcal{V}_2$. The edges are drawn independently with probability α if both endpoints belong to the same community, and with probability β if they belong to different communities.

Consider the following two extreme scenarios. First, $\alpha = 1$ and $\beta = 0$ as in Figure 1.4 (a), then two disjoint cliques are drawn and it is trivial to achieve strong consistency. Second, when $\alpha = \beta = 0.5$, no algorithm could outperform random guess, since the communities themselves are indistinguishable as in Figure 1.4 (b). From this point of view, once the probability parameters α and β are determined, for any algorithm, there exists certain mathematical limit on clustering accuracy. On the way around, to achieve certain accuracy, the parameters of the underlying model have to satisfy certain conditions.

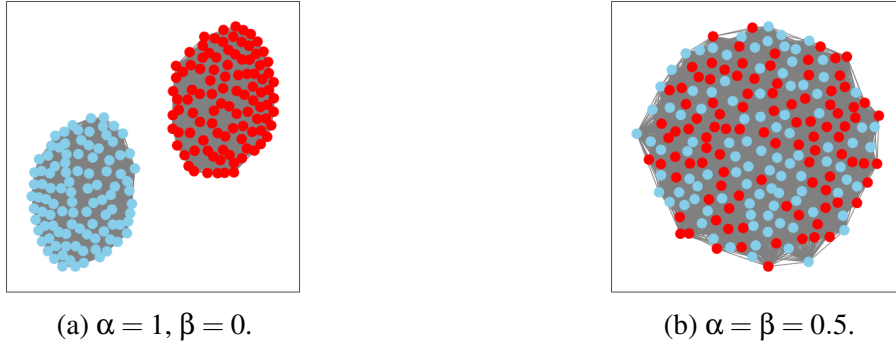


Figure 1.4. Two extreme scenarios of the binary symmetric $\text{SBM}(N, 2, \alpha, \beta)$.

We will study the IT limits on clustering accuracy for the binary symmetric HSBM in Chapter 2 and the multi-block HSBM in Chapter 3 respectively.

1.2.3 Related literature

Graphs

For the binary symmetric $\text{SBM}(N, 2, a^{\frac{\log(N)}{N}}, b^{\frac{\log(N)}{N}})$, the threshold is $\mathbb{D}_H(a, b) := (\sqrt{a} - \sqrt{b})^2/2$, which can be obtained from (3.1.1) by taking $\mathcal{L} = \{2\}$ and $K = 2$. Exact recovery can be achieved efficiently if $\mathbb{D}_H(a, b) \geq 1$, but is impossible when $\mathbb{D}_H(a, b) < 1$ [2, 169]. For multi-block $\text{SBM}(N, \boldsymbol{\alpha}, \mathbf{P}^{\frac{\log(N)}{N}})$, exact recovery can be achieved efficiently when $\mathbb{D}_{\text{CH}}(\boldsymbol{\alpha}, \mathbf{P}) \geq 1$, but is impossible when $\mathbb{D}_{\text{CH}}(\boldsymbol{\alpha}, \mathbf{P}) < 1$ [5, 8, 212], where

$$\mathbb{D}_{\text{CH}}(\boldsymbol{\alpha}, \mathbf{P}) = \min_{i, j \in [K], i \neq j} \mathbb{D}_{\text{CH}}(i, j),$$

$$\mathbb{D}_{\text{CH}}(i, j) := \max_{t \in [0, 1]} \sum_{k=1}^K \alpha_k \cdot [t P_{ik} + (1-t) P_{jk} - (P_{ik})^t (P_{jk})^{1-t}]$$

denotes the *Chernoff-Hellinger* (CH) divergence, which can be deduced from (3.2.5) by taking $\mathcal{L} = \{2\}$. Table 1.1 reviews the related literature where exact recovery was achieved above the threshold. For simplicity, the time complexity of singular value decomposition in our case

Table 1.1. Relevant literature for exact recovery in graph SBM

Graph	$K = 2$	$K \geq 3$	First stage	Second stage	Exact recovery	Time complexity	No prior
[2]	✓	✗	Spectral	Majority	✓	$2N^2 + N$	✓
	✓	✗	MLE	✗	✓	NP-hard	✗
[118]	✓	✗	SDP	✗	✓	$N^{3.5}$ [138]	✓
[5]	✓	✓	Sphere Comparison	Degree	✓	$N^{1+1/\log(c)}$	✗
[8]	✓	✓		Profiling	✓	$N^{1+1/\log(c)}$	✓
[169]	✓	✗	Spectral	Majority	✓	$2N^2 + N$	✓
[212]	✓	✓	Spectral	KL	✓	$KN^2 + KN \log(N)$	✓
[104]	✓	✓	Spectral	MLE	✓	$KN^2 + KN^3$	✓
[4]	✓	✗	Spectral	✗	✓	$2N^2$	✓

is $O(N^3)$. When referring to the complexity of SDP, a factor of $\log^{O(1)}(N/\varepsilon)$ is hidden, with ε being the accuracy of the output. If the algorithm has two stages, the entry (e.g., $N^3 + N$) represents the sum of the complexities for the first ($O(N^3)$) and second ($O(N)$) stages. If K is

needed as input but $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ are not, the entry in column 'No prior' is marked as ✓. The second stage algorithms in [2, 169] are based on the "Majority vote" principle. In [212], the estimator in second stage is aiming at minimizing *Kullback-Leibler* (KL) divergence between estimated and true distribution.

Hypergraphs

Our results in Chapter 2 and Chapter 3 are most general to date, covering non-uniform hypergraphs with multiple communities. We now illustrate how they fit into existing literature. So far, all thresholds obtained in existing literature apply only to uniform hypergraphs. Our results establish, for the first time, the threshold for the non-uniform case, in the sense of showing both achievability above and impossibility below. See Table 1.2 for related literature.

For the binary symmetric uniform model $\text{HSBM}(N, 2, a_\ell \frac{\log(N)}{\binom{N-1}{\ell-1}}, b_\ell \frac{\log(N)}{\binom{N-1}{\ell-1}})$, the threshold for exact recovery is given by $\mathbb{D}_{\text{GH}}^{(\ell)} := 2^{-(\ell-1)}(\sqrt{a_\ell} - \sqrt{b_\ell})^2$ [146], which is a special case of (3.1.1) by taking $\mathcal{L} = \{\ell\}$, $K = 2$ and symmetries among edges crossing two communities. Exact recovery is impossible if $\mathbb{D}_{\text{GH}}^{(\ell)} < 1$ [146], but can be achieved efficiently if $\mathbb{D}_{\text{GH}}^{(\ell)} > 1$ [146, 106]. For the multi-block case, the model formulation of [216] is a bit different from our model 1.1.5, since multisets are allowed as edges there. Our proofs work for their case as well if some steps are skipped. After proper scaling, their threshold for uniform $\text{HSBM}(N, \boldsymbol{\alpha}, \mathcal{P}^{(\ell)} \log(N) / \binom{N-1}{\ell-1})$ can be written as $\mathbb{D}_{\text{GCH}}^{(\ell)} = \min_{j,k \in [K], j \neq k} \mathbb{D}_{\text{GCH}}^{(\ell)}(j, k)$, with

$$\mathbb{D}_{\text{GCH}}^{(\ell)}(j, k) := \max_{t \in [0,1]} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \frac{\bar{N}_{\mathbf{w}}}{\binom{N-1}{\ell-1}} [t \mathcal{P}_{j \oplus \mathbf{w}}^{(\ell)} + (1-t) \mathcal{P}_{k \oplus \mathbf{w}}^{(\ell)} - (\mathcal{P}_{j \oplus \mathbf{w}}^{(\ell)})^t \cdot (\mathcal{P}_{k \oplus \mathbf{w}}^{(\ell)})^{1-t}],$$

which agrees with (3.2.5) by taking $\mathcal{L} = \{\ell\}$. Exact recovery is impossible when $\mathbb{D}_{\text{GCH}}^{(\ell)} < 1$, but can be achieved with the full knowledge of tensor $\mathcal{P}^{(\ell)}$ and number of communities K when $\mathbb{D}_{\text{GCH}}^{(\ell)} > 1$ [216].

Table 1.2. Relevant literature for exact recovery in HSBM

Hyper-graph	Uni-form	Non-uniform	$K = 2$	$K \geq 3$	First stage	Second stage	Exact recovery	Time complexity	No prior
[108]	✓	✓	✓	✓	Spectral	✗	✗	KN^2	✓
[146]	✓	✗	✓	✗	SDP	✗	✓	$N^{3.5}$	✓
[61]	✓	✗	✓	✓	Spectral	MLE	✓	$KN^2 + KN^3$	✓
[106]	✓	✗	✓	✗	Spectral	✗	✓	$2N^2$	✓
	✓	✗	✓	✗	SDP	✗	✓	$N^{3.5}$	✓
[216]	✓	✗	✓	✓	Spectral	MAP	✓	$KN^2 + N$	✗
[198]	✓	✓	✓	✗	Spectral	✗	✓	$2N^2$	✓
Alg 3.2.1	✓	✓	✓	✓	Spectral	KL	✓	$KN^2 + KN \log(N)$	✓
Alg 3.2.4	✓	✓	✓	✓	Spectral	MAP	✓	KN^3	✗

1.3 Summary of Contributions

The main results in each chapter are summarized as follows.

1.3.1 Chapter 2

Consider the unsupervised classification problem in random hypergraphs under the non-uniform *Hypergraph Stochastic Block Model* (HSBM) with two equal-sized communities, where each edge appears independently with some probability depending only on the labels of its vertices. In this paper, we establish the threshold for strong consistency and the *Information-Theoretic* (IT) limits on the clustering accuracy, which are expressed in terms of the *generalized Hellinger* distance. Below the threshold, it is impossible to assign all vertices to their own communities, and the lower bound of the expected *mismatch ratio* is derived. Above the threshold, the problem space is (often) divided into two disjoint subspaces. When only the adjacency matrix is given, we prove that with high probability, *one-stage* spectral algorithms succeed in assigning every vertex correctly in the subspace far away from the threshold but fail in the one closer to the threshold. We propose two refinement algorithms to improve the clustering accuracy based on the initial partition, which attain the lowest possible mismatch ratio previously

derived from the IT perspective. Moreover, for non-uniform hypergraphs, strong consistency is achievable by aggregating information from all uniform layers, even if it is impossible when each layer is considered alone.

1.3.2 Chapter 3

We consider the community detection problem in random hypergraphs under the non-uniform hypergraph stochastic block model (HSBM), where each hyperedge appears independently with some given probability depending only on the labels of its vertices. We establish, for the first time in the literature, a sharp threshold for exact recovery for the non-uniform, multiple-community case, subject to only minor constraints. One crucial point here is that, by aggregating information from all the uniform layers, we may obtain exact recovery even in cases when this is impossible if each layer were considered alone. In addition, for the almost exact recovery regime, we prove a wide-ranging, information-theoretic lower bound on the number of misclassified vertices *for any algorithm*. The bound depends on a *generalized Chernoff-Hellinger* divergence involving model parameters. We provide two efficient algorithms (one for the minimal information case, another for the full information case) which successfully achieve exact recovery when above the threshold, and attain the lowest possible mismatch ratio when the exact recovery is impossible. The theoretical analysis of our algorithms relies on the concentration and regularization of the adjacency matrix for non-uniform random hypergraphs, which could be of independent interest. We also address some open problems regarding parameter knowledge and estimation.

1.3.3 Chapter 4

We consider the community detection problem in sparse random hypergraphs under the non-uniform hypergraph stochastic block model (HSBM), a general model of random networks with community structure and higher-order interactions. When the random hypergraph has bounded expected degrees, we provide a spectral algorithm that outputs a partition with at least a

γ fraction of the vertices classified correctly, where $\gamma \in (0.5, 1)$ depends on the signal-to-noise ratio (SNR) of the model. When the SNR grows slowly as the number of vertices goes to infinity, our algorithm achieves weak consistency, which improves the previous results in [108] for non-uniform HSBMs.

Our spectral algorithm consists of three major steps: (1) Hyperedge selection: select hyperedges of certain sizes to provide the maximal signal-to-noise ratio for the induced sub-hypergraph; (2) Spectral partition: construct a regularized adjacency matrix and obtain an approximate partition based on singular vectors; (3) Correction and merging: incorporate the hyperedge information from adjacency tensors to upgrade the error rate guarantee. The theoretical analysis of our algorithm relies on the concentration and regularization of the adjacency matrix for sparse non-uniform random hypergraphs, which can be of independent interest.

1.4 Notations

In this section, we summarize the notations used throughout the dissertation. We will further specify the notations used in each chapter.

1.4.1 General notations

Let $\mathbb{N}^+ = \{1, 2, \dots\}$ denote the set of positive integers. For any $N \in \mathbb{N}^+$, let $[N] = \{1, 2, \dots, N\}$. Let $\mathbb{R}$ denote the set of real numbers. For any $k \in \mathbb{N}^+$, let $\mathbb{R}^k$ denote the k -dimensional Euclidean space.

For any $a \in \mathbb{R}$, let $\text{sign}(a) = 1$ if $a > 0$, $\text{sign}(a) = -1$ if $a < 0$, and $\text{sign}(0) = 0$, and denote $a^+ = \max\{0, a\}$. For any $a \in \mathbb{R}$, let $\lfloor a \rfloor$ and $\lceil a \rceil$ denote the largest integer less than or equal to a and the smallest integer greater than or equal to a respectively. For any $a, b \in \mathbb{R}$, denote $a \wedge b = \min\{a, b\}$ and $a \vee b = \max\{a, b\}$, and denote $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$ and $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$.

For any two sequences of numbers $\{a_N\}$, $\{b_N\}$, denote $a_N = O(b_N)$ or $a_N \lesssim b_N$ (resp. $a_N = \Omega(b_N)$ or $a_N \gtrsim b_N$) if there exists some constants C and N_0 such that $a_N \leq C b_N$ (resp.

$a_N \geq Cb_N$ for all $N \geq N_0$. Denote $a_N = \Theta(b_N)$ or $a_N \asymp b_N$ if both $a_N \lesssim b_N$ and $a_N \gtrsim b_N$. Denote $a_N = o(b_N)$ or $a_N \ll b_N$ (resp. $a_N = \omega(b_N)$ or $a_N \gg b_N$) if for any $\varepsilon > 0$, there exists $N_0 \in \mathbb{N}^+$ s.t. $a_N < \varepsilon b_N$ (resp. $a_N > \varepsilon b_N$) for all $N \geq N_0$. For any two sequences of numbers $\{a_N\}$ and $\{b_N\}$, we write $a_N \sim b_N$ if $\lim_{N \rightarrow \infty} a_N/b_N = 1$.

Let $\mathbb{1}_{\{A\}}$ denote the indicator function, and $\mathbb{1}_{\{A\}} = 1$ when event A is true.

1.4.2 Matrices and tensors

The lowercase letters (e.g., a, b), lowercase boldface letters (e.g., $\mathbf{a}, \mathbf{w}$), uppercase boldface letters (e.g., $\mathbf{A}, \mathbf{W}$), and boldface calligraphic letters (e.g., $\mathcal{A}, Q$) are used to denote scalars, vectors, matrices, and tensors respectively. Let $\mathbf{1}_N$ denote the all one column vector of length N . For vector $\mathbf{a} \in \mathbb{R}^N$, let $\|\mathbf{a}\|_p = (\sum_{i=1}^N |\mathbf{a}_i|^p)^{1/p}$ denote the ℓ_p norm of $\mathbf{a}$ for $p \geq 1$. For matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, let A_{ij} , $\mathbf{A}_{i:}$ and $\mathbf{A}_{:j}$ denote its (i, j) th entry, i -th row and j -th column respectively. Let $\|\mathbf{A}\| := \max_{\|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2$ and $\|\mathbf{A}\|_F := \sqrt{\sum_{i,j} A_{ij}^2}$ denote the operator and Frobenius norm respectively.

Chapter 2

Information-Theoretic limits and strong consistency on the binary non-uniform hypergraph stochastic block model

2.1 Introduction

Community detection refers to the task of partitioning the vertices of a graph into groups with similar connectivity patterns. It has attracted significant attention due to its applications in data science, social networks, and biological studies. A classic example is the Zachary's karate club problem [213]. Community detection is now a central topic in network analysis and machine learning [173, 174, 23]. Random graph models are widely used for analyzing clustering algorithms, benchmarking, and establishing theoretical guarantees. In this work, we focus on the equi-sized two-community model. Formally, let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph with N vertices. The label vector $\mathbf{y} \in \{\pm 1\}^{|\mathcal{V}|}$ assigns each vertex to a community, where $\mathcal{V}_+ = \{v \in \mathcal{V} | y_v = 1\}$ and $\mathcal{V}_- = \{v \in \mathcal{V} | y_v = -1\}$, with $|\mathcal{V}_+| = |\mathcal{V}_-| = N/2$. Given $\hat{\mathbf{y}}, \mathbf{y} \in \{\pm 1\}^{|\mathcal{V}|}$, define the *Hamming distance* as

$$\mathbb{D}_{\text{HD}}(\mathbf{y}, \hat{\mathbf{y}}) = \sum_{v \in \mathcal{V}} \mathbb{1}_{\{y_v \neq \hat{y}_v\}}, \quad (2.1.1)$$

which counts the number of entries having different values. We further define the *mismatch* ratio as

$$\eta_N := \eta_N(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N} \min_{s \in \{\pm 1\}} \mathbb{D}_{\text{HD}}(s\mathbf{y}, \hat{\mathbf{y}}), \quad (2.1.2)$$

which is often used for quantifying the accuracy of the estimator $\hat{\mathbf{y}}$. Note that a random guess estimator has expected accuracy $(1/2)^2 + (1/2)^2 = 1/2$, thus the estimator $\hat{\mathbf{y}}$ is meaningful only if $\eta_N \leq 1/2$. A natural question arises here: when is it possible to correctly classify all vertices? An estimator $\hat{\mathbf{y}}$ is said to achieve *strong consistency* (exact recovery) if

$$\lim_{N \rightarrow \infty} \mathbb{P}(\eta_N = 0) = \lim_{N \rightarrow \infty} \mathbb{P}(\hat{\mathbf{y}} = \pm \mathbf{y}) = 1. \quad (2.1.3)$$

At the same time, $\hat{\mathbf{y}}$ is said to achieve *weak consistency* (almost exact recovery) if $\eta_N = o(1)$.

In the pioneering work by [128], *Stochastic Block Model* (SBM) was first introduced for sociology studies. Motivated by exploring the phase transition behaviors in various regimes, many insightful problems were brought out. It has been successively studied in [52, 42, 89, 182, 70, 164, 37, 68, 178, 66, 10, 57, 161] over the past several decades. A major breakthrough was the establishment of the exact recovery thresholds for the binary case by [2, 169] and multi-block case by [5, 212, 9]. Readers may refer to [1] for a more detailed review. The literature abounds with different methods of approach, like spectral algorithms [109, 12, 108, 60, 69, 216], *semidefinite programming* (SDP) [146, 151, 106] and *Graph Neural Networks* (GNN) [59, 29, 30, 199]

The *Hypergraph Stochastic Block Model* (HSBM), first introduced in [109], is a generalization of SBM to hypergraphs, which models the real life social networks better since it captures higher order interactions among three or more individuals. It has been studied successively in [22, 12, 108, 146, 60, 61, 69, 176, 85, 216, 115, 82, 200] over the last decade. For strong consistency of *uniform* HSBMs, the exact thresholds were given in [146, 106, 216] by generalizing techniques

in [2, 4, 5]. So far, most results concern community detection on uniform hypergraphs, which require the same number of vertices in each edge. However, the assumption on uniformity between edges is constraining and somewhat impractical, and the non-uniform HSBM was less explored in the literature, with notable results in [108, 85, 15, 82].

In this paper, we establish the *Information-Theoretic* threshold for strong consistency of the binary non-uniform HSBM. Analysis concerning efficiency and optimality of different algorithms will be presented. In addition, we will discuss how aggregating information from different layers benefits the clustering result.

2.1.1 The model

Definition 2.1.1 (Hypergraph). *A hypergraph $\mathcal{H}$ is a pair $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the vertex set and $\mathcal{E}$ denotes the set of non-empty subsets of $\mathcal{V}$. If every hyperedge e is an ℓ -subset of $\mathcal{V}$, then $\mathcal{H}$ is called ℓ -uniform. The degree D_v of a vertex $v \in \mathcal{V}$ is the number of hyperedges in $\mathcal{E}$ containing v .*

Definition 2.1.2 (Symmetric binary uniform HSBM). *Let $\mathbf{y} \in \{\pm 1\}^{|\mathcal{V}|}$ denote the label vector on $\mathcal{V}$, where $\mathbf{y}$ is chosen uniformly at random among all the vectors satisfying $\mathbf{1}^\top \mathbf{y} = 0$. The ℓ -uniform hypergraph $\mathcal{H}_\ell = (\mathcal{V}, \mathcal{E}_\ell)$ ($\ell \geq 2$ some fixed integer) is drawn in the following manner: each ℓ -hyperedge $e := \{i_1, \dots, i_\ell\} \subset \mathcal{V}$ is sampled independently with probability α_ℓ if $y_{i_1} = \dots = y_{i_\ell}$, otherwise with probability β_ℓ .*

Definition 2.1.3 (Non-uniform HSBM). *Let $\mathbf{y} \in \{\pm 1\}^{|\mathcal{V}|}$ denote the label vector, sampled uniformly at random among all the vectors satisfying $\mathbf{1}^\top \mathbf{y} = 0$. Let $\mathcal{L} = \{\ell | \ell \geq 2\}$ be a set of finitely many integers with L denoting its largest element. The non-uniform hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ is a collection of independent uniform ones, i.e., $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$ with $\mathcal{E} := \cup_{\ell \in \mathcal{L}} \mathcal{E}_\ell$, where each $\mathcal{H}_\ell = (\mathcal{V}, \mathcal{E}_\ell)$ is sampled from the ℓ -uniform HSBM 2.1.2 independently given the same $\mathbf{y}$.*

Remark 2.1.4. *Unlike the model in [82], where each vertex is assigned to some community independently and the community sizes might fluctuate, our model 2.1.3 imposes a strict half-half constraint $\mathbf{1}^\top \mathbf{y} = 0$, ensuring exactly equal-sized communities. The proof strategies for the main results differ from those in [82], which are tailored to our model.*

One can associate an ℓ -uniform hypergraph $\mathcal{H}_\ell$ to an order- ℓ symmetric tensor $\mathcal{A}^{(\ell)}$, where the entry $\mathcal{A}_e^{(\ell)}$ denotes the presence of the ℓ -hyperedge $e = \{i_1, \dots, i_\ell\}$, i.e., $\mathcal{A}_{i_1, \dots, i_\ell}^{(\ell)} = \mathbb{1}_{\{e \in \mathcal{E}_\ell\}}$. However, it is impossible to aggregate information of each uniform hypergraph directly since tensors are of different orders. Meanwhile, most computations involving tensors are NP-hard [127]. To that end, we introduce the following *adjacency matrix* for hypergraphs, which is obtained by aggregating information from different tensors entrywisely.

Definition 2.1.5 (Aggregated adjacency matrix). *For the non-uniform hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, let $\mathcal{A}^{(\ell)}$ be the order- ℓ adjacency tensor corresponding to underlying ℓ -uniform hypergraph for each $\ell \in \mathcal{L}$. Let*

$$A_{ij} := \mathbb{1}_{\{i \neq j\}} \cdot \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell, e \supset \{i, j\}} \mathcal{A}_e^{(\ell)}, \quad (2.1.4)$$

denote the number of hyperedges containing both vertices i and j in the non-uniform hypergraph $\mathcal{H}$. The diagonal entries are defined as $A_{ii} := 0$ for each $i \in \mathcal{V}$ since each ℓ -hyperedge contains ℓ distinct vertices.

For the discussion of strong consistency, it typically requires that the expected degree of each vertex grows to infinity with respect to the number of vertices. To simplify the presentation, we adapt the following assumption on the model parameters.

Assumption 2.1.6. *For each $\ell \in \mathcal{L}$, assume that $\alpha_\ell \geq \beta_\ell$. We write $\alpha_\ell = a_\ell \cdot q_N / \binom{N-1}{\ell-1}$ and $\beta_\ell = b_\ell \cdot q_N / \binom{N-1}{\ell-1}$, where q_N denotes the magnitude parameter with $1 \ll q_N \lesssim \log(N)$ and $a_\ell \geq b_\ell \gtrsim 1$.*

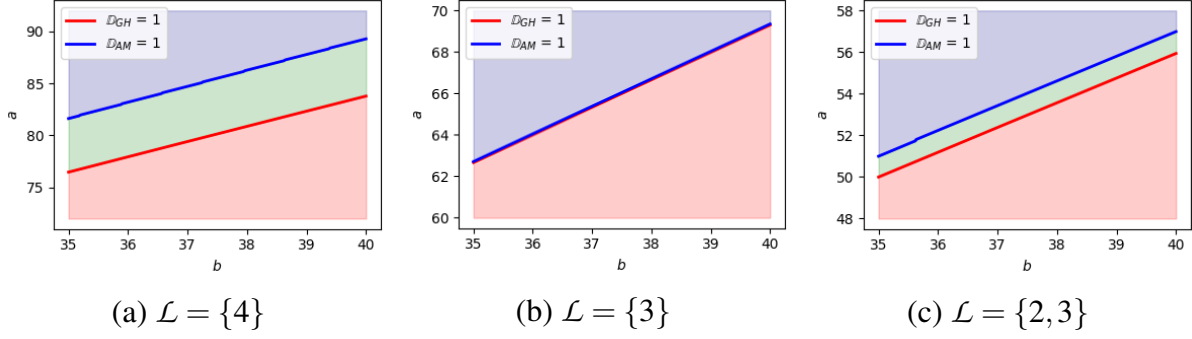


Figure 2.1. Plots for areas $\mathbb{D}_{\text{GH}} < 1$ (red), $\mathbb{D}_{\text{AM}} > 1 > \mathbb{D}_{\text{GH}}$ (green) and $\mathbb{D}_{\text{AM}} > 1$ (blue)

2.1.2 Formulation of the problems

We first define the following *Generalized Hellinger* divergence $\mathbb{D}_{\text{GH}}$

$$\mathbb{D}_{\text{GH}} := \sum_{\ell \in \mathcal{L}} \frac{1}{2^{\ell-1}} (\sqrt{a_\ell} - \sqrt{b_\ell})^2, \quad (2.1.5)$$

which quantifies the fundamental *Information-Theoretic* (IT) limits for the non-uniform HSBM model 2.1.3, when the full set of adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ is accessible. In contrast, we introduce the following function of the model parameters:

$$\mathbb{D}_{\text{AM}} := \sup_{t \geq 0} \sum_{\ell \in \mathcal{L}} \frac{1}{2^{\ell-1}} \left[a_\ell (1 - e^{-(\ell-1)t}) + b_\ell \sum_{j=1}^{\ell-1} \binom{\ell-1}{j} (1 - e^{-(\ell-1-2j)t}) \right], \quad (2.1.6)$$

which characterizes the algorithmic limits when only the *Adjacency Matrix* (AM) serves as input. Except for special cases $\mathcal{L} = \{2\}$ and $\mathcal{L} = \{3\}$ illustrated in Figure 2.1 (b), a gap emerges between $\mathbb{D}_{\text{GH}}$ and $\mathbb{D}_{\text{AM}}$, as demonstrated for $\mathcal{L} = \{4\}$ in Figure 2.1 (a) and $\mathcal{L} = \{2, 3\}$ in Figure 2.1 (c).

Phase transition threshold for exact recovery

For the uniform hypergraphs, the problem is reduced to the graph case when $\mathcal{L} = \{2\}$, where the two quantities are $\mathbb{D}_{\text{GH}} = \mathbb{D}_{\text{AM}} = (\sqrt{a_2} - \sqrt{b_2})^2/2$. It is impossible to achieve strong consistency when $\mathbb{D}_{\text{GH}} < 1$ [2], while *one-stage* algorithms, such as spectral clustering [4, 78]

and *semidefinite programming* (SDP) [118], achieve strong consistency when $\mathbb{D}_{\text{GH}} > 1$. However, when $\mathcal{L} = \{\ell\}$ for some fixed $\ell \geq 4$, the gap between $\mathbb{D}_{\text{GH}}$ and $\mathbb{D}_{\text{AM}}$ appears, and the story is not as elegant as before. According to [146], for any algorithm, it is impossible to accomplish exact recovery when $\mathbb{D}_{\text{GH}} < 1$ (red area). On the other hand, when $\mathbb{D}_{\text{AM}} > 1$ (blue area), both spectral method and SDP achieve strong consistency using the *aggregated adjacency matrix* $\mathbf{A}$ (2.1.4), as proved in [106]. When $\mathbb{D}_{\text{AM}} < 1 < \mathbb{D}_{\text{GH}}$ (green area), strong consistency can be accomplished via some *two-stage* algorithms, e.g. [61, 216], where in the second stage, the adjacency tensor $\mathcal{A}^{(\ell)}$ is required for refinement. In terms of the efficiency of information usage, we raise the following question.

1. Is it possible to achieve strong consistency when $\mathbb{D}_{\text{GH}} > 1$ using purely the adjacency matrix $\mathbf{A}$?

For the non-uniform case, the literature is less explored. [15] proved that exact recovery can be accomplished by SDP when $\mathbb{D}_{\text{AM}} > 1$ (blue in Figure 2.1 (c)) with time complexity $O(N^{3.5})$ [138]. The phase transition phenomena was presented in [82], where the authors proved that strong consistency can be accomplished if and only if $\mathbb{D}_{\text{GH}} > 1$. The authors presented some two-stage algorithms, where the requirements for tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ are essential. However, as in the uniform case, the regime $\mathbb{D}_{\text{AM}} < 1 < \mathbb{D}_{\text{GH}}$ (green area) is less explored. The algorithms in [82] are bit redundant under our binary model, since their methods were not specialized to the current setting. In terms of algorithm efficiency, several questions arise here.

2. Is there any one-stage algorithm faster than SDP such that the strong consistency can be achieved when $\mathbb{D}_{\text{GH}} > 1$?

3. When using the information from all uniform layers, is it possible to yield better results than considering each layer alone?

Information-Theoretic lower bound

We start with a simple example to illustrate the concept of the *Information-Theoretic* (IT) lower bound. The model 2.1.3 is reduced to the graph case with binary symmetric communities when taking $\mathcal{L} = \{2\}$. Consider the following two extreme scenarios. First, $\alpha_2 = 1$ and $\beta_2 = 0$,

then two disjoint cliques are drawn and it is trivial to achieve strong consistency. Second, when $\alpha_2 = \beta_2$, no algorithm could outperform random guess, since the communities themselves are indistinguishable. From this point of view, once the parameters $\{\alpha_\ell\}_{\ell \in \mathcal{L}}, \{\beta_\ell\}_{\ell \in \mathcal{L}}$ in model 2.1.3 are determined, for any algorithm, there exists a certain mathematical limit on clustering accuracy. In addition, to achieve certain accuracy, the parameters of the underlying model 2.1.3 have to satisfy certain conditions.

The IT limits are less explored in the literature. For graphs, it was studied by [5, 214, 4, 3] for the binary case and [104, 212, 215] for the multi-community case. For uniform hypergraphs, [61] proved that when restricting the model parameters on some (potential) subset (Eq.(1) therein) of the red in Figure 2.1 (a)-(c), some minimax lower bound on the expected mismatch ratio (Theorem 7.1) can be obtained. However, the question was not answered for the non-uniform hypergraphs. Concerning the IT lower bound, two questions arise here.

4. Is it possible to characterize the accuracy limit explicitly in terms of model parameters? If so, can optimal recovery be efficiently achieved ?

2.1.3 Contributions and organization of the paper

Our contribution to the non-uniform HSBM problem can be summarized as follows.

- (1) We establish the threshold $\mathbb{D}_{\text{GH}} = 1$ for strong consistency in Theorems 2.2.1 and 2.2.5.
- (2) We present the IT lower bound on expected mismatch ratio for any algorithm in Theorem 2.2.3, informally stating that $\mathbb{E}\eta_N \geq N^{-\mathbb{D}_{\text{GH}}}$.
- (3) When only the adjacency matrix $\mathbf{A}$ is used as an input, Algorithm 2.2.1 achieves strong consistency when $\mathbb{D}_{\text{AM}} > 1$.
- (4) We provide two refinement methods, Algorithm 2.2.2 and Algorithm 2.2.3, achieving strong consistency when $\mathbb{D}_{\text{AM}} < 1 < \mathbb{D}_{\text{GH}}$. Both achieve IT lower bound when $\mathbb{D}_{\text{GH}} < 1$.

2.2 Main Results

2.2.1 Information-Theoretic limits

We first present the necessary condition to achieve strong consistency.

Theorem 2.2.1 (Impossibility of exact recovery). *For model 2.1.3 under Assumption 2.1.6, suppose*

$$\limsup_{N \rightarrow \infty} \mathbb{D}_{\text{GH}} \cdot q_N / \log(N) < 1, \quad (2.2.1)$$

then with probability tending to 1, every algorithm will misclassify at least two vertices.

Remark 2.2.2. *Here, $\mathbb{D}_{\text{GH}}$ is the rate function derived from the Large Deviation Principle (LDP) analysis using the adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$. The detailed arguments are deferred to Lemmas A.1.1 and A.1.2.*

The condition (2.2.1) also covers the scenario when $\mathbb{D}_{\text{GH}} \gg 1$ ($a_\ell \gg b_\ell$) and $1 \ll q_N \ll \log(N)$, which was ignored in previous literature. In addition, when $q_N = \log(N)$ and $\mathcal{L} = \{\ell\}$ for some fixed integer $\ell \geq 2$, (2.2.1) can be simplified as $\mathbb{D}_{\text{GH}} < 1$, which is the necessary condition of strong consistency for ℓ -uniform hypergraphs derived in [146]. By taking $\mathcal{L} = \{2\}$, (2.2.1) becomes $(\sqrt{a_2} - \sqrt{b_2})^2 < 1$, as derived in [2] for graphs.

As noted in Remark 2.1.4, model 2.1.3 differs from that in [82], leading to distinct proof techniques. Interestingly, when the model in [82] is specialized to the symmetric binary case considered here, their threshold $\mathbb{D}_{\text{GCH}}$ coincides with our $\mathbb{D}_{\text{GH}}$. This serves as a mutual validation of both thresholds.

Furthermore, we establish a lower bound on the reconstruction error rate for any estimator, explicitly characterized by the model parameters.

Theorem 2.2.3 (Lower bound on the expected mismatch ratio). *Under Assumption 2.1.6, for*

any sequence of estimators $\hat{\mathbf{y}}$, the following holds:

$$\liminf_{N \rightarrow \infty} q_N^{-1} \log \mathbb{E} \eta_N(\mathbf{y}, \hat{\mathbf{y}}) \geq -\mathbb{D}_{\text{GH}}. \quad (2.2.2)$$

Informally, Theorem 2.2.3 can be interpreted as $\mathbb{E} \eta_N \geq e^{-\mathbb{D}_{\text{GH}} \cdot q_N}$, which can be further simplified as $\mathbb{E} \eta_N \geq N^{-\mathbb{D}_{\text{GH}}}$ when $q_N = \log(N)$.

2.2.2 Algorithm performance using purely the adjacency matrix

Theorem 2.2.4. *Under Assumption 2.1.6, for some absolute $\varepsilon > 0$, suppose*

$$\liminf_{N \rightarrow \infty} \mathbb{D}_{\text{AM}} \cdot q_N / \log(N) \geq 1 + \varepsilon, \quad (2.2.3)$$

then with probability tending to 1, Algorithm 2.2.1 accomplishes strong consistency.

Algorithm 2.2.1: Spectral partition via the second eigenvector

Data: The $N \times N$ matrix $\mathbf{X}$, either $\mathbf{X} = \mathbf{A}$ or $\mathbf{X} = \mathbf{L} := \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$.

- 1 Compute the eigenvector $\mathbf{u}_2$ corresponding to the second largest eigenvalue λ_2 of $\mathbf{X}$.
- 2 Construct $\hat{\mathcal{V}}_+^{(0)} = \{v \mid \hat{y}_v^{(0)} > 0\}$ and $\hat{\mathcal{V}}_-^{(0)} = \{v \mid \hat{y}_v^{(0)} < 0\}$, where $\hat{\mathbf{y}}^{(0)} = \text{sign}(\mathbf{u}_2)$.

Result: $\hat{\mathcal{V}}_+^{(0)}$ and $\hat{\mathcal{V}}_-^{(0)}$.

When $q_N = \log(N)$, (2.2.3) can be simplified as $\mathbb{D}_{\text{AM}} \geq 1 + \varepsilon$, which can be reduced to the result in [106] for the uniform hypergraph case, where only the adjacency matrix is taken as input. In addition, we show that the second eigenvector of the *normalized Laplacian* $\mathbf{L} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$ can be used for clustering. Furthermore, the case $q_N \ll \log(N)$ but $a_\ell \gg b_\ell$ is also covered by Theorem 2.2.4, since (2.2.3) could still hold.

2.2.3 Refinement algorithms

We present two refinement algorithms below, both requiring an initial estimation $\hat{\mathbf{y}}^{(0)}$ and the adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$. It is worth noting that for uniform hypergraphs, the requirement of adjacency tensors is not essential, as one can exactly reconstruct the tensor $\mathcal{A}^{(\ell)}$ with high

probability from the adjacency matrix $\mathbf{A}$ in (2.1.4) using the reconstruction method from [49]. However, for non-uniform hypergraphs, the adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ ($|\mathcal{L}| \geq 2$) are essential for the refinement process, since it remains unclear whether reconstruction of $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ for each uniform layer is possible using only the aggregated matrix $\mathbf{A}$.

Majority voting

We first introduce several important quantities. After iteration $t \geq 0$, let $\hat{\mathbf{y}}^{(t)}$ denote the estimate of $\mathbf{y}$ and $\hat{\mathcal{V}}_+^{(t)}, \hat{\mathcal{V}}_-^{(t)}$ be the corresponding partition of vertices. Let $\hat{\mathcal{E}}_{\alpha_\ell}^{(t)}$ denote the set of ℓ -hyperedges with all nodes in either $\hat{\mathcal{V}}_+^{(t)}$ or $\hat{\mathcal{V}}_-^{(t)}$, and define $\hat{\mathcal{E}}_{\beta_\ell}^{(t)} := \mathcal{E}_\ell \setminus \hat{\mathcal{E}}_{\alpha_\ell}^{(t)}$. For each $v \in \mathcal{V}$, let $\hat{\mathcal{E}}_{\alpha_\ell}^{(t)}(v)$ (resp. $\hat{\mathcal{E}}_{\beta_\ell}^{(t)}(v)$) denote the subset of $\hat{\mathcal{E}}_{\alpha_\ell}^{(t)}$ (resp. $\hat{\mathcal{E}}_{\beta_\ell}^{(t)}$) with v contained in each ℓ -hyperedge. Formally, we define

$$\hat{\mathcal{E}}_{\alpha_\ell}^{(t)} := \left\{ \{i_1, \dots, i_\ell\} \in \mathcal{E}_\ell : \hat{y}_{i_1}^{(t)} = \dots = \hat{y}_{i_\ell}^{(t)} \right\}, \quad \hat{\mathcal{E}}_{\alpha_\ell}^{(t)}(v) := \left\{ e \in \hat{\mathcal{E}}_{\alpha_\ell}^{(t)} : v \in e \right\}, \quad (2.2.4a)$$

$$\hat{\mathcal{E}}_{\beta_\ell}^{(t)} := \mathcal{E}_\ell \setminus \hat{\mathcal{E}}_{\alpha_\ell}^{(t)}, \quad \hat{\mathcal{E}}_{\beta_\ell}^{(t)}(v) := \left\{ e \in \hat{\mathcal{E}}_{\beta_\ell}^{(t)} : v \in e \right\}. \quad (2.2.4b)$$

Let $\hat{N}_{\alpha_\ell}^{(t)}$ and $\hat{N}_{\alpha_\ell}^{(t)}(v)$ denote the capacity of $\hat{\mathcal{E}}_{\alpha_\ell}^{(t)}$ and $\hat{\mathcal{E}}_{\alpha_\ell}^{(t)}(v)$ respectively, formally defined as

$$\hat{N}_{\alpha_\ell}^{(t)} := \binom{|\hat{\mathcal{V}}_+^{(t)}|}{\ell} + \binom{|\hat{\mathcal{V}}_-^{(t)}|}{\ell}, \quad \hat{N}_{\alpha_\ell}^{(t)}(v) := \binom{|\hat{\mathcal{V}}_+^{(t)}| - 1}{\ell - 1} + \binom{|\hat{\mathcal{V}}_-^{(t)}| - 1}{\ell - 1}. \quad (2.2.5)$$

Similarly, after iteration $t \geq 0$, let

$$\hat{N}_{\beta_\ell}^{(t)} := \binom{N}{\ell} - \hat{N}_{\alpha_\ell}^{(t)} \quad \text{and} \quad \hat{N}_{\beta_\ell}^{(t)}(v) := \binom{N-1}{\ell-1} - \hat{N}_{\alpha_\ell}^{(t)}(v), \quad (2.2.6)$$

denote the capacity of $\hat{\mathcal{E}}_{\beta_\ell}^{(t)}$ and $\hat{\mathcal{E}}_{\beta_\ell}^{(t)}(v)$ respectively.

For each $\ell \in \mathcal{L}$, we first compute the following two quantities

$$\hat{\alpha}_\ell = |\hat{\mathcal{E}}_{\alpha_\ell}^{(0)}| / \hat{N}_{\alpha_\ell}^{(0)}, \quad \hat{\beta}_\ell = |\hat{\mathcal{E}}_{\beta_\ell}^{(0)}| / \hat{N}_{\beta_\ell}^{(0)}. \quad (2.2.7)$$

which are the estimates of probabilities α_ℓ and β_ℓ using all the in-cluster and cross-cluster edges, respectively. Alternatively, at each iteration $t \geq 0$, α_ℓ and β_ℓ can be estimated via the following two quantities purely using edges containing v ,

$$\widehat{\alpha}_\ell^{(t)}(v) = |\widehat{\mathcal{E}}_{\alpha_\ell}^{(t)}(v)| / \widehat{N}_{\alpha_\ell}^{(t)}(v), \quad \widehat{\beta}_\ell^{(t)}(v) = |\widehat{\mathcal{E}}_{\beta_\ell}^{(t)}(v)| / \widehat{N}_{\beta_\ell}^{(t)}(v), \quad (2.2.8)$$

where $\widehat{N}_{\alpha_\ell}^{(t)}(v)$ and $\widehat{N}_{\beta_\ell}^{(t)}(v)$ are defined in (2.2.5). Denote the *cross-entropy* by

$$\mathbb{H}_{\text{CE}}(y, \widehat{y}) := -y \log \widehat{y} - (1-y) \log(1-\widehat{y}), \quad y, \widehat{y} \in [0, 1]. \quad (2.2.9)$$

Then for each $v \in \mathcal{V}$, its alignment with the estimated partition at iteration t is characterized by the following weighted cross-entropy terms:

$$\widehat{f}_{\text{in}}^{(t)}(v) = \sum_{\ell \in \mathcal{L}} \widehat{N}_{\alpha_\ell}^{(t)}(v) \cdot \mathbb{H}_{\text{CE}}(\widehat{\alpha}_\ell^{(t)}(v), \widehat{\alpha}_\ell) + \widehat{N}_{\beta_\ell}^{(t)}(v) \cdot \mathbb{H}_{\text{CE}}(\widehat{\beta}_\ell^{(t)}(v), \widehat{\beta}_\ell) \quad (2.2.10a)$$

$$\widehat{f}_{\text{cross}}^{(t)}(v) = \sum_{\ell \in \mathcal{L}} \widehat{N}_{\alpha_\ell}^{(t)}(v) \cdot \mathbb{H}_{\text{CE}}(\widehat{\alpha}_\ell^{(t)}(v), \widehat{\beta}_\ell) + \widehat{N}_{\beta_\ell}^{(t)}(v) \cdot \mathbb{H}_{\text{CE}}(\widehat{\beta}_\ell^{(t)}(v), \widehat{\alpha}_\ell) \quad (2.2.10b)$$

With all the quantities defined above, we can now present the Algorithm 2.2.2, inspired by [212, 82]. The algorithm iteratively refines the initial partition $\widehat{\mathcal{V}}_+^{(0)}$ and $\widehat{\mathcal{V}}_-^{(0)}$ by flipping the sign of vertices based on the majority voting principle, and finally achieves strong consistency after $\lceil \log(N) \rceil$ iterations. Algorithm 2.2.2 can be initialized with any weak consistent partition.

The mechanism of Algorithm 2.2.2 can be intuitively understood as follows. When estimating $\widehat{\alpha}_\ell$ and $\widehat{\beta}_\ell$ in (2.2.7), all the hyperedges (at least $N \log(N)$ many each type) are utilized. Subsequently, $\widehat{\alpha}_\ell$ and $\widehat{\beta}_\ell$ enjoy sufficient concentrations around their expectations, and serve as “ground truth” in the cross-entropy term $\mathbb{H}_{\text{CE}}(y, \widehat{y})$ in (2.2.9). However, the estimates $\widehat{\alpha}_\ell^{(t)}(v)$ and $\widehat{\beta}_\ell^{(t)}(v)$ in (2.2.8) only involve hyperedges containing v (roughly $\log(N)$ many), which would be close to their expectations α_ℓ and β_ℓ only when v is classified to its true partition. Note that $\mathbb{H}_{\text{CE}}(y, \widehat{y})$ decreases as $\widehat{y}$ approaches y . The smaller $\widehat{f}_{\text{in}}^{(t)}(v)$ (resp. $\widehat{f}_{\text{cross}}^{(t)}(v)$) is, the more neighbors

Algorithm 2.2.2: Majority voting

Data: The tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ and an initial estimation $\hat{\mathbf{y}}^{(0)}$.

- 1 Compute $\hat{\alpha}_\ell$ and $\hat{\beta}_\ell$ defined in (2.2.7).
- 2 **for** $t = 1, \dots, \lceil \log(N) \rceil$ **do**
- 3 Initialize $\hat{\mathcal{V}}_+^{(t)} \leftarrow \hat{\mathcal{V}}_+^{(t-1)}$, $\hat{\mathcal{V}}_-^{(t)} \leftarrow \hat{\mathcal{V}}_-^{(t-1)}$.
- 4 **for** $v \in \mathcal{V}$ **do**
- 5 Compute $\hat{f}_{\text{in}}^{(t)}$ and $\hat{f}_{\text{cross}}^{(t)}$ defined in (2.2.10a) and (2.2.10b) respectively.
- 6 Flip the sign of v if $\hat{f}_{\text{in}}^{(t)} > \hat{f}_{\text{cross}}^{(t)}$, otherwise keep the sign unchanged.
- 7 **end**
- 8 **end**

Result: The final partition $\hat{\mathcal{V}}_+ = \hat{\mathcal{V}}_+^{(\lceil \log(N) \rceil)}$, $\hat{\mathcal{V}}_- = \hat{\mathcal{V}}_-^{(\lceil \log(N) \rceil)}$.

that v contains in the current (resp. opposite) community.

Refinement via spectral method

To facilitate the analysis, for each vertex $v \in \mathcal{V}$ and each layer $\ell \in \mathcal{L}$, we categorize the associated edges based on the community membership of their vertices. For each $0 \leq r \leq \ell - 1$, let $\mathcal{E}_\ell^{(r)}(v)$ denote the set of ℓ -uniform hyperedges containing v such that exactly r vertices in the edge have different community membership from v . Formally, we define

$$\mathcal{E}_\ell^{(r)}(v) := \{e \in [\mathcal{V}]^\ell \mid v \in e, |\{u \in e \setminus \{v\} : y_u \neq y_v\}| = r\}. \quad (2.2.11)$$

After the first stage, we can approximate $\mathcal{E}_\ell^{(r)}(v)$ by constructing the ℓ -hyperedge set $\hat{\mathcal{E}}_\ell^{(r)}(v)$ based on the initial partition $\hat{\mathbf{y}}^{(0)}$. For each $e \in \hat{\mathcal{E}}_\ell^{(r)}(v)$, we consider its vertex $j \neq v$, and let $\hat{\mathcal{A}}_{e,v,j}^{(\ell)} = \mathcal{A}_e^{(\ell)} / (\ell - 1)$ when $r = 0, \ell - 1$; when $1 \leq r \leq \ell - 2$, we let

$$\hat{\mathcal{A}}_{e,v,j}^{(\ell)} = \mathcal{A}_e^{(\ell)} / r, \text{ if } \hat{y}_v^{(0)} = \hat{y}_j^{(0)}; \quad \hat{\mathcal{A}}_{e,v,j}^{(\ell)} = \mathcal{A}_e^{(\ell)} / (\ell - 1 - r), \text{ if } \hat{y}_v^{(0)} \neq \hat{y}_j^{(0)}. \quad (2.2.12)$$

Furthermore, for each $\ell \in \mathcal{L}$, we compute the parameter estimate

$$\hat{\Psi}_\ell = (\ell - 2) \log(2) + \log \left(\frac{\mathbf{1}_N^\top \mathbf{A}^{(\ell)} \mathbf{1}_N + (\hat{\mathbf{y}}^{(0)})^\top \mathbf{A}^{(\ell)} \hat{\mathbf{y}}^{(0)}}{\mathbf{1}_N^\top \mathbf{A}^{(\ell)} \mathbf{1}_N - (\hat{\mathbf{y}}^{(0)})^\top \mathbf{A}^{(\ell)} \hat{\mathbf{y}}^{(0)}} - 1 + 2^{-\ell+2} \right). \quad (2.2.13)$$

We then construct the weighted adjacency matrix $\hat{\mathbf{A}}$ by

$$\hat{\mathbf{A}}_{vj} := \mathbb{1}_{\{v \neq j\}} \cdot \sum_{\ell \in \mathcal{L}} \hat{\Psi}_\ell \sum_{r=0}^{\ell-1} \sum_{e \in \hat{\mathcal{E}}_\ell^{(r)}(v)} \hat{\mathcal{A}}_{e,v,j}^{(\ell)}. \quad (2.2.14)$$

This construction enables us to design the following new refinement algorithm, inspired by [3], that uses the spectral method to improve the initial partition $\hat{\mathbf{y}}^{(0)}$.

Algorithm 2.2.3: Refinement via spectral method

Data: The tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ and initial estimation $\hat{\mathbf{y}}^{(0)}$.

- 1 Construct the weighted adjacency matrix $\hat{\mathbf{A}}$ via (2.2.12) and (2.2.14).
- 2 Compute the second eigenpair $(\hat{\lambda}_2, \hat{\mathbf{u}}_2)$ of $\hat{\mathbf{A}}$.
- 3 Construct $\hat{\mathbf{y}} = \text{sign}(\hat{\mathbf{u}}_2)$ with $\hat{\mathcal{V}}_+ = \{v \mid \hat{y}_v > 0\}$ and $\hat{\mathcal{V}}_- = \{v \mid \hat{y}_v < 0\}$.

Result: The refined partition $\hat{\mathcal{V}}_+$ and $\hat{\mathcal{V}}_-$.

2.2.4 Strong consistency after second-stage refinement

In this section, we present a two-stage algorithm to achieve strong consistency when $\mathbb{D}_{\text{AM}} < 1 < \mathbb{D}_{\text{GH}}$ (green area in Figure 2.1 (a) (c)). The first stage is the same as Algorithm 2.2.1, which provides an initial partition of vertices. The second stage refines the initial partition using the information from all uniform layers.

Theorem 2.2.5 (Achievability). *Under Assumption 2.1.6, for some absolute $\varepsilon > 0$, suppose*

$$\liminf_{N \rightarrow \infty} \mathbb{D}_{\text{GH}} \cdot q_N / \log(N) \geq 1 + \varepsilon. \quad (2.2.15)$$

Then with probability tending to 1, Algorithm 2.2.2 attains strong consistency.

When $q_N = \log(N)$, (2.2.15) can be reduced to $\mathbb{D}_{\text{GH}} > 1 + \varepsilon$ for any absolute small $\varepsilon > 0$. Theorem 2.2.1 and Theorem 2.2.5 establish the sharp phase transition for the achievability of exact recovery. The same threshold was established for the graph case $\mathcal{L} = \{2\}$ in [2]. For the ℓ -uniform hypergraph, the necessity was proved by [146], and the regime $\mathbb{D}_{\text{AM}} > 1$ (blue) was solved by [106]. We fill the gap when $\mathbb{D}_{\text{AM}} < 1 < \mathbb{D}_{\text{GH}}$ (green) and establish the threshold for a more general case.

Benefits of aggregation

Consider the model 2.1.3 with $\mathcal{L} = \{2, 3\}$, where $a_2 = 4$, $b_2 = 1$, and the divergence for the graph layer $\ell = 2$ can be calculated as $\mathbb{D}^{(2)}(a_2, b_2) = (\sqrt{a_2} - \sqrt{b_2})^2/2 = 1/2$. When the divergence for the layer $\ell = 3$ satisfies $\mathbb{D}^{(3)}(a_3, b_3) > 1/2$, the entire non-uniform hypergraph satisfies $\mathbb{D}_{\text{GH}} > 1$ according to (2.1.5), and the strong consistency can be efficiently achieved. Compared with the 3-uniform HSBM, where the strong consistency can be achieved if and only if $\mathbb{D}^{(3)}(a_3, b_3) > 1$, the difficulty of the task is reduced when aggregating the different layers.

2.2.5 Accomplishment of optimal accuracy

Theorem 2.2.6 (Optimality). *Suppose Assumption 2.1.6 and the condition (2.2.1).*

(1) *Take $q_N = \log(N)$. After running Algorithm 2.2.1, the following holds with high probability*

$$\limsup_{N \rightarrow \infty} q_N^{-1} \log \mathbb{E} \eta_N(\mathbf{y}, \hat{\mathbf{y}}^{(0)}) \leq -\mathbb{D}_{\text{AM}}. \quad (2.2.16)$$

(2) *After applying Algorithm 2.2.2, the following holds with high probability*

$$\limsup_{N \rightarrow \infty} q_N^{-1} \mathbb{E} \log \eta_N(\mathbf{y}, \hat{\mathbf{y}}) \leq -\mathbb{D}_{\text{GH}}. \quad (2.2.17)$$

Theorem 2.2.6 (1) only gives the upper bound for the expected mismatch ratio $\mathbb{E} \eta_N$, while Theorem 2.2.6 (2) indicates that the mismatch ratio η_N can be decreased after Algorithm 2.2.2 since $\mathbb{D}_{\text{AM}} < \mathbb{D}_{\text{GH}}$ as in Figure 2.1. At the same time, the so obtained accuracy matches

the lower bound in Theorem 2.2.3, attaining the lowest possible mismatch ratio. Regarding the input requirements, as long as $\eta_N(\mathbf{y}, \hat{\mathbf{y}}^{(0)}) = o(1)$, Algorithm 2.2.2 could improve the accuracy to the best possible.

2.3 Proof outline of Information-Theoretic limits

We present the proof outlines of Theorems 2.2.1 and 2.2.3. The proofs of Lemmas are deferred to Appendix A.1.

2.3.1 Impossibility of the exact recovery

Let $\hat{\mathbf{y}}$ denote an estimation of $\mathbf{y}$ based on the observed hypergraph $\mathcal{H}$ where $\mathbb{H}$ denotes its law. The probability of failing exact recovery is given by

$$\mathbb{P}_{\text{fail}} := \mathbb{P}(\hat{\mathbf{y}} \neq \pm \mathbf{y}) = \sum_{\mathcal{H}} [1 - \mathbb{P}(\hat{\mathbf{y}} = \pm \mathbf{y} \mid \mathbb{H} = \mathcal{H})] \cdot \mathbb{P}(\mathbb{H} = \mathcal{H}). \quad (2.3.1)$$

The *Maximum A Posteriori* (MAP) estimator is the best possible estimator in the sense that it minimizes the failure probability $\mathbb{P}_{\text{fail}}$ in (2.3.1) among all estimators, which is defined as

$$\hat{\mathbf{y}}_{\text{MAP}} := \arg \max_{\mathbf{z} \in \{\pm 1\}^{|\mathcal{V}|}, \mathbf{1}^\top \mathbf{z} = 0} \mathbb{P}(\mathbf{y} = \mathbf{z} \mid \mathbb{H} = \mathcal{H}). \quad (2.3.2)$$

Note that $\hat{\mathbf{y}}_{\text{MAP}}$ is the most likely assignment of $\mathbf{y}$ given the observed hypergraph $\mathcal{H}$. Consequently, no other estimator could succeed exact recovery if $\hat{\mathbf{y}}_{\text{MAP}}$ fails. Furthermore, $\hat{\mathbf{y}}_{\text{MAP}}$

corresponds to the *Maximum Likelihood Estimation* (MLE) since by Bayes' Theorem,

$$\begin{aligned}
\hat{\mathbf{y}}_{\text{MLE}} &= \arg \max_{\mathbf{z} \in \{\pm 1\}^{|\mathcal{V}|}, \mathbf{1}^\top \mathbf{z} = 0} \mathbb{P}(\mathbb{H} = \mathcal{H} \mid \mathbf{y} = \mathbf{z}) \\
&= \arg \max_{\mathbf{z} \in \{\pm 1\}^{|\mathcal{V}|}, \mathbf{1}^\top \mathbf{z} = 0} \frac{\mathbb{P}(\mathbb{H} = \mathcal{H} \mid \mathbf{y} = \mathbf{z}) \mathbb{P}(\mathbf{y} = \mathbf{z})}{\sum_{\mathbf{x} \in \{\pm 1\}^{|\mathcal{V}|}, \mathbf{1}^\top \mathbf{x} = 0} \mathbb{P}(\mathbb{H} = \mathcal{H} \mid \mathbf{y} = \mathbf{x}) \mathbb{P}(\mathbf{y} = \mathbf{x})} \\
&= \arg \max_{\mathbf{z} \in \{\pm 1\}^{|\mathcal{V}|}, \mathbf{1}^\top \mathbf{z} = 0} \mathbb{P}(\mathbf{y} = \mathbf{z} \mid \mathbb{H} = \mathcal{H}) = \hat{\mathbf{y}}_{\text{MAP}}
\end{aligned} \tag{2.3.3}$$

where the second equality holds since the denominator in the second line is a constant irrelevant to any specific assignment $\mathbf{z}$ which can be factored out, and the prior distribution of $\mathbf{y}$ is uniform with $\mathbb{P}(\mathbf{y} = \mathbf{z}) = 1/\binom{|\mathcal{V}|}{|\mathcal{V}|/2}$.

While the MLE provides the optimal estimator in theory, it remains computationally intractable in practice. To make the objective function of the MLE more explicit, we introduce the notion of an *in-cluster* edge.

Definition 2.3.1 (In-cluster edge). *The edge $e = \{i_1, \dots, i_\ell\}$ is in-cluster with respect to $\mathbf{y}$ if $y_{i_1} = \dots = y_{i_\ell}$.*

Lemma 2.3.2. *Let $\mathcal{H}$ be a hypergraph sampled from model 2.1.3. For each $\ell \in \mathcal{L}$, let $[\mathcal{V}]^\ell := \{\mathcal{S} \mid \mathcal{S} \subseteq \mathcal{V}, |\mathcal{S}| = \ell\}$ denote the set of ℓ -subsets of $\mathcal{V}$, and suppose $\alpha_\ell > \beta_\ell$. Then, among all the vectors $\mathbf{z} \in \{\pm 1\}^{|\mathcal{V}|}$ with $\mathbf{1}^\top \mathbf{z} = 0$, $\hat{\mathbf{y}}_{\text{MLE}}$ maximizes the following quantity*

$$f(\mathbf{z} \mid \mathcal{H}) := \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell (1 - \beta_\ell)}{\beta_\ell (1 - \alpha_\ell)} \right) \cdot \sum_{e \in [\mathcal{V}]^\ell} \mathbb{1}_{\{e \text{ is in-cluster w.r.t. } \mathbf{z}\}} \cdot \mathcal{A}_e^{(\ell)} \tag{2.3.4}$$

Alternatively, $\hat{\mathbf{y}}_{\text{MLE}}$ minimizes $f(\mathbf{z} \mid \mathcal{H})$ when $\alpha_\ell < \beta_\ell$ (disassortative) for each $\ell \in \mathcal{L}$.

According to the discussions in (2.3.2), (2.3.3) and (2.3.4), MLE fails exact recovery when $\hat{\mathbf{y}}_{\text{MLE}}$ returns some $\tilde{\mathbf{y}}$, such that $f(\tilde{\mathbf{y}} \mid \mathcal{H}) \geq f(\mathbf{y} \mid \mathcal{H})$ but $\tilde{\mathbf{y}} \neq \pm \mathbf{y}$. Formally,

$$\mathbb{P}_{\text{fail}} := \mathbb{P}(\hat{\mathbf{y}}_{\text{MLE}} \neq \pm \mathbf{y}) = \mathbb{P}(\exists \tilde{\mathbf{y}} \neq \mathbf{y}, \text{ s.t. } f(\tilde{\mathbf{y}} \mid \mathcal{H}) \geq f(\mathbf{y} \mid \mathcal{H})). \tag{2.3.5}$$

The desired $\tilde{\mathbf{y}}$ can be constructed from the true label vector $\mathbf{y}$ by flipping the signs of two vertices $a \in \mathcal{V}_+$ and $b \in \mathcal{V}_-$, but keeping remaining signs unchanged. For any $a \in \mathcal{V}_+$, $b \in \mathcal{V}_-$, define the vector $\tilde{\mathbf{y}} \in \{\pm 1\}^{|\mathcal{V}|}$ via

$$\tilde{y}_v = \begin{cases} y_v, & v \in \mathcal{V} \setminus \{a, b\}, \\ -y_v, & v \in \{a, b\} \end{cases} \quad (2.3.6)$$

We will show that, under condition (2.2.1), $\hat{\mathbf{y}}_{\text{MLE}}$ returns $\tilde{\mathbf{y}}$ rather than $\mathbf{y}$ with probability tending to 1.

Remark 2.3.3. *In contrast, the approach in [82] does not directly analyze the failure of the optimal estimator. Instead, it establishes the existence of $2q_N$ pairwise disconnected ambiguous vertices (half from $\mathcal{V}_+$ and half from $\mathcal{V}_-$) under condition (2.2.1). These vertices have identical degree profiles, making them indistinguishable; thus, no algorithm can outperform random guessing, resulting in the failure of exact recovery.*

Proof of Theorem 2.2.1. Recall the definition of $\mathcal{E}_\ell^{(s)}$ in (2.2.11). According to (2.3.4) and (2.3.6), the following holds:

$$\begin{aligned} & f(\mathbf{y}|\mathcal{H}) - f(\tilde{\mathbf{y}}|\mathcal{H}) \\ &= \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \cdot \left[\sum_{e \in \mathcal{E}_\ell} \mathbb{1}_{\{e \text{ is in-cluster w.r.t. } \mathbf{y}\}} - \sum_{e \in \mathcal{E}_\ell} \mathbb{1}_{\{e \text{ is in-cluster w.r.t. } \tilde{\mathbf{y}}\}} \right] \\ &= \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \cdot \left[(|\mathcal{E}_\ell^{(0)}(a)| + |\mathcal{E}_\ell^{(0)}(b)|) - (|\mathcal{E}_\ell^{(\ell-1)}(a)| + |\mathcal{E}_\ell^{(\ell-1)}(b)|) \right] \\ &= W_a + W_b, \end{aligned} \quad (2.3.7)$$

where for each $v \in \mathcal{V}$, the random variable W_v is defined as

$$W_v := \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \cdot \left(\sum_{e \in \mathcal{E}_\ell^{(0)}(v)} \mathcal{A}_e^{(\ell)} - \sum_{e \in \mathcal{E}_\ell^{(\ell-1)}(v)} \mathcal{A}_e^{(\ell)} \right). \quad (2.3.8)$$

Note that the event $\{W_a \leq 0\} \cap \{W_b \leq 0\}$ guarantees $f(\tilde{\mathbf{y}}|\mathcal{H}) \geq f(\mathbf{y}|\mathcal{H})$, which yields the following lower bound on the failure probability $\mathbb{P}_{\text{fail}}$ in (2.3.5):

$$\begin{aligned} \mathbb{P}_{\text{fail}} &\geq \mathbb{P}(\exists a \in \mathcal{V}_+, b \in \mathcal{V}_-, \text{ s.t. } \{W_a \leq 0\} \cap \{W_b \leq 0\}) \\ &= \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+} \{W_a \leq 0\} \bigcap \bigcup_{b \in \mathcal{V}_-} \{W_b \leq 0\}\right). \end{aligned} \quad (2.3.9)$$

However, directly analysis on (2.3.9) is challenging because W_a and W_b are dependent, as there may exist hyperedges containing both $a \in \mathcal{V}_+$ and $b \in \mathcal{V}_-$. Similarly, W_a and $W_{a'}$ are dependent for $a, a' \in \mathcal{V}_+$.

To address this dependency, we further partition the hyperedges involved in W_v and introduce pairwise independent random variables. This allows us to lower bound the failure probability using combinations of these new variables. Specifically, let $\mathcal{U} \subset \mathcal{V}$ be a subset of vertices of size $\gamma_N|\mathcal{V}|$, with $|\mathcal{U} \cap \mathcal{V}_+| = |\mathcal{U} \cap \mathcal{V}_-| = \gamma_N|\mathcal{V}|/2$ and $\gamma_N = o(1)$. For $a \in \mathcal{V}_+ \cap \mathcal{U}$ and $b \in \mathcal{V}_- \cap \mathcal{U}$, define

$$\begin{aligned} &W_a[\mathcal{U}] \quad (2.3.10a) \\ &:= \sum_{\ell \in \mathcal{L}} \log \frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \cdot \left(\sum_{e \in \mathcal{E}_\ell^{(0)}(a)} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\left\{ \begin{smallmatrix} e \cap \mathcal{U} = \{a\} \\ e \setminus \{a\} \subset [\mathcal{V}_+ \setminus \mathcal{U}] \end{smallmatrix} \right\}} - \sum_{e \in \mathcal{E}_\ell^{(\ell-1)}(a)} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\left\{ \begin{smallmatrix} e \cap \mathcal{U} = \{a\} \\ e \setminus \{a\} \subset [\mathcal{V}_- \setminus \mathcal{U}] \end{smallmatrix} \right\}} \right), \end{aligned}$$

$$\begin{aligned} &W_b[\mathcal{U}] \quad (2.3.10b) \\ &:= \sum_{\ell \in \mathcal{L}} \log \frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \cdot \left(\sum_{e \in \mathcal{E}_\ell^{(0)}(b)} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\left\{ \begin{smallmatrix} e \cap \mathcal{U} = \{b\} \\ e \setminus \{b\} \subset [\mathcal{V}_- \setminus \mathcal{U}] \end{smallmatrix} \right\}} - \sum_{e \in \mathcal{E}_\ell^{(\ell-1)}(b)} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\left\{ \begin{smallmatrix} e \cap \mathcal{U} = \{b\} \\ e \setminus \{b\} \subset [\mathcal{V}_+ \setminus \mathcal{U}] \end{smallmatrix} \right\}} \right). \end{aligned}$$

Clearly, after introducing $\mathcal{U}$, the edge sets involved in $W_a[\mathcal{U}]$ and $W_b[\mathcal{U}]$ are disjoint, thus independent. Furthermore, we define the edge set

$$\mathcal{W}_\ell^{(\geq 2)} := \{e \in \mathcal{E}_\ell \mid e \text{ contains at least 2 vertices in } \mathcal{U}\}. \quad (2.3.11)$$

It is clear that $\mathcal{W}_\ell^{(\geq 2)} \subset \mathcal{E}_\ell$. For each $v \in \mathcal{U}$, define the random variable $J_v[\mathcal{U}]$ by

$$J_v[\mathcal{U}] := \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1 - \beta_\ell)}{\beta_\ell(1 - \alpha_\ell)} \right) \cdot \sum_{e \in \mathcal{W}_\ell^{(\geq 2)}} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\{e \ni v, v \in \mathcal{U}\}}. \quad (2.3.12)$$

Then for some $\zeta_N = o(1)$, we define the random variable $J[\mathcal{U}]$ as

$$J[\mathcal{U}] := \bigcap_{v \in \mathcal{U}} \mathbb{1}_{\{J_v[\mathcal{U}] \leq \zeta_N q_N\}}. \quad (2.3.13)$$

Lemma 2.3.4 proves that $W_a[\mathcal{U}]$, $W_b[\mathcal{U}]$ and $J[\mathcal{U}]$ are pairwise independent.

Lemma 2.3.4. *For $a \in \mathcal{V}_+ \cap \mathcal{U}$ and $b \in \mathcal{V}_- \cap \mathcal{U}$, the random variables $W_a[\mathcal{U}]$ and $W_b[\mathcal{U}]$, defined in (2.3.10a) and (2.3.10b) respectively, are independent. Meanwhile, $J[\mathcal{U}]$, defined in (2.3.13), is independent of W_a and W_b . For distinct $a, a' \in \mathcal{V}_+ \cap \mathcal{U}$, W_a and $W_{a'}$ are independent. Furthermore, the following holds*

$$\begin{aligned} \{W_a[\mathcal{U}] \leq -\zeta_N q_N\} \cap \{J[\mathcal{U}] = 1\} &\implies \{W_a \leq 0\}, \\ \{W_b[\mathcal{U}] \leq -\zeta_N q_N\} \cap \{J[\mathcal{U}] = 1\} &\implies \{W_b \leq 0\}, \end{aligned}$$

where the random variables W_v for $v \in \mathcal{V}$ is defined in (2.3.8).

Combining the discussion in (2.3.9) and Lemma 2.3.4, $\mathbb{P}_{\text{fail}}$ can be further bounded by

$$\begin{aligned}
\mathbb{P}_{\text{fail}} &\geq \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+} \{\mathbf{W}_a \leq 0\} \bigcap \bigcup_{b \in \mathcal{V}_-} \{\mathbf{W}_b \leq 0\}\right) \\
&\geq \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+ \cap \mathcal{U}} \{\mathbf{W}_a \leq 0\} \bigcap \bigcup_{b \in \mathcal{V}_- \cap \mathcal{U}} \{\mathbf{W}_b \leq 0\}\right) \\
&\geq \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+ \cap \mathcal{U}} \{\mathbf{W}_a \leq 0\} \bigcap \bigcup_{b \in \mathcal{V}_- \cap \mathcal{U}} \{\mathbf{W}_b \leq 0\} \mid \mathbf{J}[\mathcal{U}] = 1\right) \cdot \mathbb{P}(\mathbf{J}[\mathcal{U}] = 1) \\
&\geq \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+ \cap \mathcal{U}} \{\mathbf{W}_a[\mathcal{U}] \leq -\zeta_N q_N\} \bigcap \bigcup_{b \in \mathcal{V}_- \cap \mathcal{U}} \{\mathbf{W}_b[\mathcal{U}] \leq -\zeta_N q_N\} \mid \mathbf{J}[\mathcal{U}] = 1\right) \cdot \mathbb{P}(\mathbf{J}[\mathcal{U}] = 1) \\
&\geq \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+ \cap \mathcal{U}} \{\mathbf{W}_a[\mathcal{U}] \leq -\zeta_N q_N\}\right) \cdot \mathbb{P}\left(\bigcup_{b \in \mathcal{V}_- \cap \mathcal{U}} \{\mathbf{W}_b[\mathcal{U}] \leq -\zeta_N q_N\}\right) \cdot \mathbb{P}(\mathbf{J}[\mathcal{U}] = 1) \\
&= \left(1 - \prod_{a \in \mathcal{V}_+ \cap \mathcal{U}} \mathbb{P}(\mathbf{W}_a[\mathcal{U}] > -\zeta_N q_N)\right) \cdot \\
&\quad \left(1 - \prod_{b \in \mathcal{V}_- \cap \mathcal{U}} \mathbb{P}(\mathbf{W}_b[\mathcal{U}] > -\zeta_N q_N)\right) \cdot \left(1 - \mathbb{P}(\mathbf{J}[\mathcal{U}] = 0)\right). \tag{2.3.14}
\end{aligned}$$

Below, Lemma 2.3.5 provides the upper bounds for the desired probabilities, which allows us to continue the analysis.

Lemma 2.3.5. *Recall $\mathbb{D}_{\text{GH}}$ in (2.1.5) and $\mathbf{W}_a[\mathcal{U}], \mathbf{J}[\mathcal{U}]$ in (2.3.10a), (2.3.13) respectively. The following holds*

$$\prod_{a \in \mathcal{V}_+ \cap \mathcal{U}} \mathbb{P}(\mathbf{W}_a[\mathcal{U}] > -\zeta_N q_N) \leq \exp\left(-\frac{\gamma_N \cdot |\mathcal{V}|}{2} \cdot e^{-\mathbb{D}_{\text{GH}} \cdot q_N}\right), \tag{2.3.15}$$

$$\mathbb{P}(\mathbf{J}[\mathcal{U}] = 0) \leq \exp\left(\log(\gamma_N N) - q_N \cdot [\zeta_N \log(\zeta_N / \gamma_N) - (\zeta_N - \gamma_N)]\right). \tag{2.3.16}$$

The probability $\prod_{b \in \mathcal{V}_- \cap \mathcal{U}} \mathbb{P}(\mathbf{W}_b[\mathcal{U}] > -\zeta_N q_N)$ admits the same upper bound as in (2.3.15).

Under the condition (2.2.1), there exists some absolute small $\varepsilon > 0$ such that $\mathbb{D}_{\text{GH}} \cdot q_N / \log(N) \leq 1 - \varepsilon$, thus $N e^{-\mathbb{D}_{\text{GH}} \cdot q_N} \geq N^\varepsilon$. Then by choosing $\zeta_N = 1 / \log(q_N)$, $\gamma_N = N^{-\varepsilon/2}$ and

substituting (2.3.15) and (2.3.16) into (2.3.14), we obtain

$$\begin{aligned}
& \mathbb{P}_{\text{fail}} \\
& \geq 1 - 2 \prod_{a \in \mathcal{V}_+ \cap \mathcal{U}} \mathbb{P}(\mathbf{W}_a[\mathcal{U}] > -\zeta_N q_N) \\
& \quad - \mathbb{P}(\mathbf{J}[\mathcal{U}] = 0) - \mathbb{P}(\mathbf{J}[\mathcal{U}] = 0) \cdot \prod_{a \in \mathcal{V}_+ \cap \mathcal{U}} [\mathbb{P}(\mathbf{W}_a[\mathcal{U}] > -\zeta_N q_N)]^2 \\
& \geq 1 - 2 \exp\left(-\frac{\gamma_N \cdot N}{2} \cdot e^{-\mathbb{D}_{\text{GH}} \cdot q_N}\right) - \exp\left(\log(\gamma_N N) - q_N \cdot [\zeta_N \log(\zeta_N/\gamma_N) - (\zeta_N - \gamma_N)]\right) \\
& \quad - \exp\left(\log(\gamma_N N) - q_N \cdot [\zeta_N \log(\zeta_N/\gamma_N) - (\zeta_N - \gamma_N)] - \gamma_N N \cdot e^{-\mathbb{D}_{\text{GH}} \cdot q_N}\right) \\
& \geq 1 - 2 \exp(-N^{\varepsilon/2}/2) \\
& \quad - 2 \exp\left(-q_N \cdot \left[-\left(1 - \frac{\varepsilon}{2}\right) \frac{\log(N)}{q_N} + \frac{\varepsilon \log(N)}{2 \log(q_N)} - \frac{\log \log(q_N) + 1}{\log(q_N)} + N^{-\varepsilon/2}\right]\right) \\
& = 1 - 2 \exp(-N^{\varepsilon/2}/2) \\
& \quad - 2 \exp\left(-q_N \cdot \left[\frac{\varepsilon q_N - (2 - \varepsilon) \log(q_N)}{2 q_N \log(q_N)} \log(N) - \frac{\log \log(q_N) + 1}{\log(q_N)} + N^{-\varepsilon/2}\right]\right)
\end{aligned}$$

where the last term vanishes since $\varepsilon q_N - (2 - \varepsilon) \log(q_N) > 0$ for large N and $1 \ll q_N \lesssim \log(N)$ by Assumption 2.1.6. Consequently, $\mathbb{P}_{\text{fail}} \rightarrow 1$ as $N \rightarrow \infty$, implying that $\hat{\mathbf{y}}_{\text{MLE}}$ in (2.3.3) fails to recover the true assignment $\mathbf{y}$ exactly. Therefore, with probability tending to 1, it is impossible to finish exact recovery, since no other estimator could outperform $\hat{\mathbf{y}}_{\text{MLE}}$. $\square$

2.3.2 Information-Theoretic lower bound on mismatch ratio

The proof of Theorem 2.2.1 analyzes the optimal estimator $\hat{\mathbf{y}}_{\text{MLE}}$ for the full label vector $\mathbf{y}$. In contrast, the proof of Theorem 2.2.3 considers the optimal estimator for each individual label y_v , and derives the expected number of misclassified vertices by estimating the corresponding error probability. For each vertex $v \in \mathcal{V}$, let $\mathbf{y}_{-v} \in \{\pm 1\}^{|\mathcal{V}|-1}$ denote the label vector of nodes other than v . Following the approach in [3], we consider the *genie-aided* estimator, defined as

$$\hat{y}_v^{\text{genie}} = \arg \max_{s=\pm 1} \mathbb{P}(y_v = s | \{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}, \mathbf{y}_{-v}), \quad (2.3.17)$$

where $\mathbb{P}(y_v = s | \{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}, \mathbf{y}_{-v})$ denotes the *posterior* probability conditioned on $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ and $\mathbf{y}_{-v}$. However, the estimator in (2.3.17) is not directly computable or analyzable, since evaluating the posterior probabilities requires access to the true label vector $\mathbf{y}$ in order to compute the likelihood $\mathbb{P}(\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}} | y_v = s, \mathbf{y}_{-v})$, which is unavailable in practice. To address this challenge, Lemma 2.3.6 reformulates the problem in terms of the *log-odds* ratio, which is more tractable for analysis.

Lemma 2.3.6. *Under Assumption 2.1.6, the following holds for each $v \in \mathcal{V}$*

$$\log \left(\frac{\mathbb{P}(y_v = s | \{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}, \mathbf{y}_{-v})}{\mathbb{P}(y_v = -s | \{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}, \mathbf{y}_{-v})} \right) = W_v, \quad s \in \{\pm 1\}, \quad (2.3.18)$$

where the variable W_v is defined in (2.3.8).

Lemma 2.3.7 (Fundamental limit via genie-aided approach, [3, Lemma F.3]). *Let $\mathcal{X}$ be a Borel space and $(\mathbf{y}, \mathbf{X})$ be a random element in $\{\pm 1\}^{|\mathcal{V}|} \times \mathcal{X}$. Let $\mathcal{F}$ be a family of Borel mappings from $\mathcal{X}$ to $\{\pm 1\}^{|\mathcal{V}|}$. Recall the mismatch ratio η_N defined in (2.1.2) for any two $\mathbf{y}, \hat{\mathbf{y}} \in \{\pm 1\}^{|\mathcal{V}|}$. For each $v \in [N]$, let $\mathbb{P}(y_v = s | \mathbf{X}, \mathbf{y}_{-v})$ with $s \in \{\pm 1\}$ denote the posterior probability conditioned on $\mathbf{X} \in \mathcal{X}$ and $\mathbf{y}_{-v} \in \{\pm 1\}^{N-1}$. Then for any estimator $\hat{\mathbf{y}} \in \mathcal{F}$, the following holds*

$$\mathbb{E} \eta_N(\hat{\mathbf{y}}, \mathbf{y}) \geq \frac{N-1}{3N-1} \cdot \frac{1}{N} \sum_{v \in [N]} \mathbb{P} \left(\frac{\mathbb{P}(y_v = s | \mathbf{X}, \mathbf{y}_{-v})}{\mathbb{P}(y_v = -s | \mathbf{X}, \mathbf{y}_{-v})} < 1 \right). \quad (2.3.19)$$

Proof of Theorem 2.2.3. According to Lemmas 2.3.6, 2.3.7, and the fact that vertices are interchangeable, for any estimator $\hat{\mathbf{y}}$, the following holds

$$\mathbb{E} \eta_N(\hat{\mathbf{y}}, \mathbf{y}) \geq \frac{N-1}{3N-1} \cdot \frac{1}{N} \sum_{v \in [N]} \mathbb{P} \left(\frac{\mathbb{P}(y_v = s | \mathbf{X}, \mathbf{y}_{-v})}{\mathbb{P}(y_v = -s | \mathbf{X}, \mathbf{y}_{-v})} < 1 \right) = \frac{N-1}{3N-1} \cdot \mathbb{P}(W_v < 0). \quad (2.3.20)$$

We now consider the probability of the event $\{W_v \leq -\zeta_N q_N\}$, where $\zeta_N = 1/\log(q_N)$. Following

the same LDP analysis as in Lemma A.1.1 and Lemma A.1.2, we have

$$\liminf_{N \rightarrow \infty} q_N^{-1} \log (\mathbb{E} \eta_N(\widehat{\mathbf{y}}, \mathbf{y})) \geq \liminf_{N \rightarrow \infty} q_N^{-1} \log (\mathbb{P}(\mathbf{W}_v < 0)) \quad (2.3.21)$$

$$\geq \liminf_{N \rightarrow \infty} q_N^{-1} \log (\mathbb{P}(\mathbf{W}_v < -\zeta_N q_N)) = -\mathbb{D}_{\text{GH}}, \quad (2.3.22)$$

which completes the proof. $\square$

2.4 Performance analysis of the spectral method

This section is devoted to the proof outlines of Theorems 2.2.4 and 2.2.6 (1) with details deferred to Appendix A.2. We present some preliminaries first.

2.4.1 Preliminary results

For each $\ell \in \mathcal{L}$, let $\mathbf{A}^{(\ell)}$ denote the adjacency matrix obtained from tensor $\mathcal{A}^{(\ell)}$. For $u, v \in \mathcal{V}$, the expected number of ℓ -edges containing u and v can be expressed as

- $y_u = y_v$: $\mathbb{E} \mathbf{A}_{uv}^{(\ell)} = \alpha_\ell \binom{N/2-2}{\ell-2} + \beta_\ell \left[\binom{N-2}{\ell-2} - \binom{N/2-2}{\ell-2} \right]$.
- $y_u \neq y_v$: $\mathbb{E} \mathbf{A}_{uv}^{(\ell)} = \beta_\ell \binom{N/2-2}{\ell-2}$.

For convenience, we denote

$$\alpha = \sum_{\ell \in \mathcal{L}} \left[\alpha_\ell \binom{N/2-2}{\ell-2} + \beta_\ell \left[\binom{N-2}{\ell-2} - \binom{N/2-2}{\ell-2} \right] \right], \quad \beta = \sum_{\ell \in \mathcal{L}} \beta_\ell \binom{N-2}{\ell-2}. \quad (2.4.1)$$

Let $\mathbf{A}^*$ denote the *expected adjacency matrix*, which can be expressed as

$$\mathbf{A}^* := \mathbb{E}[\mathbf{A}] = \sum_{\ell \in \mathcal{L}} \mathbb{E}[\mathbf{A}^{(\ell)}] = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix} \otimes (\mathbf{1}_{N/2} \mathbf{1}_{N/2}^\top) - \alpha \mathbf{I}_N, \quad (2.4.2)$$

Let $\mathbf{u}_2$ and $\mathbf{u}_2^*$ denote the second eigenpair of $\mathbf{A}$ and $\mathbf{A}^*$ respectively. For the proof of Theorem 2.2.4, the central idea is to show that $\sqrt{N} \mathbf{u}_2$ is bounded below from zero by some absolute

constant $\varepsilon > 0$ entrywisely, i.e., $\sqrt{N}y_v u_{2,v} > \varepsilon$ for each $v \in \mathcal{V}$. Similar ideas appeared in [93, 4, 106]. Note that $\mathbf{u}_2^* = N^{-1/2}[\mathbf{1}_{N/2}^\top, -\mathbf{1}_{N/2}^\top]^\top$ and $\text{sign}(u_{2,v}^*) = \text{sign}(y_v)$ for each $v \in \mathcal{V}$. However, $\mathbf{u}_2^*$ is not the ideal candidate for direct comparison, due to the possibility that $\sqrt{N}\|\mathbf{u}_2 - \mathbf{u}_2^*\|_\infty > 1$ as shown in [4, Theorem 3.3]. Instead, Proposition 2.4.1 shows that the ideal pivot vector is $\mathbf{A}\mathbf{u}_2^*/\lambda_2^*$, which is a key step in the proof of Theorem 2.2.4.

Proposition 2.4.1. *Let $(\lambda_2, \mathbf{u}_2)$ (resp. $(\lambda_2^*, \mathbf{u}_2^*)$) be the second eigenpair of $\mathbf{A}$ (resp. $\mathbf{A}^*$). For some absolute constant $C > 0$, the following holds with probability at least $1 - O(N^{-2})$*

$$\sqrt{N} \cdot \min_{s \in \{\pm 1\}} \|s\mathbf{u}_2 - \mathbf{A}\mathbf{u}_2^*/\lambda_2^*\|_\infty \leq C/\log(\rho_N). \quad (2.4.3)$$

Here, ρ_N is the maximum expected degree of the model 2.1.3, formally,

$$\rho_N := \max_{v \in \mathcal{V}} \mathbb{E}[D_v]. \quad (2.4.4)$$

where $D_v := \sum_{j \in \mathcal{V}, j \neq v} A_{vj}$ denotes the degree of vertex $v \in \mathcal{V}$.

Lemma 2.4.2 presents a *Large Deviation Principle* (LDP) analysis of the $\sqrt{N}y_v(\mathbf{A}\mathbf{u}_2^*)_v$, where $\mathbb{D}_{\text{AM}}$ in (2.1.6) serves as the rate function. The LDP analysis is crucial for proving the effectiveness of the spectral method when only the adjacency matrix $\mathbf{A}$ is available.

Lemma 2.4.2. *Define the following function*

$$\mathbb{D}_{\text{AM}}(t) := \sum_{\ell \in \mathcal{L}} \frac{1}{2^{\ell-1}} \left[a_\ell (1 - e^{-(\ell-1)t}) + b_\ell \sum_{r=1}^{\ell-1} \binom{\ell-1}{r} (1 - e^{-(\ell-1-2r)t}) \right] \quad (2.4.5)$$

For any $\tilde{\delta} > 0$, there exists a large N_0 such that when $N \geq N_0$, $\varepsilon_N < \varepsilon$,

$$\mathbb{P}\left(\sqrt{N}y_v(\mathbf{A}\mathbf{u}_2^*)_v \leq \lambda_2^*(\varepsilon_N + C/\log(\rho_N))\right) \leq \exp\left(-q_N \cdot \left(\sup_{t \geq 0} [\delta t + \mathbb{D}_{\text{AM}}(t)] - \tilde{\delta}\right)\right). \quad (2.4.6)$$

Proof of Lemma 2.4.2. For each $v \in \mathcal{V}$, recall A_{vj} in (2.1.4), then

$$\begin{aligned}
\sqrt{N}y_v(\mathbf{A}u_2^*)_v &= y_v \sum_{j \neq v, y_j = +1} A_{vj} - y_v \sum_{j \neq v, y_j = -1} A_{vj} \\
&= \sum_{j \neq v, y_j = y_v} A_{vj} - \sum_{j \neq v, y_j \neq y_v} A_{vj} \\
&= \sum_{j \neq v, y_j = y_v} \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell, e \supset \{v, j\}} \mathcal{A}_e^{(\ell)} - \sum_{j \neq v, y_j \neq y_v} \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell, e \supset \{v, j\}} \mathcal{A}_e^{(\ell)} \\
&= \sum_{\ell \in \mathcal{L}} \sum_{r=0}^{\ell-1} \sum_{e \in \mathcal{E}_\ell^{(r)}(v)} \left((\ell-1-r) \cdot \mathcal{A}_e^{(\ell)} - r \cdot \mathcal{A}_e^{(\ell)} \right) \\
&= \sum_{\ell \in \mathcal{L}} \left(\sum_{e \in \mathcal{E}_\ell^{(0)}(v)} (\ell-1) \cdot \mathcal{A}_e^{(\ell)} + \sum_{r=1}^{\ell-1} \sum_{e \in \mathcal{E}_\ell^{(r)}(v)} (\ell-1-2r) \cdot \mathcal{A}_e^{(\ell)} \right), \\
&= \sum_{\ell \in \mathcal{L}} \left((\ell-1) \cdot \sum_{e \in \mathcal{E}_\ell^{(0)}(v)} \mathcal{A}_e^{(\ell)} + \sum_{r=1}^{\ell-1} (\ell-1-2r) \cdot \sum_{e \in \mathcal{E}_\ell^{(r)}(v)} \mathcal{A}_e^{(\ell)} \right), \tag{2.4.7}
\end{aligned}$$

where the third to last equality follows from the edge classification in (2.2.11), since for each edge $e \in \mathcal{E}_\ell^{(r)}(v)$, it contains r vertices of the opposite label as v with each contributing -1 , and $\ell-1-r$ vertices of the same label with each contributing $+1$.

We then apply Lemma A.1.2 on (2.4.7) to control (2.4.6). Note that $\mathbb{E}\mathcal{A}_e^{(\ell)} = \alpha_\ell$ for $e \in \mathcal{E}_\ell^{(0)}(v)$ and $\mathbb{E}\mathcal{A}_e^{(\ell)} = \beta_\ell$ for $e \in \mathcal{E}_\ell^{(r)}(v)$ with $1 \leq r \leq \ell-1$. Meanwhile, according to (2.4.2) and Assumption 2.1.6, we write

$$\lambda_2^* = \frac{(\alpha - \beta)N}{2} - \alpha \asymp q_N. \tag{2.4.8}$$

We choose $g_N = 1$, $\rho_i^{(\ell)} = 2^{-\ell+1}$ and values of α_i^ℓ, c_i^ℓ as in (2.4.7). According to Lemma A.1.2, for any $0 < \delta < \sum_{\ell \in \mathcal{L}} 2^{-\ell+1} ((\ell-1)a_\ell + \sum_{r=1}^{\ell-1} (\ell-1-2r)b_\ell)$ and $\tilde{\delta} > 0$, there exists some $N_0 > 0$ such that the following holds for all $N \geq N_0$, $\delta_N < \delta$

$$\mathbb{P}\left(\sqrt{N}y_v(\mathbf{A}u_2^*)_v \leq \delta_N q_N\right) = \exp\left(-q_N \left[\sup_{t \geq 0} (-t\delta + \mathbb{D}_{\text{AM}}(t)) - \tilde{\delta}\right]\right).$$

Here, $\mathbb{D}_{\text{AM}}(t)$ is defined in (2.4.5) and we choose $\delta = \lambda_2^* \varepsilon / q_N$ for some absolute small constant $\varepsilon > 0$. $\square$

2.4.2 Proof of Theorem 2.2.4

Let $\hat{s} = \arg \min_{s \in \{\pm 1\}} \|\mathbf{s}\mathbf{u}_2 - \mathbf{A}\mathbf{u}_2^* / \lambda_2^*\|_\infty$. Define the indicator function $F_N = \mathbb{1}_{\{(2.4.3) \text{ holds}\}}$.

Then we have

$$\begin{aligned}
& \mathbb{P}(\eta_N(\mathbf{y}, \text{sign}(\hat{s}\mathbf{u}_2)) = 0) = \mathbb{P}(\text{sign}(\hat{s}\mathbf{u}_2) = \mathbf{y}) \\
& \geq \mathbb{P}(\sqrt{N} \min_{v \in \mathcal{V}} (\hat{s}\mathbf{u}_{2,v} \cdot \mathbf{y}_v) \geq \varepsilon \mid F_N = 1) \cdot \mathbb{P}(F_N = 1) \\
& \geq \mathbb{P}(\sqrt{N} \min_{v \in \mathcal{V}} (\hat{s}\mathbf{u}_{2,v} \cdot \mathbf{y}_v) \geq \varepsilon) - \mathbb{P}(F_N = 0) \\
& \geq \mathbb{P}(\sqrt{N} \min_{v \in \mathcal{V}} (\mathbf{A}\mathbf{u}_2^*)_v / \lambda_2^* - C / \log(\rho_N) \geq \varepsilon) - \mathbb{P}(F_N = 0) \\
& \geq 1 - \sum_{v \in \mathcal{V}} \mathbb{P}(\sqrt{N} (\mathbf{A}\mathbf{u}_2^*)_v \leq \lambda_2^* (\varepsilon + C / \log(\rho_N))) - \mathbb{P}(F_N = 0)
\end{aligned}$$

According to Proposition 2.4.1, $\mathbb{P}(F_N = 0) \lesssim O(N^{-2})$. We take $\delta = \varepsilon \cdot \lambda_2^* / q_N$ from now on. Combining the conclusion from Lemma 2.4.2 where we take $q_N = \log(N)$, for any $\tilde{\delta} > 0$, there exists some large $N_0 > 0$ such that the following holds for all $N \geq N_0$,

$$\mathbb{P}(\eta_N(\mathbf{y}, \text{sign}(\hat{s}\mathbf{u}_2)) = 0) \geq 1 - N^{1 - \sup_{t \geq 0} [-\delta t + \mathbb{D}_{\text{AM}}(t)] + \tilde{\delta}} - N^{-2}, \quad (2.4.9)$$

where $\mathbb{D}_{\text{AM}}(t)$ is defined in (2.4.5). By choosing ε small enough, one can ensure that $1 - \sup_{t \geq 0} [-\delta t + \mathbb{D}_{\text{AM}}(t)] + \tilde{\delta} \leq -\tilde{\delta}$. Therefore, it follows that $\mathbb{P}(\eta_N(\mathbf{y}, \text{sign}(\hat{s}\mathbf{u}_2)) = 0) \rightarrow 1$.

2.4.3 Proof of Theorem 2.2.6 (1)

Let $\hat{s} = \arg \min_{s \in \{\pm 1\}} \|s\mathbf{u}_2 - \mathbf{A}\mathbf{u}_2^*/\lambda_2^*\|_\infty$. By definition,

$$\mathbb{E}\eta_N(\mathbf{y}, \text{sign}(\hat{s}\mathbf{u}_2)) \leq \frac{1}{N} \sum_{v \in \mathcal{V}} \mathbb{P}(\mathbf{y}_v \cdot \hat{s}\mathbf{u}_{2,v} < 0)$$

Define $F_N = \mathbb{1}_{\{(2.4.3) \text{ holds}\}}$. Similarly, according to Proposition 2.4.1, we have

$$\begin{aligned} \mathbb{P}(\mathbf{y}_v \cdot \hat{s}\mathbf{u}_{2,v} < 0) &= \mathbb{P}(\mathbf{y}_v \cdot \hat{s}\mathbf{u}_{2,v} < 0 \mid F_N = 1) \cdot \mathbb{P}(F_N = 1) + \mathbb{P}(\mathbf{y}_v \cdot \hat{s}\mathbf{u}_{2,v} < 0 \mid F_N = 0) \cdot \mathbb{P}(F_N = 0) \\ &\leq \mathbb{P}(\sqrt{N}(\mathbf{A}\mathbf{u}_2^*)_v \leq \lambda_2^*(\epsilon + C/\log(\rho_N))) + O(N^{-2}) \end{aligned}$$

Again, we choose $\delta = \epsilon \cdot \lambda_2^*/q_N$. Then by Lemma 2.4.2 where we take $q_N = \log(N)$, for any $\tilde{\delta} > 0$, there exists some large $N_0 > 0$ such that the following holds for all $N \geq N_0$,

$$\mathbb{P}(\sqrt{N}(\mathbf{A}\mathbf{u}_2^*)_v \leq \lambda_2^*(\epsilon + C/\log(\rho_N))) \leq N^{-\sup_{t \geq 0} [-\delta t + \mathbb{D}_{\text{AM}}(t)] + \tilde{\delta}},$$

where $\mathbb{D}_{\text{AM}}(t)$ is defined in (2.4.5). Combining the above results, we have

$$\mathbb{E}\eta_N(\mathbf{y}, \text{sign}(\hat{s}\mathbf{u}_2)) \leq N^{-\sup_{t \geq 0} [-\delta t + \mathbb{D}_{\text{AM}}(t)] + \tilde{\delta}} + O(N^{-2}).$$

By letting $\epsilon, \tilde{\delta}$ go to zero, the desired result follows since $\mathbb{D}_{\text{AM}} = \sup_{t \geq 0} \mathbb{D}_{\text{AM}}(t)$.

2.5 Refinement algorithms

2.5.1 Majority voting refinement Algorithm 2.2.2

Proof of Theorem 2.2.5. According to Section 4 in [82], the entire vertex set $\mathcal{V}$ can be split into two disjoint subsets $\mathcal{S}$ and $\mathcal{V} \setminus \mathcal{S}$. The vertices in $\mathcal{S}$ enjoy three good properties, such that they can be corrected to their own communities in each step. After $\log(N)$ many iterations, there will be no wrongly classified vertex in $\mathcal{S}$. As for the other subset $\mathcal{V} \setminus \mathcal{S}$, it is proved that $|\mathcal{V} \setminus \mathcal{S}| < 1$

with probability tending to 1 when (2.2.15) holds, thus completing the proof. $\square$

Proof of Theorem 2.2.6 (2). Suppose (2.2.1). The vertices in $\mathcal{S}$ can be corrected by Algorithm 2.2.2 as well. However the upper bound on the size of $\mathcal{V} \setminus \mathcal{S}$ is not as tight as before, bounded by $N^{1-\mathbb{D}_{\text{GH}}}$. Therefore the upper bound on the mismatch ratio $\eta_N(\mathbf{y}, \hat{\mathbf{y}})$ becomes $N^{-\mathbb{D}_{\text{GH}}}$, where in the worst scenario all nodes in $\mathcal{V} \setminus \mathcal{S}$ were wrongly classified. $\square$

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Chapter 3

Optimal and exact recovery on the general non-uniform hypergraph stochastic block model

3.1 Introduction

3.1.1 Formulation of the problems

The *Hypergraph Stochastic Block Model* (HSBM), as a generalization of the SBM, was first introduced in [109], where each edge of the uniform hypergraph appears independently with some given probability. In recent years, many efforts have been made to study community detection problems on random hypergraphs. For exact recovery of uniform HSBMs, the phase transition was shown to occur when the expected degree for each vertex is at least $\log(N)$, and the thresholds were given in [146, 106, 216] by generalizing techniques in [2, 4, 5]. Meanwhile, results on almost exact and partial recovery were shown in [108, 61, 143, 85]. For detection of uniform HSBM, it is conjectured in [22] that the phase transition occurs in the regime of constant expected degrees. [176] and [186] provided algorithms that output a partition better than a random guess when above the *Kesten-Stigum* threshold. [115, 114] proved that detection is impossible and the *Kesten-Stigum* threshold is tight for ℓ -uniform hypergraphs with binary communities when $\ell = 3, 4$, while the KS threshold is not tight when $\ell \geq 7$, and some regimes remain unknown.

It should be noted that most of the results concern community detection on uniform hypergraphs, which require the same number of vertices per edge. This is a constraining and somewhat impractical assumption. As will be demonstrated after Theorem 3.2.9, the thresholds for the non-uniform case show that using the information from all uniform layers yields strictly better results than considering each layer alone. However, the non-uniform HSBM was less explored in the literature, with notable results in [108, 85, 15, 65, 198]. Generally, the results here only considered sufficiency and did not establish fundamental thresholds.

3.1.2 Notation and preliminaries

Let $\mathcal{H} = (\mathcal{H}, \mathcal{E})$ denote the hypergraph, where $\mathcal{V}$ and $\mathcal{E}$ denote the set of vertices and (hyper-)edges respectively. For vertex $v \in \mathcal{V}$, let d_v denote the observed number of hyperedges containing v , with D_v denoting its law. Note that each ℓ -hyperedge $e \ni v$ can be classified into different categories according to the distribution of the other $\ell - 1$ nodes among $\mathcal{V}_1, \dots, \mathcal{V}_K$, then D_v can be written as a summation of independent random variables due to edge-wise independence,

$$D_v := \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W} C_K^{(\ell-1)}} D_{v, \mathbf{w}}^{(\ell)}, \quad (3.1.1)$$

where $D_{v, \mathbf{w}}^{(\ell)}$ denotes the law of $d_{v, \mathbf{w}}^{(\ell)}$, counting the number of ℓ -hyperedges containing v with the other $\ell - 1$ nodes distributed as $\mathbf{w}$ among $\mathcal{V}_1, \dots, \mathcal{V}_K$. Assume that v belongs to the community $\mathcal{V}_k$, that is, $Y_v = k$, then the distribution of membership among the blocks in e is denoted by $k \oplus \mathbf{w} := (w_1, \dots, w_{k-1}, w_k + 1, w_{k+1}, \dots, w_K)$. According to model 1.1.8, $D_{v, \mathbf{w}}^{(\ell)}|_{Y_v=k} \sim \text{Bin}(N_{\mathbf{w}}, Q_{k \oplus \mathbf{w}}^{(\ell)})$, where for each $\mathbf{w} = (w_1, \dots, w_K) \in \mathcal{W} C_K^{(\ell-1)}$, and the capacity of such ℓ -hyperedges is denoted by

$$N_{\mathbf{w}} := \prod_{k=1}^K \binom{|\mathcal{V}_k|}{w_k} \quad (3.1.2)$$

Lemma 3.1.1 indicates that $\bar{N}_{\mathbf{w}}$ in (3.2.4) is a good approximation of $N_{\mathbf{w}}$.

Lemma 3.1.1. *With probability at least $1 - N^{-\log(N)}$, $||\mathcal{V}_j| - \alpha_j N| \leq \sqrt{N} \log(N)$ for each $j \in [K]$.*

Recall $N_{\mathbf{w}}$ from (3.1.2) and $\bar{N}_{\mathbf{w}}$ from (3.2.4). Then for each for $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$, it follows

$$(1 - \log(N)/\sqrt{N})^{\ell-1} \leq N_{\mathbf{w}}/\bar{N}_{\mathbf{w}} \leq (1 + \log(N)/\sqrt{N})^{\ell-1}. \quad (3.1.3)$$

3.1.3 Organization

We present our main results in Section 3.2. Section 3.3 contains the proof of Theorem 3.2.4. The proofs of Theorems 3.2.6, 3.2.12, Theorems 3.2.6, 3.2.10 and Theorems 3.2.13, 3.2.14 are in Sections 3.4, 3.5 and 3.6 respectively. Some open problems will be addressed in Section 3.8.

3.2 Main results

Our results are multi-fold, and work for the general, non-uniform HSBM case. All these results are new and optimal for non-uniform HSBM.

1. We prove a wide-ranging, Information-Theoretic lower bound on the number of misclassified vertices *for any algorithm*, depending on a generalized Chernoff-Hellinger distance involving model parameters (Theorem 3.2.4 (2)).
2. The necessary condition and threshold for exact recovery are derived in Theorem 3.2.4 (1).
3. We present an efficient Algorithm 3.2.4 which, above this threshold, with full information about the model, *achieves* exact recovery for all but a negligible subset of the space of problems (Theorem 3.2.13).
4. We present an efficient Algorithm 3.2.1 which, also above the threshold, *achieves* exact recovery when only the number of communities is known, provided that the edge probabilities are proportional (Theorem 3.2.9).

5. Below this threshold, we prove that the provided algorithms are *optimal* in Theorems 3.2.10 and 3.2.14, meaning that the number of vertices they mislabeled reaches is *at most* the lower bound determined by the Information-Theoretical limits.

3.2.1 Regime of interests

For ease of presentation, we rewrite the notations to accommodate different regimes. Define a *finite* family $\mathcal{F}$ of growth rates $q : \mathbb{N} \mapsto \mathbb{R}_+$ with $\lim_{N \rightarrow \infty} q(N) = \infty$, such that

$$\forall q_1, q_2 \in \mathcal{F}, \quad \lim_{N \rightarrow \infty} q_1(N)/q_2(N) \in \{0, +\infty\}. \quad (3.2.1)$$

This means that $\mathcal{F}$ is a family of growth rates of different magnitudes.

Definition 3.2.1 (Edge density). *For model 1.1.8, we write the probability of sampling edge $e = \{i_1, \dots, i_\ell\}$ as $\mathbb{P}(\mathcal{A}_e^{(\ell)} = 1) = Q_{\mathbf{y}(e)}^{(\ell)} = Q_{\mathbf{w}}^{(\ell)}$, where $\mathbf{y}(e) := [y_{i_1}, \dots, y_{i_\ell}]$ is the membership vector of the vertices in e , and $\mathbf{w} \in \mathcal{WC}_K^{(\ell)}$ denotes the weak composition formulated by $\mathbf{y}(e)$. Furthermore, $Q_{\mathbf{w}}^{(\ell)}$ can be factorized as*

$$\mathbb{P}(\mathcal{A}_e^{(\ell)} = 1) = Q_{\mathbf{w}}^{(\ell)} = \mathcal{P}_{\mathbf{w}}^{(\ell)} \cdot q_{\mathbf{w}}^{(\ell)} / \binom{N-1}{\ell-1}, \quad (3.2.2)$$

where $\mathcal{P}_{\mathbf{w}}^{(\ell)}$ is some constant independent of N , and $q_{\mathbf{w}}^{(\ell)} \in \mathcal{F}$. Let q_N denote the slowest rate function among all $\{q_{\mathbf{w}}^{(\ell)}\}_{\mathbf{w} \in \mathcal{WC}_K^{(\ell)}, \ell \in \mathcal{L}}$.

Assumption 3.2.2. *There exists some absolute constant $C_{(3.2.3)} > 1$ such that for every N , the probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ in model 1.1.8 satisfy*

$$C_{(3.2.3)} := \max_{\ell \in \mathcal{L}} \max_{\mathbf{w}, \mathbf{w}' \in \mathcal{WC}_K^{(\ell)}} Q_{\mathbf{w}}^{(\ell)} / Q_{\mathbf{w}'}^{(\ell)}, \quad (3.2.3)$$

which further implies $q_{\mathbf{w}}^{(\ell)} = q_N$ for each $\mathbf{w} \in \mathcal{WC}_K^{(\ell)}$ and $\ell \in \mathcal{L}$.

3.2.2 Information-Theoretic limits

We introduce the *generalized Chernoff-Hellinger* (GCH) divergence to establish the *Information-Theoretic* limits.

Definition 3.2.3. For each $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$, denote $k \oplus \mathbf{w} := (w_1, \dots, w_{k-1}, w_k + 1, w_{k+1}, \dots, w_K)$ and define the following quantity

$$\bar{N}_{\mathbf{w}} := \prod_{j=1}^K \binom{\lfloor \alpha_j N \rfloor}{w_j}. \quad (3.2.4)$$

Under Assumption 3.2.2, define the Generalized Chernoff-Hellinger (GCH) divergence by

$$\begin{aligned} \mathbb{D}_{\text{GCH}} &= \min_{1 \leq j < k \leq K} \mathbb{D}_{\text{GCH}}(j, k), \\ \mathbb{D}_{\text{GCH}}(j, k) &:= \max_{t \in [0, 1]} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \frac{\bar{N}_{\mathbf{w}}}{\binom{N-1}{\ell-1}} [t \mathcal{P}_{j \oplus \mathbf{w}}^{(\ell)} + (1-t) \mathcal{P}_{k \oplus \mathbf{w}}^{(\ell)} - (\mathcal{P}_{j \oplus \mathbf{w}}^{(\ell)})^t \cdot (\mathcal{P}_{k \oplus \mathbf{w}}^{(\ell)})^{1-t}]. \end{aligned} \quad (3.2.5)$$

For the more general scenario, where Assumption 3.2.2 is no longer true, we define

$$\begin{aligned} \mathbb{D}_{\text{GCH}} &= \min_{1 \leq j < k \leq K} \mathbb{D}_{\text{GCH}}(j, k) \\ \mathbb{D}_{\text{GCH}}(j, k) &:= q_N^{-1} \max_{t \in [0, 1]} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} [t Q_{j \oplus \mathbf{w}}^{(\ell)} + (1-t) Q_{k \oplus \mathbf{w}}^{(\ell)} - (Q_{j \oplus \mathbf{w}}^{(\ell)})^t \cdot (Q_{k \oplus \mathbf{w}}^{(\ell)})^{1-t}]. \end{aligned} \quad (3.2.6)$$

Note that $\mathbb{D}_{\text{GCH}}$ in (3.2.5) is a special case of (3.2.6) under Assumption 3.2.2. When $C_{(3.2.3)} = 1$, leading to $\mathbb{D}_{\text{GCH}} = 0$, HSBM is indistinguishable from a non-uniform Erdős-Rényi hypergraph. Therefore, the partition is undetectable.

The success of exact recovery is hindered by isolated vertices, which appear when $q_N = O(\log(N))$, as no algorithm could outperform random guess in the presence of isolated vertices in multiple communities. The necessary condition for exact recovery and Information-Theoretic lower bound for any algorithm are presented below.

Theorem 3.2.4. *Let $\{\kappa_N\}_N$ be a sequence such that $\kappa_N \in (0, 1]$ for each N , and suppose*

$$\lim_{N \rightarrow \infty} \mathbb{D}_{\text{GCH}} \cdot q_N / (\kappa_N \cdot \log(N)) < 1. \quad (3.2.7)$$

Then, given any community detection algorithm $\mathfrak{A}$, the following holds.

(1) With probability at least $1 - 2^{-q_N}$, $\mathfrak{A}$ will misclassify at least one vertex.

(2) When $\kappa_N \log(N) \rightarrow \infty$ as $N \rightarrow \infty$, it follows that $\lim_{N \rightarrow \infty} (N^{\kappa_N} \cdot \mathbb{E}[\eta_N]) \geq 1$.

Remark 3.2.5. *Informally, (2) states that any algorithm $\mathfrak{A}$ is expected to misclassify at least $N^{1-\kappa_N}$ nodes, with expectation over the sample hypergraph.*

The necessary condition for exact recovery becomes $\mathbb{D}_{\text{GCH}} < 1$ by taking $q_N = \log(N)$ and $\kappa_N = 1$ in (3.2.7). For SBM (graph) and uniform HSBM, it was shown that exact recovery is impossible when $\mathbb{D}_{\text{GCH}} < 1$ in [5] and [216], respectively, both under Assumption 3.2.2. By contrast, Theorem 3.2.4 (1) is broader and its proof does not rely on Assumption 3.2.2.

Theorem 3.2.4 (2) represents a real breakthrough in the literature of HSBM. It states the lower bound on the expected mismatch ratio in regimes where exact recovery is impossible. Here are two examples: first, when $\kappa_N = 1$, $q_N = \log(N)$, $\mathbb{E}[\eta_N] \geq N^{-(1+o(1))\mathbb{D}_{\text{GCH}}}$; second, when $1 \ll \mathbb{D}_{\text{GCH}} \cdot q_N \ll \log(N)$ while $\kappa_N \rightarrow 0$ but $\kappa_N \log(N) \rightarrow \infty$, a lower bound on weak consistency is established. Similar results were previously shown for SBM in the binary [214, 4, 3] and multi-community [212] cases. Note also that [104, 105] (SBM) and [61] (uniform HSBM) established minimax rates for mismatch ratio when certain constraints on the probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ are satisfied, which describes a subset of the entire problem space. By contrast, Theorem 3.2.4 (2) is free of such constraints.

3.2.3 Agnostic partition

Under Assumption 3.2.2, Algorithm 3.2.1 could achieve the exact recovery and reach the lowest expected mismatch ratio possible without knowing the probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$. The number of communities K is currently required as input. However, as will be discussed

in [83, Section 3.7], one could estimate K and turn Algorithm 3.2.1 into an algorithm *entirely agnostic*.

There are two stages in Algorithm 3.2.1: first, Algorithm 3.2.2 assigns all but a vanishing fraction of vertices correctly (weak consistency), then Algorithm 3.2.3 refines the partition iteratively to reach the lowest possible mismatch ratio. We present Algorithms 3.2.2 and 3.2.3 here, while Algorithm 3.4.1 is deferred to Section 3.4.

Algorithm 3.2.1: Agnostic partition

Data: The adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, number of communities K .

- 1 Run **Algorithm 3.4.1 (Trimming)** on $\mathcal{H}$ with $\mathcal{J} := \mathcal{J}_1$ in Theorem 3.4.4 to obtain $\mathbf{A}_{\mathcal{J}_1}$;
- 2 Run **Algorithm 3.2.2 (Initialization)** with input $\mathbf{A}_{\mathcal{J}_1}, K, \bar{d}$ to obtain the initial estimate $\hat{\mathbf{y}}^{(0)}$;
- 3 Run **Algorithm 3.2.3 (Refinement)** with input $\hat{\mathbf{y}}^{(0)}, K, \{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ and obtain the output $\hat{\mathbf{y}}$;

Result: $\hat{\mathbf{y}}$

Stage I

Let d_v denote the observed number of hyperedges containing v , and $\bar{d}$ denote the average degree of $\mathcal{H}$. Figure 3.1 is an intuitive explanation of Algorithm 3.2.2.

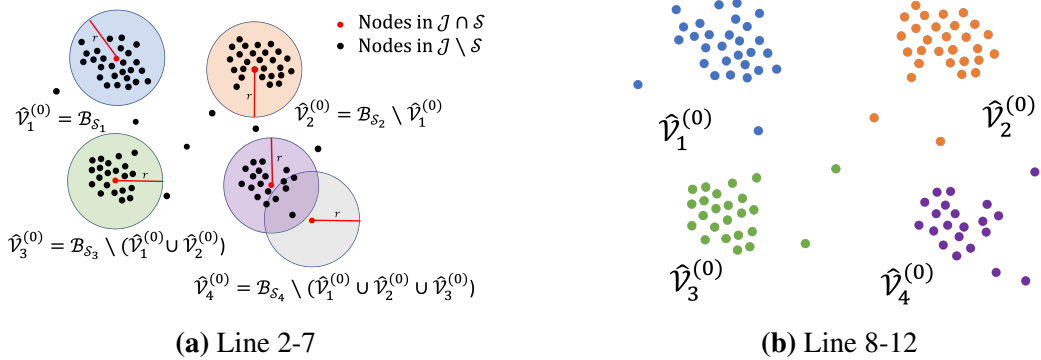


Figure 3.1. Illustration of Algorithm 3.2.2

Theorem 3.2.6 (Agnostic weak consistency). *Let ρ_N denote the maximum expected degree among all vertices, see (3.4.1). Suppose Assumptions 3.2.2, 3.2.7 and $\rho_N \gtrsim \log(N)$, then with probability at least $1 - 2(e/2)^{-N} - 2N^{-c}$ for some constant $c > 0$, Algorithm 3.2.2 achieves weak consistency by applying regularization $\mathcal{J}_1$ in Theorem 3.4.4.*

Algorithm 3.2.2: Spectral initialization

Data: $\mathbf{A}_g$, number of communities K , radius $r = \bar{r}$, where $\bar{r} = [N \log(\bar{d})]^{-1} \bar{d}^2$ with

$$\bar{d} = \sum_{v=1}^N d_v / N.$$

- 1 Compute the rank- K approximation $\mathbf{A}_g^{(K)} = \sum_{i=1}^K \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$ of $\mathbf{A}_g$.
- 2 Let $\mathcal{S}$ be a set of $\lceil 2 \log^2(N) \rceil$ nodes randomly sampled from $\mathcal{J}$ without replacement.
For each $s \in \mathcal{S}$, construct the ball centered at s by

$$\mathcal{B}_r(s) = \{w \in \mathcal{J} : \|(\mathbf{A}_g^{(K)})_{s:} - (\mathbf{A}_g^{(K)})_{w:}\|_2^2 \leq r\}.$$
- 3 Take $\hat{\mathcal{V}}_1^{(0)} = \mathcal{B}_r(s_1)$ to be the ball with most vertices, i.e., $s_1 = \arg \max_{s \in \mathcal{S}} |\mathcal{B}_r(s)|$.

Break ties arbitrarily.

- 4 **while** $2 \leq k \leq K$ **do**
 - 5 $s_k = \arg \max_{s \in \mathcal{S}} |\mathcal{B}_r(s) \setminus (\bigcup_{j=1}^{k-1} \hat{\mathcal{V}}_j^{(0)})|;$
 - 6 $\hat{\mathcal{V}}_k^{(0)} = \mathcal{B}_r(s_k) \setminus (\bigcup_{j=1}^{k-1} \hat{\mathcal{V}}_j^{(0)})$
- 7 **end**
- 8 **while** $v \in \mathcal{J} \setminus (\bigcup_{k=1}^K \hat{\mathcal{V}}_k^{(0)})$ **do**
 - 9 $k = \arg \min_{k \in [K]} \|(\mathbf{A}_g^{(K)})_{s_k:} - (\mathbf{A}_g^{(K)})_{v:}\|_2^2;$
 - 10 $\hat{\mathcal{V}}_k^{(0)} \leftarrow \hat{\mathcal{V}}_k^{(0)} \cup \{v\}$
- 11 **end**

- 12 Randomly assign the remaining vertices $v \in \mathcal{V} \setminus \mathcal{J}$ to one of the communities

$$\hat{\mathcal{V}}_1^{(0)}, \dots, \hat{\mathcal{V}}_K^{(0)};$$

- 13 Obtain the initial estimate of the membership vector $\hat{\mathbf{y}}^{(0)}$ based on $\hat{\mathcal{V}}_1^{(0)}, \dots, \hat{\mathcal{V}}_K^{(0)}$.

Result: $\hat{\mathbf{y}}^{(0)}$

Assumption 3.2.7. For each $\mathbf{w} \in \mathcal{WC}_K^{(\ell-2)}$, let

$$j \oplus i \oplus \mathbf{w} := (w_1, \dots, w_j + 1, \dots, w_i + 1, \dots, w_K) \quad (3.2.8)$$

denote a weak composition in $\mathcal{WC}_K^{(\ell)}$. For every distinct pair $j, k \in [K]$, there exists some $i \in [K]$ such that $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ satisfies

$$\lim_{N \rightarrow \infty} \frac{N}{\rho_N} \cdot \left| \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-2)}} \bar{N}_{\mathbf{w}} \cdot (Q_{j \oplus i \oplus \mathbf{w}}^{(\ell)} - Q_{k \oplus i \oplus \mathbf{w}}^{(\ell)}) \right| \neq 0. \quad (3.2.9)$$

Remark 3.2.8. Denote $\alpha^{\mathbf{w}} := \prod_{j=1}^K (\alpha_j)^{w_j}$ for $\mathbf{w} = (w_1, \dots, w_K)$. Under Assumption 3.2.2, if there exists some distinct pair $j, k \in [K]$ such that for every $i \in [K]$,

$$\sum_{\ell \in \mathcal{L}} (\ell - 1) \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-2)}} \binom{\ell-2}{\mathbf{w}} \cdot \alpha^{\mathbf{w}} \cdot (\mathcal{P}_{j \oplus i \oplus \mathbf{w}}^{(\ell)} - \mathcal{P}_{k \oplus i \oplus \mathbf{w}}^{(\ell)}) = 0, \quad (3.2.10)$$

where $\binom{\ell-2}{\mathbf{w}} := (\ell-2)! / \prod_{j=1}^K w_j!$ with $\sum_{j=1}^K w_j = \ell-2$ denotes the multinomial coefficient, then Assumption 3.2.7 is violated. The set of tensors satisfying (3.2.10) is a linear proper subspace of the entire parameter space. In the absence of Assumption 3.2.2, Assumption 3.2.7 says that the expected degrees of each community, as well as the differences between communities, have to be proportional to the highest expected degree (potentially with the exception of one community). It remains true that, under HSBM, the set of problems that cannot be solved by Algorithm 3.2.2 is asymptotically negligible in the entire problem space.

Assumption 3.2.7 is a natural extension of the condition in [5] for the graph case $\mathcal{L} = \{2\}$, and it also appears in [216] under Assumption 3.2.2. However, while in the graph case this condition is truly necessary for achievability (since its absence creates two classes that are indistinguishable from each other), it is not clear whether this is the hypergraph case for Assumption 3.2.7. It is certainly needed for any method using the adjacency matrix to succeed, but, as shown in [216], due to the many parameters involved, one may construct a model with

distinguishable classes via probability tensors (Figure 1.2 (b)), but for which the expected $K \times K$ adjacency matrix has identical rows (this uniform case is yet to be solved).

Stage II

Let $\{\hat{\mathcal{V}}_k^{(t)}\}_{k=1}^K$ denote the partitioned communities in iteration t with the initial partition $\{\hat{\mathcal{V}}_k^{(0)}\}_{k=1}^K$ obtained from Algorithm 3.2.2. For each $\ell \in \mathcal{L}$ and $\mathbf{w} \in \mathcal{WC}_K^{(\ell)}$, let

$$\hat{Q}_{\mathbf{w}}^{(\ell)} := \frac{1}{\hat{N}_{\mathbf{w}}^{(0)}} \sum_{\hat{\mathbf{y}}^{(0)}(e)=\mathbf{w}} \mathcal{A}_e^{(\ell)}, \quad \text{where } \hat{N}_{\mathbf{w}}^{(t)} = \prod_{j=1}^K \binom{|\hat{\mathcal{V}}_j^{(t)}|}{w_j}, \quad \forall t \geq 0, \quad (3.2.11)$$

be an estimation of $Q_{\mathbf{w}}^{(\ell)}$ in (3.2.2). Let $\hat{D}_{v,\mathbf{w}}^{(t)}$ denote the number of ℓ -hyperedges containing v with the rest nodes distributed as $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$ at iteration $t \geq 0$, formally,

$$\hat{D}_{v,\mathbf{w}}^{(t)} := |\{e \subset [\mathcal{V}]^\ell \mid e \ni v, \mathbf{y}(e \setminus \{v\}) = \mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}, w_j \text{ nodes in } \hat{\mathcal{V}}_j^{(t)}\}|. \quad (3.2.12)$$

We provide an intuitive explanation of Algorithm 3.2.3 here. Suppose $v \in \mathcal{V}_k$ and define $\hat{\mu}_{v,\mathbf{w}}^{(t)} := \hat{D}_{v,\mathbf{w}}^{(t)} / \hat{N}_{\mathbf{w}}^{(t)}$, which is an estimation of $Q_{k \oplus \mathbf{w}}^{(\ell)}$ only using edges containing v . In contrast, all edges of type $\hat{\mathbf{y}}^{(0)}(e) = k \oplus \mathbf{w}$ are used when computing $\hat{Q}_{\mathbf{w}}^{(\ell)}$, which is a good approximation of $Q_{\mathbf{w}}^{(\ell)}$ by Lemma 3.5.1. Let $\mathbb{H}_{\text{CE}}(y, \hat{y}) := -y \log \hat{y} - (1-y) \log(1-\hat{y})$ denote the *cross entropy* between the two probability distributions with means y and $\hat{y}$, respectively. Then (3.2.15) can be reformulated as

$$k^* = \arg \min_{1 \leq k \leq K} \left\{ \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \hat{N}_{\mathbf{w}}^{(t)} \cdot \left[-\hat{\mu}_{v,\mathbf{w}}^{(t)} \cdot \log \hat{Q}_{k \oplus \mathbf{w}}^{(\ell)} - (1 - \hat{\mu}_{v,\mathbf{w}}^{(t)}) \cdot \log(1 - \hat{Q}_{k \oplus \mathbf{w}}^{(\ell)}) \right] \right\} \quad (3.2.13)$$

$$= \arg \min_{1 \leq k \leq K} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \hat{N}_{\mathbf{w}}^{(t)} \cdot \mathbb{H}_{\text{CE}}(\hat{\mu}_{v,\mathbf{w}}^{(t)}, \hat{Q}_{k \oplus \mathbf{w}}^{(\ell)}). \quad (3.2.14)$$

Note that $\mathbb{H}_{\text{CE}}(y, \hat{y}) = 0$ if and only if $y = \hat{y}$, and it would be relatively large when v is misclassified. Thus, the correct assignment of v can be found by minimizing the weighted cross-entropy terms.

The theoretical guarantee is provided below.

Algorithm 3.2.3: Agnostic refinement

Data: $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}, \hat{\mathbf{y}}^{(0)}$.

- 1 For each $\ell \in \mathcal{L}$, $\mathbf{w} \in \mathcal{WC}_K^{(\ell)}$, calculate $\hat{Q}_{\mathbf{w}}^{(\ell)}$ in (3.2.11).
- 2 **while** $0 \leq t \leq \lceil \log(N) \rceil$ **do**
- 3 $\hat{\mathcal{V}}_k^{(t+1)} \leftarrow \emptyset$ for all $k \in [K]$;
- 4 **while** $v \in \mathcal{V}$ **do**
- 5 For each $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$, calculate $\hat{D}_{v,\mathbf{w}}^{(t)}$ in (3.2.12). Find

$$k^* = \arg \max_{1 \leq k \leq K} \left\{ \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \hat{D}_{v,\mathbf{w}}^{(t)} \cdot \log \hat{Q}_{k \oplus \mathbf{w}}^{(\ell)} + (\hat{N}_{\mathbf{w}}^{(t)} - \hat{D}_{v,\mathbf{w}}^{(t)}) \cdot \log(1 - \hat{Q}_{k \oplus \mathbf{w}}^{(\ell)}) \right\},$$

(3.2.15)
- 6 Tie broken uniformly at random;
- 7 $\hat{\mathcal{V}}_{k^*}^{(t+1)} \leftarrow \hat{\mathcal{V}}_{k^*}^{(t+1)} \cup \{v\}$.
- 8 **end**
- 9 $t \leftarrow t + 1$
- 10 **end**
- 11 Take $\hat{\mathcal{V}}_k = \hat{\mathcal{V}}_k^{(\lceil \log(N) \rceil)}$ for all $k \in [K]$ and obtain $\hat{\mathbf{y}}$.
- Result:** $\hat{\mathbf{y}}$

Theorem 3.2.9 (Agnostic strong consistency). *For some absolute $\epsilon > 0$, suppose*

$$\lim_{N \rightarrow \infty} \mathbb{D}_{\text{GCH}} \cdot q_N / \log(N) \geq 1 + \epsilon, \quad (3.2.16)$$

together with Assumptions 3.2.2 and 3.2.7. Then with probability at least $1 - 6N^{-\epsilon}$, Algorithm 3.2.1 achieves exact recovery without prior knowledge of $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$.

When $q_N = \log(N)$, i.e. $\mathbb{D}_{\text{GCH}} > 1$, [8, 212] (SBM) achieved exact recovery agnostically. By contrast, [216] completed exact recovery on uniform HSBM by requiring prior knowledge of probability tensors. Therefore, our Theorem 3.2.9 is more general than any other result in the current HSBM literature.

In addition, (3.2.16) indicates that the structural information of the non-uniform hypergraph is stronger than the uniform one. Exact recovery may be achievable when all uniform

layers are aggregated, even if it is impossible when any of the uniform layers is used in isolation. An example with numerical experiments is discussed in [198].

Optimality

When (3.2.16) does not hold, strong consistency is impossible as per Theorem 3.2.4 (1), and one can only expect weak consistency where the lower bound for the expected mismatch ratio was derived in Theorem 3.2.4 (2). A natural question arises here: can any algorithm achieve this lower bound? The answer is yes: Algorithm 3.2.3 can refine the first stage result to reach the lowest possible expected mismatch ratio.

Theorem 3.2.10 (Agnostic optimality). *For the sequence $\kappa_N \in (0, 1]$ with $\kappa_N \log(N) \rightarrow \infty$ as $N \rightarrow \infty$ and some absolute (small) $\varepsilon > 0$, suppose Assumptions 3.2.2, 3.2.7 hold, $\rho_N \gtrsim \log(N)$ and*

$$\lim_{N \rightarrow \infty} \mathbb{D}_{\text{GCH}} \cdot q_N / (\kappa_N \cdot \log(N)) \geq 1 + \varepsilon. \quad (3.2.17)$$

Then with probability at least $1 - 6e^{-\varepsilon \cdot \kappa_N \log(N)}$, Algorithm 3.2.1 satisfies $\lim_{N \rightarrow \infty} (N^{\kappa_N} \cdot \eta_N) \leq 1$.

3.2.4 Partition with prior knowledge

When the probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ are known, the regime beyond Assumption 3.2.2 can be handled by Algorithm 3.2.4, where the optimal membership assignment for each $v \in \mathcal{V}$ is determined through a *leave-one-out* pipeline.

The essential idea of Algorithm 3.2.4 is to make full use of the Maximum A Posteriori (MAP) estimation. We provide an intuitive explanation of the algorithm below. Let $\mathbf{y}_{-v} \in [K]^{N-1}$ denote the membership vector of the vertices other than v and $\hat{\mathbf{y}}_{-v}$ be an estimation of $\mathbf{y}_{-v}$. Let $\mathbb{H}$ and $\mathbf{Y}_{-v}$ denote the laws of $\mathcal{H}$ and $\mathbf{y}_{-v}$ respectively. Conditioned on the events $\{\mathbb{H} = \mathcal{H}\}$ and $\{\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}\}$, one can prove (deferred to Section 3.6) that the probability of v being misclassified

Algorithm 3.2.4: Partition with prior knowledge

Data: $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, K , probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$, $\{\alpha_k\}_{k=1}^K$.
1 while $v \in \mathcal{V}$ **do**
 2 Obtain $\mathcal{H}_v$ by removing all edges containing v from $\mathcal{H}$;
 3 Run **Algorithm 3.2.2 (Initialization)** on $\mathcal{H}_v$ to obtain the estimation $\hat{\mathbf{y}}_{-v}^{(0)}$ for $\mathbf{y}_{-v}$;
 4 $\hat{\mathbf{y}}_v = \arg \max_{k \in [K]} \mathbb{P}(\mathbf{D}_v = d_v | \mathbf{Y}_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}^{(0)}) \cdot \alpha_k$.
5 end
Result: $\hat{\mathbf{y}}$

is minimized when the following MAP estimation is applied

$$\hat{\mathbf{y}}_v^{\text{MAP}} := \arg \max_{k \in [K]} \mathbb{P}(\mathbf{Y}_v = k \mid \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}). \quad (3.2.18)$$

Similar ideas appeared in [5, 1] for graph SBM, and [61, 216] for uniform HSBM. However, the posterior probability in (3.2.18) cannot be directly computed from the definition. Recall that d_v is the observed number of hyperedges in $\mathcal{H}$ containing v with $\mathbf{D}_v$ denoting its law. We propose (3.2.19), an estimator equivalent to (3.2.18) (as proved by Lemma 3.2.11), which can be computed directly when knowing $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ and $\{\alpha_k\}_{k=1}^K$.

Lemma 3.2.11. *When the estimation $\hat{\mathbf{y}}_{-v}$ is independent of the membership assignment $\mathbf{y}_v$, the MAP in (3.2.18) can be reformulated as follows*

$$\hat{\mathbf{y}}_v^{\text{MAP}} = \arg \max_{k \in [K]} \mathbb{P}(\mathbf{D}_v = d_v \mid \mathbf{Y}_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbf{Y}_v = k). \quad (3.2.19)$$

In Line 3, the initial estimation $\hat{\mathbf{y}}_{-v}^{(0)}$ is obtained by running Algorithm 3.2.2 on $\mathcal{H}_v$. It is independent of the membership assignment for the vertex v since no edges containing v were involved throughout the process. Therefore by Lemma 3.2.11, the $\hat{\mathbf{y}}_v$ in Line 4 is equivalent to the best estimator MAP in (3.2.19).

In [5, 1], the authors presented a two-stage strategy, where the graph $\mathcal{H}$ is divided into

$\mathcal{H}^{(0)}$ and $\mathcal{H}^{(1)}$. In the first stage, the sparser graph $\mathcal{H}^{(0)}$ is used to obtain the initial estimation $\hat{\mathbf{y}}^{(0)}$, then $\hat{\mathbf{y}}_{-v}^{(0)}$ is obtained by removing $\hat{\mathbf{y}}_v^{(0)}$. In the second stage, (3.2.18) is applied on the denser graph $\mathcal{H}^{(1)}$ for local refinement. We have found their proof difficult to duplicate for the hypergraph case. In [216], the authors presented a similar strategy (complexity KN^2) and claimed (without proof) that it can be modified to work for the hypergraph. For simplicity and completion, we have chosen instead to go with a slightly slower Algorithm 3.2.4 (complexity KN^3), which forgoes the splitting but ensures the independence between $\mathbf{y}_v$ and $\hat{\mathbf{y}}_{-v}$.

Stage I

The agnostic result Theorem 3.2.6 requires $\rho_N \gtrsim \log(N)$. However, with prior knowledge, Algorithm 3.2.2 can still achieve weak consistency as long as $\rho_N = \omega(1)$.

Theorem 3.2.12 (Weak consistency with prior knowledge). *Suppose Assumption 3.2.7 and $\rho_N = \omega(1)$. If we know $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ and apply $\mathcal{J}_2$ with $r = [N \log(\rho_N)]^{-1} \rho_N^2$ in Theorem 3.4.4, Algorithm 3.2.2 achieves weak consistency with probability $1 - 2(e/2)^{-N} - N^{-10}$.*

With knowledge of the probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$, Assumption 3.2.7 can be weakened. It is equivalent to saying that no two rows of the expected $K \times K$ adjacency matrix for the communities, when scaled by the highest degree, are asymptotically equal, as in Remark 3.2.8. In fact, this can be boiled down to an "all the layers" condition, where (3.2.9) happens on a uniform layer by uniform layer basis. The set satisfying (3.2.10) can be shrunk as follows. When Assumption 3.2.7 fails for the non-uniform hypergraph but holds true for at least one uniform layer, we can construct a weighted adjacency matrix $\tilde{\mathbf{A}} = \sum_{\ell \in \mathcal{L}} w_\ell \mathbf{A}^{(\ell)}$ as an input for Algorithm 3.2.2 under some properly chosen weights $\{w_\ell\}_{\ell \in \mathcal{L}}$ to meet Assumption 3.2.7, and use it to obtain an almost exact labeling in the first stage. Importantly, the second stage refinement, which gives the threshold, reverts to using tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ and does not rely on Assumption 3.2.7. Therefore at the end of stage two, provided that the problem is above the threshold, the algorithm will output an exact labeling with high probability despite the

introduction of weights. Thus, the only cases not covered by Theorem 3.2.12 lie at the intersection of these unsolvable (so far) uniform cases.

Stage II

Compared to Algorithm 3.2.3, $\hat{y}_v$ in Line 4 is able to handle the regime beyond Assumption 3.2.2. For each $\ell \in \mathcal{L}$, denote $Q_{\max}^{(\ell)} := \max_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell)}} Q_{\mathbf{w}}^{(\ell)}$, $Q_{\min}^{(\ell)} := \min_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell)}} Q_{\mathbf{w}}^{(\ell)}$, and we write $Q_{\max}^{(\ell)} = \mathcal{P}_{\max}^{(\ell)} \cdot q_{\max}^{(\ell)} / \binom{N-1}{\ell-1}$, $Q_{\min}^{(\ell)} = \mathcal{P}_{\min}^{(\ell)} \cdot q_{\min}^{(\ell)} / \binom{N-1}{\ell-1}$. The theoretical guarantee of Algorithm 3.2.4 is provided below.

Theorem 3.2.13 (Strong consistency with prior knowledge). *Suppose Assumption 3.2.7,*

$$\log(q_{\max}^{(\ell)} / q_{\min}^{(\ell)}) / \log(q_{\max}^{(\ell)}) \ll 1, \quad (3.2.20)$$

and (3.2.16) for some absolute $\varepsilon > 0$. When the tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ are known, Algorithm 3.2.4 achieves exact recovery with probability at least $1 - N^{-\varepsilon} - N^{-10}$.

An example satisfying (3.2.20) but violating Assumption 3.2.2 is $q_{\mathbf{w}_1}^{(\ell)} = N^{1/6}$ while $q_{\mathbf{w}}^{(\ell)} = N^{1/6} / \log(N)$ for any $\mathbf{w} \neq \mathbf{w}_1$.

When $q_N = \log(N)$, the condition (3.2.16) can be simplified as $\mathbb{D}_{\text{GCH}} > 1$. However, (3.2.16) could still be true when $1 \ll q_N \ll \log(N)$ and $\mathbb{D}_{\text{GCH}} \gg 1$ as in (3.2.6). Compared with Theorem 3.2.9, the additional situation covered by Theorem 3.2.13 happens when one of the communities has some sub-critical connectivity probabilities ($o(\log(N) / \binom{N-1}{\ell-1})$) either within itself or with respect to the other communities, while all the others are super-critically connected to each other, which will means that $\mathbb{D}_{\text{GCH}} \cdot q_N = \Omega(\log(N))$ satisfying (3.2.16) by Lemma D.1.1. In graph SBM [5] and uniform HSBM [216], exact recovery was achieved only under Assumption 3.2.2 when knowing the probability tensors. Theorem 3.2.13 removes this constraint and proves that Algorithm 3.2.4 works well under a broader regime.

Optimality

Suppose that we have the access to a weakly consistent community partition, the extra knowledge on the probability tensors helps Algorithm 3.2.4 to achieve the lowest possible expected mismatch ratio if $\rho_N = \omega(1)$. Compared with Theorem 3.2.10, Theorem 3.2.14 covers a wider scenario including $\log^{-1}(N) \ll \kappa_N \ll 1$ with a better convergence rate.

Theorem 3.2.14 (Optimality with prior knowledge). *Suppose Assumption 3.2.7, $\rho_N = \omega(1)$, (3.2.20) and (3.2.17), where the sequence $\kappa_N \in (0, 1]$ satisfies $\kappa_N \log(N) \rightarrow \infty$ as $N \rightarrow \infty$. Then with probability at least $1 - N^{-10}$, Algorithm 3.2.4 satisfies $\lim_{N \rightarrow \infty} (N^{\kappa_N} \cdot \mathbb{E}[\eta_N]) \leq 1$.*

3.2.5 Comparison with existing literature

The contribution of our results consists of the following aspects.

Information-Theoretic limits

Theorem 3.2.4 (1) is the first result in the literature establishing the necessary condition for exact recovery under general non-uniform HSBM. Previously, the phase transition phenomenon for exact recovery was only characterized for graphs [2, 5, 212, 4, 3] and uniform hypergraphs [146, 216]. Different from [5, 216], our proof of necessity does not rely on the additional Assumption 3.2.2. Furthermore, for the first time in the literature of hypergraphs, Theorem 3.2.4 (2) establishes the lower bound on the expected mismatch ratio for non-uniform HSBM. Previously, such results were obtained only in graph SBM for binary [4, 3] and multiple [212] communities. Minimax rates for the mismatch ratio were established for [104, 105] (graph SBM) and [61] (uniform HSBM), under specific constraints on the probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$. By contrast, Theorem 3.2.4 (2) is general.

Weak consistency

Theorem 3.2.12 is the first result showing that weak consistency can be achieved under general non-uniform HSBM as long as $\rho_N = \omega(1)$. Previously, [85] achieved weak consistency

when $1 \ll \rho_N \ll \log(N)$, under a special case of model 1.1.8 . In [217], weak consistency was achieved when $\rho_N \gtrsim \omega(\log N)$, a regime covered by our results. In [143], the authors studied the degree-corrected uniform HSBM by using a tensor power iteration method, and achieved weak consistency when the average degree is $\omega(\log^2 n)$, again covered by our results. A way to generalize their algorithm to non-uniform hypergraphs was discussed, but the theoretical analysis remains open.

The setting in [108] is most similar to our model 1.1.8 , but it only achieved weak consistency when the minimum expected degree is of at least the order $\Omega(\log^2(N))$. Their algorithm cannot be applied to the regime $\rho_N \lesssim \log(N)$ due to the lack of concentration for the normalized Laplacian because of the existence of isolated vertices. We were able to overcome this issue by focusing on the adjacency matrix instead. Similar to [100, 150], we regularized the adjacency matrix by zeroing out rows and columns corresponding to vertices with large degree (Algorithm 3.4.1) and proved a concentration result for the regularized matrix down to the bounded expected degree regime (Theorem 3.4.4).

Strong consistency

Theorems 3.2.9 and 3.2.13 are the first results to provide efficient algorithms for the general non-uniform case. Previously, strong consistency was only achieved under uniform HSBM for binary [146, 106], multi-community [61, 216] case, and non-uniform binary case [198], under Assumption 3.2.2. The method in [119] was not specialized to HSBM, and it does not contain any HSBM thresholds. The refinement stage is crucial for multi-block case, which brings in the sharp threshold [5, 212, 61, 216]. In Algorithm 3.2.4 and [216], the second stage is based on *Maximum A Posteriori* (MAP) estimation, where the MAP achieves the minimum error and full knowledge of $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ required. In [61], the agnostic refinement is based on *local Maximum Likelihood Estimation* (MLE), which works well for uniform hypergraphs. It remains to be seen whether this method can be generalized for non-uniform hypergraphs. Additionally, the time complexity of [61] is $O(N^4)$, since the spectral initialization is run N times during

the refinement stage. So far, Algorithm 3.2.1 is the only one that achieves strong consistency agnostically with the lowest time complexity.

Optimality

Although, as noted before, many works provided algorithms achieving weak consistency, the optimality of the methods was less explored. For the first time in the literature, we are able to provide Algorithms 3.2.1 and 3.2.4 which are provably optimal in achieving weak consistency below the threshold for exact recovery, as in Theorems 3.2.10 and 3.2.14. In particular, it provides the upper bound on the expected number of misclassified nodes for non-uniform HSBM, covering the most general scenario so far. Previously, spectral methods were proved to be optimal for binary-community clustering [214, 4, 3] with the optimal mismatch ratio being $N^{-(\sqrt{a}-\sqrt{b})^2/2}$. The optimal mismatch ratio was also explored under multi-community SBM[104, 105, 215], labeled SBM [212] and uniform HSBM [61].

Comparison with existing tensor methods

Many tensor-based methods contain the assumption that the K th smallest eigenvalue of a ℓ -uniform tensor (perhaps resulting from Tucker decomposition) is bounded away from 0 (in fact, the lower bounds tend to be more constraining); e.g., Assumption 1 in [11] and (4.7) in [143]. In general, Assumption 3.2.7 is strictly weaker than saying that the K th smallest eigenvalue is nonzero: in [83, Section 3.7], we provide an example satisfying the Assumption 3.2.7, but for which the least eigenvalue for the $K \times K$ “reduced” expectation matrix is 0, thus violating the conditions identified in [11] for tensor methods. In [119, Section 4.1], the authors involves the parameter Δ_f^2 , which represents the minimum square 2-norm differences between different rows of various matricizations of the expectation tensor. Their exact recovery condition, $\Delta_{\min}^2/\sigma^2 \geq C(p^{-d/2} \vee p^{-(d-1)} \log p)$, is more restrictive than ours, since it imposes a lower bound, whereas for us the lower bound is 0. Those tensor based methods cannot be generalized to the non-uniform case directly, since an efficient method to incorporate tensors of different orders has not yet been

devised. It remains to be seen if some other (non-spectral) method can be used to eliminate the remaining edge cases violating Assumption 3.2.7.

3.3 Information-Theoretic limits

This section is devoted to the proof of Theorem 3.2.4 with the proofs of Lemmas deferred to [83].

3.3.1 Impossibility of exact recovery

The proof of Theorem 3.2.4 (1) is presented in this section. Without loss of generality, let $j^*, k^* \in [K]$ be the indices where $\mathbb{D}_{\text{GCH}} \cdot q_N$ in (3.2.6) attains its minimum, i.e., $(j^*, k^*) = \arg \min_{j, k \in [K]} \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N$. The main idea is to confirm the existence of many pairwise disconnected *ambiguous* vertices (deferred to Definition 3.3.1). With high probability, those disconnected vertices maintain the same degree profile, but half from $\mathcal{V}_{j^*}$ and half from $\mathcal{V}_{k^*}$. Consequently, no algorithm performs better than random guess, which leads to the failure of exact recovery. Similar ideas appeared in [5]. By contrast, the central idea of [198] is to explicitly construct some configuration where the optimal estimator Maximum A Posteriori (MAP) fails. Note that $\mathbb{D}_{\text{GCH}}$ in (3.2.5) can be reduced to $\mathbb{D}_{\text{GH}}$ [198] when the general model is restricted to the binary symmetric model. Therefore, those two distinct proofs can be viewed as a mutual verification for the correctness of the thresholds.

Regime of interest

Before elaborating the proof sketch above, we establish the proof of impossibility for some easier cases quickly, which would simplify our discussion for the more difficult case later.

First, it suffices to consider the case $\kappa_N = 1$, i.e., $\mathbb{D}_{\text{GCH}} \cdot q_N < \log(N)$ when under (3.2.7).

Second, we restrict our attention to a sub-hypergraph described below. Denote

$$C_{j^*, k^*, \mathbf{w}}^{(\ell)} = \max\{Q_{j^* \oplus \mathbf{w}}^{(\ell)} / Q_{k^* \oplus \mathbf{w}}^{(\ell)}, Q_{k^* \oplus \mathbf{w}}^{(\ell)} / Q_{j^* \oplus \mathbf{w}}^{(\ell)}\} \quad (3.3.1)$$

for each $\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}$. When $C_{j^*,k^*,\mathbf{w}}^{(\ell)} = 1$, it follows that $Q_{j^* \oplus \mathbf{w}}^{(\ell)} = Q_{k^* \oplus \mathbf{w}}^{(\ell)}$. The edges with the membership counts $j^* \oplus \mathbf{w}$ or $k^* \oplus \mathbf{w}$ have zero contribution to $\mathbb{D}_{\text{GCH}}$ in (3.2.6), and they can not be used to distinguish vertices in $\mathcal{V}_{j^*}$ from vertices in $\mathcal{V}_{k^*}$. Consequently, we can exclude them from consideration. For each $\ell \in \mathcal{L}$, define the following set

$$\mathcal{W}_{j^*,k^*}^{(\ell)} = \{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)} \mid C_{j^*,k^*,\mathbf{w}}^{(\ell)} > 1\}. \quad (3.3.2)$$

Throughout the proof, we assume $1 \ll \mathbb{D}_{\text{GCH}} \cdot q_N < \log(N)$, and we focus on the type of edges with membership counts $j^* \oplus \mathbf{w}$ or $k^* \oplus \mathbf{w}$, where $\mathbf{w} \in \mathcal{W}_{j^*,k^*}^{(\ell)}$ for each $\ell \in \mathcal{L}$.

Proof of Necessity

Let $t^* \in [0, 1]$ be the point where the maximum of $\mathbb{D}_{\text{GCH}}(j^*, k^*) \cdot q_N$ is attained, then t^* should be the critical point for the function we maximize, i.e.,

$$\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{j^*,k^*}^{(\ell)}} \bar{N}_{\mathbf{w}} [Q_{j^* \oplus \mathbf{w}}^{(\ell)} - Q_{k^* \oplus \mathbf{w}}^{(\ell)} - (Q_{j^* \oplus \mathbf{w}}^{(\ell)})^{t^*} (Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{1-t^*} \log \frac{Q_{j^* \oplus \mathbf{w}}^{(\ell)}}{Q_{k^* \oplus \mathbf{w}}^{(\ell)}}] = 0. \quad (3.3.3)$$

The t^* is unique since the function to the left of the equal sign in (3.3.3) is strictly decreasing. For each $\mathbf{w} \in \mathcal{W}_{j^*,k^*}^{(\ell)}$, define *degree profile* as

$$d_{j^*,k^*}^*(\mathbf{w}) := \left[(Q_{j^* \oplus \mathbf{w}}^{(\ell)})^{t^*} \cdot (Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{1-t^*} \cdot \bar{N}_{\mathbf{w}} \right]. \quad (3.3.4)$$

Intuitively, $\mathcal{V}_{j^*}$ and $\mathcal{V}_{k^*}$ are most similar among all communities. The degree profile $d_{j^*,k^*}^*(\mathbf{w})$ interpolates between the expected degree profiles of the vertices in $\mathcal{V}_{j^*}$ and $\mathcal{V}_{k^*}$, and, if $\mathcal{V}_{j^*}$ and $\mathcal{V}_{k^*}$ are similar enough, vertices with this degree profile are very likely to exist in both communities. If they do exist in both, they are indistinguishable: there is no way to be sure which community they belong to. We introduce the concept of *ambiguity* below.

Definition 3.3.1 (Ambiguity). For $S \subset \mathcal{V}$, let $D_{v,\mathbf{w}}^{(\ell)}|_S$ be the number of edges containing v , with

the remaining $\ell - 1$ vertices all in $\mathcal{V} \setminus \mathcal{S}$. A vertex $v \in \mathcal{S}$ is said to be $(j^*, k^*, \mathcal{S})$ -ambiguous between $\mathcal{V}_{j^*} \setminus \mathcal{S}$ and $\mathcal{V}_{k^*} \setminus \mathcal{S}$ if $D_{v, \mathbf{w}}^{(\ell)}|_{\mathcal{S}} = d_{j^*, k^*}^*(\mathbf{w})$ for each $\mathbf{w} \in \mathcal{W}_{j^*, k^*}^{(\ell)}$, $\ell \in \mathcal{L}$.

For distinct vertices $v, u \in \mathcal{S}$, the events $\{v \text{ is } (j^*, k^*, \mathcal{S})\text{-ambiguous}\}$ and $\{u \text{ is } (j^*, k^*, \mathcal{S})\text{-ambiguous}\}$ are independent, since the random variables involved are distinct. Lemma 3.3.2 characterizes the probability of v being ambiguous.

Lemma 3.3.2. Assume that vertex $v \in \mathcal{V}_{k^*}$, i.e., $Y_v = k^*$. Assume (3.2.7) with $\kappa_N = 1$ and let $j^*, k^* \in [K]$ be the pair achieving $\mathbb{D}_{\text{GCH}}$ in (3.2.6), then

$$\prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}_{j^*, k^*}^{(\ell)}} \mathbb{P}(D_{v, \mathbf{w}}^{(\ell)} = d_{j^*, k^*}^*(\mathbf{w})) = \exp(-(1 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N). \quad (3.3.5)$$

Proof of Theorem 3.2.4 (1). Let $\mathcal{S} \subset \mathcal{V}$ be a set of vertices with size $\lfloor Nq_N^{-2} \log^{-1}(N) \rfloor$, where each vertex is chosen randomly from $\mathcal{V}$. By Lemma 3.1.1, with probability at least $1 - O(N^{-3})$, $|\mathcal{S} \cap \mathcal{V}_l| = \alpha_l Nq_N^{-2} \log^{-1}(N)$ with deviation smaller than $O(\sqrt{N})$ for each $l \in [K]$. Let $v^* \in \mathcal{S} \cap \mathcal{V}_{k^*}$ be a $(j^*, k^*, \mathcal{S})$ -ambiguous vertex. Since $|\mathcal{S}|/|\mathcal{V}| = o(1)$, the extra condition of having the other $\ell - 1$ vertices all in $\mathcal{V} \setminus \mathcal{S}$ will erase only a negligible number of edges. Assume (3.2.7) for sufficiently large N with $\kappa_N = 1$, then $\mathbb{D}_{\text{GCH}} \cdot q_N / \log(N) < 1 - 2\varepsilon < 1$ for some absolute constant $\varepsilon > 0$. By following the proof of Lemma 3.3.2, it yields that

$$\mathbb{P}(v^* \text{ is } (j^*, k^*, \mathcal{S})\text{-ambiguous}) = \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}_{j^*, k^*}^{(\ell)}} \mathbb{P}(D_{v^*, \mathbf{w}}^{(\ell)}|_{\mathcal{S}} = d_{j^*, k^*}^*(\mathbf{w})) \quad (3.3.6)$$

$$= \exp(-(1 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N) \geq N^{-1+2\varepsilon+o(1)} \geq N^{-1+\varepsilon}. \quad (3.3.7)$$

By independence between ambiguous nodes and the fact $1 \ll q_N \lesssim \log(N)$, we have

$$\begin{aligned} \mathbb{E}[\# (j^*, k^*, \mathcal{S})\text{-ambiguous nodes in } \mathcal{S} \cap \mathcal{V}_{k^*}] &\gtrsim \alpha_{k^*} \lfloor Nq_N^{-2} \log^{-1}(N) \rfloor \cdot N^{-1+\varepsilon} \\ &\asymp N^\varepsilon q_N^{-2} \log^{-1}(N) \gg q_N. \end{aligned} \quad (3.3.8)$$

Then by Bernstein's inequality, with probability at least $1 - e^{-N^\epsilon \log^{-3}(N)}$, the number of $(j^\star, k^\star, \mathcal{S})$ -ambiguous nodes will concentrate around its expectation with deviation less than $N^\epsilon \log^{-3}(N)$. Therefore, there exists at least q_N many $(j^\star, k^\star, \mathcal{S})$ -ambiguous vertices.

We will prove that $v^\star$ is isolated in $\mathcal{S}$ with high probability in the considered sub-hypergraph. Recall that $Q_{k^\star \oplus \mathbf{w}}^{(\ell)}$ denotes the probability of sampling a ℓ -hyperedge containing $v^\star \in \mathcal{V}_{k^\star}$ with the other $\ell - 1$ vertices distributed as $\mathbf{w}$ among $\{\mathcal{V}_l\}_{l=1}^K$. If one more constraint is added, i.e., containing $v^\star$ and at least one other vertex from $\mathcal{S}$, then the number of such hyperedges is denoted by $\bar{N}_{\mathbf{w}}|_{\mathcal{S}} = (1 + o(1))q_N^{-2} \log^{-1}(N) \cdot \bar{N}_{\mathbf{w}}$, where $\bar{N}_{\mathbf{w}}$ is defined in (3.2.4). The probability that $v^\star$ is isolated in $\mathcal{S}$ in the considered sub-hypergraph is

$$\begin{aligned} & \mathbb{P}(v^\star \text{ isolated in } \mathcal{S}) \\ &= \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}_{j^\star, k^\star}^{(\ell)}} (1 - Q_{k^\star \oplus \mathbf{w}}^{(\ell)})^{\bar{N}_{\mathbf{w}}|_{\mathcal{S}}} = \exp \left(- (1 + o(1)) \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{j^\star, k^\star}^{(\ell)}} \frac{Q_{k^\star \oplus \mathbf{w}}^{(\ell)} \cdot \bar{N}_{\mathbf{w}}}{q_N^2 \cdot \log(N)} \right), \end{aligned} \quad (3.3.9)$$

where we applied $e^{-x} = 1 - x$ when $x = o(1)$ since $Q_{k^\star \oplus \mathbf{w}}^{(\ell)} = o(1)$. Note that we only focus on the weak compositions in $\mathcal{W}_{j^\star, k^\star}^{(\ell)}$ and the fact that $\mathbf{w} \notin \mathcal{W}_{j^\star, k^\star}^{(\ell)}$ contributes 0 to $\mathbb{D}_{\text{GCH}}$. It follows by Lemma D.1.1 that $\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{j^\star, k^\star}^{(\ell)}} Q_{k^\star \oplus \mathbf{w}}^{(\ell)} \cdot \bar{N}_{\mathbf{w}} \lesssim \mathbb{D}_{\text{GCH}} \cdot q_N$. Consequently, there exists some constant $C > 0$ such that

$$(3.3.9) = \exp \left(- C \cdot \mathbb{D}_{\text{GCH}} \cdot q_N \cdot [q_N^2 \cdot \log(N)]^{-1} \right) > e^{-C/q_N^2} = 1 - C/q_N^2, \quad (3.3.10)$$

where the second inequality holds since $\mathbb{D}_{\text{GCH}} \cdot q_N / \log(N) < 1$ by (3.2.7).

From (3.3.8), we can pick at least q_N pairs of $(j^\star, k^\star, \mathcal{S})$ -ambiguous vertices, where in each pair, one from $\mathcal{S} \cap \mathcal{V}_{j^\star}$ and the other one from $\mathcal{S} \cap \mathcal{V}_{k^\star}$. The probability that these q_N pairs of vertices are all isolated from $\mathcal{S}$ in the considered sub-hypergraph is at least $(1 - Cq_N^{-2})^{2q_N} \geq 1 - 4C/q_N$, which tends to 1 since $q_N = \omega(1)$. Consequently, those q_N pairs of vertices are indistinguishable. It follows that no algorithm can perform better than random guess for *each* of

these vertices, and by independence, no algorithm can have probability of success better than $[(\frac{2^{qN}}{q_N})]^{-1} \asymp 2^{-qN}$, which goes to 0. Therefore, every algorithm will misclassify at least one vertex with probability $1 - 2^{-qN}$. $\square$

3.3.2 Information-Theoretic lower bound

To establish the proof of Theorem 3.2.4 (2), we first introduce the following *Generalized Kullback-Leibler* (GKL) divergence.

Definition 3.3.3. For each $\ell \in \mathcal{L}$, denote the sequence of numbers by $\boldsymbol{\mu}^{(\ell)} = (\mu_{\mathbf{w}}^{(\ell)})_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \in [0, 1]^{|\mathcal{W}C_K^{(\ell-1)}|}$. Define the generalized Kullback-Leibler (GKL) divergence

$$\mathbb{D}_{\text{GKL}} = \min_{1 \leq j < k \leq K} \mathbb{D}_{\text{GKL}}(j, k), \quad (3.3.11)$$

$$\mathbb{D}_{\text{GKL}}(j, k) := \min_{\{\boldsymbol{\mu}^{(\ell)}\}_{\ell \in \mathcal{L}}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \max \{ \mathbb{D}_{\text{KL}}(\mu_{\mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}), \mathbb{D}_{\text{KL}}(\mu_{\mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) \},$$

where $\bar{N}_{\mathbf{w}}$ is defined in (3.2.4), and

$$\mathbb{D}_{\text{KL}}(\mu_{\mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}) := \mu_{\mathbf{w}}^{(\ell)} \log \left(\frac{\mu_{\mathbf{w}}^{(\ell)}}{Q_{j \oplus \mathbf{w}}^{(\ell)}} \right) + (1 - \mu_{\mathbf{w}}^{(\ell)}) \log \left(\frac{1 - \mu_{\mathbf{w}}^{(\ell)}}{1 - Q_{j \oplus \mathbf{w}}^{(\ell)}} \right) \quad (3.3.12)$$

denotes the KL divergence between two Bernoulli distributions with means $\mu_{\mathbf{w}}^{(\ell)}$ and $Q_{j \oplus \mathbf{w}}^{(\ell)}$.

Lemma 3.3.4 builds the connection between $\mathbb{D}_{\text{GCH}}$ in (3.2.5) and $\mathbb{D}_{\text{GKL}}$ in (3.3.11).

Lemma 3.3.5 establishes the existence of some sequence $\{\mathbf{p}^{(\ell)}\}_{\ell \in \mathcal{L}}$ achieving $\mathbb{D}_{\text{GKL}}$.

Lemma 3.3.4. When $Q_{\mathbf{w}}^{(\ell)} = o(1)$ for all $\mathbf{w} \in \mathcal{W}C_K^{(\ell)}$, $\ell \in \mathcal{L}$, we have $\mathbb{D}_{\text{GKL}}(j, k) = (1 + o(1)) \cdot \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N$.

Lemma 3.3.5. Let $(j^*, k^*) = \arg \min_{j < k} \mathbb{D}_{\text{GKL}}(j, k)$. There exists some probability measure

sequence $\{\mathbf{p}^{(\ell)}\}_{\ell \in \mathcal{L}}$ such that

$$\begin{aligned} \mathbb{D}_{\text{GKL}} &= \mathbb{D}_{\text{GKL}}(j^*, k^*) \\ &= \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mathbf{p}_{\mathbf{w}}^{(\ell)} \parallel Q_{j^* \oplus \mathbf{w}}^{(\ell)}) = \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mathbf{p}_{\mathbf{w}}^{(\ell)} \parallel Q_{k^* \oplus \mathbf{w}}^{(\ell)}). \end{aligned} \quad (3.3.13)$$

Let Φ denote the true model 1.1.8 and Ψ denote the perturbed stochastic model 3.3.6, below. The lower bound of the expected mismatch ratio in Theorem 3.2.4 (2) is established by the *coupling* between Φ and Ψ . Here, in our analysis, the generalized KL divergence $\mathbb{D}_{\text{GKL}}$ plays a fundamental role, similar to [148].

Definition 3.3.6 (Perturbed model Ψ). *For j^*, k^* in Lemma 3.3.5, let v^* denote node with the smallest index belonging to $\mathcal{V}_{j^*} \cup \mathcal{V}_{k^*}$. The two models Φ and Ψ are coupled as follows.*

1. *The clusters $\mathcal{V}_1, \dots, \mathcal{V}_K$ are generated first, thus the same under Φ and Ψ .*
2. *For each ℓ -edge $e \not\ni v^*$, it is sampled with probability $Q_{\mathbf{y}(e)}^{(\ell)}$, both under Φ and Ψ .*
3. *For each $e \ni v^*$ with the other $\ell - 1$ nodes distributed as $\mathbf{w}$, it is sampled with probability $Q_{\mathbf{y}_{v^*} \oplus \mathbf{w}}^{(\ell)}$ under Φ , while with probability $\mathbf{p}_{\mathbf{w}}^{(\ell)}$ (Lemma 3.3.5) under Ψ respectively.*

Let $\mathbb{P}_{\Phi}, \mathbb{E}_{\Phi}$ (resp. $\mathbb{P}_{\Psi}, \mathbb{E}_{\Psi}$) denote the probability measure, expectation under Φ (resp. Ψ). Let $\mathcal{V}_{\text{err}}$ denote the set of misclassified nodes. Those nodes are interchangeable under Φ , then $\mathbb{E}_{\Phi}[\eta_N] = N^{-1} \mathbb{E}_{\Phi}[|\mathcal{V}_{\text{err}}|]$. Let $\Lambda := \log(d\mathbb{P}_{\Psi}/d\mathbb{P}_{\Phi})$ denote the log-likelihood ratio, and

$$\Lambda = \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_{\ell}, e \ni v^* \\ \mathbf{y}(e \setminus v^*) = \mathbf{w}}} \left[\mathcal{A}_e^{(\ell)} \log \left(\frac{\mathbf{p}_{\mathbf{w}}^{(\ell)}}{Q_{\mathbf{y}_{v^*} \oplus \mathbf{w}}^{(\ell)}} \right) + (1 - \mathcal{A}_e^{(\ell)}) \log \left(\frac{1 - \mathbf{p}_{\mathbf{w}}^{(\ell)}}{1 - Q_{\mathbf{y}_{v^*} \oplus \mathbf{w}}^{(\ell)}} \right) \right]. \quad (3.3.14)$$

Assume $\alpha_1 \geq \dots \geq \alpha_K$ without loss of generality. Note that $j^* < k^*$ from Lemma 3.3.5. The connection between Ψ and Φ can be described quantitatively by Lemma 3.3.7.

Lemma 3.3.7. *Let $f(N)$ be some function of N , then the following holds*

$$\mathbb{P}_\Psi(\Lambda \leq f(N)) \leq (e^{f(N)} \cdot \mathbb{E}_\Phi[\eta_N] + \alpha_{j^*}) / (\alpha_{j^*} + \alpha_{k^*}). \quad (3.3.15)$$

Proof of Theorem 3.2.4 (2). First, by applying $f(N) = \log(1/\mathbb{E}_\Phi[\eta_N]) - \log(2/\alpha_{k^*})$ and the fact $\alpha_{j^*} + \alpha_{k^*} < 1$, Lemma 3.3.7 yields that

$$\mathbb{P}_\Psi(\Lambda \leq \log(1/\mathbb{E}_\Phi[\eta_N]) - \log(2/\alpha_{k^*})) \leq 1 - (\alpha_{j^*} + \alpha_{k^*})^{-1} \alpha_{k^*} / 2 < 1 - \alpha_{k^*} / 4. \quad (3.3.16)$$

Furthermore, by Chebyshev's inequality, we have

$$\mathbb{P}_\Psi(\Lambda \leq \mathbb{E}_\Psi[\Lambda] + \sqrt{4\text{Var}_\Psi(\Lambda)/\alpha_{k^*}}) \geq 1 - \alpha_{k^*} / 4. \quad (3.3.17)$$

Consequently, the following comparison of the two upper bounds of Λ should always be true:

$$\log(1/\mathbb{E}_\Phi[\eta_N]) - \log(2/\alpha_{k^*}) \leq \mathbb{E}_\Psi[\Lambda] + \sqrt{4\text{Var}_\Psi(\Lambda)/\alpha_{k^*}}. \quad (3.3.18)$$

Lemma 3.3.8. *Recall $1 \ll \mathbb{D}_{\text{GCH}} \cdot q_N \lesssim \log(N)$ in (3.2.6), then the following holds:*

$$\mathbb{E}_\Psi[\Lambda] + \sqrt{4\text{Var}_\Psi(\Lambda)/\alpha_{k^*}} = (1 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N. \quad (3.3.19)$$

Recall (3.2.7) where $\kappa_N \in (0, 1]$ and $\kappa_N \log(N) \rightarrow \infty$ when $N \rightarrow \infty$. According Lemma 3.3.8, when $\mathbb{E}_\Phi[\eta_N] < N^{-\kappa_N}$, the left hand side of (3.3.18) can be further bounded by

$$\kappa_N \cdot \log(N) - \log(2/\alpha_{k^*}) \leq (1 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N. \quad (3.3.20)$$

Since α_{k^*} is some fixed constant, the necessary condition for $\lim_{N \rightarrow \infty} (N^{\kappa_N} \mathbb{E}_\Phi[\eta_N]) < 1$ is

$$\lim_{N \rightarrow \infty} \mathbb{D}_{\text{GCH}} \cdot q_N / (\kappa_N \cdot \log(N)) \geq 1. \quad (3.3.21)$$

Therefore, Theorem 3.2.4 (2) follows since we proved its contrapositive. $\square$

3.4 Stage I: Weak consistency by spectral initialization

In this section, we will show that Algorithm 3.2.2, inspired by [68, 211, 212, 216], achieves almost exact recovery, i.e., $\eta(\mathbf{y}, \hat{\mathbf{y}}^{(0)}) = o(1)$. The proofs of Lemmas are deferred to [83].

3.4.1 Spectral initialization

Recall that d_v denotes the number of hyperedges containing v with $\mathbf{D}_v$ in (3.1.1) denoting its law. Let $\bar{d} = \sum_{v=1}^N d_v / N$ (resp. $\tilde{d} = \max_{v \in \mathcal{N}} d_v$) denote the *observed average* (resp. *maximum*) degree. Let ξ_N (resp. ρ_N) denote the *average* (resp. *maximum*) *expected* degree, defined by

$$\xi_N := N^{-1} \mathbb{E} \left(\sum_{v \in \mathcal{V}} \mathbf{D}_v \right), \quad \rho_N := \max_{v \in \mathcal{V}} \mathbb{E}(\mathbf{D}_v). \quad (3.4.1)$$

Lemma 3.4.1 states the fact that $\bar{d}$ is always a good approximation of ξ_N , while $\tilde{d}$ preserves the order of ρ_N when $\rho_N \gtrsim \log(N)$.

Lemma 3.4.1. *Deterministically, $\rho_N \asymp \xi_N$. With high probability, $\bar{d} - \xi_N \leq \xi_N / \sqrt{N}$. When $\rho_N \gtrsim \log(N)$, we have $\tilde{d} \asymp \rho_N$ with probability at least $1 - O(N^{-c})$ for some constant $c > 0$.*

3.4.2 Concentration inequalities

Our analysis relies on the concentration results Theorems 3.4.2 and 3.4.4. In fact, the results provided below holds for *inhomogeneous Erdős-Rényi* hypergraphs 1.1.10, more general than the HSBM 1.1.8, which could be of independent interest.

Theorem 3.4.2. Let $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$ be the inhomogeneous non-uniform Erdős-Rényi hypergraph in Definition 1.1.10, associated with the probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$. For each $\ell \in \mathcal{L}$, we rescale the tensor by $\mathcal{D}^{(\ell)} := \binom{N-1}{\ell-1} Q^{(\ell)}$, and denote $\mathcal{D}_{\max}^{(\ell)} := \max_{i_1, \dots, i_{\ell} \in \mathcal{V}} \mathcal{D}_{i_1, \dots, i_{\ell}}^{(\ell)}$. Denote $\mathcal{D}_{\max} := \sum_{\ell \in \mathcal{L}} \mathcal{D}_{\max}^{(\ell)}$. For some constant $C_{(3.4.2)} > 0$, suppose that

$$\mathcal{D}_{\max} := \sum_{\ell \in \mathcal{L}} \mathcal{D}_{\max}^{(\ell)} \geq C_{(3.4.2)} \cdot \log(N). \quad (3.4.2)$$

Then with probability at least $1 - 2N^{-10} - 2e^{-N}$, the adjacency matrix $\mathbf{A}$ of $\mathcal{H}$ satisfies

$$\|\mathbf{A} - \mathbb{E}\mathbf{A}\| \leq C_{(3.4.3)} \cdot \sqrt{\mathcal{D}_{\max}}, \quad (3.4.3)$$

where the constants $C_{(3.4.3)} := 10L^2 + 2\beta$, with $\beta = \beta_0 \sqrt{\beta_1} + L$ and β_0, β_1 satisfying

$$\begin{aligned} \beta_0 &= 16 + 32L(1 + e^2) + 1792(1 + e^{-2})L^2, \\ L^{-1}\beta_1 \log(L^{-1}\beta_1) - L^{-1}\beta_1 + 1 &> 11/C_{(3.4.2)}. \end{aligned} \quad (3.4.4)$$

Similarly matrix concentration results for the graph case has appeared in the literature several times, e.g., [153, Theorem 5.2], [150, Theorem 5.1]. The main obstacle for the generalization from graph to hypergraph 1.1.10 is that, the entries of the adjacency matrix $\mathbf{A}$ in (1.1.6) have long-range dependencies, and they are no longer zero/one entries. The proofs for Theorem 3.4.2 and Theorem 3.4.4 are deferred to [83].

Furthermore, Algorithm 3.2.2 could still achieve weak consistency $1 \ll \rho_N \ll \log(N)$, which is beyond regime of exact recovery. However, Theorem 3.4.2 is not reliable in this case, and it cannot be applied directly. Instead, the concentration could be proved for some regularized versions of $\mathbf{A}$ (Definition 3.4.3) following the same proof strategy, leading to Theorem 3.4.4. This explains why the trimming stage Algorithm 3.4.1 is necessary.

Definition 3.4.3 (Regularization). Given $\mathbf{A} \in \mathbb{R}^{N \times N}$ and index set $\mathcal{J}$, let $\mathbf{A}_{\mathcal{J}} \in \mathbb{R}^{N \times N}$ be the

Algorithm 3.4.1: Trimming

Data: Hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, regularization strategy $\mathcal{J}$

- 1 Obtain the adjacency matrix $\mathbf{A}$ of the hypergraph $\mathcal{H}$ by (1.1.6).
- 2 Construct $\mathbf{A}_{\mathcal{J}} \in \mathbb{R}^{N \times N}$ by zeroing out the columns $\mathbf{A}_{\cdot i}$ and rows $\mathbf{A}_{i \cdot}$, where $i \notin \mathcal{J}$;

Result: $\mathbf{A}_{\mathcal{J}}$

matrix obtained from $\mathbf{A}$ by zeroing out the rows and columns not indexed by $\mathcal{J}$. Namely,

$$(\mathbf{A}_{\mathcal{J}})_{ij} = \mathbb{1}_{\{i, j \in \mathcal{J}\}} \cdot \mathbf{A}_{ij}. \quad (3.4.5)$$

Theorem 3.4.4. *Following the conventions in Theorem 3.4.2, let $\mathbf{A}_{\mathcal{J}}$ be the regularized version of $\mathbf{A}$, where $\mathcal{J} \in \{\mathcal{J}_1, \mathcal{J}_2\}$ defined as below.*

1. *Let $\mathcal{J}_1$ be the vertex set obtained by removing $\lfloor N \exp(-\bar{d}) \rfloor$ many nodes having the largest d_v , where $\bar{d} = N^{-1} \sum_{v \in \mathcal{V}} d_v$ is the sample average degree. If $\rho_N \gtrsim \log(N)$, there exists some constant $c > 0$ such that (3.4.6) holds with probability at least $1 - 2(e/2)^{-N} - 2N^{-c}$.*
2. *Define $\mathcal{J}_2 := \{v \in \mathcal{V}, d_v \leq 20L\mathcal{D}_{\max}\}$, then the inequality (3.4.6) holds with probability at least $1 - 2(e/2)^{-N} - N^{-10}$ for some constant $C_{(3.4.6)}$.*

$$\|\mathbf{A}_{\mathcal{J}} - \mathbb{E}\mathbf{A}_{\mathcal{J}}\| \leq C_{(3.4.6)} \sqrt{\mathcal{D}_{\max}}, \quad (3.4.6)$$

where $C_{(3.4.6)} := 10L^2 + 2\beta$ with $\beta = \beta_0\sqrt{\beta_1} + L$, $\beta_1 = L$ and

$$\beta_0 = 16 + 32L(1 + e^2) + 1792(1 + e^{-2})L^2.$$

=

The construction of $\mathcal{J}_1$ is purely agnostic and does not require prior knowledge. By contrast, an upper bound on the vertex degree is pre-determined for $\mathcal{J}_2$, requiring $\{Q(\ell)\}_{\ell \in \mathcal{L}}$. Therefore, $\mathcal{J}_1$ is suitable for Algorithm 3.2.1, while $\mathcal{J}_2$ is suitable for Algorithm 3.2.4.

3.4.3 Proofs of weak consistency

Without loss of generality, we assume $\alpha_1 \geq \dots \geq \alpha_K$. We present some Lemmas first.

Lemma 3.4.5. *Under the conditions of Theorem 3.4.4, $\|\mathbf{A}_j^{(K)} - \mathbb{E}\mathbf{A}_j\|_F^2 \lesssim \rho_N + \rho_N^2/N$ with probability at least $1 - O(N^{-c})$ for some constant $c > 0$.*

Lemma 3.4.6. *Suppose Assumption 3.2.7 and conditions in Theorem 3.4.4, then $\|(\mathbb{E}\mathbf{A}_j)_{u\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2 \gtrsim N^{-1}\rho_N^2$ with probability at least $1 - O(N^{-c})$ for some constant $c > 0$ when u and v are from different communities, otherwise $\|(\mathbb{E}\mathbf{A}_j)_{u\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2 \lesssim N^{-2}\rho_N^2$.*

Lemma 3.4.7. *Recall that in Algorithm 3.2.2, $\bar{r} = [N \log(\bar{d})]^{-1} \bar{d}^2$, and S is a set of $\lceil 2 \log^2(N) \rceil$ nodes randomly sampled from $\mathcal{J}$ without replacement. Define the vertex sets*

$$I_k := \{v \in \mathcal{V}_k \cap \mathcal{J} : \|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2 \leq \bar{r}/4\} \quad (3.4.7)$$

$$\mathcal{U}_k := \{v \in \mathcal{V}_k \cap \mathcal{J} : \|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2 \leq 4\bar{r}\}. \quad (3.4.8)$$

(a) *For all $k \in [K]$, $|S \cap I_k| \geq 1$ with probability at least $1 - O(N^{-c})$ for some $c > 0$.*

(b) *For each $k \in [K]$, the s_k constructed in Algorithm 3.2.2 satisfies $s_k \in \mathcal{U}_k$ and $|\mathcal{B}_r(s_k)| \geq \alpha_k N(1 - o(1))$.*

(c) *The centers $\{s_k\}_{k=1}^K$ constructed in Algorithm 3.2.2 are pairwise far away from each other,*

$$\|(\mathbf{A}_j^{(K)})_{s_j\cdot} - (\mathbf{A}_j^{(K)})_{s_k\cdot}\|_2^2 \gtrsim N^{-1}\rho_N^2, \quad \text{for any pair } j \neq k, j, k \in [K]. \quad (3.4.9)$$

Lemma 3.4.8. *In Line 4, 8 of Algorithm 3.2.2, if $v \in \mathcal{V}_i \cap \mathcal{J}$ is misclassified to $\widehat{\mathcal{V}}_j^{(0)}$ for some $j \neq i$, then v is far away from its expected center, i.e., $\|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2 \gtrsim N^{-1}\rho_N^2$.*

Proof of Theorem 3.2.6. For each $k \in [K]$, Lemma 3.4.7 (a) ensures that at least one vertex will be picked in each $\mathcal{V}_k$, (b) ensures that the constructed ball $\mathcal{B}_r(s_k)$ contains most of the

vertices of $\mathcal{V}_k$. Furthermore, Lemma 3.4.6 proves that the expected centers are pairwise far away from each other, while Lemma 3.4.7 (c) proves that the same argument holds as well for the centers $\{s_k\}_{k=1}^K$ of balls $\{\mathcal{B}(s_k)\}_{k=1}^K$. Throughout the Algorithm 3.2.2, vertices would be misclassified during lines 4, 8, 12. Lemma 3.4.8 indicates that the misclassified vertices in lines 4, 8 must be far away from their expected centers. The number of vertices outside $\mathcal{J}_1$ is upper bounded by $\lfloor N \exp(-\bar{d}) \rfloor$, which is at most of constant order since $\bar{d} \gtrsim \log(N)$ according to Lemma 3.4.1. Then with probability at least $1 - 2(e/2)^{-N} - 2N^{-c}$ for some $c > 0$, the mismatch ratio is bounded by

$$\begin{aligned}
\eta_N = \eta(\mathbf{y}, \hat{\mathbf{y}}^{(0)}) &\leq \frac{1}{N} \left(\frac{\sum_{v \text{ misclassified}} \|(\mathbf{A}_j^{(K)})_{v:} - (\mathbb{E}\mathbf{A}_j)_{v:}\|_2^2}{\min_{v \text{ misclassified}} \|(\mathbf{A}_j^{(K)})_{v:} - (\mathbb{E}\mathbf{A}_j)_{v:}\|_2^2} + \lfloor N \exp(-\bar{d}) \rfloor \right) \\
&\leq \|\mathbf{A}_j^{(K)} - \mathbb{E}\mathbf{A}_j\|_F^2 / \left(N \cdot \min_{v \text{ misclassified}} \|(\mathbf{A}_j^{(K)})_{v:} - (\mathbb{E}\mathbf{A}_j)_{v:}\|_2^2 \right) + \exp(-\bar{d}) \\
&\lesssim (\rho_N + \rho_N^2/N) / (N \cdot \rho_N^2/N) + N^{-1} \asymp \rho_N^{-1} = o(1), \quad (\text{Lemmas 3.4.5, 3.4.8}) \quad (3.4.10)
\end{aligned}$$

where the last line holds since $\rho_N \gtrsim \log(N)$. □

Proof of Theorem 3.2.12. The regularization method $\mathcal{J}_1$ relies on two key facts:

1. only $\lfloor N \exp(-\bar{d}) \rfloor = o(N)$ many nodes are removed, since $\bar{d} \gtrsim \log(N)$ by Lemma 3.4.1;
2. the degree of node in $\mathcal{J}_1$ is upper bounded by $\mathcal{D}_{\max}$ up to some constant when $\rho_N \gtrsim \log(N)$, according to Lemma B.1.9.

However, the second fact is no longer valid when $1 \ll \rho_N \ll \log(N)$. To get rid of that, one could refer to the other trimming process $\mathcal{J}_2$ in Theorem 3.4.4. Deterministically, the degree of each vertex in $\mathcal{J}_2$ is upper bounded by $20L\mathcal{D}_{\max}$, where the construction of $\mathcal{J}_2$ relies on the prior knowledge of probability tensors $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$. According to (B.1.12), the number of vertices outside $\mathcal{J}_2$ is upper bounded by $N \cdot e^{-c\rho_N} = o(N)$ for some constant $c > 0$. Then by following the same analysis above, with probability at least $1 - 2(e/2)^{-N} - N^{-10}$, the mismatch ratio is

bounded by

$$\eta_N \lesssim \frac{\rho_N + \rho_N^2/N}{N\rho_N^2/N} + N^{-1} \cdot (N/e^{c\rho_N}) \asymp \rho_N^{-1} + N^{-1} + e^{-c\rho_N} \asymp \rho_N^{-1} = o(1). \quad (3.4.11)$$

Therefore, Algorithm 3.2.2 achieves weak consistency by applying $\mathcal{J}_2$ as long as $\rho_N = \omega(1)$. $\square$

3.5 Stage II: Agnostic iterative refinement

This section is devoted to the proofs of Theorem 3.2.9 and Theorem 3.2.10, while the proofs of all the Lemmas are deferred to [83]. We first introduce Algorithm 3.2.3, which achieves the optimal recovery by iteratively refining the initial estimate $\hat{\mathbf{y}}^{(0)}$ obtained from Algorithm 3.2.2.

3.5.1 Tensor probability reconstruction

The difference between $Q_{\mathbf{w}}^{(\ell)}$ and $\hat{Q}_{\mathbf{w}}^{(\ell)}$ is controlled by Lemma 3.5.1. It is proved that $\hat{Q}_{\mathbf{w}}^{(\ell)}$ is a good estimator of $Q_{\mathbf{w}}^{(\ell)}$ under Assumption 3.2.2.

Lemma 3.5.1. *For each $\ell \in \mathcal{L}$ and $\mathbf{w} \in \mathcal{WC}_K^{(\ell)}$, recall $Q_{\mathbf{w}}^{(\ell)}$ in (3.2.2) and $\hat{Q}_{\mathbf{w}}^{(\ell)}$ in (3.2.11). Define $Q_{\max}^{(\ell)} := \max_{\mathbf{w} \in \mathcal{WC}_K^{(\ell)}} Q_{\mathbf{w}}^{(\ell)}$ and denote $Q_{\max}^{(\ell)} = \mathcal{P}_{\max}^{(\ell)} q_{\max}^{(\ell)} / \binom{N-1}{\ell-1}$ using (3.2.2) where $\mathcal{P}_{\max}^{(\ell)} \asymp 1$ and $1 \ll q_{\max}^{(\ell)} \ll N$. The following holds with probability at least $1 - \exp(-N \log q_{\max}^{(\ell)} / q_{\max}^{(\ell)})$,*

$$|\hat{Q}_{\mathbf{w}}^{(\ell)} - Q_{\mathbf{w}}^{(\ell)}| \lesssim \log q_{\max}^{(\ell)} / N^{\ell-1}. \quad (3.5.1)$$

Furthermore, $|\hat{Q}_{\mathbf{w}}^{(\ell)} - Q_{\mathbf{w}}^{(\ell)}| / Q_{\mathbf{w}}^{(\ell)} = \log q_N / q_N = o(1)$ when under Assumption 3.2.2.

The precision of the estimation $\hat{Q}_{\mathbf{w}}^{(\ell)}$ is based on the fact that the number of edges corresponding to partition $\mathbf{w}$ is q_N many for each $\mathbf{w} \in \mathcal{WC}_K^{(\ell)}$ and $\ell \in \mathcal{L}$, while the number of different types $\mathbf{w}$ is finite. As long as all $Q_{\mathbf{w}}^{(\ell)}$ are comparable (Assumption 3.2.2), this estimation should be correct. In the absence of Assumption 3.2.2, e.g., $q_{\max}^{(\ell)} = N^\beta$ for some $\beta \in (0, 1)$ while $q_{\mathbf{w}}^{(\ell)} = \log \log(N)$, $|\hat{Q}_{\mathbf{w}}^{(\ell)} - Q_{\mathbf{w}}^{(\ell)}| = o(1)$ is not true. Thus, $\hat{Q}_{\mathbf{w}}^{(\ell)}$ cannot be accurate.

We present briefly a discussion of Assumption 3.2.2 here. According to Lemma 3.5.1, Assumption 3.2.2 is crucial for $\widehat{Q}_{\mathbf{w}}^{(\ell)}$ being an accurate estimation of $Q_{\mathbf{w}}^{(\ell)}$. In fact, when a mixture of Bernoulli random variables is collected where their expectations satisfy Assumption 3.2.2, it is possible to estimate each expectation accurately, since the sample average for each Bernoulli random variable (in this scenario, all but a negligible fraction of samples are chosen from their true law) would be well concentrated around its expectation with lower order deviations. However, there is no hope to estimate the lower order expectations accurately if their expectations vary in magnitude, since the fluctuation of the sample average with higher order expectation could be significantly larger than the lower order expectations.

3.5.2 Proofs of strong consistency and optimality

We first show that $\mathcal{V}$ can be partitioned into two disjoint subsets $\mathcal{G}$ and $\mathcal{V} \setminus \mathcal{G}$, where each $v \in \mathcal{G}$ enjoys the following three properties:

(G1) Denote $Q_{\max}^{(\ell)} := \max_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell)}}} Q_{\mathbf{w}}^{(\ell)}$, $\mathcal{D}_{\max}^{(\ell)} := \binom{N-1}{\ell-1} Q_{\max}^{(\ell)}$ and $\mathcal{D}_{\max} := \sum_{\ell \in \mathcal{L}} \mathcal{D}_{\max}^{(\ell)}$. The degree of v is upper bounded by

$$D_v \leq 10\mathcal{D}_{\max}. \quad (\text{G1})$$

(G2) Let $\mu_{v,\mathbf{w}}^{(\ell)} := D_{v,\mathbf{w}}^{(\ell)}/N_{\mathbf{w}}$ and let $\mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)})$ be the KL divergence in (3.3.12). If $v \in \mathcal{V}_k \cap \mathcal{G}$, then the weighted sum of $\mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)})$ is much smaller than the weighted sum of $\mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)})$ for any other $j \neq k$. Formally,

$$\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} N_{\mathbf{w}} \cdot [\mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}) - \mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)})] \geq \mathcal{D}_{\max}/\log q_N. \quad (\text{G2})$$

(G3) Let $\mathcal{E}(v, \mathcal{V} \setminus \mathcal{G})$ denote the set of hyperedges containing v with all the remaining nodes in $\mathcal{V} \setminus \mathcal{G}$. Let L denote the maximum element of $\mathcal{L}$. Let $\delta > 0$ be some absolute constant.

The number of edges containing v is bounded by

$$|\mathcal{E}(v, \mathcal{V} \setminus \mathcal{G})| \leq L(\mathcal{D}_{\max})^{1+\delta}. \quad (\text{G3})$$

We explain how this construction can be done in the proof of Lemma 3.5.2. Briefly, we show that most of vertices satisfy (G1) and (G2), and proceed to eliminate the nodes that do not satisfy (G3). The number of vertices we need to eliminate is small.

Lemma 3.5.2 (Size of $\mathcal{V} \setminus \mathcal{G}$). *For the sequence $\kappa_N \in (0, 1]$ with $\kappa_N \log(N) \rightarrow \infty$ as $N \rightarrow \infty$, assume (3.2.17) for some absolute (small) constant $\varepsilon > 0$. Then $|\mathcal{V} \setminus \mathcal{G}| < N^{1-\kappa_N}$ with probability at least $1 - 6e^{-\varepsilon \kappa_N \log(N)}$.*

Let $\widehat{\mathcal{N}}_{jk}^{(t)} := (\widehat{\mathcal{V}}_j^{(t)} \cap \mathcal{V}_k) \cap \mathcal{G}$ denote the set of vertices in $\mathcal{G}$ which belongs to $\mathcal{V}_k$ but misclassified to $\widehat{\mathcal{V}}_j^{(t)}$, then the misclassified vertices in $\widehat{\mathcal{V}}_j^{(t)} \cap \mathcal{G}$ can be represented as $\widehat{\mathcal{N}}_j^{(t)} := \cup_{k \neq j} \widehat{\mathcal{N}}_{jk}^{(t)} = (\widehat{\mathcal{V}}_j^{(t)} \setminus \mathcal{V}_j) \cap \mathcal{G}$. Let $\widehat{\mathcal{N}}^{(t)} := \cup_{j=1}^K \widehat{\mathcal{N}}_j^{(t)} = \cup_{j=1}^K (\widehat{\mathcal{V}}_j^{(t)} \setminus \mathcal{V}_j) \cap \mathcal{G}$ denote the set of misclassified vertices in $\mathcal{G}$ at step t . Lemma 3.5.3 shows that the size of $|\widehat{\mathcal{N}}^{(t)}|$ decays over consecutive iterations.

Lemma 3.5.3. *In outer for-loop of Algorithm 3.2.3, the number of misclassified nodes decays*

$$|\widehat{\mathcal{N}}^{(t+1)}|/|\widehat{\mathcal{N}}^{(t)}| \leq e^{-1} \cdot \log C_{(3.2.3)}/C_{(3.2.3)}, \quad (3.5.2)$$

with probability at least $1 - 2(e/2)^{-N} - 2N^{-c}$ for some constant $c > 0$.

Proof of Theorem 3.2.10. Recall that $\eta_N = \eta(\mathbf{y}, \widehat{\mathbf{y}}^{(0)}) \lesssim (\rho_N)^{-1}$ according to (3.4.10). Note that the agnostic result Theorem 3.2.6 requires $\rho_N \gtrsim \log(N)$, then by Lemma 3.5.3, the number of misclassified nodes in $\mathcal{G}$ after $\lceil \log(N) \rceil$ iterations will be at most

$$\begin{aligned} |\widehat{\mathcal{N}}^{\lceil \log(N) \rceil}| &\leq \eta_N N \cdot (e^{-1} \log C_{(3.2.3)}/C_{(3.2.3)})^{\lceil \log(N) \rceil} \\ &\lesssim N \cdot N^{-1-\log C_{(3.2.3)}/\log(N)} = o(1), \end{aligned}$$

where the inequality holds since $C_{(3.2.3)} > 1$. Consequently, $\widehat{\mathcal{N}}^{\lceil \log(N) \rceil} = \emptyset$, indicating that all nodes in $\mathcal{G}$ are correctly assigned. Thus, the misclassified nodes are all in the set $\mathcal{V} \setminus \mathcal{G}$, leading to $N \cdot \eta_N \leq |\mathcal{V} \setminus \mathcal{G}| \leq N^{1-\kappa_N}$. Therefore, the desired argument follows with probability at least $1 - 6e^{-\varepsilon \cdot \kappa_N \log(N)} - (1 + o(1)) \lceil \log(N) \rceil N^{-c}$. $\square$

Proof of Theorem 3.2.9. Denote $\kappa_N := \mathbb{D}_{\text{GCH}} \cdot q_N / \log(N)$. In addition, assume that the limit of the sequence $\{\kappa_N\}_N$ exists, denoted by $\kappa = \lim_{N \rightarrow \infty} \kappa_N$. There are three scenarios.

(a) $\kappa \in (0, 1]$. By Theorem 3.2.4 (2), $\mathbb{E}[\eta_N] \geq N^{-(1+o(1)) \cdot \kappa}$. According to Theorem 3.2.10, $\eta_N \leq N^{-\kappa_N}$ with probability at least $1 - 6e^{-\varepsilon \cdot \kappa_N \log(N)} - (1 + o(1)) \lceil \log(N) \rceil N^{-c}$. Therefore, we have $\mathbb{E}[\eta_N] = N^{-\kappa} = o(1)$, reaching the lowest expected mismatch ratio possible.

(b) $\kappa = 0$ with $\kappa_N \gg \log^{-1}(N)$; we still have $\mathbb{E}[\eta_N] \geq N^{-(1+o(1)) \kappa_N} = o(1)$ by Theorem 3.2.4 (2), and Algorithm 3.2.3 achieving the lowest possible expected mismatch ratio. Otherwise when $\kappa_N \lesssim \log^{-1}(N)$, we have $N^{-\kappa_N} = O(1)$, which represents the regime of partial or weak recovery, not the focus of this paper.

(c) $\kappa > 1$, equivalently (3.2.16) holds. Following the same analysis above, $\mathcal{V} \setminus \mathcal{G} = \emptyset$ with probability at least $1 - 6e^{-\varepsilon \log(N)}$ by Lemma 3.5.2. All nodes in $\mathcal{G} = \mathcal{V}$ will be classified correctly by Lemma 3.5.3, achieving exact recovery. Note that when $\kappa = 1$, it is not clear whether the exact recovery can be achieved or not.

Furthermore, if the sequence $\{\kappa_N\}_N$ goes to infinity as $N \rightarrow \infty$, then clearly, Algorithm 3.2.1 accomplishes exact recovery. $\square$

We note that Algorithm 3.2.3 utilizes the adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$. The proofs of Theorem 3.2.10 and Theorem 3.2.9 do not rely on Assumption 3.2.7. For the set of problems violating Assumption 3.2.7, if a weak consistent partition is obtained, it can be improved by Algorithm 3.2.3, achieving the lowest expected mismatch ratio possible.

3.6 Partition with prior knowledge

This section is devoted to the proofs of Theorems 3.2.13 and 3.2.14, with the proofs of Lemmas deferred to [83].

3.6.1 Optimality of MAP

Let $\mathbb{P}_{\text{err}}(v)$ denote the probability of $v \in \mathcal{V}$ being misclassified. Let $\hat{y}_v, \hat{\mathbf{y}}_{-v}$ be some estimations of y_v and $\mathbf{y}_{-v}$ respectively. Let $\mathbb{P}(\cdot \mid \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})$ denote the conditional probability when observing $\mathcal{H}$ and $\hat{\mathbf{y}}_{-v}$, where $\mathbb{H}$ and $\mathbf{Y}_{-v}$ denote the laws of $\mathcal{H}$ and $\mathbf{y}_{-v}$, respectively. Then by definition,

$$\begin{aligned} \mathbb{P}_{\text{err}}(v) &:= \mathbb{P}(\hat{y}_v \neq y_v) = \sum_{\mathcal{H}, \hat{\mathbf{y}}_{-v}} \mathbb{P}(\hat{y}_v \neq y_v \mid \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \\ &= \sum_{\mathcal{H}, \hat{\mathbf{y}}_{-v}} [1 - \mathbb{P}(\hat{y}_v = y_v \mid \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})] \cdot \mathbb{P}(\mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}). \end{aligned} \quad (3.6.1)$$

By Bayes, the term $\mathbb{P}(\mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})$ is irrelevant to any specific assignment of y_v , since

$$\mathbb{P}(\mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) = \mathbb{P}(\mathbb{H} = \mathcal{H} \mid \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \quad (3.6.2)$$

$$= \mathbb{P}(\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \sum_{j \in [K]} \mathbb{P}(\mathbb{H} = \mathcal{H} \mid Y_v = j, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = j) \quad (3.6.3)$$

Therefore, $\mathbb{P}_{\text{err}}(v)$ is minimized only if the MAP estimator $\hat{y}_v^{\text{MAP}}$ in (3.2.18) is substituted into (3.6.1). However, $\hat{y}_v^{\text{MAP}}$ in (3.2.18) cannot be computed directly in practice as explained. Instead, Algorithm 3.2.4 utilized the simplified version (3.2.19) inspired by Lemma 3.2.11. Recall $D_v = \sum_{\ell, \mathbf{w}} D_{v, \mathbf{w}}^{(\ell)}$ in (3.1.1) and $d_v = \sum_{\ell, \mathbf{w}} d_{v, \mathbf{w}}^{(\ell)}$, where $D_{v, \mathbf{w}}^{(\ell)} \mid Y_v = k \sim \text{Bin}(N_{\mathbf{w}}, Q_{k \oplus \mathbf{w}}^{(\ell)})$ and $d_{v, \mathbf{w}}^{(\ell)}$ denote

its realization. From simplicity, we denote

$$\widehat{\mathbb{P}}_k(\mathbf{D}_v = d_v) := \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \mathbb{P}(\mathbf{D}_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)} | \mathbf{Y}_v = k, \mathbf{Y}_{-v} = \widehat{\mathbf{y}}_{-v}), \quad (3.6.4a)$$

$$\mathbb{P}_k(\mathbf{D}_v = d_v) := \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \mathbb{P}(\mathbf{D}_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)} | \mathbf{Y}_v = k, \mathbf{Y}_{-v} = \widehat{\mathbf{y}}_{-v}). \quad (3.6.4b)$$

According to Lemma 3.2.11 and edgewise independence, the probabilities in (3.6.1) that are irrelevant to v cancels out. By substituting $\widehat{\mathbf{y}}_v^{\text{MAP}}$ in (3.2.19) into (3.6.1), it follows that¹

$$\begin{aligned} \mathbb{P}_{\text{err}}(v) &= \sum \mathbb{P}(\widehat{\mathbf{y}}_v^{\text{MAP}} \neq k | \mathbf{D}_v = d_v, \mathbf{Y}_{-v} = \widehat{\mathbf{y}}_{-v}) \cdot \mathbb{P}_k(\mathbf{D}_v = d_v) \\ &= \sum \mathbb{P}_k(\mathbf{D}_v = d_v) \cdot \mathbb{P}(\exists j \neq k, \text{s.t. } \alpha_j \widehat{\mathbb{P}}_j(\mathbf{D}_v = d_v) > \alpha_k \widehat{\mathbb{P}}_k(\mathbf{D}_v = d_v)). \end{aligned} \quad (3.6.5)$$

Furthermore, $\mathbb{P}_{\text{err}}(v)$ admits the upper and lower bounds presented in Lemma 3.6.1.

Lemma 3.6.1. *Assume $v \in \mathcal{V}_k$ for some $k \in [K]$. Define the following quantity*

$$\widehat{\mathbf{R}}_k := \sum_{j \in [K], j \neq k} \mathbb{P}(\exists j \neq k, \text{s.t. } \alpha_j \widehat{\mathbb{P}}_j(\mathbf{D}_v = d_v) > \alpha_k \widehat{\mathbb{P}}_k(\mathbf{D}_v = d_v)). \quad (3.6.6)$$

Then the upper and lower bounds of $\mathbb{P}_{\text{err}}(v)$ are given by

$$(K-1)^{-1} \sum \mathbb{P}_k(\mathbf{D}_v = d_v) \cdot \widehat{\mathbf{R}}_k \leq \mathbb{P}_{\text{err}}(v) \leq \sum \mathbb{P}_k(\mathbf{D}_v = d_v) \cdot \widehat{\mathbf{R}}_k. \quad (3.6.7)$$

3.6.2 Error decomposition

As discussed previously, $\mathbb{P}_{\text{err}}(v)$ can be characterized from both direction, as long as $\widehat{\mathbf{R}}_k$ is evaluated accurately. The difficulty lies on the fact that (3.6.4a) is not directly accessible for analysis due to non-equivalence between $\widehat{\mathbf{y}}_{-v}$ and $\mathbf{y}_{-v}$, and we do not know how far it deviates from the true probability (3.6.4b). Fortunately, the ratio (3.6.4a)/(3.6.4b) can be bounded from

¹The summation in (3.6.5), (3.6.7) and (3.6.11) is over all possible realizations $\{d_{v,\mathbf{w}}^{(\ell)}\}_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}, \ell \in \mathcal{L}}$.

both sides in almost all cases with some allowable errors, thus (3.6.4a) can be replaced by (3.6.4b) when evaluating $\widehat{\mathbf{R}}_k$ in (3.6.6).

We now present the proof sketch (with details provided in [83]). Let $\mathcal{E}_\ell(v)$ denote the set of ℓ -edges containing v , while $\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})$ be its subset where each edge contains v and at least one node from $\widehat{\mathcal{N}}^{(0)}$, where $\widehat{\mathcal{N}}^{(0)}$ in Lemma 3.5.3 denotes the set of misclassified nodes after initial estimation. Note that $\mathbf{y}(e) = \widehat{\mathbf{y}}^{(0)}(e)$ for $e \notin \mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})$, then the desired ratio is

$$\frac{(3.6.4a)}{(3.6.4b)} = \prod_{\ell \in \mathcal{L}} \prod_{e \in \mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})} \left[\frac{\widehat{\mathbb{P}}_k(\mathcal{A}_e^{(\ell)} = 1)}{\mathbb{P}_k(\mathcal{A}_e^{(\ell)} = 1)} \right]^{\mathcal{A}_e^{(\ell)}} \cdot \left[\frac{\widehat{\mathbb{P}}_k(\mathcal{A}_e^{(\ell)} = 0)}{\mathbb{P}_k(\mathcal{A}_e^{(\ell)} = 0)} \right]^{1 - \mathcal{A}_e^{(\ell)}}, \quad (3.6.8)$$

since the probabilities for edges $e \notin \mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})$ cancel out. We say that the realization $\mathcal{E}_\ell(v)$ is *good* if the cardinality of its subset $\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})$ is relatively small. We say that the edge set $\mathcal{E}(v) = \cup_{\ell \in \mathcal{L}} \mathcal{E}_\ell(v)$ is good if $\mathcal{E}_\ell(v)$ is good for each $\ell \in \mathcal{L}$.

Definition 3.6.2 (Good realization). Recall $Q_{\max}^{(\ell)}, q_{\max}^{(\ell)}$ defined in Lemma 3.5.1. We say that the realization $\mathcal{E}(v)$ is good if $G(v) := \prod_{\ell \in \mathcal{L}} G_\ell(v) = 1$, where for each $\ell \in \mathcal{L}$

$$G_\ell(v) = \mathbb{1}_{\left\{ |\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})| \leq 4q_{\max}^{(\ell)} / \log(q_{\max}^{(\ell)}) \right\}}. \quad (3.6.9)$$

Conversely, $\mathcal{E}(v)$ is Not good if $G(v) = 0$, i.e., $G_\ell(v) = 0$ for at least one $\ell \in \mathcal{L}$.

According to Bayes, we have $\widehat{\mathbf{R}}_k = \widehat{\mathbf{R}}_{k|0} \cdot \mathbb{P}(G(v) = 0) + \widehat{\mathbf{R}}_{k|1} \cdot \mathbb{P}(G(v) = 1)$, where

$$\widehat{\mathbf{R}}_{k|\ell} := \sum_{j \in [K], j \neq k} \mathbb{P}[\exists j \neq k, \text{s.t. } \alpha_j \widehat{\mathbb{P}}_j(\mathbf{D}_v = d_v) > \alpha_k \widehat{\mathbb{P}}_k(\mathbf{D}_v = d_v) | G(v) = \ell], \quad (3.6.10)$$

for $\ell \in \{0, 1\}$. Consequently, $\mathbb{P}_{\text{err}}(v)$ in (3.6.5) is upper bounded by $\mathbb{P}_{\text{err}}(v) \leq \textcircled{0} + \textcircled{1}$, where

$$\textcircled{\ell} = \sum \mathbb{P}_k(\mathbf{D}_v = d_v) \cdot \widehat{\mathbf{R}}_{k|\ell} \cdot \mathbb{P}(G(v) = \ell). \quad (3.6.11)$$

Bound for $\textcircled{0}$ The probability of $\mathcal{E}(v)$ being *Not good* can be controlled by $\mathbb{D}_{\text{GCH}} \cdot q_N$.

Lemma 3.6.3. For $v \in \mathcal{V}'_k$, we have $\mathbb{P}_k(G(v) = 0) \lesssim \exp(-(4 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N)$.

Since $\widehat{\mathbf{R}}_{k|0} \leq K - 1$ as defined in (3.6.10) and $\sum_{d_v} \mathbb{P}_k(\mathbf{D}_v = d_v) = 1$, it then follows that

$$\textcircled{0} \lesssim \exp(-(4 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N). \quad (3.6.12)$$

Bound for $\textcircled{1}$. When $G(v) = 1$, the ratio (3.6.4a)/(3.6.4b) can be bounded from both sides as proved by Lemma 3.6.4, which further leads to an upper bound on $\widehat{\mathbf{R}}_{k|1}$, as well as $\textcircled{1}$.

Lemma 3.6.4. Suppose (3.2.20). Conditioned on $v \in \mathcal{V}'_k$ and $G(v) = 1$, there exists some $U_k \in \mathbb{R}^+$ with $\log U_k \ll \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N$ for any $j \neq k$, such that $U_k^{-1} \leq \widehat{\mathbb{P}}_k(\mathbf{D}_v = d_v) / \mathbb{P}_k(\mathbf{D}_v = d_v) \leq U_k$ with probability at least $1 - N^{-10}$.

With some arguments deferred to [83], with probability $1 - N^{-10}$, $\textcircled{1}$ is upper bounded by

$$\textcircled{1} \lesssim \exp(-(1 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N). \quad (3.6.13)$$

Bound for $\mathbb{P}_{\text{err}}(v)$. Combining (3.6.7) and upper bounds above for $\textcircled{0}$ and $\textcircled{1}$, it follows that

$$\mathbb{P}_{\text{err}}(v) \leq \textcircled{0} + \textcircled{1} \lesssim (1 + o(1)) \cdot e^{-(1+o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N}, \quad (3.6.14)$$

with probability at least $1 - N^{-10}$, where the last inequality holds since $\mathbb{D}_{\text{GCH}} \cdot q_N \rightarrow \infty$ as $N \rightarrow \infty$.

The term $\mathbb{D}_{\text{GCH}} \cdot q_N$ plays a crucial role in understanding the limit of Algorithm 3.2.4.

3.6.3 Algorithm correctness

The complete proofs are presented in this subsection.

Proof of Theorem 3.2.14. Suppose (3.2.17) where the sequence $\kappa_N \in (0, 1]$ with $\kappa_N \log(N) \rightarrow \infty$, then there exists some universal (small) constant $\varepsilon > 0$ such that $\mathbb{D}_{\text{GCH}} \cdot q_N \geq (1 + \varepsilon) \kappa_N \log(N)$.

According to (3.6.14), $\mathbb{P}_{\text{err}}(v) \lesssim (1 + o(1)) \cdot N^{-(1+\varepsilon)\kappa_N}$ for each $v \in \mathcal{V}$, then with probability at least $1 - N^{-10}$, the expected mismatch ratio is upper bounded by

$$\mathbb{E}[\eta_N] = \mathbb{E}[\# \text{ of misclassified nodes}]/N = \sum \mathbb{P}_{\text{err}}(v)/N \lesssim (1 + o(1)) \cdot N^{-(1+\varepsilon)\kappa_N}. \quad (3.6.15)$$

An argument similar to the proof of Theorem 3.2.9, together with Theorem 3.2.4 (2), shows that Algorithm 3.2.4 reaches the lowest possible expected mismatch ratio. $\square$

Proof of Theorem 3.2.13. Note that $\mathbb{D}_{\text{GCH}} \cdot q_N \geq (1 + \varepsilon) \log(N)$ when taking $\kappa_N = 1$. Then the probability of $\hat{y}_v^{\text{MAP}}$ in (3.2.19) failing exact recovery is upper bounded by

$$\mathbb{P}(\exists v \in \mathcal{V}, \hat{y}_v^{\text{MAP}} \neq y_v) \leq \sum \mathbb{P}_{\text{err}}(v) \leq N \cdot (1 + o(1)) \cdot N^{-(1+\varepsilon)\kappa_N} \leq (1 + o(1)) \cdot N^{-\varepsilon}. \quad (3.6.16)$$

Therefore, Algorithm 3.2.4 achieves exact recovery with probability at least $1 - N^{-\varepsilon} - N^{-10}$. $\square$

3.7 Estimation for number of communities

For the practical purpose, it is better to assume as little prior knowledge as possible. To make Algorithm 3.2.1 completely agnostic, we propose the following Algorithm 3.7.1 for the estimation of the number of communities.

Algorithm 3.7.1: Number of Communities

Data: $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$

- 1 Compute the degree d_v for each vertex $v \in \mathcal{V}$ and find $\tilde{d} = \max_{v \in \mathcal{V}} d_v$;
- 2 Construct the adjacency matrix $\mathbf{A}$ and compute the eigenvalues $\lambda_1(\mathbf{A}) \geq \dots \geq \lambda_N(\mathbf{A})$;
- 3 Find the first eigenvalue $\lambda_r(\mathbf{A})$ such that $\lambda_r \leq (\tilde{d})^{3/4}$;

Result: $\hat{K} = r - 1$

The functionality of Algorithm 3.7.1 is based on Assumption 3.7.1 (*eigenvalue separation*) and Assumption 3.2.2 (*proportional degree*) under the exact recovery regime (3.2.2). On the other hand, one may refer to [185] for estimating the number of communities under

the bounded degree regime. Recall that ρ_N, ξ_N in (3.4.1) represent the maximum and average expected degrees, respectively. Let $\zeta_N := \min_{v \in \mathcal{V}} \mathbb{E}(D_v)$ denote the minimum expected degree.

Assumption 3.7.1. Define the matrix $\mathbb{E}\tilde{\mathbf{A}}$ where $\mathbb{E}\tilde{\mathbf{A}}_{ij} = \mathbb{E}\mathbf{A}_{ij}$ for $i \neq j$ and $\mathbb{E}\tilde{\mathbf{A}}_{ii} = \mathbb{E}\mathbf{A}_{ij}$ when i and j are from the same block. The eigenvalues $\lambda_1(\mathbb{E}\tilde{\mathbf{A}}) \geq \dots \geq \lambda_N(\mathbb{E}\tilde{\mathbf{A}})$ satisfy $\lambda_K(\mathbb{E}\tilde{\mathbf{A}}) \gtrsim \zeta_N$ and $\lambda_{K+1}(\mathbb{E}\tilde{\mathbf{A}}) = 0$.

Lemma 3.7.2. Under Assumptions 3.2.2, 3.7.1, $\lambda_k(\mathbb{E}\mathbf{A}) \gtrsim \rho_N$ for $k \in [K]$, and $\lambda_k(\mathbb{E}\mathbf{A}) \lesssim N^{-1}\rho_N$ for $k \in \mathcal{V} \setminus [K]$.

Proof of Lemma 3.7.2. Define the diagonal matrix $\mathbf{D} := \mathbb{E}\tilde{\mathbf{A}} - \mathbb{E}\mathbf{A}$, then $\lambda_k(\mathbf{D}) = \mathbb{E}\tilde{\mathbf{A}}_{kk} \lesssim N^{-1}\rho_N$. Note the facts that $\zeta_N \asymp \rho_N$ by Assumption 3.2.2, and $\lambda_K(\mathbb{E}\tilde{\mathbf{A}}) \gtrsim \zeta_N \gg N^{-1}\rho_N$ by Assumption 3.7.1, then the desired result follows by Weyl's inequality D.2.1, since $\lambda_k(\mathbb{E}\tilde{\mathbf{A}}) - \|\mathbf{D}\| \leq \lambda_k(\mathbb{E}\mathbf{A}) \leq \lambda_k(\mathbb{E}\tilde{\mathbf{A}}) + \|\mathbf{D}\|$ for all $k \in \mathcal{V}$. $\square$

Correctness of Algorithm 3.7.1. By Weyl's inequality D.2.1, for any $k \in \mathcal{V}$, the eigenvalue $\lambda_k(\mathbb{E}\mathbf{A})$ satisfies

$$\lambda_k(\mathbb{E}\mathbf{A}) - \|\mathbf{A} - \mathbb{E}\mathbf{A}\| \leq \lambda_k(\mathbf{A}) \leq \lambda_k(\mathbb{E}\mathbf{A}) + \|\mathbf{A} - \mathbb{E}\mathbf{A}\|. \quad (3.7.1)$$

By Lemma 3.4.1 $\tilde{d} \asymp \mathcal{D}_{\max} \asymp \rho_N$, then for sufficiently large N , by Lemma 3.7.2 and Theorem 3.4.2

$$\lambda_K(\mathbf{A}) \gtrsim \rho_N - C_{(3.4.3)} \cdot \sqrt{\mathcal{D}_{\max}} \gtrsim \rho_N - \sqrt{\rho_N} \gg (\rho_N)^{3/4} \asymp (\tilde{d})^{3/4}, \quad (3.7.2)$$

$$\lambda_{K+1}(\mathbf{A}) \leq N^{-1}\rho_N + C_{(3.4.3)} \cdot \sqrt{\mathcal{D}_{\max}} \lesssim \sqrt{\rho_N} \ll (\rho_N)^{3/4} \asymp (\tilde{d})^{3/4}. \quad (3.7.3)$$

Therefore, the accuracy of $\hat{K}$ in Algorithm 3.7.1 is guaranteed by the separation between $\lambda_K(\mathbf{A})$ and $\lambda_{K+1}(\mathbf{A})$. $\square$

We have seen that Algorithm 3.7.1 works when all the degrees are proportional. One

may be interested in exploring modifications of Algorithm 3.7.1 for scenarios not covered by Assumptions 3.2.2, 3.7.1, such as the presence of communities with outstandingly large degrees.

3.7.1 Eigenvalue separation

In particular, we want to emphasize that Assumption 3.7.1 is stronger than Assumption 3.2.7. Consider the following example, where $K = 3$, $L = 2$ with $\alpha_1 = \alpha_2 = \alpha_3 = 1/3$, and

$$Q^{(2)} = \frac{\log(N)}{N} \begin{bmatrix} 4 & 3 & 7 \\ 3 & 5 & 8 \\ 7 & 8 & 15 \end{bmatrix}, \quad \mathbb{E}\tilde{\mathbf{A}} = Q^{(2)} \otimes (\mathbf{1}_{N/3} \mathbf{1}_{N/3}^T), \quad (3.7.4)$$

where $\otimes$ denotes the *Kronecker* product. Obviously, Assumption 3.2.7 is satisfied. However, $\lambda_3(\mathbb{E}\tilde{\mathbf{A}}) = 0$ since $Q_{3\cdot}^{(2)} = Q_{1\cdot}^{(2)} + Q_{2\cdot}^{(2)}$. As we can see, a mixture of assortative ($Q_{3\cdot}^{(2)}$) and disassortative ($Q_{1\cdot}^{(2)}, Q_{2\cdot}^{(2)}$) clusters may result in violating Assumption 3.7.1 but satisfying Assumption 3.2.7. We further note that this would never happen when all communities are purely assortative (diagonally dominant) or purely disassortative due to the symmetry requirement for probability tensors.

As far as we know, even in the graph case, Assumption 3.7.1 is crucial in most algorithms aiming to estimate for the number of communities. For example, [56, Theorem 1] applies a *Network cross-validation* algorithm. This implicitly requires Assumption 3.7.1, as it relies on Davis-Kahan [208, Theorem 4] to obtain an upper bound for the singular subspace perturbation, which is meaningful only if the K th largest singular value is at least of order $\log(N)$.

As we just mentioned, Assumption 3.2.7, which Algorithm 3.2.1 relies on, is weaker than Assumption 3.7.1. It would be interesting to close the gap between the those two assumptions, i.e., to find whether it is possible to give an accurate estimation the number of communities when the underlying generating model is a mixture of assortative and disassortative clusters.

3.7.2 Proportional degrees pitfalls and possible fixes

Assumption 3.2.2 is crucial for Algorithm 3.7.1. When this is dropped, and we are under the regime $\rho_N \gg \zeta_N \gtrsim \log(N)$, Algorithm 3.7.1 tends to underestimate the number of communities due to the possibility that the upper bound $(\tilde{d})^{3/4}$ is much larger than $\lambda_K(\mathbf{A})$, leading to the problem that sparse clusters would be absorbed by denser clusters. One could strengthen Assumption 3.7.1 to satisfy $\lambda_K(\mathbb{E}\tilde{\mathbf{A}}) \gtrsim \rho_N$ to make full use of Algorithm 3.7.1, though not so realistic in practice.

Instead, one could start with a lax overestimation of the number of communities, and then slightly modify Algorithm 3.2.2 to correct the overestimate. The intuition is that all but a vanishing fraction of the rows corresponding to vertices in $\mathcal{V}_k$ would concentrate around their expected center, thus their corresponding vertices will be located in the ball $\mathcal{B}_r(s_k)$, while $\hat{\mathcal{V}}_i^{(0)} = o(N)$ for $i > K$ by Lemma 3.4.7. Also, Algorithm 3.2.2 works agnostically without Assumption 3.2.2 when $\rho_N \gtrsim \log(N)$ as proved in Section 3.4. Thus, we propose Algorithm 3.7.2 as an informal alternative; here we pick some large enough i as an initial guess and iteratively decrease i until the partition has all classes proportional. However, we leave the criterion for deciding if a class is too small up to the reader, and we offer no guarantees. Note also that Algorithm 3.7.2 is computationally inefficient: it has time complexity $O(N^2 \log^2(N))$ since at least $\log(N)$ eigenpairs should be computed.

3.8 Concluding remarks

We mention a few other interesting directions below.

First, Assumption 3.2.7 is crucial for any spectral algorithm based on $\mathbf{A}$, since it is required for separation between expected centers. It will be interesting to see whether this condition can be removed, or at least weakened, such that the weak consistency could be achieved by some other non-spectral method.

Second, it is known that exact recovery is achievable *at* the threshold in the case of graph

Algorithm 3.7.2: Number of Communities

Data: $\mathbf{A}_g$, radius $r = [N \log(\rho_N)]^{-1} \rho_N^2$ with $\rho_N = \tilde{d} = \max_{v \in \mathcal{V}} d_v$, $i = \log(N)$

```

1 while  $i \geq 2$  do
2   Compute the rank- $i$  approximation  $\mathbf{A}_g^{(i)} = \sum_{j=1}^i \lambda_j \mathbf{u}_j^\top \mathbf{u}_j$  of  $\mathbf{A}_g$ .
3   Construct balls of vertices centered at  $s \in \mathcal{S}$  by
       $\mathcal{B}_r(s) = \{w \in \mathcal{J} : \|(\mathbf{A}_g^{(K)})_s - (\mathbf{A}_g^{(K)})_w\|_2^2 \leq r\}$ , where  $\mathcal{S}$  is randomly sampled
      from  $\mathcal{J}$  with  $|\mathcal{S}| = \log^2(N)$ .
4   Let  $\mathcal{B}_r(s_1)$  denote the ball with most vertices and take  $\hat{\mathcal{V}}_1^{(0)} = \mathcal{B}_r(s_1)$ .
5   while  $2 \leq k \leq i$  do
6      $s_k = \arg \max_{s \in \mathcal{S}} |\mathcal{B}_r(s) \setminus (\bigcup_{j=1}^{k-1} \hat{\mathcal{V}}_j^{(0)})|$ ;
7      $\hat{\mathcal{V}}_k^{(0)} = \mathcal{B}_r(s_k) \setminus (\bigcup_{j=1}^{k-1} \hat{\mathcal{V}}_j^{(0)})$ ;
8   end
9   if  $|\hat{\mathcal{V}}_i^{(0)}| \ll N$ ; // Informal condition, should be replaced with an
      explicit criterion.
10  then
11     $i \leftarrow i - 1$ 
12  else
13    Quit the loop;
14  end
15 end

Result:  $\hat{K} = i$ 

```

SBM, i.e., when $\mathbb{D}_H(a, b) = (\sqrt{a} - \sqrt{b})^2/2 = 1$ [169] and $\mathbb{D}_{CH}(\boldsymbol{\alpha}, \mathbf{P}) = 1$ [5]. However, it is still unclear whether exact recovery can be achieved when $\mathbb{D}_{GCH} = 1$ for model 1.1.8.

Last but not least, it will be interesting to see if model 1.1.8 can be generalized to have other degree distributions [105, 130], different edge labels [212] and contextual feature vectors [57, 161], which would be of particular interest for real-world applications.

3.9 Acknowledgments

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Chapter 4

Partial recovery and weak consistency in the non-uniform hypergraph stochastic block model

4.1 Introduction

Clustering is one of the central problems in network analysis and machine learning [173, 181, 174]. Many clustering algorithms make use of graph models, which represent pairwise relationships among data. A well-studied probabilistic model is the *Stochastic Block Model* (SBM), which was first introduced in [128] as a random graph model that generates community structure with given ground truth for clusters so that one can study algorithm accuracy. The past decades have brought many notable results in the analysis of different algorithms and fundamental limits for community detection in SBMs in different settings [68, 197, 116, 166]. A major breakthrough was the proof of phase transition behaviors of community detection algorithms in various connectivity regimes [163, 44, 170, 167, 169, 2, 7]. See the survey [1] for more references.

Hypergraphs can represent more complex relationships among data [36, 31], including recommendation systems [51, 155], computer vision [113, 202], and biological networks [165, 190], and they have been shown empirically to have advantages over graphs [218]. Besides community detection problems, sparse hypergraphs and their spectral theory have also found

applications in data science [135, 219, 120], combinatorics [88, 102, 183], and statistical physics [53, 180].

With the motivation from a broad set of applications, many efforts have been made in recent years to study community detection on random hypergraphs. The *Hypergraph Stochastic Block Model* (HSBM), as a generalization of graph SBM, was first introduced and studied in [109]. In this model, we observe a random uniform hypergraph where each hyperedge appears independently with some given probability depending on the community structure of the vertices in the hyperedge.

Succinctly put, the HSBM recovery problem is to find the ground truth clusters either approximately or exactly, given a sample hypergraph and estimates of model parameters. We may ask the following questions about the quality of the solutions (see [1] for further details in the graph case).

1. **Exact recovery (strong consistency):** With high probability, find all clusters exactly (up to permutation).
2. **Almost exact recovery (weak consistency):** With high probability, find a partition of the vertex set such that at most $o(N)$ vertices are misclassified.
3. **Partial recovery:** Given a fixed $\gamma \in (0.5, 1)$, with high probability, find a partition of the vertex set such that at least a fraction γ of the vertices are clustered correctly.
4. **Weak recovery (detection):** With high probability, find a partition correlated with the true partition.

A faithful representation of a hypergraph is its adjacency tensor. However, most of the computations involving tensors are NP-hard [127]. Instead, they considered spectral algorithms for exact recovery using hypergraph Laplacians. Subsequently, the analysis was extended to sparse, non-uniform hypergraphs [107, 110, 108].

For exact recovery of uniform HSBMs, it was shown that the phase transition occurs in the regime of logarithmic expected degrees in [156, 60, 61]. The thresholds are given for binary [146, 106] and multiple [216] community cases, by generalizing the techniques in [2, 5, 4]. After our work appeared on arXiv, thresholds for exact recovery on non-uniform HSBMs were given by [82, 198]. Strong consistency on the degree-corrected non-uniform HSBM was studied in [77].

In [109], the authors used spectral clustering based on the hypergraph's Laplacian to recover HSBMs that are dense and uniform. Subsequently, they extended their results to sparse, non-uniform hypergraphs in [107, 110, 108].

Spectral methods were considered in [60, 12, 69, 216, 210, 106], while *semidefinite programming* (SDP) methods were analyzed in [146, 151, 15]. Weak consistency for HSBMs was studied in [60, 61, 108, 111, 143].

For detection of the HSBM, the authors of [22] proposed a conjecture that the phase transition occurs in the regime of constant expected degrees. The positive part of the conjecture for the binary and multi-block case was solved in [176] and [186], respectively. Their algorithms can output a partition better than a random guess when above the *Kesten-Stigum* threshold, but can not guarantee the correctness ratio. [115, 114] proved that detection is impossible and the *Kesten-Stigum* threshold is tight for ℓ -uniform hypergraphs with binary communities when $\ell = 3, 4$, while KS threshold is not tight when $\ell \geq 7$, and some regimes remain unknown.

4.1.1 Non-uniform hypergraph stochastic block model

The non-uniform HSBM was first studied in [108], which removed the uniform hypergraph assumption in previous works, and it is a more realistic model to study higher-order interaction on networks [160, 202]. It can be seen as a superposition of several uniform HSBMs with different model parameters. We first define the uniform HSBM in our setting and extend it to non-uniform hypergraphs.

Definition 4.1.1 (Uniform HSBM). *Let $\mathcal{V} = [N]$ be the vertex set and $\ell \in \mathbb{N}^+$ be some fixed inte-*

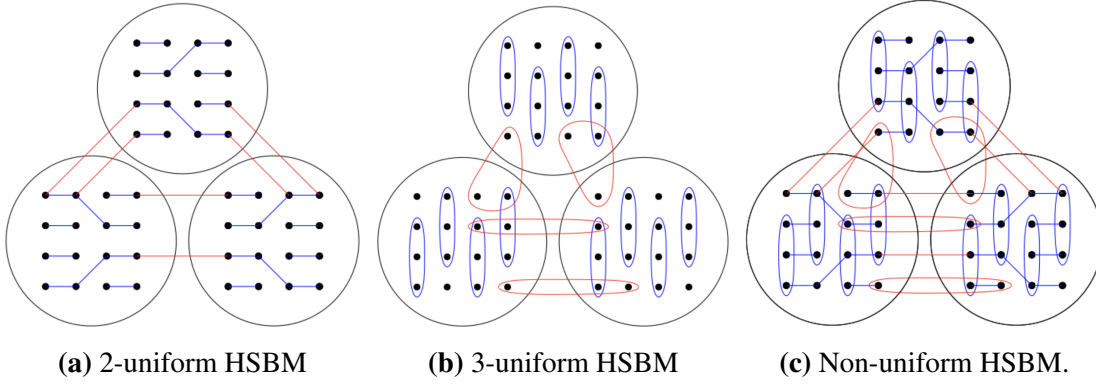


Figure 4.1. Uniform HSBM and non-uniform HSBM.

ger. Let $\mathcal{V}_1, \dots, \mathcal{V}_K$ be a partition of $\mathcal{V}$ into K disjoint communities, each of size N/K (assuming N is divisible by K). For any set of ℓ distinct vertices $i_1, \dots, i_\ell$, a hyperedge $\{i_1, \dots, i_\ell\}$ is generated with probability $a_\ell / \binom{N-1}{\ell-1}$ if all the vertices $i_1, \dots, i_\ell$ are from the same community; otherwise with probability $b_\ell / \binom{N-1}{\ell-1}$. We denote this distribution on the set of ℓ -uniform hypergraphs as

$$\mathcal{H}_\ell \sim \text{HSBM}_\ell \left(N, K, \frac{a_\ell}{\binom{N-1}{\ell-1}}, \frac{b_\ell}{\binom{N-1}{\ell-1}} \right). \quad (4.1.1)$$

Definition 4.1.2 (Non-uniform HSBM). Let $\mathcal{L} = \{\ell : \ell \geq 2, \ell \in \mathbb{N}\}$ be a set of integers with finite cardinality. Let $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ be a non-uniform random hypergraph, which can be considered as a collection of ℓ -uniform hypergraphs, i.e., $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$ with each $\mathcal{H}_\ell$ sampled from (4.1.1).

Examples of 2-uniform and 3-uniform HSBM, and an example of non-uniform HSBM with $\mathcal{L} = \{2, 3\}$ and $K = 3$ is displayed in Figure 4.1 (a), Figure 4.1 (b) and Figure 4.1 (c) respectively.

4.1.2 Main results

To illustrate our main results, we first introduce the concepts γ -correctness and *Signal-to-Noise Ratio* (SNR) to measure the accuracy of the obtained partitions.

Definition 4.1.3 (γ -correctness). Let $\mathcal{V}_1, \dots, \mathcal{V}_K$ be a partition of the vertex set $\mathcal{V}$. A collection of subsets $\hat{\mathcal{V}}_1, \dots, \hat{\mathcal{V}}_K$ of $\mathcal{V}$ is γ -correct if $|\mathcal{V}_k \cap \hat{\mathcal{V}}_k| \geq \gamma |\mathcal{V}_k|$ for all $k \in [K]$.

Definition 4.1.4. For model 4.1.2 under Assumption 4.1.5, we define the signal-to-noise ratio as

$$\text{SNR}_{\mathcal{L}}(K) := \frac{\left[\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_{\ell} - b_{\ell})K^{-\ell+1} \right]^2}{\sum_{\ell \in \mathcal{L}} (\ell - 1) \left[(a_{\ell} - b_{\ell})K^{-\ell+1} + b_{\ell} \right]}. \quad (4.1.2)$$

Let L denote the maximum element in the set $\mathcal{L}$. The following constant $C_{\mathcal{L}}(K, \mathbf{v})$ is used to characterize the accuracy of the clustering result,

$$C_{\mathcal{L}}(K, \mathbf{v}) := \frac{[\mathbf{v}^{L-1} - (1 - \mathbf{v})^{L-1}]^2}{2^3 \cdot (L - 1)^2} \cdot \left(\mathbb{1}_{\{K=2\}} + \frac{1}{4L} \cdot \mathbb{1}_{\{K \geq 3\}} \right) \quad (4.1.3)$$

Note that a non-uniform HSBM can be seen as a collection of observations for the same underlying community structure through several uniform HSBMs of different orders. A possible issue is that some uniform hypergraphs with small SNR might not be informative (if we observe an ℓ -uniform hypergraph with parameters $a_{\ell} = b_{\ell}$, including hyperedge information from it ultimately increases the noise). To improve our error rate guarantees, we start by adding a pre-processing step (Algorithm 4.4.1) for hyperedge selection according to SNR and then apply the algorithm on the sub-hypergraph with maximal SNR.

We state the following assumption that will be used in our analysis of Algorithms 4.1.1 ($K = 2$) and 4.1.2 ($K \geq 3$).

Assumption 4.1.5. For each $\ell \in \mathcal{L}$, assume a_{ℓ}, b_{ℓ} are constants independent of N , and $a_{\ell} \geq b_{\ell}$. Let L denote the maximum element of the set $\mathcal{L}$. Given $\mathbf{v} \in (1/K, 1)$, assume that there exists a universal constant $C_{(4.1.4a)}$ and some $\mathbf{v}$ -dependent constant $C_{\mathbf{v}} > 0$, such that

$$d := \sum_{\ell \in \mathcal{L}} (\ell - 1)a_{\ell} \geq C_{(4.1.4a)}, \quad (4.1.4a)$$

$$\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_{\ell} - b_{\ell}) \geq C_{\mathbf{v}} \sqrt{d} \cdot K^{L-1} \cdot \left(2^3 \cdot \mathbb{1}_{\{K=2\}} + \sqrt{\log(K/(1 - \mathbf{v}))} \cdot \mathbb{1}_{\{K \geq 3\}} \right). \quad (4.1.4b)$$

The constant $C_{(4.1.4a)}$ is not necessarily large; for example, $C_{(4.1.4a)} = (2^{1/L} - 1)^{-1/3}$

should suffice, but smaller $C_{(4.1.4a)}$ may work as well. Both two inequalities above prevent the hypergraph from being too sparse, while (4.1.4b) also requires that the difference between *in-block* and *across-blocks* densities is large enough. The choices of $C_{(4.1.4a)}$, C_v and their relationship will be discussed in Remark 4.5.17.

The binary-community case

We start with Algorithm 4.1.1, which outputs a γ -correct partition when the non-uniform HSBM $\mathcal{H}$ is sampled from model 4.1.2 with only 2 communities. Inspired by the innovative graph algorithm in [64], we generalize it to non-uniform hypergraphs while we provide a complete and detailed analysis at the same time.

Algorithm 4.1.1: Binary Partition

Data: The adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, the number of communities K and $\{a_\ell\}_{\ell \in \mathcal{L}}$, $\{b_\ell\}_{\ell \in \mathcal{L}}$.

- 1 Run **Algorithm 4.4.1 Pre-processing** to obtain subset $\mathcal{S}$ which achieves maximal SNR.
- 2 Randomly color all hyperedges red or blue with equal probability.
- 3 Run **Algorithm 4.4.5 Spectral Partition** on the red hypergraph and output $\hat{\mathcal{V}}_1^{(0)}, \hat{\mathcal{V}}_2^{(0)}$.
- 4 Run **Algorithm 4.4.6 Correction** on the blue hypergraph and output $\hat{\mathcal{V}}_1, \hat{\mathcal{V}}_2$.

Result: The estimated partition $\hat{\mathcal{V}}_1, \hat{\mathcal{V}}_2$

Theorem 4.1.6 ($K = 2$). *Let $v \in (0.5, 1)$ and $\rho = 2 \exp(-C_{\mathcal{L}}(2, v) \cdot \text{SNR}_{\mathcal{L}}(2))$ with $\text{SNR}_{\mathcal{L}}(K)$, $C_{\mathcal{L}}(K, v)$ defined in (4.1.2), (4.1.3), and let $\gamma = \max\{v, 1 - 2\rho\}$. Then under Assumption 4.1.5, Algorithm 4.1.1 outputs a γ -correct partition for sufficiently large N with probability at least $1 - O(N^{-2})$.*

The multi-community case

For the multi-community case ($K \geq 3$), another algorithm with more subroutines is developed in Algorithm 4.1.2, which outputs a γ -correct partition with high probability. We state the result as follows.

Algorithm 4.1.2: General Partition

Data: The adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, the number of communities K and $\{a_\ell\}_{\ell \in \mathcal{L}}$, $\{b_\ell\}_{\ell \in \mathcal{L}}$.

- 1 Run **Algorithm 4.4.1 Pre-processing** to obtain subset $\mathcal{S}$ which achieves maximal SNR.
- 2 Randomly color all hyperedges red or blue with equal probability.
- 3 Randomly partition the vertex $\mathcal{V}$ into two disjoint subsets $\mathcal{Y}$ and $\mathcal{Z}$ by assigning $+1$ or -1 to each vertex with equal probability.
- 4 Let $\mathbf{B}$ denote the adjacency matrix of the bipartite hypergraph between $\mathcal{Y}$ and $\mathcal{Z}$ consisting only of the red hyperedges, with rows indexed by $\mathcal{Z}$ and columns indexed by $\mathcal{Y}$.
- 5 Run **Algorithm 4.4.2 Spectral Partition** on the red hypergraph and output $\hat{\mathcal{U}}_1^{(0)}, \dots, \hat{\mathcal{U}}_K^{(0)}$.
This step only uses the red hyperedges between vertices in $\mathcal{Y}$ and $\mathcal{Z}$ and outputs approximate clusters for $\mathcal{U}_k := \mathcal{V}_k \cap \mathcal{Z}$ for each $k = 1, \dots, K$
- 6 Run **Algorithm 4.4.3 Correction** on the red hypergraph and output $\hat{\mathcal{U}}_1, \dots, \hat{\mathcal{U}}_K$.
- 7 Run **Algorithm 4.4.4 Merging** on the blue hypergraph and output $\hat{\mathcal{V}}_1, \dots, \hat{\mathcal{V}}_K$.
This step only uses the blue hyperedges between vertices in $\mathcal{Y}$ and $\mathcal{Z}$ and assigns the vertices in $\mathcal{Y}$ to an appropriate approximate cluster.

Result: The estimated partition $\hat{\mathcal{V}}_1, \dots, \hat{\mathcal{V}}_K$.

Theorem 4.1.7 ($K \geq 3$). Let $\nu \in (1/K, 1)$ and $\rho = \exp(-C_{\mathcal{L}}(K, \nu) \cdot \text{SNR}_{\mathcal{L}}(K))$ with $\text{SNR}_{\mathcal{L}}(K)$, $C_{\mathcal{L}}(K, \nu)$ defined in (4.1.2), (4.1.3), and let $\gamma = \max\{\nu, 1 - K\rho\}$. Then under Assumption 4.1.5, Algorithm 4.1.2 outputs a γ -correct partition for sufficiently large N with probability at least $1 - O(N^{-2})$.

The time complexities of Algorithms 4.1.1 and 4.1.2 are $O(N^3)$, with the bulk of time spent in Stage 1 by the spectral method.

To the best of our knowledge, Theorems 4.1.6 and 4.1.7 are the first results for partial recovery of non-uniform HSBMs. When the number of blocks is 2, Algorithm 4.1.1 guarantees a better error rate for partial recovery as in Theorem 4.1.6. This happens because Algorithm 4.1.1 does not need the merging routine in Algorithm 4.1.2: if one of the communities is obtained, then the other one is also obtained via the complement.

Remark 4.1.8. Taking $\mathcal{L} = \{2\}$, Theorem 4.1.7 can be reduced to [64, Lemma 9] for the graph

case. The failure probability $O(N^{-2})$ can be decreased to $O(N^{-p})$ for any $p > 0$, as long as one is willing to pay the price by increasing the constants $C_{(4.1.4a)}$, C_v in (4.1.4a), (4.1.4b) respectively.

Our Algorithms 4.1.1 and 4.1.2 can be summarized in 3 steps:

1. **Hyperedge selection:** select hyperedges of certain sizes to provide the maximal signal-to-noise ratio (SNR) for the induced sub-hypergraph.
2. **Spectral partition:** construct a regularized adjacency matrix and obtain an approximate partition based on singular vectors (first approximation).
3. **Correction and merging:** incorporate the hyperedge information from adjacency tensors to upgrade the error rate guarantee (second, better approximation).

The algorithm requires the input of model parameters a_ℓ, b_ℓ , which can be estimated by counting cycles in hypergraphs as shown in [170, 209]. Estimation of the number of blocks can be done by counting the outliers in the spectrum of the non-backtracking operator, e.g., as shown (for different regimes and different parameters) in [179, 22, 149, 186].

Weak consistency

Throughout the proofs for Theorem 4.1.6 and Theorem 4.1.7, we make only one assumption on the growth or finiteness of d and $\text{SNR}_\mathcal{L}(K)$, and it happens in estimating the failure probability as noted in Remark 4.1.10. Consequently, the corollary below follows, which covers the case when d and $\text{SNR}_\mathcal{L}(K)$ grow with N .

Corollary 4.1.9 (Weak consistency). *Let $\mathcal{L}$ be a set of finite integers and K be some fixed integer. If $\text{SNR}_\mathcal{L}(K)$ defined in (4.1.2) goes to infinity as $N \rightarrow \infty$ and $\text{SNR}_\mathcal{L}(K) = o(\log(N))$, then with probability $1 - O(N^{-2})$, Algorithms 4.1.1 and 4.1.2 output a partition with only $o(N)$ misclassified vertices.*

The paper [108] also proves weak consistency for non-uniform HSBMs, but in a much denser regime than we do here ($d = \Omega(\log^2(N))$, instead of $d = \omega(1)$, as in Corollary 4.1.9). In fact, we now know that strong consistency should be achievable in this denser regime, as [82] shows. When restricting to the uniform HSBM case, Corollary 4.1.9 achieves weak consistency under the same sparsity condition as in [13].

Remark 4.1.10. *To be precise, Algorithms 4.1.1 and 4.1.2 work optimally in the $\text{SNR}_L = o(\log(N))$ regime. When $\text{SNR}_L(K) = \Omega(\log(N))$, it implies that $\rho = N^{-\Omega(1)}$, and one may have $e^{-N\rho} = \Omega(1)$ in (4.5.26), which may not decrease to 0 as $N \rightarrow \infty$. Therefore the theoretical guarantees of Algorithms 4.4.3 and 4.4.4 may not remain valid. This, however, should not matter: in the regime when $\text{SNR}_L(K) = \Omega(\log(N))$, strong (rather than weak) consistency is expected, as per [82]. Therefore, the regime of interest for weak consistency is $\text{SNR}_L = o(\log(N))$.*

4.1.3 Comparison with existing results

Although many algorithms and theoretical results have been developed for hypergraph community detection, most of them are restricted to uniform hypergraphs, and few results are known for non-uniform ones. We will discuss the most relevant results.

In [143], the authors studied the degree-corrected HSBM with general connection probability parameters by using a tensor power iteration algorithm and Tucker decomposition. Their algorithm achieves weak consistency for uniform hypergraphs when the average degree is $\omega(\log^2(N))$, which is the regime complementary to the regime we studied here. They discussed a way to generalize the algorithm to non-uniform hypergraphs, but the theoretical analysis remains open. The recent paper [217] analyzed non-uniform hypergraph community detection by using hypergraph embedding and optimization algorithms and obtained weak consistency when the expected degrees are of $\omega(\log(N))$, again a complementary regime to ours. Results on spectral norm concentration of sparse random tensors were obtained in [72, 175, 135, 152, 219], but no provable tensor algorithm in the bounded expected degree case is known. Testing for the

community structure for non-uniform hypergraphs was studied in [209, 140], which is a problem different from community detection.

In our approach, we relied on knowing the tensors for each uniform hypergraph. However, in computations, we only ran the spectral algorithm on the adjacency matrix of the entire hypergraph since the stability of tensor algorithms does not yet come with guarantees due to the lack of concentration, and for non-uniform hypergraphs, $\ell - 1$ adjacency tensors would be needed. This approach presented the challenge that, unlike for graphs, the adjacency matrix of a random non-uniform hypergraph has dependent entries, and the concentration properties of such a random matrix were previously unknown. We overcame this issue and proved concentration bounds from scratch down to the bounded degree regime. Similar to [100, 150], we provided here a regularization analysis by removing rows in the adjacency matrix with large row sums (suggestive of large degree vertices) and proving a concentration result for the regularized matrix down to the bounded expected degree regime (see Theorem 4.3.4).

In terms of partial recovery for hypergraphs, our results are new, even in the uniform case. In [13, Theorem 1], for uniform hypergraphs, the authors showed detection (not partial recovery) is possible when the average degree is $\Omega(1)$; in addition, the error rate is not exponential in the model parameters, but only polynomial. Here, we mention two more results for the graph case. In the arbitrarily slowly growing degrees regime, it was shown in [214, 99] that the error rate in (4.1.2) is optimal up to a constant in the exponent. In the bounded expected degrees regime, the authors in [168, 63] provided algorithms that can asymptotically recover the optimal fraction of vertices, when the signal-to-noise ratio is large enough. It's an open problem to extend their analysis to obtain a minimax error rate for hypergraphs.

In [108], the authors considered weak consistency in a non-uniform HSBM model with a spectral algorithm based on the hypergraph Laplacian matrix, and showed that weak consistency is achievable if the expected degree is of $\Omega(\log^2(N))$ with high probability [109, Theorem 4.2]. Their algorithm can't be applied to sparse regimes straightforwardly since the normalized Laplacian is not well-defined due to the existence of isolated vertices in the bounded degree case.

In addition, our weak consistency results obtained here are valid as long as the expected degree is $\omega(1)$ and $o(\log(N))$, which is the entire set of problems on which weak consistency is expected. By contrast, in [108], weak consistency is shown only when the minimum expected degree is $\Omega(\log^2(N))$, which is a regime complementary to ours and where exact recovery should (in principle) be possible: for example, this is known to be an exact recovery regime in the uniform case [61, 146, 151, 216].

Finally, although our analysis does not provide a way to identify the exact threshold for the phase transition of detection in the non-uniform hypergraph, we conjecture that (similar to [22] in the uniform case) $\text{SNR}_{\mathcal{L}}(K) = 1$ in (4.1.2) is the exact threshold for detection.

In subsequent works [82, 198] we proposed algorithms to achieve weak consistency. However, their methods can not cover the regime when the expected degree is $\Omega(1)$ due to the lack of concentration. Additionally, [200] proposed *Projected Tensor Power Method* as the refinement stage to achieve strong consistency, as long as the first stage partition is partially correct, as ours.

4.1.4 Organization of the paper

In Section 4.2, we include the definitions of adjacency matrices of hypergraphs. The concentration results for the adjacency matrices are provided in Section 4.3. The algorithms for partial recovery are presented in Section 4.4. The proof for the correctness of our algorithms for Theorem 4.1.7 and Corollary 4.1.9 are given in Section 4.5. The proof of Theorem 4.1.6, as well as the proofs of many auxiliary lemmas and useful lemmas in the literature, are provided in the supplemental materials.

4.2 Preliminaries

Definition 4.2.1 (Adjacency tensor). *An ℓ -uniform hypergraph $\mathcal{H}_\ell = (\mathcal{V}, \mathcal{E}_\ell)$ can be associated to an order- ℓ adjacency tensor $\mathcal{A}^{(\ell)}$. For each ℓ -hyperedge $e = \{i_1, \dots, i_\ell\}$, define the corresponding*

entry by

$$\mathcal{A}_e^{(\ell)} := \mathcal{A}_{[i_1, \dots, i_\ell]}^{(\ell)} = \mathbb{1}_{\{e \in \mathcal{E}_\ell\}}. \quad (4.2.1)$$

Definition 4.2.2 (Adjacency matrix). *For some $\ell \in \mathbb{N}^+$, let $\mathcal{A}^{(\ell)}$ denote the adjacency tensor associated to the uniform HSBM $\mathcal{H}_\ell = (\mathcal{V}, \mathcal{E}_\ell)$ sampled from model (4.1.1). The adjacency matrix $\mathbf{A}^{(\ell)} := [\mathbf{A}_{ij}^{(\ell)}]_{1 \leq i, j \leq N}$ corresponding to $\mathcal{H}_\ell$ is defined by*

$$\mathbf{A}_{ij}^{(\ell)} = \mathbb{1}_{\{i \neq j\}} \cdot \sum_{\substack{e \in \mathcal{E}_\ell \\ \{i, j\} \subset e}} \mathcal{A}_e^{(\ell)}. \quad (4.2.2)$$

For the non-uniform hypergraph $\mathcal{H}$ sampled from model 4.1.2 with $\mathcal{L}$ be a finite set of integers, its corresponding adjacency matrix $\mathbf{A} := [\mathbf{A}_{ij}]_{1 \leq i, j \leq N}$ is defined entrywisely as

$$\mathbf{A}_{ij} = \sum_{\ell \in \mathcal{L}} \mathbf{A}_{ij}^{(\ell)}. \quad (4.2.3)$$

We compute the expectation of $\mathbf{A}$ first. In each ℓ -uniform hypergraph $\mathcal{H}_\ell$, two distinct vertices $i, j \in \mathcal{V}$ with $i \neq j$ are picked arbitrarily since our model does not allow for loops. Assume for a moment $(N/K) \in \mathbb{N}$, then the expected number of ℓ -hyperedge containing i and j can be computed as follows.

- If i and j are from the same block, the ℓ -hyperedge is sampled with probability $a_\ell / \binom{N-1}{\ell-1}$ when the other $\ell - 2$ vertices are from the same block as i, j , otherwise with probability $b_\ell / \binom{N-1}{\ell-1}$. Then

$$\alpha_\ell := \mathbb{E} \mathbf{A}_{ij}^{(\ell)} = \binom{\frac{N}{K} - 2}{\ell - 2} \frac{a_\ell}{\binom{N-1}{\ell-1}} + \left[\binom{N-2}{\ell-2} - \binom{\frac{N}{K} - 2}{\ell-2} \right] \frac{b_\ell}{\binom{N-1}{\ell-1}}.$$

- If i and j are not from the same block, we sample the ℓ -hyperedge with probability

$b_\ell / \binom{N-1}{\ell-1}$, and

$$\beta_\ell := \mathbb{E}A_{ij}^{(\ell)} = \binom{N-2}{\ell-2} \frac{b_\ell}{\binom{N-1}{\ell-1}}.$$

By assumption $a_\ell \geq b_\ell$, then $\alpha_\ell \geq \beta_\ell$ for each $\ell \in \mathcal{L}$. Summing over ℓ , the *expected adjacency* matrix under the K -block non-uniform HSBM can be written as

$$\mathbb{E}\mathbf{A} = \begin{bmatrix} \alpha \mathbf{J}_{\frac{N}{K}} & \beta \mathbf{J}_{\frac{N}{K}} & \cdots & \beta \mathbf{J}_{\frac{N}{K}} \\ \beta \mathbf{J}_{\frac{N}{K}} & \alpha \mathbf{J}_{\frac{N}{K}} & \cdots & \beta \mathbf{J}_{\frac{N}{K}} \\ \vdots & \vdots & \ddots & \vdots \\ \beta \mathbf{J}_{\frac{N}{K}} & \beta \mathbf{J}_{\frac{N}{K}} & \cdots & \alpha \mathbf{J}_{\frac{N}{K}} \end{bmatrix} - \alpha \mathbf{I}_N, \quad (4.2.4)$$

where $\mathbf{J}_{\frac{N}{K}} \in \mathbb{R}^{\frac{N}{K} \times \frac{N}{K}}$ denotes the all-one matrix and

$$\alpha := \sum_{\ell \in \mathcal{L}} \alpha_\ell, \quad \beta := \sum_{\ell \in \mathcal{L}} \beta_\ell. \quad (4.2.5)$$

Lemma 4.2.3. *The eigenvalues of $\mathbb{E}\mathbf{A}$ are given below:*

$$\begin{aligned} \lambda_1(\mathbb{E}\mathbf{A}) &= \frac{N}{K}(\alpha + (K-1)\beta) - \alpha, \\ \lambda_j(\mathbb{E}\mathbf{A}) &= \frac{N}{K}(\alpha - \beta) - \alpha, \quad 2 \leq j \leq K, \\ \lambda_j(\mathbb{E}\mathbf{A}) &= -\alpha, \quad K+1 \leq j \leq N. \end{aligned}$$

Lemma 4.2.3 can be verified via direct computation. Lemma 4.2.4 is used for approximately equi-partitions, meaning that eigenvalues of $\widetilde{\mathbb{E}\mathbf{A}}$ can be approximated by eigenvalues of $\mathbb{E}\mathbf{A}$ when N is sufficiently large.

Lemma 4.2.4. *For any partition $(\mathcal{V}_1, \dots, \mathcal{V}_K)$ of $\mathcal{V}$ where $N_i := |\mathcal{V}_i|$, consider the following*

matrix

$$\widetilde{\mathbb{E}\mathbf{A}} = \begin{bmatrix} \alpha \mathbf{J}_{N_1} & \beta \mathbf{J}_{N_1 \times N_2} & \cdots & \beta \mathbf{J}_{N_1 \times N_{K-1}} & \beta \mathbf{J}_{N_1 \times N_K} \\ \beta \mathbf{J}_{N_2 \times N_1} & \alpha \mathbf{J}_{N_2} & \cdots & \beta \mathbf{J}_{N_2 \times N_{K-1}} & \beta \mathbf{J}_{N_2 \times N_K} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \beta \mathbf{J}_{N_{K-1} \times N_1} & \beta \mathbf{J}_{N_{K-1} \times N_2} & \cdots & \alpha \mathbf{J}_{N_{K-1}} & \beta \mathbf{J}_{N_{K-1} \times N_K} \\ \beta \mathbf{J}_{N_K \times N_1} & \beta \mathbf{J}_{N_K \times N_2} & \cdots & \beta \mathbf{J}_{N_K \times N_{K-1}} & \alpha \mathbf{J}_{N_K} \end{bmatrix} - \alpha \mathbf{I}_N.$$

Assume that $N_i = \frac{N}{K} + O(\sqrt{N} \log(N))$ for all $k \in [K]$. Then, for all $1 \leq i \leq K$,

$$\frac{|\lambda_i(\widetilde{\mathbb{E}\mathbf{A}}) - \lambda_i(\mathbb{E}\mathbf{A})|}{|\lambda_i(\mathbb{E}\mathbf{A})|} = O\left(N^{-\frac{1}{4}} \log^{\frac{1}{2}}(N)\right).$$

Note that both $(\widetilde{\mathbb{E}\mathbf{A}} + \alpha \mathbf{I}_N)$ and $(\mathbb{E}\mathbf{A} + \alpha \mathbf{I}_N)$ are rank K matrices, then $\lambda_i(\widetilde{\mathbb{E}\mathbf{A}}) = \lambda_i(\mathbb{E}\mathbf{A}) = -\alpha$ for all $(K+1) \leq i \leq N$. At the same time, SNR in (4.1.2) is related to the following quantity

$$\frac{[\lambda_2(\mathbb{E}\mathbf{A})]^2}{\lambda_1(\mathbb{E}\mathbf{A})} = \frac{[(N-K)\alpha - N\beta]^2}{K[(N-K)\alpha + N(K-1)\beta]} = (1 + o(1)) \cdot \frac{[\sum_{\ell \in \mathcal{L}} (\ell-1)(a_\ell - b_\ell)K^{-\ell+1}]^2}{\sum_{\ell \in \mathcal{L}} (\ell-1)((a_\ell - b_\ell)K^{-\ell+1} + b_\ell)}.$$

When $\mathcal{L} = \{2\}$ and K is fixed, SNR in (4.1.2) is equal to $\frac{(a-b)^2}{K[a+(K-1)b]}$, which corresponds to the SNR for the undirected graph in [64], see also [1, Section 6].

4.3 Spectral norm concentration

The correctness of Algorithm 4.1.2 and Algorithm 4.1.1 relies on the concentration of the adjacency matrix of $\mathcal{H}$. The following two concentration results for general random hypergraphs are included, which are independent of HSBM model. The proofs are deferred to Appendix C.1.

Inhomogeneous Erdős-Rényi random hypergraphs

Definition 4.3.1 (Inhomogeneous Erdős-Rényi hypergraph). *Let $Q^{(\ell)} \in ([0, 1]^N)^{\otimes \ell}$ be a symmetric probability tensor, i.e., $Q_{i_1, \dots, i_\ell} = Q_{\pi(1), \dots, \pi(\ell)}$ for any permutation π on $[\ell]$, where $\ell \geq 2$ is some finite integer. Let $\mathcal{H}_\ell = (\mathcal{V}, \mathcal{E}_\ell)$ denote inhomogeneous ℓ -uniform Erdős-Rényi hypergraph associated with the probability tensor $Q^{(\ell)}$. Let $\mathcal{A}^{(\ell)}$ denote the adjacency tensor of $\mathcal{H}_\ell$, where each ℓ -hyperedge $e = \{i_1, \dots, i_\ell\} \subset \mathcal{V}$ appears with probability $\mathbb{P}(\mathcal{A}_e^{(\ell)} = 1) = Q_{i_1, \dots, i_\ell}^{(\ell)}$. Let $\mathcal{L} = \{\ell \mid \ell \geq 2, \ell \in \mathbb{N}\}$ be a finite set of integers, where L denotes its maximum element. An inhomogeneous non-uniform Erdős-Rényi hypergraph is the union of uniform ones, i.e., $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$.*

Theorem 4.3.2. *Let $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$ be an inhomogeneous non-uniform Erdős-Rényi hypergraph sampled from model 4.3.1. Let $\{Q^{(\ell)}\}_{\ell \in \mathcal{L}}$ denote the probability tensors such that $Q_{i_1, \dots, i_\ell}^{(\ell)} = d_{i_1, \dots, i_\ell} / \binom{N-1}{\ell-1}$ for each $\ell \in \mathcal{L}$, and denote $d_\ell = \max_{[i_1, \dots, i_\ell]} d_{i_1, \dots, i_\ell}$. Let $\mathbf{A}$ be the adjacency matrix of $\mathcal{H}$ constructed from the adjacency tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$. For some constant $C_{(4.3.1)} > 0$, suppose that the following assumption hold*

$$d := \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot d_\ell \geq C_{(4.3.1)} \log(N). \quad (4.3.1)$$

Then for any $\theta > 0$, there exists a constant $C_{(4.3.2)} = 512L(\ell - 1)(\theta + 6) [2 + (\ell - 1)(1 + \theta) / C_{(4.3.1)}]$ such that with probability at least $1 - 2N^{-\theta} - 2e^{-N}$, the adjacency matrix $\mathbf{A}$ of $\mathcal{H}$ satisfies

$$\|\mathbf{A} - \mathbb{E}\mathbf{A}\| \leq C_{(4.3.2)} \sqrt{d}. \quad (4.3.2)$$

The inequality (4.3.2) can be reduced to the result for graph case obtained in [100, 153] by taking $\mathcal{L} = \{2\}$. The result for a uniform hypergraph is obtained in [151]. Note that d is a fixed constant in our community detection problem, thus the Assumption 4.3.1 does not hold and the inequality (4.3.2) cannot be directly applied. However, we can still prove a concentration bound for a regularized version of $\mathbf{A}$, following the same strategy of the proof for Theorem 4.3.2.

Definition 4.3.3 (Regularized matrix). *Given any $N \times N$ matrix $\mathbf{A}$ and an index set I , let $\mathbf{A}_I$ be the $N \times N$ matrix obtained from $\mathbf{A}$ by zeroing out the rows and columns not indexed by I . Namely,*

$$(\mathbf{A}_I)_{ij} = \mathbb{1}_{\{i,j \in I\}} \cdot \mathbf{A}_{ij}. \quad (4.3.3)$$

Note that every hyperedge of size ℓ containing i is counted $(\ell - 1)$ times in the i -th row of $\mathbf{A}$. We define the i -th row sum of $\mathbf{A}$ as

$$\text{row}(i) := \sum_j \mathbf{A}_{ij} := \sum_j \mathbb{1}_{\{i \neq j\}} \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ \{i,j\} \subset e}} \mathcal{A}_e^{(\ell)} = \sum_{\ell \in \mathcal{L}} (\ell - 1) \sum_{e \in \mathcal{E}_\ell: i \in e} \mathcal{A}_e^{(\ell)}.$$

Theorem 4.3.4 is the concentration result for the regularized $\mathbf{A}_I$, by zeroing out rows and columns corresponding to vertices with high row sums.

Theorem 4.3.4. *Following all the notations in Theorem 4.3.2, for any constant $\tau > 1$, define the index set*

$$I = \{i \in \mathcal{V} : \text{row}(i) \leq \tau d\}. \quad (4.3.4)$$

Let $\mathbf{A}_I$ be the regularized version of $\mathbf{A}$, as in Definition 4.3.3. Then for any $\theta > 0$, there exists a constant $C_\tau = 2((5L + 1)(\ell - 1) + \alpha_0 \sqrt{\tau})$ with $\alpha_0 = 16 + \frac{32}{\tau}(1 + e^2) + 128L(\ell - 1)(\theta + 4)(1 + \frac{1}{e^2})$, such that

$$\|\mathbf{A}_I - \mathbb{E}\mathbf{A}_I\| \leq C_\tau \sqrt{d} \quad (4.3.5)$$

holds with probability at least $1 - 2(e/2)^{-N} - N^{-\theta}$.

4.4 Algorithms blocks

In this section, we are going to present the algorithmic blocks constructing our main partition method (Algorithm 4.1.2): pre-processing (Algorithm 4.4.1), initial result by spectral

method (Algorithm 4.4.2), correction of blemishes via majority rule (Algorithm 4.4.3), and merging (Algorithm 4.4.4).

Algorithm 4.4.1: Pre-processing

Data: The set of integers $\mathcal{L}$, and the parameters $\{a_\ell\}_{\ell \in \mathcal{L}}, \{b_\ell\}_{\ell \in \mathcal{L}}$.

1 **for** each subset $\mathcal{S} \subset \mathcal{L}$ **do**

2 Let $\mathcal{H}_\mathcal{S} = \bigcup_{\ell \in \mathcal{S}} \mathcal{H}_\ell$ denote the restriction of the non-uniform hypergraph $\mathcal{H}$ on $\mathcal{S}$.
 Compute

$$\text{SNR}_\mathcal{S} := \frac{[\sum_{\ell \in \mathcal{S}} (\ell - 1)(a_\ell - b_\ell)K^{-\ell+1}]^2}{\sum_{\ell \in \mathcal{S}} (\ell - 1)((a_\ell - b_\ell)K^{-\ell+1} + b_\ell)}$$

3 **end**

4 Among all the $\mathcal{S}$, find the subset $\mathcal{L}^*$ which maximizes $\text{SNR}_\mathcal{S}$, i.e.,

$$\mathcal{L}^* := \arg \max_{\mathcal{S} \subset \mathcal{L}} \text{SNR}_\mathcal{S},$$

where $\mathcal{L}^*$ denotes its maximal element.

Result: $\mathcal{L}^*$

4.4.1 Three or more communities ($K \geq 3$)

The proof of Theorem 4.1.7 is structured as follows.

Lemma 4.4.1. *Under the assumptions of Theorem 4.1.7, Algorithm 4.4.2 outputs a $\mathbf{v}$ -correct partition $\widehat{\mathcal{U}}_1^{(0)}, \dots, \widehat{\mathcal{U}}_K^{(0)}$ of $\mathcal{Z} = (\mathcal{Z} \cap \mathcal{V}_1) \cup \dots \cup (\mathcal{Z} \cap \mathcal{V}_K)$ with probability at least $1 - O(N^{-2})$.*

Lines 4 and 6 contribute most complexity in Algorithm 4.4.2, requiring $O(N^3)$ and $O(N^2 \log^2(N))$ each (technically, one should be able to get away with $O(N^2 \log(1/\epsilon))$ in line 4, for some desired accuracy ϵ to get the singular vectors). We will conservatively estimate the time complexity of Algorithm 4.4.2 as $O(N^3)$.

Lemma 4.4.2. *Under the assumptions of Theorem 4.1.7, for any $\mathbf{v}$ -correct partition $\widehat{\mathcal{U}}_1^{(0)}, \dots, \widehat{\mathcal{U}}_K^{(0)}$ of $\mathcal{Z} = (\mathcal{Z} \cap \mathcal{V}_1) \cup \dots \cup (\mathcal{Z} \cap \mathcal{V}_K)$ and the red hypergraph over $\mathcal{Z}$, Algorithm 4.4.3 computes a $\gamma_{\mathcal{C}}$ -correct partition $\widehat{\mathcal{U}}_1, \dots, \widehat{\mathcal{U}}_K$ with probability $1 - O(e^{-N^\rho})$, while $\gamma_{\mathcal{C}} = \max\{\mathbf{v}, 1 - K\rho\}$ with $\rho := k \exp(-C_{\mathcal{L}}(K, \mathbf{v}) \cdot \text{SNR}_{\mathcal{L}}(K))$ where $\mathcal{L}$ is obtained from Algorithm 4.4.1, and $\text{SNR}_{\mathcal{L}}(K)$ and $C_{\mathcal{L}}(K, \mathbf{v})$ are defined in (4.1.2), (4.1.3).*

Algorithm 4.4.2: Spectral Partition

- Data:** The tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, the matrix $\mathbf{A}$, and the parameters $\{a_\ell\}_{\ell \in \mathcal{L}}$, $\{b_\ell\}_{\ell \in \mathcal{L}}$.
- 1 Randomly label vertices in $\mathcal{Y}$ with $+1$ and -1 sign with equal probability, and partition $\mathcal{Y}$ into 2 disjoint subsets $\mathcal{Y}_1$ and $\mathcal{Y}_2$;
 - 2 Let $\mathbf{B}_1$ (resp. $\mathbf{B}_2$) denote the adjacency matrices with all vertices in $\mathcal{Z} \cup \mathcal{Y}_1$ (resp. $\mathcal{Z} \cup \mathcal{Y}_2$), with rows indexed by $\mathcal{Z}$ and columns indexed by $\mathcal{Y}_1$ (resp. $\mathcal{Y}_2$);
 - 3 Pad $\mathbf{B}_1, \mathbf{B}_2$ with zeros to obtain the $N \times N$ matrices $\mathbf{A}_1$ and $\mathbf{A}_2$;
 - 4 Let $d := \sum_{\ell \in \mathcal{L}} (\ell - 1)a_\ell$. Zero out all the rows and columns of $\mathbf{A}_1$ corresponding to vertices whose row sum is bigger than $20Ld$, to obtain the matrix $(\mathbf{A}_1)_{I_1}$;
 - 5 Find the space $\mathbf{U}$, spanned by the first K left singular vectors of $(\mathbf{A}_1)_{I_1}$;
 - 6 Randomly sample $s = 2K \log^2(N)$ vertices from $\mathcal{Y}_2$ without replacement. Denote the corresponding columns in $\mathbf{A}_2$ by $\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_s}$. For each $i \in \{i_1, \dots, i_s\}$, project $\mathbf{a}_i - \bar{\mathbf{a}}$ onto $\mathbf{U}$, where the elements in vector $\bar{\mathbf{a}} \in \mathbb{R}^N$ is defined by $\bar{\mathbf{a}}(j) = \mathbb{1}_{\{j \in \mathcal{Z}\}} \cdot (\bar{\alpha} + \bar{\beta})/2$, with $\bar{\alpha}, \bar{\beta}$ defined in (4.5.6);
 - 7 For each projected vector $\mathbf{P}_{\mathbf{U}}(\mathbf{a}_i - \bar{\mathbf{a}})$, identify the top $N/(2K)$ coordinates in value as a set $\mathcal{U}_i$. Discard half of the s sets $\mathcal{U}_i$, those with the lowest blue hyperedge density in them;
 - 8 Sort the remaining sets according to blue hyperedge density and identify first K distinct subsets $\hat{\mathcal{U}}_1^{(0)}, \dots, \hat{\mathcal{U}}_K^{(0)}$ such that $|\hat{\mathcal{U}}_j^{(0)} \cap \hat{\mathcal{U}}_k^{(0)}| < \lceil (1 - \nu)N/K \rceil$ if $j \neq k$;
- Result:** The partition $\hat{\mathcal{U}}_1^{(0)}, \dots, \hat{\mathcal{U}}_K^{(0)}$
-

Lemma 4.4.3. *Given any ν -correct partition $\hat{\mathcal{U}}_1, \dots, \hat{\mathcal{U}}_K$ of $\mathcal{Z} = (\mathcal{Z} \cap \mathcal{V}_1) \cup \dots \cup (\mathcal{Z} \cap \mathcal{V}_K)$ and the blue hypergraph between $\mathcal{Y}$ and $\mathcal{Z}$, with probability $1 - O(e^{-N\rho})$, Algorithm 4.4.4 outputs a γ -correct partition $\hat{\mathcal{V}}_1, \dots, \hat{\mathcal{V}}_K$ of $\mathcal{V}_1 \cup \mathcal{V}_2 \cup \dots \cup \mathcal{V}_K$, while $\gamma = \max\{\nu, 1 - K\rho\}$.*

The time complexities of Algorithms 4.4.3 and 4.4.4 are $O(N)$, since each vertex is adjacent to only constant many hyperedges.

4.4.2 Binary communities ($K = 2$)

The spectral partition step is given in Algorithm 4.4.5, and the correction step is given in Algorithm 4.4.6.

Lemma 4.4.4. *Under the conditions of Theorem 4.1.6, the Algorithm 4.4.5 outputs a ν -correct partition $\hat{\mathcal{V}}_1^{(0)}, \hat{\mathcal{V}}_2^{(0)}$ of $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ with probability at least $1 - O(N^{-2})$.*

Algorithm 4.4.3: Correction

Data: The initial partition $\{\widehat{\mathcal{U}}_k^{(0)}\}_{k \in [K]}$, the tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, and the parameters $\{a_\ell\}_{\ell \in \mathcal{L}}, \{b_\ell\}_{\ell \in \mathcal{L}}$.

- 1 **for** each $u \in \mathcal{Z}$ **do**
- 2 Add u to $\widehat{\mathcal{U}}_k$ if the number of red hyperedges, which contains u with the remaining vertices located in vertex set $\widehat{\mathcal{U}}_k^{(0)}$, is at least μ_C (4.5.24);
- 3 Break ties arbitrarily;
- 4 **end**

Result: $\widehat{\mathcal{U}}_1, \dots, \widehat{\mathcal{U}}_K$

Algorithm 4.4.4: Merging

Data: The partition $\{\widehat{\mathcal{U}}_k\}_{k \in [K]}$, the tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, and the parameters $\{a_\ell\}_{\ell \in \mathcal{L}}, \{b_\ell\}_{\ell \in \mathcal{L}}$.

- 1 **for** all $u \in \mathcal{Y}$ **do**
- 2 Add u to $\widehat{\mathcal{V}}_k$ if the number of blue hyperedges, which contains u with the remaining vertices located in vertex set $\widehat{\mathcal{U}}_K$, is at least μ_M (4.5.32);
- 3 Label the conflicts arbitrarily;
- 4 **end**

Result: The estimated sets $\widehat{\mathcal{V}}_1, \dots, \widehat{\mathcal{V}}_K$

Lemma 4.4.5. *Given any $\mathbf{v}$ -correct partition $\mathcal{V}_1^{(0)}, \mathcal{V}_2^{(0)}$ of $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$, with probability at least $1 - O(e^{-N\rho})$, the Algorithm 4.4.6 computes a γ -correct partition $\widehat{\mathcal{V}}_1, \widehat{\mathcal{V}}_2$ with $\gamma = \{\mathbf{v}, 1 - 2\rho\}$ and $\rho = 2 \exp(-C_{\mathcal{L}}(2, \mathbf{v}) \cdot \text{SNR}_{\mathcal{L}}(2))$, where $\text{SNR}_{\mathcal{L}}(2)$ and $C_{\mathcal{L}}(2, \mathbf{v})$ are defined in (4.1.2) and (4.1.3) respectively.*

4.5 Algorithm's correctness

We are going to present the correctness of Algorithm 4.1.2 in this section. The correctness of Algorithm 4.1.1 is deferred to Appendix C.3. We first introduce some definitions.

Vertex set splitting and adjacency matrix

In Algorithm 4.1.2, we first randomly partition the vertex set $\mathcal{V}$ into two disjoint subsets $\mathcal{Z}$ and $\mathcal{Y}$ by assigning $+1$ and -1 to each vertex independently with equal probability. Let $\mathbf{B} \in \mathbb{R}^{|\mathcal{Z}| \times |\mathcal{Y}|}$ denote the submatrix of $\mathbf{A}$, while $\mathbf{A}$ was defined in (4.2.3), where rows and columns

Algorithm 4.4.5: Spectral Partition

Data: The tensors $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, the matrix $\mathbf{A}$, and the parameters $\{a_\ell\}_{\ell \in \mathcal{L}}$, $\{b_\ell\}_{\ell \in \mathcal{L}}$.

- 1 Zero out all the rows and the columns of $\mathbf{A}$ corresponding to vertices with row sums greater than $20Ld$, to obtain the regularized matrix $\mathbf{A}_I$;
- 2 Find the subspace $\mathbf{U}$, which is spanned by the eigenvectors corresponding to the largest two eigenvalues of $\mathbf{A}_I$;
- 3 Compute $\mathbf{P}_\mathbf{U}\mathbf{1}_N$, the projection of all-ones vector onto $\mathbf{U}$;
- 4 Let $\mathbf{v}$ be the unit vector in $\mathbf{U}$ perpendicular to $\mathbf{P}_\mathbf{U}\mathbf{1}_N$;
- 5 Sort the vertices according to their values in $\mathbf{v}$. Let $\widehat{\mathcal{V}}_1^{(0)} \subset V$ be the corresponding top $N/2$ vertices, and $\widehat{\mathcal{V}}_2^{(0)} \subset V$ be the remaining $N/2$ vertices;

Result: $\widehat{\mathcal{V}}_1^{(0)}, \widehat{\mathcal{V}}_2^{(0)}$

of $\mathbf{B}$ correspond to vertices in $\mathcal{Z}$ and $\mathcal{Y}$ respectively. Let N_j denote the number of vertices in $\mathcal{Z} \cap \mathcal{V}_j$, where $\mathcal{V}_j$ denotes the true partition with $|\mathcal{V}_j| = \frac{N}{K}$ for all $j \in [K]$, then N_j can be written as a sum of independent Bernoulli random variables, i.e.,

$$N_j = |\mathcal{Z} \cap \mathcal{V}_j| = \sum_{v \in \mathcal{V}_j} \mathbb{1}_{\{v \in \mathcal{Z}\}}, \quad (4.5.1)$$

and $|\mathcal{Y} \cap \mathcal{V}_j| = |\mathcal{V}_j| - |\mathcal{Z} \cap \mathcal{V}_j| = \frac{N}{K} - N_j$ for each $j \in [K]$.

Definition 4.5.1. *The splitting $\mathcal{V} = \mathcal{Z} \cup \mathcal{Y}$ is perfect if $|\mathcal{Z} \cap \mathcal{V}_j| = |\mathcal{Y} \cap \mathcal{V}_j| = N/(2K)$ for all $j \in [K]$. And the splitting $\mathcal{Y} = \mathcal{Y}_1 \cup \mathcal{Y}_2$ is perfect if $|\mathcal{Y}_1 \cap \mathcal{V}_j| = |\mathcal{Y}_2 \cap \mathcal{V}_j| = N/(4K)$ for all $j \in [K]$.*

However, the splitting is *imperfect* in most cases since the size of $\mathcal{Z}$ and $\mathcal{Y}$ would not be exactly the same under the independence assumption. The random matrix $\mathbf{B}$ is parameterized by $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ and $\{N_j\}_{j=1}^K$. If we take expectation over $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ given the block size information

Algorithm 4.4.6: Correction

```

1 for  $v \in \widehat{\mathcal{V}}_1^{(0)}$  do
2   | Label  $v$  as “bad” if the number of blue-hyperedges which contain  $v$  with the
   |   remaining vertices in  $\widehat{\mathcal{V}}_2^{(0)}$  is at least  $\mu_C$ ; otherwise label  $v$  as “good”.
3 end
4 for  $v \in \widehat{\mathcal{V}}_2^{(0)}$  do
5   | Label  $v$  as “bad” if the number of blue-hyperedges which contain  $v$  with the
   |   remaining vertices in  $\widehat{\mathcal{V}}_1^{(0)}$  is at least  $\mu_C$ ; otherwise label  $v$  as “good”;
6 end
7 for  $v \in \mathcal{V}$  do
8   | if  $v \in \widehat{\mathcal{V}}_i^{(0)}$  and  $v$  is good then
9   |   | Assign  $v$  to  $\widehat{\mathcal{V}}_i$ ;
10  | end
11  | else
12  |   | Assign  $v$  to  $\widehat{\mathcal{V}}_{3-i}$ ;
13  | end
14 end

Result:  $\widehat{\mathcal{V}}_1, \widehat{\mathcal{V}}_2$ 

```

$\{N_j\}_{j=1}^K$, then it gives rise to the expectation of the *imperfect* splitting, denoted by $\widetilde{\mathbf{B}}$,

$$\widetilde{\mathbf{B}} := \begin{bmatrix} \alpha \mathbf{J}_{N_1 \times (\frac{N}{K} - N_1)} & \beta \mathbf{J}_{N_1 \times (\frac{N}{K} - N_2)} & \cdots & \beta \mathbf{J}_{N_1 \times (\frac{N}{K} - N_K)} \\ \beta \mathbf{J}_{N_2 \times (\frac{N}{K} - N_1)} & \alpha \mathbf{J}_{N_2 \times (\frac{N}{K} - N_2)} & \cdots & \beta \mathbf{J}_{N_2 \times (\frac{N}{K} - N_K)} \\ \vdots & \vdots & \ddots & \vdots \\ \beta \mathbf{J}_{N_K \times (\frac{N}{K} - N_1)} & \beta \mathbf{J}_{N_K \times (\frac{N}{K} - N_2)} & \cdots & \alpha \mathbf{J}_{N_K \times (\frac{N}{K} - N_K)} \end{bmatrix},$$

where α, β are defined in (4.2.5). In the *perfect* splitting case, the dimension of each block is $N/(2K) \times N/(2K)$ since $\mathbb{E}N_j = N/(2K)$ for all $j \in [K]$, and the expectation matrix $\overline{\mathbf{B}}$ can be written as

$$\overline{\mathbf{B}} := \begin{bmatrix} \alpha \mathbf{J}_{\frac{N}{2K}} & \beta \mathbf{J}_{\frac{N}{2K}} & \cdots & \beta \mathbf{J}_{\frac{N}{2K}} \\ \beta \mathbf{J}_{\frac{N}{2K}} & \alpha \mathbf{J}_{\frac{N}{2K}} & \cdots & \beta \mathbf{J}_{\frac{N}{2K}} \\ \vdots & \vdots & \ddots & \vdots \\ \beta \mathbf{J}_{\frac{N}{2K}} & \beta \mathbf{J}_{\frac{N}{2K}} & \cdots & \alpha \mathbf{J}_{\frac{N}{2K}} \end{bmatrix}.$$

In Algorithm 4.4.2, $\mathcal{Y}_1$ is a random subset of $\mathcal{Y}$ obtained by selecting each element with probability $1/2$ independently, and $\mathcal{Y}_2 = \mathcal{Y} \setminus \mathcal{Y}_1$. Let $\tilde{N}_j$ denote the number of vertices in $\mathcal{Y}_1 \cap \mathcal{V}_j$, then $\tilde{N}_j$ can be written as a sum of independent Bernoulli random variables,

$$\tilde{N}_j = |\mathcal{Y}_1 \cap \mathcal{V}_j| = \sum_{v \in \mathcal{V}_j} \mathbb{1}_{\{v \in \mathcal{Y}_1\}}, \quad (4.5.2)$$

and $|\mathcal{Y}_2 \cap \mathcal{V}_j| = |\mathcal{V}_j| - |\mathcal{Z} \cap \mathcal{V}_j| - |\mathcal{Y}_1 \cap \mathcal{V}_j| = N/K - N_j - \tilde{N}_j$ for all $j \in [K]$.

Similarly, $\mathbb{E}\tilde{N}_j = N/(4K)$, which means that this splitting is expected to be perfect. However, we still obtain *imperfect* splitting in most cases due to the independence assumption.

Remark 4.5.2. In [64], the authors assumed the perfect splitting throughout the proof, which is not necessarily true. Our work provides a rigorous analysis of the algorithm provided in [64], without the perfect splitting assumption.

Induced sub-hypergraph

For the non-uniform hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, consider the vertex subset $(\mathcal{Y}_1 \cup \mathcal{Z}) \subset \mathcal{V}$, which leads our attention to the corresponding induced sub-hypergraph.

Definition 4.5.3 (Induced sub-hypergraph). Let $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ be a non-uniform random hypergraph and $\mathcal{S} \subset \mathcal{V}$ be any subset of the vertices of $\mathcal{H}$. Then the induced sub-hypergraph $\mathcal{H}[\mathcal{S}]$ is the hypergraph whose vertex set is $\mathcal{S}$ and whose hyperedge set $\mathcal{E}[\mathcal{S}]$ consists of all of the edges in $\mathcal{E}$ that have all endpoints located in $\mathcal{S}$.

Remark 4.5.4. Note that $\mathcal{H}$ can be considered as a collection of ℓ -uniform hypergraphs for varying ℓ , a.k.a., $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$. The decomposition works for $\mathcal{H}[\mathcal{S}]$ as well, a.k.a. $\mathcal{H}[\mathcal{S}] = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell[\mathcal{S}]$ and $\mathcal{E}[\mathcal{S}] = \cup_{\ell \in \mathcal{L}} \mathcal{E}_\ell[\mathcal{S}]$

Let $\mathcal{H}[\mathcal{Y}_1 \cup \mathcal{Z}]$ (resp. $\mathcal{H}[\mathcal{Y}_2 \cup \mathcal{Z}]$) denote the induced sub-hypergraph on vertex set $\mathcal{Y}_1 \cup \mathcal{Z}$ (resp. $\mathcal{Y}_2 \cup \mathcal{Z}$), and $\mathbf{B}_1 \in \mathbb{R}^{|\mathcal{Z}| \times |\mathcal{Y}_1|}$ (resp. $\mathbf{B}_2 \in \mathbb{R}^{|\mathcal{Z}| \times |\mathcal{Y}_2|}$) denote the adjacency matrix corresponding to the sub-hypergraph, where the rows and columns of $\mathbf{B}_1$ (resp. $\mathbf{B}_2$) are corresponding to

elements in $\mathcal{Z}$ and $\mathcal{Y}_1$ (resp., $\mathcal{Z}$ and $\mathcal{Y}_2$). Therefore, $\mathbf{B}_1$ and $\mathbf{B}_2$ are parameterized by $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$, $\{N_j\}_{j=1}^K$ and $\{\tilde{N}_j\}_{j=1}^K$, and the entries in $\mathbf{B}_1$ are independent of the entries in $\mathbf{B}_2$, due to the independence of hyperedges. If we take expectation over $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ conditioning on $\{N_j\}_{j=1}^K$ and $\{\tilde{N}_j\}_{j=1}^K$, then it gives rise to the expectation of the *imperfect* splitting, denoted by $\tilde{\mathbf{B}}_1$,

$$\tilde{\mathbf{B}}_1 := \begin{bmatrix} \tilde{\alpha}_{11} \mathbf{J}_{N_1 \times \tilde{N}_1} & \cdots & \tilde{\beta}_{1k} \mathbf{J}_{N_1 \times \tilde{N}_K} \\ \vdots & \ddots & \vdots \\ \tilde{\beta}_{k1} \mathbf{J}_{N_K \times \tilde{N}_1} & \cdots & \tilde{\alpha}_{kk} \mathbf{J}_{N_K \times \tilde{N}_K} \end{bmatrix}, \quad (4.5.3)$$

where for $1 \leq i, j \leq K$, the entries $\tilde{\alpha}_{ij}$ and $\tilde{\beta}_{ij}$ are defined as follows:

$$\tilde{\alpha}_{ii} := \sum_{\ell \in \mathcal{L}} \left\{ \binom{N_i + \tilde{N}_i - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \left(\sum_{k=1}^K (N_k + \tilde{N}_k) - 2 \right) \frac{b_\ell}{\binom{N-1}{\ell-1}} \right\}, \quad (4.5.4a)$$

$$\tilde{\beta}_{ij} := \sum_{\ell \in \mathcal{L}} \left(\sum_{k=1}^K (N_k + \tilde{N}_k) - 2 \right) \frac{b_\ell}{\binom{N-1}{\ell-1}}, \quad i \neq j, \quad (4.5.4b)$$

The expectation of the *perfect* splitting, denoted by $\bar{\mathbf{B}}_1$, can be written as

$$\bar{\mathbf{B}}_1 := \begin{bmatrix} \bar{\alpha} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} & \bar{\beta} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} & \cdots & \bar{\beta} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} \\ \bar{\beta} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} & \bar{\alpha} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} & \cdots & \bar{\beta} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{\beta} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} & \bar{\beta} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} & \cdots & \bar{\alpha} \mathbf{J}_{\frac{N}{2K} \times \frac{N}{4K}} \end{bmatrix}, \quad (4.5.5)$$

where

$$\bar{\alpha} := \sum_{\ell \in \mathcal{L}} \left\{ \binom{\frac{3N}{4K} - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \left(\frac{3N}{4} - 2 \right) \frac{b_\ell}{\binom{N-1}{\ell-1}} \right\}, \quad \bar{\beta} := \sum_{\ell \in \mathcal{L}} \left(\frac{3N}{4} - 2 \right) \frac{b_\ell}{\binom{N-1}{\ell-1}}. \quad (4.5.6)$$

The matrices $\tilde{\mathbf{B}}_2, \bar{\mathbf{B}}_2$ can be defined similarly, since dimensions of $|\mathcal{Y}_2 \cap \mathcal{V}_j|$ are also determined by N_j and $\tilde{N}_j$. Obviously, $\bar{\mathbf{B}}_2 = \bar{\mathbf{B}}_1$ since $\mathbb{E}\tilde{N}_i = \mathbb{E}(N/K - N_j - \tilde{N}_j) = N/(4K)$ for all $j \in [K]$.

Fixing Dimensions

The dimensions of $\tilde{\mathbf{B}}_1$ and $\tilde{\mathbf{B}}_2$, as well as blocks they consist of, are not deterministic—since N_i and $\tilde{N}_j$, defined in (4.5.1) and (4.5.2) respectively, are sums of independent random variables. As such, we cannot directly compare them. In order to overcome this difficulty, we embed $\mathbf{B}_1$ and $\mathbf{B}_2$ into the following $N \times N$ matrices:

$$\mathbf{A}_1 := \begin{bmatrix} \mathbf{0}_{|Z| \times |Z|} & \mathbf{B}_1 & \mathbf{0}_{|Z| \times |\mathcal{Y}_2|} \\ \mathbf{0}_{|\mathcal{Y}| \times |Z|} & \mathbf{0}_{|\mathcal{Y}| \times |\mathcal{Y}_1|} & \mathbf{0}_{|\mathcal{Y}| \times |\mathcal{Y}_2|} \end{bmatrix}, \quad \mathbf{A}_2 := \begin{bmatrix} \mathbf{0}_{|Z| \times |Z|} & \mathbf{0}_{|Z| \times |\mathcal{Y}_1|} & \mathbf{B}_2 \\ \mathbf{0}_{|\mathcal{Y}| \times |Z|} & \mathbf{0}_{|\mathcal{Y}| \times |\mathcal{Y}_1|} & \mathbf{0}_{|\mathcal{Y}| \times |\mathcal{Y}_2|} \end{bmatrix}. \quad (4.5.7)$$

Note that $\mathbf{A}_1$ and $\mathbf{A}_2$ have the same size. Also by definition, the entries in $\mathbf{A}_1$ are independent of the entries in $\mathbf{A}_2$. If we take expectation over $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ conditioning on $\{N_j\}_{j=1}^K$ and $\{\tilde{N}_j\}_{j=1}^K$, then we obtain the expectation matrices of the *imperfect* splitting, denoted by $\tilde{\mathbf{A}}_1$ (resp. $\tilde{\mathbf{A}}_2$), written as

$$\tilde{\mathbf{A}}_1 := \begin{bmatrix} \mathbf{0}_{|Z| \times |Z|} & \tilde{\mathbf{B}}_1 & \mathbf{0}_{|Z| \times |\mathcal{Y}_2|} \\ \mathbf{0}_{|\mathcal{Y}| \times |Z|} & \mathbf{0}_{|\mathcal{Y}| \times |\mathcal{Y}_1|} & \mathbf{0}_{|\mathcal{Y}| \times |\mathcal{Y}_2|} \end{bmatrix}, \quad \tilde{\mathbf{A}}_2 := \begin{bmatrix} \mathbf{0}_{|Z| \times |Z|} & \mathbf{0}_{|Z| \times |\mathcal{Y}_1|} & \tilde{\mathbf{B}}_2 \\ \mathbf{0}_{|\mathcal{Y}| \times |Z|} & \mathbf{0}_{|\mathcal{Y}| \times |\mathcal{Y}_1|} & \mathbf{0}_{|\mathcal{Y}| \times |\mathcal{Y}_2|} \end{bmatrix}. \quad (4.5.8)$$

The expectation matrix of the *perfect* splitting, denoted by $\bar{\mathbf{A}}_1$ (resp. $\bar{\mathbf{A}}_2$), can be written as

$$\bar{\mathbf{A}}_1 := \begin{bmatrix} \mathbf{0}_{\frac{N}{2} \times \frac{N}{2}} & \bar{\mathbf{B}}_1 & \mathbf{0}_{\frac{N}{2} \times \frac{N}{4}} \\ \mathbf{0}_{\frac{N}{2} \times \frac{N}{2}} & \mathbf{0}_{\frac{N}{2} \times \frac{N}{4}} & \mathbf{0}_{\frac{N}{2} \times \frac{N}{4}} \end{bmatrix}, \quad \bar{\mathbf{A}}_2 := \begin{bmatrix} \mathbf{0}_{\frac{N}{2} \times \frac{N}{2}} & \mathbf{0}_{\frac{N}{2} \times \frac{N}{4}} & \bar{\mathbf{B}}_2 \\ \mathbf{0}_{\frac{N}{2} \times \frac{N}{2}} & \mathbf{0}_{\frac{N}{2} \times \frac{N}{4}} & \mathbf{0}_{\frac{N}{2} \times \frac{N}{4}} \end{bmatrix}. \quad (4.5.9)$$

Obviously, $\tilde{\mathbf{A}}_i$ and $\tilde{\mathbf{B}}_i$ (resp. $\bar{\mathbf{A}}_i$ and $\bar{\mathbf{B}}_i$) have the same non-zero singular values for $i = 1, 2$. In the remaining of this section, we will deal with $\tilde{\mathbf{A}}_i$ and $\bar{\mathbf{A}}_i$ instead of $\tilde{\mathbf{B}}_i$ and $\bar{\mathbf{B}}_i$ for $i = 1, 2$.

4.5.1 Spectral Partition: Proof of Lemma 4.4.1

Proof Outline

Recall that $\mathbf{A}_1$ is defined as the adjacency matrix of the induced sub-hypergraph $\mathcal{H}[\mathcal{Y}_1 \cup \mathcal{Z}]$ in Section 4.5. Consequently, the index set should contain information only from $\mathcal{H}[\mathcal{Y}_1 \cup \mathcal{Z}]$. Define the index sets

$$I = \left\{ i \in \mathcal{V} : \text{row}(i) \leq 20Ld \right\}, \quad I_1 = \left\{ i \in \mathcal{V} : \text{row}(i)|_{\mathcal{Y}_1 \cup \mathcal{Z}} \leq 20Ld \right\}, \quad (4.5.10)$$

where $d = \sum_{\ell \in \mathcal{L}} (\ell - 1)a_\ell$, and $\text{row}(i)|_{\mathcal{Y}_1 \cup \mathcal{Z}}$ is the row sum of i on $\mathcal{H}[\mathcal{Y}_1 \cup \mathcal{Z}]$. We say $\text{row}(i)|_{\mathcal{Y}_1 \cup \mathcal{Z}} = 0$ if $i \notin \mathcal{Y}_1 \cup \mathcal{Z}$, and for vertex $i \in \mathcal{Y}_1 \cup \mathcal{Z}$, and let

$$\text{row}(i)|_{\mathcal{Y}_1 \cup \mathcal{Z}} := \sum_{j=1}^N \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell[\mathcal{Y}_1 \cup \mathcal{Z}] \\ \{i,j\} \subseteq e}} \mathcal{A}_e^{(\ell)} = \sum_{\ell \in \mathcal{L}} (\ell - 1) \sum_{\substack{e \in \mathcal{E}_\ell[\mathcal{Y}_1 \cup \mathcal{Z}] \\ \{i,j\} \subseteq e}} \mathcal{A}_e^{(\ell)}.$$

As a result, the matrix $(\mathbf{A}_1)_{I_1}$ is obtained by restricting $\mathbf{A}_1$ on index set I_1 . The next 4 steps guarantee that Algorithm 4.4.2 outputs a $\mathbf{v}$ -correct partition.

- (i) Find the singular subspace $\mathbf{U}$ spanned by the first K left singular vectors of $(\mathbf{A}_1)_{I_1}$.
- (ii) Randomly pick $s = 2K \log^2(N)$ vertices from $\mathcal{Y}_2$ and denote the corresponding columns in $\mathbf{A}_2$ by $\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_s}$. Project each vector $\mathbf{a}_i - \bar{\mathbf{a}}$ onto the singular subspace $\mathbf{U}$, with $\bar{\mathbf{a}} \in \mathbb{R}^N$ defined by $\bar{\mathbf{a}}(j) = \mathbb{1}_{\{j \in \mathcal{Z}\}} \cdot (\bar{\alpha} + \bar{\beta})/2$, where $\bar{\alpha}, \bar{\beta}$ were defined in (4.5.6).
- (iii) For each projected vector $\mathbf{P}_{\mathbf{U}}(\mathbf{a}_i - \bar{\mathbf{a}})$, identify the top $N/(2K)$ coordinates in value and place the corresponding vertices into a set $\hat{\mathcal{U}}_i^{(0)}$. Discard half of the obtained s subsets, those with the lowest *blue* edge densities.
- (iv) Sort the remaining sets according to blue hyperedge density and identify K distinct subsets $\hat{\mathcal{U}}_1^{(0)}, \dots, \hat{\mathcal{U}}_K^{(0)}$ such that $|\hat{\mathcal{U}}_i^{(0)} \cap \hat{\mathcal{U}}_j^{(0)}| < \lceil (1 - \mathbf{v})N/K \rceil$ if $i \neq j$.

Based on the 4 steps above in Algorithm 4.4.2, the proof of Lemma 4.4.1 is structured in 4 parts.

- (i) Let $\tilde{\mathbf{U}}$ denote the subspace spanned by first K left singular vectors of $\tilde{\mathbf{A}}_1$ defined in (4.5.8). Section 4.5.1 shows that the subspace angle between $\mathbf{U}$ and $\tilde{\mathbf{U}}$ is smaller than any $c \in (0, 1)$ as long as a_ℓ, b_ℓ satisfy certain conditions depending on c .
- (ii) The vector $\tilde{\mathbf{\delta}}_i$, defined in (4.5.14), reflects the underlying true partition $\mathcal{Z} \cap \mathcal{V}_{\pi(i)}$ for each $i \in [S]$, where $\pi(i)$ denotes the membership of vertex i . Section 4.5.1 shows that $\bar{\mathbf{\delta}}_i$, an approximation of $\tilde{\mathbf{\delta}}_i$ defined in (4.5.15), can be recovered by the projected vector $\mathbf{P}_{\mathbf{U}}(\mathbf{a}_i - \bar{\mathbf{a}})$, since projection error $\|\mathbf{P}_{\mathbf{U}}(\mathbf{a}_i - \bar{\mathbf{a}}) - \bar{\mathbf{\delta}}_i\|_2 < c\|\tilde{\mathbf{\delta}}_i\|_2$ for any $c \in (0, 1)$ if a_ℓ, b_ℓ satisfy the desired property in part (i).
- (iii) Section 4.5.1 indicates that the coincidence ratio between the remaining sets and the true partition is at least v , after discarding half of the sets with the lowest blue edge densities.
- (iv) Lemma 4.5.15 proves that we can find K distinct subsets $\hat{\mathcal{U}}_i^{(0)}$ within $K \log^2(N)$ trials with high probability.

Bounding the angle between $\mathbf{U}$ and $\tilde{\mathbf{U}}$

The angle between subspaces $\mathbf{U}$ and $\tilde{\mathbf{U}}$ is defined as

$$\sin \angle(\mathbf{U}, \tilde{\mathbf{U}}) := \|\mathbf{P}_{\mathbf{U}} - \mathbf{P}_{\tilde{\mathbf{U}}}\|.$$

A natural idea is to apply Wedin's $\sin \Theta$ Theorem (Lemma D.2.6). Lemma 4.5.5 indicates that the difference of $\sigma_j(\tilde{\mathbf{A}}_1)$ and $\sigma_j(\bar{\mathbf{A}}_1)$ is relatively small, compared to $\sigma_j(\bar{\mathbf{A}}_1)$.

Lemma 4.5.5. *Let $\sigma_j(\bar{\mathbf{A}}_1)$ (resp. $\sigma_j(\tilde{\mathbf{A}}_1)$) denote the singular values of $\bar{\mathbf{A}}_1$ (resp. $\tilde{\mathbf{A}}_1$) for all*

$j \in [K]$, where the matrices $\bar{\mathbf{A}}_1$ and $\tilde{\mathbf{A}}_1$ are defined in (4.5.9) and (4.5.8) respectively. Then

$$\begin{aligned}\sigma_1(\bar{\mathbf{A}}_1) &= \frac{N \left[\bar{\alpha} + (K-1)\bar{\beta} \right]}{2\sqrt{2}K} = \frac{N}{2\sqrt{2}K} \sum_{\ell \in \mathcal{L}} \left[\binom{\frac{3N}{4K} - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + K \binom{\frac{3N}{4} - 2}{\ell - 2} \frac{b_\ell}{\binom{N-1}{\ell-1}} \right], \\ \sigma_j(\bar{\mathbf{A}}_1) &= \frac{N(\bar{\alpha} - \bar{\beta})}{2\sqrt{2}K} = \frac{N}{2\sqrt{2}K} \sum_{\ell \in \mathcal{L}} \binom{\frac{3N}{4K} - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}}, \quad 2 \leq j \leq K, \\ \sigma_j(\bar{\mathbf{A}}_1) &= 0, \quad K+1 \leq j \leq N.\end{aligned}$$

with $\bar{\alpha}, \bar{\beta}$ defined in (4.5.6). Moreover, with probability at least $1 - 2K \exp(-K \log^2(N))$, for each $j \in [K]$,

$$\frac{|\sigma_j(\bar{\mathbf{A}}_1) - \sigma_j(\tilde{\mathbf{A}}_1)|}{\sigma_j(\bar{\mathbf{A}}_1)} = O\left(N^{-\frac{1}{4}} \log^{\frac{1}{2}}(N)\right).$$

Therefore, with Lemma 4.5.5, we can write $\sigma_j(\tilde{\mathbf{A}}_1) = \sigma_j(\bar{\mathbf{A}}_1)(1 + o(1))$. Define $\mathbf{E}_1 := \mathbf{A}_1 - \tilde{\mathbf{A}}_1$ and its restriction on I_1 (4.5.10) as

$$(\mathbf{E}_1)_{I_1} := (\mathbf{A}_1 - \tilde{\mathbf{A}}_1)_{I_1} = (\mathbf{A}_1)_{I_1} - (\tilde{\mathbf{A}}_1)_{I_1}, \quad (4.5.11)$$

as well as $\boldsymbol{\delta}_1 := (\tilde{\mathbf{A}}_1)_{I_1} - \tilde{\mathbf{A}}_1$. Then $(\mathbf{A}_1)_{I_1} - \tilde{\mathbf{A}}_1$ is decomposed as

$$(\mathbf{A}_1)_{I_1} - \tilde{\mathbf{A}}_1 = [(\mathbf{A}_1)_{I_1} - (\tilde{\mathbf{A}}_1)_{I_1}] + [(\tilde{\mathbf{A}}_1)_{I_1} - \tilde{\mathbf{A}}_1] = (\mathbf{E}_1)_{I_1} + \boldsymbol{\delta}_1.$$

Lemma 4.5.6. *Let $d = \sum_{\ell \in \mathcal{L}} (\ell - 1)a_\ell$, where $\mathcal{L}$ is obtained from Algorithm 4.4.1. There exists a constant $C_1 \geq (2^{1/L} - 1)^{-1/3}$ such that if $d \geq C_1$, then with probability at least $1 - \exp(-d^{-2}N/L)$, no more than $d^{-3}N$ vertices have row sums greater than $20Ld$.*

Lemma 4.5.6 shows that the number of high-degree vertices is relatively small. Consequently, Corollary 4.5.7 indicates $\|\boldsymbol{\delta}_1\| \leq \sqrt{d}$ with high probability.

Corollary 4.5.7. *Assume $d \geq \max\{C_1, \sqrt{2}\}$, where C_1 is the constant in Lemma 4.5.6, then*

$\|\boldsymbol{\delta}_1\| \leq \sqrt{d}$ with probability at least $1 - \exp(-d^{-2}N/L)$.

Proof of Corollary 4.5.7. Note that $N - |I| \leq d^{-3}N$ and $I \subset I_1$, then $N - |I_1| \leq d^{-3}N$. From Lemma 4.5.6, there are at most $d^{-3}N$ vertices with row sum greater than $20Ld$ in the adjacency matrix $\mathbf{A}_1$, then the matrix $\boldsymbol{\delta}_1 = (\tilde{\mathbf{A}}_1)_{I_1} - \tilde{\mathbf{A}}_1$ has at most $2d^{-3}N^2$ non-zero entries. Every entry of $\tilde{\mathbf{A}}_1$ in (4.5.8) is bounded by α , then,

$$\begin{aligned} \|\boldsymbol{\delta}_1\| &\leq \|\boldsymbol{\delta}_1\|_F = \|(\tilde{\mathbf{A}}_1)_{I_1} - \tilde{\mathbf{A}}_1\|_F \\ &\leq \sqrt{2d^{-3}N^2} \alpha = N\sqrt{2d^{-3}} \sum_{\ell \in \mathcal{L}} \left[\binom{\frac{N}{K} - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \binom{N-2}{\ell-2} \frac{b_\ell}{\binom{N-1}{\ell-1}} \right] \\ &\leq \sqrt{2d^{-3}} \sum_{\ell \in \mathcal{L}} (\ell - 1)a_\ell \leq \sqrt{2d^{-1}} \leq \sqrt{d}. \end{aligned}$$

□

Moreover, by taking $\tau = 20L$, $\theta = 3$ in Theorem 4.3.4, with probability at least $1 - N^{-2}$

$$\|(\mathbf{E}_1)_{I_1}\| \leq C_3\sqrt{d}, \quad (4.5.13)$$

where constant $C_3 > 0$ depends on L . Together with upper bounds for $\|(\mathbf{E}_1)_{I_1}\|$ and $\|\boldsymbol{\delta}_1\|$, Lemma 4.5.8 shows that the angle between $\mathbf{U}$ and $\tilde{\mathbf{U}}$ is relatively small with high probability.

Lemma 4.5.8. *For any $c \in (0, 1)$, there exists some constant $C_2 > 0$ such that, if*

$$\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell) \geq C_2 K^{L-1} \sqrt{d},$$

then $\sin \angle(\mathbf{U}, \tilde{\mathbf{U}}) \leq c$ with probability $1 - N^{-2}$. Here $\angle(\mathbf{U}, \tilde{\mathbf{U}})$ is the angle between $\mathbf{U}$ and $\tilde{\mathbf{U}}$.

Proof of Lemma 4.5.8. From (4.5.13) and Corollary 4.5.7, with probability at least $1 - N^{-2}$,

$$\|(\mathbf{A}_1)_{I_1} - \tilde{\mathbf{A}}_1\| \leq \|(\mathbf{E}_1)_{I_1}\| + \|\boldsymbol{\delta}_1\| \leq (C_3 + 1)\sqrt{d}.$$

Since $\sigma_{K+1}(\tilde{\mathbf{A}}_1) = 0$, using Lemma 4.5.5 to approximate $\sigma_K(\tilde{\mathbf{A}}_1)$, we obtain

$$\begin{aligned} \sigma_K(\tilde{\mathbf{A}}_1) - \sigma_{K+1}(\tilde{\mathbf{A}}_1) &= \sigma_K(\tilde{\mathbf{A}}_1) = (1 + o(1)) \cdot \sigma_K(\bar{\mathbf{A}}_1) \geq \frac{1}{2} \sigma_K(\bar{\mathbf{A}}_1) \\ &\geq \frac{N}{4\sqrt{2}K} \sum_{\ell \in \mathcal{L}} \binom{\frac{3N}{4K} - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} \geq \frac{1}{8K} \sum_{\ell \in \mathcal{L}} \left(\frac{3}{4K}\right)^{\ell-2} (\ell-1)(a_\ell - b_\ell) \\ &\geq \frac{1}{8K} \left(\frac{1}{2K}\right)^{L-2} \sum_{\ell \in \mathcal{L}} (\ell-1)(a_\ell - b_\ell) \geq \frac{C_2 \sqrt{d}}{2^{L+1}}. \end{aligned}$$

Then for any $c \in (0, 1)$, we can find $C_2 = \lceil 2^{L+2}(C_3 + 1)/c \rceil$ such that $\|(\mathbf{A}_1)_{I_1} - \tilde{\mathbf{A}}_1\| \leq (1 - 1/\sqrt{2})\sigma_K(\tilde{\mathbf{A}}_1)$. By Wedin's Theorem (Lemma D.2.6), the angle $\angle(\mathbf{U}, \tilde{\mathbf{U}})$ is bounded by

$$\sin \angle(\mathbf{U}, \tilde{\mathbf{U}}) := \|\mathbf{P}_{\mathbf{U}} - \mathbf{P}_{\tilde{\mathbf{U}}}\| \leq \frac{\sqrt{2}\|(\mathbf{A}_1)_{I_1} - \tilde{\mathbf{A}}_1\|}{\sigma_K(\tilde{\mathbf{A}}_1)} \leq \frac{\sqrt{2}(C_3 + 1)\sqrt{d}}{C_2 \sqrt{d}/2^{L+1}} = \frac{\sqrt{2}}{2} c < c.$$

□

Bound the projection error

Randomly pick $s = 2K \log^2(N)$ vertices from $\mathcal{V}_2$. Let $\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_s}, \tilde{\mathbf{a}}_{i_1}, \dots, \tilde{\mathbf{a}}_{i_s}, \bar{\mathbf{a}}_{i_1}, \dots, \bar{\mathbf{a}}_{i_s}$ and $\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_s}$ be the corresponding columns of $\mathbf{A}_2, \tilde{\mathbf{A}}_2, \bar{\mathbf{A}}_2$ and $\mathbf{E}_2 := \mathbf{A}_2 - \tilde{\mathbf{A}}_2$ respectively, where $\mathbf{A}_2, \tilde{\mathbf{A}}_2$ and $\bar{\mathbf{A}}_2$ were defined in (4.5.7), (4.5.8) and (4.5.9). Let $\pi(i)$ denote the membership of vertex i . Note that entries of vector $\tilde{\mathbf{a}}_i$ are $\tilde{\alpha}_{ii}, \tilde{\beta}_{ij}$ or 0, according to the membership of vertices in $\mathcal{Z}$, where $\tilde{\alpha}_{ii}, \tilde{\beta}_{ij}$ were defined in (4.5.4a), (4.5.4b). Then the corresponding vector $\tilde{\mathbf{d}}_i \in \mathbb{R}^N$ with the entries given by

$$\tilde{\mathbf{a}}_i(j) = \begin{cases} \tilde{\alpha}_{ii}, & \text{if } j \in \mathcal{Z} \cap \mathcal{V}_{\pi(i)} \\ \tilde{\beta}_{ij}, & \text{if } j \in \mathcal{Z} \setminus \mathcal{V}_{\pi(i)} \\ 0, & \text{if } j \in \mathcal{Y} \end{cases}, \quad \tilde{\mathbf{d}}_i(j) = \begin{cases} (\tilde{\alpha}_{ii} - \tilde{\beta}_{ij})/2 > 0, & \text{if } j \in \mathcal{Z} \cap \mathcal{V}_{\pi(i)} \\ (\tilde{\beta}_{ij} - \tilde{\alpha}_{ii})/2 < 0, & \text{if } j \in \mathcal{Z} \setminus \mathcal{V}_{\pi(i)} \\ 0, & \text{if } j \in \mathcal{Y} \end{cases}, \quad (4.5.14)$$

can be used to recover the vertex set $\mathcal{Z} \cap \mathcal{V}_{\pi(i)}$ based on the sign of elements in $\tilde{\mathbf{d}}_i$. However, it is hard to handle with $\tilde{\mathbf{d}}_i$ due to the randomness of $\tilde{\alpha}_{ii}, \tilde{\beta}_{ij}$ originated from N_i and $\tilde{N}_j$. Note that N_i

and $\tilde{N}_j$ concentrate around $N/(2K)$ and $N/(4K)$ respectively as shown in Lemma 4.5.5. Thus a good approximation of $\tilde{\mathbf{\delta}}_i$, which rules out randomness of N_i and $\tilde{N}_j$, was given by $\bar{\mathbf{\delta}}_i := \bar{\mathbf{a}}_i - \bar{\mathbf{a}}$, with entries given by $\bar{\mathbf{a}}(j) := \mathbb{1}_{\{j \in Z\}} \cdot (\bar{\alpha} + \bar{\beta})/2$, where $\bar{\alpha}$ and $\bar{\beta}$ were defined in (4.5.6), and

$$\bar{\mathbf{a}}_i(j) = \begin{cases} \bar{\alpha}, & \text{if } j \in Z \cap \mathcal{V}_{\pi(i)} \\ \bar{\beta}, & \text{if } j \in Z \setminus \mathcal{V}_{\pi(i)} \\ 0, & \text{if } j \in \mathcal{J} \end{cases}, \quad \bar{\mathbf{\delta}}_i(j) = \begin{cases} (\bar{\alpha} - \bar{\beta})/2 > 0, & \text{if } j \in Z \cap \mathcal{V}_{\pi(i)} \\ (\bar{\beta} - \bar{\alpha})/2 < 0, & \text{if } j \in Z \setminus \mathcal{V}_{\pi(i)} \\ 0, & \text{if } j \in \mathcal{J} \end{cases}. \quad (4.5.15)$$

By construction, $\bar{\mathbf{\delta}}_i$ identifies vertex set $Z \cap \mathcal{V}_{\pi(i)}$ in the case of *perfect splitting* for any $i \in \{i_1, \dots, i_s\} \cap \mathcal{J}_2 \cap \mathcal{V}_{\pi(i)}$. However, the access to $\bar{\mathbf{\delta}}_i$ is limited in practice, thus the projection $\mathbf{P}_U(\mathbf{a}_i - \bar{\mathbf{a}})$ is used instead as an approximation of $\bar{\mathbf{\delta}}_i$. Lemma 4.5.9 proves that at least half of the projected vectors have small projection errors.

Lemma 4.5.9. *For any $c \in (0, 1)$, there exist constants C_1 and C_2 such that if $d > C_1$ and*

$$\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell) > C_2 K^L \sqrt{d},$$

then among all projected vectors $\mathbf{P}_U(\mathbf{a}_i - \bar{\mathbf{a}})$ for $i \in \{i_1, \dots, i_s\} \cap \mathcal{J}_2$, with probability $1 - O(N^{-K})$, at least half of them satisfy

$$\|\mathbf{P}_U(\mathbf{a}_i - \bar{\mathbf{a}}) - \bar{\mathbf{\delta}}_i\|_2 < c \|\bar{\mathbf{\delta}}_i\|_2. \quad (4.5.16)$$

Proof Lemma 4.5.9. Note that $\bar{\mathbf{\delta}}_i = \mathbf{P}_{\bar{U}} \bar{\mathbf{\delta}}_i$, where $\bar{U}$ is spanned by the first K left singular vectors of $\bar{\mathbf{A}}_1$ with $\text{rank}(\bar{\mathbf{A}}_1) = K$, and $\bar{\mathbf{A}}_1, \bar{\mathbf{A}}_2$ preserve the same eigen-subspace. The approximation error between $\mathbf{P}_U(\mathbf{a}_i - \bar{\mathbf{a}})$ and $\bar{\mathbf{\delta}}_i$ can be decomposed as

$$\begin{aligned} \mathbf{P}_U(\mathbf{a}_i - \bar{\mathbf{a}}) - \bar{\mathbf{\delta}}_i &= \mathbf{P}_U[(\mathbf{a}_i - \tilde{\mathbf{a}}_i) + (\tilde{\mathbf{a}}_i - \bar{\mathbf{a}}_i) + (\bar{\mathbf{a}}_i - \bar{\mathbf{a}})] - \mathbf{P}_{\bar{U}} \bar{\mathbf{\delta}}_i \\ &= \mathbf{P}_U \mathbf{e}_i + \mathbf{P}_U(\tilde{\mathbf{a}}_i - \bar{\mathbf{a}}_i) + (\mathbf{P}_U - \mathbf{P}_{\bar{U}}) \bar{\mathbf{\delta}}_i. \end{aligned}$$

Then by triangle inequality,

$$\|\mathbf{P}_U(\mathbf{a}_i - \bar{\mathbf{a}}) - \bar{\boldsymbol{\delta}}_i\|_2 \leq \|\mathbf{P}_U \mathbf{e}_i\|_2 + \|\mathbf{P}_U(\tilde{\mathbf{a}}_i - \bar{\mathbf{a}}_i)\|_2 + \|\mathbf{P}_U - \mathbf{P}_{\bar{U}}\| \cdot \|\bar{\boldsymbol{\delta}}_i\|_2.$$

Note that $\|\bar{\boldsymbol{\delta}}_i\| = O(N^{-\frac{1}{2}})$ and $\tilde{N}_i$ concentrates around $N/(4K)$ for each $i \in [K]$ with deviation at most $\sqrt{N} \log(N)$, then by definitions of $\bar{\alpha}$ and $\bar{\beta}$ in (4.5.6),

$$\|\mathbf{P}_U(\tilde{\mathbf{a}}_i - \bar{\mathbf{a}}_i)\|_2 \leq \|\tilde{\mathbf{a}}_i - \bar{\mathbf{a}}_i\|_2 = O\left([K\sqrt{N} \log(N)(\bar{\alpha} - \bar{\beta})^2]^{\frac{1}{2}}\right) = O[N^{-\frac{3}{4}} \log^{\frac{1}{2}}(N)] = o(\|\bar{\boldsymbol{\delta}}_i\|_2).$$

Meanwhile, by an argument similar to Lemma 4.5.8, it can be proved that $\sin \angle(\mathbf{U}, \bar{\mathbf{U}}) < c/2$ for any $c \in (0, 1)$, if constants C_1, C_2 are chosen properly, hence $\|\mathbf{P}_U - \mathbf{P}_{\bar{U}}\| \cdot \|\bar{\boldsymbol{\delta}}_i\|_2 < \frac{c}{2} \|\bar{\boldsymbol{\delta}}_i\|_2$. Lemma 4.5.10 shows that at least half of the indices from $\{i_1, \dots, i_s\} \cap \mathcal{Y}_2$ satisfy $\|\mathbf{P}_U \mathbf{e}_i\|_2 < \frac{c}{2} \|\bar{\boldsymbol{\delta}}_i\|_2$, which completes the proof. $\square$

Lemma 4.5.10. *Let $d = \sum_{\ell \in \mathbf{L}} (\ell - 1)a_\ell$. For any $c \in (0, 1)$, with probability $1 - O(N^{-K \log(N)})$, at least $\frac{s}{2}$ of the vectors $\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_s}$ satisfy*

$$\|\mathbf{P}_U \mathbf{e}_i\|_2 \leq 2\sqrt{dK(\mathbf{L} + 2)/N} < \frac{c}{2} \|\bar{\boldsymbol{\delta}}_i\|_2, \quad i \in \{i_1, \dots, i_s\} \subset \mathcal{Y}_2.$$

Definition 4.5.11. *The vector $\mathbf{a}_i$ satisfying (4.5.16) is referred as good vector. The index of the good vector is hence referred to as a good vertex.*

To avoid introducing extra notations, let $\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_{s_1}}$ denote *good* vectors with $i_1, \dots, i_{s_1}$ denoting good indices. Lemma 4.5.10 indicates that the number of good vectors is at least $s_1 \geq \frac{s}{2} = K \log^2(N)$.

Accuracy

We are going to prove the accuracy of the initial partition obtained from Algorithm 4.4.2. Lemmas 4.5.12, 4.5.13 and 4.5.15 are crucial in proving our results. We present the proof logic first and defer the Lemma statements later.

For each projected vector $\mathbf{P}_U(\mathbf{a}_i - \bar{\mathbf{a}})$, let $\widehat{\mathcal{U}}_i^{(0)}$ denote the set of its largest $\frac{N}{2K}$ coordinates, where $i \in \{i_1, \dots, i_s\}$ and $s = 2K \log^2(N)$. Note that vector $\bar{\boldsymbol{\delta}}_{i_j}$ in (4.5.14) only identifies blocks $\mathcal{V}_{\pi(i_j)}$ and $\mathcal{V} \setminus \mathcal{V}_{\pi(i_j)}$, which can be regarded as clustering two blocks with different sizes. By Lemma 4.5.9, good vectors satisfy $\|\mathbf{P}_U(\mathbf{a}_{i_j} - \bar{\mathbf{a}}) - \bar{\boldsymbol{\delta}}_{i_j}\|_2 < c \|\bar{\boldsymbol{\delta}}_{i_j}\|_2$ for any $c \in (0, 1)$. Then by Lemma 4.5.12 (after proper normalization), for a good index i_j , the number of vertices in $\widehat{\mathcal{U}}_{i_j}^{(0)}$ clustered correctly is at least $(1 - \frac{4K}{3}c^2)\frac{N}{K}$. By choosing $c = \sqrt{3(1-\nu)/(8K)}$, the condition $|\widehat{\mathcal{U}}_{i_j}^{(0)} \cap \mathcal{V}_j| > (1+\nu)/2 |\widehat{\mathcal{U}}_{i_j}^{(0)}|$ in part (ii) of Lemma 4.5.13 is satisfied. In Lemma 4.5.8, we choose

$$C_2 = 2^{L+2}(C_3 + 1)/c = 2^{L+2}(C_3 + 1)\sqrt{8K/(3-3\nu)}, \quad (4.5.18)$$

where C_3 defined in (4.5.13). Hence, with high probability, all good vectors have at least μ_T blue hyperedges (we call this “*high blue hyperedge density*”). From Lemma 4.5.10, at least half of the selected vectors are good. Then, in Algorithm 4.4.2, throwing out half of the obtained sets $\widehat{\mathcal{U}}_i^{(0)}$ (those with the lowest blue hyperedge density) guarantees that the remaining sets are good.

Recall that, by choosing constant appropriately, we can make the subspace angle $\sin \angle(\mathbf{U}, \bar{\mathbf{U}}) < c$ for any $c \in (0, 1)$ ($\bar{\mathbf{U}}$ is spanned by the first K left singular vectors of $\bar{\mathbf{A}}_1$). Then for vectors $\bar{\boldsymbol{\delta}}_{i_1}, \dots, \bar{\boldsymbol{\delta}}_{i_K}$ with each i_j selected from different vertex set $\mathcal{V}_j$, there is a vector $\mathbf{P}_U(\mathbf{a}_{i_j} - \bar{\mathbf{a}})$ in $\mathbf{U}$ arbitrarily close to $\bar{\boldsymbol{\delta}}_{i_j}$, which was proved by Lemma 4.5.9. From (i) of Lemma 4.5.13, so obtained $\widehat{\mathcal{U}}_{i_j}^{(0)}$ must satisfy $|\widehat{\mathcal{U}}_{i_j}^{(0)} \cap \mathcal{V}_j| \geq \nu |\widehat{\mathcal{U}}_{i_j}^{(0)}|$ for each $j \in [K]$. The remaining thing is to select K different $\widehat{\mathcal{U}}_{i_j}^{(0)}$ with each of them concentrating around distinct $\mathcal{V}_j$ for each $j \in [K]$. This problem is equivalent to finding K vertices in $\mathcal{Y}_2$, each from a different partition class, which can be done with $K \log^2(N)$ samplings as shown in Lemma 4.5.15.

To summarize, this section is a more precise and quantitative version of the following

argument: with high probability, the following holds for the good vectors $\mathbf{a}_{i_j}$, $j \in [K]$:

$$\begin{aligned} \left\{ i_j : \exists i \text{ s.t. } |\widehat{\mathcal{U}}_{i_j}^{(0)} \cap \mathcal{V}_i| \geq \frac{1+\mathbf{v}}{2} \frac{N}{K} \right\} &\subset \left\{ i_j : \widehat{\mathcal{U}}_{i_j}^{(0)} \text{ has more than } \mu_T \text{ blue hyperedges} \right\} \\ &\subset \left\{ i_j : \exists i \text{ s.t. } |\widehat{\mathcal{U}}_{i_j}^{(0)} \cap \mathcal{V}_i| \geq \mathbf{v} \frac{N}{K} \right\}. \end{aligned}$$

Lemma 4.5.12 (Adapted from Lemma 23 in [64]). *Assume $N/K \in \mathbb{N}$. Let $\mathbf{v}, \bar{\mathbf{v}} \in \mathbb{R}^N$ be two unit vectors, where $\bar{\mathbf{v}}$ has $\frac{N}{K}$ of its entries being $\frac{1}{\sqrt{N}}$ with the rest being $-\frac{1}{\sqrt{N}}$. If $\sin \angle(\bar{\mathbf{v}}, \mathbf{v}) < c \leq 0.5$, then $\mathbf{v}$ contains at least $(1 - \frac{4K}{3}c^2)\frac{N}{K}$ positive entries v_i such that the corresponding entries $\bar{v}_i$ are positive as well.*

Lemma 4.5.13. *For the set $\mathcal{X} \subset \mathcal{Z}$ with size $|\mathcal{X}| = N/(2K)$. Define*

$$\begin{aligned} \mu_1 &:= \frac{1}{2} \sum_{\ell \in \mathcal{L}} \ell(\ell-1) \left\{ \left[\binom{\frac{\mathbf{v}N}{2K}}{\ell} + \binom{\frac{(1-\mathbf{v})N}{2K}}{\ell} \right] \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \binom{\frac{N}{2K}}{\ell} \frac{b_\ell}{\binom{N-1}{\ell-1}} \right\}, \\ \mu_2 &:= \frac{1}{2} \sum_{\ell \in \mathcal{L}} \ell(\ell-1) \left\{ \left[\binom{\frac{(1+\mathbf{v})N}{4K}}{\ell} + (K-1) \binom{\frac{(1-\mathbf{v})N}{4K(K-1)}}{\ell} \right] \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \binom{\frac{N}{2K}}{\ell} \frac{b_\ell}{\binom{N-1}{\ell-1}} \right\}, \end{aligned}$$

and let $\mu_T := (\mu_1 + \mu_2)/2 \in [\mu_1, \mu_2]$. *There exists some constant $c > 0$ depending on K , $\{a_\ell\}_{\ell \in \mathcal{L}}$ and $\mathbf{v}$ such that for sufficiently large N ,*

- (i) *If $|\mathcal{X} \cap \mathcal{V}_j| \leq \mathbf{v}|\mathcal{X}|$ for each $j \in [K]$, then with probability $1 - e^{-cN}$, the number of blue hyperedges in the hypergraph induced by $\mathcal{X}$ is at most μ_T .*
- (ii) *Conversely, if $|\mathcal{X} \cap \mathcal{V}_j| \geq \frac{1+\mathbf{v}}{2}|\mathcal{X}|$ for some $j \in [K]$, then with probability $1 - e^{-cN}$, the number of blue hyperedges in the hypergraph induced by $\mathcal{X}$ is at least μ_T .*

Remark 4.5.14. *Lemma 4.5.13 is reduced to [64, Lemma 31] when $\mathcal{L} = \{2\}$.*

Let $\widehat{\mathcal{U}}_i^{(0)}$ denote the set of $N/(2K)$ largest coordinates of the projected vector $\mathbf{P}_U(\mathbf{a}_i - \bar{\mathbf{a}})$ with $i \in \mathcal{V}_{\pi(i)} \cap \mathcal{Y}_2$. According to Lemma 4.5.13, when the blue hyperedge density inside $\widehat{\mathcal{U}}_i^{(0)}$ is at least μ_T , it guarantees that $|\widehat{\mathcal{U}}_i^{(0)} \cap \mathcal{V}_{\pi(i)}| \geq \mathbf{v}|\widehat{\mathcal{U}}_i^{(0)}|$. Therefore, it is enough to consider half of the sets with highest blue edge densities in Algorithm 4.4.2.

Lemma 4.5.15. *Through random sampling without replacement in Step 6 of Algorithm 4.4.2, with probability $1 - N^{-\Omega(\log(N))}$, there exists at least K indices $i_1, \dots, i_K$ in $\mathcal{V}_2$ among the $K \log^2(N)$ samples such that*

$$|\widehat{\mathcal{U}}_{i_j}^{(0)} \cap \widehat{\mathcal{U}}_{i_k}^{(0)}| \leq (1 - \nu)N/K, \text{ for any } j, k \in [K] \text{ with } j \neq k.$$

4.5.2 Local Correction: Proof of Lemma 4.4.2

For notation convenience, let $\mathcal{U}_i := \mathcal{Z} \cap \mathcal{V}_i$ denote the intersection of $\mathcal{Z}$ and true partition $\mathcal{V}_i$ for all $i \in [K]$. In Algorithm 4.1.2, we first color the hyperedges with red and blue with equal probability. By running Algorithm 4.4.2 on the red hypergraph, we obtain a ν -correct partition $\widehat{\mathcal{U}}_1^{(0)}, \dots, \widehat{\mathcal{U}}_K^{(0)}$, i.e.,

$$|\widehat{\mathcal{U}}_k \setminus \widehat{\mathcal{U}}_k^{(0)}| \leq (1 - \nu) \cdot |\widehat{\mathcal{U}}_k^{(0)}| = (1 - \nu) \cdot \frac{N}{2K}, \quad \forall k \in [K]. \quad (4.5.20)$$

In the rest of this subsection, we condition on the event that (4.5.20) holds true.

Consider a hyperedge $e = \{i_1, \dots, i_\ell\}$ in the underlying ℓ -uniform hypergraph. If vertices $i_1, \dots, i_\ell$ are from the same block, then e is a red hyperedge with probability $a_\ell/2 \binom{N-1}{\ell-1}$; if vertices $i_1, \dots, i_\ell$ are not from the same block, then e is a red hyperedge with probability $b_\ell/2 \binom{N-1}{\ell-1}$. The presence of those two types of hyperedges can be denoted by

$$\mathbf{T}_e^{(a_\ell)} \sim \text{Bernoulli} \left(\frac{a_\ell}{2 \binom{N-1}{\ell-1}} \right), \quad \mathbf{T}_e^{(b_\ell)} \sim \text{Bernoulli} \left(\frac{b_\ell}{2 \binom{N-1}{\ell-1}} \right),$$

respectively. For any finite set $\mathcal{S}$, let $[\mathcal{S}]^l$ denote the family of l -subsets of $\mathcal{S}$, i.e., $[\mathcal{S}]^l = \{\mathcal{Z} | \mathcal{Z} \subseteq \mathcal{S}, |\mathcal{Z}| = l\}$. Consider a vertex $u \in \mathcal{U}_1 := \mathcal{Z} \cap \mathcal{V}_1$. The weighted number of red hyperedges,

which contains $u \in \mathcal{U}_1$ with the remaining vertices in $\widehat{\mathcal{U}}_j^{(0)}$, can be written as

$$\widehat{\mathbf{S}}_{1j}^{(0)}(u) := \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left\{ \sum_{e \in \mathcal{E}_{1,j}^{(a_\ell)}} \mathbf{T}_e^{(a_\ell)} + \sum_{e \in \mathcal{E}_{1,j}^{(b_\ell)}} \mathbf{T}_e^{(b_\ell)} \right\}, \quad u \in \mathcal{U}_1, \quad (4.5.21)$$

where $\mathcal{E}_{1,j}^{(a_\ell)} := \mathcal{E}_\ell([\mathcal{U}_1]^1, [\mathcal{U}_1 \cap \widehat{\mathcal{U}}_j^{(0)}]^{\ell-1})$ denotes the set of ℓ -hyperedges with one vertex from $[\mathcal{U}_1]^1$ and the other $\ell - 1$ from $[\mathcal{U}_1 \cap \widehat{\mathcal{U}}_j^{(0)}]^{\ell-1}$, and

$$\mathcal{E}_{1,j}^{(b_\ell)} := \mathcal{E}_\ell([\mathcal{U}_1]^1, [\widehat{\mathcal{U}}_j^{(0)}]^{\ell-1} \setminus [\mathcal{U}_1 \cap \widehat{\mathcal{U}}_j^{(0)}]^{\ell-1}) \quad (4.5.22)$$

denotes the set of ℓ -hyperedges with one vertex in $[\mathcal{U}_1]^1$ while the remaining $\ell - 1$ vertices in $[\widehat{\mathcal{U}}_j^{(0)}]^{\ell-1} \setminus [\mathcal{U}_1 \cap \widehat{\mathcal{U}}_j^{(0)}]^{\ell-1}$ (not all ℓ vertices are from $\mathcal{V}_1$) with their cardinalities

$$|\mathcal{E}_{1,j}^{(a_\ell)}| = \binom{|\mathcal{U}_1 \cap \widehat{\mathcal{U}}_j^{(0)}|}{\ell - 1}, \quad |\mathcal{E}_{1,j}^{(b_\ell)}| = \left[\binom{|\widehat{\mathcal{U}}_j^{(0)}|}{\ell - 1} - \binom{|\mathcal{U}_1 \cap \widehat{\mathcal{U}}_j^{(0)}|}{\ell - 1} \right].$$

We multiply $(\ell - 1)$ in (4.5.21) as weight since the rest $\ell - 1$ vertices are all located in $\widehat{\mathcal{U}}_j^{(0)}$, which can be regarded as the neighbors of vertex u in $\widehat{\mathcal{U}}_j^{(0)}$. According to the facts that $|\widehat{\mathcal{U}}_j^{(0)} \cap \mathcal{U}_j| \geq \nu N / (2K)$ in (4.5.20) and $|\widehat{\mathcal{U}}_j^{(0)}| = N / (2K)$ for each $j \in [K]$, we have

$$|\mathcal{E}_{1,1}^{(a_\ell)}| \geq \binom{\frac{\nu N}{2K}}{\ell - 1}, \quad |\mathcal{E}_{1,j}^{(a_\ell)}| \leq \binom{\frac{(1-\nu)N}{2K}}{\ell - 1}, \quad j \neq 1.$$

To simplify the calculation, we take the lower and upper bound of $|\mathcal{E}_{1,1}^{(a_\ell)}|$ and $|\mathcal{E}_{1,j}^{(a_\ell)}|$ ($j \neq 1$) respectively. By taking expectation with respect to $\mathbf{T}_e^{(a_\ell)}$ and $\mathbf{T}_e^{(b_\ell)}$, for any $u \in \mathcal{U}_1$, the following

holds:

$$\begin{aligned}\mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u) &= \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left[\binom{\frac{\nu N}{2K}}{\ell - 1} \frac{a_\ell - b_\ell}{2^{\binom{N-1}{\ell-1}}} + \binom{\frac{N}{2K}}{\ell - 1} \frac{b_\ell}{2^{\binom{N-1}{\ell-1}}} \right], \\ \mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u) &= \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left[\binom{\frac{(1-\nu)N}{2K}}{\ell - 1} \frac{a_\ell - b_\ell}{2^{\binom{N-1}{\ell-1}}} + \binom{\frac{N}{2K}}{\ell - 1} \frac{b_\ell}{2^{\binom{N-1}{\ell-1}}} \right], \quad j \neq 1.\end{aligned}$$

According to the assumptions in Theorem 4.1.7, $\mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u) - \mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u) = \Omega(1)$. Define

$$\mu_C := \frac{1}{2} \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left\{ \left[\binom{\frac{\nu N}{2K}}{\ell - 1} + \binom{\frac{(1-\nu)N}{2K}}{\ell - 1} \right] \frac{a_\ell - b_\ell}{2^{\binom{N-1}{\ell-1}}} + 2 \cdot \binom{\frac{N}{2K}}{\ell - 1} \frac{b_\ell}{2^{\binom{N-1}{\ell-1}}} \right\}. \quad (4.5.24)$$

In Algorithm 4.4.3, vertex u is assigned to $\widehat{\mathcal{U}}_i$ if it has the maximal number of neighbors in $\widehat{\mathcal{U}}_i^{(0)}$.

If $u \in \mathcal{U}_1$ is mislabeled, then one of the following events must happen:

- $\widehat{\mathbf{S}}_{11}^{(0)}(u) \leq \mu_C$, meaning that u fails to have enough neighbors in $\widehat{\mathcal{U}}_1^{(0)}$ to be assigned to $\widehat{\mathcal{U}}_1$.
- $\widehat{\mathbf{S}}_{1j}^{(0)}(u) \geq \mu_C$ for some $j \neq 1$, meaning that u survived Algorithm 4.4.3 without being corrected.

Lemma 4.5.16 shows that the probabilities of those two events can be bounded in terms of the SNR.

Lemma 4.5.16. *For sufficiently large N and any $u \in \mathcal{U}_1 = \mathcal{Z} \cap \mathcal{V}'_1$, we have*

$$\widehat{\rho}_1^{(0)} := \mathbb{P}\left(\widehat{\mathbf{S}}_{11}^{(0)}(u) \leq \mu_C\right) \leq \rho, \quad \widehat{\rho}_j^{(0)} := \mathbb{P}\left(\widehat{\mathbf{S}}_{1j}^{(0)}(u) \geq \mu_C\right) \leq \rho, \quad (j \neq 1), \quad (4.5.25)$$

where $\rho := \exp(-C_{\mathcal{L}}(K, \nu) \cdot \text{SNR}_{\mathcal{L}}(K))$ with $C_{\mathcal{L}}(K, \nu)$ and $\text{SNR}_{\mathcal{L}}(K)$ defined in (4.1.3) and (4.1.2) respectively.

As a result, the probability that either of those events happened is bounded by ρ . For vertex $v \in \mathcal{V}'$, let Γ_v and Λ_v denote the i.i.d indicator random variables with means $\widehat{\rho}_1^{(0)}$ and

$\hat{\rho}_j^{(0)}$ ($j \neq 1$) respectively. In the true community $\mathcal{U}_1$, the number of mislabeled vertices is upper bounded by

$$\hat{\mathbf{R}}_1^{(0)} = \sum_{v \in \mathcal{U}_1} \Gamma_v + \sum_{j=2}^K \sum_{v \in \mathcal{U}_1 \cap \hat{\mathcal{U}}_j^{(0)}} \Lambda_t.$$

Together with Lemma 4.5.16, we have

$$\mathbb{E} \hat{\mathbf{R}}_1^{(0)} \leq \frac{N}{2K} \hat{\rho}_1^{(0)} + \sum_{j=2}^K \frac{(1-v)N}{2K} \hat{\rho}_j^{(0)} \leq \frac{N}{2K} \cdot K\rho = \frac{N\rho}{2}.$$

Let $t_1 := N\rho/2$, where v denotes the correctness after Algorithm 4.4.2, then by Chernoff bound (Lemma D.3.3),

$$\mathbb{P}(\hat{\mathbf{R}}_1^{(0)} \geq N\rho) = \mathbb{P}(\hat{\mathbf{R}}_1^{(0)} - N\rho/2 \geq t_1) \leq \mathbb{P}(\hat{\mathbf{R}}_1^{(0)} - \mathbb{E} \hat{\mathbf{R}}_1^{(0)} \geq t_1) \leq e^{-ct_1} = O(e^{-N\rho}). \quad (4.5.26)$$

Then with probability $1 - O(e^{-N\rho})$, the fraction of mislabeled vertices in $\mathcal{U}_1$ is smaller than $K\rho$, i.e., the correctness of $\mathcal{U}_1$ is at least $\gamma_C := \max\{v, 1 - K\rho\}$. Therefore, Algorithm 4.4.3 outputs a γ_C -correct partition $\hat{\mathcal{U}}_1, \dots, \hat{\mathcal{U}}_K$ with probability $1 - O(e^{-N\rho})$.

4.5.3 Merging: Proof of Lemma 4.4.3

By running Algorithm 4.4.3 on the red hypergraph, we obtain a γ_C -correct partition $\hat{\mathcal{U}}_1, \dots, \hat{\mathcal{U}}_K$ where $\gamma_C := \max\{v, 1 - K\rho\} \geq v$, i.e.,

$$|\mathcal{U}_j \cap \hat{\mathcal{U}}_j| \geq v \cdot |\hat{\mathcal{U}}_j| = \frac{vN}{2K}, \quad \forall j \in [K]. \quad (4.5.27)$$

In the rest of this subsection, we shall condition on the event (4.5.27) and denote $\mathcal{W}_j := \mathcal{Y} \cap \mathcal{V}_j$. The failure probability of Algorithm 4.4.4 is estimated by the presence of hyperedges between vertex sets $\mathcal{Y}$ and $\mathcal{Z}$.

Consider a hyperedge $e = \{i_1, \dots, i_\ell\}$ in the underlying ℓ -uniform hypergraph. If vertices $i_1, \dots, i_\ell$ are all from the same cluster $\mathcal{V}_j$, then the probability that e is an existing blue edge conditioning on the event that e is not a red edge is

$$\psi_\ell := \mathbb{P} \left[e \text{ is a blue edge} \mid e \text{ is not a red edge} \right] = \frac{\frac{a_\ell}{2^{\binom{N-1}{\ell-1}}}}{1 - \frac{a_\ell}{2^{\binom{N-1}{\ell-1}}}} = (1 + o(1)) \cdot \frac{a_\ell}{2^{\binom{N-1}{\ell-1}}}, \quad (4.5.28)$$

and the presence of e can be represented by an indicator random variable $\zeta_e^{(a_\ell)} \sim \text{Bernoulli}(\psi_\ell)$. Similarly, if not all the vertices $i_1, \dots, i_\ell$ are from the same cluster $\mathcal{V}_j$, the probability that e is an existing blue edge conditioning on the event that e is not red

$$\phi_\ell := \mathbb{P} \left[e \text{ is a blue edge} \mid e \text{ is not a red edge} \right] = \frac{\frac{b_\ell}{2^{\binom{N-1}{\ell-1}}}}{1 - \frac{b_\ell}{2^{\binom{N-1}{\ell-1}}}} = (1 + o(1)) \cdot \frac{b_\ell}{2^{\binom{N-1}{\ell-1}}}, \quad (4.5.29)$$

and the presence of e can be represented by an indicator variable $\xi_e^{(b_\ell)} \sim \text{Bernoulli}(\phi_\ell)$.

For any vertex $w \in \mathcal{W}_j := \mathcal{Y} \cap \mathcal{V}_j$ with fixed $j \in [K]$, we want to compute the number of hyperedges containing w with all remaining vertices located in vertex set $\widehat{\mathcal{U}}_j$ for some fixed $k \in [K]$. Following a similar argument given in Section 4.5.2, this number can be written as

$$\widehat{\mathcal{S}}_{jk}(w) := \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left\{ \sum_{e \in \widehat{\mathcal{E}}_{j,k}^{(a_\ell)}} \zeta_e^{(a_\ell)} + \sum_{e \in \widehat{\mathcal{E}}_{j,k}^{(b_\ell)}} \xi_e^{(b_\ell)} \right\}, \quad w \in \mathcal{W}_j, \quad (4.5.30)$$

where $\widehat{\mathcal{E}}_{j,k}^{(a_\ell)} := \mathcal{E}_\ell([\mathcal{W}_j]^1, [\mathcal{U}_j \cap \widehat{\mathcal{U}}_j]^{\ell-1})$ denotes the set of ℓ -hyperedges with 1 vertex from $[\mathcal{W}_j]^1$ and the other $\ell - 1$ vertices from $[\mathcal{U}_j \cap \widehat{\mathcal{U}}_j]^{\ell-1}$, while $\widehat{\mathcal{E}}_{j,k}^{(b_\ell)} := \mathcal{E}_\ell([\mathcal{W}_j]^1, [\widehat{\mathcal{U}}_j]^{\ell-1} \setminus [\mathcal{U}_j \cap \widehat{\mathcal{U}}_j]^{\ell-1})$ denotes the set of ℓ -hyperedges with 1 vertex in $[\mathcal{W}_j]^1$ while the remaining $\ell - 1$ vertices are in $[\widehat{\mathcal{U}}_j]^{\ell-1} \setminus [\mathcal{U}_j \cap \widehat{\mathcal{U}}_j]^{\ell-1}$, with their cardinalities

$$|\widehat{\mathcal{E}}_{j,k}^{(a_\ell)}| = \binom{|\mathcal{U}_j \cap \widehat{\mathcal{U}}_j|}{\ell - 1}, \quad |\widehat{\mathcal{E}}_{j,k}^{(b_\ell)}| = \left[\binom{|\widehat{\mathcal{U}}_j|}{\ell - 1} - \binom{|\mathcal{U}_j \cap \widehat{\mathcal{U}}_j|}{\ell - 1} \right].$$

Similarly, we multiply $(\ell - 1)$ in (4.5.30) as weight since the rest $\ell - 1$ vertices can be regarded as the neighbors of w in $\widehat{\mathcal{U}}_j$. By accuracy of Algorithm 4.4.3 in (4.5.27), $|\widehat{\mathcal{U}}_j \cap \mathcal{U}_j| \geq \nu N / (2K)$, then

$$|\widehat{\mathcal{E}}_{j,j}^{(a_\ell)}| \geq \binom{\frac{\nu N}{2K}}{\ell-1}, \quad |\widehat{\mathcal{E}}_{j,k}^{(a_\ell)}| \leq \binom{\frac{(1-\nu)N}{2K}}{\ell-1}, \quad j \neq l.$$

Taking expectation with respect to $\zeta_e^{(a_\ell)}$ and $\xi_e^{(b_\ell)}$, for any $w \in \mathcal{W}_j$, we have

$$\begin{aligned} \mathbb{E}\widehat{\mathbf{S}}_{jj}(w) &= \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left[\binom{\frac{\nu N}{2K}}{\ell-1} (\Psi_\ell - \Phi_\ell) + \binom{\frac{N}{2K}}{\ell-1} \Phi_\ell \right], \\ \mathbb{E}\widehat{\mathbf{S}}_{jk}(w) &= \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left[\binom{\frac{(1-\nu)N}{2K}}{\ell-1} (\Psi_\ell - \Phi_\ell) + \binom{\frac{N}{2K}}{\ell-1} \Phi_\ell \right], \quad j \neq l. \end{aligned}$$

By assumptions in Theorem 4.1.7, $\mathbb{E}\widehat{\mathbf{S}}_{jj}(w) - \mathbb{E}\widehat{\mathbf{S}}_{jk}(w) = \Omega(1)$. We define

$$\mu_M := \frac{1}{2} \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left\{ \left[\binom{\frac{\nu N}{2K}}{\ell-1} + \binom{\frac{(1-\nu)N}{2K}}{\ell-1} \right] (\Psi_\ell - \Phi_\ell) + 2 \binom{\frac{N}{2K}}{\ell-1} \Phi_\ell \right\}. \quad (4.5.32)$$

After Algorithm 4.4.4, if a vertex $w \in \mathcal{W}_j$ is mislabelled, one of the following events must happen

- $\widehat{\mathbf{S}}_{jj}(w) \leq \mu_M$, which implies that w fails to have enough neighbors in $\widehat{\mathcal{U}}_j$ to be assigned to $\widehat{\mathcal{U}}_j$.
- $\widehat{\mathbf{S}}_{jk}(w) \geq \mu_M$ for some $j \neq k$, which implies that w survived Algorithm 4.4.4 without being corrected.

By an argument similar to Lemma 4.5.16, we can prove that for any $w \in \mathcal{W}_j$,

$$\widehat{\rho}_j := \mathbb{P}(\widehat{\mathbf{S}}_{jj}(w) \leq \mu_M) \leq \rho, \quad \widehat{\rho}_k := \mathbb{P}(\widehat{\mathbf{S}}_{jk}(w) \geq \mu_M) \leq \rho, \quad (j \neq l),$$

where $\rho := \exp(-C_{\mathcal{L}}(K, \nu) \cdot \text{SNR}_{\mathcal{L}}(K))$ with $C_{\mathcal{L}}(K, \nu)$ and $\text{SNR}_{\mathcal{L}}(K)$ defined in (4.1.3) and (4.1.2) respectively. The misclassified probability for $w \in \mathcal{W}_j$ is then upper bounded by $\sum_{j=1}^K \widehat{\rho}_j$.

Let Γ_l be the i.i.d indicator random variables with mean $K\rho$. Then, the number of mislabelled vertices in $\mathcal{W}_j$ is at most

$$\widehat{\mathbf{R}}_j = \sum_{v \in \mathcal{W}_j} \Gamma_v. \quad (4.5.33)$$

Consequently, $\mathbb{E}\widehat{\mathbf{R}}_j \leq N/(2K) \cdot K\rho = N\rho/2$. Let $t_j := N\rho/2$, by Chernoff bound (Lemma D.3.3),

$$\mathbb{P}\left(\widehat{\mathbf{R}}_j \geq N\rho\right) = \mathbb{P}\left(\widehat{\mathbf{R}}_j - N\rho/2 \geq t_j\right) \leq \mathbb{P}\left(\widehat{\mathbf{R}}_j - \mathbb{E}\widehat{\mathbf{R}}_j \geq t_j\right) \leq e^{-ct_j} = O(e^{-N\rho}).$$

Hence with probability $1 - O(e^{-N\rho})$, the fraction of mislabeled vertices in $\mathcal{W}_j$ is smaller than $K\rho$, i.e., the correctness in $\mathcal{W}_j$ is at least $\gamma_M := \max\{\mathbf{v}, 1 - K\rho\}$.

4.5.4 Proof of Theorem 4.1.7

Now we are ready to prove Theorem 4.1.7. The correctness of Algorithm 4.4.3 and Algorithm 4.4.4 are denoted by γ_C and γ_M respectively, then with probability at least $1 - O(e^{-N\rho})$, the correctness γ of Algorithm 4.1.2 is $\gamma := \min\{\gamma_C, \gamma_M\} = \max\{\mathbf{v}, 1 - K\rho\}$. We will have $\gamma = 1 - K\rho$ if $\mathbf{v} \leq 1 - K\rho$, equivalently,

$$\text{SNR}_{\mathcal{L}}(K) \geq \frac{1}{C_{\mathcal{L}}(K, \mathbf{v})} \log\left(\frac{K}{1 - \mathbf{v}}\right), \quad (4.5.34)$$

otherwise $\gamma = \mathbf{v}$. The inequality (4.5.34) holds since

$$\begin{aligned} \text{SNR}_{\mathcal{L}}(K) &= \frac{[\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_{\ell} - b_{\ell})K^{-\ell+1}]^2}{\sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot ((a_{\ell} - b_{\ell})K^{-\ell+1} + b_{\ell})} \\ &\geq \frac{[\sum_{\ell \in \mathcal{L}} (a_{\ell} - b_{\ell})]^2}{K^{2L-2}(L-1)d} \geq \frac{(C_v)^2}{L-1} \log\left(\frac{K}{1 - \mathbf{v}}\right) \geq \frac{1}{C_{\mathcal{L}}(K, \mathbf{v})} \log\left(\frac{K}{1 - \mathbf{v}}\right). \end{aligned}$$

where the first two inequalities hold since $d := \sum_{\ell \in \mathcal{L}} (\ell - 1)a_{\ell}$ and condition (4.1.4b), while the last inequality holds by taking $C_v \geq \max\{\sqrt{(L-1)/C_{\mathcal{L}}(K, \mathbf{v})}, C_2\}$ with C_2 defined in (4.5.18).

Remark 4.5.17. The lower bound $C_{(4.1.4a)}$ in (4.1.4a) comes from the requirement in Lemma 4.5.6 that only a few high degree vertices be deleted. The constant C_v in (4.1.4b) comes from the requirement in Lemma 4.5.8 that the subspace angle is small. When $C_{(4.1.4a)}$ is not so large (or the hypergraph is too sparse), one could still achieve good accuracy γ if C_v is large enough (the difference between a_ℓ and b_ℓ is large enough).

Remark 4.5.18. The condition (4.5.34) indicates that the improvement of accuracy from local refinement (Algorithm 4.4.3 and Algorithm 4.4.4) will be guaranteed when $\text{SNR}_L(K)$ is large enough. If $\text{SNR}_L(K)$ is small, we use correctness of Algorithm 4.4.2 instead, i.e., $\gamma = v$, to represent the correctness of Algorithm 4.1.2.

4.5.5 Proof of Corollary 4.1.9

For any fixed $v \in (1/K, 1)$, $\text{SNR}_L(K) \rightarrow \infty$ implies $\rho \rightarrow 0$ and

$$d := \sum_{\ell \in \mathcal{L}} (\ell - 1) a_\ell \rightarrow \infty. \quad (4.5.36)$$

We now check the condition (4.1.4b) in Assumption 4.1.5. Note that $\ell \geq 2$ for each $\ell \in \mathcal{L}$, then

$$\begin{aligned} & \frac{[\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell)]^2}{d} \\ &= K^{L-1} \cdot K^2 \frac{[\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell) K^{-2+1}]^2}{\sum_{\ell \in \mathcal{L}} (\ell - 1) a_\ell K^{-L+1}} \\ &\geq K^{L+1} \frac{[\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell) K^{-\ell+1}]^2}{\sum_{\ell \in \mathcal{L}} (\ell - 1) a_\ell K^{-\ell+1}} \\ &\geq \frac{[\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell) K^{-\ell+1}]^2}{\sum_{\ell \in \mathcal{L}} (\ell - 1) ((a_\ell - b_\ell) K^{-\ell+1} + b_\ell)} = \text{SNR}_L(K) \rightarrow \infty. \end{aligned}$$

Consequently, there exists some v close to 1 and some constant $C_v > 0$ such that condition (4.1.4b) is satisfied. Applying Theorem 4.1.7, we find $\gamma = 1 - o(1)$, which implies weak consistency. The constraint of $\text{SNR}_L(K) = o(\log(N))$ is used in the proof of Lemma 4.4.2, see Remark 4.1.10.

4.6 Acknowledgements

Chapter 4, in full, is a reprint of the material “Ioana Dumitriu, Hai-Xiao Wang and Yizhe Zhu. Partial recovery and weak consistency in the non-uniform hypergraph stochastic block model, *Combinatorics, Probability and Computing* 34.1 (2025): pp. 1-51. arXiv: 2112.11671”. The dissertation author is the co-author of this paper.

Appendix A

Deferred proofs in Chapter 2

A.1 Proofs in Section 2.3

Proof of Lemma 2.3.2. Recall $[\mathcal{V}]^\ell := \{\mathcal{S} \subseteq \mathcal{V} : |\mathcal{S}| = \ell\}$. Consider the following log-likelihood

$$\begin{aligned}
& \log \mathbb{P}(\mathbb{H} = \mathcal{H} \mid \mathbf{y} = \mathbf{z}) \\
&= \log \left(\prod_{\ell \in \mathcal{L}} \prod_{e \in [\mathcal{V}]^\ell} [\mathbb{P}(e \in \mathcal{E}_\ell \mid \mathbf{y} = \mathbf{z})]^{\mathcal{A}_e^{(\ell)}} \cdot [\mathbb{P}(e \notin \mathcal{E}_\ell \mid \mathbf{y} = \mathbf{z})]^{1 - \mathcal{A}_e^{(\ell)}} \right) \\
&= \sum_{\ell \in \mathcal{L}} \left(\sum_{e \in [\mathcal{V}]^\ell} \mathbb{1}_{\{e \text{ is in-cluster}\}} \cdot [\mathcal{A}_e^{(\ell)} \log \alpha_\ell + (1 - \mathcal{A}_e^{(\ell)}) \log(1 - \alpha_\ell)] \right. \\
&\quad \left. + \sum_{e \in [\mathcal{V}]^\ell} (1 - \mathbb{1}_{\{e \text{ is in-cluster}\}}) \cdot [\mathcal{A}_e^{(\ell)} \log \beta_\ell + (1 - \mathcal{A}_e^{(\ell)}) \log(1 - \beta_\ell)] \right) \\
&= \sum_{\ell \in \mathcal{L}} \left(\log \left(\frac{\alpha_\ell(1 - \beta_\ell)}{\beta_\ell(1 - \alpha_\ell)} \right) \cdot \sum_{e \in [\mathcal{V}]^\ell} \mathbb{1}_{\{e \text{ is in-cluster}\}} \cdot \mathcal{A}_e^{(\ell)} + C_\ell \right),
\end{aligned}$$

where $C_\ell > 0$ is some constant irrelevant to the membership assignment $\mathbf{y}$. Note that the logarithm map preserves the monotocity and C_ℓ can be factored out, then MLE in (2.3.3) can be further simplified as

$$\begin{aligned}
\hat{\mathbf{y}}_{\text{MLE}} &= \arg \max_{\mathbf{z} \in \{\pm 1\}^{|\mathcal{V}|}, \mathbf{1}^\top \mathbf{z} = 0} \sum_{\ell \in \mathcal{L}} \left(\log \left(\frac{\alpha_\ell(1 - \beta_\ell)}{\beta_\ell(1 - \alpha_\ell)} \right) \cdot \sum_{e \in [\mathcal{V}]^\ell} \mathbb{1}_{\{e \text{ is in-cluster}\}} \cdot \mathcal{A}_e^{(\ell)} \right) \\
&= \arg \max_{\mathbf{z} \in \{\pm 1\}^{|\mathcal{V}|}, \mathbf{1}^\top \mathbf{z} = 0} f(\mathbf{z} \mid \mathcal{H}).
\end{aligned}$$

Therefore, the desired argument is proved since each $\ell \in \mathcal{L}$, the following holds

$$\log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \begin{cases} > 0, & \text{if } \alpha_\ell > \beta_\ell, \\ < 0, & \text{if } \alpha_\ell < \beta_\ell. \end{cases} \quad (\text{A.1.1})$$

□

Proof of Lemma 2.3.4. First, for any $a \in \mathcal{V}_+ \cap \mathcal{U}$, $b \in \mathcal{V}_- \cap \mathcal{U}$, the independence between $W_a[\mathcal{U}]$ and $J[\mathcal{U}]$ follows directly. The reason is that each hyperedge in $J[\mathcal{U}]$ contains at least two vertices in $\mathcal{U}$, while hyperedge in W_a contains exactly one vertex $a \in \mathcal{U}$, thus no common edges. The independence between $W_b[\mathcal{U}]$ and $J[\mathcal{U}]$ follows similarly. The independence between $W_a[\mathcal{U}]$ and $W_b[\mathcal{U}]$ is also clear since they do not share any common hyperedge by construction. At the same time, we have

$$\{W_a[\mathcal{U}] \leq -\zeta_N q_N\} \cap \{J[\mathcal{U}] = 1\} \implies \{W_a[\mathcal{U}] \leq -\zeta_N q_N\} \cap \{J_a \leq \zeta_N q_N\}, \quad (\text{A.1.2})$$

$$\{W_b[\mathcal{U}] \leq -\zeta_N q_N\} \cap \{J[\mathcal{U}] = 1\} \implies \{W_b[\mathcal{U}] \leq -\zeta_N q_N\} \cap \{J_b \leq \zeta_N q_N\}. \quad (\text{A.1.3})$$

Thus it suffices to prove the following relationship

$$\{W_a[\mathcal{U}] \leq -\zeta_N q_N\} \cap \{J_a \leq \zeta_N q_N\} \implies \{W_a \leq 0\}. \quad (\text{A.1.4})$$

To this end, we expand W_a and plug in the condition on the left hand side of the above equation, then

$$W_a = \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \cdot \left(\sum_{e \in \mathcal{E}_\ell^{(0)}(a)} \mathcal{A}_e^{(\ell)} - \sum_{e \in \mathcal{E}_\ell^{(\ell-1)}(a)} \mathcal{A}_e^{(\ell)} \right) \quad (\text{A.1.5})$$

$$\leq W_a[\mathcal{U}] + J_a \leq 0, \quad (\text{A.1.6})$$

which completes the proof. □

Proof of Lemma 2.3.5. For $a \in \mathcal{V}_+ \cap \mathcal{U}$, according to Lemma A.1.1, $\mathbb{P}(\mathbf{W}_a[\mathcal{U}] \leq -\zeta_N \cdot q_N) = e^{-\mathbb{D}_{\text{GH}} \cdot q_N}$. According to Lemma 2.3.4, $\mathbf{W}_a[\mathcal{U}]$ and $\mathbf{W}_{a'}[\mathcal{U}]$ are independent for distinct $a, a' \in \mathcal{V}_+ \cap \mathcal{U}$, thus

$$\prod_{a \in \mathcal{V}_+ \cap \mathcal{U}} \mathbb{P}(\mathbf{W}_a[\mathcal{U}] > -\zeta_N \cdot q_N) = \prod_{a \in \mathcal{V}_+ \cap \mathcal{U}} \left(1 - \mathbb{P}(\mathbf{W}_a[\mathcal{U}] \leq -\zeta_N \cdot q_N)\right) \quad (\text{A.1.7})$$

$$= (1 - e^{-\mathbb{D}_{\text{GH}} \cdot q_N})^{\gamma_N |\mathcal{V}|/2} \leq \exp\left(-\frac{\gamma_N \cdot |\mathcal{V}|}{2} \cdot e^{-\mathbb{D}_{\text{GH}} \cdot q_N}\right), \quad (\text{A.1.8})$$

where we used the fact that $1 - x \leq e^{-x}$ for $x = o(1)$ in the last inequality. The upper bound for $\prod_{b \in \mathcal{V}_- \cap \mathcal{U}} \mathbb{P}(\mathbf{W}_b[\mathcal{U}] > -\zeta_N \cdot q_N)$ can be obtained similarly.

For the second part of the proof, through a union bound, the problem is transferred to seeking an upper bound on $\mathbb{P}(\mathbf{J}_v[\mathcal{U}] > \zeta_N q_N)$ for any $v \in \mathcal{U}$, since

$$\mathbb{P}(\mathbf{J}[\mathcal{U}] = 0) = \mathbb{P}(\exists v \in \mathcal{U} \text{ s.t. } \mathbb{1}_{\{\mathbf{J}_v[\mathcal{U}] \leq \zeta_N q_N\}} = 0) \leq \sum_{v \in \mathcal{U}} \mathbb{P}(\mathbf{J}_v[\mathcal{U}] > \zeta_N q_N).$$

Recall $\mathcal{W}_\ell^{(\geq 2)}$ in (2.3.11), where each edge $e \in \mathcal{W}_\ell^{(\geq 2)}$ contains at least 2 vertices in $\mathcal{U}$. Thus for edge $e \in \mathcal{W}_\ell^{(\geq 2)}$ with $e \ni v$, at least one more vertex in $e \setminus \{v\}$ is from $\mathcal{U}$. We now calculate the expectation of $\mathbf{J}_v[\mathcal{U}]$ defined in (2.3.12). Note that the cardinality of the set $\mathcal{W}_\ell^{(\geq 2)}$ is upper bounded by $\binom{N-1}{\ell-1} - \binom{(1-\gamma_N)N}{\ell-1}$, then

$$\mathbb{E}\mathbf{J}_v[\mathcal{U}] := \sum_{\ell \in \mathcal{L}} \log\left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)}\right) \cdot \left[\binom{N-1}{\ell-1} - \binom{(1-\gamma_N)N}{\ell-1}\right] \cdot \mathbb{E}\mathcal{A}_e^{(\ell)}.$$

where $\mathbb{E}\mathcal{A}_e^{(\ell)} = \alpha_\ell$ if e is in-cluster, otherwise $\mathbb{E}\mathcal{A}_e^{(\ell)} = \beta_\ell$. Since $\ell \in \mathcal{L}$ is some fixed integer and $\ell \ll N$, we then have

$$\binom{N-1}{\ell-1} [1 - (1-\gamma_N)^{\ell-1}] \leq \binom{N-1}{\ell-1} - \binom{(1-\gamma_N)N}{\ell-1} \leq \binom{N-1}{\ell-1} [1 - (1-\gamma_N) \cdot (1-2\gamma_N)^{\ell-2}].$$

The expectation of $J_v[\mathcal{U}]$ is then bounded both from above and below, as follows

$$\begin{aligned}\mathbb{E}J_v[\mathcal{U}] &\geq \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \cdot [1 - (1-\gamma_N)^{\ell-1}] \cdot b_\ell \\ \mathbb{E}J_v[\mathcal{U}] &\leq \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \cdot [1 - (1-2\gamma_N)^{\ell-1}] \cdot a_\ell.\end{aligned}$$

Note that $a_\ell \gtrsim b_\ell \gtrsim 1$ under Assumption 2.1.6, and $\gamma_N = o(1)$ by construction of $\mathcal{U}$. Since $1 - x + O(x^2) = e^{-x}$ for $x = o(1)$, we have $1 - (1-\gamma_N)^{\ell-1} = (\ell-1)\gamma_N$ and $1 - (1-2\gamma_N)^{\ell-1} = (\ell-1)\gamma_N$. Consequently, $\mathbb{E}J_v[\mathcal{U}] = \Theta(\gamma_N q_N)$ for any $v \in \mathcal{U}$, and $J_v[\mathcal{U}]$ concentrates around its expectation with high probability. According to Lemma D.3.6, the following holds

$$\mathbb{P}(J_v[\mathcal{U}] \geq \zeta_N q_N) \leq \exp \left(-q_N \cdot [\zeta_N \log(\zeta_N/\gamma_N) - (\zeta_N - \gamma_N)] \right).$$

Recall that $|\mathcal{U}| = \gamma_N |\mathcal{V}|$, then by a union bound

$$\mathbb{P}(J[\mathcal{U}] = 0) \leq \sum_{v \in \mathcal{U}} \mathbb{P}(J_v[\mathcal{U}] \geq \zeta_N q_N) \leq \exp \left(\log(\gamma_N N) - q_N \cdot [\zeta_N \log(\zeta_N/\gamma_N) - (\zeta_N - \gamma_N)] \right).$$

which completes the proof. $\square$

Proof of Lemma 2.3.6. The likelihood for observing $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ conditioned on $y_v = s \in \{\pm 1\}$ and $\mathbf{y}_{-v}$ is

$$\mathbb{P}(\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}} \mid y_v = s, \mathbf{y}_{-v}) \propto \prod_{\ell \in \mathcal{L}} \prod_{e \ni v} [\mathbb{P}(e \in \mathcal{E}_\ell \mid y_v = s, \mathbf{y}_{-v})]^{\mathcal{A}_e^{(\ell)}} \cdot [\mathbb{P}(e \notin \mathcal{E}_\ell \mid y_v = s, \mathbf{y}_{-v})]^{1-\mathcal{A}_e^{(\ell)}}.$$

where $\propto$ only hides factors that are irrelevant to v . Note that the label vector is sampled uniformly at random from all vectors $\mathbf{z} \in \{\pm 1\}^{|\mathcal{V}|}$ satisfying $\mathbf{1}^\top \mathbf{z} = 0$. According to the Bayes Theorem

and the analysis in (2.3.7), for $s \in \{\pm 1\}$, the log-odd ratio can be expressed as

$$\begin{aligned}
& \log \left(\frac{\mathbb{P}(\mathbf{y}_v = s | \{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}, \mathbf{y}_{-v})}{\mathbb{P}(\mathbf{y}_v = -s | \{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}, \mathbf{y}_{-v})} \right) = \log \left(\frac{\mathbb{P}(\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}} | \mathbf{y}_v = 1, \mathbf{y}_{-v})}{\mathbb{P}(\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}} | \mathbf{y}_v = -1, \mathbf{y}_{-v})} \right) \\
&= \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1 - \beta_\ell)}{\beta_\ell(1 - \alpha_\ell)} \right) \cdot \left[|\mathcal{E}_\ell^{(0)}(v)| - |\mathcal{E}_\ell^{(\ell-1)}(v)| \right] \\
&= \mathbf{W}_v,
\end{aligned}$$

which completes the proof. $\square$

Proof of Lemma 2.3.7. We first define the sign between two vectors $\mathbf{y}$ and $\hat{\mathbf{y}}$ as

$$s(\mathbf{y}, \hat{\mathbf{y}}) := \arg \min_{s \in \{\pm 1\}} \|\mathbf{y} - s\hat{\mathbf{y}}\|_1. \quad (\text{A.1.9})$$

Consequently, $s(\mathbf{y}, \hat{\mathbf{y}}) = \text{sign}(\langle \mathbf{y}, \hat{\mathbf{y}} \rangle)$ if $\langle \mathbf{y}, \hat{\mathbf{y}} \rangle \neq 0$. At the same time, when $|\langle \mathbf{y}, \hat{\mathbf{y}} \rangle| > 1$, $s(\mathbf{y}, \hat{\mathbf{y}}) = s(\mathbf{y}_v, \hat{\mathbf{y}}_v)$ for all $v \in [N]$. Thus for any $\hat{\mathbf{y}} \in \mathcal{F}$, the following holds

$$\mathbb{E} \eta_N(\hat{\mathbf{y}}, \mathbf{y}) = \mathbb{E} \left[\frac{1}{N} \mathbb{D}_{\text{HD}}(s(\mathbf{y}, \hat{\mathbf{y}}) \cdot \mathbf{y}, \hat{\mathbf{y}}) \right] = \mathbb{E} \left[\frac{1}{N} \sum_{v=1}^N \mathbb{1}_{\{\mathbf{y}_v \neq \hat{\mathbf{y}}_v\}} \right]. \quad (\text{A.1.10})$$

$\square$

Lemma A.1.1 (Generalized version of [146, Lemma 2]). *According to the Assumptions in Lemma 2.3.5, the following holds for $\zeta_N = o(1)$ and $q_N \gg 1$*

$$\mathbb{P}(\mathbf{W}_a[\mathcal{U}] \leq -\zeta_N \cdot q_N) = e^{-\mathbb{D}_{\text{GH}} \cdot q_N}, \quad \mathbb{P}(\mathbf{W}_b[\mathcal{U}] \leq -\zeta_N \cdot q_N) = e^{-\mathbb{D}_{\text{GH}} \cdot q_N}, \quad (\text{A.1.11})$$

where the rate function is

$$\mathbb{D}_{\text{GH}} = \sum_{\ell \in \mathcal{L}} \frac{1}{2^{\ell-1}} (\sqrt{a_\ell} - \sqrt{b_\ell})^2. \quad (\text{A.1.12})$$

Proof of Lemma A.1.1. For vertex $a \in \mathcal{V}_+ \cap \mathcal{U}$, we write $\mathbf{W}_a[\mathcal{U}] = \mathbf{Y}_a - \mathbf{Z}_a$, where $\mathbf{Y}_a$ and $\mathbf{Z}_a$

are defined as

$$Y_a := \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \cdot \sum_{e \in \mathcal{E}_\ell^{(0)}(a)} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\left\{ \begin{array}{l} e \cap \mathcal{U} = \{a\} \\ e \setminus \{a\} \subset [\mathcal{V}_+ \setminus \mathcal{U}] \end{array} \right\}}, \quad (\text{A.1.13})$$

$$Z_a := \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) \cdot \sum_{e \in \mathcal{E}_\ell^{(\ell-1)}(a)} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\left\{ \begin{array}{l} e \cap \mathcal{U} = \{a\} \\ e \setminus \{a\} \subset [\mathcal{V}_- \setminus \mathcal{U}] \end{array} \right\}}. \quad (\text{A.1.14})$$

According to Assumption 2.1.6, hyperedges e in Y_a and Z_a are sampled independently with probabilities $p_1^{(\ell)} = \alpha_\ell = a_\ell \cdot q_N / \binom{N-1}{\ell-1}$ and $p_2^{(\ell)} = b_\ell \cdot q_N / \binom{N-1}{\ell-1}$. We consider apply Lemma A.1.2. For each $\ell \in \mathcal{L}$, we take $n_\ell = 2$, $\alpha_1^{(\ell)} = a_\ell$, $\alpha_2^{(\ell)} = b_\ell$ with $a_\ell > b_\ell$, and

$$c_1^{(\ell)} = \log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) = \log \left(\frac{a_\ell}{b_\ell} \right), \quad c_2^{(\ell)} = -\log \left(\frac{\alpha_\ell(1-\beta_\ell)}{\beta_\ell(1-\alpha_\ell)} \right) = \log \left(\frac{b_\ell}{a_\ell} \right). \quad (\text{A.1.15})$$

At the same time, $|\mathcal{V}_+| = |\mathcal{V}_-| = N/2$, $|\mathcal{U} \cap \mathcal{V}_+| = |\mathcal{U} \cap \mathcal{V}_-| = \gamma_N N/2$ and $\gamma_N = o(1)$, we have

$$N_1^{(\ell)} = N_2^{(\ell)} = \binom{(1-\gamma_N)N/2}{\ell-1} = (1+o(1)) \cdot \frac{1}{2^{\ell-1}} \binom{N-1}{\ell-1}. \quad (\text{A.1.16})$$

Thus $g_N = 1$ and $\rho_1^{(\ell)} = \rho_2^{(\ell)} = 2^{-(\ell-1)}$ for each $\ell \in \mathcal{L}$. Consequently,

$$\sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} c_i^{(\ell)} \alpha_i^{(\ell)} \rho_i^{(\ell)} = \sum_{\ell \in \mathcal{L}} 2^{-(\ell-1)} (a_\ell - b_\ell) \log(a_\ell/b_\ell) > 0.$$

Then all the conditions of Lemma A.1.2 are satisfied. By taking $\delta = 0$, $\zeta_N = o(1)$ and $g_N = 1$, Lemma A.1.2 indicates that

$$\mathbb{P}(Y_a - Z_a \leq -\zeta_N \cdot q_N) = \exp \left(-[1 + o(1)] \cdot \mathbb{D}_{\text{GH}} \cdot g_N \cdot q_N \right) = e^{-\mathbb{D}_{\text{GH}} \cdot q_N}, \quad (\text{A.1.17})$$

where the rate function is

$$\mathbb{D}_{\text{GH}} := \max_{t \geq 0} \sum_{\ell \in \mathcal{L}} \frac{1}{2^{\ell-1}} [a_\ell + b_\ell - (a_\ell)^t (b_\ell)^{1-t} - (a_\ell)^{1-t} (b_\ell)^t]. \quad (\text{A.1.18})$$

Note that by union bound

$$\mathbb{D}_{\text{GH}} \leq \sum_{\ell \in \mathcal{L}} \frac{1}{2^{\ell-1}} \max_{t \geq 0} [a_\ell + b_\ell - (a_\ell)^t (b_\ell)^{1-t} - (a_\ell)^{1-t} (b_\ell)^t] = \sum_{\ell \in \mathcal{L}} \frac{1}{2^{\ell-1}} (\sqrt{a_\ell} - \sqrt{b_\ell})^2$$

On the other way around, the lower bound can be achieved by simply taking $t = 1/2$.

For vertex $b \in \mathcal{V}'_- \cap \mathcal{U}$, we write $W_b[\mathcal{U}] = Y_b - Z_b$, where Y_b and Z_b are defined as

$$\begin{aligned} Y_b &:= \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell (1 - \beta_\ell)}{\beta_\ell (1 - \alpha_\ell)} \right) \cdot \sum_{e \in \mathcal{E}_\ell^{(0)}(b)} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\left\{ \begin{array}{l} e \cap \mathcal{U} = \{b\} \\ e \setminus \{b\} \subset [\mathcal{V}'_- \setminus \mathcal{U}] \end{array} \right\}}, \\ Z_b &:= \sum_{\ell \in \mathcal{L}} \log \left(\frac{\alpha_\ell (1 - \beta_\ell)}{\beta_\ell (1 - \alpha_\ell)} \right) \cdot \sum_{e \in \mathcal{E}_\ell^{(\ell-1)}(b)} \mathcal{A}_e^{(\ell)} \mathbb{1}_{\left\{ \begin{array}{l} e \cap \mathcal{U} = \{b\} \\ e \setminus \{b\} \subset [\mathcal{V}'_+ \setminus \mathcal{U}] \end{array} \right\}}. \end{aligned}$$

The proof is completed by following the same steps as above. $\square$

Lemma A.1.2 (Generalized version of [146, Theorem 10]). *Let g_N be some non-decreasing function with respect to N satisfying $1 \lesssim g_N \ll N \log^{-2}(N)$. Let $\mathcal{L} = \{\ell | \ell \geq 2, \ell \in \mathbb{N}\}$ be a set of integers with finite cardinality. For each $\ell \in \mathcal{L}$, let $n_\ell \geq 2$ denote some positive integer. For each $\ell \in \mathcal{L}$ and $i \in [n_\ell]$, let $Y_i^{(\ell)}$ denote the random variable sampled from the binomial distribution $\text{Bin}(N_i^{(\ell)}, p_i^{(\ell)})$, where*

$$N_i^{(\ell)} = [1 + o(1)] \cdot \rho_i^{(\ell)} \cdot g_N \binom{N-1}{\ell-1} \quad \text{and} \quad p_i^{(\ell)} = [1 + o(1)] \cdot \alpha_i^{(\ell)} \cdot \frac{q_N}{\binom{N-1}{\ell-1}}, \quad (\text{A.1.19})$$

where $\rho_i^{(\ell)}$ and $\alpha_i^{(\ell)}$ are some positive constants independent of N . Let $\mathbf{X}_N = \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} c_i^{(\ell)} Y_i^{(\ell)}$ be a summation of random variables such that

1. at least one of the $c_i^{(\ell)}$'s is negative;
2. $\sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} c_i^{(\ell)} \alpha_i^{(\ell)} \rho_i^{(\ell)} > 0$.

Let $\delta \in (-\infty, \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} c_i^{(\ell)} \alpha_i^{(\ell)} \rho_i^{(\ell)})$ and define the rate function $\mathbb{D}$ by

$$\mathbb{D} := \sup_{t \geq 0} \left(-t\delta + \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} \rho_i^{(\ell)} (1 - e^{-c_i^{(\ell)} t}) \right). \quad (\text{A.1.20})$$

Let $x_N = (\delta + o(1)) \cdot g_N \cdot q_N$, then we have

$$\lim_{N \rightarrow \infty} [g_N \cdot q_N]^{-1} \log \mathbb{P}(X_N \leq x_N) = -\mathbb{D}. \quad (\text{A.1.21})$$

Consequently, for any $\delta \in (-\infty, \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} c_i^{(\ell)} \alpha_i^{(\ell)} \rho_i^{(\ell)})$ and $\varepsilon > 0$, there exists $N_0 > 0$ such that for all $N > N_0$, the following holds

$$\mathbb{P}(X_N \leq \delta g_N q_N) \leq e^{-g_N \cdot q_N \cdot [\mathbb{D} - \varepsilon]}. \quad (\text{A.1.22})$$

Proof of Lemma A.1.2. We first prove the upper bound of (A.1.21) by calculating moment generating function. By Markov inequality and independence between $Y_i^{(\ell)}$'s, for any $t \geq 0$, we have

$$\begin{aligned} \mathbb{P}(X_N \leq x_N) &= \mathbb{P}(e^{-tX_N} \geq e^{-tx_N}) \\ &\leq \frac{\mathbb{E} e^{-tX_N}}{e^{-tx_N}} = e^{tx_N} \prod_{\ell \in \mathcal{L}} \prod_{i=1}^{n_\ell} \mathbb{E} e^{-tc_i^{(\ell)} Y_i^{(\ell)}} = e^{tx_N} \prod_{\ell \in \mathcal{L}} \prod_{i=1}^{n_\ell} \left(1 - p_i^{(\ell)} (1 - e^{-tc_i^{(\ell)} t}) \right)^{N_i^{(\ell)}} \\ &\leq \exp \left(tx_N - \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} N_i^{(\ell)} p_i^{(\ell)} (1 - e^{-c_i^{(\ell)} t}) \right), \quad (\text{since } 1 - x \leq e^{-x}) \\ &= \exp \left(- (1 + o(1)) \cdot g_N \cdot q_N \cdot \left(-t\delta + \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} \rho_i^{(\ell)} (1 - e^{-c_i^{(\ell)} t}) \right) \right), \end{aligned}$$

where we applied $x_N = (\delta + o(1)) \cdot g_N \cdot q_N$ in the last equality. By taking the supremum of

both sides with respect to t , the desired upper bound of (A.1.21) follows as

$$\limsup_{N \rightarrow \infty} \frac{\log \mathbb{P}(X_N \leq x_N)}{g_N \cdot q_N} \leq \sup_{t \geq 0} \left(-t\delta + \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} \rho_i^{(\ell)} (1 - e^{-c_i^{(\ell)} t}) \right) = \mathbb{D}.$$

One more thing to verify is the existence and uniqueness of the extreme point. Define the function $f(t) := -t\delta + \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} \rho_i^{(\ell)} (1 - e^{-c_i^{(\ell)} t})$. For any $t \geq 0$, we have

$$f'(t) = -\delta + \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} c_i^{(\ell)} \rho_i^{(\ell)} e^{-c_i^{(\ell)} t}, \quad f''(t) = - \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} (c_i^{(\ell)})^2 \rho_i^{(\ell)} e^{-c_i^{(\ell)} t} < 0.$$

Hence, $f(t)$ is strictly concave, while $f'(t)$ is continuous and strictly decreasing. By assumption,

$$f'(0) = \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} c_i^{(\ell)} \rho_i^{(\ell)} - \delta > 0,$$

and $\lim_{t \rightarrow \infty} f'(t) = -\infty$ since at least one of the $c_i^{(\ell)}$'s is negative. Consequently, there exists a unique $t^* > 0$ such that $f'(t^*) = 0$. Note that $f(0) = 0$, thus $\mathbb{D} := f(t^*) > 0$, as defined in (A.1.20).

We now turn to the lower bound. For each $\ell \in \mathcal{L}$, let $\{y_i^{(\ell)}\}_{i \in [n_\ell]}$ be a collection of realizations sampled from the random variables $\{Y_i^{(\ell)}\}_{i \in [n_\ell]}$, satisfying

$$\sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} c_i^{(\ell)} y_i^{(\ell)} \leq x_N = \delta \cdot (1 + o(1)) \cdot g_N \cdot q_N. \quad (\text{A.1.23})$$

Such a collection of $\{y_i^{(\ell)}\}_{i \in [n_\ell]}$ do exists. Recall that $f'(t^*) = 0$ for some $t^* > 0$, which is equivalent to

$$\sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} c_i^{(\ell)} \alpha_i^{(\ell)} \rho_i^{(\ell)} e^{-c_i^{(\ell)} t^*} = \delta.$$

For each $\ell \in \mathcal{L}$ and $i \in [n_\ell]$, let $y_i^{(\ell)} = (1 - o(1)) \cdot \tau_i^{(\ell)} g_N q_N$ where we denote $\tau_i^{(\ell)} := \alpha_i^{(\ell)} \rho_i^{(\ell)} e^{-c_i^{(\ell)} t^*}$.

Obviously, the so constructed collection of samples satisfies (A.1.23), and we have

$$\mathbb{P}(\mathbf{X}_N \leq x_N) \geq \prod_{\ell \in \mathcal{L}} \prod_{i=1}^{n_\ell} \mathbb{P}(\mathbf{Y}_i^{(\ell)} = y_i^{(\ell)}) = \prod_{\ell \in \mathcal{L}} \prod_{i=1}^{n_\ell} \binom{N_i^{(\ell)}}{y_i^{(\ell)}} (p_i^{(\ell)})^{y_i^{(\ell)}} (1 - p_i^{(\ell)})^{N_i^{(\ell)} - y_i^{(\ell)}}. \quad (\text{A.1.24})$$

Note that $y_i^{(\ell)} \ll \sqrt{N_i^{(\ell)}}$ since $g_N \ll N \log^{-2}(N)$, then by Lemma D.1.3,

$$\begin{aligned} & \log \left[\binom{N_i^{(\ell)}}{y_i^{(\ell)}} (p_i^{(\ell)})^{y_i^{(\ell)}} (1 - p_i^{(\ell)})^{N_i^{(\ell)} - y_i^{(\ell)}} \right] \\ & \geq y_i^{(\ell)} \log \left(\frac{e N_i^{(\ell)}}{y_i^{(\ell)}} \right) - \frac{1}{2} \log(y_i^{(\ell)}) - \frac{1}{12 y_i^{(\ell)}} - \log(4\sqrt{2\pi}) \\ & \quad + y_i^{(\ell)} \log(p_i^{(\ell)}) + (N_i^{(\ell)} - y_i^{(\ell)}) \log(1 - p_i^{(\ell)}) \\ & = y_i^{(\ell)} \log \left(\frac{e N_i^{(\ell)} p_i^{(\ell)}}{(1 - p_i^{(\ell)}) y_i^{(\ell)}} \right) + N_i^{(\ell)} \log(1 - p_i^{(\ell)}) - \frac{1}{2} \log(y_i^{(\ell)}) - \frac{1}{12 y_i^{(\ell)}} - \log(4\sqrt{2\pi}). \end{aligned}$$

Moreover, we know that

$$\begin{aligned} y_i^{(\ell)} \log \left(\frac{e N_i^{(\ell)} p_i^{(\ell)}}{(1 - p_i^{(\ell)}) y_i^{(\ell)}} \right) &= (1 - o(1)) g_N \cdot q_N \cdot \tau_i^{(\ell)} \log \left(\frac{e \alpha_i^{(\ell)} \rho_i^{(\ell)}}{\tau_i^{(\ell)}} \right), \\ N_i^{(\ell)} \log(1 - p_i^{(\ell)}) &= -(1 + o(1)) g_N \cdot q_N \cdot \alpha_i^{(\ell)} \rho_i^{(\ell)}, \quad \left(\log(1 - x) = -x \text{ for } x = o(1) \right), \\ \frac{1}{2} \log(y_i^{(\ell)}) - \frac{1}{12 y_i^{(\ell)}} - \log(4\sqrt{2\pi}) &= o(g_N \cdot q_N). \end{aligned}$$

Consequently, by taking logarithm on both side of (A.1.24), we have

$$\log \mathbb{P}(\mathbf{X}_N \leq x_N) \geq -(1 + o(1)) g_N \cdot q_N \left(\sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \left[\alpha_i^{(\ell)} \rho_i^{(\ell)} - \tau_i^{(\ell)} \log \left(\frac{e \alpha_i^{(\ell)} \rho_i^{(\ell)}}{\tau_i^{(\ell)}} \right) \right] \right).$$

Plugging the facts $\tau_i^{(\ell)} = \alpha_i^{(\ell)} \rho_i^{(\ell)} e^{-c_i^{(\ell)} t^*}$ and $\sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} c_i^{(\ell)} \alpha_i^{(\ell)} \rho_i^{(\ell)} e^{-c_i^{(\ell)} t^*} = \delta$, we have

$$\begin{aligned} & \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \left[\alpha_i^{(\ell)} \rho_i^{(\ell)} - \tau_i^{(\ell)} \log \left(\frac{e \alpha_i^{(\ell)} \rho_i^{(\ell)}}{\tau_i^{(\ell)}} \right) \right] = \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} \rho_i^{(\ell)} \left[1 - (1 + c_i^{(\ell)} t^*) e^{-c_i^{(\ell)} t^*} \right] \\ &= - \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} \rho_i^{(\ell)} c_i^{(\ell)} e^{-c_i^{(\ell)} t^*} \cdot t^* + \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} \rho_i^{(\ell)} \cdot (1 - e^{-c_i^{(\ell)} t^*}) \\ &= -\delta t^* + \sum_{\ell \in \mathcal{L}} \sum_{i=1}^{n_\ell} \alpha_i^{(\ell)} \rho_i^{(\ell)} (1 - e^{-c_i^{(\ell)} t^*}) = f(t^*) = \mathbb{D}. \end{aligned}$$

Therefore, by taking infimum of both sides, we have

$$\liminf_{N \rightarrow \infty} \frac{\log \mathbb{P}(X_N \leq x_N)}{g_N \cdot q_N} \geq -\mathbb{D},$$

which complete the proof for the desired lower bound. $\square$

A.2 Proofs in Section 2.4

A.2.1 Preliminary results

The analysis of Theorem 2.2.4 relies heavily on Lemmas 2.4.2, A.2.1 and Theorem 3.4.2, which are the key ingredients to generalize the result from uniform hypergraphs to non-uniform hypergraphs.

Lemma A.2.1 (Row concentration). *Following the conventions in model 2.1.3, let $p_e^{(\ell)}$ denote the probability of the ℓ -hyperedge e being sampled. Denote*

$$p^{(\ell)} = \max_{e \in [\mathcal{V}]^\ell} p_e^{(\ell)}, \quad \rho_N := N \sum_{\ell \in \mathcal{L}} p^{(\ell)} \binom{N-2}{\ell-2}. \quad (\text{A.2.1})$$

Denote $(1 \vee x) = \max\{1, x\}$. For some constant $c > 0$, define the function

$$\varphi(x) = (5 + c) \cdot [1 \vee \log(1/x)]^{-1}. \quad (\text{A.2.2})$$

For any non-zero $\mathbf{w} \in \mathbb{R}^N$, define the map $\zeta(\mathbf{w}, c) : \mathbb{R}^N \times \mathbb{R} \mapsto \mathbb{R}$ as

$$\zeta(\mathbf{w}, c) = \varphi\left(\frac{\|\mathbf{w}\|_2}{\sqrt{N}\|\mathbf{w}\|_\infty}\right) \cdot \|\mathbf{w}\|_\infty \rho_N. \quad (\text{A.2.3})$$

Then for any $v \in \mathcal{V}$ and $\mathbf{w} \in \mathbb{R}^N$ independent of $(\mathbf{A} - \mathbf{A}^*)_{v\cdot}$, when $\rho_N \gtrsim \log(N)$, the following event holds with probability at least $1 - 2N^{-3-c}$

$$|(\mathbf{A} - \mathbf{A}^*)_{v\cdot} \mathbf{w}| \leq \zeta(\mathbf{w}, c). \quad (\text{A.2.4})$$

Proof of Lemma A.2.1. Without loss of generality, we assume $\|\mathbf{w}\|_\infty = 1$ since rescaling $\mathbf{w}$ does not change the event (A.2.4). Let $[\mathcal{V}]^\ell$ denote the family of ℓ -subsets of V , i.e., $[\mathcal{V}]^\ell = \{\mathcal{T} | \mathcal{T} \subseteq \mathcal{V}, |\mathcal{T}| = \ell\}$. By definition (2.1.4) of the adjacency matrix $\mathbf{A}$, we have

$$\mathbf{S}_v := (\mathbf{A} - \mathbf{A}^*)_{v\cdot} \mathbf{w} = \sum_{j=1}^{|\mathcal{V}|} (\mathbf{A}_{vj} - \mathbb{E} \mathbf{A}_{vj}) \mathbf{w}_j = \sum_{j=1}^{|\mathcal{V}|} \mathbf{w}_j \mathbb{1}_{\{v \neq j\}} \cdot \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in [\mathcal{V}]^\ell \\ e \supset \{v, j\}}} (\mathcal{A}_e^{(\ell)} - \mathbb{E} \mathcal{A}_e^{(\ell)})$$

Due to the independence between hyperedges and Markov inequality, for any $t > 0$, we have

$$\mathbb{P}(\mathbf{S}_v \geq \delta) = \mathbb{P}(e^{t\mathbf{S}_v} \geq e^{t\delta}) \leq e^{-t\delta} \prod_{j=1, j \neq v}^{|\mathcal{V}|} \prod_{\ell \in \mathcal{L}} \prod_{\substack{e \in [\mathcal{V}]^\ell \\ e \supset \{v, j\}}} \mathbb{E} \exp[t \mathbf{w}_j (\mathcal{A}_e^{(\ell)} - \mathbb{E} \mathcal{A}_e^{(\ell)})]. \quad (\text{A.2.5})$$

For simplicity, let $p_e^{(\ell)}$ denote the probability of the ℓ -hyperedge e being generated, then

$$\begin{aligned} & \mathbb{E} \exp[t \mathbf{w}_j (\mathcal{A}_e^{(\ell)} - \mathbb{E} \mathcal{A}_e^{(\ell)})] \\ &= p_e^{(\ell)} e^{t \mathbf{w}_j (1 - p_e^{(\ell)})} + (1 - p_e^{(\ell)}) e^{-t \mathbf{w}_j p_e^{(\ell)}} = e^{-t \mathbf{w}_j p_e^{(\ell)}} [1 - p_e^{(\ell)} + p_e^{(\ell)} e^{t \mathbf{w}_j}] \\ &\leq \exp(-t \mathbf{w}_j p_e^{(\ell)} + p_e^{(\ell)} (e^{t \mathbf{w}_j} - 1)) = \exp(-p_e^{(\ell)} (1 + t \mathbf{w}_j - e^{t \mathbf{w}_j})) \leq \exp\left(\frac{e^{t\|\mathbf{w}\|_\infty}}{2} t^2 \mathbf{w}_j^2 p_e^{(\ell)}\right) \end{aligned}$$

where we use the facts $1 + x \leq e^x \leq 1 + x + \frac{e^r}{2} x^2$ for $|x| \leq r$ in the last two inequalities. Taking

the logarithm of $\mathbb{P}(S \geq \delta)$ and plugging in $\|\mathbf{w}\|_\infty = 1$, we have

$$\log \mathbb{P}(S_v \geq \delta) \leq -t\delta + \sum_{j=1}^{|\mathcal{V}|} \mathbb{1}_{\{v \neq j\}} \cdot \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in [\mathcal{V}]^\ell \\ e \supset \{v, j\}}} \frac{e^{t\|\mathbf{w}\|_\infty}}{2} t^2 w_j^2 p_e^{(\ell)} \quad (\text{A.2.6})$$

$$\leq -t\delta + \frac{e^t}{2} t^2 \sum_{\ell \in \mathcal{L}} p^{(\ell)} \sum_{\substack{e \in [\mathcal{V}]^\ell \\ e \supset \{v, j\}}} \sum_{j=1, j \neq v}^{|\mathcal{V}|} w_j^2 \quad (\text{A.2.7})$$

$$\leq -t\delta + \frac{e^t}{2} t^2 \|\mathbf{w}\|_2^2 \cdot \sum_{\ell \in \mathcal{L}} p^{(\ell)} \binom{N-2}{\ell-2} \leq -t\delta + \frac{e^t}{2} t^2 \|\mathbf{w}\|_2^2 \cdot \rho_N, \quad (\text{A.2.8})$$

where the last two lines hold since the cardinality of ℓ -hyperedges with $e \supset \{v, j\}$ is at most $\binom{N-2}{\ell-2}$ and $\sum_{j=1, j \neq v}^{|\mathcal{V}|} w_j^2 \leq \|\mathbf{w}\|_2^2$ with the definition of ρ_N (2.4.4). We take $t = 1 \vee \log(\sqrt{N}/\|\mathbf{w}\|_2)$ and note that $\log(\sqrt{N}/\|\mathbf{w}\|_2) \geq 0$ since $\|\mathbf{w}\|_2 \leq \sqrt{N}\|\mathbf{w}\|_\infty = \sqrt{N}$, then $t \leq 1 + \log(\sqrt{N}/\|\mathbf{w}\|_2)$. Also, $(1 \vee \log x)^2 \leq x$ for $x \geq 1$, then

$$\frac{e^t}{2} t^2 \|\mathbf{w}\|_2^2 \leq \frac{e\sqrt{N}}{2} t^2 \|\mathbf{w}\|_2 = \frac{e\sqrt{N}}{2} \|\mathbf{w}\|_2 \left[1 \vee \log \left(\frac{\sqrt{N}}{\|\mathbf{w}\|_2} \right) \right]^2 \leq \frac{en}{2}. \quad (\text{A.2.9})$$

To conclude the result, for some constant $c > 0$, we take

$$\delta = \varphi \left(\frac{\|\mathbf{w}\|_2}{\sqrt{N}} \right) \cdot N \cdot \sum_{\ell \in \mathcal{L}} p^{(\ell)} \binom{N-2}{\ell-2} = \varphi \left(\frac{\|\mathbf{w}\|_2}{\sqrt{N}} \right) \rho_N. \quad (\text{A.2.10})$$

Therefore,

$$\log \mathbb{P}(S_v \geq \delta) \leq \left(-(5+c) + \frac{e}{2} \right) \rho_N \leq -(3+c) \rho_N \lesssim -(3+c) q_N. \quad (\text{A.2.11})$$

By replacing $\mathbf{w}$ with $-\mathbf{w}$, the similar bound for lower tail could be obtained. $\square$

A.2.2 Leave-one-out analysis

Leave-one-out analysis is a powerful tool in the area of random matrix theory when analyzing the resolvent of matrices by applying Schur complement formula [26]. We adapt

this idea to our problem. For each $\ell \in \mathcal{L}$, define N auxiliary tensors $\mathcal{A}^{(\ell, -1)}, \dots, \mathcal{A}^{(\ell, -N)}$, where $\mathcal{A}^{(\ell, -v)} \in (\mathbb{R}^{|\mathcal{V}|})^{\otimes \ell}$ is obtained by zeroing out the hyperedges containing v . Let $\mathbf{A}^{(\ell, -v)}$ denote the adjacency matrix obtained from $\mathcal{A}^{(\ell, -v)}$, and define

$$\mathbf{A}^{(-v)} = \sum_{\ell \in \mathcal{L}} \mathbf{A}^{(\ell, -v)}. \quad (\text{A.2.12})$$

The rest of this section is devoted to presenting results concerning $\mathbf{A}, \mathbf{A}^*$ and $\mathbf{A}^{(-v)}$, in preparation for the proof of Proposition 2.4.1 in the next subsection.

Lemma A.2.2. *Deterministically, $\|\mathbf{A}^*\|_{2 \rightarrow \infty} \lesssim \rho_N / \sqrt{N}$. With probability at least $1 - O(N^{-10})$,*

$$\|\mathbf{A}\|_{2 \rightarrow \infty} \lesssim \sqrt{\rho_N} + \rho_N / \sqrt{N}. \quad (\text{A.2.13})$$

For each $v \in \mathcal{V}$, the following holds with probability at least $1 - O(N^{-10})$

$$\|\mathbf{A} - \mathbf{A}^{(-v)}\| \lesssim \|\mathbf{A}\|_{2 \rightarrow \infty}, \quad \|\mathbf{A}^{(-v)} - \mathbf{A}^*\| \lesssim 2\sqrt{\rho_N} + \rho_N / \sqrt{N}. \quad (\text{A.2.14})$$

Proof of Lemma A.2.2. Let $p^{(\ell)} = \max_e p_e^{(\ell)}$, then $\mathbf{A}_{ij}^* = \mathbb{E}[\mathbf{A}_{ij}] \leq \sum_{\ell \in \mathcal{L}} p^{(\ell)} \binom{N-2}{\ell-2}$ for each $i \neq j$. According to definition of ρ_N in (2.4.4), we have $\rho_N \asymp N \sum_{\ell \in \mathcal{L}} p^{(\ell)} \binom{N-2}{\ell-2}$. The proof of the first sentence is finished by definition, since

$$\|\mathbf{A}^*\|_{2 \rightarrow \infty} = \max_{\|\mathbf{x}\|_2=1} \|\mathbf{A}^* \mathbf{x}\|_\infty = \max_{i \in \mathcal{V}} \|\mathbf{A}_{i:}^*\|_2 \leq \sqrt{N} \sum_{\ell \in \mathcal{L}} p^{(\ell)} \binom{N-2}{\ell-2} \asymp \frac{\rho_N}{\sqrt{N}}. \quad (\text{A.2.15})$$

For the second argument, note that $\|\mathbf{A}\|_{2 \rightarrow \infty} \leq \|\mathbf{A}\|$, then by Theorem 3.4.2, the following hold with probability at least $1 - O(N^{-10})$

$$\|\mathbf{A}\|_{2 \rightarrow \infty} \leq \|\mathbf{A} - \mathbf{A}^*\|_{2 \rightarrow \infty} + \|\mathbf{A}^*\|_{2 \rightarrow \infty} \leq \|\mathbf{A} - \mathbf{A}^*\| + \|\mathbf{A}^*\|_{2 \rightarrow \infty} \lesssim \sqrt{\rho_N} + \rho_N / \sqrt{N}. \quad (\text{A.2.16})$$

For the last part, recall the definition of $\mathbf{A}^{(-v)}$ in (A.2.12), then $(\mathbf{A} - \mathbf{A}^{(-v)})_{v:} = \mathbf{A}_{v:}$ while for

$i \neq v$,

$$(\mathbf{A} - \mathbf{A}^{(-v)})_{ij} = \mathbb{1}_{\{i \neq j\}} \cdot \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ e \ni v, e \supset \{i, j\}}} \mathcal{A}_e^{(\ell)}. \quad (\text{A.2.17})$$

Also, the hyperedge $e \supset \{i, v\}$ is counted $\ell - 1$ times in i th row, then

$$\|(\mathbf{A} - \mathbf{A}^{(-v)})_{i:}\|_2 \leq \|(\mathbf{A} - \mathbf{A}^{(-v)})_{i:}\|_1 = \sum_{\ell \in \mathcal{L}} (\ell - 1) \mathbf{A}_{iv}. \quad (\text{A.2.18})$$

Since $\mathcal{L}$ is finite, then

$$\begin{aligned} \|\mathbf{A} - \mathbf{A}^{(-v)}\|^2 &\leq \|\mathbf{A} - \mathbf{A}^{(-v)}\|_{\text{F}}^2 = \sum_{i=1}^{|\mathcal{V}|} \|(\mathbf{A} - \mathbf{A}^{(-v)})_{i:}\|_2^2 = \|\mathbf{A}_{v:}\|_2^2 + \sum_{i \neq v} \|(\mathbf{A} - \mathbf{A}^{(-v)})_{i:}\|_2^2 \\ &\leq \|\mathbf{A}_{v:}\|_2^2 + \sum_{i \neq v} \left(\sum_{\ell \in \mathcal{L}} (\ell - 1) \mathbf{A}_{iv} \right)^2 = \|\mathbf{A}_{v:}\|_2^2 + \left(\sum_{\ell \in \mathcal{L}} (\ell - 1) \right)^2 \cdot \sum_{i \neq v} \mathbf{A}_{iv}^2 \\ &\leq \left(\left(\sum_{\ell \in \mathcal{L}} (\ell - 1) \right)^2 + 1 \right) \cdot \|\mathbf{A}_{v:}\|_2^2 \lesssim \|\mathbf{A}\|_{2 \rightarrow \infty}. \end{aligned}$$

Consequently, with probability at least $1 - O(N^{-10})$,

$$\|\mathbf{A}^{(-v)} - \mathbf{A}^*\| \leq \|\mathbf{A}^{(-v)} - \mathbf{A}\| + \|\mathbf{A} - \mathbf{A}^*\| \lesssim \|\mathbf{A}\|_{2 \rightarrow \infty} + \|\mathbf{A} - \mathbf{A}^*\| \lesssim \rho_N / \sqrt{N} + 2\sqrt{\rho_N}.$$

□

Let $\{\lambda_i\}_{i=1}^{|\mathcal{V}|}$, $\{\lambda_i^{(-v)}\}_{i=1}^{|\mathcal{V}|}$ and $\{\lambda_i^*\}_{i=1}^{|\mathcal{V}|}$ denote the eigenvalues of $\mathbf{A}$, $\mathbf{A}^{(-v)}$ and $\mathbf{A}^*$ respectively, where the eigenvalues are sorted in the decreasing order.

Corollary A.2.3. *For each $i \in \mathcal{V}$, the following holds with probability at least $1 - O(N^{-10})$,*

$$|\lambda_i - \lambda_i^{(-v)}| \lesssim \sqrt{\rho_N} + \rho_N / \sqrt{N}, \quad |\lambda_i^* - \lambda_i^{(-v)}| \lesssim 2\sqrt{\rho_N} + \rho_N / \sqrt{N}, \quad |\lambda_i - \lambda_i^*| \lesssim \sqrt{\rho_N}.$$

In particular in our model 2.1.3, with probability at least $1 - O(N^{-10})$,

$$|\lambda_i| \asymp |\lambda_i^*| \asymp |\lambda_i^{(-v)}| \asymp \rho_N, \quad i = 1, 2.$$

Proof of Corollary A.2.3. By Weyl's inequality D.2.1 and Lemma A.2.2, for any $i \in \mathcal{V}$,

$$\begin{aligned} |\lambda_i - \lambda_i^{(-v)}| &\leq \|\mathbf{A} - \mathbf{A}^{(-v)}\| \lesssim \sqrt{\rho_N} + \rho_N/\sqrt{N}, \quad |\lambda_i - \lambda_i^*| \leq \|\mathbf{A} - \mathbf{A}^*\| \lesssim \sqrt{\rho_N}, \\ |\lambda_i^* - \lambda_i^{(-v)}| &\leq \|\mathbf{A}^* - \mathbf{A}^{(-v)}\| \lesssim 2\sqrt{\rho_N} + \rho_N/\sqrt{N}. \end{aligned}$$

For the second part, $\rho_N \gtrsim \log(N)$, then the result follows since $\sqrt{\rho_N} \ll \rho_N$ and $\rho_N/\sqrt{N} \ll \rho_N$. $\square$

Lemma A.2.4. Under model (2.1.3), the following holds with probability at least $1 - O(N^{-10})$,

$$\max_{v \in \mathcal{V}} \|s^{(-v)} \mathbf{u}^{(-v)} - s\mathbf{u}\|_2 \lesssim \|\mathbf{u}\|_\infty, \quad (\text{A.2.19a})$$

$$\max_{v \in \mathcal{V}} \|s^{(-v)} \mathbf{u}^{(-v)} - \mathbf{u}^*\|_2 \lesssim \frac{1}{\sqrt{\rho_N}} + \frac{1}{\sqrt{N}}, \quad (\text{A.2.19b})$$

$$\max_{v \in \mathcal{V}} \|s^{(-v)} \mathbf{u}^{(-v)} - \mathbf{u}^*\|_\infty \lesssim \|\mathbf{u}\|_\infty + \frac{1}{\sqrt{N}}. \quad (\text{A.2.19c})$$

Proof of Lemma A.2.4. First, let $\delta = \min_{i \neq 2} |\lambda_i^{(-v)} - \lambda|$ denote the absolute eigen gap, then by generalized Davis-Kahan Lemma D.2.4 where we take $\mathbf{D} = \mathbf{I}_N$, $\Lambda = \lambda_2^{(-v)}$, $\mathbf{U} = s^{(-v)} \mathbf{u}^{(-v)}$, $\mathbf{P} = \mathbf{I}_N - \mathbf{U}\mathbf{U}^\top$, $\hat{\lambda} = \lambda$, $\hat{\mathbf{u}} = s\mathbf{u}$, we have

$$\|s^{(-v)} \mathbf{u}^{(-v)} - s\mathbf{u}\|_2 \leq \sqrt{2} \|\mathbf{P}\mathbf{u}\|_2 \leq \frac{\sqrt{2} \|(\mathbf{A}^{(-v)} - \lambda \mathbf{I}_N) \mathbf{u}\|_2}{\delta} = \frac{\sqrt{2} \|(\mathbf{A}^{(-v)} - \mathbf{A}) \mathbf{u}\|_2}{\delta} \quad (\text{A.2.20})$$

By Weyl's inequality Lemma D.2.1 and Corollary A.2.3, the following holds with probability at

least $1 - O(N^{-10})$,

$$\begin{aligned}
\delta &= \min_{i \neq 2} |\lambda_i^{(-v)} - \lambda| = \min \{ |\lambda_3^{(-v)} - \lambda|, |\lambda_1^{(-v)} - \lambda| \} \\
&\geq \min \{ |\lambda_3 - \lambda|, |\lambda_1 - \lambda| \} - \|\mathbf{A} - \mathbf{A}^{(-v)}\| \\
&\geq \min \{ |\lambda_3^* - \lambda^*|, |\lambda_1^* - \lambda^*| \} - 2\|\mathbf{A} - \mathbf{A}^*\| - \|\mathbf{A} - \mathbf{A}^{(-v)}\| \gtrsim \rho_N,
\end{aligned} \tag{A.2.21}$$

where the last line holds by Theorem 3.4.2 and Lemma A.2.2. Let $\mathbf{w} = (\mathbf{A} - \mathbf{A}^{(-v)})\mathbf{u}$. Note that

$$\|(\mathbf{A} - \mathbf{A}^{(-v)})_{i\cdot}\|_1 = \sum_{\ell \in \mathcal{L}} (\ell - 1)A_{i\ell} \text{ for } i \neq v, \text{ then}$$

$$\begin{aligned}
|\mathbf{w}_v| &= |(\mathbf{A}\mathbf{u})_v| = |\lambda| \cdot |\mathbf{u}_v| \leq |\lambda| \cdot \|\mathbf{u}\|_\infty, \\
|\mathbf{w}_i| &= |[(\mathbf{A} - \mathbf{A}^{(-v)})\mathbf{u}]_i| \leq \|(\mathbf{A} - \mathbf{A}^{(-v)})_{i\cdot}\|_1 \cdot \|\mathbf{u}\|_\infty = \sum_{\ell \in \mathcal{L}} (\ell - 1)A_{i\ell} \cdot \|\mathbf{u}\|_\infty, \quad i \neq v.
\end{aligned}$$

Again by Lemmas A.2.2, A.2.3, with probability at least $1 - O(N^{-10})$,

$$\begin{aligned}
\|(\mathbf{A} - \mathbf{A}^{(-v)})\mathbf{u}\|_2^2 &\leq \left(|\lambda|^2 + \sum_{i \neq v} \left(\sum_{\ell \in \mathcal{L}} (\ell - 1)A_{i\ell} \right)^2 \right) \cdot \|\mathbf{u}\|_\infty^2 \\
&\lesssim (|\lambda^*|^2 + \|\mathbf{A}\|_{2 \rightarrow \infty}) \cdot \|\mathbf{u}\|_\infty^2 \lesssim \rho_N^2 \cdot \|\mathbf{u}\|_\infty^2.
\end{aligned}$$

Therefore, by plugging in the lower bound of δ (A.2.21) into (A.2.20), we have

$$\|s^{(-v)}\mathbf{u}^{(-v)} - s\mathbf{u}\|_2 \lesssim \|\mathbf{u}\|_\infty.$$

For the second part, $s^{(-v)}(\mathbf{u}^{(-v)})^\top \mathbf{u}^* \geq 0$ by (A.2.30). Let $\delta^* = \min\{|\lambda_3^* - \lambda_2^*|, |\lambda_1^* - \lambda_2^*|\} \asymp \rho_N$, then by Davis-Kahan Lemma D.2.3 and Lemma A.2.2, for any $v \in \mathcal{V}$, the following holds with probability at least $1 - O(N^{-10})$,

$$\|s^{(-v)}\mathbf{u}^{(-v)} - \mathbf{u}^*\|_2 \leq \frac{4\|\mathbf{A}^{(-v)} - \mathbf{A}^*\|}{\delta^*} \lesssim \frac{2\sqrt{\rho_N} + \frac{\rho_N}{\sqrt{N}}}{\rho_N} \lesssim \frac{1}{\sqrt{\rho_N}} + \frac{1}{\sqrt{N}},$$

Note that $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2$ for any $\mathbf{x} \in \mathbb{R}^N$, then

$$\begin{aligned} \|s^{(-v)}\mathbf{u}^{(-v)} - \mathbf{u}^\star\|_\infty &\leq \|s^{(-v)}\mathbf{u}^{(-v)} - s\mathbf{u}\|_\infty + \|\mathbf{u}\|_\infty + \|\mathbf{u}^\star\|_\infty \\ &\leq \|s^{(-v)}\mathbf{u}^{(-v)} - s\mathbf{u}\|_2 + \|\mathbf{u}\|_\infty + \|\mathbf{u}^\star\|_\infty \lesssim \|\mathbf{u}\|_\infty + \|\mathbf{u}^\star\|_\infty, \end{aligned}$$

and the desired result follows since $\|\mathbf{u}^\star\|_\infty = 1/\sqrt{N}$ by definition of our model. $\square$

Lemma A.2.5. *The following holds with probability at least $1 - O(N^{-3})$,*

$$\max_{v \in \mathcal{V}} |\mathbf{A}_{v:}(s^{(-v)}\mathbf{u}^{(-v)} - \mathbf{u}^\star)| \lesssim \frac{\rho_N}{\log(\rho_N)} \cdot \left(\|\mathbf{u}\|_\infty + \frac{1}{\sqrt{N}} \right).$$

Proof of Lemma A.2.5. By definition of $\mathbf{A}^{(-v)}$ in (A.2.12), the vector $\mathbf{u}^{(-v)}$ and the row $\mathbf{A}_{v:}$ are independent. Also, $\mathbf{u}^\star$ is independent of $\mathbf{A}$ as well since it is a constant vector. For ease of presentation, we denote $\mathbf{w}^{(-v)} := s^{(-v)}\mathbf{u}^{(-v)} - \mathbf{u}^\star$. Then $\mathbf{w}^{(-v)}$ is independent of $\mathbf{A}_{v:}$. By triangle inequality,

$$|\mathbf{A}_{v:}\mathbf{w}^{(-v)}| \leq \underbrace{|\mathbf{A}_{v:}^\star \mathbf{w}^{(-v)}|}_{\textcircled{1}} + \underbrace{|(\mathbf{A}_{v:} - \mathbf{A}_{v:}^\star) \mathbf{w}^{(-v)}|}_{\textcircled{2}}. \quad (\text{A.2.22})$$

The first term $\textcircled{1}$ can be bounded by Lemmas A.2.2, A.2.4 as follows

$$\textcircled{1} \leq \|\mathbf{A}^\star\|_{2 \rightarrow \infty} \cdot \|\mathbf{w}^{(-v)}\|_2 \leq \frac{\rho_N}{\sqrt{N}} \|\mathbf{u}\|_\infty. \quad (\text{A.2.23})$$

The second term $\textcircled{2}$ can be bounded by row concentration Lemma A.2.1,

$$|(\mathbf{A}_{v:} - \mathbf{A}_{v:}^\star) \mathbf{w}^{(-v)}| \leq \zeta(\mathbf{w}, c) = \varphi\left(\frac{\|\mathbf{w}^{(-v)}\|_2}{\sqrt{N}\|\mathbf{w}^{(-v)}\|_\infty}\right) \cdot \|\mathbf{w}^{(-v)}\|_\infty \rho_N. \quad (\text{A.2.24})$$

By the union bound, the event (A.2.4) holds simultaneously for all $v \in \mathcal{V}$ with probability at

least $1 - O(N^{-2})$. Define

$$\gamma_N = \frac{C_{(3.4.3)}}{\sqrt{\rho_N}}. \quad (\text{A.2.25})$$

We present the discussions based on the magnitude of $\frac{\|\mathbf{w}^{(-v)}\|_2}{\sqrt{N}\|\mathbf{w}^{(-v)}\|_\infty}$. Recall function $\varphi(x)$ in (A.2.2).

- $\frac{\|\mathbf{w}^{(-v)}\|_2}{\sqrt{N}\|\mathbf{w}^{(-v)}\|_\infty} \leq \gamma_N$. Note that $\varphi(x) : \mathbb{R}_+ \mapsto \mathbb{R}_+$ is non-decreasing, then

$$|(\mathbf{A}_{v:} - \mathbf{A}_{v:}^*)\mathbf{w}^{(-v)}| \leq \varphi(\gamma_N) \cdot \|\mathbf{w}^{(-v)}\|_\infty \cdot \rho_N. \quad (\text{A.2.26})$$

- $\frac{\|\mathbf{w}^{(-v)}\|_2}{\sqrt{N}\|\mathbf{w}^{(-v)}\|_\infty} > \gamma_N$. Note that $\varphi(x)/x : \mathbb{R}_+ \mapsto \mathbb{R}_+$ is non-increasing, then $\varphi(x) \leq x\varphi(\gamma_N)/\gamma_N$ for any $x > \gamma_N$, thus

$$|(\mathbf{A}_{v:} - \mathbf{A}_{v:}^*)\mathbf{w}^{(-v)}| \leq \frac{\varphi(\gamma_N)}{\gamma_N} \cdot \frac{\|\mathbf{w}^{(-v)}\|_2}{\sqrt{N}} \cdot \rho_N \quad (\text{A.2.27})$$

Combining the bounds above and Lemma A.2.4, for each $v \in \mathcal{V}$, the second term can be bounded as

$$\begin{aligned} \textcircled{2} &= |(\mathbf{A}_{v:} - \mathbf{A}_{v:}^*)\mathbf{w}^{(-v)}| \leq \varphi(\gamma_N)\rho_N \cdot \max\left\{\|\mathbf{w}^{(-v)}\|_\infty, \frac{\|\mathbf{w}^{(-v)}\|_2}{\gamma_N\sqrt{N}}\right\} \\ &\lesssim \varphi(\gamma_N)\rho_N \cdot \max\left\{\|\mathbf{u}\|_\infty + \|\mathbf{u}^*\|_\infty, \frac{1}{\sqrt{N}}\right\} \lesssim \frac{\rho_N}{\log(\rho_N)} \cdot (\|\mathbf{u}\|_\infty + \|\mathbf{u}^*\|_\infty), \end{aligned}$$

where the last inequality holds since $\sqrt{N}\|\mathbf{u}\|_\infty \geq \|\mathbf{u}\|_2 = 1$ and $\varphi(\gamma_N) \asymp [1 \vee \log(\rho_N)]^{-1}$. Consequently, combining the terms $\textcircled{1}$ and $\textcircled{2}$, we have

$$\max_{v \in \mathcal{V}} |\mathbf{A}_{v:}(s^{(-v)}\mathbf{u}^{(-v)} - \mathbf{u}^*)| \lesssim \left(\frac{\rho_N}{\sqrt{N}} + \frac{\rho_N}{\log(\rho_N)}\right) \cdot (\|\mathbf{u}\|_\infty + \|\mathbf{u}^*\|_\infty),$$

and the desired result follows since $\|\mathbf{u}^*\|_\infty = 1/\sqrt{N}$. $\square$

Lemma A.2.6. *With probability at least $1 - O(N^{-2})$,*

$$\|\mathbf{A}\mathbf{u}^*\|_\infty \lesssim \frac{\rho_N}{\sqrt{N}}. \quad (\text{A.2.28})$$

Proof of Lemma A.2.6. We first prove $\|(\mathbf{A} - \mathbf{A}^*)\mathbf{u}^*\|_\infty \lesssim \rho_N/\sqrt{N}$ as an intermediate result. By row concentration Lemma A.2.1, for each $v \in \mathcal{V}$, with probability at least $1 - O(N^{-3})$,

$$|(\mathbf{A} - \mathbf{A}^*)_{v:} \mathbf{u}^*| \leq \varphi\left(\frac{\|\mathbf{u}^*\|_2}{\sqrt{N}\|\mathbf{u}^*\|_\infty}\right) \cdot \|\mathbf{u}^*\|_\infty \rho_N \lesssim \frac{\rho_N}{\sqrt{N}},$$

where the last equality holds since $\|\mathbf{u}^*\|_2 = 1$, $\|\mathbf{u}^*\|_\infty = 1/\sqrt{N}$ and $\varphi(1) = 5 + c = O(1)$ for some constant $c > 0$. Then it follows by the union bound. Note that $\|\mathbf{A}^*\|_{2 \rightarrow \infty} \lesssim \rho_N/\sqrt{N}$ by Lemma A.2.2, then by triangle inequality, with probability at least $1 - O(N^{-2})$,

$$\|\mathbf{A}\mathbf{u}^*\|_\infty \leq \|(\mathbf{A} - \mathbf{A}^*)\mathbf{u}^*\|_\infty + \|\mathbf{A}^*\mathbf{u}^*\|_\infty \lesssim \frac{\rho_N}{\sqrt{N}} + \|\mathbf{A}^*\|_{2 \rightarrow \infty} \|\mathbf{u}^*\|_2 \lesssim \frac{\rho_N}{\sqrt{N}}.$$

□

A.2.3 Proof of Proposition 2.4.1

Proof of Proposition 2.4.1. Recall that $\{(\lambda_i, \mathbf{u}_i)\}_{i=1}^{|\mathcal{V}|}$, $\{(\lambda_i^*, \mathbf{u}_i^*)\}_{i=1}^{|\mathcal{V}|}$ and $\{(\lambda_i^{(-v)}, \mathbf{u}_i^{(-v)})\}_{i=1}^{|\mathcal{V}|}$ are eigenpairs of $\mathbf{A}$, $\mathbf{A}^*$ and $\mathbf{A}^{(-v)}$ respectively, with the eigenvalues sorted in the decreasing order. We are in particular interested in the second eigenpairs. For notation convenience, we denote $\lambda = \lambda_2$, $\lambda^* = \lambda_2^*$, $\lambda^{(-v)} = \lambda_2^{(-v)}$ and $\mathbf{u} = \mathbf{u}_2$, $\mathbf{u}^* = \mathbf{u}_2^*$, $\mathbf{u}^{(-v)} = \mathbf{u}_2^{(-v)}$. We first prove that the eigenvector $\mathbf{u}$ is delocalized, i.e.,

$$\|\mathbf{u}\|_\infty \lesssim 1/\sqrt{N}. \quad (\text{A.2.29})$$

Note that $\langle \mathbf{u}^*, \mathbf{u}^* \rangle = 1 > 0$. Define the following sign functions for $\mathbf{u}$ and $\mathbf{u}^{(-\nu)}$,

$$s = \text{sign}(\langle \mathbf{u}, \mathbf{u}^* \rangle), \quad s^{(-\nu)} = \text{sign}(\langle \mathbf{u}^{(-\nu)}, \mathbf{u}^* \rangle), \quad (\text{A.2.30})$$

Then $s\mathbf{u}^\top \mathbf{u}^* \geq 0$ and $s^{(-\nu)}(\mathbf{u}^{(-\nu)})^\top \mathbf{u}^* \geq 0$. Meanwhile $(\lambda^*)^{-1} \asymp (\lambda)^{-1}$ by Corollary A.2.3, then

$$\begin{aligned} \|\mathbf{u}\|_\infty &= \left\| s \frac{\mathbf{A}\mathbf{u}}{\lambda} \right\|_\infty \leq \left\| \frac{\mathbf{A}(s\mathbf{u} - \mathbf{u}^*)}{\lambda} \right\|_\infty + \left\| \frac{\mathbf{A}\mathbf{u}^*}{\lambda} \right\|_\infty \\ &\lesssim |\lambda^*|^{-1} \left(\max_{v \in \mathcal{V}'} |\mathbf{A}_v: (s\mathbf{u} - \mathbf{u}^*)| + \|\mathbf{A}\mathbf{u}^*\|_\infty \right) \\ &\leq |\lambda^*|^{-1} \left(\max_{v \in \mathcal{V}'} |\mathbf{A}_v: (s\mathbf{u} - s^{(-\nu)}\mathbf{u}^{(-\nu)})| + \max_{v \in \mathcal{V}'} |\mathbf{A}_v: (s^{(-\nu)}\mathbf{u}^{(-\nu)} - \mathbf{u}^*)| + \|\mathbf{A}\mathbf{u}^*\|_\infty \right) \\ &\leq |\lambda^*|^{-1} \left(\|\mathbf{A}\|_{2 \rightarrow \infty} \|s\mathbf{u} - s^{(-\nu)}\mathbf{u}^{(-\nu)}\|_2 + \max_{v \in \mathcal{V}'} |\mathbf{A}_v: (s^{(-\nu)}\mathbf{u}^{(-\nu)} - \mathbf{u}^*)| + \|\mathbf{A}\mathbf{u}^*\|_\infty \right) \\ &\lesssim \rho_N^{-1} \left[(\sqrt{\rho_N} + \frac{\rho_N}{\sqrt{N}}) \|\mathbf{u}\|_\infty + \frac{\rho_N}{\log(\rho_N)} \cdot \left(\|\mathbf{u}\|_\infty + \frac{1}{\sqrt{N}} \right) + \frac{\rho_N}{\sqrt{N}} \right], \end{aligned}$$

where the last inequality holds by Lemmas A.2.3, A.2.4, A.2.5, A.2.6. By rearranging the terms on both sides,

$$\left(1 - \frac{1}{\sqrt{\rho_N}} - \frac{1}{\sqrt{N}} - \frac{1}{\log(\rho_N)} \right) \cdot \|\mathbf{u}\|_\infty \lesssim \left(1 + \frac{1}{\log(\rho_N)} \right) \frac{1}{\sqrt{N}}. \quad (\text{A.2.31})$$

Then (A.2.29) follows since $\rho_N \gtrsim \log(N)$. Following the same strategy, we have

$$\left\| s\mathbf{u} - \frac{\mathbf{A}\mathbf{u}^*}{\lambda^*} \right\|_\infty = \left\| s \frac{\mathbf{A}\mathbf{u}}{\lambda} - \frac{\mathbf{A}\mathbf{u}^*}{\lambda} + \frac{\mathbf{A}\mathbf{u}^*}{\lambda} - \frac{\mathbf{A}\mathbf{u}^*}{\lambda^*} \right\|_\infty \leq \frac{\|\mathbf{A}(s\mathbf{u} - \mathbf{u}^*)\|_\infty}{|\lambda|} + \frac{|\lambda - \lambda^*|}{|\lambda| \cdot |\lambda^*|} \cdot \|\mathbf{A}\mathbf{u}^*\|_\infty,$$

where the first term can be bounded as before. For the second term, by Corollary A.2.3, one has

$$\frac{|\lambda - \lambda^*|}{|\lambda| \cdot |\lambda^*|} \lesssim \frac{\sqrt{\rho_N}}{\rho_N^2} = (\rho_N)^{-3/2}. \quad (\text{A.2.32})$$

Then it follows that with probability at least $1 - O(N^{-2})$,

$$\begin{aligned} \left\| s\mathbf{u} - \frac{\mathbf{A}\mathbf{u}^*}{\lambda^*} \right\|_\infty &\lesssim \rho_N^{-1} \left[(\sqrt{\rho_N} + \frac{\rho_N}{\sqrt{N}}) \|\mathbf{u}\|_\infty + \frac{\rho_N}{\log(\rho_N)} \cdot \left(\|\mathbf{u}\|_\infty + \frac{1}{\sqrt{N}} \right) \right] + (\rho_N)^{-3/2} \frac{\rho_N}{\sqrt{N}} \\ &\lesssim \left(\frac{1}{\sqrt{\rho_N}} + \frac{1}{\sqrt{N}} + \frac{2}{\log(\rho_N)} + \frac{1}{\sqrt{\rho_N}} \right) \cdot \frac{1}{\sqrt{N}} \lesssim \frac{1}{\sqrt{N} \cdot \log(\rho_N)}, \end{aligned}$$

where the last inequality holds since the term $[\log(\rho_N)]^{-1}$ dominates. $\square$

A.3 Analysis of spectral method via normalized graph Laplacian

This section is devoted to the performance analysis of Algorithm 2.2.1 when the input is normalized graph Laplacian $\mathbf{L} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$. We will show that the spectral method can achieve the same performance as the adjacency matrix $\mathbf{A}$ in Appendix A.2. For the sake of simplicity, we only outline the differences in the proofs.

A.3.1 Generalized eigenvalue problem

We define $D_v = \sum_{j=1}^{|\mathcal{V}|} A_{vj}$ for each $v \in \mathcal{V}$, and the diagonal matrix $\mathbf{D} = \text{diag}\{D_1, \dots, D_N\}$.

By definition, $D_v \geq 0$ for each $v \in \mathcal{V}$. We say $(\lambda, \mathbf{u})$ with $\|\mathbf{u}\|_2 = 1$ is an eigenpair of $(\mathbf{A}, \mathbf{D})$ if

$$\mathbf{A}\mathbf{u} = \lambda \mathbf{D}\mathbf{u}. \quad (\text{A.3.1})$$

With a slightly abuse of notation, let $\mathbf{D}^{-1}$ denote the *Moore-Penrose* inverse of $\mathbf{D}$, that is, $(\mathbf{D}^{-1})_{vv} = (D_v)^{-1}$ if $D_v \neq 0$, otherwise $(\mathbf{D}^{-1})_{vv} = 0$, which happens only if $A_{vj} = 0$ for each $j \in \mathcal{V}$.

The normalized Laplacian is defined as $\mathbf{L} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$, where similarly, $(\mathbf{D}^{-1/2})_{vv} = (D_v)^{-1/2}$ if $D_v \neq 0$, otherwise $(\mathbf{D}^{-1/2})_{vv} = 0$. Meanwhile,

$$\mathbf{A}\mathbf{u} = \lambda \mathbf{D}\mathbf{u} \iff \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2} \mathbf{D}^{1/2} \mathbf{u} = \lambda \mathbf{D}^{1/2} \mathbf{u} \iff \mathbf{L} \mathbf{D}^{1/2} \mathbf{u} = \lambda \mathbf{D}^{1/2} \mathbf{u}, \quad (\text{A.3.2})$$

where $\mathbf{D}^{1/2}\mathbf{u}$ and $\mathbf{u}$ share the same sign entrywisely, then it suffices to study the generalized eigenvalue problem (A.3.1) if one is interested in the spectral properties of graph Laplacian $\mathbf{L}$.

Analogously, following the ideas in Appendix A.2, the expected and the leave-one-out versions are

$$\begin{aligned} \mathbf{D}_{vv}^* &= \sum_{j=1}^{|\mathcal{V}|} A_{vj}^*, \quad \mathbf{D}^* = \text{diag}(\mathbf{D}_{11}^*, \dots, \mathbf{D}_{nn}^*), \quad \mathbf{L}^* = (\mathbf{D}^*)^{-1/2} \mathbf{A}^* (\mathbf{D}^*)^{-1/2}, \\ \mathbf{D}_{uu}^{(-v)} &= \sum_{j=1}^{|\mathcal{V}|} A_{uj}^{(-v)}, \quad \mathbf{D}^{(-v)} = \text{diag}(\mathbf{D}_{11}^{(-v)}, \dots, \mathbf{D}_{nn}^{(-v)}), \quad \mathbf{L}^{(-v)} = (\mathbf{D}^{(-v)})^{-1/2} \mathbf{A}^{(-v)} (\mathbf{D}^{(-v)})^{-1/2}. \end{aligned}$$

In this section, let $\{(\lambda_i, \mathbf{u}_i)\}_{i=1}^{|\mathcal{V}|}$, $\{(\lambda_i^*, \mathbf{u}_i^*)\}_{i=1}^{|\mathcal{V}|}$ and $\{(\lambda_i^{(-v)}, \mathbf{u}_i^{(-v)})\}_{i=1}^{|\mathcal{V}|}$ denote the eigenpairs of the following generalized eigenvalue problems, respectively

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{D}\mathbf{u}, \quad \mathbf{A}^*\mathbf{u}^* = \lambda\mathbf{D}^*\mathbf{u}, \quad \mathbf{A}^{(-v)}\mathbf{u} = \lambda\mathbf{D}^{(-v)}\mathbf{u}, \quad (\text{A.3.3})$$

where the eigenvalues are sorted in the decreasing order. We then introduce the following analogous results.

Lemma A.3.1. *Suppose that $\mathbb{D}_{\text{AM}} = 1 + \delta$ for some constant $\delta > 0$, then $D_v \geq \varepsilon \rho_N$ for each $v \in \mathcal{V}$ with probability at least $1 - N^{-\mathbb{D}_{\text{AM}} + \delta/2}$ for some constant $\varepsilon > 0$ depending on δ , α and β in (2.4.1). Furthermore, $D_{\min} = \min_{v \in \mathcal{V}} D_v \gtrsim \rho_N$.*

Proof of Lemma A.3.1. The proof follows similarly as the proof for Lemma 2.4.2. For some $\varepsilon > 0$, $\mathbb{P}(D_v < \varepsilon \rho_N) \leq \mathbb{P}(\sqrt{N} \mathbf{y}_v(\mathbf{A} \mathbf{u}_2^*)_v \leq \delta \rho_N) \leq N^{-\mathbb{D}_{\text{AM}} + \delta/2}$ when $\delta = \mathbb{D}_{\text{AM}} - 1 > 0$. $\square$

Corollary A.3.2. *Following the conventions in Theorem 3.4.2, suppose that $\mathbb{D}_{\text{AM}} = 1 + \delta$ for some constant $\delta > 0$, then with probability at least $1 - N^{-\mathbb{D}_{\text{AM}} + \delta/2}$,*

$$\|\mathbf{D}^{-1} \mathbf{A} - (\mathbf{D}^*)^{-1} \mathbf{A}^*\| \leq C_{(\text{A.3.2})} / \sqrt{\rho_N}. \quad (\text{A.3.4})$$

Proof of Corollary A.3.2. By triangle inequality

$$\begin{aligned} \|\mathbf{D}^{-1}\mathbf{A} - (\mathbf{D}^\star)^{-1}\mathbf{A}^\star\| &\leq \|\mathbf{D}^{-1}\mathbf{A} - \mathbf{D}^{-1}\mathbf{A}^\star + \mathbf{D}^{-1}\mathbf{A}^\star - (\mathbf{D}^\star)^{-1}\mathbf{A}^\star\| \\ &\leq \|\mathbf{D}^{-1}\| \cdot \|\mathbf{A} - \mathbf{A}^\star\| + \|(\mathbf{D}^{-1} - (\mathbf{D}^\star)^{-1})\mathbf{A}^\star\| \end{aligned}$$

where $\|\mathbf{D}^{-1}\| = (\mathbf{D}_{\min})^{-1} \lesssim \rho_N^{-1}$ by Lemma A.3.1 and $\|\mathbf{A} - \mathbf{A}^\star\| \lesssim \sqrt{\rho_N}$ by Theorem 3.4.2. For the second term, recall the definitions of α, β in (2.4.1) and ρ_N in (2.4.4), then with probability at least $1 - N^{-\mathbb{D}_{\text{GH}} + \delta/2}$,

$$\begin{aligned} \|(\mathbf{D}^{-1} - (\mathbf{D}^\star)^{-1})\mathbf{A}^\star\|^2 &\leq \|(\mathbf{D}^{-1} - (\mathbf{D}^\star)^{-1})\mathbf{A}^\star\|_{\text{F}}^2 \\ &\leq \sum_{i=1}^{|\mathcal{V}|} (\mathbf{D}_{ii}^{-1} - (\mathbf{D}_{ii}^\star)^{-1})^2 \sum_{j=1}^{|\mathcal{V}|} \alpha^2 \leq \frac{N\alpha^2}{\mathbf{D}_{\min}^2 \cdot (\mathbf{D}_{ii}^\star)^2} \sum_{i=1}^{|\mathcal{V}|} (\mathbf{D}_{ii} - \mathbf{D}_{ii}^\star)^2 \\ &\lesssim \frac{N\alpha^2}{\mathbf{D}_{\min}^2 \cdot (\mathbf{D}_{ii}^\star)^2} \sum_{i=1}^{|\mathcal{V}|} \sum_{k=1}^{|\mathcal{V}|} (A_{ik} - A_{ik}^\star)^2 \leq \frac{N\alpha^2}{\mathbf{D}_{\min}^2 \cdot (\mathbf{D}_{ii}^\star)^2} \|\mathbf{A} - \mathbf{A}^\star\|_{\text{F}}^2 \\ &\leq \frac{N^2\alpha^2}{\mathbf{D}_{\min}^2 \cdot (\mathbf{D}_{ii}^\star)^2} \|\mathbf{A} - \mathbf{A}^\star\|_2^2 \lesssim \frac{1}{\rho_N}, \end{aligned}$$

where the last inequality holds by Theorem 3.4.2 since $\mathbf{D}_{\min} \gtrsim \rho_N$ by Lemma A.3.1, $\mathbf{D}_{ii}^\star = \rho_N$ and $N\alpha \asymp \rho_N$. The desired result follows by combining the two estimates above. $\square$

Lemma A.3.3. *Deterministically, $\|(\mathbf{D}^\star)^{-1}\mathbf{A}^\star\|_{2 \rightarrow \infty} \lesssim 1/\sqrt{N}$. Suppose that $\mathbb{D}_{\text{AM}} = 1 + \delta$ for some $\delta > 0$, then with probability at least $1 - N^{-\mathbb{D}_{\text{AM}} + \delta/2}$,*

$$\|\mathbf{D}^{-1}\mathbf{A}\|_{2 \rightarrow \infty} \lesssim 1/\sqrt{\rho_N} + 1/\sqrt{N}. \quad (\text{A.3.5})$$

For each $v \in \mathcal{V}$, the following holds with probability at least $1 - N^{-\mathbb{D}_{\text{AM}} + \delta/2}$

$$\begin{aligned} \|\mathbf{D}^{-1}\mathbf{A} - (\mathbf{D}^{(-v)})^{-1}\mathbf{A}^{(-v)}\| &\lesssim \|\mathbf{D}^{-1}\mathbf{A}\|_{2 \rightarrow \infty}, \\ \|(\mathbf{D}^{(-v)})^{-1}\mathbf{A}^{(-v)} - (\mathbf{D}^\star)^{-1}\mathbf{A}^\star\| &\leq 2/\sqrt{\rho_N} + 1/\sqrt{N}. \end{aligned}$$

Proof of Lemma A.3.3. By Lemma A.2.2, $\|\mathbf{A}^*\|_{2 \rightarrow \infty} \leq \rho_N / \sqrt{N}$, the first result then follows since $\mathbf{D}^* = \text{diag}(\rho_N, \dots, \rho_N)$. The remaining results follows the same arguments as in the proof of Lemma A.2.2, and the lower bound of the minimum degree in $\mathbf{D}$, as well as the lower bound of non-zero entries in $\mathbf{D}^{(-v)}$ (for the sake of *Moore-Penrose* inverse). $\square$

Corollary A.3.4. *For each $i \in \mathcal{V}$, the following holds with probability at least $1 - N^{-\mathbb{D}_{\text{AM}} + \delta/2}$,*

$$|\lambda_i - \lambda_i^*| \lesssim \frac{1}{\sqrt{\rho_N}}, \quad |\lambda_i - \lambda_i^{(-v)}| \lesssim \frac{1}{\sqrt{\rho_N}} + \frac{1}{\sqrt{N}}, \quad |\lambda_i^* - \lambda_i^{(-v)}| \lesssim \frac{2}{\sqrt{\rho_N}} + \frac{1}{\sqrt{N}}. \quad (\text{A.3.6})$$

In particular in our model 2.1.3, with probability at least $1 - N^{-\mathbb{D}_{\text{GH}} + \delta/2}$,

$$|\lambda_i| \asymp |\lambda_i^*| \asymp |\lambda_i^{(-v)}| \asymp 1, \quad i = 1, 2. \quad (\text{A.3.7})$$

$$|\lambda_i| \asymp |\lambda_i^*| \asymp |\lambda_i^{(-v)}| \asymp \frac{1}{\sqrt{\rho_N}}, \quad i = 3, \dots, N. \quad (\text{A.3.8})$$

Proof of Corollary A.3.4. By Weyl's inequality D.2.1 and Corollary A.3.2, for any $i \in \mathcal{V}$,

$$|\lambda_i - \lambda_i^*| \leq \|\mathbf{D}^{-1} \mathbf{A} - (\mathbf{D}^*)^{-1} \mathbf{A}^*\| \lesssim 1/\sqrt{\rho_N}. \quad (\text{A.3.9})$$

The remaining result follows as well by Lemma A.3.3 and Lemma D.2.1. $\square$

A.3.2 Proof of algorithm functionality

Following the approach in Section 2.4, we use $\frac{1}{\lambda_2} (\mathbb{E} \mathbf{D})^{-1} \mathbf{A} \mathbf{u}_2$ as the pivot vector. Proposition A.3.5 provides an analogous result to Proposition 2.4.1. Using the analytical framework established for proving Theorem 2.2.4, we show that Algorithm 2.2.1 achieves identical performance regardless of whether the input is the normalized graph Laplacian $\mathbf{L}$ or the adjacency matrix $\mathbf{A}$.

Proposition A.3.5. *Suppose $\mathbb{D}_{\text{AM}} = 1 + \delta$ for some constant $\delta > 0$, then with probability at least*

$$1 - N^{-\mathbb{D}_{\text{AM}} + \delta/2}$$

$$\sqrt{N} \cdot \min_{s \in \{\pm 1\}} \|s\mathbf{u}_2 - \frac{1}{\lambda_2}(\mathbb{E}\mathbf{D})^{-1}\mathbf{A}\mathbf{u}_2\|_\infty \leq C/\log(\rho_N). \quad (\text{A.3.10})$$

Proof of Proposition A.3.5. We first show that the eigenvector $\mathbf{u}_2$ is delocalized, i.e., $\|\mathbf{u}_2\|_\infty \lesssim N^{-1/2}$. For convenience in notation, we denote $\lambda = \lambda_2$, $\lambda^\star = \lambda_2^\star$, $\lambda^{(-v)} = \lambda_2^{(-v)}$ and $\mathbf{u} = \mathbf{u}_2$, $\mathbf{u}^\star = \mathbf{u}_2^\star$, $\mathbf{u}^{(-v)} = \mathbf{u}_2^{(-v)}$. By Corollary A.3.4, we then have

$$\begin{aligned} \|\mathbf{u}\|_\infty &= \|s\mathbf{D}^{-1}\mathbf{A}\mathbf{u}/\lambda\|_\infty \\ &\leq |\lambda|^{-1} \cdot \left(\|\mathbf{D}^{-1}\mathbf{A}(s\mathbf{u} - \mathbf{u}^\star)\|_\infty + \|\mathbf{D}^{-1}\mathbf{A}\mathbf{u}^\star\|_\infty \right) \\ &\lesssim |\lambda^\star|^{-1} \cdot \left(\max_{v \in \mathcal{V}} |\mathbf{D}^{-1}\mathbf{A}_v \cdot (s\mathbf{u} - \mathbf{u}^\star)| + \|\mathbf{D}^{-1}\mathbf{A}\mathbf{u}^\star\|_\infty \right) \\ &\lesssim |\lambda^\star|^{-1} \cdot \left(\|\mathbf{D}^{-1}\mathbf{A}\|_{2 \rightarrow \infty} \cdot \|s\mathbf{u} - s^{(-v)}\mathbf{u}^{(-v)}\|_2 + \max_{v \in \mathcal{V}} |\mathbf{D}^{-1}\mathbf{A}_v \cdot (s^{(-v)}\mathbf{u}^{(-v)} - \mathbf{u}^\star)| + \|\mathbf{D}^{-1}\mathbf{A}\mathbf{u}^\star\|_\infty \right) \\ &\lesssim \left[\left(\frac{1}{\sqrt{\rho_N}} + \frac{1}{\sqrt{N}} \right) \|\mathbf{u}\|_\infty + \frac{1}{\log(\rho_N)} \cdot \left(\|\mathbf{u}\|_\infty + \frac{1}{\sqrt{N}} \right) + \frac{1}{\sqrt{N}} \right] \end{aligned}$$

where in the second to last line, the first term could be bounded by Lemma A.3.6 and Lemma A.3.3, while the last two terms can still be bounded by arguments combining lower bound of entries in $\mathbf{D}$ and Lemmas A.2.5, A.2.6, respectively. By rearranging terms, one could prove

$\|\mathbf{u}\|_\infty \lesssim 1/\sqrt{N}$. Similarly, with the help from Lemmas A.2.4, A.2.5, A.2.6

$$\begin{aligned}
& \left\| s\mathbf{u} - \frac{(\mathbb{E}\mathbf{D})^{-1}\mathbf{A}\mathbf{u}^*}{\lambda^*} \right\|_\infty \\
&= \left\| s \frac{\mathbf{D}^{-1}\mathbf{A}\mathbf{u}}{\lambda} - \frac{\mathbf{D}^{-1}\mathbf{A}\mathbf{u}^*}{\lambda} + \frac{\mathbf{D}^{-1}\mathbf{A}\mathbf{u}^*}{\lambda} - \frac{(\mathbb{E}\mathbf{D})^{-1}\mathbf{A}\mathbf{u}^*}{\lambda^*} \right\|_\infty \\
&\lesssim |\lambda^*|^{-1} \|\mathbf{D}^{-1}\mathbf{A}(s\mathbf{u} - \mathbf{u}^*)\|_\infty + |\lambda^*|^{-1} \|\mathbf{D}^{-1} - (\mathbb{E}\mathbf{D})^{-1}\| \cdot \|\mathbf{A}\mathbf{u}^*\| \\
&\lesssim |\lambda^*|^{-1} \cdot \left(\|\mathbf{D}^{-1}\mathbf{A}\|_{2 \rightarrow \infty} \cdot \|s\mathbf{u} - s^{(-v)}\mathbf{u}^{(-v)}\|_2 + \max_{v \in \mathcal{V}} |\mathbf{D}^{-1}\mathbf{A}_v| (s^{(-v)}\mathbf{u}^{(-v)} - \mathbf{u}^*) \right) \\
&\quad + |\lambda^*|^{-1} \frac{|\mathbf{D}_{\min} - \mathbb{E}\mathbf{D}_v|}{|\mathbf{D}_{\min}| \cdot |\mathbb{E}\mathbf{D}_v|} \cdot \|\mathbf{A}\mathbf{u}^*\| \\
&\lesssim \left[\left(\frac{1}{\sqrt{\rho_N}} + \frac{1}{\sqrt{N}} \right) \|\mathbf{u}\|_\infty + \frac{1}{\log(\rho_N)} \cdot \left(\|\mathbf{u}\|_\infty + \frac{1}{\sqrt{N}} \right) + \frac{1}{\sqrt{N}} \right] + \frac{\sqrt{\rho_N}}{\rho_N \cdot \rho_N} \cdot \frac{\rho_N}{\sqrt{N}} \\
&\lesssim \frac{1}{\sqrt{N} \log(\rho_N)},
\end{aligned}$$

where in the second to last line, the difference $|\lambda - \lambda^*|$ is controlled by Corollary A.3.4, and a simple application of Chernoff proves that $|\mathbf{D}_v - \mathbb{E}\mathbf{D}_v| \leq \sqrt{\rho_N}$ with high probability for each $v \in \mathcal{V}$. $\square$

The thing remaining is to show that the Algorithm 2.2.1 achieves exact recovery when $\mathbb{D}_{\text{AM}} > 1$, which follows exactly as the proof of Theorem 2.2.4.

Lemma A.3.6. *Under the setting in Proposition A.3.5,*

$$\|s\mathbf{u} - s^{(-v)}\mathbf{u}^{(-v)}\|_2 \lesssim \|\mathbf{u}\|_\infty. \quad (\text{A.3.11})$$

Proof of Lemma A.3.6. Let $\delta = \min_{i \neq 2} |\lambda_i^{(-v)} - \lambda|$ denote the absolute eigen gap, then by generalized Davis-Kahan Lemma D.2.4 where we take $\mathbf{A}^{(-v)}$, $\mathbf{D}^{(-v)}$, $\Lambda = \lambda_2^{(-v)}$, $\mathbf{U} = s^{(-v)}\mathbf{u}^{(-v)}$, $\mathbf{P} = \mathbf{I}_N - \mathbf{U}\mathbf{U}^\top$, $\hat{\lambda} = \lambda$, $\hat{\mathbf{u}} = s\mathbf{u}$, we have

$$\|s^{(-v)}\mathbf{u}^{(-v)} - s\mathbf{u}\|_2 \leq \sqrt{2} \|\mathbf{P}\mathbf{u}\|_2 \quad (\text{A.3.12})$$

$$\leq \frac{\sqrt{2} \cdot \|(\mathbf{D}^{(-v)})^{-1}\mathbf{A}^{(-v)} - \lambda\mathbf{I}_N\|_2}{\delta} = \frac{\sqrt{2} \cdot \|[(\mathbf{D}^{(-v)})^{-1}\mathbf{A}^{(-v)} - \mathbf{D}^{-1}\mathbf{A}]\mathbf{u}\|_2}{\delta} \quad (\text{A.3.13})$$

By Lemma D.2.1 and Corollary A.2.3, the following holds with probability at least $1 - O(N^{-10})$,

$$\begin{aligned}
\delta &= \min_{i \neq 2} |\lambda_i^{(-v)} - \lambda| = \min \{ |\lambda_3^{(-v)} - \lambda|, |\lambda_1^{(-v)} - \lambda| \} \\
&\geq \min \{ |\lambda_3 - \lambda|, |\lambda_1 - \lambda| \} - \|(\mathbf{D}^{(-v)})^{-1} \mathbf{A}^{(-v)} - \mathbf{D}^{-1} \mathbf{A}\| \\
&\geq \min \{ |\lambda_3^* - \lambda^*|, |\lambda_1^* - \lambda^*| \} - 2\|\mathbf{D}^{-1} \mathbf{A} - (\mathbf{D}^*)^{-1} \mathbf{A}^*\| - \|(\mathbf{D}^{(-v)})^{-1} \mathbf{A}^{(-v)} - \mathbf{D}^{-1} \mathbf{A}\| \\
&\gtrsim 1,
\end{aligned}$$

where the last line holds by Corollary A.3.2. Let $\mathbf{w} = [\mathbf{D}^{-1} \mathbf{A} - (\mathbf{D}^{(-v)})^{-1} \mathbf{A}^{(-v)}] \mathbf{u}$. For $i \neq v$,

$$\|[\mathbf{D}^{-1} \mathbf{A} - (\mathbf{D}^{(-v)})^{-1} \mathbf{A}^{(-v)}]_i\|_1 = \mathbb{1}_{\{\mathbf{D}_{ii} \neq 0\}} \sum_{\ell \in \mathcal{L}} (\ell - 1) \mathbf{A}_{i\ell} \cdot (\mathbf{D}_{ii})^{-1}. \quad (\text{A.3.14})$$

Then

$$\begin{aligned}
|w_v| &= |(\mathbf{D}^{-1} \mathbf{A} \mathbf{u})_v| = |\lambda| \cdot |u_v| \leq |\lambda| \cdot \|\mathbf{u}\|_\infty, \\
|w_i| &= |[\mathbf{D}^{-1} \mathbf{A} \mathbf{u} - (\mathbf{D}^{(-v)})^{-1} \mathbf{A}^{(-v)} \mathbf{u}]_i| \leq \|[\mathbf{D}^{-1} \mathbf{A} - (\mathbf{D}^{(-v)})^{-1} \mathbf{A}^{(-v)}]_i\|_1 \cdot \|\mathbf{u}\|_\infty
\end{aligned}$$

Again by Corollary A.3.2, Corollary A.3.4, with probability at least $1 - O(N^{-10})$,

$$\begin{aligned}
\|(\mathbf{A} - \mathbf{A}^{(-v)}) \mathbf{u}\|_2^2 &\leq \left(|\lambda|^2 + \mathbb{1}_{\{\mathbf{D}_{ii} \neq 0\}} \sum_{\ell \in \mathcal{L}} (\ell - 1) \mathbf{A}_{i\ell} \cdot (\mathbf{D}_{ii})^{-1} \right) \cdot \|\mathbf{u}\|_\infty^2 \\
&\lesssim (|\lambda^*|^2 + \|\mathbf{D}^{-1} \mathbf{A}\|_{2 \rightarrow \infty}) \cdot \|\mathbf{u}\|_\infty^2 \lesssim \|\mathbf{u}\|_\infty^2.
\end{aligned}$$

Therefore, by plugging in the lower bound for δ (A.2.21) into Equation (A.2.20), we have

$$\|s^{(-v)} \mathbf{u}^{(-v)} - s \mathbf{u}\|_2 \lesssim \|\mathbf{u}\|_\infty. \quad (\text{A.3.15})$$

□

A.4 Performance analysis of SDP

This section is devoted to the performance analysis of the following *semidefinite programming* (SDP).

$$\begin{aligned} & \max \langle \mathbf{A}, \mathbf{X} \rangle \\ & \text{subject to } \mathbf{X} \succeq 0, \langle \mathbf{X}, \mathbf{1}\mathbf{1}^\top \rangle = 0, X_{ii} = 1, \quad \forall i \in \mathcal{V}. \end{aligned} \tag{A.4.1}$$

Algorithm A.4.1: SDP via adjacency matrix

Data: The adjacency matrix $\mathbf{A}$

- 1 Solve the SDP (A.4.1) with $\hat{\Sigma}$ denoting its solution.
- 2 Let $\hat{\mathbf{u}}$ denote the eigenvector corresponding to the largest eigenvalue of $\hat{\Sigma}$.

Result: $\hat{\mathcal{V}}_+ = \{v : \text{sign}(\hat{u}_v) > 0\}$ and $\hat{\mathcal{V}}_- = \{v : \text{sign}(\hat{u}_v) < 0\}$.

Theorem A.4.1. *When the adjacency matrix $\mathbf{A}$ is used as the input, Algorithm A.4.1 achieves exact recovery if and only if $\mathbb{D}_{\text{AM}} > 1$.*

The proof of Theorem A.4.1 is included here for completeness, so that one could witness the appearance of $\mathbb{D}_{\text{AM}}$, while the discrepancy between $\mathbb{D}_{\text{AM}}$ and $\mathbb{D}_{\text{GH}}$ could be seen straightforwardly.

A.4.1 Necessity

The proof strategy follows the same idea as [146, Theorem 8]. Consider the Min-Bisection problem

$$\begin{aligned} & \max \sum_{1 \leq i < j \leq N} x_i A_{ij} x_j \\ & \text{subject to } \mathbf{x} \in \{\pm 1\}^{|\mathcal{V}|}, \mathbf{1}^\top \mathbf{x} = 0. \end{aligned} \tag{A.4.2}$$

Note that any feasible $\mathbf{x}$ to (A.4.2) corresponds to a feasible solution $\mathbf{X} = \mathbf{x}\mathbf{x}^\top$ to (A.4.1). Consequently, (A.4.1) is a relaxation of (A.4.2) [112]. Let $\widehat{\Sigma}$ and $\widehat{\mathbf{y}}_{\text{MB}}$ denote the solutions to (A.4.1) and (A.4.2) respectively. The probability that SDP (A.4.1) fails exact recovery is then lower bounded by

$$\mathbb{P}_{\text{fail}} = \mathbb{P}(\widehat{\Sigma} \neq \mathbf{y}\mathbf{y}^\top) \geq \mathbb{P}(\widehat{\mathbf{y}}_{\text{MB}} \neq \pm \mathbf{y}). \quad (\text{A.4.3})$$

Then it suffices to consider the lower bound of the probability that $\widehat{\mathbf{y}}_{\text{MB}}$ fails exact recovery. Given the adjacency matrix $\mathbf{A}$ and the assignment vector $\mathbf{x} \in \{\pm 1\}^{|\mathcal{V}|}$ with $\mathbf{1}^\top \mathbf{y} = 0$, define

$$g(\mathbf{x}) = \sum_{1 \leq i < j \leq N} x_i A_{ij} x_j. \quad (\text{A.4.4})$$

The failure of $\widehat{\mathbf{y}}_{\text{MB}}$ implies the existence of some $\widetilde{\mathbf{y}}$ such that $g(\widetilde{\mathbf{y}}) \geq g(\mathbf{y})$, i.e.,

$$\mathbb{P}(\widehat{\mathbf{y}}_{\text{MB}} \neq \pm \mathbf{y}) = \mathbb{P}(\exists \widetilde{\mathbf{y}} \text{ s.t. } g(\widetilde{\mathbf{y}}) \geq g(\mathbf{y})). \quad (\text{A.4.5})$$

Following the same idea as the proof of Theorem 2.2.1, one could construct such $\widetilde{\mathbf{y}}$ by swapping signs of $a \in \mathcal{V}_+$ and $b \in \mathcal{V}_-$ but fixing others in $\mathbf{y}$. Then we have the decomposition

$$g(\mathbf{y}) - g(\widetilde{\mathbf{y}}) = \sum_{1 \leq i < j \leq N} A_{ij} y_i y_j - \sum_{1 \leq i < j \leq N} A_{ij} \widetilde{y}_i \widetilde{y}_j \quad (\text{A.4.6})$$

$$= \sum_{1 \leq i < j \leq N} \sum_{\ell \in \mathcal{L}} \sum_{e \supset \{i, j\}} \mathcal{A}_e^{(\ell)} (y_i y_j - \widetilde{y}_i \widetilde{y}_j) = \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathcal{A}_e^{(\ell)} \sum_{\{i, j\} \subset e} (y_i y_j - \widetilde{y}_i \widetilde{y}_j) \quad (\text{A.4.7})$$

$$= \underbrace{\sum_{\ell \in \mathcal{L}} \sum_{e \ni a, e \not\ni b} \mathcal{A}_e^{(\ell)} \sum_{\{i, j\} \subset e} (y_i y_j - \widetilde{y}_i \widetilde{y}_j)}_{X_a} + \underbrace{\sum_{\ell \in \mathcal{L}} \sum_{e \not\ni a, e \ni b} \mathcal{A}_e^{(\ell)} \sum_{\{i, j\} \subset e} (y_i y_j - \widetilde{y}_i \widetilde{y}_j)}_{X_b} \quad (\text{A.4.8})$$

$$+ \underbrace{\sum_{\ell \in \mathcal{L}} \sum_{e \supset \{a, b\}} \mathcal{A}_e^{(\ell)} \sum_{\{i, j\} \subset e} (y_i y_j - \widetilde{y}_i \widetilde{y}_j)}_{X_{ab}}. \quad (\text{A.4.9})$$

Furthermore, the contributions from X_a , X_b and X_{ab} are independent due to the edge indepen-

dence. For X_{ab} , note that the terms irrelevant to a and b cancel out, since $\mathbf{y}$ and $\tilde{\mathbf{y}}$ are the same except a, b . Then for each $e \in \mathcal{E}_\ell^{(r)}(a)$ and $e \supset \{a, b\}$,

$$\begin{aligned}
& \sum_{\{i,j\} \subset e} (y_i y_j - \tilde{y}_i \tilde{y}_j) \\
&= (y_a y_b - \tilde{y}_a \tilde{y}_b) + \sum_{\{i,j\} \subset e \setminus \{a,b\}} (y_i y_j - \tilde{y}_i \tilde{y}_j) \\
&= (y_a y_b - \tilde{y}_a \tilde{y}_b) + y_a \sum_{j \in e \setminus \{a,b\}} y_j - \tilde{y}_a \sum_{j \in e \setminus \{a,b\}} \tilde{y}_j + y_b \sum_{j \in e \setminus \{a,b\}} y_j - \tilde{y}_b \sum_{j \in e \setminus \{a,b\}} \tilde{y}_j \\
&= (y_a y_b - \tilde{y}_a \tilde{y}_b) + 2y_a \sum_{i \in e \setminus \{a,b\}} y_i + 2y_b \sum_{j \in e \setminus \{a,b\}} y_j \\
&= 0 + 2y_a(\ell - 2 - 2r) + 2y_b(\ell - 2 - 2r) = 0.
\end{aligned}$$

where in the last line, we used the facts that $y_a y_b = \tilde{y}_a \tilde{y}_b$ and $y_a + y_b = 0$.

For X_a , the contributions of the terms irrelevant to a cancels, since $\mathbf{y}$ and $\tilde{\mathbf{y}}$ are the same except a, b . If $e \in \mathcal{E}_\ell^{(r)}(a)$, then

$$\sum_{\{i,j\} \subset e} (y_i y_j - \tilde{y}_i \tilde{y}_j) = 2y_a \sum_{i \in e \setminus \{a\}} y_i = 2y_a(\ell - 1 - 2r). \quad (\text{A.4.10})$$

The same argument holds for X_b similarly. Denote $F_v := \mathbb{1}\{X_v \leq 0\}$, then $\{F_a = 1\} \cap \{F_b = 1\} \iff g(\mathbf{y}) \leq g(\tilde{\mathbf{y}})$, and it suffices to show that $\{F_a = 1\}$ happens with high probability. However, F_a and $F_{a'}$ are not independent for distinct $a, a' \in \mathcal{V}_+$ due to the existence of edges containing a, a' simultaneously. Following the same strategy in Appendix A.1, let $\mathcal{U} \subset \mathcal{V}$ be a subset of vertices of size $\gamma_N N$ where $|\mathcal{U} \cap \mathcal{V}_+| = |\mathcal{U} \cap \mathcal{V}_-| = \gamma_N N/2$ with $\gamma_N = q_N^{-3}$ and denote

$\mathcal{W}_\ell^{(\geq 2)} := \{e \in [\mathcal{V}]^\ell \mid e \text{ contains at least 2 vertices in } \mathcal{U}\} \subset \mathcal{E}_\ell$. We define

$$\begin{aligned} Y_a &= \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \cap \mathcal{U} = \{a\} \\ e \not\ni b}} \mathcal{A}_e^{(\ell)} \sum_{\{i,j\} \subset e} (y_i y_j - \tilde{y}_i \tilde{y}_j), & Z_a &= \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{W}_\ell^{(\geq 2)} \\ e \ni a, e \not\ni b}} \mathcal{A}_e^{(\ell)} \sum_{\{i,j\} \subset e} (y_i y_j - \tilde{y}_i \tilde{y}_j), \\ Y_b &= \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \cap \mathcal{U} = \{b\} \\ e \not\ni a}} \mathcal{A}_e^{(\ell)} \sum_{\{i,j\} \subset e} (y_i y_j - \tilde{y}_i \tilde{y}_j), & Z_b &= \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{W}_\ell^{(\geq 2)} \\ e \ni b, e \not\ni a}} \mathcal{A}_e^{(\ell)} \sum_{\{i,j\} \subset e} (y_i y_j - \tilde{y}_i \tilde{y}_j). \end{aligned}$$

Note that Y_a, Y_b, Z_a, Z_b are pairwise independent since there is no common edges. Denote events

$$\begin{aligned} A_a &= \mathbb{1}\{Y_a \leq -\zeta_N q_N\}, & J_a &= \mathbb{1}\{Z_a \leq \zeta_N q_N\}, \\ B_b &= \mathbb{1}\{Y_b \leq -\zeta_N q_N\}, & J_b &= \mathbb{1}\{Z_b \leq \zeta_N q_N\}, \end{aligned}$$

where we take $\zeta = (\log q_N)^{-1}$. Obviously,

$$\{A_a = 1\} \cap \{J_a = 1\} \implies \{F_a = 1\}, \quad \{B_b = 1\} \cap \{J_b = 1\} \implies \{F_b = 1\}.$$

Let $\mathbb{D}_{\text{AM}} = 1 - \varepsilon$ for some absolute $\varepsilon > 0$ and $q_N = \log(N)$. With proofs deferred later, we have the following inequalities

$$\prod_{a \in \mathcal{V}_+ \cap \mathcal{U}} \mathbb{P}(A_a = 0) \leq \exp\left(-\frac{1}{2} N^{\varepsilon + o(1)}\right), \quad \mathbb{P}(J_a = 0) \leq N^{-3 + o(1)}. \quad (\text{A.4.11})$$

Then the probability of SDP failing exact recovery is lower bounded by

$$\begin{aligned}
& \mathbb{P}_{\text{fail}} \\
&= \mathbb{P}(\widehat{\Sigma} \neq \mathbf{y}\mathbf{y}^\top) \geq \mathbb{P}(\widehat{\mathbf{y}}_{\text{MB}} \neq \pm \mathbf{y}) = \mathbb{P}(\exists \tilde{\mathbf{y}} \text{ s.t. } g(\tilde{\mathbf{y}}) \geq g(\mathbf{y})) \\
&\geq \mathbb{P}(\exists a \in \mathcal{V}_+, b \in \mathcal{V}_- \text{ s.t. } \{\mathbf{F}_a = 1\} \cap \{\mathbf{F}_b = 1\}) = \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+} \{\mathbf{F}_a = 1\} \cap \bigcap_{b \in \mathcal{V}_-} \{\mathbf{F}_b = 1\}\right) \\
&= \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+} \{\mathbf{F}_a = 1\} \cap \bigcap_{b \in \mathcal{V}_-} \{\mathbf{F}_b = 1\} \mid \{\mathbf{J}_a = 1\} \cap \{\mathbf{J}_b = 1\}\right) \cdot \mathbb{P}(\{\mathbf{J}_a = 1\} \cap \{\mathbf{J}_b = 1\}) \\
&\geq \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+ \cap \mathcal{U}} \{\mathbf{A}_a = 1\} \cap \bigcap_{b \in \mathcal{V}_- \cap \mathcal{U}} \{\mathbf{B}_b = 1\} \mid \{\mathbf{J}_a = 1\} \cap \{\mathbf{J}_b = 1\}\right) \cdot \mathbb{P}(\{\mathbf{J}_a = 1\} \cap \{\mathbf{J}_b = 1\}) \\
&\geq \mathbb{P}\left(\bigcup_{a \in \mathcal{V}_+ \cap \mathcal{U}} \{\mathbf{A}_a = 1\}\right) \cdot \mathbb{P}\left(\bigcup_{b \in \mathcal{V}_- \cap \mathcal{U}} \{\mathbf{B}_b = 1\}\right) \cdot \mathbb{P}(\mathbf{J}_a = 1) \cdot \mathbb{P}(\mathbf{J}_b = 1) \\
&= \left(1 - \prod_{a \in \mathcal{V}_+ \cap \mathcal{U}} \mathbb{P}(\mathbf{A}_a = 0)\right) \cdot \left(1 - \prod_{b \in \mathcal{V}_- \cap \mathcal{U}} \mathbb{P}(\mathbf{B}_b = 0)\right) \cdot \left(1 - \mathbb{P}(\mathbf{J}_a = 0)\right) \cdot \left(1 - \mathbb{P}(\mathbf{J}_b = 0)\right) \\
&\geq 1 - 2 \exp\left(-\frac{N^{\varepsilon+o(1)}}{2}\right) - 2N^{-3+o(1)} \\
&\quad - 2N^{-3+o(1)} \cdot \exp(-N^{\varepsilon+o(1)}) - 2N^{-6+o(1)} \cdot \exp\left(-\frac{N^{\varepsilon+o(1)}}{2}\right).
\end{aligned}$$

Therefore, as N goes to infinity, with probability tending to 1, SDP (A.4.1) fails exact recovery when $\mathbb{D}_{\text{AM}} < 1$.

Proof of (A.4.11). Note that $\mathbf{Y}_a$ admits the decomposition

$$\mathbf{Y}_a = \sum_{\ell \in \mathcal{L}} \sum_{r=0}^{\ell-1} c_r^{(\ell)} \mathbf{Y}_r^{(\ell)}, \tag{A.4.12}$$

where $c_r^{(\ell)} = 2y_a(\ell - 1 - 2r)$ and $\mathbf{Y}_r^{(\ell)} \sim \text{Bin}(N_r^{(\ell)}, p_r^{(\ell)})$ with $p_r^{(\ell)} = \alpha_\ell$ if $r = 0$, otherwise $p_r^{(\ell)} = \beta_\ell$, and

$$N_r^{(\ell)} = \left(\frac{N}{2} - \frac{\gamma_N N}{2} - 1\right) \binom{\frac{N}{2} - \frac{\gamma_N N}{2} - 1}{\ell - 1 - r} = (1 + o(1)) \cdot \frac{1}{2^{\ell-1}} \binom{\ell-1}{r} \binom{N-1}{\ell-1}, \tag{A.4.13}$$

Apply Lemma A.1.2 with $g_N = 1$, $\delta = 0$, $\rho_r^{(\ell)} = \frac{1}{2^{\ell-1}} \binom{\ell-1}{r}$, and $\alpha_r^{(\ell)} = a_\ell$ if $r = 0$, otherwise

$\alpha_r^{(\ell)} = b_\ell$. We then have

$$\mathbb{P}(A_a = 1) = \exp\left(-(1 + o(1)) \cdot \mathbb{D} \cdot q_N\right), \quad (\text{A.4.14})$$

where the rate function $\mathbb{D}$ there becomes

$$\begin{aligned} \mathbb{D} &= \max_{t \geq 0} \left(-t\delta + \sum_{\ell \in \mathcal{L}} \sum_{r=0}^{\ell-1} \alpha_r^{(\ell)} \rho_r^{(\ell)} (1 - e^{-c_r^{(\ell)} t}) \right) \\ &= \max_{t \geq 0} \sum_{\ell \in \mathcal{L}} \frac{1}{2^{\ell-1}} \left[a_\ell (1 - e^{-(\ell-1)2t}) + b_\ell \sum_{r=1}^{\ell-1} \binom{\ell-1}{r} (1 - e^{-(\ell-1-2r)2t}) \right] \\ &= \mathbb{D}_{\text{AM}} \text{ in (2.1.6).} \end{aligned}$$

We take $q_N = \log(N)$. Since $\mathbb{D}_{\text{AM}} = 1 - \varepsilon$ for some absolute $\varepsilon > 0$, by applying $1 - x \leq e^{-x}$, we have

$$\prod_{a \in \mathcal{V}_+} \mathbb{P}(A_a = 0) = \prod_{a \in \mathcal{V}_+} \left(1 - \mathbb{P}(A_a = 1) \right) \leq (1 - N^{-(1+o(1)) \cdot \mathbb{D}_{\text{AM}}})^{\frac{N}{2}} \quad (\text{A.4.15})$$

$$\leq \exp\left(-\frac{1}{2} N^{1-(1+o(1)) \cdot \mathbb{D}_{\text{AM}}}\right) = \exp\left(-\frac{1}{2} N^{\varepsilon+o(1)}\right). \quad (\text{A.4.16})$$

The bound for $\mathbb{P}(J_a = 0)$ could be obtained similarly by following the same argument in Lemma 2.3.5. □

A.4.2 Sufficiency

The proof strategy for sufficiency follows the same idea as [118]. For some $\mathbf{v} \in \mathbb{R}$, define

$$\mathbf{S} = \mathbf{D} - \mathbf{A} + \mathbf{v} \mathbf{1} \mathbf{1}^\top \quad (\text{A.4.17})$$

Lemma A.4.2 ([118, Lemma 3]). *Suppose that there exists some diagonal matrix $\mathbf{D} \in \mathbb{R}^{N \times N}$ and $\mathbf{v} \in \mathbb{R}$ such that $\lambda_{N-1}(\mathbf{S}) > 0$, $\mathbf{S} \mathbf{y} = \mathbf{0}$ and $\mathbf{S}$ being positive semidefinite, then $\hat{\Sigma} = \mathbf{y} \mathbf{y}^\top$ is the unique solution to (A.4.1).*

For each $v \in \mathcal{V}$, define

$$\mathbf{D}_v = \sum_{j=1}^{|\mathcal{V}|} A_{vj} y_v y_j. \quad (\text{A.4.18})$$

Then $\mathbf{D}_v y_v = \sum_{j=1}^{|\mathcal{V}|} A_{vj} y_j$, leading to $\mathbf{D} \mathbf{y} = \mathbf{A} \mathbf{y}$. Consequently, $\mathbf{S} \mathbf{y} = \mathbf{0}$ since $\mathbf{1}^\top \mathbf{y} = 0$. Then it suffices to verify that $\mathbf{S}$ is positive definite in the subspace orthogonal to $\mathbf{y}$ with high probability, i.e.,

$$\mathbb{P} \left(\inf_{\mathbf{x} \perp \mathbf{y}, \|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{S} \mathbf{x} > 0 \right) \geq 1 - N^{-\Omega(1)}. \quad (\text{A.4.19})$$

Note that $\mathbf{A}^* = \mathbb{E}[\mathbf{A}] = \frac{\alpha+\beta}{2} \mathbf{1}\mathbf{1}^\top + \frac{\alpha-\beta}{2} \mathbf{y}\mathbf{y}^\top - \alpha \mathbf{I}_N$ with α, β defined in (2.4.1), then one has

$$\mathbf{x}^\top \mathbf{S} \mathbf{x} = \mathbf{x}^\top \mathbf{D} \mathbf{x} - \frac{\alpha-\beta}{2} \mathbf{x}^\top \mathbf{y} \mathbf{y}^\top \mathbf{x} + \left(v - \frac{\alpha+\beta}{2} \right) \mathbf{x}^\top \mathbf{1} \mathbf{1}^\top \mathbf{x} + \alpha - \mathbf{x}^\top (\mathbf{A} - \mathbf{A}^*) \mathbf{x} \quad (\text{A.4.20})$$

$$\geq \mathbf{x}^\top \mathbf{D} \mathbf{x} + \alpha - \mathbf{x}^\top (\mathbf{A} - \mathbf{A}^*) \mathbf{x} \geq \min_{v \in \mathcal{V}} \mathbf{D}_v + \alpha - \|\mathbf{A} - \mathbf{A}^*\|, \quad (\text{A.4.21})$$

where the second to last inequality holds by choosing some proper $v \geq \frac{\alpha+\beta}{2}$. Note that $\|\mathbf{A} - \mathbf{A}^*\| \lesssim \sqrt{\rho_N}$ by Theorem 3.4.2, then the lower bound above is determined by $\min_{v \in \mathcal{V}} \mathbf{D}_v$. Following the same strategy in the proof of Theorem 2.2.4, one could do the following decomposition

$$\begin{aligned} \sqrt{N} y_v \mathbf{A} \mathbf{y} &= y_v \sum_{j \neq v, y_j = +1} A_{vj} - y_v \sum_{j \neq v, y_j = -1} A_{vj} = \sum_{j \neq v, y_j = y_v} A_{vj} - \sum_{j \neq v, y_j \neq y_v} A_{vj} \\ &= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell^{(0)}(v)} (\ell - 1) \cdot \mathcal{A}_e^{(\ell)} - \sum_{e \in \mathcal{E}_\ell^{(\ell-1)}(v)} (\ell - 1) \cdot \mathcal{A}_e^{(\ell)} \\ &\quad + \sum_{\ell \in \mathcal{L}} \sum_{r=1}^{\ell-2} \sum_{e \in \mathcal{E}_\ell^{(r)}} \left((\ell - 1 - r) \cdot \mathcal{A}_e^{(\ell)} - r \cdot \mathcal{A}_e^{(\ell)} \right), \end{aligned} \quad (\text{A.4.22})$$

where the contribution of (A.4.22) does not vanish without the weighting trick, which impairs

the way of obtaining the correct threshold $\mathbb{D}_{\text{GH}}$. By an simple application of Lemma A.1.2, one could prove that $\mathbb{P}(\sqrt{N}y_v \mathbf{A} \mathbf{y} > \delta \rho_N) \geq 1 - N^{-\mathbb{D}_{\text{AM}} + \delta/2}$ when $\mathbb{D}_{\text{AM}} = 1 + \delta$ for some $\delta > 0$. Therefore, Algorithm A.4.1 achieves exact recovery with high probability when $\mathbb{D}_{\text{AM}} > 1$.

Appendix B

Deferred proofs in Chapter 3

B.1 The Kahn-Szemerédi approach and proofs of concentration

B.1.1 Proof outline of Theorem 3.4.2

We follow the approach in [101, 153, 71] to establish the proof of Theorem 3.4.2. According to Courant-Fischer-Weyl minimax principle, the spectral norm of any $N \times N$ Hermitian matrix $\mathbf{W}$ can be reformulated as

$$\|\mathbf{W}\| := \sup_{\mathbf{x} \in \mathbb{S}^{N-1}} \|\mathbf{W}\mathbf{x}\|_2 = \sup_{\mathbf{x} \in \mathbb{S}^{N-1}} |\mathbf{x}^\top \mathbf{W}\mathbf{x}|, \quad (\text{B.1.1})$$

where $\mathbb{S}^{N-1} := \{\mathbf{x} \in \mathbb{R}^N : \|\mathbf{x}\|_2 = 1\}$ denotes the unit sphere, and the second equality makes use of the Hermitian property of $\mathbf{W}$. As indicated by a short continuity argument (Lemma B.1.1), if one wants to bound the supremum in (B.1.1), it suffices to control $\mathbf{x}^\top \mathbf{W}\mathbf{x}$ for all $\mathbf{x}$ in a suitable ε -net $\mathcal{N}_\varepsilon$ of $\mathbb{S}^{N-1}$ with cardinality bounded by $|\mathcal{N}_\varepsilon| \leq (1 + \frac{2}{\varepsilon})^N$, while the existence of such $\mathcal{N}_\varepsilon$ is guaranteed by compactness of $\mathbb{S}^{N-1}$ and [195, Corollary 4.2.13].

Lemma B.1.1 ([195, Lemma 4.4.1]). *Let $\mathbf{W}$ be any Hermitian $N \times N$ matrix and $\mathcal{N}_\varepsilon$ be an ε -net on the unit sphere $\mathbb{S}^{N-1}$ with $\varepsilon \in (0, 1)$, then $\|\mathbf{W}\| \leq \frac{1}{1-\varepsilon} \sup_{\mathbf{x} \in \mathcal{N}_\varepsilon} |\langle \mathbf{W}\mathbf{x}, \mathbf{x} \rangle|$.*

Proof of Lemma B.1.1. An ε -net for a compact metric space (X, d) is a finite subset $\mathcal{N}$ of X such that for each point $\mathbf{x} \in X$, there is a point $\mathbf{x} \in \mathcal{N} \subset X$ with $d(\mathbf{x}, \mathbf{x}) \leq \varepsilon$. Note that $\mathcal{N}_\varepsilon$ is

an ε -net on the unit sphere $\mathbb{S}^{N-1}$, then for any $\mathbf{x} \in \mathbb{S}^{N-1}$, there exists some $\mathbf{x}_0 \in \mathcal{N}_\varepsilon$ such that $\|\mathbf{x} - \mathbf{x}_0\|_2 \leq \varepsilon$, and we have

$$\|\mathbf{W}\mathbf{x}\|_2 - \|\mathbf{W}\mathbf{x}_0\|_2 \leq \|\mathbf{W}\mathbf{x} - \mathbf{W}\mathbf{x}_0\|_2 \leq \|\mathbf{W}\|_2 \cdot \|\mathbf{x} - \mathbf{x}_0\|_2 \leq \varepsilon \|\mathbf{W}\|,$$

leading to $\|\mathbf{W}\mathbf{x}\|_2 - \varepsilon \|\mathbf{W}\| \leq \|\mathbf{W}\mathbf{x}_0\|_2$. If the supremum is taken over $\mathbf{x}$ on the left hand side, then

$$\sup_{\mathbf{x} \in \mathbb{S}^{N-1}} (\|\mathbf{W}\mathbf{x}\|_2 - \varepsilon \|\mathbf{W}\|) = (1 - \varepsilon) \|\mathbf{W}\| \leq \|\mathbf{W}\mathbf{x}_0\|_2 \leq \sup_{\mathbf{y} \in \mathcal{N}_\varepsilon} \|\mathbf{W}\mathbf{y}\|_2.$$

Consequently, $\|\mathbf{W}\| \leq \frac{1}{1-\varepsilon} \sup_{\mathbf{x} \in \mathcal{N}_\varepsilon} \|\mathbf{W}\mathbf{x}\|_2 = \frac{1}{1-\varepsilon} \sup_{\mathbf{x} \in \mathcal{N}_\varepsilon} |\langle \mathbf{W}\mathbf{x}, \mathbf{x} \rangle|$. $\square$

Define $\mathbf{W} := \mathbf{A} - \mathbb{E}\mathbf{A}$ and take $\mathcal{N} := \mathcal{N}_{\frac{1}{2}}$ with its size bounded by $|\mathcal{N}| \leq 5^N$, then

$$\|\mathbf{A} - \mathbb{E}\mathbf{A}\| =: \|\mathbf{W}\| \leq 2 \sup_{\mathbf{x} \in \mathcal{N}} |\mathbf{x}^\top \mathbf{W}\mathbf{x}| \leq 2 \cdot 5^N \cdot |\mathbf{x}^\top \mathbf{W}\mathbf{x}|, \text{ for some } \mathbf{x} \in \mathcal{N}. \quad (\text{B.1.2})$$

Therefore, upper bound of $\|\mathbf{A} - \mathbb{E}\mathbf{A}\|$ will be obtained once the concentration of random variable $\mathbf{x}^\top \mathbf{W}\mathbf{x}$ is demonstrated for each fixed $\mathbf{x} := [x_1, x_2, \dots, x_N]^\top \in \mathcal{N}$. Roughly speaking, one might seek upper bound on $|\mathbf{x}^\top \mathbf{W}\mathbf{x}|$ of order $O(\sqrt{\mathcal{D}_{\max}})$ which holds with high probability, then a union bound over $\mathcal{N}$ establishes Theorem 3.4.2, as long as concentration of $\mathbf{x}^\top \mathbf{W}\mathbf{x}$ is sufficient to beat the cardinality of the net $\mathcal{N}$.

The problem now reduces to prove $|\mathbf{x}^\top \mathbf{W}\mathbf{x}| = |\sum_i \sum_j W_{ij} x_i x_j| = O(\sqrt{\mathcal{D}_{\max}})$ with high probability. However, two main difficulties lie on

1. the dependence between matrix elements W_{ij} ,
2. not every pair $W_{ij} x_i x_j$ enjoys the sufficient concentration to beat the cardinality of $\mathcal{N}$.

To get rid of those difficulties, we follow Kahn-Szemerédi's argument [101, Theorem 2.2] to

split $\mathbf{x}^\top \mathbf{W} \mathbf{x}$ into *light* and *heavy* pairs according to $\max_{(i,j) \in [N]^2} |\mathbf{x}_i \mathbf{x}_j|$,

$$\mathcal{L}(\mathbf{x}) := \left\{ (i, j) : |\mathbf{x}_i \mathbf{x}_j| \leq \frac{\sqrt{\mathcal{D}_{\max}}}{N} \right\}, \quad \mathcal{H}(\mathbf{x}) := \left\{ (i, j) : |\mathbf{x}_i \mathbf{x}_j| > \frac{\sqrt{\mathcal{D}_{\max}}}{N} \right\}, \quad (\text{B.1.3})$$

where $\mathcal{D}_{\max} := \sum_{\ell \in \mathcal{L}} \mathcal{D}_{\max}^{(\ell)}$. By triangle inequality, (B.1.2) adapts the following decomposition, where contribution from *light* and *heavy* couples can be bounded separately,

$$\begin{aligned} \|\mathbf{A} - \mathbb{E}\mathbf{A}\| &\leq 2 \sup_{\mathbf{x} \in \mathcal{N}} |\langle \mathbf{W} \mathbf{x}, \mathbf{x} \rangle| = 2 \sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in [N]^2} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \\ &\leq 2 \left(\sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| + \sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \right). \end{aligned} \quad (\text{B.1.4})$$

As will be shown later, the contribution of *light* pairs is bounded by $\alpha \sqrt{\mathcal{D}_{\max}}$ with probability at least $1 - 2e^{-N}$ according to (B.1.7), while contribution of *heavy* pairs is bounded by $\beta \sqrt{\mathcal{D}_{\max}}$ with probability at least $1 - 2N^{-10}$ if $\mathcal{D}_{\max} \geq c \log(N)$ for some constant $c > 0$, as in Lemma B.1.8. Therefore,

$$\|\mathbf{A} - \mathbb{E}\mathbf{A}\| \leq C_{(3.4.3)} \sqrt{\mathcal{D}_{\max}},$$

holds with probability at least $1 - 2e^{-N} - 2N^{-10}$, where $C_{(3.4.3)} := 2(\alpha + \beta)$ with $\alpha = 5L^2$, $\beta = \beta_0 \sqrt{\beta_1} + L$,

$$\beta_0 = 16 + 32L(1 + e^2) + 1792(1 + e^{-2})L^2, \quad \frac{\beta_1}{L} \log\left(\frac{\beta_1}{L}\right) - \frac{\beta_1}{L} + 1 > \frac{11}{c},$$

which completes the proof of Theorem 3.4.2. The remaining of this section is organized as follows. The bounds for contributions of *light* and *heavy* pairs are provided in Appendix B.1.2, Appendix B.1.3, and the proof of Theorem 3.4.4 will be established in Appendix B.1.4.

B.1.2 Light couples

The contribution from *light* pairs can be decomposed into summation of contributions from independent hyperedges. An application of Bernstein's inequality (Lemma D.3.4) provides us with the concentration we need to beat the cardinality $|\mathcal{N}| = 5^N$.

For each ℓ -hyperedge $e \in \mathcal{E}_\ell$, define $\mathcal{W}_e^{(\ell)} := \mathcal{A}_e^{(\ell)} - \mathbb{E}\mathcal{A}_e^{(\ell)}$. Then for each fixed $\mathbf{x} \in \mathbb{S}^{N-1}$, the contribution from light pair can be reformulated as

$$\begin{aligned} \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{w}_{ij} \mathbf{x}_i \mathbf{x}_j &= \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \left(\sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ e \supset \{i,j\}}} \mathcal{W}_e^{(\ell)} \right) \mathbf{x}_i \mathbf{x}_j = \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathcal{W}_e^{(\ell)} \cdot \sum_{\substack{(i,j) \in \mathcal{L}(\mathbf{x}) \\ \{i,j\} \subseteq e}} \mathbf{x}_i \mathbf{x}_j \\ &= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathcal{Y}_e^{(\ell)}, \end{aligned} \quad (\text{B.1.5})$$

where we denote $\mathcal{Y}_e^{(\ell)} := \mathcal{W}_e^{(\ell)} \left(\sum_{\substack{(i,j) \in \mathcal{L}(\mathbf{x}) \\ \{i,j\} \subseteq e}} \mathbf{x}_i \mathbf{x}_j \right)$. Note that $e \supset (i, j)$ is an ℓ -hyperedge, then the number of choices for the pair (i, j) under some fixed e is at most ℓ^2 . By the definition of light pair in (B.1.3), $\mathcal{Y}_e^{(\ell)}$ is upper bounded by $|\mathcal{Y}_e^{(\ell)}| \leq \ell^2 \frac{\sqrt{\mathcal{D}_{\max}^{(\ell)}}}{N} \leq L^2 \frac{\sqrt{\mathcal{D}_{\max}}}{N}$ for all $\ell \in \mathcal{L}$. Moreover, (B.1.5) is a sum of independent, mean-zero ($\mathbb{E}\mathcal{Y}_e^{(\ell)} = 0$) random variables, and its second moment is bounded by

$$\begin{aligned} \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathbb{E}[(\mathcal{Y}_e^{(\ell)})^2] &:= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \left[\mathbb{E}[(\mathcal{W}_e^{(\ell)})^2] \left(\sum_{\substack{(i,j) \in \mathcal{L}(\mathbf{x}) \\ \{i,j\} \subseteq e}} \mathbf{x}_i \mathbf{x}_j \right)^2 \right] \\ &\leq \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \left[\mathbb{E}[\mathcal{A}_e^{(\ell)}] \cdot \ell^2 \left(\sum_{\substack{(i,j) \in \mathcal{L}(x) \\ \{i,j\} \subseteq e}} \mathbf{x}_i^2 \mathbf{x}_j^2 \right) \right] \leq \sum_{\ell \in \mathcal{L}} \binom{N-2}{\ell-2} \cdot \frac{\mathcal{D}_{\max}^{(\ell)}}{\binom{N-1}{\ell-1}} \cdot \ell^2 \sum_{(i,j) \in \mathcal{L}(x)} \mathbf{x}_i^2 \mathbf{x}_j^2 \\ &\leq \sum_{\ell \in \mathcal{L}} \frac{\ell^2(\ell-1)}{N-\ell+2} \cdot \mathcal{D}_{\max}^{(\ell)} \leq \frac{2}{N} \sum_{\ell \in \mathcal{L}} \ell^3 \cdot \mathcal{D}_{\max}^{(\ell)} \leq \frac{2L^3}{N} \mathcal{D}_{\max}, \end{aligned}$$

where in the last line, we use the facts $\mathcal{D}_{\max} = \sum_{\ell \in \mathcal{L}} \mathcal{D}_{\max}^{(\ell)}$, $\mathbf{x} \in \mathbb{S}^{N-1}$, $\mathbf{x}^\top \mathbf{x} = 1$ and

$$\sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{x}_i^2 \mathbf{x}_j^2 \leq \sum_{(i,j) \in [N]^2} \mathbf{x}_i^2 \mathbf{x}_j^2 = \sum_{i=1}^N \mathbf{x}_i^2 \left(\sum_{j=1}^N \mathbf{x}_j^2 \right) = \sum_{i=1}^N \mathbf{x}_i^2 = 1.$$

Then Bernstein's inequality (Lemma D.3.4) implies that for any $\alpha > 0$,

$$\begin{aligned} & \mathbb{P} \left(\left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \geq \alpha \sqrt{\mathcal{D}_{\max}} \right) = \mathbb{P} \left(\left| \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathcal{Y}_e^{(\ell)} \right| \geq \alpha \sqrt{\mathcal{D}_{\max}} \right) \\ & \leq 2 \exp \left(- \frac{\frac{1}{2} \alpha^2 \mathcal{D}_{\max}}{\frac{2\mathbf{L}^3}{N} \mathcal{D}_{\max} + \frac{1}{3} \mathbf{L}^2 \frac{\sqrt{\mathcal{D}_{\max}}}{N} \cdot \alpha \sqrt{\mathcal{D}_{\max}}} \right) \leq 2 \exp \left(- \frac{\alpha^2 N}{4\mathbf{L}^3 + \frac{2\alpha}{3} \mathbf{L}^2} \right). \end{aligned}$$

Consequently by a union bound,

$$\begin{aligned} & \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \geq \alpha \sqrt{\mathcal{D}_{\max}} \right) \\ & \leq |\mathcal{X}| \cdot \mathbb{P} \left(\left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \geq \alpha \sqrt{\mathcal{D}_{\max}} \right) \\ & \leq 2 \exp \left(\log(5) \cdot N - \frac{\alpha^2 N}{4\mathbf{L}^3 + \frac{2\alpha}{3} \mathbf{L}^2} \right) \leq 2e^{-N}, \end{aligned} \tag{B.1.7}$$

where the last inequality holds by taking $\alpha = 5\mathbf{L}^2$.

B.1.3 Heavy couples

General Strategy

The contribution of *heavy* couples can be bounded separately,

$$\left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \leq \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \mathbf{A}_{ij} \mathbf{x}_i \mathbf{x}_j \right| + \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} (\mathbb{E} \mathbf{A})_{ij} \mathbf{x}_i \mathbf{x}_j \right|. \tag{B.1.8}$$

Meanwhile, for any pair $(i, j) \in [N]^2$,

$$(\mathbb{E}\mathbf{A})_{ij} \leq \sum_{\ell \in \mathcal{L}} \binom{N-2}{\ell-2} \frac{\mathcal{D}_{\max}^{(\ell)}}{\binom{N-1}{\ell-1}} \leq \sum_{\ell \in \mathcal{L}} \frac{(\ell-1)}{N-1} \cdot \mathcal{D}_{\max}^{(\ell)} = \frac{L\mathcal{D}_{\max}}{N-1}, \quad (\text{B.1.9})$$

and by definition of *heavy* pair

$$\begin{aligned} \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} (\mathbb{E}\mathbf{A})_{ij} x_i x_j \right| &= \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} (\mathbb{E}\mathbf{A})_{ij} \frac{x_i^2 x_j^2}{x_i x_j} \right| \leq \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \frac{L\mathcal{D}_{\max}}{N-1} \frac{x_i^2 x_j^2}{|x_i x_j|} \\ &\leq L\sqrt{\mathcal{D}_{\max}} \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} x_i^2 x_j^2 \leq L\sqrt{\mathcal{D}_{\max}}, \end{aligned} \quad (\text{B.1.10})$$

where we use the fact $\sum_{(i,j) \in \mathcal{H}(\mathbf{x})} x_i^2 x_j^2 \leq 1$ again. Therefore it suffices to show that, with high probability,

$$\left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \mathbf{A}_{ij} x_i x_j \right| = O(\sqrt{\mathcal{D}_{\max}}). \quad (\text{B.1.11})$$

Unfortunately, the contribution of *heavy* couples does not enjoy sufficient concentration to beat the cardinality of the net $\mathcal{N}$, which was the strategy we used in Appendix B.1.2. Instead, the key idea here is to prove the *discrepancy property* (Definition B.1.2) holds with high probability for the associated random regular graph. Essentially, the edge counts $\mathcal{E}_{\mathbf{A}}(\mathcal{S}, \mathcal{T}) := \sum_{u \in \mathcal{S}} \sum_{v \in \mathcal{T}} = \mathbf{1}_{\mathcal{S}}^{\top} \mathbf{A} \mathbf{1}_{\mathcal{T}}$ are not much larger than their expectation, uniformly over choices of $\mathcal{S}, \mathcal{T} \subset [N]$, which can be proved by using tail estimates of random variables $\mathcal{E}_{\mathbf{A}}(\mathcal{S}, \mathcal{T})$. Conditioning on the event that *discrepancy* property holds, one can show that the contribution of *heavy* couples is deterministically of order $O(\sqrt{\mathcal{D}_{\max}})$ by Lemma B.1.3, as long as row and column sums of $\mathbf{A}$ are bounded by $\sqrt{\mathcal{D}_{\max}}$ up to some constant.

Definition B.1.2 (Discrepancy property, **DP**). *Let $\mathbf{A}$ be an $N \times N$ matrix with non-negative*

entries. For subsets $\mathcal{S}, \mathcal{T} \subset [N]$, define

$$\mathcal{E}_{\mathbf{A}}(\mathcal{S}, \mathcal{T}) := \sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{T}} A_{ij}.$$

We say that $\mathbf{A}$ has the discrepancy property with parameter $\delta > 0$, $\kappa_1 > 1, \kappa_2 \geq 0$, denoted by $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$, if for all non-empty $\mathcal{S}, \mathcal{T} \subset [N]$, at least one of the following hold:

1. $\mathcal{E}_{\mathbf{A}}(\mathcal{S}, \mathcal{T}) \leq \kappa_1 \delta |\mathcal{S}| |\mathcal{T}|$;
2. $\mathcal{E}_{\mathbf{A}}(\mathcal{S}, \mathcal{T}) \cdot \log \left(\frac{\mathcal{E}_{\mathbf{A}}(\mathcal{S}, \mathcal{T})}{\delta |\mathcal{S}| |\mathcal{T}|} \right) \leq \kappa_2 (|\mathcal{S}| \vee |\mathcal{T}|) \cdot \log \left(\frac{eN}{|\mathcal{S}| \vee |\mathcal{T}|} \right)$.

Lemma B.1.3 ([71, Lemma 6.6], $\mathbf{DP} \implies$ heavy couples are small). *Let $\mathbf{A}$ be a non-negative symmetric $N \times N$ matrix. Suppose that*

1. *all row and column sums of $\mathbf{A}$ are bounded by d ,*
2. *$\mathbf{A}$ has $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ with $\delta = \frac{Cd}{N}$ for some $C > 0, \kappa_1 > 1, \kappa_2 \geq 0$,*

then for any $x \in \mathbb{S}^{N-1}$, deterministically,

$$\left| \sum_{(i,j) \in \mathcal{H}(x)} A_{ij} x_i x_j \right| \leq \beta_0 \sqrt{d},$$

where $\beta_0 = 16 + 32C(1 + \kappa_1) + 64\kappa_2(1 + \frac{2}{\kappa_1 \log \kappa_1})$.

Our model

To prove the *discrepancy* property in our model, we follow the discrepancy analysis in [100, 71], and consider the weighted graph associated with the adjacency matrix $\mathbf{A}$. By Lemma B.1.5, the *discrepancy* property follows with high probability if *uniform upper tail* property (Definition B.1.4) holds. Therefore, our tasks reduce to prove that row sums of $\mathbf{A}$ are bounded and **UUTP** holds with high probability, which are shown in Lemma B.1.6 and Lemma B.1.7 respectively.

Definition B.1.4 (Uniform upper tail property, **UUTP**). Let $\mathbf{A}$ be a random symmetric $N \times N$ matrix with non-negative entries. An $N \times N$ matrix $\mathbf{Q}$ is associated to $\mathbf{A}$ through $f_{\mathbf{Q}}(\mathbf{A}) := \sum_{i,j=1}^N Q_{ij} A_{ij}$. Define

$$\mu := f_{\mathbf{Q}}(\mathbb{E}\mathbf{A}) = \sum_{i,j=1}^N Q_{ij} \mathbb{E}A_{ij}, \quad \tilde{\sigma}^2 := f_{\mathbf{Q} \circ \mathbf{Q}}(\mathbb{E}\mathbf{A}) = \sum_{i,j=1}^N Q_{ij}^2 \mathbb{E}A_{ij},$$

where $\circ$ denotes the Hadamard (entrywise) matrix product. We say that $\mathbf{A}$ satisfies the uniform upper tail property **UUTP**(c_0, γ_0) with $c_0 > 0, \gamma_0 \geq 0$, if for any $a, t > 0$ and any symmetric $N \times N$ matrix $\mathbf{Q}$ satisfying $Q_{ij} \in [0, a]$ for all $i, j \in [N]$, we have

$$\mathbb{P}\left(f_{\mathbf{Q}}(\mathbf{A}) \geq (1 + \gamma_0)\mu + t\right) \leq \exp\left(-c_0 \frac{\tilde{\sigma}^2}{a^2} h\left(\frac{at}{\tilde{\sigma}^2}\right)\right).$$

where the function $h(x) := (1 + x) \log(1 + x) - x$ for all $x > -1$.

Lemma B.1.5 ([71, Lemma 6.4], **UUTP** $\implies$ **DP**). Let $\mathbf{A}$ be a symmetric $N \times N$ random matrix with non-negative entries. Assume that

1. $\mathbb{E}A_{ij} \leq \delta$ for all $i, j \in [N]$ for some $\delta > 0$,
2. $\mathbf{A}$ has **UUTP**(c_0, γ_0) with parameter $c_0, \gamma_0 > 0$.

Then for any $\theta > 0$, the discrepancy property **DP**($\delta, \kappa_1, \kappa_2$) holds for $\mathbf{A}$ with probability at least $1 - N^{-\theta}$ with $\kappa_1 = e^2(1 + \gamma_0)^2, \kappa_2 = \frac{2}{c_0}(1 + \gamma_0)(\theta + 4)$.

Lemma B.1.6. Denote $\mathcal{D}_{\max} := cq_N$ for some constant $c > 0$, where q_N is the magnitude parameter in (3.2.2). Then for some large enough constant $\beta_1 > 0$ ¹, satisfying $L^{-1}\beta_1 \log(L^{-1}\beta_1) - L^{-1}\beta_1 + 1 > 11/c$, there exists some $\tilde{c} > 0$ such that

$$\mathbb{P}(\mathbf{D}_v > \beta_1 \mathcal{D}_{\max}) \leq e^{-\tilde{c}\tilde{p}_N}. \quad (\text{B.1.12})$$

¹The existence of such β_1 is guaranteed since $g(x) = x \log(x) - x + 1$ takes its minimum at $x = 1$ and increases when $x > 1$.

At the same time, the following holds with probability at least $1 - Ne^{-11q_N}$,

$$\max_{1 \leq i \leq n} \left(\sum_{j=1}^N A_{ij} \right) \leq \beta_1 \mathcal{D}_{\max}. \quad (\text{B.1.13})$$

In particular, when $q_N = \log(N)$, (B.1.13) holds with probability at least $1 - N^{-10}$.

Proof of Lemma B.1.6. Recall D_v in (3.1.1), and for each fixed $v \in [N]$, we have

$$\sum_{j=1}^N A_{vj} = \sum_{j=1}^N \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ e \supset \{v, j\}}} \mathcal{A}_e^{(\ell)} = \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ e \ni v}} (\ell - 1) \mathcal{A}_e^{(\ell)} \asymp \sum_{\ell \in \mathcal{L}} \sum_{e \ni v} \mathcal{A}_e^{(\ell)} = D_v, \quad (\text{B.1.14})$$

since each $\ell \in \mathcal{L}$ is constant. From now on, we work on $\sum_{j=1}^N A_{vj}$ for convenience. By Markov inequality (Lemma D.3.1) and independence between hyperedges, (B.1.12) can be proved as follows

$$\begin{aligned} \mathbb{P} \left(\sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max} \right) &= \mathbb{P} \left(\sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ e \ni v}} (\ell - 1) \mathcal{A}_e^{(\ell)} > \beta_1 \mathcal{D}_{\max} \right) \\ &\leq \inf_{\theta > 0} \left(\exp(-\theta \beta_1 \mathcal{D}_{\max}) \prod_{\ell \in \mathcal{L}} \prod_{\substack{e \in \mathcal{E}_\ell \\ e \ni i}} \mathbb{E} \exp[\theta (\ell - 1) \mathcal{A}_e^{(\ell)}] \right) \\ &\leq \inf_{\theta > 0} \left(\exp(-\theta \beta_1 \mathcal{D}_{\max}) \prod_{\ell \in \mathcal{L}} \prod_{\substack{e \in \mathcal{E}_\ell \\ e \ni i}} \left[\frac{\mathcal{D}_{\max}^{(\ell)}}{\binom{N-1}{\ell-1}} e^{\theta(\ell-1)} + 1 - \frac{\mathcal{D}_{\max}^{(\ell)}}{\binom{N-1}{\ell-1}} \right] \right) \\ &\leq \inf_{\theta > 0} \left(\exp(-\theta \beta_1 \mathcal{D}_{\max}) \prod_{\ell \in \mathcal{L}} \prod_{\substack{e \in \mathcal{E}_\ell \\ e \ni i}} \exp \left[\frac{\mathcal{D}_{\max}^{(\ell)}}{\binom{N-1}{\ell-1}} (e^{\theta(\ell-1)} - 1) \right] \right), \quad (1 + x \leq e^x) \\ &\leq \inf_{\theta > 0} \left(\exp(-\theta \beta_1 \mathcal{D}_{\max}) \cdot \exp \left[\sum_{\ell \in \mathcal{L}} \mathcal{D}_{\max}^{(\ell)} (e^{\theta(\ell-1)} - 1) \right] \right), \quad (\text{at most } \binom{N-1}{\ell-1} \text{ edges contains } i) \\ &\leq \inf_{\theta > 0} \exp[-\mathcal{D}_{\max}(\theta \beta_1 - e^{\theta L} + 1)] = \exp \left[-\mathcal{D}_{\max} \left(\frac{\beta_1}{L} \log \left(\frac{\beta_1}{L} \right) - \frac{\beta_1}{L} + 1 \right) \right] \\ &\leq \exp \left[-q_N \cdot c \left(\frac{\beta_1}{L} \log \left(\frac{\beta_1}{L} \right) - \frac{\beta_1}{L} + 1 \right) \right] \leq e^{-11q_N} = e^{-\tilde{c}\rho_N}, \end{aligned}$$

where the last line holds since $\mathcal{D}_{\max} \asymp \rho_N$ as proved in Lemma 3.4.1. The proof of (B.1.13) is

completed by a union bound using (B.1.12)

$$\mathbb{P}\left(\max_{1 \leq i \leq n} \left(\sum_{j=1}^N A_{ij}\right) > \beta_1 \mathcal{D}_{\max}\right) \leq \sum_{i=1}^N \mathbb{P}\left(\sum_{j=1}^N A_{ij} > \beta_1 \mathcal{D}_{\max}\right) \leq N e^{-11q_N}.$$

The last part follows directly by plugging $q_N = \log(N)$. $\square$

Lemma B.1.7 (UUTP from Non-uniform Inhomogeneous Hypergraph). *Let $\mathbf{A}$ be the adjacency matrix of $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_m$, then $\mathbf{A}$ satisfies **UUTP**(c_0, γ_0) with $c_0 = L^{-2}, \gamma_0 = 0$.*

Proof of Lemma B.1.7. Note that

$$\begin{aligned} f_{\mathbf{Q}}(\mathbf{A}) - \mu &= \sum_{i,j=1}^N Q_{ij}(A_{ij} - \mathbb{E}A_{ij}) = \sum_{i,j=1}^N Q_{ij}W_{ij} = \sum_{i,j=1}^N Q_{ij} \left(\sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ e \supset \{i,j\}}} \mathcal{W}_e^{(\ell)} \right) \\ &= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathcal{W}_e^{(\ell)} \left(\sum_{\{i,j\} \subset e} Q_{ij} \right) = \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} Z_e^{(\ell)}, \end{aligned}$$

where $Z_e^{(\ell)} = \mathcal{W}_e^{(\ell)} \left(\sum_{\{i,j\} \subset e} Q_{ij} \right)$ are independent centered random variables upper bounded by $|Z_e^{(\ell)}| \leq \sum_{\{i,j\} \subset e} Q_{ij} \leq L^2 a$ for each $\ell \in \mathcal{L}$ since $Q_{ij} \in [0, a]$. Moreover, the variance can be written as

$$\begin{aligned} \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathbb{E}(Z_e^{(\ell)})^2 &:= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathbb{E}(\mathcal{W}_e^{(\ell)})^2 \left(\sum_{\{i,j\} \subset e} Q_{ij} \right)^2 \\ &\leq \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathbb{E}[\mathcal{A}_e^{(\ell)}] \cdot \ell^2 \sum_{\{i,j\} \subset e} Q_{ij}^2 \leq L^2 \sum_{i,j=1}^N Q_{ij}^2 \mathbb{E}A_{ij} = L^2 \tilde{\sigma}^2. \end{aligned}$$

where the last inequality holds since by definition $\mathbb{E}A_{ij} = \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ \{i,j\} \subset e}} \mathbb{E}[\mathcal{A}_e^{(\ell)}]$. Then by Bennett's inequality (Lemma D.3.5), we obtain

$$\mathbb{P}(f_{\mathbf{Q}}(\mathbf{A}) - \mu \geq t) \leq \exp\left(-\frac{\tilde{\sigma}^2}{L^2 a^2} h\left(\frac{at}{\tilde{\sigma}^2}\right)\right),$$

where the last line holds since $xh(\frac{1}{x}) = (1+x)\log(1+\frac{1}{x}) - 1$ decreases when $x > 0$. $\square$

Contribution of (B.1.11)

Lemma B.1.8. *If $\mathcal{D}_{\max} \geq c \log(N)$ for some constant $c > 0$, there exists some constant β such that with probability at least $1 - 2N^{-10}$,*

$$\left| \sum_{(i,j) \in \mathcal{H}(x)} (A_{ij} - \mathbb{E}A_{ij})x_i x_j \right| \leq \beta \sqrt{\mathcal{D}_{\max}}. \quad (\text{B.1.15})$$

Proof of Lemma B.1.8. Note that $\mathbb{E}A_{ij}$ is bounded ((B.1.9)) and $\mathbf{A}$ satisfies **UUTP**($L^{-2}, 0$) (Lemma B.1.7), then by Lemma B.1.5, for any $\theta > 0$, the *discrepancy* property **DP**($\delta, \kappa_1, \kappa_2$) holds for $\mathbf{A}$ with probability at least $1 - N^{-\theta}$, where

$$\delta = \frac{L}{N} \mathcal{D}_{\max}, \quad \kappa_1 = e^2(1 + \gamma_0)^2 = e^2, \quad \kappa_2 = \frac{2}{c_0}(1 + \gamma_0)(\theta + 4) = 2L^2(\theta + 4).$$

Let $\mathcal{E}_1$ be the event that all row sums of $\mathbf{A}$ are bounded by $\beta_1 \mathcal{D}_{\max}$, and $\mathcal{E}_2$ be the event that **DP**($\delta, \kappa_1, \kappa_2$) holds for $\mathbf{A}$, then $\mathbb{P}(\mathcal{E}_1 \cap \mathcal{E}_2) \geq 1 - 2N^{-10}$ where we take $\theta = 10$ for convenience. On the event $\mathcal{E}_1 \cap \mathcal{E}_2$, by Lemma B.1.3, (B.1.15) holds with $\beta = \beta_0 \sqrt{\beta_1} + L$, where β_0, β_1 satisfying

$$\begin{aligned} \beta_0 &= 16 + 32L(1 + \kappa_1) + 64\kappa_2 \left(1 + \frac{2}{\kappa_1 \log(\kappa_1)}\right) = 16 + 32L(1 + e^2) + 1792(1 + e^{-2})L^2, \\ \frac{\beta_1}{L} \log\left(\frac{\beta_1}{L}\right) - \frac{\beta_1}{L} + 1 &> \frac{11}{c}. \end{aligned}$$

□

B.1.4 Proof of Theorem 3.4.4

Following the strategy in Appendix B.1.1, for any $\mathcal{J} \subset [N]$,

$$\begin{aligned} & \|(\mathbf{A} - \mathbb{E}\mathbf{A})_{\mathcal{J}}\| \\ & \leq 2 \left(\sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} [(\mathbf{A} - \mathbb{E}\mathbf{A})_{\mathcal{J}}]_{ij} x_i x_j \right| + \sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} [(\mathbf{A} - \mathbb{E}\mathbf{A})_{\mathcal{J}}]_{ij} x_i x_j \right| \right), \end{aligned} \quad (\text{B.1.16})$$

we want to bound the contribution of *light* and *heavy* pairs separately.

Light couples

By (B.1.7), with probability at least $1 - 2e^{-N}$,

$$\sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} [(\mathbf{A} - \mathbb{E}\mathbf{A})_{\mathcal{J}}]_{ij} x_i x_j \right| \leq 5L^2 \sqrt{\mathcal{D}_{\max}}. \quad (\text{B.1.17})$$

A union bound proves that (B.1.17) holds for any given subset $\mathcal{J} \subset [N]$ with probability at least $1 - 2(e/2)^{-N}$, since the number of possible choices for $\mathcal{J}$ is at most 2^N , including the options $\mathcal{J} := \mathcal{J}_1, \mathcal{J}_2$ of our interests.

Heavy couples

Our goal is to prove

$$\sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} [(\mathbf{A} - \mathbb{E}\mathbf{A})_{\mathcal{J}}]_{ij} x_i x_j \right| \leq \beta \sqrt{\mathcal{D}_{\max}}, \quad (\text{B.1.18})$$

Similar to (B.1.10), deterministically for $\mathcal{J} := \mathcal{J}_1, \mathcal{J}_2$.

$$\left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} [(\mathbb{E}\mathbf{A})_{\mathcal{J}}]_{ij} \right| \leq L \sqrt{\mathcal{D}_{\max}}. \quad (\text{B.1.19})$$

To control the contribution of *heavy* pairs, it suffices to show that *discrepancy* holds for $\mathbf{A}_{\mathcal{J}}$ and all row sums of $\mathbf{A}_{\mathcal{J}}$ are bounded by $\beta_1 \mathcal{D}_{\max}$, according to Lemma B.1.3.

To verify the *discrepancy* property, it suffices to show that $\mathbf{UUTP}(L^{-2}, 0)$ holds for $\mathbf{A}_J$, which is true since $\mathbf{A}_J$ is obtained from $\mathbf{A}$ by restricting rows and columns to J . Then by Lemma B.1.5, for any $\theta > 0$, the *discrepancy* property $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ holds for $\mathbf{A}_J$ with probability at least $1 - N^{-\theta}$, where

$$\delta = \frac{L}{N} \mathcal{D}_{\max}, \quad \kappa_1 = e^2(1 + \gamma_0)^2 = e^2, \quad \kappa_2 = \frac{2}{c_0}(1 + \gamma_0)(\theta + 4) = 2L^2(\theta + 4).$$

The remaining is to verify the boundedness of row sums of $\mathbf{A}_J$.

- For J_1 , Lemma B.1.9 proves that row sums of each $v \in J_1$ is bounded by $\beta_1 \mathcal{D}_{\max}$ with probability at least $1 - N^{-c}$ for some constant $c > 0$ under the regime $\rho_N \gtrsim \log(N)$.
- Deterministically, it is true for $v \in J_2 := \{i \in [N], \sum_{j=1}^N A_{ij} \leq \beta_1 \mathcal{D}_{\max}\}$ by taking $\beta_1 = L$.

Let $\mathcal{E}_1$ be the event that all row sums of $\mathbf{A}_J$ are bounded by $\beta_1 \mathcal{D}_{\max}$, while $\mathcal{E}_2$ be the event that $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ holds for $\mathbf{A}_J$, then

- For J_1 , $\mathbb{P}(\mathcal{E}_1 \cap \mathcal{E}_2) \geq 1 - 2N^{-c}$, where we take $\theta = c$ for convenience.
- For J_2 , $\mathbb{P}(\mathcal{E}_1 \cap \mathcal{E}_2) \geq 1 - N^{-10}$, where we take $\theta = 10$ for convenience.

On the event $\mathcal{E}_1 \cap \mathcal{E}_2$, (B.1.18) holds with $\beta = \beta_0 \sqrt{\beta_1} + L$, $\beta_1 = L$,

$$\beta_0 = 16 + 32L(1 + \kappa_1) + 64\kappa_2(1 + \frac{2}{\kappa_1 \log(\kappa_1)}) = 16 + 32L(1 + e^2) + 1792(1 + e^{-2})L^2.$$

Lemma B.1.9. *If $\rho_N \gtrsim \log(N)$, then there exists some constant $c > 0$ such that row sums of each $v \in J_1$ is bounded by $\beta_1 \mathcal{D}_{\max}$ with probability at least $1 - N^{-c}$.*

Proof of Lemma B.1.9. Recall D_v in (3.1.1), which can be further written as

$$D_v = \sum_{\ell \in \mathcal{L}} \sum_{e \ni v} \mathcal{A}_e^{(\ell)} \asymp \sum_{\ell \in \mathcal{L}} \sum_{e \ni v} (\ell - 1) \mathcal{A}_e^{(\ell)} = \sum_{j=1}^N A_{vj}, \quad (\text{B.1.20})$$

since $\ell \geq 2$ is some constant for each $\ell \in \mathcal{L}$. From now on, we work on $\sum_{j=1}^N A_{vj}$ for convenience. In the following, we calculate the probability that the maximum vertex degree in $\mathcal{J}_1$ is greater than $\beta_1 \mathcal{D}_{\max}$. By the definition of $\mathcal{J}_1$,

$$\begin{aligned} \mathbb{P}\left(\max_{v \in \mathcal{J}_1} \left(\sum_{j=1}^N A_{vj}\right) > \beta_1 \mathcal{D}_{\max}\right) &= \mathbb{P}\left(\#\left\{v : \sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max}\right\} > \lfloor N \exp(-\bar{d}) \rfloor\right) \\ &\stackrel{(\text{Markov})}{\leq} \frac{1}{\lfloor N \exp(-\bar{d}) \rfloor + 1} \mathbb{E}\left(\#\left\{v : \sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max}\right\}\right) = \frac{\sum_{v \in [N]} \mathbb{E} \mathbb{1}\left\{v : \sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max}\right\}}{\lfloor N \exp(-\bar{d}) \rfloor + 1} \\ &\leq \frac{N}{\lfloor N \exp(-\bar{d}) \rfloor + 1} \max_{v \in [N]} \mathbb{P}\left(\sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max}\right) \leq \frac{1}{\exp(-\bar{d})} \max_{v \in [N]} \mathbb{P}\left(\sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max}\right). \end{aligned}$$

Note that $\mathcal{D}_{\max} \asymp \rho_N$ as proved in Lemma 3.4.1, then $\rho_N \gtrsim \log(N)$ is equivalent to say $\mathcal{D}_{\max} \geq c \log(N)$ for some constant $c > 0$.

- For the numerator, by going through the same procedure in Lemma B.1.6 and a union bound, one can prove that

$$\begin{aligned} \max_{v \in [N]} \mathbb{P}\left(\sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max}\right) &\leq \sum_{v \in [N]} \mathbb{P}\left(\sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max}\right) \\ &\leq \exp\left[-\log(N) \cdot c \left(\frac{\beta_1}{L} \log\left(\frac{\beta_1}{L}\right) - \frac{\beta_1}{L} + 1\right) + \log(N)\right]. \end{aligned}$$

- For the denominator, $\bar{d} \leq \xi_N + \xi_N/\sqrt{N}$ with probability tending to 1 according to Lemma 3.4.1.

Consequently, when $\rho_N \gtrsim \log(N)$, by taking β_1 large enough satisfying $\frac{\beta_1}{L} \log\left(\frac{\beta_1}{L}\right) - \frac{\beta_1}{L} + 1 - \frac{1}{c} - \frac{\xi_N}{c \log(N)} \geq 1$, we then have

$$\begin{aligned} \mathbb{P}\left(\max_{v \in \mathcal{J}_1} \left(\sum_{j=1}^N A_{vj}\right) > \beta_1 \mathcal{D}_{\max}\right) &\leq \exp(\bar{d}) \cdot \max_{v \in [N]} \mathbb{P}\left(\sum_{j=1}^N A_{vj} > \beta_1 \mathcal{D}_{\max}\right) \\ &\leq \exp\left[-\log(N) \cdot c \left(\frac{\beta_1}{L} \log\left(\frac{\beta_1}{L}\right) - \frac{\beta_1}{L} + 1 - \frac{1}{c} - \frac{\xi_N}{c \log(N)}\right)\right] \leq N^{-c}, \end{aligned}$$

which completes the proof. The existence of such β_1 is guaranteed since the function $f(x) =$

$x \log(x) - x$ takes its minimum at $x = 1$ with minimum $f(1) = -1$ and $c^{-1} = O(1)$, $\xi_N/[c \log(N)] \lesssim \xi_N/\rho_N = O(1)$. The requirement $\rho_N \gtrsim \log(N)$ is crucial since one needs to beat the union bound as in the proof of (B.1.13). $\square$

Putting together

Following the discussions above, (B.1.16) adapts the following upper bound

$$\|(\mathbf{A} - \mathbb{E}\mathbf{A})_J\| \leq C_{(3.4.4)} \sqrt{\mathcal{D}_{\max}},$$

where the constant $C_{(3.4.4)} := 2(\alpha + \beta)$, with $\alpha = 5L^2$, $\beta = \beta_0 \sqrt{\beta_1} + L$, $\beta_0 = 16 + 32L(1 + e^2) + 1792(1 + e^{-2})L^2$, and $\beta_1 = L$.

B.2 Proofs in Section 3.3

Proof of Lemma 3.1.1. This can be established by Hoeffding Lemma D.3.2. $\square$

Proof of Lemma 3.3.2. Note that $Y_v = k^*$. Without loss of generality, we assume $Q_{k^* \oplus \mathbf{w}}^{(\ell)} \lesssim Q_{j^* \oplus \mathbf{w}}^{(\ell)}$, then $Q_{k^* \oplus \mathbf{w}}^{(\ell)} \lesssim \mathbb{D}_{\text{GCH}} \cdot q_N$ according to Lemma D.1.1. Also, $N_{\mathbf{w}} \asymp N^{\ell-1}$ and $Q_{k^* \oplus \mathbf{w}}^{(\ell)} \asymp q_{k^* \oplus \mathbf{w}}^{(\ell)}/N^{\ell-1} = o(1)$, then $N_{\mathbf{w}} \log(1 - Q_{k^* \oplus \mathbf{w}}^{(\ell)}) = -(1 + o(1)) \cdot N_{\mathbf{w}} Q_{k^* \oplus \mathbf{w}}^{(\ell)}$. At the same time, $1 \ll$

$d_{j^*,k^*}^*(\mathbf{w}) \ll N$ in (3.3.4), then by Stirling's approximation Lemma D.1.3, for large enough N ,

$$\begin{aligned}
\mathbb{P}[D_{v,\mathbf{w}}^{(\ell)} = d_{j^*,k^*}^*(\mathbf{w})] &= \binom{N_{\mathbf{w}}}{d_{j^*,k^*}^*(\mathbf{w})} (Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{d_{j^*,k^*}^*(\mathbf{w})} (1 - Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{N_{\mathbf{w}} - d_{j^*,k^*}^*(\mathbf{w})} \\
&= \exp \left\{ \log \binom{N_{\mathbf{w}}}{d_{j^*,k^*}^*(\mathbf{w})} + d_{j^*,k^*}^*(\mathbf{w}) \cdot \log(Q_{k^* \oplus \mathbf{w}}^{(\ell)}) + [N_{\mathbf{w}} - d_{j^*,k^*}^*(\mathbf{w})] \cdot \log(1 - Q_{k^* \oplus \mathbf{w}}^{(\ell)}) \right\} \\
&= \exp \left\{ d_{j^*,k^*}^*(\mathbf{w}) \log \left(\frac{N_{\mathbf{w}}}{d_{j^*,k^*}^*(\mathbf{w})} \right) + d_{j^*,k^*}^*(\mathbf{w}) + d_{j^*,k^*}^*(\mathbf{w}) \log \left(\frac{Q_{k^* \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{k^* \oplus \mathbf{w}}^{(\ell)}} \right) \right. \\
&\quad \left. + N_{\mathbf{w}} \cdot \log(1 - Q_{k^* \oplus \mathbf{w}}^{(\ell)}) + o(q_{k^* \oplus \mathbf{w}}^{(\ell)}) \right\} \\
&= \exp \left\{ d_{j^*,k^*}^*(\mathbf{w}) \cdot \log \left(\frac{N_{\mathbf{w}} Q_{k^* \oplus \mathbf{w}}^{(\ell)}}{d_{j^*,k^*}^*(\mathbf{w}) \cdot (1 - Q_{k^* \oplus \mathbf{w}}^{(\ell)})} \right) - N_{\mathbf{w}} \cdot Q_{k^* \oplus \mathbf{w}}^{(\ell)} + d_{j^*,k^*}^*(\mathbf{w}) + o(q_{k^* \oplus \mathbf{w}}^{(\ell)}) \right\} \\
&= \exp \left\{ -\bar{N}_{\mathbf{w}} \left[t^*(Q_{j^* \oplus \mathbf{w}}^{(\ell)})^{t^*} (Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{1-t^*} \log \frac{Q_{j^* \oplus \mathbf{w}}^{(\ell)}}{Q_{k^* \oplus \mathbf{w}}^{(\ell)}} + Q_{k^* \oplus \mathbf{w}}^{(\ell)} - (Q_{j^* \oplus \mathbf{w}}^{(\ell)})^{t^*} (Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{1-t^*} + o(Q_{k^* \oplus \mathbf{w}}^{(\ell)}) \right] \right\}
\end{aligned}$$

where in the last inequality, we plug in $d_{j^*,k^*}^*(\mathbf{w})$ in (3.3.4), and replace $N_{\mathbf{w}}$ with $\bar{N}_{\mathbf{w}}$ according to Lemma 3.1.1. Therefore, by hyperedge independence, we have

$$\begin{aligned}
&\prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}_{j^*,k^*}^{(\ell)}} \mathbb{P}[D_{v,\mathbf{w}}^{(\ell)} = d_{j^*,k^*}^*(\mathbf{w})] \\
&= \exp \left\{ - \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{j^*,k^*}^{(\ell)}} \bar{N}_{\mathbf{w}} \left[t^*(Q_{j^* \oplus \mathbf{w}}^{(\ell)})^{t^*} (Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{1-t^*} \log \left(\frac{Q_{j^* \oplus \mathbf{w}}^{(\ell)}}{Q_{k^* \oplus \mathbf{w}}^{(\ell)}} \right) \right. \right. \\
&\quad \left. \left. + Q_{k^* \oplus \mathbf{w}}^{(\ell)} - (Q_{j^* \oplus \mathbf{w}}^{(\ell)})^{t^*} (Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{1-t^*} + o(Q_{k^* \oplus \mathbf{w}}^{(\ell)}) \right] \right\} \\
&= \exp \left\{ - \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{j^*,k^*}^{(\ell)}} \bar{N}_{\mathbf{w}} \cdot \left[t^* Q_{j^* \oplus \mathbf{w}}^{(\ell)} + (1 - t^*) Q_{k^* \oplus \mathbf{w}}^{(\ell)} - (Q_{j^* \oplus \mathbf{w}}^{(\ell)})^{t^*} (Q_{k^* \oplus \mathbf{w}}^{(\ell)})^{1-t^*} + o(Q_{k^* \oplus \mathbf{w}}^{(\ell)}) \right] \right\} \\
&= \exp(-(1 + o(1)) \cdot \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N),
\end{aligned}$$

where the second to last equality holds according to the first order condition (3.3.3). $\square$

Proof of Lemma 3.3.4. We rewrite $\mathbb{D}_{\text{GKL}}(j, k)$ as

$$\begin{aligned} \min_{\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \left\{ y_{\mathbf{w}}^{(\ell)} \log \frac{y_{\mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + (1 - y_{\mathbf{w}}^{(\ell)}) \log \frac{1 - y_{\mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \right\} \\ \text{s.t. } \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \left\{ \mathbb{D}_{\text{KL}}(y_{\mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}) - \mathbb{D}_{\text{KL}}(y_{\mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) \right\} \leq 0. \end{aligned} \quad (\text{B.2.2})$$

By [47], the associated Lagrangian is defined as

$$\begin{aligned} L(\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}, \lambda) := \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \left\{ y_{\mathbf{w}}^{(\ell)} \log \frac{y_{\mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + (1 - y_{\mathbf{w}}^{(\ell)}) \log \frac{1 - y_{\mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \right\} \\ + \lambda \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \left\{ y_{\mathbf{w}}^{(\ell)} \log \frac{Q_{k \oplus \mathbf{w}}^{(\ell)}}{Q_{j \oplus \mathbf{w}}^{(\ell)}} + (1 - y_{\mathbf{w}}^{(\ell)}) \log \frac{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{j \oplus \mathbf{w}}^{(\ell)}} \right\}. \end{aligned}$$

The optimal choice of $\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}$, which minimizes $L(\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}, \lambda)$, satisfies the first order condition

$$\frac{\partial L(\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}, \lambda)}{\partial y_{\mathbf{w}}^{(\ell)}} = \bar{N}_{\mathbf{w}} \cdot \left[\left(\log \frac{y_{\mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} - \log \frac{1 - y_{\mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \right) + \lambda \left(\log \frac{Q_{k \oplus \mathbf{w}}^{(\ell)}}{Q_{j \oplus \mathbf{w}}^{(\ell)}} - \log \frac{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{j \oplus \mathbf{w}}^{(\ell)}} \right) \right] = 0.$$

Note that $Q_{k \oplus \mathbf{w}}^{(\ell)} = o(1)$ in (3.2.2), then the solution to (B.2.2) has to satisfy $y_{\mathbf{w}}^{(\ell)} = o(1)$ for each $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$, otherwise optimal value of (B.2.2) goes to infinity since $y_{\mathbf{w}}^{(\ell)} \log \frac{y_{\mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}}$ would diverge. Thus $\log \frac{1 - y_{\mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}}$ and $\log \frac{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{j \oplus \mathbf{w}}^{(\ell)}}$ converge to 0 as N grows. Then (B.2.2) is minimized at

$$y_{\mathbf{w}}^{(\ell)} = (Q_{j \oplus \mathbf{w}}^{(\ell)})^{\lambda} \cdot (Q_{k \oplus \mathbf{w}}^{(\ell)})^{1-\lambda} \cdot (1 + o(1)), \text{ for each } \mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}, \ell \in \mathcal{L}.$$

Consider the Lagrange dual function

$$\begin{aligned}
g(\lambda) &:= \min_{\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}} L(\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}, \lambda) \\
&= \min_{\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \left\{ y_{\mathbf{w}}^{(\ell)} \log \frac{(Q_{k \oplus \mathbf{w}}^{(\ell)})^{\lambda-1} y_{\mathbf{w}}^{(\ell)}}{(Q_{j \oplus \mathbf{w}}^{(\ell)})^{\lambda}} + (1 - y_{\mathbf{w}}^{(\ell)}) \log \frac{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})^{\lambda-1} (1 - y_{\mathbf{w}}^{(\ell)})}{(1 - Q_{j \oplus \mathbf{w}}^{(\ell)})^{\lambda}} \right\} \\
&= \min_{\{\mathbf{y}^{(\ell)}\}_{\ell \in \mathcal{L}}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot (1 - y_{\mathbf{w}}^{(\ell)}) \cdot \left[(\lambda - 1) \log(1 - Q_{k \oplus \mathbf{w}}^{(\ell)}) + \log(1 - y_{\mathbf{w}}^{(\ell)}) - \lambda \log(1 - Q_{j \oplus \mathbf{w}}^{(\ell)}) \right] \\
&= \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \left[\lambda Q_{j \oplus \mathbf{w}}^{(\ell)} + (1 - \lambda) Q_{k \oplus \mathbf{w}}^{(\ell)} - (Q_{j \oplus \mathbf{w}}^{(\ell)})^{\lambda} \cdot (Q_{k \oplus \mathbf{w}}^{(\ell)})^{1-\lambda} \right],
\end{aligned}$$

where in the last two equalities we use the facts $y_{\mathbf{w}}^{(\ell)} = o(1)$, $Q_{j \oplus \mathbf{w}}^{(\ell)} = o(1)$ and $\lim_{x \rightarrow 0} \log(1+x) = x$. By taking maximum of $g(\lambda)$, we have

$$\begin{aligned}
\mathbb{D}_{\text{GKL}}(j, k) &= (\text{B.2.2}) = \max_{\lambda \in [0,1]} g(\lambda) \\
&= \max_{\lambda \in [0,1]} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \left[\lambda Q_{j \oplus \mathbf{w}}^{(\ell)} + (1 - \lambda) Q_{k \oplus \mathbf{w}}^{(\ell)} - (Q_{j \oplus \mathbf{w}}^{(\ell)})^{\lambda} \cdot (Q_{k \oplus \mathbf{w}}^{(\ell)})^{1-\lambda} \right] \\
&\stackrel{(3.2.2)}{=} q_N \cdot \max_{\lambda \in [0,1]} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \frac{\bar{N}_{\mathbf{w}}}{\binom{N-1}{\ell-1}} \cdot \left[\lambda \mathcal{P}_{j \oplus \mathbf{w}}^{(\ell)} + (1 - \lambda) \mathcal{P}_{k \oplus \mathbf{w}}^{(\ell)} - (\mathcal{P}_{j \oplus \mathbf{w}}^{(\ell)})^{\lambda} \cdot (\mathcal{P}_{k \oplus \mathbf{w}}^{(\ell)})^{1-\lambda} \right] \\
&= (1 + o(1)) \cdot \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N.
\end{aligned}$$

□

Proof of Lemma 3.3.5. Proof by contradiction. Suppose that there is no such $\{\mathbf{p}^{(\ell)}\}_{\ell \in \mathcal{L}}$ satisfying (3.3.13). Then for each $\{\mathbf{p}^{(\ell)}\}_{\ell \in \mathcal{L}}$, without loss of generality, we assume

$$\begin{aligned}
\mathbb{D}_{\text{GKL}}(j^*, k^*) &= \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mathbf{p}_{\mathbf{w}}^{(\ell)} \parallel Q_{j^* \oplus \mathbf{w}}^{(\ell)}) \\
&> \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mathbf{p}_{\mathbf{w}}^{(\ell)} \parallel Q_{k^* \oplus \mathbf{w}}^{(\ell)}).
\end{aligned}$$

We pick a sequence $\{\mathbf{p}^{(\ell)}\}_{\ell \in \mathcal{L}}$, where $\mathbf{p}^{(\ell)} = (\mathbf{p}_{\mathbf{w}}^{(\ell)})_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}}$ for each ℓ . There exists some $\mathbf{w}_0 \in \mathcal{W}C_K^{(\ell-1)}$ such that $\mathbb{D}_{\text{KL}}(\mathbf{p}_{\mathbf{w}_0}^{(\ell)} \parallel \mathbf{Q}_{j^* \oplus \mathbf{w}_0}^{(\ell)}) > \mathbb{D}_{\text{KL}}(\mathbf{p}_{\mathbf{w}_0}^{(\ell)} \parallel \mathbf{Q}_{k^* \oplus \mathbf{w}_0}^{(\ell)}) \geq 0$, due to the non-negativity of KL divergence, with their difference $f(\mathbf{p}_{\mathbf{w}_0}^{(\ell)}) > 0$ defined as

$$f(\mathbf{p}) := \mathbb{D}_{\text{KL}}(\mathbf{p} \parallel \mathbf{Q}_{j^* \oplus \mathbf{w}_0}^{(\ell)}) - \mathbb{D}_{\text{KL}}(\mathbf{p} \parallel \mathbf{Q}_{k^* \oplus \mathbf{w}_0}^{(\ell)}) = p \cdot \log \frac{Q_{k^* \oplus \mathbf{w}_0}^{(\ell)}}{Q_{j^* \oplus \mathbf{w}_0}^{(\ell)}} + (1 - p) \cdot \log \frac{1 - Q_{k^* \oplus \mathbf{w}_0}^{(\ell)}}{1 - Q_{j^* \oplus \mathbf{w}_0}^{(\ell)}}.$$

By continuity of $f(\mathbf{p})$ and $\mathbb{D}_{\text{KL}}(\mathbf{p} \parallel \mathbf{Q}_{j^* \oplus \mathbf{w}_0}^{(\ell)})$, one is able to find some $\mathbf{t}_{\mathbf{w}_0}^{(\ell)}$ such that $\mathbb{D}_{\text{KL}}(\mathbf{t}_{\mathbf{w}_0}^{(\ell)} \parallel \mathbf{Q}_{j^* \oplus \mathbf{w}_0}^{(\ell)}) < \mathbb{D}_{\text{KL}}(\mathbf{p}_{\mathbf{w}_0}^{(\ell)} \parallel \mathbf{Q}_{j^* \oplus \mathbf{w}_0}^{(\ell)})$ but $f(\mathbf{t}_{\mathbf{w}_0}^{(\ell)})$ remains positive. As a result, for this fixed $\ell \in \mathcal{L}$, there exists some sequence $(\mathbf{t}_{\mathbf{w}}^{(\ell)})_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}}$ with $\mathbf{t}_{\mathbf{w}}^{(\ell)} = \mathbf{p}_{\mathbf{w}}^{(\ell)}$ for all $\mathbf{w} \neq \mathbf{w}_0$, and $\mathbf{t}_{\mathbf{w}_0}^{(\ell)}$ satisfying the property as before. Consequently, one could construct a new sequence $\{\mathbf{t}^{(\ell)}\}_{\ell \in \mathcal{L}}$ satisfying

$$\begin{aligned} \mathbb{D}_{\text{GKL}}(j^*, k^*) &= \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mathbf{p}_{\mathbf{w}}^{(\ell)} \parallel \mathbf{Q}_{j^* \oplus \mathbf{w}}^{(\ell)}) \\ &> \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mathbf{t}_{\mathbf{w}}^{(\ell)} \parallel \mathbf{Q}_{j^* \oplus \mathbf{w}}^{(\ell)}) \\ &> \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mathbf{t}_{\mathbf{w}}^{(\ell)} \parallel \mathbf{Q}_{k^* \oplus \mathbf{w}}^{(\ell)}), \end{aligned}$$

which contradicts to the minimum property in definition of $\mathbb{D}_{\text{GKL}}(j^*, k^*)$ in (3.3.11). $\square$

Proof of Lemma 3.3.7. Note that the nodes are interchangeable under Φ , and

$$\mathbb{P}_{\Psi}(\Lambda \leq f(N)) = \mathbb{P}_{\Psi}(\Lambda \leq f(N), v^* \in \mathcal{V}_{\text{err}}) + \mathbb{P}_{\Psi}(\Lambda \leq f(N), v^* \notin \mathcal{V}_{\text{err}}). \quad (\text{B.2.3})$$

The first term on the right hand side of (B.2.3) can be upper bounded by the following *change of*

measure argument,

$$\begin{aligned}\mathbb{P}_\Psi(\Lambda \leq f(N), v^* \in \mathcal{V}_{\text{err}}) &= \int_{\Lambda \leq f(N), v^* \in \mathcal{V}_{\text{err}}} d\mathbb{P}_\Psi = \int_{\Lambda \leq f(N), v^* \in \mathcal{V}_{\text{err}}} \exp(\Lambda) d\mathbb{P}_\Phi \\ &\leq \exp(f(N)) \cdot \mathbb{P}_\Phi(\Lambda \leq f(N), v^* \in \mathcal{V}_{\text{err}}) \leq \exp(f(N)) \cdot \mathbb{P}_\Phi(v^* \in \mathcal{V}_{\text{err}}) \leq \frac{e^{f(N)} \cdot \mathbb{E}_\Phi[\eta_N]}{\alpha_{j^*} + \alpha_{k^*}},\end{aligned}$$

where the last inequality holds since we can not distinguish v^* and any other node $v \in \mathcal{V}_{y_{v^*}}$, indeed

$$\begin{aligned}\mathbb{P}_\Phi(v^* \in \mathcal{V}_{\text{err}}) &= \mathbb{P}_\Phi(v \in \mathcal{V}_{\text{err}} | v \in \mathcal{V}_{j^*} \cup \mathcal{V}_{k^*}) \\ &\leq \frac{\mathbb{P}_\Phi(v \in \mathcal{V}_{\text{err}})}{\mathbb{P}_\Phi(v \in \mathcal{V}_{j^*} \cup \mathcal{V}_{k^*})} = \frac{\mathbb{E}_\Phi[|\mathcal{V}_{\text{err}}|]}{(\alpha_{j^*} + \alpha_{k^*})n} = \frac{\mathbb{E}_\Phi[\eta_N]}{\alpha_{j^*} + \alpha_{k^*}}.\end{aligned}$$

Now, we focus on the second term on the right hand side of (B.2.3). Under the perturbed model Ψ in 3.3.6, the observed edges do not depend on whether v^* belongs to $\mathcal{V}_{j^*}$ or $\mathcal{V}_{k^*}$, since the generation of edge $e \ni v^*$ only relies on the distribution of $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$ for the rest $\ell - 1$ nodes. To determine the community membership of v^* , one can only use the information from the edges containing v^* . Therefore, it doesn't matter whether v^* truly belongs to $\mathcal{V}_{j^*}$ or $\mathcal{V}_{k^*}$ when assigning v^* to $\widehat{\mathcal{V}}_{j^*}$. By symmetry, the two conditional probabilities should be the same, and the following holds

$$\begin{aligned}\mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_{j^*} | v^* \in \mathcal{V}_{j^*}) &= \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_{j^*} | v^* \in \mathcal{V}_{k^*}), \\ \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_{k^*} | v^* \in \mathcal{V}_{j^*}) &= \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_{k^*} | v^* \in \mathcal{V}_{k^*}).\end{aligned}\tag{B.2.4}$$

Then the second term on the right hand side of (B.2.3) can be bounded as follows

$$\begin{aligned}
& \mathbb{P}_\Psi(\Lambda \leq f(N), v^* \notin \mathcal{V}_{\text{err}}) \\
& \leq \mathbb{P}_\Psi(v^* \notin \mathcal{V}_{\text{err}}) \\
& = \frac{\alpha_{j^*}}{\alpha_{j^*} + \alpha_{k^*}} \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_j | v^* \in \mathcal{V}_{j^*}) + \frac{\alpha_{k^*}}{\alpha_{j^*} + \alpha_{k^*}} \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_k | v^* \in \mathcal{V}_{k^*}) \\
& \stackrel{\text{(B.2.4)}}{=} \frac{\alpha_{j^*}}{\alpha_{j^*} + \alpha_{k^*}} \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_j | v^* \in \mathcal{V}_{j^*}) + \frac{\alpha_{k^*}}{\alpha_{j^*} + \alpha_{k^*}} \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_k | v^* \in \mathcal{V}_{j^*}) \\
& \leq \frac{\alpha_{j^*}}{\alpha_{j^*} + \alpha_{k^*}} \left(\mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_j | v^* \in \mathcal{V}_{j^*}) + \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_k | v^* \in \mathcal{V}_{j^*}) \right) \leq \frac{\alpha_{j^*}}{\alpha_{j^*} + \alpha_{k^*}},
\end{aligned}$$

where the last line holds since $j^* < k^*$ and $\alpha_1 \geq \dots \geq \alpha_K$ implies $\alpha_{j^*} \geq \alpha_{k^*}$, together with the fact $\mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_{j^*} | v^* \in \mathcal{V}_{j^*}) + \mathbb{P}_\Psi(v^* \in \widehat{\mathcal{V}}_{k^*} | v^* \in \mathcal{V}_{j^*}) \leq 1$. Therefore the desired result follows. $\square$

Proof of Lemma 3.3.8. Since Λ in (3.3.14) is a summation of Bernoulli random variables, a standard concentration argument shows that $\sqrt{\text{Var}_\Psi(\Lambda)}/\mathbb{E}_\Psi(\Lambda) = o(1)$, thus $\mathbb{E}_\Psi(\Lambda)$ dominates. We present the details by evaluating $\mathbb{E}_\Psi[\Lambda]$ and $\text{Var}_\Psi(\Lambda)$. First, the randomness of Λ comes from edges $\mathcal{A}_e^{(\ell)}$, then by linearity of expectation,

$$\begin{aligned}
\mathbb{E}_\Psi[\Lambda] &= \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell, e \ni v^* \\ \mathbf{y}(e \setminus v^*) = \mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}}} \mathbb{E} \left(\mathcal{A}_e^{(\ell)} \log \left(\frac{\mathbf{p}_\mathbf{w}^{(\ell)}}{Q_{\mathbf{y}_{v^*} \oplus \mathbf{w}}^{(\ell)}} \right) + (1 - \mathcal{A}_e^{(\ell)}) \log \left(\frac{1 - \mathbf{p}_\mathbf{w}^{(\ell)}}{1 - Q_{\mathbf{y}_{v^*} \oplus \mathbf{w}}^{(\ell)}} \right) \right) \\
&= \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell, e \ni v^* \\ \mathbf{y}(e \setminus v^*) = \mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}}} \left(\mathbf{p}_\mathbf{w}^{(\ell)} \log \left(\frac{\mathbf{p}_\mathbf{w}^{(\ell)}}{Q_{\mathbf{y}_{v^*} \oplus \mathbf{w}}^{(\ell)}} \right) + (1 - \mathbf{p}_\mathbf{w}^{(\ell)}) \log \left(\frac{1 - \mathbf{p}_\mathbf{w}^{(\ell)}}{1 - Q_{\mathbf{y}_{v^*} \oplus \mathbf{w}}^{(\ell)}} \right) \right) \\
&= \sum_{\ell \in \mathcal{L}} \sum_{\ell \in \mathcal{W}C_K^{(\ell-1)}} N_\mathbf{w} \cdot \mathbb{D}_{\text{KL}}(\mathbf{p}_\mathbf{w}^{(\ell)} \parallel Q_{\mathbf{y}_{v^*} \oplus \mathbf{w}}^{(\ell)}) = (1 + o(1)) \cdot \mathbb{D}_{\text{GKL}} = (1 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N,
\end{aligned}$$

where the last two equalities hold by Lemmas 3.1.1, 3.3.4, 3.3.5. On the other hand, using the facts $\text{Var}(c\mathcal{A}_e^{(\ell)}) = c^2 \text{Var}(\mathcal{A}_e^{(\ell)})$ and $\text{Var}(\mathcal{A}_e^{(\ell)} + c) = \text{Var}(\mathcal{A}_e^{(\ell)})$ for any deterministic $c \in \mathbb{R}$, we

have

$$\mathbb{V}\text{ar}_\Psi(\Lambda) = \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} p_{\mathbf{w}}^{(\ell)} (1 - p_{\mathbf{w}}^{(\ell)}) \cdot \left[\log \left(\frac{p_{\mathbf{w}}^{(\ell)} \cdot (1 - Q_{\mathbf{y}^* \oplus \mathbf{w}}^{(\ell)})}{Q_{\mathbf{y}^* \oplus \mathbf{w}}^{(\ell)} \cdot (1 - p_{\mathbf{w}}^{(\ell)})} \right) \right]^2.$$

Note that $Q_{\mathbf{w}}^{(\ell)} = o(1)$ for each $\mathbf{w} \in \mathcal{W}C_K^{(\ell)}$, $\ell \in \mathcal{L}$, then by the argument above, the so constructed $\{\mathbf{p}^{(\ell)}\}_{\ell \in \mathcal{L}}$ in Lemma 3.3.5 satisfies $p_{\mathbf{w}}^{(\ell)} = o(1)$ and $N_{\mathbf{w}} p_{\mathbf{w}}^{(\ell)} \gg 1$. Then the desired result follows since

$$\sqrt{\mathbb{V}\text{ar}_\Psi(\Lambda)} / \mathbb{E}_\Psi[\Lambda] \lesssim \frac{1}{\sqrt{\mathbb{D}_{\text{GCH}} \cdot q_N}} = o(1).$$

□

B.3 Proofs in Section 3.4

Proof of Lemma 3.4.1. For the first part, $\xi_N \asymp \rho_N$ is trivially true when Assumption 3.2.2 holds. On the other hand, due to the fact $\alpha_1, \dots, \alpha_K = \Omega(1)$, the conclusion $\xi_N \asymp \rho_N$ still holds since the dense part inside the summation $\xi_N = N^{-1} \sum_{v=1}^N \mathbb{E}[D_v]$ distinguishes itself.

For the second part, by Bernstein D.3.4, we have

$$\begin{aligned} \mathbb{P}(\bar{d} - \xi_N \geq \xi_N / \sqrt{N}) &= \mathbb{P}\left(\sum_{v \in \mathcal{V}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} (D_{v,\mathbf{w}}^{(\ell)} - \mathbb{E}D_{v,\mathbf{w}}^{(\ell)}) \geq \sqrt{N} \xi_N \right) \\ &\leq 2 \exp\left(- \frac{N \cdot \xi_N^2 / 2}{M \binom{K+L-1}{\ell-1} \xi_N + \sqrt{N} \xi_N / 3} \right). \end{aligned}$$

Then $\bar{d} - \xi_N \leq \xi_N / \sqrt{N}$ with probability at least $1 - e^{-N^{1/2}}$ as long as $\xi_N = \omega(1)$.

For the last part, we assume $q_N \geq \log(N)$ in (3.2.2) from now on. We are going to present some large deviation results. Let $\tilde{d} = \max_{v \in \mathcal{V}} d_v$ denote realized maximum degree. Define the random variable $\tilde{D}$ by $\tilde{D} = \max_{v \in \mathcal{V}} D_v$, then $\mathbb{E}\tilde{D} = \mathbb{E} \max_{v \in \mathcal{V}} D_v$. Define $W_v := D_v - \mathbb{E}D_v$, then following (3.1.1), $W_v = \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} W_{v,\mathbf{w}}^{(\ell)}$ where $W_{v,\mathbf{w}}^{(\ell)}$ are mean zero random variables,

and define

$$\tilde{W} := \max_{v \in \mathcal{V}} W_v \leq \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} \max_{v \in \mathcal{V}} W_{v, \mathbf{w}}^{(\ell)}. \quad (\text{B.3.1})$$

Note that $D_{v, \mathbf{w}}^{(\ell)} \sim \text{Bin}(N_{\mathbf{w}}, Q_{\mathbf{k} \oplus \mathbf{w}}^{(\ell)})$ when $Y_v = k$, then $\sigma^2 = \mathbb{E}(W_{v, \mathbf{w}}^{(\ell)})^2 = N_{\mathbf{w}} Q_{\mathbf{k} \oplus \mathbf{w}}^{(\ell)} (1 - Q_{\mathbf{k} \oplus \mathbf{w}}^{(\ell)}) \leq \mathbb{E} D_{v, \mathbf{w}}^{(\ell)} < \mathbb{E} \tilde{D}$. By Bernstein D.3.4, for any $\varepsilon > 0$, we have

$$\mathbb{P}[W_{v, \mathbf{w}}^{(\ell)} \geq (1 - \varepsilon) \mathbb{E} \tilde{D}] \leq \exp\left(-\frac{3(1 - \varepsilon)^2 (\mathbb{E} \tilde{D})^2}{6\sigma^2 + 2\sqrt{(1 - \varepsilon) \mathbb{E} \tilde{D}}}\right) \leq \exp\left(-\frac{1}{2}(1 - \varepsilon)^2 \mathbb{E} \tilde{D}\right) \quad (\text{B.3.2})$$

Note that $\mathbb{E} \tilde{D} \geq c q_N$ for some $c > 0$ under regime (3.2.2). By the union bounds

$$\begin{aligned} \mathbb{P}(\tilde{W} \geq (1 - \varepsilon) \mathbb{E} \tilde{D}) &\leq \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} \mathbb{P}(\max_{v \in \mathcal{V}} W_{v, \mathbf{w}}^{(\ell)} \geq (1 - \varepsilon) \mathbb{E} \tilde{D}) \\ &\leq \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} \sum_{v=1}^N \mathbb{P}[W_{v, \mathbf{w}}^{(\ell)} \geq (1 - \varepsilon) \mathbb{E} \tilde{D}] \leq (\ell - 1) \cdot \binom{\ell + K - 1}{K - 1} \exp\left(\log(N) - \frac{1}{2}(1 - \varepsilon)^2 \mathbb{E} \tilde{D}\right) \\ &\lesssim \exp\left\{-\log(N) \cdot [(1 - \varepsilon)^2 c^2 / 2 - 1]\right\} \xrightarrow{N \rightarrow \infty} 0, \quad \text{when } c > \sqrt{2}/(1 - \varepsilon), \end{aligned}$$

where the last inequality holds when we choose $q_N = \log(N)$ and $c > \sqrt{2}/(1 - \varepsilon)$. Note that $\mathbb{D}_{\text{GCH}} > 1$ implies $c > 2^{\ell-1}$ for $\ell \in \mathcal{L}$, then it just a matter of choosing proper ε . Similarly, $\tilde{W} \geq -(1 - \varepsilon) \mathbb{E} \tilde{D}$ with probability tending to 1. Note that $|\tilde{D} - \mathbb{E} \tilde{D}| \leq |\tilde{W}| \leq (1 - \varepsilon) \mathbb{E} \tilde{D}$ with high probability, then

$$\varepsilon \mathbb{E} \tilde{D} \leq \tilde{D} \leq 2 \mathbb{E} \tilde{D}. \quad (\text{B.3.3})$$

We now refer to the magnitude of $\mathbb{E} \tilde{D}$. By Jensen's inequality, for any $t > 0$, we have

$$\exp(t \mathbb{E} \tilde{D}) \leq \mathbb{E} \exp(t \tilde{D}) = \mathbb{E} \exp(t \max_{v \in \mathcal{V}} D_v) \leq \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} \mathbb{E} \exp(t \max_{v \in \mathcal{V}} D_{v, \mathbf{w}}^{(\ell)}), \quad (\text{B.3.4})$$

since $\tilde{D} \leq \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} \max_{v \in \mathcal{V}} D_{v, \mathbf{w}}^{(\ell)}$. By the union bound,

$$\mathbb{E} \exp(t \max_{v \in \mathcal{V}} D_{v, \mathbf{w}}^{(\ell)}) \leq \sum_{v=1}^N \mathbb{E} \exp(t D_{v, \mathbf{w}}^{(\ell)}) = \sum_{v=1}^N [Q_{\mathbf{y}(v) \oplus \mathbf{w}}^{(\ell)} e^t + (1 - Q_{\mathbf{y}(v) \oplus \mathbf{w}}^{(\ell)})]^{N_{\mathbf{w}}} \quad (\text{B.3.5})$$

$$\leq N[1 + Q_{\max}^{(\ell)}(e^t - 1)]^{N_{\mathbf{w}}} \leq \exp[\log(N) + N_{\mathbf{w}} Q_{\max}^{(\ell)}(e^t - 1)], \quad (\text{B.3.6})$$

since by definition $Q_{\max}^{(\ell)} = \max_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell)}}} Q_{\mathbf{w}}^{(\ell)}$ and $1 + x \leq e^x$. We adapt the notation in (3.2.2) and (3.4.2), then $\mathcal{D}_{\max} = \sum_{\ell \in \mathcal{L}} N_{\mathbf{w}} Q_{\max}^{(\ell)}$. Note that $N_{\mathbf{w}} Q_{\max}^{(\ell)} \leq \mathcal{P}_{\max}^{(\ell)} q_N$, By taking $t = 1$, we have

$$\mathbb{E} \tilde{D} \leq \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} (\log(N) + (e - 1) N_{\mathbf{w}} Q_{\max}^{(\ell)}) \quad (\text{B.3.7})$$

$$\leq \sum_{\ell \in \mathcal{L}} |\mathcal{W}_{C_K^{(\ell-1)}}| \cdot (\log(N) + (e - 1) N_{\mathbf{w}} Q_{\max}^{(\ell)}) \lesssim \mathcal{D}_{\max}, \quad (\text{B.3.8})$$

where $\sum_{\ell \in \mathcal{L}} N_{\mathbf{w}} Q_{\max}^{(\ell)} \gtrsim \log(N)$ since $q_N = \log(N)$ and $|\mathcal{W}_{C_K^{(\ell-1)}}| = \binom{\ell+K-1}{K-1} = O(1)$. At the same time, $\mathbb{E} \tilde{D} \geq \max_{\ell \in \mathcal{L}} N_{\mathbf{w}} Q_{\max}^{(\ell)} \asymp \mathcal{D}_{\max}$. Therefore $\tilde{d} \asymp \mathcal{D}_{\max}$ with high probability. The desired result follows if $\mathcal{D}_{\max} \asymp \rho_N$, which is obviously true when Assumption 3.2.2 holds. When the assumption is not true, where some specific configuration $\mathbf{w} \in \mathcal{W}_{C_K^{(\ell)}}$ dominates for some $\ell \in \mathcal{L}$, then the vertex with maximum expected degree should certainly contain this type of edge. At the same time, $\mathcal{D}_{\max}$ should contain the dominant type $\mathbf{w}$ as well. Therefore $\mathcal{D}_{\max} \asymp \rho_N \asymp \tilde{d}$ with high probability. $\square$

Proof of Lemma 3.4.5. Define the matrix $\mathbb{E} \tilde{\mathbf{A}}$ where $\mathbb{E} \tilde{\mathbf{A}}_{ij} = \mathbb{E} \mathbf{A}_{ij}$ for $i \neq j$ and $\mathbb{E} \tilde{\mathbf{A}}_{ii} = \mathbb{E} \mathbf{A}_{ij}$ for some $j \in \mathcal{V}$ in the same community as i , i.e., $y_i = y_j$, then $\text{rank}(\mathbb{E} \tilde{\mathbf{A}}) = K$. By triangle inequality and $(a + b)^2 \leq 2a^2 + 2b^2$,

$$\|\mathbf{A}_j^{(K)} - \mathbb{E} \mathbf{A}_j\|_{\text{F}}^2 \leq 2\|\mathbf{A}_j^{(K)} - \mathbb{E} \tilde{\mathbf{A}}_j\|_{\text{F}}^2 + 2\|\mathbb{E} \mathbf{A}_j - \mathbb{E} \tilde{\mathbf{A}}_j\|_{\text{F}}^2 \quad (\text{B.3.9})$$

The matrices $\mathbb{E} \mathbf{A}_j$ and $\mathbb{E} \tilde{\mathbf{A}}_j$ only differ in diagonal elements while $\mathbb{E} \tilde{\mathbf{A}}_{ii} \lesssim N^{-1} \mathcal{D}_{\max}$, then $\|\mathbb{E} \mathbf{A}_j - \mathbb{E} \tilde{\mathbf{A}}_j\|_{\text{F}}^2 \lesssim N^{-1} \mathcal{D}_{\max}^2$. Meanwhile, the rank of $\mathbf{A}_j^{(K)} - \mathbb{E} \tilde{\mathbf{A}}_j$ is at most $2K$, then by rank

inequality

$$\|\mathbf{A}_j^{(K)} - \mathbb{E}\tilde{\mathbf{A}}_j\|_F^2 \leq 2K\|\mathbf{A}_j^{(K)} - \mathbb{E}\tilde{\mathbf{A}}_j\|_2^2 \leq 4K(\|\mathbf{A}_j^{(K)} - \mathbf{A}_j\|_2^2 + \|\mathbf{A}_j - \mathbb{E}\tilde{\mathbf{A}}_j\|_2^2). \quad (\text{B.3.10})$$

By Eckart–Young–Mirsky Lemma D.2.2, $\|\mathbf{A}_j - \mathbf{A}_j^{(K)}\|_2 = \lambda_{K+1} \leq \|\mathbf{A}_j - \mathbb{E}\tilde{\mathbf{A}}_j\|_2$. Also, $\mathbf{A}_j - \mathbb{E}\tilde{\mathbf{A}}_j = \mathbf{A}_j - \mathbb{E}\mathbf{A}_j - \text{diag}(\mathbb{E}\mathbf{A}_{ii})$. Then desired the argument holds since

$$\begin{aligned} \|\mathbf{A}_j^{(K)} - \mathbb{E}\mathbf{A}_j\|_F^2 &\lesssim 8K(\|\mathbf{A}_j^{(K)} - \mathbf{A}_j\|_2^2 + \|\mathbf{A}_j - \mathbb{E}\tilde{\mathbf{A}}_j\|_2^2) + 2\mathcal{D}_{\max}^2/N \\ &\leq 16K \cdot \|\mathbf{A}_j - \mathbb{E}\tilde{\mathbf{A}}_j\|_2^2 + 2\mathcal{D}_{\max}^2/N \lesssim 32K \cdot \|\mathbf{A}_j - \mathbb{E}\mathbf{A}_j\|_2^2 + (32K + 2)\mathcal{D}_{\max}^2/N \\ &\stackrel{\text{Theorem 3.4.4}}{\leq} 32KC_{(3.4.6)}^2 \mathcal{D}_{\max} + (32K + 2)\mathcal{D}_{\max}^2/N \asymp \rho_N + \rho_N^2/N, \end{aligned}$$

where the last step holds since $\rho_N \asymp \mathcal{D}_{\max}$, as proved in Lemma 3.4.1. $\square$

Proof of Lemma 3.4.6. First, we discuss the case both $u, v \in \mathcal{J}$, and this is a high probability event. By construction, the number of vertices not in $\mathcal{J}_1$ is upper bounded by $\lfloor N \exp(-\bar{d}) \rfloor \lesssim 1$ since $\rho_N \gtrsim \log(N)$ and $\bar{d} \asymp \log(N)$ with high probability by Lemma 3.4.1. Thus vertex $l \in \mathcal{J}_1$ with probability at least $1 - N^{-1}$. Also, by taking $\beta_1 = 20L$ in (B.1.12), the number of vertices not in $\mathcal{J}_2$ is at most $Ne^{-\tilde{c}\rho_N}$ for some constant $\tilde{c} > 0$, thus $l \in \mathcal{J}_2$ with probability at least $1 - e^{-\tilde{c}\rho_N}$. By definition,

$$\|(\mathbb{E}\mathbf{A}_j)_{u\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2 = \sum_{l \in \mathcal{V}} [(\mathbb{E}\mathbf{A}_j)_{ul} - (\mathbb{E}\mathbf{A}_j)_{vl}]^2. \quad (\text{B.3.11})$$

For vertex l with $z_l = t$ for some $t \in [K]$, entry $(\mathbb{E}\mathbf{A}_j)_{ul}$ denotes the expected number of hyperedges containing u and l . Hyperedges can be classified into different categories according to the distribution of the remaining $\ell - 2$ nodes among K communities. Assuming that $u \in \mathcal{V}_j$ and $v \in \mathcal{V}_k$ with $j \neq k$, the membership distribution of e can be represented by $j \oplus l \oplus \mathbf{w}$ for each $\mathbf{w} \in \mathcal{W}C_K^{(\ell-2)}$ with $N_{\mathbf{w}} \asymp N^{-(\ell-2)}$. For each $l \in \mathcal{J}$, the difference between $(\mathbb{E}\mathbf{A}_j)_{ul}$ and $(\mathbb{E}\mathbf{A}_j)_{vl}$

is

$$(\mathbb{E}\mathbf{A}_J)_{ul} - (\mathbb{E}\mathbf{A}_J)_{vl} = \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-2)}} N_{\mathbf{w}} \cdot (Q_{j \oplus l \oplus \mathbf{w}}^{(\ell)} - Q_{k \oplus l \oplus \mathbf{w}}^{(\ell)}). \quad (\text{B.3.12})$$

At the same time, $N_{\mathbf{w}}$ can be replaced by $\bar{N}_{\mathbf{w}}$ with high probability by Lemma 3.1.1. Then according to Assumption 3.2.7, it follows that with probability at least $1 - N^{-c}$,

$$\|(\mathbb{E}\mathbf{A}_J)_u - (\mathbb{E}\mathbf{A}_J)_v\|_2^2 \gtrsim N [(\mathbb{E}\mathbf{A}_J)_{ul} - (\mathbb{E}\mathbf{A}_J)_{vl}]^2 \gtrsim N^{-1} \rho_N^2. \quad (\text{B.3.13})$$

On the other hand, when u and v are from the same cluster, i.e., $u \in \mathcal{V}_j$ and $v \in \mathcal{V}_k$ with $j = k$, since $(\mathbb{E}\mathbf{A}_J)_{uu} = (\mathbb{E}\mathbf{A}_J)_{vv} = 0$ due to the fact that self loops are not allowed in model 1.1.8, it then follows

$$\|(\mathbb{E}\mathbf{A}_J)_u - (\mathbb{E}\mathbf{A}_J)_v\|_2^2 = [(\mathbb{E}\mathbf{A}_J)_{uv}]^2 + [(\mathbb{E}\mathbf{A}_J)_{vu}]^2 \lesssim N^{-2} \rho_N^2. \quad (\text{B.3.14})$$

We now turn to the discussion when $u \in \mathcal{J}$ but $v \notin \mathcal{J}$, and the conclusion holds similarly. As for the case $u \notin \mathcal{J}$, $v \notin \mathcal{J}$, it follows directly from the definition of $\mathbb{E}\mathbf{A}_J$ that $\|(\mathbb{E}\mathbf{A}_J)_u - (\mathbb{E}\mathbf{A}_J)_v\|_2^2 = 0$. $\square$

Proof of Lemma 3.4.7. The algorithm inputs $\bar{r} = \lfloor N \log(\bar{d}) \rfloor^{-1} \bar{d}^2$. Note that $\bar{d} \asymp \rho_N$ with high probability at least $1 - e^{-N^{1/2}}$ by Lemma 3.4.1, then $\bar{r} \asymp r := \lfloor N \log(\rho_N) \rfloor^{-1} \rho_N^2$. For now on, we will work on r instead of $\bar{r}$.

Define $O_k := \{v \in \mathcal{V}_k \cap \mathcal{J} : \|(\mathbf{A}_J^{(K)})_v - (\mathbb{E}\mathbf{A}_J)_v\|_2^2 > 4r\}$, while I_k , $\mathcal{U}_k$, O_k and $\mathcal{B}_r(v)$ enjoy the following properties.

- (i) For each $k \in [K]$, $I_k \subset \mathcal{B}_r(v) \subset \mathcal{U}_k$ for any $v \in I_k \cap \mathcal{S}$, since $(\mathbb{E}\mathbf{A}_J)_v = (\mathbb{E}\mathbf{A}_J)_w$ for all $w \in I_k$,

$$\|(\mathbf{A}_J^{(K)})_v - (\mathbf{A}_J^{(K)})_w\|_2^2 \leq 2\|(\mathbf{A}_J^{(K)})_v - (\mathbb{E}\mathbf{A}_J)_v\|_2^2 + 2\|(\mathbb{E}\mathbf{A}_J)_w - (\mathbf{A}_J^{(K)})_w\|_2^2 \leq r.$$

(ii) For each $k \in [K]$, $\mathcal{B}_r(v) \subset \mathcal{U}_k$ for any $v \in I_k \cap \mathcal{S}$.

(iii) $|I_k| \geq \alpha_k N(1 - o(1))$ for each $k \in [K]$, which follows from the fact that almost all nodes in $\mathcal{V}_k \cap \mathcal{J}$ are $r/4$ -close to its expected center, since the number of vertices outside I_k is bounded by

$$\begin{aligned} \frac{|\mathcal{V}_k \cap \mathcal{J} \setminus I_k|}{N} &\leq \frac{1}{N} \cdot \frac{\sum_{v \in (\mathcal{V}_k \cap \mathcal{J}) \setminus I_k} \|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2}{\min_{v \in (\mathcal{V}_k \cap \mathcal{J}) \setminus I_k} \|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2} \leq \frac{1}{N} \frac{\|\mathbf{A}_j^{(K)} - \mathbb{E}\mathbf{A}_j\|_F^2}{r/4} \\ &\lesssim \frac{1}{N} \frac{\rho_N + \rho_N^2/N}{[N \log(\rho_N)]^{-1} \rho_N^2} = \frac{\log(\rho_N)}{\rho_N} + \frac{\log(\rho_N)}{N} = o(1), \end{aligned}$$

where the last two equalities hold since $\|\mathbf{A}_j^{(K)} - \mathbb{E}\mathbf{A}_j\|_F^2 \lesssim \rho_N + \rho_N^2/N$ (Lemma 3.4.5), $r = [N \log(\rho_N)]^{-1} \rho_N^2$ and $1 \ll \rho_N \ll N$. Thus $|I_k| = |\mathcal{V}_k \cap \mathcal{J}| - |(\mathcal{V}_k \cap \mathcal{J}) \setminus I_k| \geq \alpha_k N(1 - o(1))$.

(iv) Following from (iii), for all $k \in [K]$,

$$|\mathcal{U}_k| \leq N - |\cup_{j \neq k} \mathcal{U}_j| \leq N - \sum_{j \neq k} |I_j| \leq \alpha_k N \left[1 + O\left(\frac{\log(\rho_N)}{\rho_N}\right) \right] = \alpha_k N(1 + o(1)).$$

(v) $|\cup_{k=1}^K \mathcal{O}_k| \leq o(N)$, since most nodes are in $\cup_{k=1}^K I_k$, and

$$|\cup_{k=1}^K \mathcal{O}_k| \leq \frac{\sum_{v \in \mathcal{J}} \|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2}{\min_{v \in (\cup_{k=1}^K \mathcal{O}_k)} \|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2} \leq \frac{\|\mathbf{A}_j^{(K)} - \mathbb{E}\mathbf{A}_j\|_F^2}{4r} = o(N)$$

(vi) $|\mathcal{B}_r(v)| = o(N)$ for all $v \in \mathcal{O}_k \cap \mathcal{S}$. First, $\mathcal{B}_r(v) \cap \mathcal{U}_k = \emptyset$ because the distance between $v \in \mathcal{O}_k$ and $w \in I_k$ is larger than r , due to the facts $(\mathbb{E}\mathbf{A}_j)_{v\cdot} = (\mathbb{E}\mathbf{A}_j)_{w\cdot}$, $(x - y)^2 \geq \frac{1}{2}(x - z)^2 - (y - z)^2$, and

$$\|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbf{A}_j^{(K)})_{w\cdot}\|_2^2 \geq \frac{1}{2} \|(\mathbf{A}_j^{(K)})_{v\cdot} - (\mathbb{E}\mathbf{A}_j)_{v\cdot}\|_2^2 - \|(\mathbf{A}_j^{(K)})_{w\cdot} - (\mathbb{E}\mathbf{A}_j)_{w\cdot}\|_2^2 > r.$$

Then as a consequence of (v), $|\mathcal{B}_r(v)| \leq |\cup_{k=1}^K \mathcal{O}_k| = o(N)$.

(vii) $\mathcal{B}_r(u) \cap \mathcal{B}_r(v) = \emptyset$ for any $u \in \mathcal{U}_j$ and $v \in \mathcal{U}_k$ with $j \neq k$. This follows the fact that $|\mathcal{B}_r(v) \cap \mathcal{U}_j| = 0$ when $|\mathcal{B}_r(v) \cap \mathcal{U}_k| \geq 1$, since the distance between expected centers $\|(\mathbb{E}\mathbf{A}_j)_v - (\mathbb{E}\mathbf{A}_j)_u\|_2^2 \gtrsim N^{-1}\rho_N^2$ (Lemma 3.4.6) is much larger than the radius $4r \asymp [N \log(\rho_N)]^{-1}\rho_N^2$ where $\rho_N = \omega(1)$.

We now prove those three arguments using the properties above.

(a). For each $k \in [K]$, the probability that a randomly selected node $s \in \mathcal{V}$ does not belong to I_k is

$$1 - \frac{|I_k|}{N} \stackrel{(iii)}{\leq} 1 - \alpha_k(1 - o(1)) = \sum_{j \neq k} \alpha_j + o(1) < 1, \quad (\text{B.3.15})$$

where $1 > \alpha_1 \geq \dots \geq \alpha_K > 0$ are some constants by Definition 1.1.5, then the probability that there exists at least one node $s \in \mathcal{S} \cap I_k$ is

$$1 - \left(1 - \frac{|I_k|}{N}\right)^{\log(N)} \geq 1 - \left(\sum_{j \neq k} \alpha_j + o(1)\right)^{\log(N)} = 1 - N^{-\log\left(\frac{1}{\sum_{j \neq k} \alpha_j + o(1)}\right)}, \quad (\text{B.3.16})$$

A simple union bound completes the argument.

(b). We prove part (b) by induction.

(1) $k = 1$. By part (a), there exists a node v_1 in $I_1 \cap \mathcal{S}$, satisfying $\mathcal{B}_r(v_1) \supset I_1$ due to (i).

By algorithm procedure, $|\mathcal{B}_r(s_1)|$ is at least $|\mathcal{B}_r(v_1)|$, then $|\mathcal{B}_r(s_1)| \geq |\mathcal{B}_r(v_1)| \geq |I_1| \geq \alpha_1 N(1 - o(1))$ thanks to (iii). To prove $s_1 \in \mathcal{U}_1$, one should verify the failure of other possibilities, namely, for $v \notin \mathcal{U}_1$, the cardinality of $\mathcal{B}_r(v)$ is too small to make v selected as a center.

- For any $v \in \mathcal{O}_k \cap \mathcal{S}$ for each $k \in [K]$, $|\mathcal{B}_r(v)| = o(N) < |\mathcal{B}_r(s_1)|$ by property (vi).
- For any $v \in \mathcal{U}_k \cap \mathcal{S}$ with $k \neq 1$, $|\mathcal{B}_r(v)| \stackrel{(ii)}{\leq} |\mathcal{U}_k| \stackrel{(iv)}{\leq} \alpha_k N(1 + o(1)) < \alpha_1 N$ since $\alpha_1 > \alpha_k$.

(2) $2 \leq k \leq K$. Suppose $s_j \in \mathcal{U}_j$ for all $j \in [k-1]$. Similarly by (i), there exists a node $v_k \in \mathcal{U}_k \cap \mathcal{S}$ satisfying $\mathcal{B}_r(v_k) \supset \mathcal{U}_k$. By induction hypothesis $|\widehat{\mathcal{V}}_j^{(0)}| = |\mathcal{B}_r(v_j) \setminus (\cup_{l=1}^{j-1} \widehat{\mathcal{V}}_l^{(0)})| \geq \alpha_j N(1 - o(1))$ for $j \in [k-1]$. Then by (vii), $\mathcal{B}_r(v_k) \setminus (\cup_{l=1}^{k-1} \widehat{\mathcal{V}}_l^{(0)}) = \mathcal{B}_r(v_k)$, and $|\mathcal{B}_r(s_k)| \geq |\mathcal{B}_r(v_k)| \geq |\mathcal{U}_k| \geq \alpha_k N(1 - o(1))$ thanks to (iii). To prove $s_k \in \mathcal{U}_k$, one should verify the failure of other possibilities, namely, for $v \notin \mathcal{U}_k$, the cardinality of so obtained set $\mathcal{B}_r(v) \setminus (\cup_{l=1}^{k-1} \widehat{\mathcal{V}}_l^{(0)})$ is too small to make v selected as a center.

- For any $v \in O_k \cap \mathcal{S}$ for all $k \in [K]$, $|\mathcal{B}_r(v)| = o(N) < |\mathcal{B}_r(s_k)|$ by property (vi).
- For any $v \in \mathcal{U}_j \cap \mathcal{S}$ where $j < k$, note that $|\mathcal{B}_r(s_j)| \geq \alpha_j N(1 - o(1))$ by induction, then

$$|\mathcal{B}_r(v) \setminus (\cup_{l=1}^{k-1} \widehat{\mathcal{V}}_l^{(0)})| \stackrel{(vii)}{\leq} |\mathcal{B}_r(v) \setminus \widehat{\mathcal{V}}_j^{(0)}| \stackrel{(ii)}{\leq} |\mathcal{U}_j| - |\mathcal{B}_r(s_j)| \stackrel{(iv)}{=} o(N) \quad (\text{B.3.17})$$

$$< |\mathcal{B}_r(s_k)| = \alpha_k N(1 + o(1)). \quad (\text{B.3.18})$$

- For any $v \in \mathcal{U}_j \cap \mathcal{S}$ where $j > k$,

$$|\mathcal{B}_r(v) \setminus (\cup_{l=1}^{k-1} \widehat{\mathcal{V}}_l^{(0)})| \stackrel{(vii)}{\leq} |\mathcal{B}_r(v)| \stackrel{(ii)}{\leq} |\mathcal{U}_j| \stackrel{(iv)}{=} \alpha_j N(1 + o(1)) \quad (\text{B.3.19})$$

$$< \alpha_k N(1 + o(1)) = |\mathcal{B}_r(s_k)|, \quad (\text{B.3.20})$$

where the last inequality holds since $\alpha_k > \alpha_j$.

(c). Consequently, $\{s_k\}_{k=1}^K$ are pairwise far from each other, i.e., for any pair $j \neq k$

$$\begin{aligned} \|(\mathbf{A}_g^{(K)})_{s_j} - (\mathbf{A}_g^{(K)})_{s_k}\|_2^2 &\geq \frac{1}{2} \|(\mathbf{A}_g^{(K)})_{s_j} - (\mathbb{E} \mathbf{A}_g)_{s_k}\|_2^2 - \|(\mathbb{E} \mathbf{A}_g)_{s_k} - (\mathbf{A}_g^{(K)})_{s_k}\|_2^2 \\ &\geq \frac{1}{2} \left(\frac{1}{2} \|(\mathbb{E} \mathbf{A}_g)_{s_j} - (\mathbb{E} \mathbf{A}_g)_{s_k}\|_2^2 - \|(\mathbf{A}_g^{(K)})_{s_j} - (\mathbb{E} \mathbf{A}_g)_{s_j}\|_2^2 \right) - \|(\mathbb{E} \mathbf{A}_g)_{s_k} - (\mathbf{A}_g^{(K)})_{s_k}\|_2^2 \gtrsim \frac{\rho_N^2}{N}, \end{aligned}$$

where the last equality holds by since Lemma 3.4.6 $\|(\mathbb{E} \mathbf{A}_g)_{s_j} - (\mathbb{E} \mathbf{A}_g)_{s_k}\|_2^2 \gtrsim N^{-1} \rho_N^2 \gg [N \log(\rho_N)]^{-1} \rho_N^2 \asymp 4r \geq \|(\mathbb{E} \mathbf{A}_g)_{s_j} - (\mathbf{A}_g)_{s_j}\|_2^2$ with $\rho_N = \omega(1)$ by the property that $s_j \in \mathcal{U}_j$ for

each $j \in [K]$. □

Proof of Lemma 3.4.8. Note that $(\mathbb{E}\mathbf{A}_g)_{v:} = (\mathbb{E}\mathbf{A}_g)_{s_i:}$ when $v \in \mathcal{V}_i \cap \mathcal{J}$ where s_i is the center of $\widehat{\mathcal{V}}_i^0$ obtained in Line 4. There are two scenarios where $v \in \mathcal{V}_i \cap \mathcal{J}$ would be misclassified into $\widehat{\mathcal{V}}_j^{(0)}$ ($j \neq i$).

1. In Line 4 of Algorithm 3.2.2, if v is r -close to center of $\widehat{\mathcal{V}}_j^{(0)}$, i.e., $\|(\mathbf{A}_g^{(K)})_{v:} - (\mathbf{A}_g^{(K)})_{s_j:}\|_2^2 \leq r$, then

$$\begin{aligned} \|(\mathbf{A}_g^{(K)})_{v:} - (\mathbb{E}\mathbf{A}_g)_{s_i:}\|_2^2 &\geq \frac{1}{2} \|(\mathbb{E}\mathbf{A}_g)_{s_i:} - (\mathbb{E}\mathbf{A}_g)_{s_j:}\|_2^2 - \|(\mathbf{A}_g^{(K)})_{v:} - (\mathbb{E}\mathbf{A}_g)_{s_j:}\|_2^2 \\ &\geq \frac{1}{2} \|(\mathbb{E}\mathbf{A}_g)_{s_i:} - (\mathbb{E}\mathbf{A}_g)_{s_j:}\|_2^2 - 2\|(\mathbf{A}_g^{(K)})_{v:} - (\mathbf{A}_g^{(K)})_{s_j:}\|_2^2 - 2\|(\mathbf{A}_g^{(K)})_{s_j:} - (\mathbb{E}\mathbf{A}_g)_{s_j:}\|_2^2 \gtrsim \frac{\rho_N^2}{N}, \end{aligned}$$

where $\|(\mathbb{E}\mathbf{A}_g)_{s_i:} - (\mathbb{E}\mathbf{A}_g)_{s_j:}\|_2^2 \gtrsim N^{-1}\rho_N^2$ by Lemma 3.4.6, and $\|(\mathbf{A}_g^{(K)})_{s_j:} - (\mathbb{E}\mathbf{A}_g)_{s_j:}\|_2^2 \leq 4r \asymp [N \log(\rho_N)]^{-1}\rho_N^2 \ll \rho_N^2/N$ since $\rho_N = \omega(1)$ and $s_j \in \mathcal{U}_j$ as proved in Lemma 3.4.7.

2. In Line 8 of Algorithm 3.2.2, when v is closer to the center of $\widehat{\mathcal{V}}_j^{(0)}$ than $\widehat{\mathcal{V}}_i^{(0)}$, i.e., $\|(\mathbf{A}_g^{(K)})_{s_j:} - (\mathbf{A}_g^{(K)})_{v:}\|_2^2 \leq \|(\mathbf{A}_g^{(K)})_{s_i:} - (\mathbf{A}_g^{(K)})_{v:}\|_2^2$. One can verify that v is far from its expected center, namely,

$$\|(\mathbf{A}_g^{(K)})_{v:} - (\mathbb{E}\mathbf{A}_g)_{s_i:}\|_2^2 \geq \frac{1}{2} \|(\mathbf{A}_g^{(K)})_{s_i:} - (\mathbf{A}_g^{(K)})_{v:}\|_2^2 - \|(\mathbb{E}\mathbf{A}_g)_{s_i:} - (\mathbf{A}_g^{(K)})_{s_i:}\|_2^2 \gtrsim \frac{\rho_N^2}{N},$$

where $\|(\mathbb{E}\mathbf{A}_g)_{s_i:} - (\mathbf{A}_g^{(K)})_{s_i:}\|_2^2 \leq 4r \asymp [N \log(\rho_N)]^{-1}\rho_N^2$ and

$$\begin{aligned} \|(\mathbf{A}_g^{(K)})_{s_i:} - (\mathbf{A}_g^{(K)})_{v:}\|_2^2 &\geq \frac{1}{2} \left(\|(\mathbf{A}_g^{(K)})_{s_i:} - (\mathbf{A}_g^{(K)})_{v:}\|_2^2 + \|(\mathbf{A}_g^{(K)})_{s_j:} - (\mathbf{A}_g^{(K)})_{v:}\|_2^2 \right) \\ &\geq \frac{1}{4} \|(\mathbf{A}_g^{(K)})_{s_i:} - (\mathbf{A}_g^{(K)})_{s_j:}\|_2^2 \stackrel{\text{Lemma 3.4.7 (c)}}{\gtrsim} \frac{\rho_N^2}{N}, \end{aligned}$$

where the last line holds since $(x-y)^2 \leq 2(x-z)^2 + 2(y-z)^2$. □

B.4 Proofs in Section 3.5

Proof of Lemma 3.5.1. At step 0, let $\widehat{D}_{\mathbf{w}}^{(0)}$ denote the number of ℓ -hyperedges with

$$\widehat{\mathbf{y}}^{(0)}(e) = \mathbf{w} = (w_1, \dots, w_K),$$

meaning that w_k nodes in $\widehat{\mathcal{V}}_k^{(0)}$ for each $k \in [K]$. Recall $\widehat{Q}_{\mathbf{w}}^{(\ell)} = \widehat{D}_{\mathbf{w}}^{(0)} / \widehat{N}_{\mathbf{w}}^{(0)}$ in (3.2.11). Consequently, for any $\mathbf{w} \in \mathcal{WC}_K^{(\ell)}$, we have

$$|\widehat{Q}_{\mathbf{w}}^{(\ell)} - Q_{\mathbf{w}}^{(\ell)}| = \frac{|\widehat{D}_{\mathbf{w}}^{(0)} - Q_{\mathbf{w}}^{(\ell)} \cdot \widehat{N}_{\mathbf{w}}^{(0)}|}{\widehat{N}_{\mathbf{w}}^{(0)}} \leq \frac{1}{\widehat{N}_{\mathbf{w}}^{(0)}} \left(|\widehat{D}_{\mathbf{w}}^{(0)} - \mathbb{E} \widehat{D}_{\mathbf{w}}^{(0)}| + |\mathbb{E} \widehat{D}_{\mathbf{w}}^{(0)} - Q_{\mathbf{w}}^{(\ell)} \cdot \widehat{N}_{\mathbf{w}}^{(0)}| \right). \quad (\text{B.4.1})$$

We will bound the two terms in the numerator separately. For the first term $|\widehat{D}_{\mathbf{w}}^{(0)} - \mathbb{E} \widehat{D}_{\mathbf{w}}^{(0)}|$, denote $Q_{\min}^{(\ell)} := \min_{\mathbf{w} \in \mathcal{WC}_K^{(\ell)}} Q_{\mathbf{w}}^{(\ell)}$, $Q_{\max}^{(\ell)} := \max_{\mathbf{w} \in \mathcal{WC}_K^{(\ell)}} Q_{\mathbf{w}}^{(\ell)}$ for each $\ell \in \mathcal{L}$. Using (3.2.2), we write $Q_{\min}^{(\ell)} = \mathcal{P}_{\min}^{(\ell)} q_{\min}^{(\ell)} / \binom{N-1}{\ell-1}$, $Q_{\max}^{(\ell)} = \mathcal{P}_{\max}^{(\ell)} q_{\max}^{(\ell)} / \binom{N-1}{\ell-1}$ where $\mathcal{P}_{\min}^{(\ell)} \asymp \mathcal{P}_{\max}^{(\ell)} \asymp 1$ and $q_{\min}^{(\ell)} \lesssim q_{\max}^{(\ell)}$. Since $\widehat{N}_{\mathbf{w}}^{(0)} \asymp N^\ell$, we then have

$$Nq_{\min}^{(\ell)} \asymp \widehat{N}_{\mathbf{w}}^{(0)} \cdot Q_{\max}^{(\ell)} \leq \mathbb{E} \widehat{D}_{\mathbf{w}}^{(0)} \leq \widehat{N}_{\mathbf{w}}^{(0)} \cdot Q_{\max}^{(\ell)} \asymp Nq_{\max}^{(\ell)} \quad (\text{B.4.2})$$

Let $\widehat{\mathcal{V}}$ denote the collection of partitions $\{\widehat{\mathcal{V}}_k\}_{k=1}^K$ such that

$$\left| \bigcup_{k=1}^K (\mathcal{V}_k \setminus \widehat{\mathcal{V}}_k) \right| \leq \eta_N \cdot N, \quad \forall \{\widehat{\mathcal{V}}_k\}_{k=1}^K \in \widehat{\mathcal{V}}, \quad (\text{B.4.3})$$

where η_N is the mismatch ratio defined in (1.2.1) and $\eta_N \lesssim (\rho_N)^{-1}$ with probability at least $1 - N^{-c}$ for some constant $c > 0$ by (3.4.10). For any $\{\widehat{\mathcal{V}}_k^{(0)}\}_{k=1}^K \in \widehat{\mathcal{V}}$, thanks to edgewise

independence, Bernstein (Lemma D.3.4) shows that

$$\begin{aligned}\mathbb{P}(|\widehat{\mathbf{D}}_{\mathbf{w}}^{(0)} - \mathbb{E}\widehat{\mathbf{D}}_{\mathbf{w}}^{(0)}| \geq N \log q_{\max}^{(\ell)}) &\leq 2 \exp\left(-\frac{(N \log q_{\max}^{(\ell)})^2/2}{N q_{\max}^{(\ell)} + N/3 \cdot \log q_{\max}^{(\ell)}}\right) \\ &= 2 \exp\left(-N(\log q_{\max}^{(\ell)})^2/(2q_{\max}^{(\ell)})\right),\end{aligned}$$

Meanwhile, the upper bound on the cardinality of $\widehat{\mathcal{V}}$ can be computed as follows: first $\eta_N N$ many vertices (wrongly classified) are selected, then each of them can be assigned to K different communities. Thus by Lemma D.1.3, for some universal constant $C > 0$, the following holds

$$|\widehat{\mathcal{V}}| \leq \binom{N}{\eta_N N} K^{\eta_N N} \leq \left(\frac{eNK}{\eta_N N}\right)^{\eta_N N} = \exp\left(\eta_N N \log\left(\frac{eK}{\eta_N}\right)\right) \lesssim \exp\left(C \frac{N \log \rho_N}{\rho_N}\right) \quad (\text{B.4.4})$$

Note that $\rho_N \asymp \sum_{\ell \in \mathcal{L}} q_{\max}^{(\ell)}$ and the function $h(x) = x/\log(x)$ increases when $x \geq e$, then a simple union bound shows that the following holds with probability at least $1 - N^{-c}$

$$\mathbb{P}\left(\exists \{\widehat{\mathcal{V}}_k^{(0)}\}_{k=1}^K \in \widehat{\mathcal{V}} \text{ s.t. } |\widehat{\mathbf{D}}_{\mathbf{w}}^{(0)} - \mathbb{E}\widehat{\mathbf{D}}_{\mathbf{w}}^{(0)}| \geq N \log q_{\max}^{(\ell)}\right) \quad (\text{B.4.5})$$

$$\leq |\widehat{\mathcal{V}}| \binom{K}{2} \cdot 2 \exp\left(-N \frac{[\log q_{\max}^{(\ell)}]^2}{2q_{\max}^{(\ell)}}\right) \lesssim \exp\left(-N \frac{\log q_{\max}^{(\ell)}}{q_{\max}^{(\ell)}}\right). \quad (\text{B.4.6})$$

Consider the second term $\mathbb{E}\widehat{\mathbf{D}}_{\mathbf{w}}^{(0)} = \sum_{\widehat{\mathbf{y}}^{(0)}(e)=\mathbf{w}} Q_{\mathbf{y}(e)}^{(\ell)}$. For hyperedge $e \subset \mathcal{V} \setminus \widehat{\mathcal{N}}^{(0)}$, i.e., $\widehat{\mathbf{y}}^{(0)}(e) = \mathbf{w} = \mathbf{y}(e)$, the contribution of such e inside the $|\mathbb{E}\widehat{\mathbf{D}}_{\mathbf{w}}^{(0)} - Q_{\mathbf{w}}^{(\ell)} \cdot \widehat{N}_{\mathbf{w}}^{(0)}|$ cancels out. Thus it suffices to consider contributions from e which contains at least one node from $\widehat{\mathcal{N}}^{(0)}$, and the number of such hyperedges is at most $\eta_N N \cdot \binom{N-1}{\ell-1}$. According to (3.4.10), $\eta_N \lesssim (\rho_N)^{-1}$ where $\rho_N \asymp \sum_{\ell \in \mathcal{L}} q_{\max}^{(\ell)}$, then the following holds with probability at least $1 - N^{-c}$ for some $c > 0$

$$|\mathbb{E}\widehat{\mathbf{D}}_{\mathbf{w}}^{(0)} - Q_{\mathbf{w}}^{(\ell)} \cdot \widehat{N}_{\mathbf{w}}^{(0)}| \leq Q_{\max}^{(\ell)} \cdot \eta_N N \binom{N-1}{\ell-1} \lesssim q_{\max}^{(\ell)} N / \rho_N \lesssim N. \quad (\text{B.4.7})$$

Therefore, with probability at least $1 - \exp(-N \log q_{\max}^{(\ell)} / q_{\max}^{(\ell)})$,

$$|\hat{Q}_{\mathbf{w}}^{(\ell)} - Q_{\mathbf{w}}^{(\ell)}| \lesssim \frac{N \log q_{\max}^{(\ell)} + N}{\hat{N}_{\mathbf{w}}^{(0)}} \lesssim \frac{\log q_{\max}^{(\ell)}}{N^{\ell-1}}. \quad (\text{B.4.8})$$

The second part follows easily since $Q_{\mathbf{w}}^{(\ell)} \asymp q_N N^{-m+1}$ for each $\ell \in \mathcal{L}$ when under Assumption 3.2.2. $\square$

Lemma B.4.1. *Recall that (3.2.17) holds for some absolute constant $\varepsilon > 0$, where $\kappa_N \log(N) \rightarrow \infty$ as $N \rightarrow \infty$. Then the number of nodes which don't satisfy either (G1) or (G2) is at most $N^{1-\kappa_N}/3$ with probability at least $1 - 6e^{-\varepsilon \kappa_N \log(N)}$.*

Proof of Lemma B.4.1. Probability of satisfying (G1). Recall D_v in (3.1.1). For each $\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}$ where $N_{\mathbf{w}}$ in (3.1.2), $D_{v,\mathbf{w}}^{(\ell)}|_{Y_v=k} \sim \text{Bin}(N_{\mathbf{w}}, Q_{k \oplus \mathbf{w}}^{(\ell)})$. Denote $Q_{\max}^{(\ell)} = \max_{\mathbf{w} \in \mathcal{W}C_K^{(\ell)}} Q_{\mathbf{w}}^{(\ell)}$. Note that

$$\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} Q_{k \oplus \mathbf{w}}^{(\ell)} \lesssim \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} Q_{\max}^{(\ell)} \lesssim \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \binom{N-1}{\ell-1} \cdot Q_{\max}^{(\ell)} = \mathcal{D}_{\max}.$$

Thanks to hyperedges independence and Markov inequality, for each $v \in \mathcal{G}$, we have

$$\begin{aligned} \mathbb{P}(D_v > 10\mathcal{D}_{\max}) &= \mathbb{P}\left(\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} D_{v,\mathbf{w}}^{(\ell)} > 10\mathcal{D}_{\max}\right) \\ &\leq \inf_{\theta > 0} \left[\exp(-10\theta \mathcal{D}_{\max}) \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \mathbb{E} \exp(\theta \cdot D_{v,\mathbf{w}}^{(\ell)}) \right] \\ &\leq \inf_{\theta > 0} \exp(-10\theta \mathcal{D}_{\max}) \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \exp\left[(1 - Q_{k \oplus \mathbf{w}}^{(\ell)} + Q_{k \oplus \mathbf{w}}^{(\ell)} e^{\theta}]^{N_{\mathbf{w}}}, \quad (1+x \leq e^x) \\ &\leq \inf_{\theta > 0} \exp(-10\theta \mathcal{D}_{\max}) \exp\left((e^{\theta} - 1) \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} Q_{k \oplus \mathbf{w}}^{(\ell)}\right) \\ &\leq \inf_{\theta > 0} \exp((e^{\theta} - 1 - 10\theta) \cdot \mathcal{D}_{\max}) \leq e^{-10\mathcal{D}_{\max}}, \end{aligned}$$

where the last line holds by taking $\theta = 2$, and $\mathbb{P}(D_v \leq 10\mathcal{D}_{\max}) \leq 1 - e^{-10\mathcal{D}_{\max}} \rightarrow 1$ since

$$\mathcal{D}_{\max} = \omega(1).$$

Probability of satisfying (G2) Note that $\mathbb{D}_{\text{GKL}}(j, k) = (1 + o(1))\mathbb{D}_{\text{GCH}}(j, k) \cdot q_N$ by Lemma 3.3.4. We claim that it suffices to prove the following holds with probability at least $1 - 2e^{-\frac{1}{2}\log(N/s)}$,

$$\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} N_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) \leq \left(1 - \frac{\log(N)}{\sqrt{N}}\right) \cdot \mathbb{D}_{\text{GKL}}(j, k) - \frac{\mathcal{D}_{\max}}{\log q_N}, \quad (\text{B.4.9})$$

where the proof is deferred to Lemma B.4.2. Assuming (B.4.9), subsequent logic is indicated as follows.

1. Remind that $(1 - \log(N)/\sqrt{N})^{\ell-1} \leq N_{\mathbf{w}}/\bar{N}_{\mathbf{w}}$ by Lemma 3.1.1, then by (B.4.9),

$$\begin{aligned} & \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) \\ & \leq \left(1 - \frac{\log(N)}{\sqrt{N}}\right)^{-1} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} N_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) < \mathbb{D}_{\text{GKL}}(j, k). \end{aligned}$$

2. Then by (3.3.11), $\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}) \geq \mathbb{D}_{\text{GKL}}(j, k)$, since part (1) and

$$\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \max \{ \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}), \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) \} \geq \mathbb{D}_{\text{GKL}}(j, k).$$

3. Then we use part (2) and $(1 - \log(N)/\sqrt{N})^{\ell-1} \leq N_{\mathbf{w}}/\bar{N}_{\mathbf{w}}$ again,

$$\begin{aligned} & \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} N_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}) \\ & \geq \left(1 - \frac{\log(N)}{\sqrt{N}}\right) \cdot \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K}^{(\ell-1)}} \bar{N}_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}) \geq \left(1 - \frac{\log(N)}{\sqrt{N}}\right) \cdot \mathbb{D}_{\text{GKL}}(j, k). \end{aligned}$$

4. Therefore by part (3) and (B.4.9), $\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} \cdot [\mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)}) - \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)})] \geq \mathcal{D}_{\max} / \log q_N$, which proves (G2).

Let $s = N^{1-\kappa_N}$. According to Lemma B.4.2, the probability that there are more than $s/3$ nodes in $\mathcal{V}$ not satisfying either (G1) or (G2) is upper bounded by Markov inequality as follows,

$$\begin{aligned} \frac{\mathbb{E}[\# \text{ of nodes that do not satisfy either (G1) or (G2)}]}{s/3} &\leq \frac{N \cdot 2e^{-\mathbb{D}_{\text{GCH}} \cdot q_N}}{s/3} \\ &\leq 6 \exp(-\mathbb{D}_{\text{GCH}} \cdot q_N + \log(N/s)) \leq 6 \exp(-\varepsilon \cdot \kappa_N \cdot \log(N)) \rightarrow 0, \end{aligned}$$

where the last inequality holds since (3.2.17) and $\kappa_N \cdot \log(N) \rightarrow \infty$. $\square$

Lemma B.4.2. *Under Assumption 3.2.2, with probability at least $1 - 2e^{-\mathbb{D}_{\text{GCH}} \cdot q_N}$, vertex v satisfies (G1) and (G2) simultaneously.*

Proof of Lemma B.4.2. Let $\mathcal{X}$ denote the set of degree profiles satisfying (G1), i.e.,

$$\begin{aligned} \mathcal{X} = \left\{ \{ \mathbf{x}^{(\ell)} \}_{\ell \in \mathcal{L}} \mid \mathbf{x}^{(\ell)} \in \mathbb{N}^{|\mathcal{W}C_K^{(\ell-1)}|}, \text{ s.t. } x_{\mathbf{w}}^{(\ell)} = \omega(1) \text{ for each } m, \mathbf{w}, \right. \\ \left. \text{and } \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} x_{\mathbf{w}}^{(\ell)} \leq 10\mathcal{D}_{\max}, \right\}. \end{aligned}$$

Since the cardinality $|\mathcal{W}C_K^{(\ell-1)}| = \binom{\ell+K-1}{K-1}$, for simplicity, we denote

$$T = \sum_{\ell \in \mathcal{L}} \binom{\ell+K-1}{K-1} = O(1). \quad (\text{B.4.10})$$

Let $\{\mathbf{D}_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}\}_{\mathbf{w}}$ denote the event $\mathbf{D}_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}$ for each $\ell \in \mathcal{L}$, $\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}$, then

$$\mathbb{P}(\{\mathbf{D}_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}\}_{\mathbf{w}}) = \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \binom{N_{\mathbf{w}}}{x_{\mathbf{w}}^{(\ell)}} \cdot [Q_{k \oplus \mathbf{w}}^{(\ell)}]^{x_{\mathbf{w}}^{(\ell)}} \cdot (1 - Q_{k \oplus \mathbf{w}}^{(\ell)})^{N_{\mathbf{w}} - x_{\mathbf{w}}^{(\ell)}}. \quad (\text{B.4.11})$$

Note that $\mu_{v, \mathbf{w}}^{(\ell)} := \mathbf{D}_{v, \mathbf{w}}^{(\ell)} / N_{\mathbf{w}}$ where $N_{\mathbf{w}} := \prod_{k=1}^K \binom{|\mathcal{V}_k|}{w_k} \asymp N^{\ell-1}$. For any $C > 0$, by Markov inequality,

we have

$$\begin{aligned}
& \mathbb{P}\left(\left\{\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) > \mathbf{C}\right\} \cap \{\mathbf{D}_v \leq 10\mathcal{D}_{\max}\}\right) \\
&= \sum_{\{\mathbf{x}^{(\ell)}\} \in \mathcal{X}} \mathbb{P}(\{\mathbf{D}_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}\}_{\mathbf{w}}^{\ell}) \cdot \mathbb{P}\left(\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) > \mathbf{C} \middle| \{\mathbf{D}_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}\}_{\mathbf{w}}^{\ell}\right) \\
&\leq \sum_{\{\mathbf{x}^{(\ell)}\} \in \mathcal{X}} \mathbb{P}(\{\mathbf{D}_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}\}_{\mathbf{w}}^{\ell}) \cdot e^{-\mathbf{C}} \cdot \mathbb{E}\left(\exp\left[\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)})\right] \middle| \{\mathbf{D}_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}\}_{\mathbf{w}}^{\ell}\right) \\
&\leq e^{-\mathbf{C}} \sum_{\{\mathbf{x}^{(\ell)}\} \in \mathcal{X}} \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} \mathbb{P}(\mathbf{D}_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}) \cdot \exp\left(x_{\mathbf{w}}^{(\ell)} \log \frac{x_{\mathbf{w}}^{(\ell)}/N_{\mathbf{w}}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + (N_{\mathbf{w}} - x_{\mathbf{w}}^{(\ell)}) \log \frac{1 - x_{\mathbf{w}}^{(\ell)}/N_{\mathbf{w}}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}}\right) \\
&\lesssim e^{-\mathbf{C}} \exp\left(\frac{T}{2} \log(\mathcal{D}_{\max})\right), \tag{B.4.12}
\end{aligned}$$

where the proof of (B.4.12) is deferred later. By Assumption 3.2.2, $C_{(3.2.3)} = O(1)$ and $\mathcal{D}_{\max} \asymp C_{(3.2.3)} \mathbb{D}_{\text{GCH}}(j, k) q_N$, then $\mathbb{D}_{\text{GKL}}(j, k) = \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N \gg \frac{\mathcal{D}_{\max}}{\log q_N}$ by Lemma 3.3.4. We take

$$\begin{aligned}
\mathbf{C} &= \left(1 - \frac{\log(N)}{\sqrt{N}}\right) \cdot \mathbb{D}_{\text{GKL}}(j, k) - \frac{\mathcal{D}_{\max}}{\log q_N} \\
&\stackrel{\text{Lemma 3.3.4}}{=} (1 - o(1)) \cdot \mathbb{D}_{\text{GCH}}(j, k) q_N \asymp \frac{\mathcal{D}_{\max}}{C_{(3.2.3)}} \gg \log(\mathcal{D}_{\max}),
\end{aligned}$$

leading to $e^{-\mathbf{C}} \exp\left(\frac{T}{2} \log(\mathcal{D}_{\max})\right) = e^{-(1-o(1)) \cdot \mathbf{C}}$. Then we are able to establish (B.4.9),

$$\begin{aligned}
& \mathbb{P}\left(\left\{\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} N_{\mathbf{w}} \cdot \mathbb{D}_{\text{KL}}(\mu_{v, \mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) \leq \left(1 - \frac{\log(N)}{\sqrt{N}}\right) \mathbb{D}_{\text{GKL}}(j, k) - \frac{\mathcal{D}_{\max}}{\log q_N}\right\} \right. \\
&\quad \left. \cap \{\mathbf{D}_v \leq 10\mathcal{D}_{\max}\}\right) \\
&\geq \mathbb{P}(\mathbf{D}_v \leq 10\mathcal{D}_{\max}) \cdot \left(1 - e^{-\mathbf{C}} \exp\left(\frac{T}{2} \log(\mathcal{D}_{\max})\right)\right) \\
&\geq \mathbb{P}(\mathbf{D}_v \leq 10\mathcal{D}_{\max}) - e^{-\mathbf{C}} \exp\left(\frac{T}{2} \log(\mathcal{D}_{\max})\right) \\
&\geq 1 - e^{-10\mathcal{D}_{\max}} - e^{-\mathbf{C}} = 1 - e^{-10\mathcal{D}_{\max}} - e^{-\mathbb{D}_{\text{GCH}} \cdot q_N} \geq 1 - 2e^{-\mathbb{D}_{\text{GCH}} \cdot q_N},
\end{aligned}$$

where the last line holds since $\mathbb{D}_{\text{GCH}} = \min_{1 \leq j, k \leq K} \mathbb{D}_{\text{GCH}}(j, k)$, $C \gg T \log(\mathcal{D}_{\max})$ and $\mathcal{D}_{\max} \geq \mathbb{D}_{\text{GCH}} \cdot q_N$ as shown in Lemma D.1.1. $\square$

Proof of inequality (B.4.12). Note that $D_{v, \mathbf{w}}^{(\ell)} |_{Y_v=k} \sim \text{Bin}(N_{\mathbf{w}}, Q_{k \oplus \mathbf{w}}^{(\ell)})$, then by Lemma D.1.3,

$$\begin{aligned} \mathbb{P}(D_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}) &= \binom{N_{\mathbf{w}}}{x_{\mathbf{w}}^{(\ell)}} \exp \left(x_{\mathbf{w}}^{(\ell)} \log Q_{\mathbf{w}}^{(\ell)} + (N_{\mathbf{w}} - x_{\mathbf{w}}^{(\ell)}) \log(1 - Q_{\mathbf{w}}^{(\ell)}) \right) \\ \log \binom{N_{\mathbf{w}}}{x_{\mathbf{w}}^{(\ell)}} &= -\frac{1}{2} \log(2\pi x_{\mathbf{w}}^{(\ell)}) - \frac{1}{2} \log \left(1 - \frac{x_{\mathbf{w}}^{(\ell)}}{N_{\mathbf{w}}} \right) + N_{\mathbf{w}} \log N_{\mathbf{w}} - x_{\mathbf{w}}^{(\ell)} \log x_{\mathbf{w}}^{(\ell)} \\ &\quad - (N_{\mathbf{w}} - x_{\mathbf{w}}^{(\ell)}) \log(N_{\mathbf{w}}) - (N_{\mathbf{w}} - x_{\mathbf{w}}^{(\ell)}) \log \left(1 - \frac{x_{\mathbf{w}}^{(\ell)}}{N_{\mathbf{w}}} \right) + o(1) \end{aligned}$$

Thus for each $\ell \in \mathcal{L}$ and $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$, we have

$$\begin{aligned} &\mathbb{P}(D_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}) \cdot \exp \left(x_{\mathbf{w}}^{(\ell)} \log \frac{x_{\mathbf{w}}^{(\ell)}/N_{\mathbf{w}}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + (N_{\mathbf{w}} - x_{\mathbf{w}}^{(\ell)}) \log \frac{1 - x_{\mathbf{w}}^{(\ell)}/N_{\mathbf{w}}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \right) \\ &= \exp \left(-\frac{1}{2} \log(2\pi x_{\mathbf{w}}^{(\ell)}) - \frac{1}{2} \log \left(1 - \frac{x_{\mathbf{w}}^{(\ell)}}{N_{\mathbf{w}}} \right) + o(1) \right) = \exp \left(-\frac{1}{2} \log(2\pi x_{\mathbf{w}}^{(\ell)}) + o(1) \right), \end{aligned}$$

where the last equality holds since $1 \ll x_{\mathbf{w}}^{(\ell)} \lesssim \mathcal{D}_{\max}^{(\ell)} \ll N^{\ell-1}$. Also, for sufficiently large N , the cardinality of $\mathcal{X}$ is bounded by

$$|\mathcal{X}| \leq \binom{10\mathcal{D}_{\max}}{T} \leq \frac{(10\mathcal{D}_{\max})^T}{T!} \leq \exp \left(T \log(\mathcal{D}_{\max}) \right), \quad (\text{B.4.13})$$

where $T = O(1)$, then

$$\begin{aligned} &\sum_{\{\mathbf{x}^{(\ell)}\} \in \mathcal{X}} \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \mathbb{P}(D_{v, \mathbf{w}}^{(\ell)} = x_{\mathbf{w}}^{(\ell)}) \cdot \exp \left(x_{\mathbf{w}}^{(\ell)} \log \frac{x_{\mathbf{w}}^{(\ell)}/N_{\mathbf{w}}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + (N_{\mathbf{w}} - x_{\mathbf{w}}^{(\ell)}) \log \frac{1 - x_{\mathbf{w}}^{(\ell)}/N_{\mathbf{w}}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \right) \\ &\leq |\mathcal{X}| \cdot \exp \left(-\frac{T}{2} \log(2\pi x_{\mathbf{w}}^{(\ell)}) + o(1) \right) \leq \exp \left(\frac{T}{2} \log(\mathcal{D}_{\max}) \right). \end{aligned}$$

$\square$

Proof of Lemma 3.5.2. Let $\mathcal{E}_{\ell}(v, \mathcal{S})$ denote the set of ℓ -hyperedges containing v with the rest

$\ell - 1$ nodes in $\mathcal{S} \subset \mathcal{V}$, and $\mathcal{E}_\ell(\mathcal{S}, \mathcal{S}) := \cup_{v \in \mathcal{S}} \mathcal{E}_\ell(v, \mathcal{S})$ denote the set of ℓ -hyperedges with all vertices in $\mathcal{S} \subset \mathcal{V}$, where the repeated edges are dropped, and $\mathcal{E}(\mathcal{S}, \mathcal{S}) := \cup_{\ell \in \mathcal{L}} \mathcal{E}_\ell(\mathcal{S}, \mathcal{S})$ similarly. To conclude the proof, we build the sequence of sets $\{\mathcal{S}^{(t)} \subset \mathcal{V}\}_{0 \leq t \leq t^*}$ as follows.

- $\mathcal{S}^{(0)} = \mathcal{S}$, where $\mathcal{S}$ denotes the set of nodes which do not satisfy (G1) or (G2). By Lemma B.4.1, $|\mathcal{S}| \leq N^{1-\kappa_N}/3$ with probability at least $1 - 6e^{-\varepsilon \kappa_N \log(N)}$.
- For $t \geq 1$, $\mathcal{S}^{(t)} = \mathcal{S}^{(t-1)} \cup \{v_t\}$ if there exists $v_t \in \mathcal{V}$ s.t. $|\mathcal{E}(v_t, \mathcal{S}^{(t-1)})| > L(\mathcal{D}_{\max})^{1+\delta}$.
- The sequence ends if no such vertex can be found in $\mathcal{V} \setminus \mathcal{S}^{(t)}$, and we obtain $\mathcal{S}^{(t^*)}$.

Denote $s = N^{1-\kappa_N}$. Assume that $|\mathcal{S}^{(t^*)}| \geq s$. Consider the step $\tilde{t} = s - |\mathcal{S}^{(0)}|$, then $\tilde{t} \geq 2s/3$. By construction, each of the nodes added so far contributes more than $L(\mathcal{D}_{\max})^{1+\delta}$ in $|\mathcal{E}(\mathcal{S}^{(\tilde{t})}, \mathcal{S}^{(\tilde{t})})|$. At that point, $|\mathcal{S}^{(\tilde{t})}| = s$. The number of edges $|\mathcal{E}(\mathcal{S}^{(\tilde{t})}, \mathcal{S}^{(\tilde{t})})| \geq 2s/3 \cdot L(\mathcal{D}_{\max})^{1+\delta} \geq 4s/3(\mathcal{D}_{\max})^{1+\delta} > s(\mathcal{D}_{\max})^{1+\delta}$ since $L \geq 2$. On the other hand, with high probability, there cannot be any such set $|\mathcal{S}^{(\tilde{t})}|$ by Lemma B.4.3. Therefore, $|\mathcal{V} \setminus \mathcal{G}| < s$. $\square$

Lemma B.4.3. *Let $\delta > 0$ be an absolute constant. Denote $s = N^{1-\kappa_N}$ where $\kappa_N \in (0, 1]$ satisfies (3.2.17). Recall $\mathcal{D}_{\max}$ in (G1). Then with probability at least $1 - e^{-s(\mathcal{D}_{\max})^{1+\delta}/4}$, there is no subset $\mathcal{S} \subset \mathcal{V}$ with $|\mathcal{S}| = s$ such that $|\mathcal{E}(\mathcal{S}, \mathcal{S})| \geq s(\mathcal{D}_{\max})^{1+\delta}$ when N is large enough.*

Proof of Lemma B.4.3. For any given subset $\mathcal{S} \subset \mathcal{V}$ with $|\mathcal{S}| = s$, by Markov's inequality,

$$\begin{aligned}
& \mathbb{P}(|\mathcal{E}(\mathcal{S}, \mathcal{S})| \geq s(\mathcal{D}_{\max})^{1+\delta}) \\
& \leq \inf_{\beta > 0} e^{-\beta s(\mathcal{D}_{\max})^{1+\delta}} \mathbb{E}[\exp(\beta \cdot |\mathcal{E}(\mathcal{S}, \mathcal{S})|)] = \inf_{\beta > 0} e^{-\beta s(\mathcal{D}_{\max})^{1+\delta}} \prod_{\ell \in \mathcal{L}} \mathbb{E}[\exp(\beta \cdot |\mathcal{E}_\ell(\mathcal{S}, \mathcal{S})|)] \\
& \leq \inf_{\beta > 0} e^{-\beta s(\mathcal{D}_{\max})^{1+\delta}} \prod_{\ell \in \mathcal{L}} [Q_{\max}^{(\ell)} \cdot e^\beta + (1 - Q_{\max}^{(\ell)})]^{s_\ell}, \quad (\text{edge independence and } 1 + x \leq e^x) \\
& \leq \inf_{\beta > 0} \exp \left[\sum_{\ell \in \mathcal{L}} \binom{s}{\ell} Q_{\max}^{(\ell)} (e^\beta - 1) - \beta s(\mathcal{D}_{\max})^{1+\delta} \right] \\
& = \exp \left\{ - \left[\sum_{\ell \in \mathcal{L}} \binom{s}{\ell} Q_{\max}^{(\ell)} - s(\mathcal{D}_{\max})^{1+\delta} \right] - s(\mathcal{D}_{\max})^{1+\delta} \cdot \log \left(\frac{s(\mathcal{D}_{\max})^{1+\delta}}{\sum_{\ell \in \mathcal{L}} \binom{s}{\ell} Q_{\max}^{(\ell)}} \right) \right\} \\
& \leq \exp \left[- \frac{s}{2} (\mathcal{D}_{\max})^{1+\delta} \right],
\end{aligned}$$

where the second to last inequality holds since the function $h(\beta) = a(e^\beta - 1) - b\beta$ achieves its infimum at $\beta = \log(\frac{b}{a})$, and the last inequality holds since $o(1) = \binom{s}{\ell} Q_{\max}^{(\ell)} \ll s(\mathcal{D}_{\max})^{1+\delta}$ and $\binom{N}{\ell} Q_{\max}^{(\ell)} \asymp \mathcal{D}_{\max}$. The number of subsets $\mathcal{S} \subset \mathcal{V}$ with size s is $\binom{N}{s} \leq (\frac{eN}{s})^s$. Then by Markov inequality, the probability of existing a subset $\mathcal{S} \subset \mathcal{V}$ with $|\mathcal{S}| = s$ such that $|\mathcal{E}(\mathcal{S}, \mathcal{S})| > s(\mathcal{D}_{\max})^{1+\delta}$ is upper bounded by

$$\begin{aligned} & \mathbb{P}(|\{\mathcal{S} : |\mathcal{E}(\mathcal{S}, \mathcal{S})| > s(\mathcal{D}_{\max})^{1+\delta} \text{ and } |\mathcal{S}| = s\}| \geq 1) \\ & \leq \mathbb{E}[|\{\mathcal{S} : |\mathcal{E}(\mathcal{S}, \mathcal{S})| > s(\mathcal{D}_{\max})^{1+\delta} \text{ and } |\mathcal{S}| = s\}|], \end{aligned}$$

where the term on the right hand side can be further bounded by

$$\begin{aligned} & \mathbb{E}[|\{\mathcal{S} : |\mathcal{E}(\mathcal{S}, \mathcal{S})| > s(\mathcal{D}_{\max})^{1+\delta} \text{ and } |\mathcal{S}| = s\}|] \\ & = \binom{N}{s} \cdot \mathbb{P}(|\{\mathcal{S} : |\mathcal{E}(\mathcal{S}, \mathcal{S})| \geq s(\mathcal{D}_{\max})^{1+\delta} \text{ and } |\mathcal{S}| = s\}|) \\ & \leq \left(\frac{eN}{s}\right)^s \cdot \exp\left(-\frac{s}{2}(\mathcal{D}_{\max})^{1+\delta}\right) \quad (\text{plug in } s = N^{1-\kappa_N}) \\ & = \exp\left(-\frac{s}{2}(\mathcal{D}_{\max})^{1+\delta} + s \log \frac{eN}{s}\right) = \exp\left(-\frac{N^{1-\kappa_N}}{2}(\mathcal{D}_{\max})^{1+\delta} \left[1 - \frac{2\kappa_N \log(N) + 1}{(\mathcal{D}_{\max})^{1+\delta}}\right]\right) \\ & \lesssim \exp\left(-\frac{N^{1-\kappa_N}}{4}(\mathcal{D}_{\max})^{1+\delta}\right), \end{aligned}$$

where the last inequality holds since $\mathcal{D}_{\max} \gtrsim \mathbb{D}_{\text{GCH}} \cdot q_N \geq (1 + \varepsilon)\kappa_N \log(N)$ by (3.2.17), and

$$\frac{2\kappa_N \log(N) + 1}{(\mathcal{D}_{\max})^{1+\delta}} \lesssim \frac{2\kappa_N \log(N) + 1}{(\mathbb{D}_{\text{GCH}} \cdot q_N)^{1+\delta}} \lesssim \frac{1}{[\kappa_N \log(N)]^\delta} = o(1) < \frac{1}{2},$$

since $\kappa_N \log(N) \rightarrow \infty$ as $N \rightarrow \infty$. □

Proof of Lemma 3.5.3. Let $\{\widehat{\mathcal{V}}_k^{(t)}\}_{k=1}^K$ denote the partition at time t . Remind that $\widehat{\mathcal{N}}^{(t)} := \cup_{j=1}^K \widehat{\mathcal{N}}_j^{(t)} = \cup_{j=1}^K (\widehat{\mathcal{V}}_j^{(t)} \setminus \mathcal{V}_j) \cap \mathcal{G}$, where $\widehat{\mathcal{N}}_j^{(t)} := \cup_{k \neq j} \widehat{\mathcal{N}}_{jk}^{(t)} = (\widehat{\mathcal{V}}_j^{(t)} \setminus \mathcal{V}_j) \cap \mathcal{G}$, and $\widehat{\mathcal{N}}_{jk}^{(t)} := (\widehat{\mathcal{V}}_j^{(t)} \cap \mathcal{V}_k) \cap \mathcal{G}$, representing the set of vertices in $\mathcal{G}$ which belong to $\mathcal{V}_k$ but are misclassified in $\widehat{\mathcal{V}}_j^{(t)}$.

For node $v \in \widehat{\mathcal{N}}_{jk}^{(t+1)}$ ($v \in \mathcal{V}_k$ but misclassified in $\widehat{\mathcal{V}}_j^{(t+1)}$), by Algorithm 3.2.3, it has to satisfy

$$0 \leq \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \left(\widehat{\mathbf{D}}_{v, \mathbf{w}}^{(t)} \cdot \log \frac{\widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)}}{\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)}} + [\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{\mathbf{D}}_{v, \mathbf{w}}^{(t)}] \cdot \log \frac{(1 - \widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - \widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)})} \right) \quad (\text{B.4.14})$$

where $\widehat{\mathbf{D}}_{v, \mathbf{w}}^{(t)}$ denotes the number ℓ -hyperedges containing v with the rest $\ell - 1$ nodes distributed as $\mathbf{w}$ at step t , i.e., w_j nodes from $\widehat{\mathcal{V}}_j^{(t)}$ for each $j \in [K]$. Let $\widehat{N}_{\mathbf{w}}^{(t)}$ denote the capacity of such hyperedges. Then with the proofs of (B.4.15a), (B.4.15b), (B.4.15c) (B.4.15d) deferred to next

several pages, we have,

$$\begin{aligned}
0 &\leq \sum_{1 \leq j \neq k \leq K} \sum_{v \in \widehat{\mathcal{N}}_{jk}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \left(\widehat{D}_{v,\mathbf{w}}^{(t)} \cdot \log \frac{\widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)}}{\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)}} + (\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{D}_{v,\mathbf{w}}^{(t)}) \cdot \log \frac{(1 - \widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - \widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)})} \right) \\
&= \sum_{1 \leq j \neq k \leq K} \sum_{v \in \widehat{\mathcal{N}}_{jk}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \left\{ \widehat{D}_{v,\mathbf{w}}^{(t)} \cdot \left(\log \frac{Q_{j \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + \log \frac{\widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)}}{Q_{j \oplus \mathbf{w}}^{(\ell)}} - \log \frac{\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} \right) \right. \\
&\quad \left. + (\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{D}_{v,\mathbf{w}}^{(t)}) \cdot \left[\log \frac{(1 - Q_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})} + \log \frac{(1 - \widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{j \oplus \mathbf{w}}^{(\ell)})} - \log \frac{(1 - \widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})} \right] \right\} \\
&\leq \sum_{1 \leq j \neq k \leq K} \sum_{v \in \widehat{\mathcal{N}}_{jk}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \left\{ \widehat{D}_{v,\mathbf{w}}^{(t)} \cdot \log \frac{Q_{j \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + (\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{D}_{v,\mathbf{w}}^{(t)}) \cdot \log \frac{(1 - Q_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})} \right\} \\
&\quad + |\widehat{\mathcal{N}}^{(t+1)}| \cdot C_{(B.4.15a)} \cdot \log q_N \tag{B.4.15a}
\end{aligned}$$

$$\begin{aligned}
&\leq \sum_{1 \leq j \neq k \leq K} \sum_{v \in \widehat{\mathcal{N}}_{jk}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \left\{ D_{v,\mathbf{w}}^{(\ell)} \cdot \log \frac{Q_{j \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + (N_{\mathbf{w}} - D_{v,\mathbf{w}}^{(\ell)}) \cdot \log \frac{(1 - Q_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})} \right\} \\
&\quad + 2 \log C_{(3.2.3)} \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} |\mathcal{E}_{\ell}(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})| \\
&\quad + |\widehat{\mathcal{N}}^{(t+1)}| \cdot (C_{(B.4.15a)} \cdot \log q_N + C_{(B.4.15b)}) \tag{B.4.15b}
\end{aligned}$$

$$\begin{aligned}
&\leq -|\widehat{\mathcal{N}}^{(t+1)}| \cdot \frac{\mathcal{D}_{\max}}{\log q_N} + 2 \log C_{(3.2.3)} \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} |\mathcal{E}_{\ell}(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})| \\
&\quad + |\widehat{\mathcal{N}}^{(t+1)}| \cdot (C_{(B.4.15a)} \cdot \log q_N + C_{(B.4.15b)}) \tag{B.4.15c}
\end{aligned}$$

$$\begin{aligned}
&\leq -|\widehat{\mathcal{N}}^{(t+1)}| \frac{\mathcal{D}_{\max}}{\log q_N} + C_{(B.4.15d)} \cdot \sqrt{|\widehat{\mathcal{N}}^{(t)}| \cdot |\widehat{\mathcal{N}}^{(t+1)}| \cdot \mathcal{D}_{\max}} \\
&\quad + |\widehat{\mathcal{N}}^{(t+1)}| \cdot \left(C_{(B.4.15a)} \cdot \log q_N + C_{(B.4.15b)} + 2 \log C_{(3.2.3)} \cdot \sum_{\ell \in \mathcal{L}} \binom{\ell + K - 1}{\ell} \right), \tag{B.4.15d}
\end{aligned}$$

where $|\mathcal{E}_{\ell}(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})|$ in (B.4.15b) denotes the number of ℓ -hyperedges e containing v where the rest $\ell - 1$ nodes distributed as $\mathbf{w}$ among $\{\widehat{\mathcal{V}}_k^{(t)}\}_{k=1}^K$ with at least one of those $\ell - 1$ nodes in $\widehat{\mathcal{N}}^{(t)}$,

and $C_{(B.4.15d)} := 2C_{(3.4.3)} \log C_{(3.2.3)}$. Note that $C_{(B.4.15a)}, C_{(B.4.15b)}, C_{(3.2.3)}, M, K = O(1)$, then

$$\frac{\mathcal{D}_{\max}}{\log q_N} \gg C_{(B.4.15a)} \cdot \log q_N + C_{(B.4.15b)} + 2 \log C_{(3.2.3)} \cdot \sum_{\ell \in \mathcal{L}} \binom{\ell + K - 1}{\ell}. \quad (B.4.16)$$

At the same time, $\mathcal{D}_{\max} \asymp q_N$ by the assumption of Theorem 3.4.2. Then for sufficiently large N , we have

$$\frac{|\widehat{\mathcal{N}}^{(t+1)}|}{|\widehat{\mathcal{N}}^{(t)}|} \leq 2C_{(3.4.3)} \log C_{(3.2.3)} \cdot \frac{[\log q_N]^2}{\mathcal{D}_{\max}} \leq \frac{2C_{(3.4.3)} \log C_{(3.2.3)}}{C_{(3.2.3)}} \frac{[\log q_N]^2}{q_N} \leq \frac{\log C_{(3.2.3)}}{C_{(3.2.3)}} \frac{1}{e}.$$

□

Proof of (B.4.15a). Note that for any $\widehat{q}, q > 0$ with $\widehat{q} < 2q$, the following holds

$$\left| \log \frac{\widehat{q}}{q} \right| \leq \log \left(\frac{q}{q - |\widehat{q} - q|} \right) \leq \frac{|\widehat{q} - q|}{q - |\widehat{q} - q|}, \quad (B.4.17)$$

where the first inequality follows by discussions on cases $2q > \widehat{q} > q$ and $\widehat{q} < q$, and the second comes from $\log(1+x) \leq x$. Thus one may focus on bounding $|\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)} - Q_{k \oplus \mathbf{w}}^{(\ell)}|$. According to Lemma 3.5.1, under Assumption 3.2.2, the following holds with probability at least $1 - \exp(-N \log q_N / q_N)$ for each $\ell \in \mathcal{L}$,

$$\left| \log \frac{\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} \right| \leq \frac{|\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)} - Q_{k \oplus \mathbf{w}}^{(\ell)}|}{Q_{k \oplus \mathbf{w}}^{(\ell)} - |\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)} - Q_{k \oplus \mathbf{w}}^{(\ell)}|} \lesssim \frac{\log q_N}{q_N}, \quad (B.4.18)$$

$$\left| \log \frac{(1 - \widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})} \right| \leq \frac{|\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)} - Q_{k \oplus \mathbf{w}}^{(\ell)}|}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)}) - |\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)} - Q_{k \oplus \mathbf{w}}^{(\ell)}|} \lesssim \frac{\log q_N}{N^{\ell-1}}. \quad (B.4.19)$$

Meanwhile, $\mathcal{D}_{\max} = \sum_{\ell \in \mathcal{L}} \mathcal{D}_{\max}^{(\ell)} \asymp q_N$ under Assumption 3.2.2, and $\widehat{D}_{v, \mathbf{w}}^{(t)} \lesssim \mathcal{D}_{\max} \ll N^{\ell-1}$ by

(G1) with $\widehat{N}_{\mathbf{w}}^{(t)} := \prod_{k=1}^K (|\widehat{\mathcal{V}}_{\mathbf{w}_k}^{(t)}|) \asymp N^{\ell-1}$. Then $(\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{D}_{v,\mathbf{w}}^{(t)}) \asymp N^{\ell-1}$, and

$$\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \widehat{D}_{v,\mathbf{w}}^{(t)} \cdot \left| \log \frac{\widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)}}{Q_{j \oplus \mathbf{w}}^{(\ell)}} - \log \frac{\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} \right| \lesssim \log q_N, \quad (\text{B.4.20})$$

$$\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} (\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{D}_{v,\mathbf{w}}^{(t)}) \cdot \left| \log \frac{(1 - \widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{j \oplus \mathbf{w}}^{(\ell)})} - \log \frac{(1 - \widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})} \right| \lesssim \log q_N. \quad (\text{B.4.21})$$

Consequently, with probability at least $1 - \exp(-N \log q_N / q_N)$,

$$\begin{aligned} & \sum_{j \neq k} \sum_{v \in \widehat{\mathcal{N}}_{jk}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \left\{ \widehat{D}_{v,\mathbf{w}}^{(t)} \cdot \left| \log \frac{\widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)}}{Q_{j \oplus \mathbf{w}}^{(\ell)}} - \log \frac{\widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} \right| \right. \\ & \quad \left. + (\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{D}_{v,\mathbf{w}}^{(t)}) \cdot \left| \log \frac{(1 - \widehat{Q}_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{j \oplus \mathbf{w}}^{(\ell)})} - \log \frac{(1 - \widehat{Q}_{k \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})} \right| \right\} \\ & \leq C_{(\text{B.4.15a})} |\widehat{\mathcal{N}}^{(t+1)}| \cdot \log q_N, \end{aligned}$$

where $C_{(\text{B.4.15a})} > 0$ is some constant. $\square$

Proof of (B.4.15b). We want to replace $\widehat{D}_{v,\mathbf{w}}^{(t)}$ with $D_{v,\mathbf{w}}^{(\ell)}$ in this step, where $D_{v,\mathbf{w}}^{(\ell)}$ (resp. $\widehat{D}_{v,\mathbf{w}}^{(t)}$) denotes the number of ℓ -hyperedges containing v with the rest $\ell - 1$ nodes distributed as $\mathbf{w}$, i.e., w_l nodes in $\mathcal{V}_l$ (resp. $\widehat{\mathcal{V}}_l^{(t)}$) for each $l \in [K]$. Note that $\widehat{D}_{v,\mathbf{w}}^{(t)} \leq D_{v,\mathbf{w}}^{(\ell)} + |\widehat{D}_{v,\mathbf{w}}^{(t)} - D_{v,\mathbf{w}}^{(\ell)}|$ and $(\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{D}_{v,\mathbf{w}}^{(t)}) \leq (N_{\mathbf{w}} - D_{v,\mathbf{w}}^{(\ell)}) + |(\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{D}_{v,\mathbf{w}}^{(t)}) - (N_{\mathbf{w}} - D_{v,\mathbf{w}}^{(\ell)})|$, then it suffices to control the contribution from deviation terms.

- For hyperedge $e \ni v$ with $\mathbf{y}(e) = \widehat{\mathbf{y}}^{(t)}(e)$, the contribution of such e inside $|\widehat{D}_{v,\mathbf{w}}^{(t)} - D_{v,\mathbf{w}}^{(\ell)}|$ cancels out. Thus it suffices to consider $e \in \mathcal{E}_{\ell}(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})$, where $\mathcal{E}_{\ell}(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})$ denotes the set of ℓ -hyperedges e containing v where the rest $\ell - 1$ nodes distributed as $\mathbf{w}$ among $\{\widehat{\mathcal{V}}_k^{(t)}\}_{k=1}^K$ with at least one of those $\ell - 1$ nodes in $\widehat{\mathcal{N}}^{(t)}$. Then under the regime (3.2.2)

and Assumption 3.2.2,

$$\begin{aligned} & \sum_{1 \leq j \neq k \leq K} \sum_{v \in \widehat{\mathcal{N}}_{jk}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} |\widehat{\mathbf{D}}_{v,\mathbf{w}}^{(t)} - \mathbf{D}_{v,\mathbf{w}}^{(\ell)}| \cdot \log \frac{Q_{j \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} \\ & \leq \log C_{(3.2.3)} \cdot \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} |\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})|. \end{aligned}$$

- Meanwhile, $(1 - \eta_N)^{\ell-1} \lesssim \left(\frac{|\widehat{\mathcal{V}}_k^{(t)}|}{|\mathcal{V}_k|} \right)^{\ell-1} \asymp \frac{\widehat{N}_{\mathbf{w}}^{(t)}}{N_{\mathbf{w}}} \lesssim (1 + \eta_N)^{\ell-1}$, where $\eta_N \lesssim (\rho_N)^{-1}$ with probability at least $1 - O(N^{-c})$ for some $c > 0$ according to (3.4.10), then we have

$$|\widehat{N}_{\mathbf{w}}^{(t)} - N_{\mathbf{w}}| = N_{\mathbf{w}} \cdot \left| 1 - \frac{\widehat{N}_{\mathbf{w}}^{(t)}}{N_{\mathbf{w}}} \right| \leq [1 - (1 - \eta_N)^{\ell-1}] N_{\mathbf{w}} \leq (\ell - 1) \eta_N N_{\mathbf{w}} \lesssim N^{\ell-1} (\rho_N)^{-1}.$$

Also, $\log \frac{1 - Q_{j \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} = \log \left(1 + \frac{Q_{k \oplus \mathbf{w}}^{(\ell)} - Q_{j \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \right) = \frac{Q_{k \oplus \mathbf{w}}^{(\ell)} - Q_{j \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} (1 + o(1)) \asymp N^{-(\ell-1)} \rho_N$, then there exists some constant $C_{(B.4.15b)} > 0$ such that

$$\sum_{1 \leq j \neq k \leq K} \sum_{v \in \widehat{\mathcal{N}}_{jk}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} |(\widehat{N}_{\mathbf{w}}^{(t)} - \widehat{\mathbf{D}}_{v,\mathbf{w}}^{(t)}) - (N_{\mathbf{w}} - \mathbf{D}_{v,\mathbf{w}}^{(\ell)})| \cdot \log \frac{1 - Q_{j \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \quad (\text{B.4.22})$$

$$\leq \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} |\widehat{N}_{\mathbf{w}}^{(t)} - N_{\mathbf{w}}| \cdot \frac{Q_{k \oplus \mathbf{w}}^{(\ell)} - Q_{j \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \quad (\text{B.4.23})$$

$$+ \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} |\widehat{\mathbf{D}}_{v,\mathbf{w}}^{(t)} - \mathbf{D}_{v,\mathbf{w}}^{(\ell)}| \cdot \frac{Q_{k \oplus \mathbf{w}}^{(\ell)} - Q_{j \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}} \quad (\text{B.4.24})$$

$$\leq C_{(B.4.15b)} \cdot |\widehat{\mathcal{N}}^{(t+1)}| + \log C_{(3.2.3)} \cdot \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}C_K^{(\ell-1)}} |\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})|. \quad (\text{B.4.25})$$

□

Proof of (B.4.15c). For $\mu_{v,\mathbf{w}}^{(\ell)} := D_{v,\mathbf{w}}^{(\ell)} / N_{\mathbf{w}}$, $N_{\mathbf{w}} \asymp N^{\ell-1}$ in (3.1.2) for $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$, then

$$D_{v,\mathbf{w}}^{(\ell)} \cdot \log \frac{Q_{j \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}}^{(\ell)}} + (N_{\mathbf{w}} - D_{v,\mathbf{w}}^{(\ell)}) \cdot \log \frac{(1 - Q_{j \oplus \mathbf{w}}^{(\ell)})}{(1 - Q_{k \oplus \mathbf{w}}^{(\ell)})} = N_{\mathbf{w}} \cdot [\mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) - \mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)})],$$

where $\mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)})$ denotes the KL divergence defined in (3.3.12). It suffices to show

$$\sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} N_{\mathbf{w}} \cdot [\mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{k \oplus \mathbf{w}}^{(\ell)}) - \mathbb{D}_{\text{KL}}(\mu_{v,\mathbf{w}}^{(\ell)} \parallel Q_{j \oplus \mathbf{w}}^{(\ell)})] \leq -\frac{\mathcal{D}_{\max}}{\log q_N} \quad (\text{B.4.26})$$

for each $v \in \widehat{\mathcal{N}}^{(t)}$, which holds true according to (G2) since $\widehat{\mathcal{N}}^{(t)} \subset \mathcal{G}$. $\square$

Proof of (B.4.15d). Note that $\rho_N \asymp \mathcal{D}_{\max} = \sum_{\ell \in \mathcal{L}} \mathcal{D}_{\max}^{(\ell)} \asymp q_N$ under Assumption 3.2.2, where $\mathcal{D}_{\max}^{(\ell)} = \binom{N-1}{\ell-1} Q_{\max}^{(\ell)}$, then

$$\mathbb{E}|\mathcal{E}_{\ell}(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})| \leq Q_{\max}^{(\ell)} \cdot \eta_N N \cdot \binom{N-2}{\ell-2} \leq \eta_N q_N \lesssim 1, \quad (\text{B.4.27})$$

where the last inequality holds since and $\eta_N \lesssim (\rho_N)^{-1}$ by (3.4.10). Then by a union bound, the following holds

$$\max_{v \in \mathcal{V}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \mathbb{E}|\mathcal{E}_{\ell}(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})| \leq \sum_{\ell \in \mathcal{L}} \binom{\ell + K - 1}{\ell} \quad (\text{B.4.28})$$

Therefore by Theorem 3.4.2,

$$\begin{aligned}
& \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} |\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})| \\
&= \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} [|\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})| - \mathbb{E}|\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})|] \\
&\quad + \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} \mathbb{E}|\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})| \\
&\leq \mathbf{1}_{\widehat{\mathcal{N}}^{(t+1)}}^\top \cdot (\mathbf{A} - \mathbb{E}\mathbf{A}) \cdot \mathbf{1}_{\widehat{\mathcal{N}}^{(t)}} + |\widehat{\mathcal{N}}^{(t+1)}| \cdot \sum_{\ell \in \mathcal{L}} \binom{\ell + K - 1}{\ell} \\
&\leq \|\mathbf{1}_{\widehat{\mathcal{N}}^{(t+1)}}\|_2 \cdot \|\mathbf{A} - \mathbb{E}\mathbf{A}\|_2 \cdot \|\mathbf{1}_{\widehat{\mathcal{N}}^{(t)}}\|_2 + |\widehat{\mathcal{N}}^{(t+1)}| \cdot \sum_{\ell \in \mathcal{L}} \binom{\ell + K - 1}{\ell} \\
&\leq C_{(3.4.3)} \sqrt{|\widehat{\mathcal{N}}^{(t)}| \cdot |\widehat{\mathcal{N}}^{(t+1)}| \cdot \mathcal{D}_{\max}} + |\widehat{\mathcal{N}}^{(t+1)}| \cdot \sum_{\ell \in \mathcal{L}} \binom{\ell + K - 1}{\ell}.
\end{aligned}$$

Define $C_{(B.4.15d)} := 2C_{(3.4.3)} \log C_{(3.2.3)}$, then by Theorem 3.4.2, the following holds

$$\begin{aligned}
& 2 \log C_{(3.2.3)} \sum_{v \in \widehat{\mathcal{N}}^{(t+1)}} \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w}} |\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(t)}, \mathbf{w})| + |\widehat{\mathcal{N}}^{(t+1)}| \cdot (C_{(B.4.15a)} \log q_N + C_{(B.4.15b)}) \\
&\leq C_{(B.4.15d)} \cdot \sqrt{|\widehat{\mathcal{N}}^{(t)}| \cdot |\widehat{\mathcal{N}}^{(t+1)}| \cdot \mathcal{D}_{\max}} \\
&\quad + |\widehat{\mathcal{N}}^{(t+1)}| \cdot \left(C_{(B.4.15a)} \log q_N + C_{(B.4.15b)} + 2 \log C_{(3.2.3)} \cdot \sum_{\ell \in \mathcal{L}} \binom{\ell + K - 1}{\ell} \right),
\end{aligned}$$

with probability at least $1 - 2N^{-10} - 2e^{-N}$. □

B.5 Proofs in Section 3.6

Proof of Lemma 3.2.11. Recall that $\widehat{\mathbf{y}}_{-v}$ is some estimation of $\mathbf{y}_{-v}$, while $\mathbb{H}$ and $\mathbf{Y}_{-v}$ denote the laws of $\mathcal{H}$ and $\mathbf{y}_{-v}$, respectively. By definition, the posterior probability is

$$\mathbb{P}(\mathbf{Y}_v = k \mid \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \widehat{\mathbf{y}}_{-v}) = \frac{\mathbb{P}(\mathbf{Y}_v = k, \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \widehat{\mathbf{y}}_{-v})}{\mathbb{P}(\mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \widehat{\mathbf{y}}_{-v})}. \quad (\text{B.5.1})$$

We first evaluate the numerator. Let d_v be the observed number of edges in $\mathcal{H}$ containing v with D_v denoting its law. Let $\mathcal{H}_{-v}$ be the hypergraph obtained by deleting edges containing v from $\mathcal{H}$, with $\mathbb{H}_{-v}$ denoting its law. Let $\{Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}\}$ denote the event that the membership vector is $Y_v = k$ with the others being $\hat{\mathbf{y}}_{-v}$. Thanks to the edgewise independence between $\mathcal{H}_{-v}$ and D_v , the following holds

$$\begin{aligned}
& \mathbb{P}(\mathbb{H} = \mathcal{H}, Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) = \mathbb{P}(\mathbb{H} = \mathcal{H} \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \\
& = \mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbb{H}_{-v} = \mathcal{H}_{-v} \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \\
& = \mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbb{H}_{-v} = \mathcal{H}_{-v}, Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \\
& = \mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbb{H}_{-v} = \mathcal{H}_{-v}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v} \mid Y_v = k) \cdot \mathbb{P}(Y_v = k) \\
& = \mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbb{H}_{-v} = \mathcal{H}_{-v}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k),
\end{aligned}$$

where the last equality holds due to two types of independence: first, $\mathcal{H}_{-v}$ is independent of the assignment of $\{Y_v = k\}$ according to the generating process of model 1.1.8; second, the estimation $\hat{\mathbf{y}}_{-v}$ is independent of the assignment of $\{Y_v = k\}$ as well due to assumption. Then by edgewise independence between D_v and $\mathcal{H}_{-v}$, and the derivation above, we have

$$\begin{aligned}
& \mathbb{P}(Y_v = k \mid \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) = \frac{\mathbb{P}(Y_v = k, \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})}{\mathbb{P}(\mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})} \\
& = \frac{\mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbb{H}_{-v} = \mathcal{H}_{-v}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k)}{\mathbb{P}(D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbb{H}_{-v} = \mathcal{H}_{-v}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})} \\
& = \mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k) \cdot [\mathbb{P}(D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})]^{-1}.
\end{aligned}$$

Note that the denominator $\mathbb{P}(D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})$ can be written as follows

$$\begin{aligned}
& \mathbb{P}(D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) = \mathbb{P}(D_v = d_v \mid \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \\
& = \mathbb{P}(\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \sum_{j \in [K]} \mathbb{P}(D_v = d_v \mid Y_v = j, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}),
\end{aligned}$$

which is irrelevant to any specific assignment of $\{Y_v = k\}$. Then it can be factored from the MAP, and the proof of the desired argument follows directly

$$\begin{aligned}
\hat{z}_v^{\text{MAP}} &= \arg \max_{k \in [K]} \mathbb{P}(Y_v = k \mid \mathbb{H} = \mathcal{H}, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \\
&= \arg \max_{k \in [K]} \mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k) \cdot [\mathbb{P}(D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})]^{-1} \\
&= \arg \max_{k \in [K]} \mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k).
\end{aligned}$$

Another way to verify the derivation above is to directly employ the independence between $\{Y_v = k\}$ and $\{\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}\}$, i.e., $\mathbb{P}(Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) = \mathbb{P}(Y_v = k) \cdot \mathbb{P}(\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})$, which then gives us

$$\mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k) \cdot [\mathbb{P}(D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})]^{-1} \quad (\text{B.5.2})$$

$$= \frac{\mathbb{P}(Y_v = k, D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})}{\mathbb{P}(D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})} \cdot \frac{\mathbb{P}(Y_v = k)}{\mathbb{P}(Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})} \quad (\text{B.5.3})$$

$$= \mathbb{P}(Y_v = k \mid D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot [\mathbb{P}(\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})]^{-1}, \quad (\text{B.5.4})$$

where both $\mathbb{P}(\mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})$ and $\mathbb{P}(D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v})$ on two sides of the equation can be factored out from MAP since they are not relevant to any specific assignment $\{Y_v = k\}$. Then the MAP becomes

$$\hat{z}_v^{\text{MAP}} = \arg \max_{k \in [K]} \mathbb{P}(Y_v = k \mid D_v = d_v, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \quad (\text{B.5.5})$$

$$= \arg \max_{k \in [K]} \mathbb{P}(D_v = d_v \mid Y_v = k, \mathbf{Y}_{-v} = \hat{\mathbf{y}}_{-v}) \cdot \mathbb{P}(Y_v = k). \quad (\text{B.5.6})$$

□

Proof of Lemma 3.6.1. For ease of presentation, define the event $\widehat{\mathcal{E}}_{jk}$ to be

$$\widehat{\mathcal{E}}_{jk} := \mathbb{1} \left\{ \widehat{\mathbb{P}}_j(\mathbf{D}_v = d_v) \cdot \alpha_j > \widehat{\mathbb{P}}_k(\mathbf{D}_v = d_v) \cdot \alpha_k \right\} \quad (\text{B.5.7})$$

Equivalently, assuming $Y_v = k$, the probability of v being misclassified is

$$\mathbb{P}_{\text{err}}(v) = \sum_{d_v} \mathbb{P}_k(\mathbf{D}_v = d_v) \cdot \mathbb{P}(\exists j \neq k, \text{ s.t. } \{\widehat{\mathcal{E}}_{jk} = 1\}). \quad (\text{B.5.8})$$

Then the upper bound in (3.6.7) in follows by a simple union bound, i.e.,

$$\mathbb{P}(\exists j \neq k, \text{ s.t. } \{\widehat{\mathcal{E}}_{jk} = 1\}) \leq \sum_{j=1, j \neq k}^K \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 1) = \sum_{j=1, j \neq k}^K \left(1 - \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0)\right). \quad (\text{B.5.9})$$

On the other hand, $\mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0) \geq \prod_{j=1, j \neq k}^K \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0)$ for any $j \in [K]$ with $j \neq k$, then

$$\begin{aligned} \sum_{j=1, j \neq k}^K \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0) &\geq (K-1) \prod_{j=1, j \neq k}^K \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0) \\ \iff \sum_{j=1, j \neq k}^K \left(1 - \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0)\right) &\leq (K-1) \left(1 - \prod_{j=1, j \neq k}^K \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0)\right) \end{aligned}$$

Then the lower bound in (3.6.7) follows since

$$\mathbb{P}(\exists j \neq k, \text{ s.t. } \{\widehat{\mathcal{E}}_{jk} = 1\}) = 1 - \prod_{j=1, j \neq k}^K \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0) \geq \frac{1}{K-1} \sum_{j=1, j \neq k}^K \left(1 - \mathbb{P}(\widehat{\mathcal{E}}_{jk} = 0)\right).$$

□

Proof of Lemma 3.6.4. According to (3.4.11), the mismatch ratio after the initial stage satisfies $\eta_N \lesssim \rho_N^{-1} = o(1)$ with probability at least $1 - N^{-10}$. Recall that $\widehat{\mathcal{N}}^{(0)}$ in Lemma 3.5.3 denotes the set of nodes misclassified by Algorithm 3.2.2, then $|\widehat{\mathcal{N}}^{(0)}| \lesssim \eta_N N$. Let $\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})$ denote the set of ℓ -hyperedges e containing v and at least one node in e belongs to $\widehat{\mathcal{N}}^{(0)}$, then the capacity of such ℓ -hyperedge is at most $N_v^{(\ell)} := \eta_N N \binom{N-2}{\ell-2} \lesssim N^{\ell-1} / \rho_N$. Without loss of generality, we

assume $v \in \mathcal{V}_k$. Note that $\mathcal{W}C_K^{(\ell-1)}$ is a finite set, then there exist $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4 \in \mathcal{W}C_K^{(\ell-1)}$ such that

$$C_k^{(\ell)} := \max_{\mathbf{w}, \mathbf{w}' \in \mathcal{W}C_K^{(\ell-1)}} \frac{Q_{k \oplus \mathbf{w}}^{(\ell)}}{Q_{k \oplus \mathbf{w}'}^{(\ell)}} = \frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)}}{Q_{k \oplus \mathbf{w}_2}^{(\ell)}}, \quad \frac{1 - Q_{k \oplus \mathbf{w}_3}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}} = \max_{\mathbf{w}, \mathbf{w}' \in \mathcal{W}C_K^{(\ell-1)}} \frac{1 - Q_{k \oplus \mathbf{w}}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}'}^{(\ell)}}, \quad (\text{B.5.10})$$

where we can derive the facts $Q_{k \oplus \mathbf{w}_1}^{(\ell)} \geq Q_{k \oplus \mathbf{w}_2}^{(\ell)}$ and $Q_{k \oplus \mathbf{w}_4}^{(\ell)} \geq Q_{k \oplus \mathbf{w}_3}^{(\ell)}$. Then for edge $e \in \mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})$, the ratio $\widehat{\mathbb{P}}_k(\mathcal{A}_e^{(\ell)} = 1) / \mathbb{P}_k(\mathcal{A}_e^{(\ell)} = 1)$ can be bounded as below

$$\frac{Q_{k \oplus \mathbf{w}_2}^{(\ell)}}{Q_{k \oplus \mathbf{w}_1}^{(\ell)}} \leq \frac{\widehat{\mathbb{P}}_k(\mathcal{A}_e^{(\ell)} = 1)}{\mathbb{P}_k(\mathcal{A}_e^{(\ell)} = 1)} \leq \frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)}}{Q_{k \oplus \mathbf{w}_2}^{(\ell)}}, \quad \frac{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_3}^{(\ell)}} \leq \frac{\widehat{\mathbb{P}}_k(\mathcal{A}_e^{(\ell)} = 0)}{\mathbb{P}_k(\mathcal{A}_e^{(\ell)} = 0)} \leq \frac{1 - Q_{k \oplus \mathbf{w}_3}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}}. \quad (\text{B.5.11})$$

By Definition 3.6.2, $G(v) = 1$ if $G_\ell(v) = 1$ for each $\ell \in \mathcal{L}$, i.e., $|\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})| \leq c_N^{(\ell)}$ where we denote $c_N^{(\ell)} = 4q_{\max}^{(\ell)} / \log(q_{\max}^{(\ell)})$ for simplicity. Consequently, the ratio $\widehat{\mathbb{P}}_k(D_v = d_v) / \mathbb{P}_k(D_v = d_v)$ can be bounded from both sides by U_k and $1/U_k$ respectively, where

$$\frac{\widehat{\mathbb{P}}_k(D_v = d_v)}{\mathbb{P}_k(D_v = d_v)} \leq \prod_{\ell \in \mathcal{L}} \left(\frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)}}{Q_{k \oplus \mathbf{w}_2}^{(\ell)}} \right)^{c_N^{(\ell)}} \left(\frac{1 - Q_{k \oplus \mathbf{w}_3}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}} \right)^{N_v^{(\ell)} - c_N^{(\ell)}} =: U_k, \quad (\text{B.5.12a})$$

$$\frac{\widehat{\mathbb{P}}_k(D_v = d_v)}{\mathbb{P}_k(D_v = d_v)} \geq \prod_{\ell \in \mathcal{L}} \left(\frac{Q_{k \oplus \mathbf{w}_2}^{(\ell)}}{Q_{k \oplus \mathbf{w}_1}^{(\ell)}} \right)^{c_N^{(\ell)}} \left(\frac{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_3}^{(\ell)}} \right)^{N_v^{(\ell)} - c_N^{(\ell)}} =: 1/U_k. \quad (\text{B.5.12b})$$

Furthermore, we rewrite the terms above in the following way

$$\left(\frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)}}{Q_{k \oplus \mathbf{w}_2}^{(\ell)}} \right)^{c_N^{(\ell)}} = \exp \left(\log \left(\frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)}}{Q_{k \oplus \mathbf{w}_2}^{(\ell)}} \right) \cdot c_N^{(\ell)} \right) \quad (\text{B.5.13})$$

$$\left(\frac{1 - Q_{k \oplus \mathbf{w}_3}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}} \right)^{N_v^{(\ell)} - c_N^{(\ell)}} = \exp \left(\log \left(1 + \frac{Q_{k \oplus \mathbf{w}_4}^{(\ell)} - Q_{k \oplus \mathbf{w}_3}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}} \right) \cdot (N_v^{(\ell)} - c_N^{(\ell)}) \right). \quad (\text{B.5.14})$$

Moreover, by using the facts that $\log(1+x) = x$ for $x = o(1)$, we have

$$\begin{aligned}
\log U_k &= \sum_{\ell \in \mathcal{L}} \left(\log \left(1 + \frac{Q_{k \oplus \mathbf{w}_4}^{(\ell)} - Q_{k \oplus \mathbf{w}_3}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}} \right) \cdot (N_v^{(\ell)} - c_N^{(\ell)}) + \log \left(\frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)}}{Q_{k \oplus \mathbf{w}_2}^{(\ell)}} \right) \cdot c_N^{(\ell)} \right) \\
&= \sum_{\ell \in \mathcal{L}} \left(N_v^{(\ell)} \cdot \log \left(1 + \frac{Q_{k \oplus \mathbf{w}_4}^{(\ell)} - Q_{k \oplus \mathbf{w}_3}^{(\ell)}}{1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)}} \right) + \log \left(\frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)} (1 - Q_{k \oplus \mathbf{w}_4}^{(\ell)})}{Q_{k \oplus \mathbf{w}_2}^{(\ell)} (1 - Q_{k \oplus \mathbf{w}_3}^{(\ell)})} \right) \cdot c_N^{(\ell)} \right) \\
&\lesssim \sum_{\ell \in \mathcal{L}} N_v^{(\ell)} Q_{k \oplus \mathbf{w}_4}^{(\ell)} + (1 + o(1)) \cdot \sum_{\ell \in \mathcal{L}} \log \left(\frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)}}{Q_{k \oplus \mathbf{w}_2}^{(\ell)}} \right) \cdot c_N^{(\ell)}.
\end{aligned}$$

For the first term, note that $Q_{k \oplus \mathbf{w}}^{(\ell)} \binom{N-1}{\ell-1} = \mathcal{P}_{k \oplus \mathbf{w}}^{(\ell)} \cdot q_{k \oplus \mathbf{w}}^{(\ell)} \gg 1$ for any $\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}$, and $N_v^{(\ell)} = \eta_N N \binom{N-2}{\ell-2} \lesssim N^{\ell-1} / \rho_N$. Since $(\rho_N)^{-1} = o(1)$ and $\sum_{\ell \in \mathcal{L}} q_{k \oplus \mathbf{w}_4}^{(\ell)} \lesssim \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N$ by Assumption 3.2.7, then for any $j \neq k$, the following holds with probability at least $1 - N^{-10}$

$$\sum_{\ell \in \mathcal{L}} N_v^{(\ell)} Q_{k \oplus \mathbf{w}_4}^{(\ell)} \asymp (\rho_N)^{-1} \sum_{\ell \in \mathcal{L}} q_{k \oplus \mathbf{w}_4}^{(\ell)} \ll \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N. \quad (\text{B.5.15})$$

For the second term, $Q_{k \oplus \mathbf{w}_1}^{(\ell)} / Q_{k \oplus \mathbf{w}_2}^{(\ell)} \leq q_{\max}^{(\ell)} / q_{\min}^{(\ell)}$. Since $\log(q_{\max}^{(\ell)} / q_{\min}^{(\ell)}) / \log(q_{\max}^{(\ell)}) = o(1)$ by (3.2.20) and $\sum_{\ell \in \mathcal{L}} q_{\max}^{(\ell)} \asymp \rho_N \asymp \mathbb{D}_{\text{GCH}}(j, k) q_N$ by Assumption 3.2.7, we then have

$$\log \left(\frac{Q_{k \oplus \mathbf{w}_1}^{(\ell)}}{Q_{k \oplus \mathbf{w}_2}^{(\ell)}} \right) \cdot c_N^{(\ell)} \leq \log(q_{\max}^{(\ell)} / q_{\min}^{(\ell)}) \cdot \frac{4q_{\max}^{(\ell)}}{\log(q_{\max}^{(\ell)})} \ll \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N \quad (\text{B.5.16})$$

for any $j \neq k$. Therefore $\log U_k \ll \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N$, and the desired argument follows. $\square$

Proof of inequality (3.6.13). Conditioning on $v \in \mathcal{V}_k$ and $G(v) = 1$, when v is misclassified to $\hat{\mathcal{V}}_j$ by MAP, i.e., $\{\hat{\mathbb{P}}_j(\mathbf{D}_v = d_v) \cdot \alpha_j > \hat{\mathbb{P}}_k(\mathbf{D}_v = d_v) \cdot \alpha_k\}$, then $\{U_j \cdot \alpha_j \mathbb{P}_j(\mathbf{D}_v = d_v) > 1/U_k \cdot \alpha_k \mathbb{P}_k(\mathbf{D}_v = d_v)\}$ is true as well, since the following holds by Lemma 3.6.4

$$1/U_k \cdot \alpha_k \mathbb{P}_k(\mathbf{D}_v = d_v) \leq \alpha_k \hat{\mathbb{P}}_k(\mathbf{D}_v = d_v) < \alpha_j \hat{\mathbb{P}}_j(\mathbf{D}_v = d_v) \leq U_j \cdot \alpha_j \mathbb{P}_j(\mathbf{D}_v = d_v). \quad (\text{B.5.17})$$

We then expand ① as follows, where the $\widehat{\mathbf{R}}_{k|1}$ in (3.6.10) is written as a union bound

$$\begin{aligned}
① &= \sum_{d_v} \mathbb{P}_k(\mathbf{D}_v = d_v) \cdot \widehat{\mathbf{R}}_{k|1} \cdot \mathbb{P}(\mathbf{G}(v) = 1) \\
&= \sum_{d_v} \mathbb{P}(\mathbf{G}(v) = 1) \cdot \mathbb{P}_k(\mathbf{D}_v = d_v) \\
&\quad \cdot \sum_{j=1, j \neq k}^K \mathbb{P} \left\{ \widehat{\mathbb{P}}_j(\mathbf{D}_v = d_v) \cdot \alpha_j > \widehat{\mathbb{P}}_k(\mathbf{D}_v = d_v) \cdot \alpha_k \mid \mathbf{G}(v) = 1 \right\} \\
&= \sum_{j=1, j \neq k}^K \sum_{d_v} \mathbb{P}(\mathbf{G}(v) = 1) \cdot \mathbb{P}_k(\mathbf{D}_v = d_v) \\
&\quad \cdot \mathbb{P} \left\{ \widehat{\mathbb{P}}_j(\mathbf{D}_v = d_v) \cdot \alpha_j > \widehat{\mathbb{P}}_k(\mathbf{D}_v = d_v) \cdot \alpha_k \mid \mathbf{G}(v) = 1 \right\} \\
&\leq \sum_{j=1, j \neq k}^K \sum_{d_v} \mathbb{P}(\mathbf{G}(v) = 1) \cdot \mathbb{P}_k(\mathbf{D}_v = d_v) \\
&\quad \cdot \mathbb{P} \left\{ U_j \cdot \alpha_j \mathbb{P}_j(\mathbf{D}_v = d_v) > 1/U_k \cdot \alpha_k \mathbb{P}_k(\mathbf{D}_v = d_v) \mid \mathbf{G}(v) = 1 \right\} \\
&= \sum_{j=1, j \neq k}^K \sum_{d_v} \mathbb{P}(\mathbf{G}(v) = 1) \cdot \mathbb{P}_k(\mathbf{D}_v = d_v) \cdot \mathbb{E} \mathbb{1}_{\{U_j \cdot \alpha_j \mathbb{P}_j(\mathbf{D}_v = d_v) > 1/U_k \cdot \alpha_k \mathbb{P}_k(\mathbf{D}_v = d_v)\}} \\
&\leq \sum_{j=1, j \neq k}^K \sum_{d_v} \mathbb{P}(\mathbf{G}(v) = 1) \cdot \mathbb{P}_k(\mathbf{D}_v = d_v) \cdot \mathbb{1}_{\{U_j \cdot \alpha_j \mathbb{P}_j(\mathbf{D}_v = d_v) > 1/U_k \cdot \alpha_k \mathbb{P}_k(\mathbf{D}_v = d_v)\}}
\end{aligned}$$

Suppose $\{U_j \cdot \alpha_j \mathbb{P}_j(\mathbf{D}_v = d_v) > 1/U_k \cdot \alpha_k \mathbb{P}_k(\mathbf{D}_v = d_v)\}$, we then have

$$\mathbb{P}_k(\mathbf{D}_v = d_v) \leq \alpha_j U_j U_k / \alpha_k \mathbb{P}_j(\mathbf{D}_v = d_v).$$

Note that $\mathbb{P}(\mathbf{G}(v) = 1) \leq 1$, $U_j, U_k \gtrsim 1$ and $\alpha_j, \alpha_k \asymp 1$, it then implies

$$\mathbb{P}_k(\mathbf{D}_v = d_v) \lesssim \alpha_j U_j U_k / \alpha_k \cdot \min\{\mathbb{P}_j(\mathbf{D}_v = d_v), \mathbb{P}_k(\mathbf{D}_v = d_v)\}$$

Then we can bound ① as follows

$$\begin{aligned}
\textcircled{1} &\leq \sum_{j=1, j \neq k}^K \sum_{d_v} \mathbb{P}(G(v) = 1) \cdot \frac{\alpha_j}{\alpha_k} \cdot U_j U_k \cdot \min\{\mathbb{P}_j(\mathbf{D}_v = d_v), \mathbb{P}_k(\mathbf{D}_v = d_v)\} \\
&\leq \sum_{j=1, j \neq k}^K \frac{\alpha_j}{\alpha_k} U_j U_k \sum_{d_v} \min\{\mathbb{P}_j(\mathbf{D}_v = d_v), \mathbb{P}_k(\mathbf{D}_v = d_v)\} \quad (\text{Lemma B.5.1}) \\
&\leq \sum_{j=1, j \neq k}^K \frac{\alpha_j}{\alpha_k} U_j U_k \cdot \exp(-(1 + o(1)) \cdot \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N) \quad (\text{Lemma 3.6.4}) \\
&\leq (K - 1) \cdot \frac{1 - \alpha_k}{\alpha_k} \cdot \exp\left(-(1 + o(1)) \cdot q_N \cdot \min_{j \in [K], j \neq k} \mathbb{D}_{\text{GCH}}(j, k) + \log U_k\right) \\
&\leq (K - 1) \cdot (1 - \alpha_k) / \alpha_k \cdot \exp\left(-(1 + o(1)) \cdot \mathbb{D}_{\text{GCH}} \cdot q_N\right)
\end{aligned}$$

where the last inequality holds since $\log U_k \ll \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N$ for any $j \neq k$, and $\mathbb{D}_{\text{GCH}} = \min_{1 \leq j < k \leq K} \mathbb{D}_{\text{GCH}}(j, k)$. $\square$

Lemma B.5.1. *For some $j \neq k$ and $j \in [K]$, we have*

$$\sum_{d_v} \min\{\mathbb{P}_j(\mathbf{D}_v = d_v), \mathbb{P}_k(\mathbf{D}_v = d_v)\} \leq \exp(-(1 - o(1)) \cdot \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N). \quad (\text{B.5.18})$$

Proof of Lemma B.5.1. Let $\mathcal{L}$ denote the maximum element of set $\mathcal{L}$. According to the independence between different layers, for any $t \in [0, 1]$, we have

$$\begin{aligned}
&\sum_{d_v} \min\left\{\mathbb{P}_j(\mathbf{D}_v = d_v), \mathbb{P}_k(\mathbf{D}_v = d_v)\right\} \\
&= \sum_{d_v^{(2)}} \cdots \sum_{d_v^{(1)}} \min\left\{\prod_{\ell \in \mathcal{L}} \mathbb{P}_j(\mathbf{D}_v^{(\ell)} = d_v^{(\ell)}), \prod_{\ell \in \mathcal{L}} \mathbb{P}_k(\mathbf{D}_v^{(\ell)} = d_v^{(\ell)})\right\} \\
&\leq \sum_{d_v^{(2)}} \cdots \sum_{d_v^{(1)}} \left(\prod_{\ell \in \mathcal{L}} \mathbb{P}_j(\mathbf{D}_v^{(\ell)} = d_v^{(\ell)})\right)^t \cdot \left(\prod_{\ell \in \mathcal{L}} \mathbb{P}_k(\mathbf{D}_v^{(\ell)} = d_v^{(\ell)})\right)^{1-t} \\
&= \sum_{d_v^{(2)}} \cdots \sum_{d_v^{(1)}} \prod_{\ell \in \mathcal{L}} \left(\mathbb{P}_j(\mathbf{D}_v^{(\ell)} = d_v^{(\ell)})\right)^t \left(\mathbb{P}_k(\mathbf{D}_v^{(\ell)} = d_v^{(\ell)})\right)^{1-t} \\
&= \prod_{\ell \in \mathcal{L}} \left[\sum_{d_v^{(\ell)}} \left(\mathbb{P}_j(\mathbf{D}_v^{(\ell)} = d_v^{(\ell)})\right)^t \left(\mathbb{P}_k(\mathbf{D}_v^{(\ell)} = d_v^{(\ell)})\right)^{1-t}\right], \quad \forall t \in [0, 1].
\end{aligned}$$

Note that $D_{v,\mathbf{w}}^{(\ell)}|_{Y_v=k} \sim \text{Bin}(N_{\mathbf{w}}, Q_{k \oplus \mathbf{w}}^{(\ell)})$ with $N_{\mathbf{w}}$ denoting the capacity of such ℓ -hyperedges, then

$$\mathbb{P}_k(D_v^{(\ell)} = d_v^{(\ell)}) = \prod_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \mathbb{P}_k[D_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)}]. \quad (\text{B.5.19})$$

Note that $d_v^{(\ell)} = \{d_{v,\mathbf{w}}^{(\ell)}\}_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}}$ and $d_{v,\mathbf{w}}^{(\ell)}$ will go through all elements in sample space. Consequently for any $t \in [0, 1]$, by exchanging the order of sum and product, we have the

$$\begin{aligned} & \sum_{d_v^{(\ell)}} \left(\mathbb{P}_j(D_v^{(\ell)} = d_v^{(\ell)}) \right)^t \left(\mathbb{P}_k(D_v^{(\ell)} = d_v^{(\ell)}) \right)^{1-t} \\ &= \sum_{d_v^{(\ell)}} \prod_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \left(\mathbb{P}_j[D_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)}] \right)^t \cdot \left(\mathbb{P}_k[D_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)}] \right)^{1-t} \\ &\leq \prod_{\mathbf{w} \in \mathcal{WC}_K^{(\ell-1)}} \sum_{d_{v,\mathbf{w}}^{(\ell)}} \left(\mathbb{P}_j[D_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)}] \right)^t \cdot \left(\mathbb{P}_k[D_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)}] \right)^{1-t}. \end{aligned}$$

By plugging in the probability $\mathbb{P}_k[D_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)}] = \binom{N_{\mathbf{w}}}{d_{v,\mathbf{w}}^{(\ell)}} [Q_{k \oplus \mathbf{w}}^{(\ell)}]^{d_{v,\mathbf{w}}^{(\ell)}} \cdot [1 - Q_{k \oplus \mathbf{w}}^{(\ell)}]^{N_{\mathbf{w}} - d_{v,\mathbf{w}}^{(\ell)}}$, we have

$$\begin{aligned} & \sum_{d_{v,\mathbf{w}}^{(\ell)}} \left(\mathbb{P}_j[D_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)}] \right)^t \cdot \left(\mathbb{P}_k[D_{v,\mathbf{w}}^{(\ell)} = d_{v,\mathbf{w}}^{(\ell)}] \right)^{1-t} \\ &= \sum_{d_{v,\mathbf{w}}^{(\ell)}} \binom{N_{\mathbf{w}}}{d_{v,\mathbf{w}}^{(\ell)}} \left(\frac{[Q_{j \oplus \mathbf{w}}^{(\ell)}]^t}{[Q_{k \oplus \mathbf{w}}^{(\ell)}]^{t-1}} \right)^{d_{v,\mathbf{w}}^{(\ell)}} \cdot \left(\frac{[1 - Q_{j \oplus \mathbf{w}}^{(\ell)}]^t}{[1 - Q_{k \oplus \mathbf{w}}^{(\ell)}]^{t-1}} \right)^{N_{\mathbf{w}} - d_{v,\mathbf{w}}^{(\ell)}} \\ &= \left([Q_{j \oplus \mathbf{w}}^{(\ell)}]^t \cdot [Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t} + [1 - Q_{j \oplus \mathbf{w}}^{(\ell)}]^t \cdot [1 - Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t} \right)^{N_{\mathbf{w}}}, \quad \forall t \in [0, 1], \end{aligned}$$

where the last equality holds by Binomial theorem. Then we put pieces together

$$\begin{aligned}
& \sum_{d_v} \min \left\{ \mathbb{P}_j(\mathbf{D}_v = d_v), \mathbb{P}_k(\mathbf{D}_v = d_v) \right\} \\
& \leq \prod_{\ell \in \mathcal{L}} \prod_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} \left([Q_{j \oplus \mathbf{w}}^{(\ell)}]^t \cdot [Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t} + [1 - Q_{j \oplus \mathbf{w}}^{(\ell)}]^t \cdot [1 - Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t} \right)^{N_{\mathbf{w}}} \\
& = \exp \left\{ \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} N_{\mathbf{w}} \cdot \log \left([Q_{j \oplus \mathbf{w}}^{(\ell)}]^t \cdot [Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t} + [1 - Q_{j \oplus \mathbf{w}}^{(\ell)}]^t \cdot [1 - Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t} \right) \right\}.
\end{aligned}$$

Note that for sufficiently small x and y , $\log(1-x) = -x + O(x^2)$ and

$$\begin{aligned}
(1-x)^t (1-y)^{1-t} &= \exp(t \log(1-x) + (1-t) \log(1-y)) = \exp(-tx - (1-t)y + O(x^2 + y^2)) \\
&= 1 - tx - (1-t)y + O(x^2 + y^2).
\end{aligned}$$

Note that $Q_{k \oplus \mathbf{w}}^{(\ell)} = o(1)$ under (3.2.2). Let $t^* \in [0, 1]$ be the maximizer of (3.2.6). By replacing $N_{\mathbf{w}}$ with $\bar{N}_{\mathbf{w}}$ using Lemma 3.1.1, we then have

$$\begin{aligned}
& \sum_{d_v} \min \left\{ \mathbb{P}_j(\mathbf{D}_v = d_v), \mathbb{P}_k(\mathbf{D}_v = d_v) \right\} \\
& \leq \exp \left\{ \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} N_{\mathbf{w}} \right. \\
& \quad \cdot \log \left[[Q_{j \oplus \mathbf{w}}^{(\ell)}]^{t^*} \cdot [Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t^*} + 1 - t^* Q_{j \oplus \mathbf{w}}^{(\ell)} - (1-t^*) Q_{k \oplus \mathbf{w}}^{(\ell)} + O((Q_{k \oplus \mathbf{w}}^{(\ell)})^2) \right] \Big\} \\
& = \exp \left\{ - \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} N_{\mathbf{w}} \cdot \left[t^* Q_{j \oplus \mathbf{w}}^{(\ell)} + (1-t^*) Q_{k \oplus \mathbf{w}}^{(\ell)} - [Q_{j \oplus \mathbf{w}}^{(\ell)}]^{t^*} \cdot [Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t^*} + O((Q_{k \oplus \mathbf{w}}^{(\ell)})^2) \right] \right\} \\
& = \exp \left\{ - \sum_{\ell \in \mathcal{L}} \sum_{\mathbf{w} \in \mathcal{W}_{C_K^{(\ell-1)}}} (1 + o(1)) \cdot \bar{N}_{\mathbf{w}} \cdot \left[t^* Q_{j \oplus \mathbf{w}}^{(\ell)} + (1-t^*) Q_{k \oplus \mathbf{w}}^{(\ell)} - [Q_{j \oplus \mathbf{w}}^{(\ell)}]^{t^*} \cdot [Q_{k \oplus \mathbf{w}}^{(\ell)}]^{1-t^*} \right] \right\} \\
& = \exp(-(1 + o(1)) \cdot \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N),
\end{aligned}$$

which completes the proof of the desired argument. $\square$

Proof of Lemma 3.6.3. For the sake of convenience, denote $c_N^{(\ell)} = 4q_{\max}^{(\ell)} / \log(q_{\max}^{(\ell)})$. The event $\{G(v) = 0\}$ means that for at least one of $\ell \in \mathcal{L}$, the cardinality of the edge set $\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})$ (each edge contains v and at least one node from $\widehat{\mathcal{N}}^{(0)}$) is not small, i.e.,

$$\mathbb{P}_k(G(v) = 0) \leq \sum_{\ell \in \mathcal{L}} \mathbb{P}_k(G_\ell(v) = 0) = \sum_{\ell \in \mathcal{L}} \mathbb{P}(|\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})| > c_N^{(\ell)}).$$

Since $\mathcal{L}$ is a finite set, we only need to focus on extreme event for each $\ell \in \mathcal{L}$. By (3.4.11), with probability at least $1 - N^{-10}$, the number of misclassified nodes in $\widehat{\mathbf{y}}^{(0)}$ is at most $\eta_N N$ where $\eta_N \lesssim (\rho_N)^{-1}$. Then the cardinality of the edge set $\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})$ is at most $\eta_N N \binom{N-2}{\ell-2} \lesssim N^{\ell-1} / \rho_N$, since we can choose the remaining $\ell - 2$ nodes (beside v and the misclassified one) from all $N - 2$ nodes. Since $\rho_N \asymp \sum_{\ell \in \mathcal{L}} q_{\max}^{(\ell)}$, then

$$\mathbb{E}\left(|\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})|\right) \leq \eta_N N \binom{N-2}{\ell-2} \cdot Q_{\max}^{(\ell)} \lesssim \frac{q_{\max}^{(\ell)}}{\rho_N} \lesssim \frac{1}{q_{\max}^{(\ell)}} \cdot q_{\max}^{(\ell)} \asymp 1.$$

We choose $\gamma_N^{(\ell)} := 1/q_{\max}^{(\ell)}$, and $\zeta_N^{(\ell)} := 4/\log(q_{\max}^{(\ell)})$, which satisfies $\zeta_N^{(\ell)} \geq \gamma_N^{(\ell)}$, and

$$\begin{aligned} & \zeta_N^{(\ell)} \log(\zeta_N^{(\ell)} / \gamma_N^{(\ell)}) + (\gamma_N^{(\ell)} - \zeta_N^{(\ell)}) \\ & \geq \frac{4}{\log(q_{\max}^{(\ell)})} \left(\log\left(\frac{4q_{\max}^{(\ell)}}{\log(q_{\max}^{(\ell)})}\right) + \frac{\log(q_{\max}^{(\ell)})}{4q_{\max}^{(\ell)}} - 1 \right) \\ & = 4 \left(\frac{\log(4q_{\max}^{(\ell)}) - \log \log(q_{\max}^{(\ell)})}{\log(q_{\max}^{(\ell)})} + \frac{1}{4q_{\max}^{(\ell)}} - \frac{1}{\log(q_{\max}^{(\ell)})} \right) \geq 4 + o(1). \end{aligned}$$

Consequently, according to Lemma D.3.6, for any $j \neq k$, we have

$$\mathbb{P}\left(|\mathcal{E}_\ell(v, \widehat{\mathcal{N}}^{(0)})| > c_N^{(\ell)}\right) \leq e^{-4q_{\max}^{(\ell)}} \leq \exp(-4 \cdot \mathbb{D}_{\text{GCH}}(j, k) \cdot q_N),$$

where the last inequality holds since $q_{\max}^{(\ell)} \gtrsim \mathbb{D}_{\text{GCH}} \cdot q_N$ by (3.2.6) and Assumption 3.2.7. The desired argument then follows since $\mathbb{D}_{\text{GCH}} = \min_{1 \leq j < k \leq K} \mathbb{D}_{\text{GCH}}(j, k)$. $\square$

Appendix C

Deferred proofs in Chapter 4

C.1 Proof of Theorem 4.3.2 and Theorem 4.3.4

C.1.1 Discretization

To prove Theorem 4.3.2, we start with a standard ε -net argument.

Lemma C.1.1 ([195, Lemma 4.4.1]). *Let $\mathbf{W}$ be any $N \times N$ Hermitian matrix and let $\mathcal{N}_\varepsilon$ be an ε -net on the unit sphere $\mathbb{S}^{N-1}$ with $\varepsilon \in (0, 1)$, then $\|\mathbf{W}\| \leq \frac{1}{1-\varepsilon} \sup_{\mathbf{x} \in \mathcal{N}_\varepsilon} |\langle \mathbf{W}\mathbf{x}, \mathbf{x} \rangle|$.*

By [195, Corollary 4.2.13], the size of $\mathcal{N}_\varepsilon$ is bounded by $|\mathcal{N}_\varepsilon| \leq (1 + 2/\varepsilon)^N$. We would have $\log |\mathcal{N}_\varepsilon| \leq N \log(5)$ when $\mathcal{N}_\varepsilon$ is taken as an $(1/2)$ -net of $\mathbb{S}^N$. Define $\mathbf{W} := \mathbf{A} - \mathbb{E}\mathbf{A}$, then $W_{ii} = 0$ for each $i \in [N]$ by the definition of adjacency matrix in (4.2.4), and we obtain

$$\|\mathbf{A} - \mathbb{E}\mathbf{A}\| = \|\mathbf{W}\| \leq 2 \sup_{\mathbf{x} \in \mathcal{N}} |\langle \mathbf{W}\mathbf{x}, \mathbf{x} \rangle|. \quad (\text{C.1.1})$$

For any fixed $\mathbf{x} \in \mathbb{S}^{N-1}$, consider the *light* and *heavy* pairs as follows.

$$\mathcal{L}(\mathbf{x}) = \left\{ (i, j) : |x_i x_j| \leq \frac{\sqrt{d}}{N} \right\}, \quad \mathcal{H}(\mathbf{x}) = \left\{ (i, j) : |x_i x_j| > \frac{\sqrt{d}}{N} \right\}, \quad (\text{C.1.2})$$

where $d = \sum_{\ell \in \mathcal{L}} (\ell - 1) d_\ell$. Thus by the triangle inequality,

$$|\langle \mathbf{x}, \mathbf{W}\mathbf{x} \rangle| \leq \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| + \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right|.$$

We denote $\mathcal{N} := \mathcal{N}_\mathcal{E}$ for convenience. Then by (C.1.1),

$$\|\mathbf{A} - \mathbb{E}\mathbf{A}\| \leq 2 \sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| + 2 \sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right|. \quad (\text{C.1.3})$$

C.1.2 Contribution from light pairs

For each ℓ -hyperedge $e \in \mathcal{E}_\ell$, we define $\mathcal{W}_e^{(\ell)} := \mathcal{A}_e^{(\ell)} - \mathbb{E}\mathcal{A}_e^{(\ell)}$. Then for any fixed $\mathbf{x} \in \mathbb{S}^{N-1}$, the contribution from light couples can be written as

$$\begin{aligned} \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j &= \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \left(\sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell \\ \{i,j\} \subset e}} \mathcal{W}_e^{(\ell)} \right) \mathbf{x}_i \mathbf{x}_j \\ &= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathcal{W}_e^{(\ell)} \left(\sum_{\substack{(i,j) \in \mathcal{L}(\mathbf{x}) \\ i \neq j, \{i,j\} \subset e}} \mathbf{x}_i \mathbf{x}_j \right) = \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathcal{J}_e^{(\ell)}, \end{aligned} \quad (\text{C.1.4})$$

where the constraint $i \neq j$ comes from the fact $\mathbf{W}_{ii} = 0$ and we denote

$$\mathcal{J}_e^{(\ell)} := \mathcal{W}_e^{(\ell)} \left(\sum_{\substack{(i,j) \in \mathcal{L}(\mathbf{x}) \\ i \neq j, \{i,j\} \subset e}} \mathbf{x}_i \mathbf{x}_j \right).$$

Note that $\mathbb{E}\mathcal{J}_e^{(\ell)} = 0$, and by the definition of light pair (C.1.2),

$$|\mathcal{J}_e^{(\ell)}| \leq \ell(\ell - 1)\sqrt{d}/N \leq \ell(\ell - 1)\sqrt{d}/N, \quad \forall \ell \in \mathcal{L}.$$

Moreover, (C.1.4) is a sum of independent, mean-zero random variables, and

$$\begin{aligned}
\sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathbb{E}[(\mathcal{Y}_e^{(\ell)})^2] &:= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \left[\mathbb{E}[(\mathcal{W}_e^{(\ell)})^2] \left(\sum_{\substack{(i,j) \in \mathcal{L}(\mathbf{x}) \\ i \neq j, \{i,j\} \subset e}} \mathbf{x}_i \mathbf{x}_j \right)^2 \right] \\
&\leq \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \left[\mathbb{E}[\mathcal{A}_e^{(\ell)}] \cdot \ell(\ell-1) \left(\sum_{\substack{(i,j) \in \mathcal{L}(\mathcal{X}) \\ i \neq j, \{i,j\} \subset e}} \mathbf{x}_i^2 \mathbf{x}_j^2 \right) \right] \\
&\leq \sum_{\ell \in \mathcal{L}} \frac{d_\ell \cdot \ell(\ell-1)}{\binom{N-1}{\ell-1}} \binom{N}{\ell-2} \sum_{(i,j) \in [N]^2} \mathbf{x}_i^2 \mathbf{x}_j^2 \\
&\leq \sum_{\ell \in \mathcal{L}} \frac{d_\ell \ell(\ell-1)^2}{N-\ell+2} \leq \frac{2}{N} \sum_{\ell \in \mathcal{L}} d_\ell (\ell-1)^3 \leq \frac{2d(\ell-1)^2}{N},
\end{aligned}$$

when $N \geq 2\ell-2$, where $d_\ell = \max d_{[i_1, \dots, i_\ell]}$ and $d = \sum_{\ell \in \mathcal{L}} (\ell-1)d_\ell$. Then Bernstein's (Lemma D.3.4) implies that for any $\alpha > 0$,

$$\begin{aligned}
&\mathbb{P} \left(\left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \geq \alpha \sqrt{d} \right) = \mathbb{P} \left(\left| \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell} \mathcal{Y}_e^{(\ell)} \right| \geq \alpha \sqrt{d} \right) \\
&\leq 2 \exp \left(- \frac{\frac{1}{2} \alpha^2 d}{\frac{2d}{N}(\ell-1)^2 + \frac{1}{3}(\ell-1)L \frac{\sqrt{d}}{N} \alpha \sqrt{d}} \right) \leq 2 \exp \left(- \frac{\alpha^2 N}{4(\ell-1)^2 + \frac{2\alpha(\ell-1)L}{3}} \right).
\end{aligned}$$

Therefore by taking a union bound,

$$\begin{aligned}
&\mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{N}} \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \geq \alpha \sqrt{d} \right) \leq |\mathcal{N}| \cdot \mathbb{P} \left(\left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \mathbf{W}_{ij} \mathbf{x}_i \mathbf{x}_j \right| \geq \alpha \sqrt{d} \right) \\
&\leq 2 \exp \left(\log(5) \cdot N - \frac{\alpha^2 N}{4(\ell-1)^2 + \frac{2\alpha(\ell-1)L}{3}} \right) \leq 2e^{-N}, \tag{C.1.6}
\end{aligned}$$

where we choose $\alpha = 5\ell(\ell-1)$ in the last line.

C.1.3 Contribution from heavy pairs

Note that for any $i \neq j$,

$$\mathbb{E}A_{ij} \leq \sum_{\ell \in \mathcal{L}} \binom{N-2}{\ell-2} \frac{d_\ell}{\binom{N-1}{\ell-1}} \leq \sum_{\ell \in \mathcal{L}} \frac{(\ell-1)d_\ell}{N} = \frac{d}{N}. \quad (\text{C.1.7})$$

and

$$\begin{aligned} \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \mathbb{E}A_{ij}x_i x_j \right| &= \left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \mathbb{E}A_{ij} \frac{x_i^2 x_j^2}{x_i x_j} \right| \\ &\leq \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} \frac{d}{N} \frac{x_i^2 x_j^2}{|x_i x_j|} \leq \sqrt{d} \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} x_i^2 x_j^2 \leq \sqrt{d}. \end{aligned} \quad (\text{C.1.8})$$

Therefore it suffices to show that, with high probability,

$$\sum_{(i,j) \in \mathcal{H}(\mathbf{x})} A_{ij}x_i x_j = o(\sqrt{d}). \quad (\text{C.1.9})$$

Here we use the discrepancy analysis from [100, 71]. We consider the weighted graph associated with the adjacency matrix $\mathbf{A}$.

Definition C.1.2 (Uniform upper tail property, **UUTP**). *Let $\mathbf{M}$ be an $N \times N$ random symmetric matrix with non-negative entries and $\mathbf{Q}$ be an $N \times N$ symmetric matrix with entries $Q_{ij} \in [0, a]$ for all $i, j \in [N]$. Define*

$$\mu := \sum_{i,j=1}^N Q_{ij} \mathbb{E}M_{ij}, \quad \tilde{\sigma}^2 := \sum_{i,j=1}^N Q_{ij}^2 \mathbb{E}M_{ij}.$$

We say that $\mathbf{M}$ satisfies the uniform upper tail property **UUTP** (c_0, γ_0) with $c_0 > 0, \gamma_0 \geq 0$, if for any $a, t > 0$,

$$\mathbb{P} \left(f_{\mathbf{Q}}(\mathbf{M}) \geq (1 + \gamma_0)\mu + t \right) \leq \exp \left(-c_0 \frac{\tilde{\sigma}^2}{a^2} h \left(\frac{at}{\tilde{\sigma}^2} \right) \right).$$

where function $f_{\mathbf{Q}}(\mathbf{M}) : \mathbb{R}^{N \times N} \mapsto \mathbb{R}$ is defined by $f_{\mathbf{Q}}(\mathbf{M}) := \sum_{i,j=1}^N \mathbf{Q}_{ij} \mathbf{M}_{ij}$ for $\mathbf{M} \in \mathbb{R}^{N \times N}$, and function $h(x) := (1+x) \log(1+x) - x$ for all $x > -1$.

Lemma C.1.3. *Let $\mathbf{A}$ be the adjacency matrix of non-uniform hypergraph $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_{\ell}$, then $\mathbf{A}$ satisfies **UUTP** (c_0, γ_0) with $c_0 = [\ell(\ell-1)]^{-1}, \gamma_0 = 0$.*

Proof of Lemma C.1.3. Note that

$$\begin{aligned} f_{\mathbf{Q}}(\mathbf{A}) - \mu &= \sum_{i,j=1}^N \mathbf{Q}_{ij} (\mathbf{A}_{ij} - \mathbb{E} \mathbf{A}_{ij}) = \sum_{i,j=1}^N \mathbf{Q}_{ij} \mathbf{W}_{ij} \\ &= \sum_{i,j=1}^N \mathbf{Q}_{ij} \left(\sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_{\ell} \\ i \neq j, \{i,j\} \subset e}} \mathcal{W}_e^{(\ell)} \right) = \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_{\ell}} \mathcal{W}_e^{(\ell)} \left(\sum_{\{i,j\} \subset e, i \neq j} \mathbf{Q}_{ij} \right) = \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_{\ell}} Z_e^{(\ell)}, \end{aligned}$$

where $Z_e^{(\ell)} = \mathcal{W}_e^{(\ell)} (\sum_{\{i,j\} \subset e, i \neq j} \mathbf{Q}_{ij})$ are independent centered random variables upper bounded by $|Z_e^{(\ell)}| \leq \sum_{\{i,j\} \subset e, i \neq j} \mathbf{Q}_{ij} \leq \ell(\ell-1)a$ for each $\ell \in \{2, \dots, M\}$ since $\mathbf{Q}_{ij} \in [0, a]$. Moreover, the variance of the sum can be written as

$$\begin{aligned} \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_{\ell}} \mathbb{E} (Z_e^{(\ell)})^2 &= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_{\ell}} \mathbb{E} (\mathcal{W}_e^{(\ell)})^2 \left(\sum_{\{i,j\} \subset e, i \neq j} \mathbf{Q}_{ij} \right)^2 \\ &\leq \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_{\ell}} \mathbb{E} [\mathcal{A}_e^{(\ell)}] \cdot \ell(\ell-1) \sum_{\{i,j\} \subset e, i \neq j} \mathbf{Q}_{ij}^2 \leq \ell(\ell-1) \sum_{i,j=1}^N \mathbf{Q}_{ij}^2 \mathbb{E} \mathbf{A}_{ij} = \ell(\ell-1) \tilde{\sigma}^2. \end{aligned}$$

where the last inequality holds since by definition $\mathbb{E} \mathbf{A}_{ij} = \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_{\ell} \\ \{i,j\} \subset e}} \mathbb{E} [\mathcal{A}_e^{(\ell)}]$. Then by Bennett's inequality Lemma D.3.5, we obtain

$$\mathbb{P}(f_{\mathbf{Q}}(\mathbf{A}) - \mu \geq t) \leq \exp \left(- \frac{\tilde{\sigma}^2}{\ell(\ell-1)a^2} h \left(\frac{at}{\tilde{\sigma}^2} \right) \right)$$

where the inequality holds since the function $x \cdot h(1/x) = (1+x) \log(1+1/x) - 1$ is decreasing with respect to x . $\square$

Definition C.1.4 (Discrepancy property, **DP**). *Let $\mathbf{M}$ be an $N \times N$ matrix with non-negative entries. For $S, T \subset [N]$, define $e_{\mathbf{M}}(S, T) = \sum_{i \in S, j \in T} \mathbf{M}_{ij}$. We say $\mathbf{M}$ has the discrepancy property*

with parameter $\delta > 0$, $\kappa_1 > 1$, $\kappa_2 \geq 0$, denoted by $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$, if for all non-empty $S, T \subset [N]$, at least one of the following hold:

1. $e_{\mathbf{M}}(S, T) \leq \kappa_1 \delta |S| |T|$;
2. $e_{\mathbf{M}}(S, T) \cdot \log \left(\frac{e_{\mathbf{M}}(S, T)}{\delta |S| |T|} \right) \leq \kappa_2 (|S| \vee |T|) \cdot \log \left(\frac{en}{|S| \vee |T|} \right)$.

Lemma C.1.5 shows that if a symmetric random matrix $\mathbf{A}$ satisfies the upper tail property $\mathbf{UUTP}(c_0, \gamma_0)$ with parameter $c_0 > 0$, $\gamma_0 \geq 0$, then the discrepancy property holds with high probability.

Lemma C.1.5 ([71, Lemma 6.4]). *Let $\mathbf{M}$ be an $N \times N$ symmetric random matrix with non-negative entries. Assume that for some $\delta > 0$, $\mathbb{E}M_{ij} \leq \delta$ for all $i, j \in [N]$ and $\mathbf{M}$ has $\mathbf{UUTP}(c_0, \gamma_0)$ with parameter $c_0, \gamma_0 > 0$. Then for any $\theta > 0$, the discrepancy property $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ holds for $\mathbf{M}$ with probability at least $1 - N^{-\theta}$ with $\kappa_1 = e^2(1 + \gamma_0)^2$, $\kappa_2 = \frac{2}{c_0}(1 + \gamma_0)(\theta + 4)$.*

When the discrepancy property holds, then deterministically the contribution from heavy pairs is $O(\sqrt{d})$, as shown in the following lemma.

Lemma C.1.6 ([71, Lemma 6.6]). *Let $\mathbf{M}$ be a non-negative symmetric $N \times N$ matrix with all row sums bounded by d . Suppose $\mathbf{M}$ has $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ with $\delta = Cd/N$ for some $C > 0$, $\kappa_1 > 1$, $\kappa_2 \geq 0$. Then for any $x \in \mathbb{S}^{N-1}$,*

$$\left| \sum_{(i,j) \in \mathcal{H}(X)} M_{ij} x_i x_j \right| \leq \alpha_0 \sqrt{d},$$

where $\alpha_0 = 16 + 32C(1 + \kappa_1) + 64\kappa_2(1 + \frac{2}{\kappa_1 \log \kappa_1})$.

Lemma C.1.7 proves that $\mathbf{A}$ has bounded row and column sums with high probability.

Lemma C.1.7. *For any $\theta > 0$, there is a constant $\alpha_1 > 0$ such that with probability at least $1 - N^{-\theta}$,*

$$\max_{1 \leq i \leq N} \sum_{j=1}^N A_{ij} \leq \alpha_1 d \tag{C.1.10}$$

with $\alpha_1 = 4 + \frac{2(\ell-1)(1+\theta)}{3c}$ and $d \geq c \log N$.

Proof. For a fixed $i \in [N]$,

$$\begin{aligned} \sum_{j=1}^N A_{ij} &= \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell: i \in e} (\ell-1) \mathcal{A}_e^{(\ell)}, \quad \sum_{j=1}^N (A_{ij} - \mathbb{E} A_{ij}) = \sum_{\ell \in \mathcal{L}} \sum_{e \in \mathcal{E}_\ell: i \in e} (\ell-1) \mathcal{W}_e^{(\ell)}, \\ \sum_{j=1}^N \mathbb{E} A_{ij} &\leq \sum_{\ell \in \mathcal{L}} \binom{N-1}{\ell-1} \frac{(\ell-1)d_\ell}{\binom{N-1}{\ell-1}} = d, \\ \sum_{\ell \in \mathcal{L}} (\ell-1)^2 \sum_{e \in \mathcal{E}_\ell: i \in e} \mathbb{E}[(\mathcal{W}_e^{(\ell)})^2] &\leq \sum_{\ell \in \mathcal{L}} (\ell-1)^2 \sum_{e \in \mathcal{E}_\ell: i \in e} \mathbb{E}[\mathcal{A}_e^{(\ell)}] \leq (\ell-1)d. \end{aligned}$$

Then for $\alpha_1 = 4 + \frac{2(\ell-1)(1+\theta)}{3c}$, by Bernstein's inequality, with the assumption that $d \geq c \log N$,

$$\begin{aligned} \mathbb{P}\left(\sum_{j=1}^N A_{ij} \geq \alpha_1 d\right) &\leq \mathbb{P}\left(\sum_{j=1}^N A_{ij} - \mathbb{E} A_{ij} \geq (\alpha_1 - 1)d\right) \\ &\leq \exp\left(-\frac{\frac{1}{2}(\alpha_1 - 1)^2 d^2}{(\ell-1)d + \frac{1}{3}(\ell-1)(\alpha_1 - 1)d}\right) \leq N^{-\frac{3c(\alpha_1 - 1)^2}{(\ell-1)(2\alpha_1 + 4)}} \leq N^{-1-\theta}. \end{aligned} \tag{C.1.12}$$

Taking a union bound over $i \in [N]$, then (C.1.10) holds with probability $1 - N^{-\theta}$. $\square$

Now we are ready to obtain (C.1.9).

Lemma C.1.8. *For any $\theta > 0$, there is a constant β depending on θ, c, L such that with probability at least $1 - 2N^{-\theta}$,*

$$\left| \sum_{(i,j) \in \mathcal{H}(\mathcal{X})} A_{ij} x_i x_j \right| \leq \beta \sqrt{d}. \tag{C.1.13}$$

Proof. By Lemma C.1.3, $\mathbf{A}$ satisfies $\mathbf{UUTP}(\frac{1}{\ell(\ell-1)}, 0)$. From (C.1.7) and Lemma C.1.5, the property $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ holds for $\mathbf{A}$ with probability at least $1 - N^{-\theta}$ with

$$\delta = \frac{d}{N}, \quad \kappa_1 = e^2, \quad \kappa_2 = 2\ell(\ell-1)(\theta+4).$$

Let $\mathcal{E}_1$ be the event that $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ holds for $\mathbf{A}$. Let $\mathcal{E}_2$ be the event that all row sums of $\mathbf{A}$ are bounded by $\alpha_1 d$. Then $\mathbb{P}(\mathcal{E}_1 \cap \mathcal{E}_2) \geq 1 - 2N^{-\theta}$. On the event $\mathcal{E}_1 \cap \mathcal{E}_2$, by Lemma C.1.6, (C.1.13) holds with $\beta = \alpha_0 \alpha_1$, where

$$\alpha_0 = 16 + 32(1 + e^2) + 128\ell(\ell - 1)(\theta + 4)(1 + e^{-2}), \quad \alpha_1 = 4 + \frac{2(\ell - 1)(1 + \theta)}{3c}.$$

□

C.1.4 Proof of Theorem 4.3.2

Proof. From (C.1.6), with probability at least $1 - 2e^{-N}$, the contribution from light pairs in (C.1.3) is bounded by $2\alpha\sqrt{d}$ with $\alpha = 5\ell(\ell - 1)$. From (C.1.8) and (C.1.13), with probability at least $1 - 2N^{-\theta}$, the contribution from heavy pairs in (C.1.3) is bounded by $2\sqrt{d} + 2\beta\sqrt{d}$. Therefore with probability at least $1 - 2e^{-N} - 2N^{-\theta}$,

$$\|\mathbf{A} - \mathbb{E}\mathbf{A}\| \leq C_L \sqrt{d},$$

where C_L is a constant depending only on c, θ, L such that $C_L = 2(\alpha + 1 + \beta)$. In particular, we can take $\alpha = 5\ell(\ell - 1)$, $\beta = 512\ell(\ell - 1)(\theta + 5) \left(2 + \frac{(\ell - 1)(1 + \theta)}{c}\right)$, and $C_L = 512\ell(\ell - 1)(\theta + 6) \left(2 + \frac{(\ell - 1)(1 + \theta)}{c}\right)$. This finishes the proof of Theorem 4.3.2. □

C.1.5 Proof of Theorem 4.3.4

Let $\mathcal{S} \subset [N]$ be any given subset. From (C.1.6), with probability at least $1 - 2e^{-N}$,

$$\sup_{\mathbf{x} \in \mathcal{X}} \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} \left(\mathbf{A}_{\mathcal{S}} - \mathbb{E}\mathbf{A}_{\mathcal{S}} \right)_{ij} x_i x_j \right| \leq 5\ell(\ell - 1)\sqrt{d}. \quad (\text{C.1.14})$$

Since there are at most 2^N many choices for $\mathcal{S}$, by taking a union bound, with probability at least $1 - 2(e/2)^{-N}$, we have for all $\mathcal{S} \subset [N]$, (C.1.14) holds. In particular, by taking $\mathcal{S} = I = \{i \in [N] :$

$\text{row}(i) \leq \tau d\}$, with probability at least $1 - 2(e/2)^{-N}$, we have

$$\sup_{\mathbf{x} \in \mathcal{K}} \left| \sum_{(i,j) \in \mathcal{L}(\mathbf{x})} [(\mathbf{A} - \mathbb{E}\mathbf{A})_I]_{ij} \mathbf{x}_i \mathbf{x}_j \right| \leq 5\ell(\ell-1)\sqrt{d}. \quad (\text{C.1.15})$$

Similar to (C.1.8), deterministically,

$$\left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} [(\mathbb{E}\mathbf{A})_I]_{ij} \mathbf{x}_i \mathbf{x}_j \right| \leq (\ell-1)\sqrt{d}. \quad (\text{C.1.16})$$

Next we show the contribution from heavy pairs for $\mathbf{A}_I$ is bounded.

Lemma C.1.9. *For any $\theta > 0$, there is a constant β_τ depending on θ, c, L, τ such that with probability at least $1 - N^{-\theta}$,*

$$\left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} [(\mathbf{A})_I]_{ij} \mathbf{x}_i \mathbf{x}_j \right| \leq \beta_\tau \sqrt{d}. \quad (\text{C.1.17})$$

Proof. Note that $\mathbf{A}$ satisfies $\mathbf{UUTP}\left(\frac{1}{\ell(\ell-1)}, 0\right)$ from Lemma C.1.3. According to Lemma C.1.5, with probability at least $1 - N^{-\theta}$, $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ holds for $\mathbf{A}$ with

$$\delta = \frac{d}{N}, \quad \kappa_1 = e^2, \quad \kappa_2 = 2\ell(\ell-1)(\theta+4).$$

The $\mathbf{DP}(\delta, \kappa_1, \kappa_2)$ property holds for $\mathbf{A}_I$ as well, since $\mathbf{A}_I$ is obtained from $\mathbf{A}$ by restricting to I .

Note that all row sums in $\mathbf{A}_I$ are bounded by τd . By Lemma C.1.6,

$$\left| \sum_{(i,j) \in \mathcal{H}(\mathbf{x})} [\mathbf{A}_I]_{ij} \mathbf{x}_i \mathbf{x}_j \right| \leq \alpha_0 \sqrt{\tau d}, \quad (\text{C.1.18})$$

where we can take $\alpha_0 = 16 + \frac{32}{\tau}(1 + e^2) + 128\ell(\ell-1)(\theta+4)\left(1 + \frac{1}{e^2}\right)$. $\square$

We can then take $\beta_\tau = \alpha_0 \sqrt{\tau}$ in Equation (C.1.17). Therefore, combining (C.1.15), (C.1.16), Equation (C.1.18), with probability at least $1 - 2(e/2)^{-N} - N^{-\theta}$, there exists a constant

C_τ depending only on τ, M, K such that $\|(\mathbf{A} - \mathbb{E}\mathbf{A})_I\| \leq C_\tau \sqrt{d}$, where $C_\tau = 2((5L+1)(\ell-1) + \alpha_0 \sqrt{\tau})$. This finishes the proof of Theorem 4.3.4.

C.2 Deferred proofs

C.2.1 Proofs of Lemmas in Section 4.2

Proof of Lemma 4.2.4. By Weyl's inequality (Lemma D.2.1), the difference between eigenvalues of $\widetilde{\mathbb{E}\mathbf{A}}$ and $\mathbb{E}\mathbf{A}$ can be upper bounded by

$$\begin{aligned} |\lambda_j(\widetilde{\mathbb{E}\mathbf{A}}) - \lambda_j(\mathbb{E}\mathbf{A})| &\leq \|\widetilde{\mathbb{E}\mathbf{A}} - \mathbb{E}\mathbf{A}\|_2 \leq \|\widetilde{\mathbb{E}\mathbf{A}} - \mathbb{E}\mathbf{A}\|_F \\ &\leq \left[2K \cdot \frac{N}{K} \cdot \sqrt{N} \log(N) \cdot (\alpha - \beta)^2 \right]^{1/2} = O\left(N^{3/4} \log^{1/2}(N) (\alpha - \beta)\right). \end{aligned}$$

The lemma follows, as $\lambda_j(\mathbb{E}\mathbf{A}) = \Omega(N(\alpha - \beta))$ for all $1 \leq j \leq K$. $\square$

C.2.2 Proofs of Lemmas in Section 4.5.1

Proof of Lemma 4.5.5. We first compute the singular values of $\overline{\mathbf{B}}_1$. From (4.5.5), the rank of matrix $\overline{\mathbf{B}}_1$ is K , and the least non-trivial singular value of $\overline{\mathbf{B}}_1$ is

$$\sigma_K(\overline{\mathbf{B}}_1) = \frac{N}{2\sqrt{2}K} (\overline{\alpha} - \overline{\beta}) = \frac{N}{2\sqrt{2}K} \sum_{\ell \in \mathcal{L}} \binom{\frac{3N}{4K} - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}},$$

where $\mathcal{L}$ is obtained from Algorithm 4.4.1. By the definition of $\overline{\mathbf{A}}_1$ in (4.5.9), the least non-trivial singular value of $\overline{\mathbf{A}}_1$ is

$$\sigma_K(\overline{\mathbf{A}}_1) = \sigma_K(\overline{\mathbf{B}}_1) = \frac{N}{2\sqrt{2}K} (\overline{\alpha} - \overline{\beta}) = \frac{N}{2\sqrt{2}K} \sum_{\ell \in \mathcal{L}} \binom{\frac{3N}{4K} - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}}.$$

Recall that N_i , defined in (4.5.1), denotes the number of vertices in $\mathcal{Z} \cap \mathcal{V}_i'$, which can be written as $N_i = \sum_{v \in \mathcal{V}_i'} \mathbb{1}_{\{v \in \mathcal{Z}\}}$. The expectation of N_i is $\mathbb{E}N_i = N/(2K)$. By Hoeffding's Lemma D.3.2,

$$\mathbb{P} \left(\left| N_i - \frac{N}{2K} \right| \geq \sqrt{N} \log(N) \right) \leq 2 \exp(-K \log^2(N)) .$$

Similarly, $\tilde{N}_i$, defined in (4.5.2), satisfies

$$\mathbb{P} \left(\left| \tilde{N}_i - \frac{N}{4K} \right| \geq \sqrt{N} \log(N) \right) \leq 2 \exp(-K \log^2(N)) .$$

As defined in (4.5.3) and (4.5.5), both $\tilde{\mathbf{B}}_1$ and $\bar{\mathbf{B}}_1$ are deterministic block matrices. Then with probability at least $1 - 2K \exp(-K \log^2(N))$, the dimensions of each block inside $\tilde{\mathbf{B}}_1$ and $\bar{\mathbf{B}}_1$ are approximately the same, with deviations up to $\sqrt{N} \log(N)$. Consequently, the matrix $\tilde{\mathbf{A}}_1$, which was defined in (4.5.8), can be treated as a perturbed version of $\bar{\mathbf{A}}_1$. By Weyl's inequality (Lemma D.2.1), for any $i \in [K]$,

$$\begin{aligned} |\sigma_i(\bar{\mathbf{B}}_1) - \sigma_i(\tilde{\mathbf{B}}_1)| &= |\sigma_i(\bar{\mathbf{A}}_1) - \sigma_i(\tilde{\mathbf{A}}_1)| \leq \|\bar{\mathbf{A}}_1 - \tilde{\mathbf{A}}_1\|_2 \leq \|\bar{\mathbf{A}}_1 - \tilde{\mathbf{A}}_1\|_F \\ &\leq \left[2K \cdot \frac{N}{K} \cdot \sqrt{N} \log(N) \cdot (\bar{\alpha} - \bar{\beta})^2 \right]^{1/2} = O \left(N^{3/4} \log^{1/2}(N) \cdot (\bar{\alpha} - \bar{\beta}) \right) . \end{aligned}$$

As a result, with probability at least $1 - 2K \exp(-K \log^2(N))$, we have

$$\frac{|\sigma_K(\bar{\mathbf{A}}_1) - \sigma_K(\tilde{\mathbf{A}}_1)|}{\sigma_K(\bar{\mathbf{A}}_1)} = \frac{|\sigma_K(\bar{\mathbf{B}}_1) - \sigma_K(\tilde{\mathbf{B}}_1)|}{\sigma_K(\bar{\mathbf{B}}_1)} = O \left(N^{-1/4} \log^{1/2}(N) \right) .$$

Therefore, singular values of $\tilde{\mathbf{A}}_1$ can be approximated by singular values of $\bar{\mathbf{A}}_1$ with vanishing errors. $\square$

Proof of Lemma 4.5.6. Let $\mathcal{X} \subset \mathcal{V}$ be a subset of vertices in hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ with size $|\mathcal{X}| = cN$ for some $c \in (0, 1)$ to be decided later. Suppose $\mathcal{X}$ is a set of vertices with high degrees that we want to zero out. We first count the ℓ -uniform hyperedges on $\mathcal{X}$ separately, then weight

them by $(\ell - 1)$, and finally sum over ℓ to compute the row sums in $\mathbf{A}$ corresponding to each vertex in $\mathcal{X}$. Let $\mathcal{E}_\ell(\mathcal{X})$ denote the set of ℓ -uniform hyperedges with all vertices located in $\mathcal{X}$, and $\mathcal{E}_\ell(\mathcal{X}^c)$ denote the set of ℓ -uniform hyperedges with all vertices in $\mathcal{X}^c = \mathcal{V} \setminus \mathcal{X}$, respectively. Let $\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^c)$ denote the set of ℓ -uniform hyperedges with at least 1 endpoint in $\mathcal{X}$ and 1 endpoint in $\mathcal{X}^c$. The relationship between total row sums and the number of non-uniform hyperedges in the vertex set $\mathcal{X}$ can be expressed as

$$\sum_{v \in \mathcal{X}} \text{row}(v) \leq \sum_{\ell \in \mathcal{L}} (\ell - 1) \left(\ell |\mathcal{E}_\ell(\mathcal{X})| + (\ell - 1) |\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^c)| \right) \quad (\text{C.2.3})$$

If the row sum of each vertex $v \in \mathcal{X}$ is at least $20Ld$, where $d := \sum_{\ell \in \mathcal{L}} (\ell - 1)a_\ell$, it follows

$$\sum_{\ell \in \mathcal{L}} (\ell - 1) \left(\ell |\mathcal{E}_\ell(\mathcal{X})| + (\ell - 1) |\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^c)| \right) \geq cN \cdot (20Ld). \quad (\text{C.2.4})$$

Then either

$$\sum_{\ell \in \mathcal{L}} \ell(\ell - 1) |\mathcal{E}_\ell(\mathcal{X})| \geq 4LcNd, \quad \text{or} \quad \sum_{\ell \in \mathcal{L}} (\ell - 1)^2 |\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^c)| \geq 16LcNd.$$

Concentration of $\sum_{\ell \in \mathcal{L}} \ell(\ell - 1) |\mathcal{E}_\ell(\mathcal{X})|$

Recall that $|\mathcal{E}_\ell(\mathcal{X})|$ denotes the number of ℓ -uniform hyperedges with all vertices located in $\mathcal{X}$, which can be viewed as the sum of independent Bernoulli random variables $T_e^{(a_\ell)}$ and $T_e^{(b_\ell)}$ given by

$$T_e^{(a_\ell)} \sim \text{Bernoulli} \left(\frac{a_\ell}{\binom{N-1}{\ell-1}} \right), \quad T_e^{(b_\ell)} \sim \text{Bernoulli} \left(\frac{b_\ell}{\binom{N-1}{\ell-1}} \right). \quad (\text{C.2.5})$$

Let $\{\mathcal{V}_1, \dots, \mathcal{V}_K\}$ be the true partition of $\mathcal{V}$. Suppose that there are $\eta_i cN$ vertices in block $\mathcal{V}_i \cap \mathcal{X}$ for each $i \in [K]$ with restriction $\sum_{i=1}^K \eta_i = 1$, then $|\mathcal{E}_\ell(\mathcal{X})|$ can be written as

$$|\mathcal{E}_\ell(\mathcal{X})| = \sum_{e \in \mathcal{E}_\ell(\mathcal{X}, a_\ell)} T_e^{(a_\ell)} + \sum_{e \in \mathcal{E}_\ell(\mathcal{X}, b_\ell)} T_e^{(b_\ell)},$$

where $\mathcal{E}_\ell(\mathcal{X}, a_\ell) := \cup_{i=1}^K \mathcal{E}_\ell(\mathcal{V}_i \cap \mathcal{X})$ denotes the union for sets of hyperedges with all vertices in the same block $\mathcal{V}_i \cap \mathcal{X}$ for some $i \in [K]$, and

$$\mathcal{E}_\ell(\mathcal{X}, b_\ell) := \mathcal{E}_\ell(\mathcal{X}) \setminus \mathcal{E}_\ell(\mathcal{X}, a_\ell) = \mathcal{E}_\ell(\mathcal{X}) \setminus \left(\cup_{i=1}^K \mathcal{E}_\ell(\mathcal{V}_i \cap \mathcal{X}) \right)$$

denotes the set of hyperedges with vertices crossing different $\mathcal{V}_i \cap \mathcal{X}$. We can compute the expectation of $|\mathcal{E}_\ell(\mathcal{X})|$ as

$$\mathbb{E}|\mathcal{E}_\ell(\mathcal{X})| = \sum_{i=1}^K \binom{\eta_i cN}{\ell} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \binom{cN}{\ell} \frac{b_\ell}{\binom{N-1}{\ell-1}}. \quad (\text{C.2.6})$$

Then

$$\sum_{\ell \in \mathcal{L}} \ell(\ell-1) \cdot \mathbb{E}|\mathcal{E}_\ell(\mathcal{X})| = \sum_{\ell \in \mathcal{L}} \ell(\ell-1) \left[\sum_{i=1}^K \binom{\eta_i cN}{\ell} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \binom{cN}{\ell} \frac{b_\ell}{\binom{N-1}{\ell-1}} \right]. \quad (\text{C.2.7})$$

As $\sum_{i=1}^K \eta_i = 1$, it follows that $\sum_{i=1}^K \binom{\eta_i cN}{\ell} \leq \binom{cN}{\ell}$ by induction, thus

$$\frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} \sum_{i=1}^K \binom{\eta_i cN}{\ell} + \frac{b_\ell}{\binom{N-1}{\ell-1}} \binom{cN}{\ell} = \frac{a_\ell}{\binom{N-1}{\ell-1}} \sum_{i=1}^K \binom{\eta_i cN}{\ell} + \frac{b_\ell}{\binom{N-1}{\ell-1}} \left(\binom{cN}{\ell} - \sum_{i=1}^K \binom{\eta_i cN}{\ell} \right)$$

where both terms on the right are positive numbers. Using this and taking $b_\ell = a_\ell$, we obtain the following upper bound for all N ,

$$\sum_{\ell \in \mathcal{L}} \ell(\ell-1) \mathbb{E}|\mathcal{E}_\ell(\mathcal{X})| \leq \sum_{\ell \in \mathcal{L}} \ell(\ell-1) \binom{cN}{\ell} \frac{a_\ell}{\binom{N-1}{\ell-1}} \leq cN \sum_{\ell \in \mathcal{L}} (\ell-1) a_\ell = cNd.$$

Note that $\sum_{\ell \in \mathcal{L}} \ell(\ell-1)|\mathcal{E}_\ell(\mathcal{X})|$ is a weighted sum of independent Bernoulli random variables (corresponding to hyperedges), each upper bounded by L^2 . Also, its variance is bounded by

$$\begin{aligned} \sigma^2 &:= \mathbb{V}\text{ar} \left(\sum_{\ell \in \mathcal{L}} \ell(\ell-1)|\mathcal{E}_\ell(\mathcal{X})| \right) = \sum_{\ell \in \mathcal{L}} L^2(\ell-1)^2 \mathbb{V}\text{ar}(|\mathcal{E}_\ell(\mathcal{X})|) \\ &\leq \sum_{\ell \in \mathcal{L}} L^2(\ell-1)^2 \mathbb{E}|\mathcal{E}_\ell(\mathcal{X})| \leq L^2 cNd. \end{aligned}$$

We can apply Bernstein's Lemma D.3.4 and obtain

$$\begin{aligned} &\mathbb{P} \left(\sum_{\ell \in \mathcal{L}} \ell(\ell-1)|\mathcal{E}_\ell(\mathcal{X})| \geq 4LcNd \right) \\ &\leq \mathbb{P} \left(\sum_{\ell \in \mathcal{L}} \ell(\ell-1)(|\mathcal{E}_\ell(\mathcal{X})| - \mathbb{E}|\mathcal{E}_\ell(\mathcal{X})|) \geq 3LcNd \right) \\ &\leq \exp \left(-\frac{(3LcNd)^2}{L^2 cNd + L^2 cNd/3} \right) \leq \exp(-6cNd). \end{aligned} \tag{C.2.8}$$

Concentration of $\sum_{\ell \in \mathcal{L}} (\ell-1)^2 |\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^{\mathbb{G}})|$

For any finite set $\mathcal{S}$, let $[\mathcal{S}]^j$ denote the family of j -subsets of $\mathcal{S}$, i.e., $[\mathcal{S}]^j = \{\mathcal{Z} | \mathcal{Z} \subseteq \mathcal{S}, |\mathcal{Z}| = j\}$. Let $\mathcal{E}_\ell([\mathcal{Y}]^j, [\mathcal{Z}]^{\ell-j})$ denote the set of ℓ -hyperedges, where j vertices are from $\mathcal{Y}$ and $\ell-j$ vertices are from $\mathcal{Z}$ within each ℓ -hyperedge. We want to count the number of ℓ -hyperedges between $\mathcal{X}$ and $\mathcal{X}^{\mathbb{G}}$, according to the number of vertices located in $\mathcal{X}^{\mathbb{G}}$ within each ℓ -hyperedge. Suppose that there are j vertices from $\mathcal{X}^{\mathbb{G}}$ within each ℓ -hyperedge for some $1 \leq j \leq \ell-1$.

- (i) Assume that all those j vertices are in the same $[\mathcal{V}_i \setminus \mathcal{X}]^j$. If the remaining $\ell-j$ vertices are from $[\mathcal{V}_i \cap \mathcal{X}]^{\ell-j}$, then this ℓ -hyperedge is connected with probability $a_\ell / \binom{N-1}{\ell-1}$, otherwise $b_\ell / \binom{N-1}{\ell-1}$. The number of this type ℓ -hyperedges can be written as

$$\sum_{i=1}^K \left[\sum_{e \in \mathcal{E}_{j,i}^{(a_\ell)}} T_e^{(a_\ell)} + \sum_{e \in \mathcal{E}_{j,i}^{(b_\ell)}} T_e^{(b_\ell)} \right],$$

where $\mathcal{E}_{j,i}^{(a_\ell)} := \mathcal{E}_\ell([\mathcal{V}_i \cap \mathcal{X}^\complement]^j, [\mathcal{V}_i \cap \mathcal{X}]^{\ell-j})$, and

$$\mathcal{E}_{j,i}^{(b_\ell)} := \mathcal{E}_\ell([\mathcal{V}_i \cap \mathcal{X}]^\complement]^j, [\mathcal{X}]^{\ell-j} \setminus [\mathcal{V}_i \cap \mathcal{X}]^{\ell-j})$$

denotes the set ℓ -hyperedges with j vertices in $[\mathcal{V}_i \cap \mathcal{X}^\complement]^j$ and the remaining $\ell - j$ vertices in $[\mathcal{X}]^j \setminus [\mathcal{V}_i \cap \mathcal{X}]^j$. We compute all possible choices and upper bound the cardinality of $\mathcal{E}_{j,i}^{(a_\ell)}$ and $\mathcal{E}_{j,i}^{(b_\ell)}$ by

$$|\mathcal{E}_{j,i}^{(a_\ell)}| \leq \binom{(\frac{1}{K} - \eta_i c)N}{j} \binom{\eta_i c N}{\ell - j}, \quad |\mathcal{E}_{j,i}^{(b_\ell)}| \leq \binom{(\frac{1}{K} - \eta_i c)N}{j} \left[\binom{cN}{\ell - j} - \binom{\eta_i c N}{\ell - j} \right].$$

- (ii) If those j vertices in $[\mathcal{V} \setminus \mathcal{X}]^j$ are not in the same $[\mathcal{V}_i \cap \mathcal{X}]^j$ (which only happens $j \geq 2$), then the number of this type hyperedges can be written as $\sum_{e \in \mathcal{E}_j^{(b_\ell)}} \mathbf{T}_e^{(b_\ell)}$, where

$$\begin{aligned} \mathcal{E}_j^{(b_\ell)} &:= \mathcal{E}_\ell([V \setminus \mathcal{X}]^j \setminus (\cup_{i=1}^K [\mathcal{V}_i \setminus \mathcal{X}]^j), [\mathcal{X}]^{\ell-j}), \\ |\mathcal{E}_j^{(b_\ell)}| &\leq \left[\binom{(1-c)N}{j} - \sum_{i=1}^K \binom{(\frac{1}{K} - \eta_i c)N}{j} \right] \binom{cN}{\ell - j}. \end{aligned}$$

Therefore, $|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^\complement)|$ can be written as a sum of independent Bernoulli random variables,

$$|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^\complement)| = \sum_{j=1}^{\ell-1} \sum_{i=1}^K \left[\sum_{e \in \mathcal{E}_{j,i}^{(a_\ell)}} \mathbf{T}_e^{(a_\ell)} + \sum_{e \in \mathcal{E}_{j,i}^{(b_\ell)}} \mathbf{T}_e^{(b_\ell)} \right] + \sum_{j=2}^{\ell-1} \sum_{e \in \mathcal{E}_j^{(b_\ell)}} \mathbf{T}_e^{(b_\ell)}. \quad (\text{C.2.10})$$

Note that $\binom{(1-c)N}{1} = \sum_{i=1}^K \binom{(1/k - \eta_i c)N}{1}$ and $|\mathcal{E}_1^{(b_\ell)}| = 0$. Then $\mathbb{E}(|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^\complement)|)$ can be computed as follows:

$$\begin{aligned}
& \mathbb{E}(|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^\complement)|) \tag{C.2.11} \\
&= \sum_{j=1}^{\ell-1} \sum_{i=1}^K \binom{(\frac{1}{K} - \eta_i c)N}{j} \left\{ \binom{\eta_i c N}{\ell-j} \frac{a_\ell}{\binom{N-1}{\ell-1}} + \left[\binom{cN}{\ell-j} - \binom{\eta_i c N}{\ell-j} \right] \frac{b_\ell}{\binom{N-1}{\ell-1}} \right\} \\
&\quad + \sum_{j=1}^{\ell-1} \left[\binom{(1-c)N}{j} - \sum_{i=1}^K \binom{(\frac{1}{K} - \eta_i c)N}{j} \right] \binom{cN}{\ell-j} \frac{b_\ell}{\binom{N-1}{\ell-1}} \\
&= \sum_{j=1}^{\ell-1} \sum_{i=1}^K \binom{(\frac{1}{K} - \eta_i c)N}{j} \binom{\eta_i c N}{\ell-j} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \sum_{j=1}^{\ell-1} \binom{(1-c)N}{j} \binom{cN}{\ell-j} \frac{b_\ell}{\binom{N-1}{\ell-1}} \\
&= \sum_{i=1}^K \left[\binom{\frac{N}{K}}{\ell} - \binom{\eta_i c N}{\ell} - \binom{(\frac{1}{K} - \eta_i c)N}{\ell} \right] \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \left[\binom{N}{\ell} - \binom{cN}{\ell} - \binom{(1-c)N}{\ell} \right] \frac{b_\ell}{\binom{N-1}{\ell-1}},
\end{aligned}$$

where we used the fact $\binom{(1-c)N}{1} = \sum_{i=1}^K \binom{(1/k - \eta_i c)N}{1}$ in the first equality and Vandermonde's identity $\binom{N_1 + N_2}{\ell} = \sum_{j=0}^{\ell} \binom{N_1}{j} \binom{N_2}{\ell-j}$ in last equality. Note that

$$f_c := \binom{N}{\ell} - \binom{cN}{\ell} - \binom{(1-c)N}{\ell}$$

counts the number of subsets of $\mathcal{V}$ with ℓ elements such that at least one element belongs to $\mathcal{X}$ and at least one element belongs to $\mathcal{X}^\complement$. On the other hand,

$$g_c = \sum_{i=1}^K \left[\binom{\frac{N}{K}}{\ell} - \binom{\eta_i c N}{\ell} - \binom{(\frac{1}{K} - \eta_i c)N}{\ell} \right].$$

counts the number of subsets of $\mathcal{V}$ with ℓ elements such that all elements belong to a single $\mathcal{V}_i$, and given such an i , that at least one element belongs to $\mathcal{X} \cap \mathcal{V}_i$ and at least one belongs to $\mathcal{X}^\complement \cap \mathcal{V}_i$. As in Figure C.1, g_c only counts the blue pairs while f_c counts red pairs in addition. By virtue of the fact that there are fewer conditions imposed on the sets included in the count for f_c ,

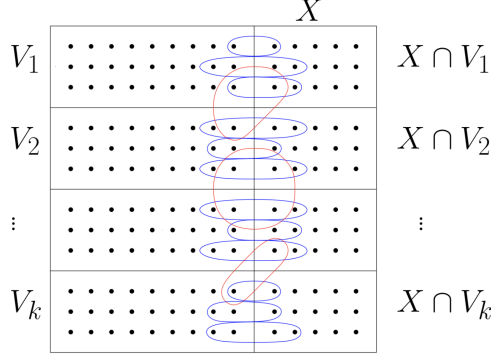


Figure C.1. Comparison of f_c and g_c

we must have $f_c \geq g_c$. Thus, rewriting (C.2.11), we obtain

$$\mathbb{E}(|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^\complement)|) = g_c \frac{a_\ell}{\binom{N-1}{\ell-1}} + (f_c - g_c) \frac{b_\ell}{\binom{N-1}{\ell-1}}.$$

Since both terms in the above sum are positive, we can upper bound by taking $a_\ell = b_\ell$ to obtain

$$\mathbb{E}(|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^\complement)|) \leq f_c \frac{a_\ell}{\binom{N-1}{\ell-1}} = \left[\binom{N}{\ell} - \binom{cN}{\ell} - \binom{(1-c)N}{\ell} \right] \frac{a_\ell}{\binom{N-1}{\ell-1}}.$$

By summing over ℓ , the expectation of $\sum_{\ell \in \mathcal{L}} (\ell-1)^2 |\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^\complement)|$ satisfies

$$\begin{aligned} \sum_{\ell \in \mathcal{L}} (\ell-1)^2 \cdot \mathbb{E}(|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^\complement)|) &\leq \sum_{\ell \in \mathcal{L}} (\ell-1)^2 \left[\binom{N}{\ell} - \binom{cN}{\ell} - \binom{(1-c)N}{\ell} \right] \frac{a_\ell}{\binom{N-1}{\ell-1}}, \\ &\leq 2N \sum_{\ell \in \mathcal{L}} (1 - c^\ell - (1-c)^\ell) (\ell-1) a_\ell \leq 8LcNd, \end{aligned}$$

where the last line holds when $c \in (0, 2^{1/L} - 1]$, since

$$\begin{aligned} &[(1-c) + c]^\ell - c^\ell - (1-c)^\ell \\ &= \binom{\ell}{1} (1-c)^{\ell-1} c + \binom{\ell}{2} (1-c)^{\ell-2} c^2 + \dots + \binom{\ell}{\ell-1} (1-c)^1 c^{\ell-1} \\ &\leq \binom{\ell}{1} c + \binom{\ell}{2} c^2 + \dots + \binom{\ell}{\ell-1} c^{\ell-1} \leq (1+c)^\ell - 1 \leq 2\ell c, \end{aligned} \tag{C.2.13}$$

where the last inequality holds by the following Claim.

Claim 1. *Let $\ell \geq 2$ be some finite integer. Then for $0 < c \leq 2^{1/\ell} - 1$, it follows that $(1+c)^\ell - 1 \leq 2\ell c$.*

Proof of the Claim. We finish the proof by induction. First, the argument $(1+c)^j - 1 \leq 2jc$ holds true for the base cases $j = 1, 2$. Suppose that the argument holds for the case $j \geq 2$. For the case $j+1 \leq \ell$, it follows that

$$(1+c)^{j+1} - 1 = (1+c)^j + c(1+c)^j - 1 \leq 2jc + c(1+c)^j \leq 2(j+1)c,$$

where the last inequality holds true if $c(1+c)^j \leq 2c$, and it holds since $c \leq 2^{1/\ell} - 1 \leq 2^{1/j} - 1$ for all $j \leq \ell$. $\square$

Similarly, we apply Bernstein Lemma D.3.4 again with $K = L^2$, $\sigma^2 \leq 8L^3 cNd$ and obtain

$$\begin{aligned} & \mathbb{P} \left(\sum_{\ell \in \mathcal{L}} (\ell-1)^2 |\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^{\mathbb{G}})| \geq 16LcNd \right) \\ & \leq \mathbb{P} \left(\sum_{\ell \in \mathcal{L}} (\ell-1)^2 (|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^{\mathbb{G}})| - \mathbb{E}|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^{\mathbb{G}})|) \geq 8LcNd \right) \leq \exp(-6cNd/L). \end{aligned} \quad (\text{C.2.14})$$

By the binomial coefficient upper bound $\binom{N}{K} \leq (\frac{eN}{K})^K$ for $1 \leq K \leq N$, there are at most

$$\binom{N}{cN} \leq \left(\frac{e}{c}\right)^{cN} = \exp(-c(\log(c) - 1)N) \quad (\text{C.2.15})$$

many subsets $\mathcal{X}$ of size $|\mathcal{X}| = cN$. Let d be sufficiently large so that $d^{-3} \leq c_0$. Substituting $c = d^{-3}$ in (C.2.15), we have

$$\binom{N}{d^{-3}N} \leq \exp[3d^{-3} \log(d)N].$$

Taking $c = d^{-3}$ in (C.2.8) and (C.2.14), we obtain

$$\mathbb{P} \left(\sum_{\ell \in \mathcal{L}} (\ell - 1)(\ell |\mathcal{E}_\ell(\mathcal{X})| + (\ell - 1)|\mathcal{E}_\ell(\mathcal{X}, \mathcal{X}^{\mathbb{L}})|) \geq 20Ld^{-2}N \right) \leq 2\exp(-2d^{-2}N/L).$$

Taking a union bound over all possible $\mathcal{X}$ with $|\mathcal{X}| = d^{-3}N$, we obtain with probability at least $1 - 2\exp(3d^{-3}\log(d)N - 2d^{-2}N/L) \leq 1 - 2\exp(-d^{-2}N/L)$, no more than $d^{-3}N$ many vertices have total row sum greater than $20Ld$. Note that we have imposed the condition that $c = d^{-3} \in (0, 2^{1/L} - 1]$ in (C.2.13), thus $d \geq (2^{1/L} - 1)^{-1/3}$, producing the lower bound in Assumption 4.1.5. $\square$

Proof of Lemma 4.5.10. Throughout the proof, let $\pi(i)$ denote the membership of vertex $i \in \mathcal{V}$. Let $\mathbf{u}_i(j)$ denote the j -th entry of the vector $\mathbf{u}_i \in \mathbb{R}^N$. Let $\langle \mathbf{u}_i, \mathbf{e}_i \rangle$ denote the inner product of two vectors.

Note that $\mathbf{U}$ is spanned by first K singular vectors of $(\mathbf{A}_1)_{I_1}$. Let $\{\mathbf{u}_i\}_{i=1}^K$ be an orthonormal basis of the $\mathbf{U}$. For each fixed $i \in \mathcal{V}_{\pi(i)} \cap \mathcal{J}_2 \cap \{i_1, \dots, i_s\}$, the projection $\mathbf{P}_\mathbf{U} := \sum_{k=1}^K \langle \mathbf{u}_k, \cdot \rangle \mathbf{u}_k$ is defined via

$$\mathbf{P}_\mathbf{U} \mathbf{e}_i = \sum_{k=1}^K \langle \mathbf{u}_k, \mathbf{e}_i \rangle \mathbf{u}_k, \quad \|\mathbf{P}_\mathbf{U} \mathbf{e}_i\|_2^2 = \sum_{k=1}^K \langle \mathbf{u}_k, \mathbf{e}_i \rangle^2.$$

As a consequence of independence between entries in $\mathbf{A}_1$ and entries in $\mathbf{A}_2$, defined in (4.5.7), it is known that $\{\mathbf{u}_k\}_{k=1}^K$ and $\mathbf{e}_i$ are independent of each other, since $\mathbf{e}_i$ are columns of $\mathbf{E}_2 := \mathbf{A}_2 - \tilde{\mathbf{A}}_2$. If the expectation is taken over $\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}$ conditioning on $\{\mathbf{u}_k\}_{k=1}^K$, then

$$\begin{aligned} \mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\langle \mathbf{u}_k, \mathbf{e}_i \rangle \middle| \{\mathbf{u}_k\}_{k=1}^K \right] &= \sum_{j=1}^N \mathbf{u}_k(j) \cdot \mathbb{E} \left(\left[(\mathbf{A}_2)_{ji} - (\mathbb{E} \mathbf{A}_2)_{ji} \right] \right) = 0, \\ \mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\|\mathbf{P}_\mathbf{U} \mathbf{e}_i\|_2^2 \middle| \{\mathbf{u}_k\}_{k=1}^K \right] &= \sum_{k=1}^K \mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\langle \mathbf{u}_k, \mathbf{e}_i \rangle^2 \middle| \{\mathbf{u}_k\}_{k=1}^K \right], \end{aligned}$$

where $\mathcal{L}$ is obtained from Algorithm 4.4.1. We expand each $\langle \mathbf{u}_k, \mathbf{e}_i \rangle^2$ and rewrite it into 2 parts,

$$\begin{aligned} \langle \mathbf{u}_k, \mathbf{e}_i \rangle^2 &= \sum_{j_1=1}^K \sum_{j_2=1}^K \mathbf{u}_k(j_1) \mathbf{e}_i(j_1) \mathbf{u}_k(j_2) \mathbf{e}_i(j_2) \\ &= \underbrace{\sum_{j=1}^K [\mathbf{u}_k(j)]^2 [\mathbf{e}_i(j)]^2}_{(a)} + \underbrace{\sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{e}_i(j_1) \mathbf{u}_k(j_2) \mathbf{e}_i(j_2)}_{(b)}, \quad \forall k \in [K]. \end{aligned} \quad (\text{C.2.16})$$

Part (a) is the contribution from graph, i.e., 2-uniform hypergraph, while part (b) is the contribution from ℓ -uniform hypergraph with $\ell \geq 3$, which only occurs in hypergraph clustering. The expectation of part (a) in (C.2.16) is upper bounded by α as defined in (4.2.5), since for any $k \in [K]$,

$$\begin{aligned} &\mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\sum_{j=1}^K [\mathbf{u}_k(j)]^2 [\mathbf{e}_i(j)]^2 \middle| \{\mathbf{u}_k\}_{k=1}^K \right] = \sum_{j=1}^N [\mathbf{u}_k(j)]^2 \cdot \text{Var}((\mathbf{A}_2)_{ji}) \\ &\leq \sum_{j=1}^N [\mathbf{u}_k(j)]^2 \cdot (\mathbb{E} \mathbf{A}_2)_{ji} \leq \alpha = \sum_{\ell \in \mathcal{L}} \left[\binom{\frac{N}{K} - 2}{\ell - 2} \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \binom{N}{\ell - 2} \frac{b_\ell}{\binom{N-1}{\ell-1}} \right] \\ &\leq \sum_{\ell \in \mathcal{L}} \binom{N}{\ell - 2} \frac{a_\ell}{\binom{N-1}{\ell-1}} \leq \frac{2}{N} \sum_{\ell \in \mathcal{L}} (\ell - 1) a_\ell = \frac{2d}{N}, \end{aligned}$$

where $\|\mathbf{u}_k\|_2^2 = \sum_{j=1}^N [\mathbf{u}_k(j)]^2 = 1$. For part (b) in (C.2.16),

$$\begin{aligned} &\mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{e}_i(j_1) \mathbf{u}_k(j_2) \mathbf{e}_i(j_2) \middle| \{\mathbf{u}_k\}_{k=1}^K \right] \\ &= \sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{u}_k(j_2) \mathbb{E} \left[\left((\mathbf{A}_2)_{j_1 i} - (\mathbb{E} \mathbf{A}_2)_{j_1 i} \right) \left((\mathbf{A}_2)_{j_2 i} - (\mathbb{E} \mathbf{A}_2)_{j_2 i} \right) \right] \\ &= \sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{u}_k(j_2) \mathbb{E} \left(\sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell[\mathcal{Y}_2 \cup \mathcal{Z}] \\ \{i, j_1\} \subset e}} (\mathcal{A}_e^{(\ell)} - \mathbb{E} \mathcal{A}_e^{(\ell)}) \right) \left(\sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell[\mathcal{Y}_2 \cup \mathcal{Z}] \\ \{i, j_2\} \subset e}} (\mathcal{A}_e^{(\ell)} - \mathbb{E} \mathcal{A}_e^{(\ell)}) \right). \end{aligned}$$

According to Definition 1.1.3 of the adjacency tensor, $\mathcal{A}_{e_1}^{(\ell)}$ and $\mathcal{A}_{e_2}^{(\ell)}$ are independent if hyperedge $e_1 \neq e_2$, then only the terms with hyperedge $e \supset \{i, j_1, j_2\}$ have nonzero contribution. Then the

expectation of part (b) can be rewritten as

$$\begin{aligned}
& \mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{e}_i(j_1) \mathbf{u}_k(j_2) \mathbf{e}_i(j_2) \middle| \{\mathbf{u}_k\}_{k=1}^K \right] \\
&= \sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{u}_k(j_2) \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell[\mathcal{Y}_2 \cup \mathcal{Z}] \\ \{i, j_1, j_2\} \subset e}} \mathbb{E}(\mathcal{A}_e^{(\ell)} - \mathbb{E}\mathcal{A}_e^{(\ell)})^2 \\
&\leq \sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{u}_k(j_2) \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell[\mathcal{Y}_2 \cup \mathcal{Z}] \\ \{i, j_1, j_2\} \subset e}} \mathbb{E}\mathcal{A}_e^{(\ell)} \\
&= \sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{u}_k(j_2) \sum_{\ell \in \mathcal{L}} \sum_{\substack{e \in \mathcal{E}_\ell[\mathcal{Y}_2 \cup \mathcal{Z}] \\ \{i, j_1, j_2\} \subset e}} \frac{a_\ell}{\binom{N-1}{\ell-1}}. \tag{C.2.18}
\end{aligned}$$

Note that $|\mathcal{Y}_2 \cup \mathcal{Z}| \leq N$, then the number of possible hyperedges e , while $e \in \mathcal{E}_\ell[\mathcal{Y}_2 \cup \mathcal{Z}]$ and $e \supset \{i, j_1, j_2\}$, is at most $\binom{N}{\ell-3}$. Thus (C.2.18) is upper bounded by

$$\begin{aligned}
& \sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{u}_k(j_2) \sum_{\ell \in \mathcal{L}} \binom{N}{\ell-3} \frac{a_\ell}{\binom{N-1}{\ell-1}} \\
&\leq \sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{u}_k(j_2) \sum_{\ell \in \mathcal{L}} \frac{(\ell-1)(\ell-2)}{(N-\ell)^2} a_\ell \leq \frac{dL}{N^2} \sum_{j_1 \neq j_2} \mathbf{u}_k(j_1) \mathbf{u}_k(j_2) \\
&\leq \frac{dL}{2N^2} \sum_{j_1 \neq j_2} \left([\mathbf{u}_k(j_1)]^2 + [\mathbf{u}_k(j_2)]^2 \right) \leq \frac{dL(N-1)}{2N^2} \left(\sum_{j_1=1}^N [\mathbf{u}_k(j_1)]^2 + \sum_{j_2=1}^N [\mathbf{u}_k(j_2)]^2 \right) \\
&\leq \frac{dL}{N},
\end{aligned}$$

where $\|\mathbf{u}_k\|_2 = 1$, $d = \sum_{\ell \in \mathcal{L}} (\ell-1)a_\ell$. With the upper bounds for part (a) and (b) in (C.2.16), the conditional expectation of $\|\mathbf{P}_\mathbf{U} \mathbf{e}_i\|_2^2$ is then bounded by

$$\mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\|\mathbf{P}_\mathbf{U} \mathbf{e}_i\|_2^2 \middle| \{\mathbf{u}_k\}_{k=1}^K \right] = \sum_{k=1}^K \mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\langle \mathbf{u}_k, \mathbf{e}_i \rangle^2 \middle| \{\mathbf{u}_k\}_{k=1}^K \right] \leq \frac{Kd}{N} (L+2),$$

Let X_i be the Bernoulli random variable defined by

$$X_i = \mathbb{1}_{\left\{\|\mathbf{P}_U \mathbf{e}_i\|_2 > 2\sqrt{Kd(L+2)/N}\right\}}, \quad i \in \{i_1, \dots, i_s\}.$$

By Markov's inequality,

$$\mathbb{E}X_i = \mathbb{P}\left(\|\mathbf{P}_U \mathbf{e}_i\|_2 > 2\sqrt{Kd(L+2)/N}\right) \leq \frac{\mathbb{E}_{\{\mathcal{A}^{(\ell)}\}_{\ell \in \mathcal{L}}} \left[\|\mathbf{P}_U \mathbf{e}_i\|_2^2 \middle| \{\mathbf{u}_k\}_{k=1}^K\right]}{4Kd(L+2)/N} \leq \frac{1}{4}.$$

Let $\delta := s[2\sum_{j=1}^s \mathbb{E}X_{i_j}]^{-1} - 1$ where $s = 2K \log^2(N)$. By Hoeffding Lemma D.3.2,

$$\begin{aligned} \mathbb{P}\left(\sum_{j=1}^s X_{i_j} \geq \frac{s}{2}\right) &= \mathbb{P}\left(\sum_{j=1}^s (X_{i_j} - \mathbb{E}X_{i_j}) \geq \delta \sum_{j=1}^s \mathbb{E}X_{i_j}\right) \\ &\leq \exp\left(-\frac{2\delta^2(\sum_{j=1}^s \mathbb{E}X_{i_j})^2}{s}\right) = O\left(\frac{1}{N^{K \log(N)}}\right). \end{aligned}$$

Therefore, with probability $1 - O(N^{-K \log(N)})$, at least $s/2$ of the vectors $\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_s}$ satisfy

$$\|\mathbf{P}_U \mathbf{e}_i\|_2 \leq 2\sqrt{Kd(L+2)/N}.$$

Meanwhile, for any $c \in (0, 2)$, there exists some large enough constant $C_2 \geq 2^{L+1} \sqrt{(L+2)/K}/c$

such that if $\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell) > C_2 K^{L-1} \sqrt{d}$, then

$$\begin{aligned} \|\bar{\mathbf{\delta}}_i\|_2 &= \frac{\sqrt{N}(\bar{\alpha} - \bar{\beta})}{2} = \frac{\sqrt{N}}{2} \sum_{\ell \in \mathcal{L}} \left(\frac{\frac{3N}{4K} - 2}{\ell - 2}\right) \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} \\ &= \frac{(1 + o(1))}{2\sqrt{N}} \sum_{\ell \in \mathcal{L}} \left(\frac{3}{4K}\right)^{\ell-2} (\ell - 1)(a_\ell - b_\ell) \geq \frac{K}{(2K)^{L-1} \sqrt{N}} \sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell), \\ &> \frac{C_2 K^L \sqrt{d}}{(2K)^{L-1} \sqrt{N}} \geq \frac{2^{L+1} \sqrt{(L+2)} K^L \sqrt{d}}{c(2K)^{L-1} \sqrt{kn}} \\ &> \frac{2}{c} 2\sqrt{\frac{Kd(L+2)}{N}} > \frac{2}{c} \|\mathbf{P}_U \mathbf{e}_i\|_2. \end{aligned}$$

□

Proof of Lemma 4.5.12. Split the vertex set $\mathcal{V}$ into $\widehat{\mathcal{V}}_+ = \{i | v_i > 0\}$ and $\widehat{\mathcal{V}}_- = \{i | v_i \leq 0\}$. Without loss of generality, assume that the first $\frac{N}{K}$ entries of $\bar{\mathbf{v}}$ are positive. We can write $\mathbf{v}$ in terms of its orthogonal projection onto $\bar{\mathbf{v}}$. Choose $\boldsymbol{\varepsilon}$ such that $\boldsymbol{\varepsilon} \perp \bar{\mathbf{v}}$ and $\|\boldsymbol{\varepsilon}\|_2 < c$ with $c_1 \geq \sqrt{1-c^2}$, then we can write

$$\mathbf{v} = c_1 \bar{\mathbf{v}} + \boldsymbol{\varepsilon} = \left[\varepsilon_1 + \frac{c_1}{\sqrt{N}}, \dots, \varepsilon_{\frac{N}{K}} + \frac{c_1}{\sqrt{N}}, \varepsilon_{\frac{N}{K}+1} - \frac{c_1}{\sqrt{N}}, \dots, \varepsilon_N - \frac{c_1}{\sqrt{N}} \right]^\top, \quad (\text{C.2.20})$$

The number of entries in $\boldsymbol{\varepsilon}$ smaller than $-\sqrt{1-c^2}/\sqrt{N}$ is at most $c^2(1-c^2)^{-1}N$. Note that $c_1 \geq \sqrt{1-c^2}$, thus at least $\frac{N}{K} - \frac{c^2}{1-c^2}N$ indices i with $\bar{v}_i = \frac{1}{\sqrt{N}}$ will have $v_i > 0$. Then the ratio we are seeking is at least

$$\frac{\frac{N}{K} - \frac{c^2}{1-c^2}N}{\frac{N}{K}} = 1 - \frac{Kc^2}{1-c^2} > 1 - \frac{4K}{3}c^2.$$

□

Proof of Lemma 4.5.13. We start with the following simple claim: for any $\ell \geq 2$ and any $\mathbf{v} \in [1/2, 1)$,

$$\mathbf{v}^\ell + (1-\mathbf{v})^\ell < \left(\frac{1+\mathbf{v}}{2} \right)^\ell. \quad (\text{C.2.21})$$

Indeed, one quick way to see this is by induction on ℓ ; we will induct from ℓ to $\ell+2$. Assume

the inequality is true for ℓ ; then

$$\begin{aligned}
v^{\ell+2} + (1-v)^{\ell+2} &= v^2 v^\ell + (1-v)^2 (1-v)^\ell \\
&= v^2 v^\ell + (1-2v+v^2)(1-v)^\ell \leq v^2 v^\ell + v^2 (1-v)^\ell \\
&= v^2 (v^\ell + (1-v)^\ell) < (v^2 + (1-v)^2)(v^\ell + (1-v)^\ell) \\
&< \left(\frac{1+v}{2}\right)^2 \left(\frac{1+v}{2}\right)^\ell = \left(\frac{1+v}{2}\right)^{\ell+2},
\end{aligned}$$

where we have used the induction hypothesis together with $1-2v \leq 0$ and $(1-v)^2 > 0$. After easily checking that the inequality works for $\ell = 2, 3$, the induction is complete.

We shall now check that the quantities defined in Lemma 4.5.13 obey the relationship $\mu_2 \geq \mu_1$ and $\mu_2 - \mu_1 = \Omega(N)$, for N large enough. First, note that the only thing we need to check is that, for sufficiently large N ,

$$\binom{\frac{vN}{2K}}{\ell} + \binom{\frac{(1-v)N}{2K}}{\ell} \leq \binom{\frac{(1+v)N}{4K}}{\ell} + (K-1) \binom{\frac{(1-v)N}{4K(K-1)}}{\ell};$$

in fact, we will show the stronger statement that for any $\ell \geq 2$ and N large enough,

$$\binom{\frac{vN}{2K}}{\ell} + \binom{\frac{(1-v)N}{2K}}{\ell} < \binom{\frac{(1+v)N}{4K}}{\ell}, \tag{C.2.22}$$

and this will suffice to see that the second part of the assertion, $\mu_2 - \mu_1 = \Omega(N)$, is also true. Asymptotically, $\binom{\frac{vN}{2K}}{\ell} \sim \frac{v^\ell}{\ell!} \left(\frac{N}{2K}\right)^\ell$, $\binom{\frac{(1-v)N}{2K}}{\ell} \sim \frac{(1-v)^\ell}{\ell!} \left(\frac{N}{2K}\right)^\ell$, and $\binom{\frac{(1+v)N}{4K}}{\ell} \sim \frac{\left(\frac{1+v}{2}\right)^\ell}{\ell!} \left(\frac{N}{2K}\right)^\ell$. Then the inequality (C.2.22) follows from the claim (C.2.21).

Let $\{\mathcal{V}_1, \dots, \mathcal{V}_K\}$ be the true partition of $\mathcal{V}$. Recall that hyperedges in $\mathcal{H} = \cup_{\ell \in \mathcal{L}} \mathcal{H}_\ell$ are colored red and blue with equal probability in Algorithm 4.1.2. Let $\mathcal{E}_\ell(\mathcal{X})$ denote the set of blue ℓ -uniform hyperedges with all vertices located in the vertex set $\mathcal{X}$. Assume $|\mathcal{X} \cap \mathcal{V}_k| = \eta_k |\mathcal{X}|$ with $\sum_{k=1}^K \eta_k = 1$. For each $\ell \in \mathcal{L}$, the presence of hyperedge $e \in \mathcal{E}_\ell(\mathcal{X})$ can be represented by

independent Bernoulli random variables

$$\mathbf{T}_e^{(a_\ell)} \sim \text{Bernoulli}\left(\frac{a_\ell}{2^{\binom{N-1}{\ell-1}}}\right), \quad \mathbf{T}_e^{(b_\ell)} \sim \text{Bernoulli}\left(\frac{b_\ell}{2^{\binom{N-1}{\ell-1}}}\right),$$

depending on whether e is a hyperedge with all vertices in the same community. Denote by

$$\mathcal{E}_\ell(\mathcal{X}, a_\ell) := \cup_{i=1}^K \mathcal{E}_\ell(\mathcal{V}_i \cap \mathcal{X})$$

the union of all ℓ -uniform sets of hyperedges with all vertices in the same $\mathcal{V}_i \cap \mathcal{X}$ for some $i \in [K]$, and by

$$\mathcal{E}_\ell(\mathcal{X}, b_\ell) := \mathcal{E}_\ell(\mathcal{X}) \setminus \mathcal{E}_\ell(\mathcal{X}, a_\ell) = \mathcal{E}_\ell(\mathcal{X}) \setminus \left(\cup_i^K \mathcal{E}_\ell(\mathcal{V}_i \cap \mathcal{X}) \right)$$

the set of ℓ -uniform hyperedges with vertices across different blocks $\mathcal{V}_i \cap \mathcal{X}$. Then the cardinality $|\mathcal{E}_\ell(\mathcal{X})|$ can be written as the

$$|\mathcal{E}_\ell(\mathcal{X})| = \sum_{e \in \mathcal{E}_\ell(\mathcal{X}, a_\ell)} \mathbf{T}_e^{(a_\ell)} + \sum_{e \in \mathcal{E}_\ell(\mathcal{X}, b_\ell)} \mathbf{T}_e^{(b_\ell)},$$

and by summing over ℓ , the weighted cardinality $|\mathcal{E}(\mathcal{X})|$ is written as

$$|\mathcal{E}(\mathcal{X})| := \sum_{\ell \in \mathcal{L}} \ell(\ell-1) |\mathcal{E}_\ell(\mathcal{X})| = \sum_{\ell \in \mathcal{L}} \ell(\ell-1) \left\{ \sum_{e \in \mathcal{E}_\ell(\mathcal{X}, a_\ell)} \mathbf{T}_e^{(a_\ell)} + \sum_{e \in \mathcal{E}_\ell(\mathcal{X}, b_\ell)} \mathbf{T}_e^{(b_\ell)} \right\},$$

with its expectation

$$\mathbb{E}|\mathcal{E}(\mathcal{X})| = \sum_{\ell \in \mathcal{L}} \ell(\ell-1) \left\{ \sum_{i=1}^K \binom{\eta_i \frac{N}{2K}}{\ell} \frac{a_\ell - b_\ell}{2^{\binom{N-1}{\ell-1}}} + \binom{\frac{N}{2K}}{\ell} \frac{b_\ell}{2^{\binom{N-1}{\ell-1}}} \right\}, \quad (\text{C.2.23})$$

since

$$|\mathcal{E}_\ell(\mathcal{X}, a_\ell)| = \sum_{i=1}^K |\mathcal{E}_\ell(\mathcal{V}_i \cap \mathcal{X})| = \sum_{i=1}^K \binom{\eta_i \frac{N}{2K}}{\ell}, \quad |\mathcal{E}_\ell(\mathcal{X}, b_\ell)| = \binom{\frac{N}{2K}}{\ell} - \sum_{i=1}^K \binom{\eta_i \frac{N}{2K}}{\ell},$$

Next, we prove the two Statements in Lemma 4.5.13 separately. First, assume that $|\mathcal{X} \cap \mathcal{V}_i| \leq \nu |\mathcal{X}|$ (i.e., $\eta_i \leq \nu$) for each $i \in [K]$. Then

$$\mathbb{E}|\mathcal{E}(\mathcal{X})| \leq \frac{1}{2} \sum_{\ell \in \mathcal{L}} \ell(\ell-1) \left\{ \left[\binom{\frac{\nu N}{2K}}{\ell} + \binom{\frac{(1-\nu)N}{2K}}{\ell} \right] \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \binom{\frac{N}{2K}}{\ell} \frac{b_\ell}{\binom{N-1}{\ell-1}} \right\} =: \mu_1.$$

To justify the above inequality, note that since $\sum_{i=1}^K \eta_i = 1$, the sum $\sum_{i=1}^K \binom{\eta_i \frac{N}{2K}}{\ell}$ is maximized when all but 2 of the η_i are 0, and since all $\eta_i \leq \nu$, this means that

$$\sum_{i=1}^K \binom{\eta_i \frac{N}{2K}}{\ell} \leq \binom{\frac{\nu N}{2K}}{\ell} + \binom{\frac{(1-\nu)N}{2K}}{\ell}.$$

Note that $\ell(\ell-1)(T_e^{(a_\ell)} - \mathbb{E}T_e^{(a_\ell)})$ and $\ell(\ell-1)(T_e^{(b_\ell)} - \mathbb{E}T_e^{(b_\ell)})$ are independent mean-zero random variables bounded by $\ell(\ell-1)$ for all $\ell \in \mathcal{L}$, and $\text{Var}(|\mathcal{E}(\mathcal{X})|) \leq L^2(\ell-1)^2 \mathbb{E}|\mathcal{E}(\mathcal{X})| = \Omega(N)$.

Recall that $\mu_T := (\mu_1 + \mu_2)/2$. Define $t = \mu_T - \mathbb{E}|\mathcal{E}(\mathcal{X})|$, then $0 < (\mu_2 - \mu_1)/2 \leq t \leq \mu_T$, hence $t = \Omega(N)$. By Bernstein's Lemma D.3.4, we have

$$\mathbb{P}\left(|\mathcal{E}(\mathcal{X})| \geq \mu_T\right) = \mathbb{P}\left(|\mathcal{E}(\mathcal{X})| - \mathbb{E}|\mathcal{E}(\mathcal{X})| \geq t\right) \leq \exp\left(-\frac{t^2/2}{\text{Var}(|\mathcal{E}(\mathcal{X})|) + \ell(\ell-1)t/3}\right) = O(e^{-cN}),$$

where $c > 0$ is some constant. On the other hand, if $|\mathcal{X} \cap \mathcal{V}_i| \geq \frac{1+\nu}{2} |\mathcal{X}|$ for some $i \in [K]$, then

$$\mathbb{E}|\mathcal{E}(\mathcal{X})| \geq \frac{1}{2} \sum_{\ell \in \mathcal{L}} \ell(\ell-1) \left\{ \left[\binom{\frac{(1+\nu)N}{4K}}{\ell} + (K-1) \binom{\frac{(1-\nu)N}{4K(K-1)}}{\ell} \right] \frac{a_\ell - b_\ell}{\binom{N-1}{\ell-1}} + \binom{\frac{N}{2K}}{\ell} \frac{b_\ell}{\binom{N-1}{\ell-1}} \right\} =: \mu_2.$$

The above can be justified by noting that at least one $|\mathcal{X} \cap \mathcal{V}_i| \geq \frac{1+\nu}{2} |\mathcal{X}|$, and that the rest of the vertices will yield a minimal binomial sum when they are evenly split between the remaining

$\mathcal{V}_j$. Similarly, define $t = \mu_T - \mathbb{E}|\mathcal{E}(\mathcal{X})|$, then $0 < (\mu_2 - \mu_1)/2 \leq -t = \Omega(N)$, and Bernstein's Lemma D.3.4 gives

$$\begin{aligned} \mathbb{P}\left(|\mathcal{E}(\mathcal{X})| \leq \mu_T\right) &= \mathbb{P}\left(|\mathcal{E}(\mathcal{X})| - \mathbb{E}|\mathcal{E}(\mathcal{X})| \leq -t\right) \\ &\leq \exp\left(-\frac{t^2/2}{\mathbb{V}\text{ar}(|\mathcal{E}(\mathcal{X})|) + \ell(\ell-1)(-t)/3}\right) = O(e^{-c'N}), \end{aligned}$$

where $c' > 0$ is some other constant. $\square$

Proof of Lemma 4.5.15. If vertex i is uniformly chosen from $\mathcal{Y}_2$, the probability that $i \notin \mathcal{V}_k$ for some $k \in [K]$ is

$$\mathbb{P}(i \notin \mathcal{V}_k | i \in \mathcal{Y}_2) = \frac{\mathbb{P}(i \notin \mathcal{V}_k, i \in \mathcal{Y}_2)}{\mathbb{P}(i \in \mathcal{Y}_2)} = 1 - \frac{|\mathcal{V}_k \cap \mathcal{Y}_2|}{|\mathcal{Y}_2|} = 1 - \frac{\frac{N}{K} - N_k - \tilde{N}_k}{N - \sum_{t=1}^K (N_t + \tilde{N}_t)}, \quad k \in [K],$$

where N_t and $\tilde{N}_t$, defined in (4.5.1) and (4.5.2), denote the cardinality of $\mathcal{Z} \cap \mathcal{V}_t$ and $\mathcal{Y}_1 \cap \mathcal{V}_t$ respectively. As proved in Lemma 4.5.5, with probability at least $1 - 2\exp(-K \log^2(N))$, we have

$$|N_t - N/(2K)| \leq \sqrt{N} \log(N), \quad |\tilde{N}_t - N/(4K)| \leq \sqrt{N} \log(N).$$

Then $\mathbb{P}(i \notin \mathcal{V}_k | i \in \mathcal{Y}_2) = 1 - \frac{1}{K} (1 + o(1))$. After $K \log^2(N)$ samples from $\mathcal{Y}_2$, the probability that there exists at least one node which belongs to $\mathcal{V}_k$ is at least

$$1 - \left(1 - \frac{1 + o(1)}{K}\right)^{K \log^2(N)} = 1 - N^{-(1+o(1))K \log(\frac{K}{K-1}) \log N}.$$

The proof is completed by a union bound over $k \in [K]$. $\square$

C.2.3 Proofs of Lemmas in Section 4.5.2

Proof of Lemma 4.5.16. We calculate $\mathbb{P}(\widehat{\mathbf{S}}_{11}^{(0)}(u) \leq \mu_C)$ first. Define $t_{1C} := \mu_C - \mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u)$, then by Bernstein Lemma D.3.4,

$$\begin{aligned} \mathbb{P}(\widehat{\mathbf{S}}_{11}^{(0)}(u) \leq \mu_C) &= \mathbb{P}(\widehat{\mathbf{S}}_{11}^{(0)}(u) - \mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u) \leq t_{1C}) \\ &\leq \exp\left(-\frac{t_{1C}^2/2}{\text{Var}[\widehat{\mathbf{S}}_{11}^{(0)}(u)] + (L-1) \cdot t_{1C}/3}\right) \leq \exp\left(-\frac{3t_{1C}^2/(L-1)}{6(L-1) \cdot \mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u) + 2t_{1C}}\right) \\ &\leq \exp\left(-\frac{[(v)^{L-1} - (1-v)^{L-1}]^2}{(L-1)^2 \cdot 2^{2L+3}} \cdot \frac{\left[\sum_{\ell \in \mathcal{L}} (\ell-1) \left(\frac{a_\ell - b_\ell}{K^{\ell-1}}\right)\right]^2}{\sum_{\ell \in \mathcal{L}} (\ell-1) \left(\frac{a_\ell - b_\ell}{K^{\ell-1}} + b_\ell\right)}\right), \end{aligned}$$

where $\mathcal{L}$ is obtained from Algorithm 4.4.1 with L denoting the maximum value in $\mathcal{L}$, and the last two inequalities hold since $\text{Var}[\widehat{\mathbf{S}}_{11}^{(0)}(u)] \leq (L-1)^2 \mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u)$, and for sufficiently large N ,

$$\begin{aligned} t_{1C} := \mu_C - \mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u) &= -\frac{1}{2} \sum_{\ell \in \mathcal{L}} (\ell-1) \cdot \left[\binom{\frac{vN}{2K}}{\ell-1} - \binom{\frac{(1-v)N}{2K}}{\ell-1} \right] \frac{a_\ell - b_\ell}{2 \binom{N-1}{\ell-1}} \\ &\leq -\frac{1}{2} \sum_{\ell \in \mathcal{L}} \frac{(v)^{\ell-1} - (1-v)^{\ell-1}}{2^\ell} \cdot (\ell-1) \frac{a_\ell - b_\ell}{K^{\ell-1}} (1+o(1)) \\ &\leq -\frac{(v)^{L-1} - (1-v)^{L-1}}{2^{L+2}} \sum_{\ell \in \mathcal{L}} (\ell-1) \cdot \frac{a_\ell - b_\ell}{K^{\ell-1}}, \end{aligned}$$

as well as the following bound on the expectation

$$\begin{aligned} 6(L-1) \mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u) + 2t_{1C} &= 2\mu_C + (6L-8) \mathbb{E}\widehat{\mathbf{S}}_{11}^{(0)}(u) \\ &= \sum_{\ell \in \mathcal{L}} (\ell-1) \left\{ \left[(6L-7) \binom{\frac{vN}{2K}}{\ell-1} + \binom{\frac{(1-v)N}{2K}}{\ell-1} \right] \frac{a_\ell - b_\ell}{2 \binom{N-1}{\ell-1}} + 6(L-1) \binom{\frac{N}{2K}}{\ell-1} \frac{b_\ell}{2 \binom{N-1}{\ell-1}} \right\} \\ &= \sum_{\ell \in \mathcal{L}} (\ell-1) \left[\frac{(6L-7) \cdot (v)^{\ell-1} + (1-v)^{\ell-1}}{2^\ell} \cdot \frac{a_\ell - b_\ell}{K^{\ell-1}} + \frac{(6L-6)b_\ell}{2^\ell K^{\ell-1}} \right] (1+o(1)) \\ &\leq \sum_{\ell \in \mathcal{L}} \frac{(6L-7) \cdot (v)^{\ell-1} + (1-v)^{\ell-1}}{2^\ell} \cdot (\ell-1) \left(\frac{a_\ell - b_\ell}{K^{\ell-1}} + b_\ell \right) (1+o(1)) \\ &\leq \frac{3(L-1)}{2} \sum_{\ell \in \mathcal{L}} (\ell-1) \left(\frac{a_\ell - b_\ell}{K^{\ell-1}} + b_\ell \right). \end{aligned}$$

Similarly, for $\mathbb{P}(\widehat{\mathbf{S}}_{1j}^{(0)}(u) \geq \mu_C)$, define $t_{jC} := \mu_C - \mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u)$ for $j \neq 1$, by Bernstein's Lemma D.3.4,

$$\begin{aligned} \mathbb{P}\left(\widehat{\mathbf{S}}_{1j}^{(0)}(u) \geq \mu_C\right) &= \mathbb{P}\left(\widehat{\mathbf{S}}_{1j}^{(0)}(u) - \mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u) \geq t_{jC}\right) \\ &\leq \exp\left(-\frac{t_{jC}^2/2}{\text{Var}[\widehat{\mathbf{S}}_{1j}^{(0)}(u)] + (\mathbf{L}-1) \cdot t_{jC}/3}\right) \leq \exp\left(-\frac{3t_{jC}^2/(\mathbf{L}-1)}{6(\mathbf{L}-1) \cdot \mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u) + 2t_{jC}}\right) \\ &\leq \exp\left(-\frac{[(\mathbf{v})^{\mathbf{L}-1} - (1-\mathbf{v})^{\mathbf{L}-1}]^2}{(\mathbf{L}-1)^2 \cdot 2^{2\mathbf{L}+3}} \cdot \frac{\left[\sum_{\ell \in \mathcal{L}} (\ell-1) \left(\frac{a_\ell - b_\ell}{K^{\ell-1}}\right)\right]^2}{\sum_{\ell \in \mathcal{L}} (\ell-1) \left(\frac{a_\ell - b_\ell}{K^{\ell-1}} + b_\ell\right)}\right). \end{aligned}$$

The last two inequalities holds since $\text{Var}[\widehat{\mathbf{S}}_{1j}^{(0)}(u)] \leq (\mathbf{L}-1)^2 \mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u)$, and for sufficiently large N ,

$$\begin{aligned} t_{jC} := \mu_C - \mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u) &= \frac{1}{2} \sum_{\ell \in \mathcal{L}} (\ell-1) \cdot \left[\binom{\frac{\mathbf{v}N}{2K}}{\ell-1} - \binom{\frac{(1-\mathbf{v})N}{2K}}{\ell-1} \right] \frac{a_\ell - b_\ell}{2 \binom{N-1}{\ell-1}} \\ &\geq \frac{(\mathbf{v})^{\mathbf{L}-1} - (1-\mathbf{v})^{\mathbf{L}-1}}{2^{\mathbf{L}+2}} \sum_{\ell \in \mathcal{L}} (\ell-1) \cdot \frac{a_\ell - b_\ell}{K^{\ell-1}}, \end{aligned}$$

as well as the following bound on the expectation

$$\begin{aligned} 6\mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u) + 2t_{jC} &= 2\mu_C + (6\mathbf{L}-8)\mathbb{E}\widehat{\mathbf{S}}_{1j}^{(0)}(u) \\ &= \sum_{\ell \in \mathcal{L}} (\ell-1) \left\{ \left[\binom{\frac{\mathbf{v}N}{2K}}{\ell-1} + (6\mathbf{L}-7) \binom{\frac{(1-\mathbf{v})N}{2K}}{\ell-1} \right] \frac{a_\ell - b_\ell}{2 \binom{N-1}{\ell-1}} + 6(\mathbf{L}-1) \binom{\frac{N}{2K}}{\ell-1} \frac{b_\ell}{2 \binom{N-1}{\ell-1}} \right\} \\ &= \sum_{\ell \in \mathcal{L}} (\ell-1) \cdot \left(\frac{(\mathbf{v})^{\ell-1} + (6\mathbf{L}-7) \cdot (1-\mathbf{v})^{\ell-1}}{2^\ell} \cdot \frac{a_\ell - b_\ell}{K^{\ell-1}} + \frac{(6\mathbf{L}-6) \cdot b_\ell}{2^\ell K^{\ell-1}} \right) (1+o(1)) \\ &\leq \sum_{\ell \in \mathcal{L}} \frac{(\mathbf{v})^{\ell-1} + (6\mathbf{L}-7) \cdot (1-\mathbf{v})^{\ell-1}}{2^\ell} \cdot (\ell-1) \left(\frac{a_\ell - b_\ell}{K^{\ell-1}} + b_\ell \right) (1+o(1)) \\ &\leq \frac{3(\mathbf{L}-1)}{2} \sum_{\ell \in \mathcal{L}} (\ell-1) \left(\frac{a_\ell - b_\ell}{K^{\ell-1}} + b_\ell \right). \end{aligned}$$

□

C.3 Algorithm correctness for the binary case

We will show the correctness of Algorithm 4.1.1 and prove Theorem 4.1.6 in this section. Without loss of generality, we assume N is even to guarantee the existence of a binary partition of size $N/2$. The method to deal with the odd N case was discussed in Lemma 4.2.4. The analysis will mainly follow from the analysis in Section 4.5. We only detail the differences.

First, with α and β defined in (4.2.5), (4.2.4) becomes

$$\bar{\mathbf{A}} := \mathbb{E}\mathbf{A} = \begin{bmatrix} \alpha \mathbf{J}_{\frac{N}{2}} & \beta \mathbf{J}_{\frac{N}{2}} \\ \beta \mathbf{J}_{\frac{N}{2}} & \alpha \mathbf{J}_{\frac{N}{2}} \end{bmatrix} - \alpha \mathbf{I}_N, \quad (\text{C.3.1})$$

Second, let the index set be $I = \{i \in \mathcal{V} : \text{row}(i) \leq 20Ld\}$, as shown in (4.3.3). Let $\mathbf{u}_i$ (resp. $\bar{\mathbf{u}}_i$) denote the eigenvector associated to $\lambda_i(\mathbf{A}_I)$ (resp. $\lambda_i(\bar{\mathbf{A}})$) for $i = 1, 2$. Define two linear subspaces $\mathbf{U} := \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ and $\bar{\mathbf{U}} := \text{Span}\{\bar{\mathbf{u}}_1, \bar{\mathbf{u}}_2\}$, then the angle between $\mathbf{U}$ and $\bar{\mathbf{U}}$ is defined as $\sin \angle(\mathbf{U}, \bar{\mathbf{U}}) := \|\mathbf{P}_{\mathbf{U}} - \mathbf{P}_{\bar{\mathbf{U}}}\|$, where $\mathbf{P}_{\mathbf{U}}$ and $\mathbf{P}_{\bar{\mathbf{U}}}$ are the orthogonal projections onto $\mathbf{U}$ and $\bar{\mathbf{U}}$, respectively.

C.3.1 Proof of Lemma 4.4.4

Bound the angle between $\mathbf{U}$ and $\bar{\mathbf{U}}$

The strategy to bound the angle is similar to Section 4.5.1, except that we apply Davis-Kahan Theorem (Lemma D.2.5) here.

Define $\mathbf{E} := \mathbf{A} - \bar{\mathbf{A}}$ and its restriction on I , namely $\mathbf{E}_I := (\mathbf{A} - \bar{\mathbf{A}})_I = \mathbf{A}_I - \bar{\mathbf{A}}_I$, as well as $\boldsymbol{\delta} := \bar{\mathbf{A}}_I - \bar{\mathbf{A}}$. Then the deviation $\mathbf{A}_I - \bar{\mathbf{A}}$ is decomposed as

$$\mathbf{A}_I - \bar{\mathbf{A}} = (\mathbf{A}_I - \bar{\mathbf{A}}_I) + (\bar{\mathbf{A}}_I - \bar{\mathbf{A}}) = \mathbf{E}_I + \boldsymbol{\delta}.$$

Theorem 4.3.4 indicates $\|\mathbf{E}_I\| \leq C_3 \sqrt{d}$ with probability at least $1 - N^{-2}$ when taking $\tau = 20L, \theta = 3$, where C_3 is a constant depending only on L . Moreover, Lemma 4.5.6 shows that

the number of vertices with high degrees is relatively small. Consequently, an argument similar to Corollary 4.5.7 leads to the conclusion $\|\boldsymbol{\delta}\| \leq \sqrt{d}$ with high probability. Together with upper bounds for $\|\mathbf{E}_I\|$ and $\|\boldsymbol{\delta}\|$, Lemma C.3.1 shows that the angle between $\mathbf{U}$ and $\bar{\mathbf{U}}$ is relatively small with high probability.

Lemma C.3.1. *For any $c \in (0, 1)$, there exists a constant C_2 depending on L and c such that if*

$$\sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell) \geq C_2 \cdot 2^{L+2} \sqrt{d},$$

then $\sin \angle(\mathbf{U}, \bar{\mathbf{U}}) \leq c$ with probability $1 - N^{-2}$.

Proof of Lemma C.3.1. First, with probability $1 - N^{-2}$, we have

$$\|\mathbf{A}_I - \bar{\mathbf{A}}\| \leq \|\mathbf{E}_I\| + \|\boldsymbol{\delta}\| \leq (C_3 + 1) \sqrt{d}.$$

According to the definitions in (4.2.5), $\alpha \geq \beta$ and $\alpha = O(1/N)$, $\beta = O(1/N)$. Meanwhile, Lemma 4.2.3 shows that $|\lambda_2(\bar{\mathbf{A}})| = [-\alpha + (\alpha - \beta)N/2]$ and $|\lambda_3(\bar{\mathbf{A}})| = \alpha$. Then

$$\begin{aligned} |\lambda_2(\bar{\mathbf{A}})| - |\lambda_3(\bar{\mathbf{A}})| &= \frac{N}{2}(\alpha - \beta) - 2\alpha \geq \frac{3}{4} \cdot \frac{N}{2}(\alpha - \beta) = \frac{3N}{8} \sum_{\ell \in \mathcal{L}} \binom{\frac{N}{2} - 2}{\ell - 2} \frac{(a_\ell - b_\ell)}{\binom{N-1}{\ell-1}} \\ &\geq \frac{1}{4} \sum_{\ell \in \mathcal{L}} \frac{(\ell - 1)(a_\ell - b_\ell)}{2^{\ell-2}} \geq \frac{1}{2^L} \sum_{\ell \in \mathcal{L}} (\ell - 1)(a_\ell - b_\ell) \geq 4C_2 \sqrt{d}. \end{aligned}$$

Then for some large enough C_2 , the following condition for Davis-Kahan Theorem (Lemma D.2.5) is satisfied

$$\|\mathbf{A}_I - \bar{\mathbf{A}}\| \leq (1 - 1/\sqrt{2}) (|\lambda_2(\bar{\mathbf{A}})| - |\lambda_3(\bar{\mathbf{A}})|).$$

Then for any $c \in (0, 1)$, we can choose $C_2 = (C_3 + 1)/c$ such that

$$\|\mathbf{P}_\mathbf{U} - \mathbf{P}_{\bar{\mathbf{U}}}\| \leq \frac{2\|\mathbf{A}_I - \bar{\mathbf{A}}\|}{|\lambda_2(\bar{\mathbf{A}})| - |\lambda_3(\bar{\mathbf{A}})|} \leq \frac{2(C_3 + 1)\sqrt{d}}{4C_2\sqrt{d}} = \frac{c}{2} \leq c.$$

□

Now, we focus on the accuracy of Algorithm 4.4.5, once the conditions in Lemma C.3.1 are satisfied.

Lemma C.3.2 ([64, Lemma 23]). *If $\sin \angle(\bar{\mathbf{U}}, \mathbf{U}) \leq c \leq \frac{1}{4}$, there exists a unit vector $\mathbf{v} \in \mathbf{U}$ such that the angle between $\bar{\mathbf{u}}_2$ and $\mathbf{v}$ satisfies $\sin \angle(\bar{\mathbf{u}}_2, \mathbf{v}) \leq 2\sqrt{c}$.*

Proof of Lemma C.3.2. Recall that $\bar{\mathbf{U}} := \text{Span}\{\bar{\mathbf{u}}_1, \bar{\mathbf{u}}_2\}$, $\mathbf{U} := \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$, where

$$\begin{aligned}\bar{\mathbf{u}}_1 &= [\frac{1}{\sqrt{N}}, \dots, \frac{1}{\sqrt{N}}]^\top, \\ \bar{\mathbf{u}}_2 &= [\frac{1}{\sqrt{N}}, \dots, \frac{1}{\sqrt{N}}, -\frac{1}{\sqrt{N}}, \dots, -\frac{1}{\sqrt{N}}]^\top.\end{aligned}$$

By assumption, the angle between $\bar{\mathbf{U}}$ and $\mathbf{U}$ is bounded by

$$\sin \angle(\bar{\mathbf{U}}, \mathbf{U}) := \|\mathbf{P}_{\bar{\mathbf{U}}} - \mathbf{P}_{\mathbf{U}}\| \leq c \leq \frac{1}{4},$$

Define $\mathbf{v}_k := \mathbf{P}_{\mathbf{U}} \bar{\mathbf{u}}_k$ and $\bar{\mathbf{x}}_k = \bar{\mathbf{u}}_k - \mathbf{v}_k$ for $k = 1, 2$. Note that $\mathbf{v}_k \in \mathbf{U}$ and $\bar{\mathbf{u}}_k \in \bar{\mathbf{U}}$, then

$$\|\bar{\mathbf{x}}_k\|_2 = \|\bar{\mathbf{u}}_k - \mathbf{v}_k\|_2 \leq \|\mathbf{P}_{\bar{\mathbf{U}}} - \mathbf{P}_{\mathbf{U}}\| \cdot \|\bar{\mathbf{u}}_k\|_2 \leq c.$$

Meanwhile, $\mathbf{v}_k = \bar{\mathbf{u}}_k - \bar{\mathbf{x}}_k$, then it implies $\|\mathbf{v}_k\|_2 \geq 1 - c$. Since $\bar{\mathbf{u}}_1^\top \bar{\mathbf{u}}_2 = 0$, we have

$$|\mathbf{v}_1^\top \bar{\mathbf{u}}_2| = |\bar{\mathbf{u}}_1^\top \bar{\mathbf{u}}_2 - \bar{\mathbf{x}}_1^\top \bar{\mathbf{u}}_2| \leq |\bar{\mathbf{u}}_1^\top \bar{\mathbf{u}}_2| + |\bar{\mathbf{x}}_1^\top \bar{\mathbf{u}}_2| \leq c.$$

Together with the fact that $|\mathbf{v}_1^\top \bar{\mathbf{x}}_2| \leq c$, $|\mathbf{v}_2^\top \bar{\mathbf{x}}_1| \leq c$, we have

$$\begin{aligned}|\mathbf{v}_1^\top \mathbf{v}_2| &= |\bar{\mathbf{u}}_1^\top \bar{\mathbf{u}}_2 - \bar{\mathbf{x}}_1^\top \mathbf{v}_2 - \mathbf{v}_1^\top \bar{\mathbf{x}}_2 + \bar{\mathbf{x}}_1^\top \bar{\mathbf{x}}_2| \\ &\leq |\bar{\mathbf{x}}_1^\top \mathbf{v}_2| + |\mathbf{v}_1^\top \bar{\mathbf{x}}_2| + |\bar{\mathbf{x}}_1^\top \bar{\mathbf{x}}_2| \leq 2c + c^2.\end{aligned}$$

Note that $|\bar{\mathbf{x}}_k^\top \bar{\mathbf{u}}_k| \leq \|\bar{\mathbf{x}}_k\|_2 \leq c$ for $k = 1, 2$, then

$$|\mathbf{v}_k^\top \bar{\mathbf{u}}_k| = |\bar{\mathbf{u}}_k^\top \bar{\mathbf{u}}_k - \bar{\mathbf{x}}_k^\top \bar{\mathbf{u}}_k| \geq |\bar{\mathbf{u}}_k^\top \bar{\mathbf{u}}_k| - |\bar{\mathbf{x}}_k^\top \bar{\mathbf{u}}_k| \geq 1 - c.$$

Let $\mathbf{v}_\perp = \mathbf{v}_2 - \mathbf{v}_1 \frac{\mathbf{v}_1^\top \mathbf{v}_2}{\|\mathbf{v}_1\|_2^2} \in \mathbf{U}$, using the estimates above, we then have

$$\begin{aligned} |\mathbf{v}_\perp^\top \bar{\mathbf{u}}_2| &= |\mathbf{v}_2^\top \bar{\mathbf{u}}_2 - \frac{\mathbf{v}_1^\top \mathbf{v}_2}{\|\mathbf{v}_1\|_2^2} \mathbf{v}_1^\top \bar{\mathbf{u}}_2| \\ &\geq |\mathbf{v}_2^\top \bar{\mathbf{u}}_2| - \left| \frac{\mathbf{v}_1^\top \mathbf{v}_2}{\|\mathbf{v}_1\|_2^2} \mathbf{v}_1^\top \bar{\mathbf{u}}_2 \right| \\ &\geq (1 - c) - \frac{(2c + c^2)c}{(1 - c)^2} \geq 1 - 2c. \end{aligned}$$

Define the unit vector $\mathbf{v} := \frac{\mathbf{v}_\perp}{\|\mathbf{v}_\perp\|_2}$, then we have

$$\sin \angle(\mathbf{v}, \bar{\mathbf{u}}_2) = \sqrt{1 - |\bar{\mathbf{u}}_2^\top \mathbf{v}|^2} \leq \sqrt{1 - (1 - 2c)^2} = 2\sqrt{c},$$

which is the desired result. $\square$

The vector $\mathbf{v}$ constructed by Algorithm 4.4.5 is the unit vector perpendicular to $\mathbf{P}_\mathbf{U} \mathbf{1}_N$, where $\mathbf{P}_\mathbf{U} \mathbf{1}_N$ is the projection of all-ones vector onto $\mathbf{U}$. Lemma C.3.1 and Lemma C.3.2 together give the following corollary.

Corollary C.3.3. *For any $c \in (0, 1)$, there exists a unit vector $\mathbf{v} \in \mathbf{U}$ such that the angle between $\bar{\mathbf{u}}_2$ and $\mathbf{v}$ satisfies $\sin \angle(\bar{\mathbf{u}}_2, \mathbf{v}) \leq c < 1$ with probability $1 - O(e^{-N})$.*

Proof. For any $c \in (0, 1)$, we could choose constants C_2, C_3 in Lemma C.3.1 such that $\sin \angle(\bar{\mathbf{U}}, \mathbf{U}) \leq \frac{c^2}{4} < 1$. Then by Lemma C.3.2, we construct $\mathbf{v}$ such that $\sin \angle(\bar{\mathbf{u}}_2, \mathbf{v}) \leq c$. $\square$

The proof of Lemma 4.4.4 is then completed by choosing C_2, C_3 in Lemma C.3.1 such that $c \leq \frac{1}{4}$, and applying Lemma 4.5.12 with $K = 2$.

C.3.2 Proof of Lemma 4.4.5

The proof strategy is similar to Section 4.5.2 and Section 4.5.3. In Algorithm 4.1.1, we first color the hyperedges with red and blue with equal probability. By running Algorithm 4.4.6 on the red graph, we obtain a $\mathbf{v}$ -correct partition $\widehat{\mathcal{V}}_1, \widehat{\mathcal{V}}_2$ of $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$, i.e., $|\mathcal{V}_k \cap \widehat{\mathcal{V}}_k| \geq \mathbf{v}N/2$ for $k = 1, 2$. We condition on this event in the rest of the proof. Meanwhile, we shall also condition the event that the maximum red degree of a vertex is at most $\log^2(N)$ with high probability, which is a simple consequence of concentration by Bernstein's inequality (Lemma D.3.4).

Similarly, we consider the probability of a hyperedge $e = \{i_1, \dots, i_\ell\}$ being blue conditioning on the event that e is not a red hyperedge in each underlying ℓ -uniform hypergraph separately. If vertices $i_1, \dots, i_\ell$ are all from the same true cluster, then the probability is ψ_ℓ , otherwise ϕ_ℓ , where ψ_ℓ and ϕ_ℓ are defined in (4.5.28) and (4.5.29), and the presence of those hyperedges are represented by random variables $\zeta_e^{(a_\ell)} \sim \text{Bernoulli}(\psi_\ell)$, $\xi_e^{(b_\ell)} \sim \text{Bernoulli}(\phi_\ell)$, respectively.

Following a similar argument in Section 4.5.2, the row sum of u can be written as

$$\widehat{\mathbf{S}}_{kj}^{(0)}(u) := \sum_{\ell \in \mathcal{L}} (\ell - 1) \cdot \left\{ \sum_{e \in \mathcal{E}_{k,j}^{(a_\ell)}} \zeta_e^{(a_\ell)} + \sum_{e \in \mathcal{E}_{k,j}^{(b_\ell)}} \xi_e^{(b_\ell)} \right\}, \quad u \in \mathcal{V}_k,$$

where $\mathcal{E}_{k,j}^{(a_\ell)} := \mathcal{E}_\ell([\mathcal{V}_k]^1, [\mathcal{V}_k \cap \widehat{\mathcal{V}}_j^{(0)}]^{\ell-1})$ denotes the set of ℓ -hyperedges with 1 vertex from $[\mathcal{V}_k]^1$ and the other $\ell - 1$ vertices from $[\mathcal{V}_k \cap \widehat{\mathcal{V}}_j^{(0)}]^{\ell-1}$, while

$$\mathcal{E}_{k,j}^{(b_\ell)} := \mathcal{E}_\ell([\mathcal{V}_k]^1, [\widehat{\mathcal{V}}_j^{(0)}]^{\ell-1} \setminus [\mathcal{V}_k \cap \widehat{\mathcal{V}}_j^{(0)}]^{\ell-1})$$

denotes the set of ℓ -hyperedges with 1 vertex in $[\mathcal{V}_k]^1$ while the remaining $\ell - 1$ vertices in

$[\widehat{\mathcal{V}}_j^{(0)}]^{\ell-1} \setminus [\mathcal{V}_k \cap \widehat{\mathcal{V}}_j^{(0)}]^{\ell-1}$, with their cardinalities

$$|\mathcal{E}_{k,j}^{(a_\ell)}| \leq \binom{|\mathcal{V}_k \cap \widehat{\mathcal{V}}_j^{(0)}|}{\ell-1}, \quad |\mathcal{E}_{k,j}^{(b_\ell)}| \leq \left[\binom{|\widehat{\mathcal{V}}_j^{(0)}|}{\ell-1} - \binom{|\mathcal{V}_k \cap \widehat{\mathcal{V}}_j^{(0)}|}{\ell-1} \right].$$

According to the facts $|\mathcal{V}_k \cap \widehat{\mathcal{V}}_k^{(0)}| \geq \nu N/2$, $|\mathcal{V}_k| = N/2$, $|\widehat{\mathcal{V}}_k^{(0)}| = N/2$ for $k = 1, 2$, we have

$$|\mathcal{E}_{k,k}^{(a_\ell)}| \geq \binom{\frac{\nu N}{2}}{\ell-1}, \quad |\mathcal{E}_{k,j}^{(a_\ell)}| \leq \binom{\frac{(1-\nu)N}{2}}{\ell-1}, \quad j \neq k.$$

To simplify the calculation, we take the lower and upper bound of $|\mathcal{E}_{k,k}^{(a_\ell)}|$ and $|\mathcal{E}_{k,j}^{(a_\ell)}| (j \neq k)$ respectively. Taking expectation with respect to $\zeta_e^{(a_\ell)}$ and $\xi_e^{(b_\ell)}$, for any $u \in \mathcal{V}_k$, we have

$$\begin{aligned} \mathbb{E}\widehat{\mathbf{S}}_{kk}^{(0)}(u) &= \sum_{\ell \in \mathcal{L}} (\ell-1) \cdot \left[\binom{\frac{\nu N}{2}}{\ell-1} (\psi_\ell - \phi_\ell) + \binom{\frac{N}{2}}{\ell-1} \phi_\ell \right], \\ \mathbb{E}\widehat{\mathbf{S}}_{kj}^{(0)}(u) &= \sum_{\ell \in \mathcal{L}} (\ell-1) \cdot \left[\binom{\frac{(1-\nu)N}{2}}{\ell-1} (\psi_\ell - \phi_\ell) + \binom{\frac{N}{2}}{\ell-1} \phi_\ell \right], \quad j \neq k. \end{aligned}$$

By assumptions in Theorem 4.1.7, $\mathbb{E}\widehat{\mathbf{S}}_{kk}^{(0)}(u) - \mathbb{E}\widehat{\mathbf{S}}_{kj}^{(0)}(u) = \Omega(1)$. We define

$$\mu_C := \frac{1}{2} \sum_{\ell \in \mathcal{L}} (\ell-1) \cdot \left\{ \left[\binom{\frac{\nu N}{2}}{\ell-1} + \binom{\frac{(1-\nu)N}{2}}{\ell-1} \right] (\psi_\ell - \phi_\ell) + 2 \binom{\frac{N}{2}}{\ell-1} \phi_\ell \right\}.$$

After Algorithm 4.4.6, if a vertex $u \in \mathcal{V}_k$ is mislabelled, one of the following events must happen

- $\widehat{\mathbf{S}}_{kk}^{(0)}(u) \leq \mu_C$, meaning that u fails to have enough neighbors in $\widehat{\mathcal{V}}_k^{(0)}$ to be correctly labeled as k ;
- $\widehat{\mathbf{S}}_{kj}^{(0)}(u) \geq \mu_C$, for some $j \neq k$, meaning that u survived Algorithm 4.4.6 without being corrected.

By an argument similar to Lemma 4.5.16, we can prove that

$$\hat{\rho}_1^{(0)} = \mathbb{P}\left(\hat{\mathbf{S}}_{kk}^{(0)}(u) \leq \mu_C\right) \leq \rho, \quad \hat{\rho}_2^{(0)} = \mathbb{P}\left(\hat{\mathbf{S}}_{kj}^{(0)}(u) \geq \mu_C\right) \leq \rho,$$

where $\rho = \exp(-C_{\mathcal{L}}(2, \mathbf{v}) \cdot \text{SNR}_{\mathcal{L}}(2))$ and

$$C_{\mathcal{L}}(2, \mathbf{v}) := \frac{[(\mathbf{v})^{L-1} - (1-\mathbf{v})^{L-1}]^2}{8(L-1)^2}, \quad \text{SNR}_{\mathcal{L}}(2) := \frac{[\sum_{\ell \in \mathcal{L}} (\ell-1) \cdot (a_{\ell} - b_{\ell}) 2^{-\ell+1}]^2}{\sum_{\ell \in \mathcal{L}} (\ell-1) ((a_{\ell} - b_{\ell}) 2^{-\ell+1} + b_{\ell})}.$$

As a result, the probability that either of those events happened is bounded by ρ . The number of mislabeled vertices in $\mathcal{V}_1'$ after Algorithm 4.4.3 is at most

$$\hat{\mathbf{R}}_k^{(0)} = \sum_{v \in \mathcal{V}_k \setminus \hat{\mathcal{V}}_k} \Gamma_v + \sum_{v \in \mathcal{V}_k \cap \hat{\mathcal{V}}_k} \Lambda_v,$$

where Γ_v (resp. Λ_v) are i.i.d indicator random variables with mean $\hat{\rho}_1^{(0)}$ (resp. $\hat{\rho}_2^{(0)}$). Then

$$\mathbb{E} \hat{\mathbf{R}}_k^{(0)} \leq \frac{N}{2} \hat{\rho}_1^{(0)} + \frac{(1-\mathbf{v})N}{2} \hat{\rho}_2^{(0)} = (1-\mathbf{v}/2)N\rho.$$

where $\mathbf{v}$ is the correctness after Algorithm 4.4.2. Let $t_k := (1+\mathbf{v}/2)N\rho$, then by Chernoff Lemma D.3.3,

$$\mathbb{P}\left(\hat{\mathbf{R}}_k^{(0)} \geq N\rho\right) = \mathbb{P}\left[\hat{\mathbf{R}}_k^{(0)} - (1-\mathbf{v}/2)N\rho \geq t_k\right] \leq \mathbb{P}\left(\hat{\mathbf{R}}_k^{(0)} - \mathbb{E} \hat{\mathbf{R}}_k^{(0)} \geq t_k\right) \leq e^{-ct_k} = O(e^{-N\rho}),$$

which means that with probability $1 - O(e^{-N\rho})$, the fraction of mislabeled vertices in $\mathcal{V}_k'$ is smaller than 2ρ , i.e., the correctness is at least $\gamma := \max\{\mathbf{v}, 1 - 2\rho\}$.

Appendix D

Technical Lemmas

D.1 Calculus

Lemma D.1.1. For $a, b > 0$, define the function $g(a, b) := \max_{t \in [0, 1]} f(t)$, where

$$f(t) = ta + (1 - t)b - a^t b^{1-t}.$$

When $a \asymp b$ but $a/b \neq 1$, $g(a, b) \asymp a$. Otherwise when $a \gg b$, $g(a, b) = a \cdot (1 - o(1))$.

Proof of Lemma D.1.1. By basic calculus, the maximizer t^* is unique, and it satisfies

$$\left. \frac{df}{dt} \right|_{t=t^*} = a - b - b(a/b)^{t^*} \log(a/b) = 0 \implies t^* = \frac{1}{\log(a/b)} \log\left(\frac{a/b - 1}{\log(a/b)}\right).$$

Consequently,

$$\begin{aligned} g(a, b) &= f(t^*) = (a - b)t^* + b(1 - (a/b)^{t^*}) \\ &= \frac{a - b}{\log(a/b)} \left(\log(a/b - 1) - \log \log(a/b) \right) + b - \frac{a - b}{\log(a/b)} \\ &= a \cdot (1 - b/a) \left(\frac{\log(a/b - 1)}{\log(a/b)} - \frac{\log \log(a/b) - 1}{\log(a/b)} \right) + b. \end{aligned}$$

The first part follows obviously. When $a \gg b$, then $b/a = o(1)$, $\log(a/b - 1)/\log(a/b) = 1 - o(1)$, $(\log \log(a/b) - 1)/\log(a/b) = o(1)$, thus $g(a, b) = a(1 - o(1))$. $\square$

Lemma D.1.2. *The function $f : \mathbb{N}^+ \mapsto \mathbb{N}^+$, defined by $f(N) = \binom{N}{\ell}$, is convex for any $\ell \geq 2$. In particular, if $N_1 + N_2 + \cdots + N_K = N$ for some $K \in \mathbb{N}^+$, then $f(N_1) + f(N_2) + \cdots + f(N_K) \leq f(N)$.*

Proof. By induction, it suffices to check when $K = 2$, which can be verified by direction calculation $\binom{N_1}{\ell} + \binom{N_2}{\ell} \leq \binom{N_1+N_2}{\ell}$. $\square$

Lemma D.1.3. *For integers $n, k \geq 1$, we have*

$$\begin{aligned} \log(n!) &= n \log(n) - n + \frac{1}{2} \log(2\pi n) + O(n^{-1}) \\ \log \binom{n}{k} &= \frac{1}{2} \log \frac{n}{2\pi k(n-k)} + n \log(n) - k \log(k) - (n-k) \log(n-k) + O\left(\frac{1}{n} + \frac{1}{k} + \frac{1}{n-k}\right) \\ \log \binom{n}{k} &= k \log \left(\frac{n}{k} - 1\right) - \frac{1}{2} \log(2\pi k) + o(k^{-1}), \text{ for } k = \omega(1) \text{ and } \frac{k}{n} = o(1). \end{aligned}$$

Moreover, for any $1 \leq k \leq \sqrt{n}$, we have

$$\frac{n^k}{4 \cdot k!} \leq \binom{n}{k} \leq \frac{n^k}{k!}, \quad \log \binom{n}{k} \geq k \log \left(\frac{en}{k}\right) - \frac{1}{2} \log(k) - \frac{1}{12k} - \log(4\sqrt{2\pi}).$$

Proof of Lemma D.1.3. According to Stirling's series [80], for any $n \geq 1$,

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}} n! \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}.$$

Then the following asymptotic expansion holds:

$$\left| \log(n!) - \left(\frac{1}{2} \log(2\pi n) + n \log(n) - n \right) \right| \leq \frac{1}{12n}.$$

Thus for any $k \geq 1$, we have $\log \binom{n}{k} = \log(n!) - \log(k!) - \log((n-k)!)$, as well as

$$\left| \log \binom{n}{k} - \left(\frac{1}{2} \log \frac{n}{2\pi k(n-k)} + n \log(n) - k \log(k) - (n-k) \log(n-k) \right) \right| \leq \frac{1}{12} \left(\frac{1}{n} + \frac{1}{k} + \frac{1}{n-k} \right).$$

When $k = \omega(1)$ but $\frac{k}{n} = o(1)$, we have $\log(1 - \frac{k}{n}) = -\frac{k}{n} + O(N^{-2})$, and

$$\begin{aligned}\log \binom{n}{k} &= -\frac{1}{2} \log(2\pi k) - \frac{1}{2} \log \left(1 - \frac{k}{n}\right) + n \log(n) - k \log(k) \\ &\quad - (n-k) \log(n) - (n-k) \log \left(1 - \frac{k}{n}\right) + o(1) \\ &= k \log(n) - k \log(k) + k + o(k).\end{aligned}$$

At the same time, we write

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} = \frac{n^k}{k!} \cdot \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) \geq \frac{n^k}{k!} \left(1 - \frac{k-1}{n}\right)^{k-1}$$

The upper bound is trivial. For the lower bound, let $f(x) = (1 - x/n)^x$ denote the function. It is easy to see that $f(x)$ is decreasing with respect to x , and

$$\left(1 - \frac{k-1}{n}\right)^{k-1} \geq \left(1 - \frac{1}{\sqrt{n}}\right)^{\sqrt{n}}.$$

Let $g(t) = (1 - 1/t)^t$ and we know that $g(t)$ is increasing when $t \geq 1$, hence $g(t) \geq g(2) = 1/4$ for any $t \geq 2$ and the desired lower bound follows. Then for any $k \leq \sqrt{n}$, we have

$$\begin{aligned}\log \binom{n}{k} &\geq \log \frac{n^k}{4 \cdot k!} \\ &\geq \log \left(\frac{n^k}{4 \cdot \sqrt{2\pi k} \left(\frac{k}{e}\right)^k e^{\frac{1}{12k}}} \right) = k \log \left(\frac{en}{k} \right) - \frac{1}{2} \log(k) - \frac{1}{12k} - \log(4\sqrt{2\pi}).\end{aligned}$$

□

D.2 Matrix inequalities

Lemma D.2.1 (Weyl's inequality, [203]). *Let $\mathbf{A}, \mathbf{E} \in \mathbb{R}^{m \times n}$ be two real $m \times n$ matrices, then $|\sigma_i(\mathbf{A} + \mathbf{E}) - \sigma_i(\mathbf{A})| \leq \|\mathbf{E}\|$ for every $1 \leq i \leq \min\{m, n\}$. Furthermore, if $m = n$ and $\mathbf{A}, \mathbf{E} \in \mathbb{R}^{n \times n}$ are real symmetric, then $|\lambda_i(\mathbf{A} + \mathbf{E}) - \lambda_i(\mathbf{A})| \leq \|\mathbf{E}\|$ for all $1 \leq i \leq n$.*

Lemma D.2.2 (Eckart–Young–Mirsky Theorem [90]). *Suppose that the matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ ($m \leq n$) admits the singular value decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, where $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_m] \in \mathbb{R}^{m \times m}$, $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_n] \in \mathbb{R}^{n \times n}$ and $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$ contains diagonal elements $\sigma_1 \geq \dots \geq \sigma_m$. Let $\mathbf{A}^{(K)} = \sum_{k=1}^K \sigma_k \mathbf{u}_k \mathbf{v}_k^\top$ be the rank- K approximation of $\mathbf{A}$. Then $\|\mathbf{A} - \mathbf{A}^{(K)}\|_2 = \sigma_{K+1} \leq \|\mathbf{A} - \mathbf{B}\|_2$ for any matrix $\mathbf{B}$ with $\text{rank}(\mathbf{B}) = K$.*

Lemma D.2.3 (Davis-Kahan Theorem, [74, 208]). *Let $\mathbf{A}$ and $\mathbf{A}^*$ be $n \times n$ symmetric matrices with eigenpairs $\{(\lambda_i, \mathbf{u}_i)\}_{i=1}^{|\mathcal{V}|}$ and $\{(\lambda_i^*, \mathbf{u}_i^*)\}_{i=1}^{|\mathcal{V}|}$ respectively, where eigenvalues are sorted in the decreasing order. We take $\lambda_0 := +\infty$ and $\lambda_{n+1} := -\infty$. For fixed $j \in \mathcal{V}$, assume that $\delta = \min\{\lambda_{j-1} - \lambda_j, \lambda_j - \lambda_{j+1}\} > 0$. Then for eigenpairs of $(\lambda_j, \mathbf{u}_j)$ of $\mathbf{A}$ and $(\lambda_j^*, \mathbf{u}_j^*)$ of $\mathbf{A}^*$, if $\langle \mathbf{u}_j, \mathbf{u}_j^* \rangle \geq 0$, one has*

$$\|\mathbf{u}_j - \mathbf{u}_j^*\|_2 \leq \frac{\sqrt{8}\|\mathbf{A} - \mathbf{A}^*\|}{\delta}.$$

Lemma D.2.4 (Generalized Davis-Kahan Theorem, [91, 78]). *Consider the generalized eigenvalue problem $\mathbf{A}\mathbf{u} = \lambda\mathbf{D}\mathbf{u}$ where $\mathbf{A} \in \mathbb{R}^{n \times n}$ being symmetric and $\mathbf{D} \in \mathbb{R}^{n \times n}$ being diagonal. Let $\mathbf{D}^{-1} \in \mathbb{R}^{n \times n}$ denote the Moore-Penrose inverse of $\mathbf{D}$, that is $(\mathbf{D}^{-1})_{vv} = (\mathbf{D}_v)^{-1}$ if $\mathbf{D}_v \neq 0$, otherwise $(\mathbf{D}^{-1})_{vv} = 0$. Then the matrix $\mathbf{D}^{-1}\mathbf{A} \in \mathbb{R}^{n \times n}$ is diagonalizable, which admits the decomposition*

$$\mathbf{D}^{-1}\mathbf{A} = \sum_{i=1}^{|\mathcal{V}|} \lambda_i \mathbf{u}_i \mathbf{u}_i^\top = \begin{bmatrix} \mathbf{U} & \mathbf{U}_\perp \end{bmatrix} \begin{bmatrix} \Lambda & \mathbf{0} \\ \mathbf{0} & \Lambda_\perp \end{bmatrix} \begin{bmatrix} \mathbf{U}^\top \\ \mathbf{U}_\perp^\top \end{bmatrix}$$

Let $\mathbf{P} = \mathbf{U}_\perp \mathbf{U}_\perp^\top = \mathbf{I} - \mathbf{U}\mathbf{U}^\top$ denote the projection onto the subspace orthogonal to the space spanned by columns of $\mathbf{U}$. Let $\delta = \min_i |(\Lambda_\perp)_{ii} - \hat{\lambda}|$ denote the absolute separation of some $\hat{\lambda}$ from $\Lambda_\perp$, then for any vector $\hat{\mathbf{u}}$,

$$\|\mathbf{P}\hat{\mathbf{u}}\| \leq \frac{\|(\mathbf{D}^{-1}\mathbf{A} - \hat{\lambda}\mathbf{I})\hat{\mathbf{u}}\|}{\delta}.$$

Proof of Lemma D.2.4. First, note that

$$(\mathbf{D}^{-1}\mathbf{A} - \hat{\lambda}\mathbf{I})\hat{\mathbf{u}} = \begin{bmatrix} \mathbf{U} & \mathbf{U}_\perp \end{bmatrix} \begin{bmatrix} \Lambda - \hat{\lambda}\mathbf{I} & \mathbf{0} \\ \mathbf{0} & \Lambda_\perp - \hat{\lambda}\mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{U}^\top \hat{\mathbf{u}} \\ \mathbf{U}_\perp^\top \hat{\mathbf{u}}, \end{bmatrix}$$

thus

$$\mathbf{U}_\perp^\top \hat{\mathbf{u}} = (\Lambda_\perp - \hat{\lambda}\mathbf{I})^{-1} \mathbf{U}_\perp^\top (\mathbf{D}^{-1}\mathbf{A} - \hat{\lambda}\mathbf{I})\hat{\mathbf{u}}$$

Note that columns of $\mathbf{U}_\perp$ are pairwise orthogonal, therefore

$$\begin{aligned} \|\mathbf{P}\hat{\mathbf{u}}\| &= \|\mathbf{U}_\perp \mathbf{U}_\perp^\top \hat{\mathbf{u}}\| = \|\mathbf{U}_\perp (\Lambda_\perp - \hat{\lambda}\mathbf{I})^{-1} \mathbf{U}_\perp^\top (\mathbf{D}^{-1}\mathbf{A} - \hat{\lambda}\mathbf{I})\hat{\mathbf{u}}\| \\ &\leq \frac{\|\mathbf{U}_\perp \mathbf{U}_\perp^\top\| \cdot \|(\mathbf{D}^{-1}\mathbf{A} - \hat{\lambda}\mathbf{I})\hat{\mathbf{u}}\|}{\delta} \leq \frac{\|(\mathbf{D}^{-1}\mathbf{A} - \hat{\lambda}\mathbf{I})\hat{\mathbf{u}}\|}{\delta}. \end{aligned}$$

□

Lemma D.2.5 (Davis-Kahan's $\sin \Theta$ Theorem, [58, Theorem 2.7], [74, 208]). *Let $\bar{\mathbf{M}}$ and $\mathbf{M} = \bar{\mathbf{M}} + \mathbf{E}$ be two real symmetric $N \times N$ matrices, with eigenvalue decompositions given respectively by*

$$\begin{aligned} \bar{\mathbf{M}} &= \sum_{i=1}^N \bar{\lambda}_i \bar{\mathbf{u}}_i \bar{\mathbf{u}}_i^\top = \begin{bmatrix} \bar{\mathbf{U}} & \bar{\mathbf{U}}_\perp \end{bmatrix} \begin{bmatrix} \bar{\Lambda} & \mathbf{0} \\ \mathbf{0} & \bar{\Lambda}_\perp \end{bmatrix} \begin{bmatrix} \bar{\mathbf{U}}^\top \\ \bar{\mathbf{U}}_\perp^\top \end{bmatrix}, \\ \mathbf{M} &= \sum_{i=1}^N \lambda_i \mathbf{u}_i \mathbf{u}_i^\top = \begin{bmatrix} \mathbf{U} & \mathbf{U}_\perp \end{bmatrix} \begin{bmatrix} \Lambda & \mathbf{0} \\ \mathbf{0} & \Lambda_\perp \end{bmatrix} \begin{bmatrix} \mathbf{U}^\top \\ \mathbf{U}_\perp^\top \end{bmatrix}. \end{aligned}$$

Here, $\{\bar{\lambda}_i\}_{i=1}^N$ (resp. $\{\lambda_i\}_{i=1}^N$) stand for the eigenvalues of $\bar{\mathbf{M}}$ (resp. $\mathbf{M}$), and $\bar{\mathbf{u}}_i$ (resp. $\mathbf{u}_i$) denotes

the eigenvector associated $\bar{\lambda}_i$ (resp. λ_i). Additionally, for some fixed integer $r \in [N]$, we denote

$$\begin{aligned}\bar{\Lambda} &:= \text{diag}\{\bar{\lambda}_1, \dots, \bar{\lambda}_r\}, \quad \bar{\Lambda}_\perp := \text{diag}\{\bar{\lambda}_{r+1}, \dots, \bar{\lambda}_N\}, \\ \bar{\mathbf{U}} &:= [\bar{\mathbf{u}}_1, \dots, \bar{\mathbf{u}}_r] \in \mathbb{R}^{N \times r}, \quad \bar{\mathbf{U}}_\perp := [\bar{\mathbf{u}}_{r+1}, \dots, \bar{\mathbf{u}}_N] \in \mathbb{R}^{N \times (N-r)}.\end{aligned}$$

The matrices $\Lambda, \Lambda_\perp, \mathbf{U}, \mathbf{U}_\perp$ are defined analogously. Assume that

$$\text{eigenvalues}(\bar{\Lambda}) \subseteq [\alpha, \beta], \quad \text{eigenvalues}(\bar{\Lambda}_\perp) \subseteq (-\infty, \alpha - \Delta] \cup [\beta + \Delta, \infty), \quad \alpha, \beta \in \mathbb{R}, \Delta > 0,$$

and the projection matrices are given by $\mathbf{P}_\mathbf{U} := \mathbf{U}\mathbf{U}^\top$, $\mathbf{P}_{\bar{\mathbf{U}}} := \bar{\mathbf{U}}\bar{\mathbf{U}}^\top$, then one has $\|\mathbf{P}_\mathbf{U} - \mathbf{P}_{\bar{\mathbf{U}}}\| \leq (2\|\mathbf{E}\|/\Delta)$. In particular, suppose that $|\bar{\lambda}_1| \geq |\bar{\lambda}_2| \geq \dots \geq |\bar{\lambda}_r| \geq |\bar{\lambda}_{r+1}| \geq \dots \geq |\bar{\lambda}_N|$ (resp. $|\lambda_1| \geq \dots \geq |\lambda_N|$). If $\|\mathbf{E}\| \leq (1 - 1/\sqrt{2})(|\bar{\lambda}|_r - |\bar{\lambda}|_{r+1})$, then one has

$$\|\mathbf{P}_\mathbf{U} - \mathbf{P}_{\bar{\mathbf{U}}}\| \leq \frac{2\|\mathbf{E}\|}{|\bar{\lambda}_r| - |\bar{\lambda}_{r+1}|}.$$

Lemma D.2.6 (Wedin's $\sin \Theta$ Theorem, [58, Theorem 2.9], [201, 208]). Let $\bar{\mathbf{M}}$ and $\mathbf{M} = \bar{\mathbf{M}} + \mathbf{E}$ be two $N_1 \times N_2$ real matrices and $N_1 \geq N_2$, with SVDs given respectively by

$$\begin{aligned}\bar{\mathbf{M}} &= \sum_{i=1}^{N_1} \bar{\sigma}_i \bar{\mathbf{u}}_i \bar{\mathbf{v}}_i^\top = \begin{bmatrix} \bar{\mathbf{U}} & \bar{\mathbf{U}}_\perp \end{bmatrix} \begin{bmatrix} \bar{\Sigma} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \bar{\Sigma}_\perp & \mathbf{0} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{V}}^\top \\ \bar{\mathbf{V}}_\perp^\top \end{bmatrix} \\ \mathbf{M} &= \sum_{i=1}^{N_1} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top = \begin{bmatrix} \mathbf{U} & \mathbf{U}_\perp \end{bmatrix} \begin{bmatrix} \Sigma & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \Sigma_\perp & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{V}^\top \\ \mathbf{V}_\perp^\top \end{bmatrix}.\end{aligned}$$

Here, $\bar{\sigma}_1 \geq \dots \geq \bar{\sigma}_{N_1}$ (resp. $\sigma_1 \geq \dots \geq \sigma_{N_1}$) stand for the singular values of $\bar{\mathbf{M}}$ (resp. $\mathbf{M}$), $\bar{\mathbf{u}}_i$ (resp. $\mathbf{u}_i$) denotes the left singular vector associated with the singular value $\bar{\sigma}_i$ (resp. σ_i), and $\bar{\mathbf{v}}_i$ (resp. $\mathbf{v}_i$) denotes the right singular vector associated with the singular value $\bar{\sigma}_i$ (resp. σ_i). In addition,

for any fixed integer $r \in [N_1]$, we denote

$$\begin{aligned}\Sigma &:= \text{diag}\{\sigma_1, \dots, \sigma_r\}, & \Sigma_\perp &:= \text{diag}\{\sigma_{r+1}, \dots, \sigma_{N_1}\}, \\ \mathbf{U} &:= [\mathbf{u}_1, \dots, \mathbf{u}_r] \in \mathbb{R}^{N_1 \times r}, & \mathbf{U}_\perp &:= [\mathbf{u}_{r+1}, \dots, \mathbf{u}_{N_1}] \in \mathbb{R}^{N_1 \times (N_1 - r)}, \\ \mathbf{V} &:= [\mathbf{v}_1, \dots, \mathbf{v}_r] \in \mathbb{R}^{N_2 \times r}, & \mathbf{V}_\perp &:= [\mathbf{v}_{r+1}, \dots, \mathbf{v}_{N_2}] \in \mathbb{R}^{N_2 \times (N_2 - r)}.\end{aligned}$$

The matrices $\bar{\Sigma}$, $\bar{\Sigma}_\perp$, $\bar{\mathbf{U}}$, $\bar{\mathbf{U}}_\perp$, $\bar{\mathbf{V}}$, $\bar{\mathbf{V}}_\perp$ are defined analogously. If $\mathbf{E} = \mathbf{M} - \bar{\mathbf{M}}$ satisfies $\|\mathbf{E}\| \leq \bar{\sigma}_r - \bar{\sigma}_{r+1}$, then with the projection matrices $\mathbf{P}_\mathbf{U} := \mathbf{U}\mathbf{U}^\top$, one has

$$\max\{\|\mathbf{P}_\mathbf{U} - \mathbf{P}_{\bar{\mathbf{U}}}\|, \|\mathbf{P}_\mathbf{V} - \mathbf{P}_{\bar{\mathbf{V}}}\|\} \leq \frac{\sqrt{2} \max\{\|\mathbf{E}^\top \bar{\mathbf{U}}\|, \|\mathbf{E} \bar{\mathbf{V}}\|\}}{\bar{\sigma}_r - \bar{\sigma}_{r+1} - \|\mathbf{E}\|}.$$

In particular, if $\|\mathbf{E}\| \leq (1 - 1/\sqrt{2})(\bar{\sigma}_r - \bar{\sigma}_{r+1})$, then one has

$$\max\{\|\mathbf{P}_\mathbf{U} - \mathbf{P}_{\bar{\mathbf{U}}}\|, \|\mathbf{P}_\mathbf{V} - \mathbf{P}_{\bar{\mathbf{V}}}\|\} \leq \frac{\sqrt{2}\|\mathbf{E}\|}{\bar{\sigma}_r - \bar{\sigma}_{r+1}}.$$

D.3 Concentration inequalities

Lemma D.3.1 (Markov's inequality, [195, Proposition 1.2.4]). *For any non-negative random variable X and $t > 0$, we have*

$$\mathbb{P}(X > t) \leq \mathbb{E}(X)/t.$$

Lemma D.3.2 (Hoeffding's inequality, [195, Theorem 2.2.6]). *Let $X_1, \dots, X_N$ be independent random variables with $X_i \in [a_i, b_i]$, then for any, $t \geq 0$, we have*

$$\mathbb{P}\left(\left|\sum_{i=1}^N (X_i - \mathbb{E}X_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^N (b_i - a_i)^2}\right).$$

Lemma D.3.3 (Chernoff's inequality, [195, Theorem 2.3.1]). *Let X_i be independent Bernoulli*

random variables with parameters p_i . Consider their sum $S_N = \sum_{i=1}^N X_i$ and denote its mean by $\mu = \mathbb{E}S_N$. Then for any $t > \mu$,

$$\mathbb{P}(S_N \geq t) \leq e^{-\mu} \left(\frac{e\mu}{t} \right)^t.$$

Lemma D.3.4 (Bernstein's inequality, [195, Theorem 2.8.4]). *Let $X_1, \dots, X_N$ be independent mean-zero random variables such that $|X_i| \leq K$ for all i . Let $\sigma^2 = \sum_{i=1}^N \mathbb{E}X_i^2$. Then for every $t \geq 0$,*

$$\mathbb{P}\left(\left|\sum_{i=1}^N X_i\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2/2}{\sigma^2 + Kt/3}\right).$$

Lemma D.3.5 (Bennett's inequality, [195, Theorem 2.9.2]). *Let $X_1, \dots, X_N$ be independent random variables. Assume that $|X_i - \mathbb{E}X_i| \leq K$ almost surely for every i . Then for any $t > 0$, we have*

$$\mathbb{P}\left(\sum_{i=1}^N (X_i - \mathbb{E}X_i) \geq t\right) \leq \exp\left(-\frac{\sigma^2}{K^2} \cdot h\left(\frac{Kt}{\sigma^2}\right)\right),$$

where $\sigma^2 = \sum_{i=1}^N \text{Var}(X_i)$ is the variance of the sum and $h(u) := (1+u) \log(1+u) - u$.

Lemma D.3.6 (Variants of Chernoff). *Let X_i be independent Bernoulli random variable with mean p_i . Consider their sum $S_N := \sum_{i=1}^N X_i$ and denote its mean by $\mu := \mathbb{E}S_N$.*

1. *Let $\mu = \gamma \cdot q_N$ for some $\gamma > 0$ and $q_N \gg 1$, then for any $\zeta > \gamma$, we have*

$$\mathbb{P}(S_N \geq \zeta q_N) \leq \exp\left(-q_N \cdot [(\gamma - \zeta) + \zeta \log(\zeta/\gamma)]\right)$$

Specifically, $\mathbb{P}(S_N \geq t) \leq N^{-[\zeta \log \frac{\zeta}{\gamma} + (\gamma - \zeta)]}$ when $q_N = \log(N)$.

2. *If $\mu = \gamma_N \cdot q_N$ for some vanishing sequence $\{\gamma_N\}_{N \geq 1}$ with $\gamma_N = o(1)$, then for any vanishing*

sequence $\{\zeta_N\}_{N \geq 1}$ with $\zeta_N = \omega(\gamma_N)$, we have

$$\mathbb{P}(S_N \geq \zeta_N \cdot q_N) \leq \exp(-q_N \cdot [\zeta_N \log \frac{\zeta_N}{\gamma_N} (1 + o(1))]).$$

Proof of Lemma D.3.6. Chernoff Lemma D.3.3 gives

$$\mathbb{P}(S_N \geq t) \leq e^{-\mu} \left(\frac{e\mu}{t} \right)^t = \exp \left(-(\mu - t) - t \log \left(\frac{t}{\mu} \right) \right)$$

Let $f(t) = t \log(\frac{t}{\mu}) + (\mu - t)$, where $f(\mu) = 0$ and $f'(t) = \log(\frac{t}{\mu}) > 0$ for $t > \mu$, thus $f(t) > 0$ for all $t > \mu$.

1. By taking $t = \zeta q_N$, we have

$$\mathbb{P}(S_N \geq t) \leq \exp \left(-(\gamma - \zeta)q_N - \zeta \log(\zeta/\gamma) \cdot q_N \right).$$

2. Note that $\gamma_N = o(1)$, $\zeta_N = o(1)$ but $\frac{\zeta_N}{\gamma_N} = \omega(1)$, then $\frac{\gamma_N - \zeta_N}{\zeta_N \log(\zeta_N/\gamma_N)} = o(1)$, hence follows.

□

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